COMPENDIUM ON
5G USE CASE LAB
DATA TRANSFER

TECHNOLOGY

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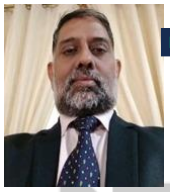

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# Message from Director General Telecom

I extend my sincere congratulations to the officers and staff associated with the compilation and publication of this Compendium of important circulars on 5G Use Case Lab.

The consolidation of various circulars issued from time to time into a single, well-structured Compendium is a significant and timely initiative. It will greatly facilitate ease of reference, ensure uniform understanding, and support effective implementation of policies and instructions across the organisation.

Such an effort reflects institutional foresight and a commitment towards strengthening administrative efficiency, transparency, and informed decision-making. I appreciate the meticulous efforts, coordination, and dedication of all those involved in compiling and presenting the circulars in a systematic and user-friendly manner.

I am confident that this Compendium will serve as a valuable reference document for officers and stakeholders and will contribute meaningfully to the effective discharge of official responsibilities.

I convey my best wishes to the team for continued success in their future endeavours.

**Ramesh Kumar Goyal**
**Director General Telecom**

# Message from Addl. Director General Telecom

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Dear Readers

I am happy to note that Technology Unit of DGT HQ and Maharashtra LSA under the able guidance of the Head of Haryana LSA has prepared a comprehensive compendium on 5G Use Case Lab.

5G Use Case Labs represent a critical step in harnessing next-generation connectivity for national development. The aim of 100 5G Use Case Labs is also to tap young minds at prominent technical institutes to involve themselves in testing, developing useful use cases for citizens. These labs function as collaborative innovation hubs where ideas are transformed into tested and scalable solutions. The unique capabilities of 5G- high bandwidth, ultra-low latency, and massive connectivity- enable breakthrough applications. Sectors such as healthcare, agriculture, education, manufacturing, and smart governance stand to benefit significantly. The labs promote indigenous innovation while strengthening the startup and MSME ecosystem.

I commend the team for their meticulous effort in collating this resource, which will serve as a useful reference. It is imperative that this compendium is continually updated and serve as a guiding beacon for 5G Use Case Lab related endeavours.

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My best wishes to the editorial team for these excellent initiatives.

**Suresh Puri**
**Addl. Director General Telecom**

# CONTENT


- Introduction
- Objective and Scope
- Institution deliverables
- Funding Support
- Roles & Responsibilities of Beneficiary Institutions
- Role of LSA
- List of 100 5G Use case Labs in India
- Installation and User Acceptance Testing (UAT)
- Training
- Gradation Framework
- 5G Innovation Hackathon 2025
- Annexure: Relevant Circulars and Documents for 5G Use Case Lab


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# 1.Introduction

The Department of Telecommunications has undertaken a landmark initiative to establish 100 5G Use Case Labs across premier educational and technical institutions in the country, guided by the visionary leadership of the Hon'ble Prime Minister of India, Shri Narendra Modi. This initiative reflects the national vision of leveraging next-generation technologies to drive innovation, foster inclusive growth, and position India as a global digital leader.

These 5G Use Case Labs were installed with the objective of creating a robust ecosystem for hands-on learning, research, and innovation in 5G technologies. The labs have been set up in IITs and other leading academic institutions across various States and Union Territories, enabling widespread geographic and institutional participation.

**Vision Statement**

*To empower students, researchers, startups, and innovators with advanced 5G capabilities, foster indigenous development of next-generation applications, and accelerate the realization of an Atmanirbhar, digitally empowered India.*

These labs serve as centres of experimentation and co-creation, enabling the development, testing, and validation of 5G-enabled applications across key socio-economic sectors such as education, agriculture, healthcare, power, smart cities, and urban governance. By bridging academia, startups, industry, and telecom service providers, the initiative aims to convert academic research into real-world, scalable solutions.

The 100 5G Use Case Labs are also envisioned as catalysts for faster 5G adoption and monetization, supporting the identification of innovative and commercially viable use cases. Collectively, this initiative strengthens India's innovation infrastructure, builds future-ready human capital, and positions the nation at the forefront of the global 5G ecosystem.

## 2. Objective & Scope

The objectives of the programme are to build competencies and foster sustained engagement in 5G technologies among students and the academic fraternity, and to enable undergraduate and postgraduate level projects through access to a real-time 5G testing and experimentation environment. The programme seeks to promote academia–industry collaboration for ideation, development, and validation of innovative 5G use cases, while also providing local and shared access to 5G test infrastructure for startups and MSMEs in the vicinity of the host institutions. Collectively, these efforts aim to strengthen the Indian academic and startup ecosystem and ensure its preparedness for future 6G technologies.

The scope of activities under the programme includes selection of eligible institutions and empanelment of domestic vendors through an Expression of Interest (EoI), followed by floating of RFPs/tenders, selection of vendors through competitive bidding, and price discovery for CAPEX and OPEX components. It encompasses coordination for installation and commissioning of 5G Use Case Labs at more than 100 institutions, along with oversight and coordination of operations, maintenance, and effective utilization of these labs for a period of five years.

The programme also provides for hosting and management of a dedicated digital portal for calling proposals, performance monitoring, remote O&M support, and release of financial instalments. It includes development of a digital networking platform to enable collaboration and optimal utilization of the labs, as well as support to institutions for organizing hackathons. Further, the scope covers creation of a knowledge dissemination platform to share learnings and outcomes with other institutions, students, startups, and stakeholders on 5G use cases being tested, developed, or demonstrated. Additionally, it envisages the creation and maintenance of a comprehensive database of domestic 5G vendors, 5G devices

(including IoT), and successful 5G projects and pilots across sectors such as healthcare, agriculture, intelligent transport systems, and Industry 4.0.

**<No. 1-TR_5G_Startup_Kit/2022-SRI Dated: 27thMarch, 2023>**
**<No. 1-TR_5G_Startup_Kit/2022-SRI-Part(1) Dated: 24thMay, 2023>**
**<No. 1-TR_5G_Startup_Kit/2022-SRI Dated: 23rdJuly, 2023>**
**<No. 1-5GLabsPMU/2023-SRI Dated: 30th September 2023>**
**<No. 1-TR_5G_Startup_Kit/2022-SRI Dated: 4thDec, 2023>**

SRI Division, DoT-HQ has been designated as Project Management Unit (PMU) and TCIL as the Implementing Agency (IA). TCIL has awarded the supply and installation work to indigenous vendors through a competitive bidding process and Zonal/State-Level 5G workshops are also organized to facilitate capacity building.

## 3. Institution Deliverables

I. At least 50 students and 10 faculty members shall utilize the lab each year for testing and project development.
II. The lab shall also engage a minimum of five startups/MSMEs annually for testing and product development activities.
III. Further, at least ten products, solutions, or intellectual properties (IPs) shall be developed and tested in a year.
IV. Efforts shall be made to contribute to the standardization process and to drive the technologies developed or under development towards adoption as part of relevant standards. Additionally, focused efforts shall be undertaken to pilot, commercialize, and deploy the tested or developed products in real-world field environments.

## 4. Funding Support

The Government shall fund 80% of the CAPEX for the 5G Use Case Lab setup and 100% of the OPEX for a period of four years. The host institution shall contribute the remaining 20% of the CAPEX, as determined through price discovery based on the financial bids.

## 5. Roles and Responsibilities of Institutions

As part of the programme, the beneficiary institution shall provide all necessary facilities, including adequate space, power supply, internet/intranet connectivity, required equipment, and technical manpower for local maintenance, to ensure successful implementation and operation of the 5G Use Case Lab. The institution shall deposit its prescribed financial contribution with the Implementing Agency (IA) within three months of final selection; shipment of equipment shall commence only after receipt of the institutional contribution.

The assets procured under the programme shall vest with the host beneficiary institution, which shall be responsible for the safety, security, and effective utilization of the lab by students, faculty, and the startup/MSME ecosystem, with necessary facilitation support from the IA. The institution may enter into separate arrangements with the IA and/or vendors for procurement of additional equipment—such as devices, network elements, and use-case-specific accessories—to enhance the lab's functionality.

Further, the institution shall ensure regular connectivity and data updates to the IA's portal regarding lab utilization and shall undertake value addition and knowledge dissemination related to 5G use cases being tested or developed, for the benefit of

other institutions, students, and startups. The institution shall also host hackathons, as required, with support from the IA on a need basis.

## 6. Role of LSAs

| SN | Activity | Role of LSA |
|---|---|---|
| 1 | Agreement with Institutions (by TCIL) | Facilitation. |
| 2 | Installation/Commissioning – 100 5G Labs | Facilitation, due observation and necessary intervention. |
| 3 | User Acceptance Testing – 100 5G Labs | Part of testing team |
| 4 | Hackathon (LSA Wise/State Wise)- 100 5G Labs | Coordination with stakeholders and extending necessary support including publicity & logistics |
| 5 | Workshops/Trainings for Institutions- 100 5G Labs | Organizing the workshops at LSA/State level and coordination with NTIPRIT for zonal level workshops including logistics support |
| 6 | Capacity Building sessions for LSAs by SRI | Proactive participation |
| 7 | Mentoring/Adoption of Institutes by OEMs/TSPs/CoEs | Coordinate with Institution, TCOE and OEMs/TSPs/CoEs |
| 8 | ITU Academia Membership | Encouraging India's academia membership in ITU |
| 9 | Sanchar Mitra Scheme/Sanchar Saathi volunteers | i. Selection of students in Coordination with Institutions.<br>ii. Guidance for activities of Sanchar Mitra volunteers in co-ordination with Institutions, LEAs, State-gov departments etc |
| 10 | SPoC for Institutions | Updating the List of Nodal officers |

| | | |
|---|---|---|
| 11 | Updating the repositories in Bharat5G/6G portal | Updating the repositories in Bharat5G/6G portal.<br>#R&D activities, Projects, Infra, thesis/abstract from Academia<br>#Use cases, products from Startups/MSMEs<br>#IPRs and standardization from Academia/Start-ups/MSME<br>#Schemes/Programs from State Agencies, Incubation Centres/ Centre of Excellence. |
| 12 | Telecom Start-ups & MSMEs (TSUM) Facilitation Cell (State gov departments, customs etc. | #State/LSA level TSUM facilitation cell in co-ordination with DoT HQ for IPR facilitation, PoC testing & certification, establishing sandbox environments, etc.<br>#Creating awareness regarding DoT initiatives/funding Schemes/ Programs- TTDF, DCIS, 5G Hackathons, 100 5G labs, 6G R&D proposals, Experts registration etc.<br>#Periodic meetings with representatives of state government, State-PSUs, VCs and other key bodies. |
| 13 | Support for 5G/4G/IoT Projects (including CNPN) | Co-ordination for the following<br>#Rural zones- 5G Model Village etc.<br>#Urban/Industrial zones- CNPN networks (Power, Mining, Ports & Airports, Factories etc.) |
| 14 | 5G Test Beds Utilisation | Facilitation & awareness for utilization of 5G test beds |
| 15 | Monthly Report | Periodic submission of progress cum action taken reports through portal |

**<No. 1-Role of LSAs (100 5G labs)/2024-SRI Dated: 15th Feb, 2024>**

## 7. List of 100 5G Use Case Labs

| Sr No. | Name of Institutes | State/UT |
|---|---|---|
| 1 | Indian Institute of Information Technology Design and Manufacturing Kurnool | Andhra Pradesh |
| 2 | Koneru Lakshmaiah Education Foundation | Andhra Pradesh |
| 3 | Vignans Foundation for Science Technology and Research | Andhra Pradesh |
| 4 | GITAM Institute of Technology and Management, Deemed to be University | Andhra Pradesh |
| 5 | Indian Institute of Technology Tirupati | Andhra Pradesh |
| 6 | National Institute of Technology Arunachal Pradesh | Arunachal Pradesh |
| 7 | Indian Institute of Technology Guwahati | Assam |
| 8 | National Institute of Technology Silchar | Assam |
| 9 | National Institute of Technology Patna | Bihar |
| 10 | Indian Institute of Technology Patna | Bihar |
| 11 | Dr Rajendra Prasad Central Agricultural University | Bihar |
| 12 | Punjab Engineering College Deemed to be University | Chandigarh |
| 13 | University Institute of Engineering and Technology Panjab University | Chandigarh |
| 14 | IIT Bhilai | Chhattisgarh |
| 15 | National Institute of Technology Raipur | Chhattisgarh |
| 16 | Delhi Technological University | Delhi |
| 17 | Indraprastha Institute of Information Technology Delhi | Delhi |
| 18 | Jamia Hamdard | Delhi |
| 19 | Indian Agricultural Research Institute | Delhi |

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|---|---|---|
| 20 | Birla Institute of Technology and Science, K. K. Birla Goa Campus | Goa |
| 21 | National Institute of Technology Goa | Goa |
| 22 | Sardar Vallabhbhai National Institute of Technology | Gujarat |
| 23 | School of Cyber Security and Digital Forensics National Forensic Sciences University Institute of National Importance Ministry of Home Affairs GoI | Gujarat |
| 24 | Indian Institute of Technology Gandhinagar | Gujarat |
| 25 | Central University of Haryana | Haryana |
| 26 | National Power Training Institute | Haryana |
| 27 | National Institute of Technology Hamirpur | Himachal Pradesh |
| 28 | Indian Institute of Technology Mandi | Himachal Pradesh |
| 29 | Institute of Technology, University of Kashmir | Jammu and Kashmir |
| 30 | Indian Institute of Technology Jammu | Jammu and Kashmir |
| 31 | Birla Institute Of Technology, Mesra | Jharkhand |
| 32 | Indian Institute of Technology, Dhanbad. | Jharkhand |
| 33 | National Institute of Technology Jamshedpur | Jharkhand |
| 34 | Siddaganga Institute of Technology | Karnataka |
| 35 | Indian Institute of Information Technology Dharwad | Karnataka |
| 36 | BMS College of Engineering | Karnataka |
| 37 | Indian Institute of Science | Karnataka |
| 38 | VTUS Centre for Post Graduate Studies ,Kalaburgi | Karnataka |
| 39 | MANIPAL INSTITUTE OF TECHNOLOGY | Karnataka |
| 40 | M. S Ramaiah Institute of Technology | Karnataka |
| 41 | National Institute of Technology Karnataka, Surathkal | Karnataka |
| 42 | Department of Electronics, Cochin University of Science and Technology | Kerala |
| 43 | National Institute of Technology Calicut | Kerala |
| 44 | Indian Institute of Space science and Technology | Kerala |
| 45 | Indian Institute of Technology Palakkad | Kerala |
| 46 | PDPM Indian Institute of Information Technology, Desing and Manufacturing Jabalpur | Madhya Pradesh |
| 47 | Indian Institute of Technology Indore | Madhya Pradesh |

| | | |
|---|---|---|
| 48 | Atal Bihari Vajpayee Indian Institute of Information Technology and Management, Gwalior, Madhya Pradesh | Madhya Pradesh |
| 49 | Maulana Azad National Institute of Technology, Bhopal, M.P. | Madhya Pradesh |
| 50 | DPU, Dr D Y Patil School of Science and Technology | Maharashtra |
| 51 | Symbiosis Institute of Technology, Pune | Maharashtra |
| 52 | Visvesvaraya National Institute of Technology, Nagpur | Maharashtra |
| 53 | Savitribai Phule Pune University, MS and Dr. Babasaheb Ambedkar Technological University, Lonere, MS Joint Proposal by 2 Universities | Maharashtra |
| 54 | COEP Technological University | Maharashtra |
| 55 | Government College of Pharmacy Amravati | Maharashtra |
| 56 | DKTEs Textile and Engineering Institute, Ichalkaranji | Maharashtra |
| 57 | National Institute of Technology Manipur | Manipur |
| 58 | Indian Institute of Information Technology Senapati, Manipur | Manipur |
| 59 | National Institute of Technology Meghalaya | Meghalaya |
| 60 | National Institute of Technology Nagaland | Nagaland |
| 61 | National Institute of Technology Mizoram | Mizoram |
| 62 | Christian Institute of Health Sciences and Research | Nagaland |
| 63 | Indian Institute of Technology Bhubaneswar | Odisha |
| 64 | Siksha O Anusandhan Deemed to be University | Odisha |
| 65 | Kalinga Institute of Industrial Technology Deemed to be University | Odisha |
| 66 | National Institute of Technology Rourkela | Odisha |
| 67 | National Institute of Technology Puducherry | Puducherry |
| 68 | Pondicherry University | Puducherry |
| 69 | Dr B R Ambedkar NIT Jalandhar | Punjab |
| 70 | Thapar Institute of Engineering and Technology | Punjab |
| 71 | Guru Nanak Dev University, Amritsar | Punjab |
| 72 | Indian Institute of Technology Ropar | Punjab |
| 73 | Banasthali Vidyapith | Rajasthan |
| 74 | Birla Institute of Technology and Science Pilani | Rajasthan |

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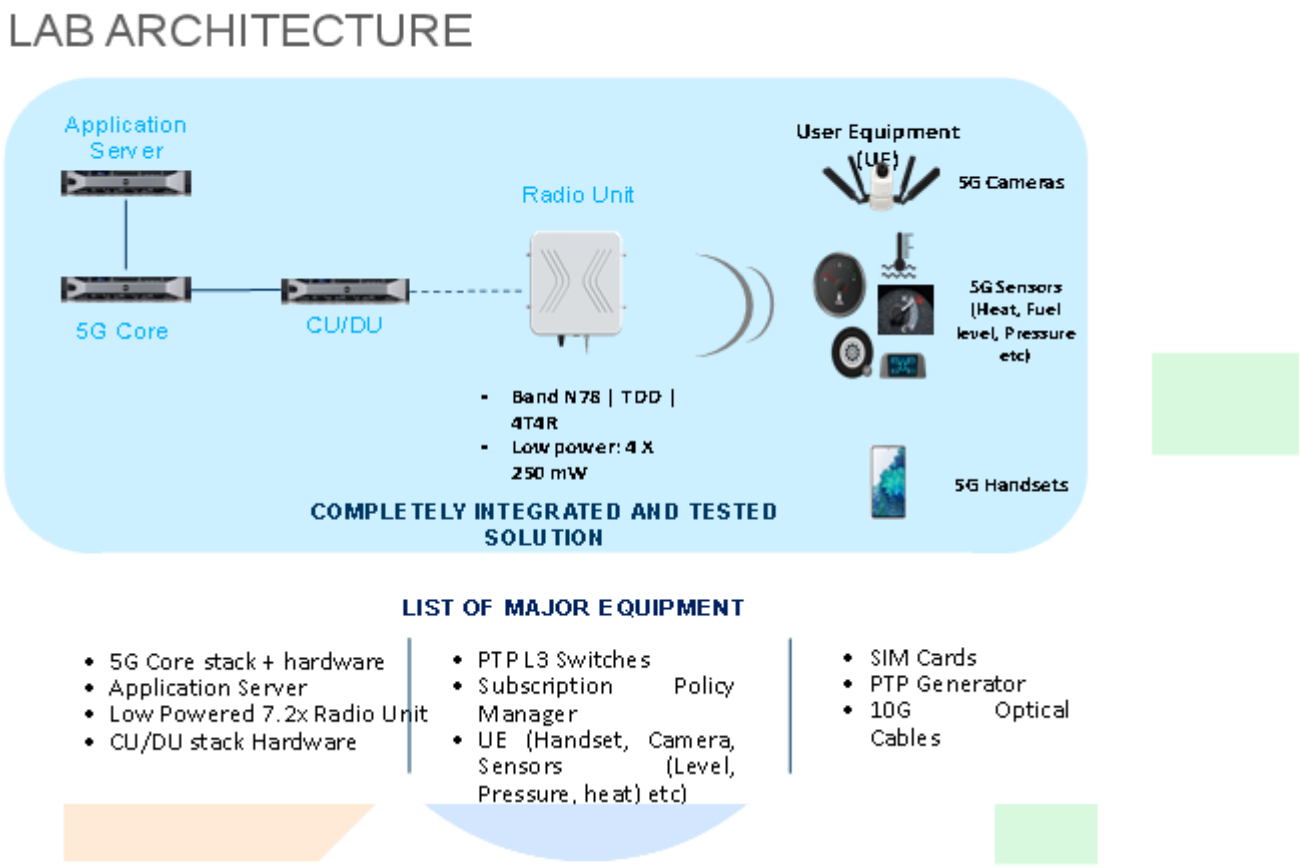
|---|---|---|
| 75 | Indian Institute of Technology Jodhpur | Rajasthan |
| 76 | Malaviya National Institute of Technology Jaipur | Rajasthan |
| 77 | MBM University, Jodhpur | Rajasthan |
| 78 | Sikkim Manipal Institute of Technology | Sikkim |
| 79 | National Institute of Technology Tiruchirappalli | Tamil Nadu |
| 80 | Vellore Institute of Technology | Tamil Nadu |
| 81 | College Of Engineering Guindy | Tamil Nadu |
| 82 | SASTRA Deemed University | Tamil Nadu |
| 83 | Sri Sivasubramaniya Nadar College of Engineering | Tamil Nadu |
| 84 | Saveetha Institute of Medical and Technical Sciences | Tamil Nadu |
| 85 | SRM Institute of Science and Technology | Tamil Nadu |
| 86 | IIIT Hyderabad | Telangana |
| 87 | National Institute of Technology Warangal | Telangana |
| 88 | University of Hyderabad | Telangana |
| 89 | Tripura Institute of Technology | Tripura |
| 90 | Jaypee Institute of Information Technology, Noida | Uttar Pradesh |
| 91 | Motilal Nehru National Institute of Technology Allahabad | Uttar Pradesh |
| 92 | Indian Institute of Information Technology Allahabad | Uttar Pradesh |
| 93 | Aligarh Muslim University | Uttar Pradesh |
| 94 | Indian Institute of Technology Kanpur | Uttar Pradesh |
| 95 | Indian Institute of Technology BHU | Uttar Pradesh |
| 96 | College of Technology, Govind Ballabh Pant University of Agriculture and Technology | Uttarakhand |
| 97 | THDC Institute of Hydropower Engineering and Technology | Uttarakhand |
| 98 | Indian Institute of Technology Roorkee | Uttarakhand |
| 99 | National Institute of Technology Durgapur | West Bengal |
| 100 | Indian Institute of Information Technology, Kalyani | West Bengal |

# 8. Installation and User Acceptance Testing (UAT)

The Lab Architecture of 5G Use case lab is as shown below.




Following equipment were supplied by vendor: -

| Sr No | Item description |
|---|---|
| 1 | 5G Core |
| 2 | 5G Radio |
| 3 | IMS Solution |
| 4 | MEC & Application Server |
| 5 | NMS with Dashboard |
| 6 | Router with Firewall |
| 7 | 5G SIMs |
| 8 | 5G Evaluation Board/ Hardware and Software Development Kit. |
| 9 | IoT Gateway |

| | |
|---|---|
| 10 | IoT sensors with analytics software |
| 11 | Sensors: -<br>(a) Temperature & Humidity sensor<br>(b) Light sensor<br>(c) Soil sensor (NPK),<br>(d) Water (TDS/ chlorine) sensor |
| 12 | 5GMini Drone |
| 13 | 5G Camera with AI enabled Video Analytics (Face/Object/ Motion detection, people counting etc.) |
| 14 | 5G XR(AR/VR) Device |
| 15 | 5G Indoor CPE |
| 16 | 5G Handsets |
| 17 | Adjustable Tripod |
| 18 | 24 U Rack |
| 19 | UPS |
| 20 | FHD Display |
| 21 | L2 Managed Switch |

**<File No. P-11014/02/2020-PP Date: 04.04.2024>**

## 9. Training

Training Manual on 5G Use Case Labs created by vendors:

1. W4S
2. Signaltron
3. VVDN
4. Coral Telecom

5G Use Case Labs Manual developed by UPW LSA.

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## 10. Gradation Framework

To ensure the effective utilization of 5G use case labs a gradation framework with 11 key parameters has been prepared. The labs will be evaluated on basis on the following Gradation parameters:

**<No:- BTU/5G use case lab/01 dated 2.1.2025>**

| Sl No. | Parameter | Target | Score |
|---|---|---|---|
| 1 | No of Students utilized the lab for product testing/ applications | 50 unique Students/ year | • 1-10 students –1 point<br>• 11 to 30 students – 2 points<br>• 31 to 50 students – 4 points<br>• >50 students – 5 points |
| 2 | No of Faculty utilized the lab for mentoring/development/testing etc. | 10 unique faculty /year | • 1 faculty – 1point<br>• 2 to 5 faculty – 2 points<br>• 6 to 10 faculty– 4 points<br>• >10 faculty – 5 points |
| 3 | No of Projects utilizing the lab | 10 projects / year | • 1 project – 1point<br>• 2 to 5 projects – 2 points<br>• 6 to 10 projects– 4 points<br>• >10 projects – 5 points |

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|---|---|---|---|
| 4 | No. of Startups/MSMEs engaged in the lab for testing/product development | 5 Startups/ MSMEs / year | 1 point per Startup/MSME (max points – 10) |
| 5 | Engagement with Atal Tinkering Labs/ Schools/ colleges and other Academic Institutions (Workshop/Mentorship etc.) | 5 Institutions/ year | 1 point per institution (max points – 10) |
| 6 | No. of Companies other than Startups/MSME/Academic Institutions) engaged in the lab for testing/product development | 5 Companies or Organizations/ year | 1 point per Company/Organization (max points – 10) |
| 7 | No of products/solutions/use case developed | 2 Products/ solutions /use cases developed | 10 points per product/solution/use case (max points – 30) |
| 8 | No of IP registered | 2 IPs in a year | 10 points /IP (max points – 30) |
| 9 | No. of developed products/use cases in the lab commercialized/ piloted/deployed | 1 Product/ year | 15 points for each product (max points – 30 points) |
| 10 | Contributions related to 5G & beyond Standards in international journals/ ITU/ 3GPP/ ISO/IEEE etc. | 2 contributions/ year | 5 points for each contribution (max points – 30) |
| 11 | Including 5G related academic course utilising the Lab in the curriculum | Completion of the course for 50 students | 2 point/10 student (max points-20) |

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**Labs will be assessed as follows based on Gradation Parameters**

| Parameter | Minimum Marks required |
|---|---|
| 1 to 4 | 18 out of 25 |
| 5 | 5 out of 10 |
| 6 to 11 | 57 out of 150 |
| Total | 130 out of 185 |

**Top 5 ranking Institutes to be felicitated every year at IMC or any other occasion**

| Sr No | Ranking | Points |
|---|---|---|
| 1 | Excellent | >130 points |
| 2 | Very Good | 101-130 points |
| 3 | Satisfactory | <80-100 points |

All engagements and outcomes will be shared and made accessible through the https://bharat5glabs.gov.in portal. The portal is made available to institutes and LSAs from 29/08/2025 and credentials have already been shared with LSAs and 100 5G Use case labs by TCIL.

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**Hon’ble PM awarding the 100 5G use Case labs to the Nation at IMC 2023**

**Shri Rohit Sharma, Member Services, DOT, along with Shri Anil Verma, Special DGT, Maharashtra LSA, Shri Vikram Malviya DDG(Technology) and other senior officers of DOT visited 5G Test Lab for a live demonstration at COEP Technological University on 17th August-2024.**

**Shri RK Goyal, Special DGT, Maharashtra LSA, Shri Gautam Kumar, DDG (Admn), Shri Ujwal Gulhane DDG(Technology) and other senior officers of DOT visited 5G Test Lab for a live demonstration at Symbiosis Technological Institute on 30th January-2025.**

**Recognising innovation in action — Shri Jyotiraditya Scindia, Hon'ble Minister of Communications, and Dr. Chandra Sekhar Pemmasani, Hon'ble Minister of State for Communications honouring the Best Performing 5G Use Case Labs under the Gradation Framework**

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**Shri Jyotiraditya Scindia, Hon'ble Minister of Communications, and Dr. Chandra Sekhar Pemmasani, Hon'ble Minister of State for Communications, launching the 5G Hackathon Booklet — fostering innovation and empowering India's digital future**

## 11. 5G Innovation Hackathon 2025

The 5G Innovation Hackathon 2025 aims to bring together students, startups, and professionals to develop cutting-edge solutions using 5G labs in 100 institutes across India. The Hackathon will facilitate solution development in key technology areas such as AI-driven network maintenance, IoT-enabled applications, smart healthcare, industrial automation, quantum communication, and more. The hackathon provides a collaborative platform to the stakeholders to innovate and develop prototypes/applications/solutions through access to 5G labs, mentorship from industry and academic experts during the hackathon, provision of seed funding to the shortlisted proposals, and support for commercialization.

**<5G Innovation Hackathon – Winners' Compendium prepared by DoT HQ>**
**<No. 1-TR_5G_Startup_Kit/2022-SRI-Part(4) Dated: 13th March 2025>**

## 12. Annexure: Relevant Circulars and Documents

| S No | Document Title/Subject | Date |
|---|---|---|
| 1 | Setting up of 100 5G Use case labs under TDIP Scheme No. 1-TR_5G_Startup_Kit/2022-SRI Dated: 27th March, 2023 | 27th Mar, 2023 |
| 2 | Update on 100 5G use case labs Project (Budget Announcement 2023-24) No. 1-TR_5G_Startup_Kit/2022-SRI-Part(1) Dated: 24th May, 2023 | 24th May, 2023 |
| 3 | Modified SOP for setting up of 100+ 5G Use case labs under TDIP Scheme 1-TR_5G_Startup_Kit/2022-SRI Dated: 23rd July, 2023 | 23rd Jul, 2023 |
| 4 | MoM of Technical Evaluation Committee chaired by Sr.DDG TEC regarding 100 5G labs RFP floated by TCIL No. 1-5GLabsPMU/2023-SRI Dated: 30.09.2023 | 30th Sep, 2023 |
| 5 | 100 5G Labs Project by DoT through TCIL under TDIP scheme – Budgetary outlay, payment structure, needful compliance by Implementation Agency(TCIL) for timely installation and optimal utilization of 5G Labs-reg 1-TR_5G_Startup_Kit/2022-SRI Dated: 4th Dec, 2023 | 4th Dec, 2023 |
| 6 | Role of DoT field units(LSAs) reg 100 5G Labs project & Beyond 1-Role of LSAs(100 5G labs)/2024-SRI Dated: 15th Feb, 2024 | 15th Feb, 2024 |
| 7 | Simplification in the procedure for obtaining Experimental Licence for 100 5G Use Case Labs –Reg File No. P-11014/02/2020-PP Date: 04.04.2024: Procedure for obtaining Experimental Licence may be hosted on DGT Webpage and link may be provided | 4th Apr, 2024 |

| | | |
|---|---|---|
| 8 | Collection and sending of regular information about Innovations and new Use Cases on 5G technology developed by the select Institutions/Startups to NTIPRIT No.: 1-20/2022-NTI.WA dated 12.07.2024 | 12th Jul, 2024 |
| 9 | 100 5G labs for higher educational institutions user acceptance test (UAT Schedule) | |
| 10 | Training Manual on 5G Use Case Labs:<br>1. W4S<br>2. Signaltron<br>3. VVDN<br>4. Coral Telecom<br>5. UPW LSA | |
| 11 | 5G Innovation Hackathon – Winners' Compendium | |
| 12 | Gradation framework for 5G lab Use case vide letter No:-BTU/5G use case lab/01 dated 2.1.2025 | 2nd Jan, 2025 |
| 13 | Launch of 5G Innovation Hackathon 2025 – Coordination and Implementation regarding No. 1-TR_5G_Startup_Kit/2022-SRI-Part(4) Dated: 13th March 2025 | 13th Mar, 2025 |
| 14 | Implementation of 5G Innovation Hackathon 2025 File No: 49-2/2022-DGT/4 Dated 20th Mar 2025 | 20th Mar, 2025 |
| 15 | Constitution of Regional Evaluation Committee for 5G Innovation Hackathon 2025 Dated: 05-05-2025 | 5th May, 2025 |

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## Disclaimer

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सत्यमेव जयते

*The Make in India logo is derived from India's national emblem. The wheel denotes peaceful progress and dynamism – a sign from India's enlightened past, pointing the way to a vibrant future. The prowling lion stands for strength, courage, tenacity and wisdom – values that are every bit as Indian today as they have ever been.*

I/3092946/2023

**Department of Telecommunications**
Standards-R&D-Innovation
(*Project Monitoring Unit*)
****

Sanchar Bhavan, 20 Ashoka Road,
New Delhi-110001

**No.** 1-TR_5G_Startup_Kit/2022-SRI **Dated:** 27th March, 2023

## OFFICE MEMORANDUM

**Sub:** Setting up of 100 5G Use case labs under TDIP Scheme

In line with the budget announcement, one hundred 5G use case labs will be deployed by DoT through M/s TCIL as the implementation agency (IA). Two committees are being constituted for the section of institutes and technical evaluation headed by DG (NTIPRIT) (with a representative from Dept. of Higher Education) and Sr. DDG(TEC) respectively. A Project Review & Steering Group (PRSG) is being constituted headed by Additional Secretary(T) for a periodic review of the project.

During the first meeting taken by Member (Technology), DCC, the CMD, TCIL has been requested to chart out necessary actions on priority and respective implementation guidelines were shared. Further, to enable effective coordination of the project, the following nodal officers from respective organizations are identified.

| S.No | Organization & Remarks | Nodal officer |
|---|---|---|
| 1. | TCIL<br>(Implementation Agency) | ED (IT&T-2), Ms. Alka Asthana,<br>Director (P), Sh. Arun Kumar Chaubey,<br>Sh. Sanjay Kumar, GM (Digital Transformation) |
| 2. | NTIPRIT -<br>(Selection Committee) | DDG (Wireless Access), NTIPRIT Sh. Ashok Kumar, |
| 3. | TEC- (Technical Evaluation Committee) | DDG (Mobile Tech), Sh. Abhay Shanker Verma, TEC |
| 4. | S-R-I, DoT HQ (PMU) | DDG (SRI), Sh. Y.G.S.C Kishore Babu,<br>Mission coordinator (TSUM)/ ADG(SRI) Sh. A Alex Vikas, |

Encl: Implementation guidelines

Signed by A Alex Vikas
Date: 27-03-2023 20:09:00
Reason: Approved

(A. Alex Vikas)
Mission Co-ordinator(TSUM)/ Asst. Director General
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**To:**
CMD (TCIL) / DG(NTIPRIT) / Sr. DDG(TEC)

**Copy to:**
1. PSO to Secretary (Telecom) & Chairman DCC - for kind information
2. Member(T)/Member(S)/Member(F)/AS(T)
3. ED (IT&T-2), TCIL/Director(P), TCIL
4. DDG(SRI), DoT HQ / DDG(MT), TEC/ DDG(WA), NTIPRIT
5. DoT Nodal for uploading OM in the DoT website & e-office

I/3092946/2023

# Implementation Guidelines for the establishment and maintenance of 100+ 5G Use case labs at Educational & Specialized Institutions/CoEs under TDIP Scheme

I/3092946/2023

CONTENTS:

1. OBJECTIVES & SCOPE : ....3

2. DELIVERABLES AND IMPACT....4

3. ELIGIBILITY CRITERION FOR INSTITUTIONS....4

4. EVALUATION CRITERION (INDICATIVE)....5

5. INDICATIVE BUDGETARY OUTLAY....5

6. FUNDING SUPPORT AND & INDICATIVE MILESTONES: ....6

7. ROLES & RESPONSIBILITIES OF BENEFICIARY INSTITUTION....7

8. ROLES & RESPONSIBILITIES OF EMPANELLED DOMESTIC VENDORS....8

9. 5G UCL ARCHITECTURE ....8

ANNEXURE-I: IMPLEMENTATION MECHANISM....12

**1.Implementation SOP....12**

**2. Implementing Agency ....13**

**3. Application Format and Procedure ....13**

**4. Project Monitoring Unit (PMU) ....13**

**5. Selection Committee to identify beneficiary institutions ....13**

**6. Empanelment Process of Vendors by IA....14**

**7. Project Review & Streeting Group (PRSG):....14**

**8. Fund Disbursement & Modalities:....15**

**9. Indicative Administrative Expenses of IA: ....16**

**10. Ownership: ....16**

ANNEXURE-II: TERMS & CONDITIONS FOR IA....17

ANNEXURE-III DRAFT TEMPLATE FOR UCL EQUIPMENT CONFIGURATION....18

# 1. Objectives & Scope :

**1.1 Objectives under the program are as under:**

i. To build competencies and engagement in 5G technologies in students and academic fraternity.
ii. To enable projects at under-graduation and post-graduation level for students using 5G environment.
iii. To Encourage academia-industry engagement to ideate and develop 5G use cases.
iv. To Provide local access to 5G test setup for Startups and MSMEs around the institution.
v. Making Indian academia & startup ecosystem **6G** ready

**1.2 Scope of activities under the program (Digital Network of 100 use case Labs)** :

i. Selection of Institutions & empanelment of domestic vendors (EoI)
ii. Floating of RFP/Tender, selection of vendors (competitive bidding) and price discovery for CAPEX & OPEX
iii. Installation coordination for setting up of Labs at selected institutions (100+)
iv. Oversight and coordination for O&M and utilization of labs(100+) for period of **5 years**
v. Hosting of dedicated portal reg call for proposals, performance monitoring, Remote O&M support, release of installments
vi. Digital platform for networking and utilization of labs,
vii. Supporting institution/institutions to host hackathons
viii. Knowledge disseminating platform for other institutions/students/startups etc., on use-cases being tested/developed/demonstrated.
ix. Database of 5G domestic vendors available, 5G devices (incl IoT), successful 5G projects/pilots related to Health, Agri, intelligent Transport systems, Industry 4.0 etc.

**Note:**

- The aforementioned scope of activates will be performed by single responsible agency in co-ordination with DoT, as per the implementation guidelines and will be termed as Implementation Agency (IA) here onwards.
- To promote proliferation of 5G use case labs with different configurations and flavors, IA may continue the partnership with empaneled vendors and other institutions to carry out similar projects (not limiting to the scope of the above project)

## 2. Deliverables and Impact

Deliverables of the program are as follows

**Program Deliverables for monitoring**

2.1 No. of Labs being established
2.2 No. of States/different geographical locations being covered

**Institution deliverables**

2.3 At least 50 no. of Students and 10 nos. faculty utilising the lab for testing/project development per year
2.4 At least 5 nos. of Start-ups/MSMEs engaged in the lab for testing/product development in a year
2.5 At least 10 no. of Products/solutions/IPs developed/tested in a year

2.6 Efforts to contribute in standardization process and drive the technologies developed/being developed to become part of the standards
2.7 Efforts to pilot/commercialise/deploy the tested/developed products got in field

*The deliverables at 2.4-2.7 are time taking considering the fact that use case development is a time taking process (ideation-development-standardization-deployment). The deliverables 2.6, 2.7 are included as aspirational targets to be monitored and respective outcomes would be captured.*

**Impact** envisaged is as under the objectives.

## 3. Eligibility Criterion for Institutions

The following Indian Entities are eligible to apply for Beneficiary Institutions (BI).

**3.1 Academic institutions** (Preferred NIRF Ranking and other aspects as mentioned in section 4. Evaluation criterion)

[*As an extension, the following categories may also be invited to host 5G Usecase labs as and when decided]*

**3.2 Other Institutions**

- *R&D institutions*
- *Centres of Excellence under Central Government/State Government / Government Agencies (incl. PPP mode)*
- *Sectoral specific institutions with domain expertise in Health, Agriculture, Transportation, Industry4.0, Fintech, Mining, Smart Cities etc. trying to leverage Emerging Technologies such as 5G/IoT/M2M* ]

## 4. Evaluation Criterion (Indicative)

Evaluation Criteria for applicants under the program may include the following:

4.1 Academic Institutions- NIRF Ranking [Research institutions (within 50/50), Engineering (within 100/200), Medical (within 25/50), Technology and Medical Universities (within 50/100), Overall ranking (50/100)] (Other disciplines are included considering the relevance of 5G usecases in sectors viz. Health, Agriculture, Education, industry4.0, Transportation etc.)

4.2 Geographic location, benefits to the local ecosystem, expertise to maintain the lab, projected utilization by no. of students/ Start-up/industry community etc.

4.3 Institution's readiness in providing necessary facilities such as: Space, power supply, other equipment, tech-manpower (for local maintenance) etc. for establishing, maintenance of the lab.

4.4 Institution's readiness and projections in the utilization of the Lab by student / faculty / start-up ecosystem.

4.5 In the case of CoEs and other institutions (Refer to section 3.II), the screening criterion may include Start-up/MSME ecosystem involved with the institution, projected utilization by Vertical specific industries/start-ups/academia, inputs from respective Government Ministries / Departments.

4.6 Not limiting to the above, Screening Committee may also consider any other aspect/parameter as deemed fit, in finalizing the list.

## 5. Indicative Budgetary Outlay

5.1 The objective is to set up 5G usecase labs with relevance across verticals. Hence, it requires 5G infrastructure, IT systems, Devices facilitating the development of use cases. The following table provides an indicative cost of the lab. The Implementing Agency (IA) may provide the final costing (+/- 25%) based on the availability and lab requirement and use cases as per the template (with necessary features) to establish the lab in turnkey basis.

*5.2 An initial estimated budget of Rs.60 crore is proposed under the TDIP scheme for 100 academic institutions for a period of 5 years (taking note of O&M period of 4 years). For other institutions (Refer to 3.II) cost may be different as applicable in view of respective vertical specializations.*

*Table 1*

| | **Type/Year** | **1** | **2** | **3** | **4** | **5** |
|---|---|---|---|---|---|---|
| **1** | Capital Cost in Rs Cr* | 40 | - | - | - | - |
| **2** | O&M Cost in Rs Cr | - | 4 | 4 | 4 | 4 |
| **3** | Miscellaneous & Administrative expenses of **Implementing Agency** (Rs. Cr) | 4 | | | | |

**5.3 Lab Cost (CAPEX):** The cost per Lab may vary subject to the outcome of RFP/Tender. It will be revised appropriately as & when IA sends funding requirements to DoT based on competitive bid.

**5.4 O&M Cost (OPEX)**: 5G being a deep-tech project, technical handholding and O&M costs may be higher than currently provisioned figure in view of the upgradation, hand-holding, operation issues considering the fact that it involves tinkering of deep technology equipment by students & startups ecosystem. As may be necessary, O&M costs, may be revised from time to time by the approval of competent authority based on IA / institution inputs.

## 6. Funding Support and & Indicative milestones:

*Table 2*

| S. No | Cost item | Price | Remarks |
|---|---|---|---|
| 1 | Cost of establishing ONE Usecase lab (excluding O&M, Administrative Expenses) | L1 | Price obtained through competitive bidding by IA |
| 2 | Total Capital cost of the Project (excluding O&M, Administrative Expenses) | X (N*L1) | N- Number of labs (beneficiary institutions) |
| 3 | Grant by DoT to institutions | 80% of X | 80:20 (DoT : Institution) |
| 4 | Total funding from beneficiary institutions | 20% of X | 80:20 (DoT : Institutions) |
| 5 | Cost to be paid by individual institution to IA | 20% of L1 | Towards capital cost |
| 6 | Total Administrative Expenses of IA + O&M Costs | Y | Funded by DoT |
| 7 | Total Funding by DoT to IA | (80% of X) + (Y) | Funded by DoT |
| 8 | Total Funding to IA | X+Y | DoT & beneficiary Institutions |

**Milestone Based Funding (Indicative)**

*Table 3*

| Year | Installment | Milestone | Release |
|---|---|---|---|
| 23-24 | 1st Installment | Signing of Agreement (Advance) | 50% of capital cost |
| | 2nd Installment | Completion of 1st Milestone | 25% of capital cost |
| | 3rd Installment | Fully functional Lab - submission of project completion certificate | 25% of capital cost |
| 24-25 | O&M | Submission of Labs utilization certificate | 100% of O&M cost (per Anum) |
| 25-26 | O&M | Submission of Labs utilization certificate | 100% of O&M cost (per Anum) |
| 26-27 | O&M | Submission of Labs utilization certificate | 100% of O&M cost (per Anum) |
| 27-28 | O&M | Submission of Labs utilization certificate | 100% of O&M cost (per Anum) |

*Note: i) After completing the due process of Empanelling vendors, Competitive bidding and Selection of Institutions, IA will send the consolidated funding requirements to DoT along with the above cost breakup, with funding milestones (Advance, supply of equipment, completion of installation, O&M etc)*
*ii) First Milestone pertains to Installation of Lab equipment by IA in the premises of beneficiary institutions, post to the recipient of payment (20% of CAPEX) from the institution.*

## 7. Roles & Responsibilities of Beneficiary Institution

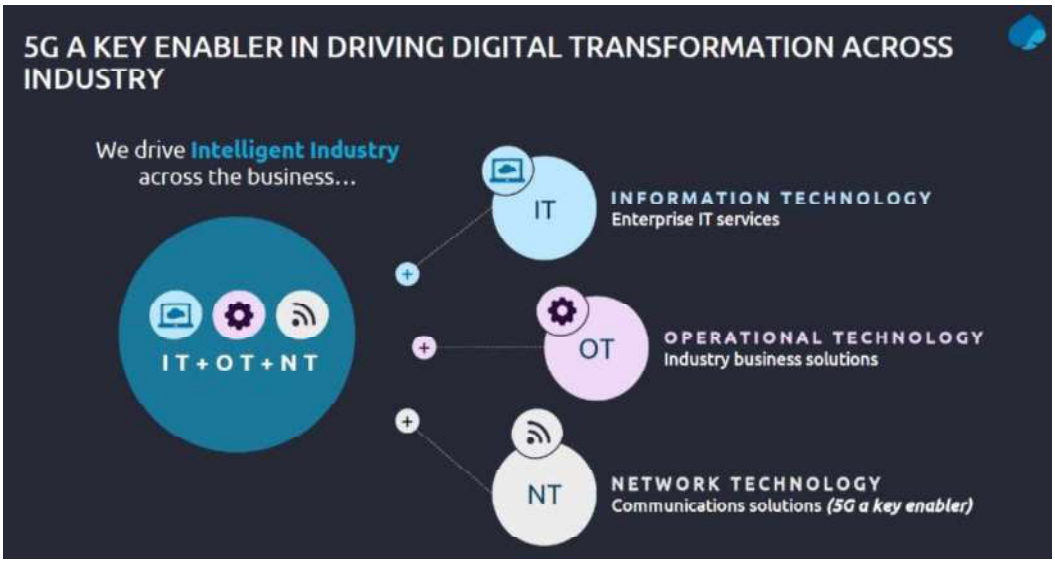
7.1 As part of the program, the beneficiary institution shall provide necessary facilities such as Space, power supply, internet/intranet connectivity, other equipment, tech-manpower (for local maintenance) etc., to ensure successful implementation of 5G usecase lab.
7.2 The institute shall pay the its prescribed contribution to the IA within 3 months of the final selection (Shipping of the equipment begins once the institution makes its payment)
7.3 Assets thus procured would belong to the host beneficiary institution.
7.4 The institution takes care of the safety, security and utilization of the Lab by students/ faculty/MSME ecosystem etc. IA will extend necessary facilitation in this regard.
7.5 The institution may have separate engagements with IA/Vendor etc., to procure other equipment (Devices/ Network elements / Usecase accessories) to enhance its functionality.
7.6 Institution shall extend connectivity to IA's Portal updating the Labs utilization and shall do value addition as part of knowledge dissemination reg 5G use cases being tested/developed for the benefit of other institutions/students/startups etc.
7.7 Hosting of Hackathons with support of IA on requirement basis.

## 8. Roles & Responsibilities of empanelled domestic vendors

1. The vendors shall supply the necessary equipment, devices etc. to set up and maintain the Lab on Turnkey basis as per the template/EoI floated by IA.
2. Vendor shall ensure that the 5G equipment to be installed, shall be indigenously designed, developed and manufactured.
3. Vendor should ensure that, IT/OT products and devices shall also be indigenously designed, developed and manufactured wherever available.
4. IA shall be responsible for overall project deliverables

## 9. 5G UCL Architecture

5G Use case Lab for Universities and CoEs Sandbox environment (Representative model)




*Architecture of a 5G usecase lab, Source (Capgemini)*

*Figure 1*

5G use case lab is an integrated setup of Network Technologies, OT & IT technologies for academic learning and experimentation, it may broadly consist of the following aspects

*Table 4*

| S.No | 5G Lab Component/s (Indicative) | Brief description |
|---|---|---|
| 1 | 5G SA Infra (Core + RAN) | On premise- Latest 3GPP Rel compliant Mini- 5G Network (5G Radio+ Core) to provide 5G experience for testing/experimentation. |
| 2 | Enablers | Management Dashboard- For configuring/controlling the Network parameters, configuring the devices; Router with Firewall; Application Server- For Hosting Applications locally & implementing MEC functionality; OT & IT Software etc. |
| 3 | 5G Student/Start-up Development kit | End devices with 5G Sims/gateways/CPEs where student or innovator can run different applications/solutions by leveraging the 5G Mini-Network Eg: 5G IoTs, 5G Drone, 5G XR device, 5G Camera etc. |
| 4 | Other Accessories for Installation | Antennas & Passive components such as cables, connectors, mounting brackets, etc.; Power/OFC cables,<br>AC-DC convertors, cabinets;<br>GPS antenna with accessories etc. |
| 5 | Supporting services | Installation, Handholding, Basic Training, standard Remote support, O&M etc. |
| 6 | Optional- Additional devices | Individual Institution may procure additional 5G enabled smart devices directly or through IA/Vendor as per the specialized requirements |

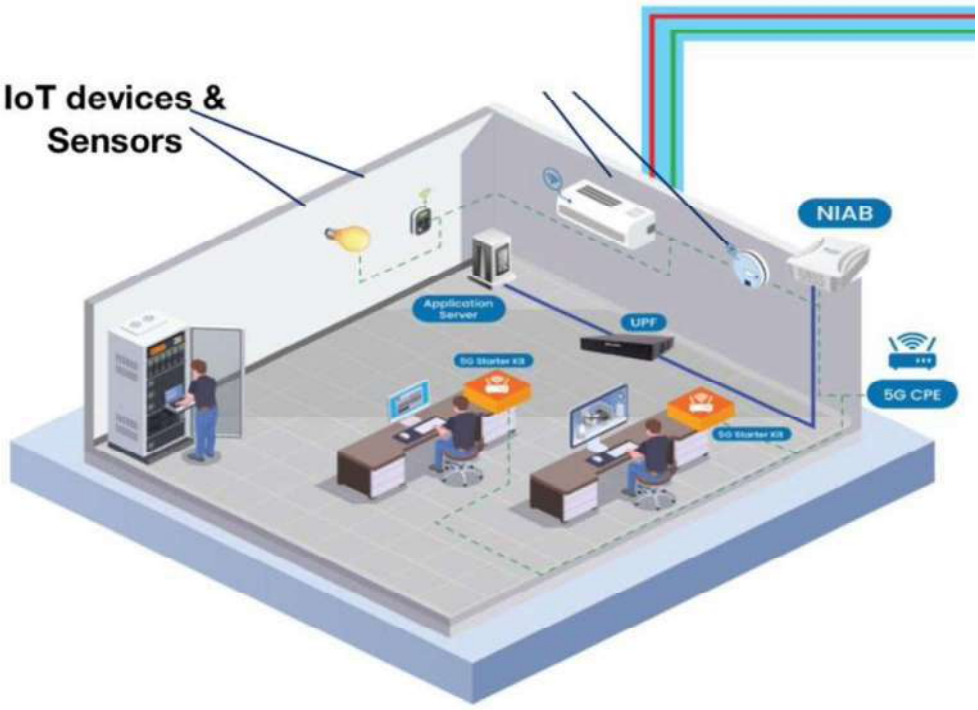



*Figure 2*

**Note:** Individual institutions are also encouraged to host 5G usecase labs with different configurations and flavors (beyond the proposed standard model)

**5G Lab (Representative Models)**

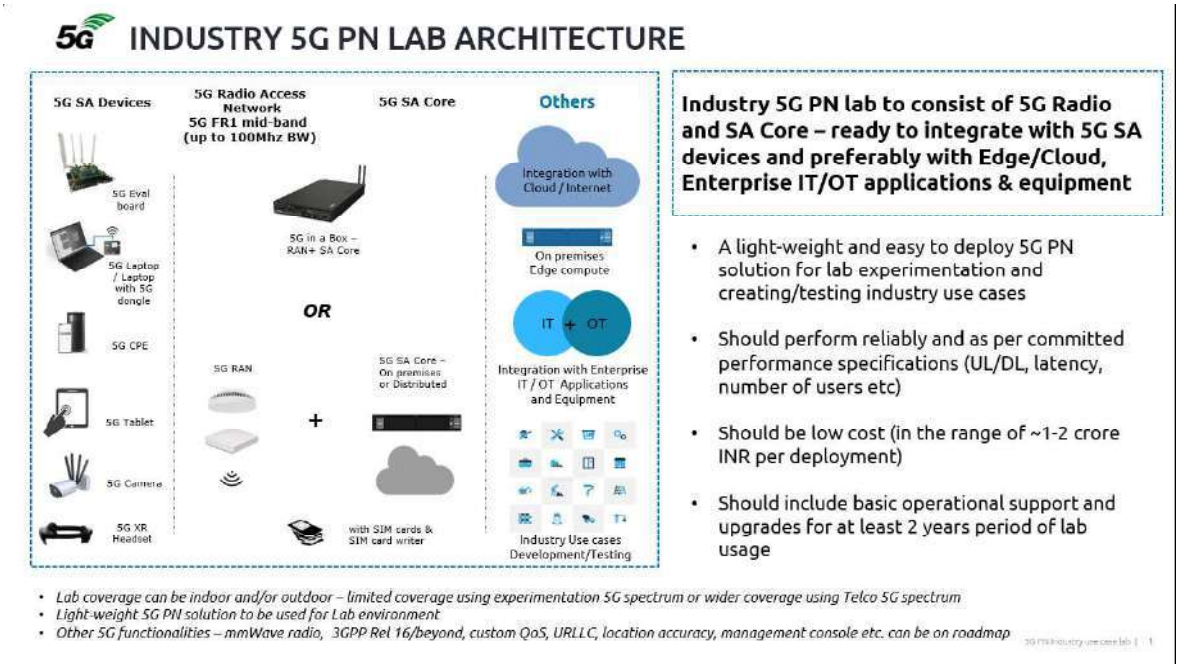



*Figure 3*

**5G Use case lab with focus on Health applications**

*Figure 4*

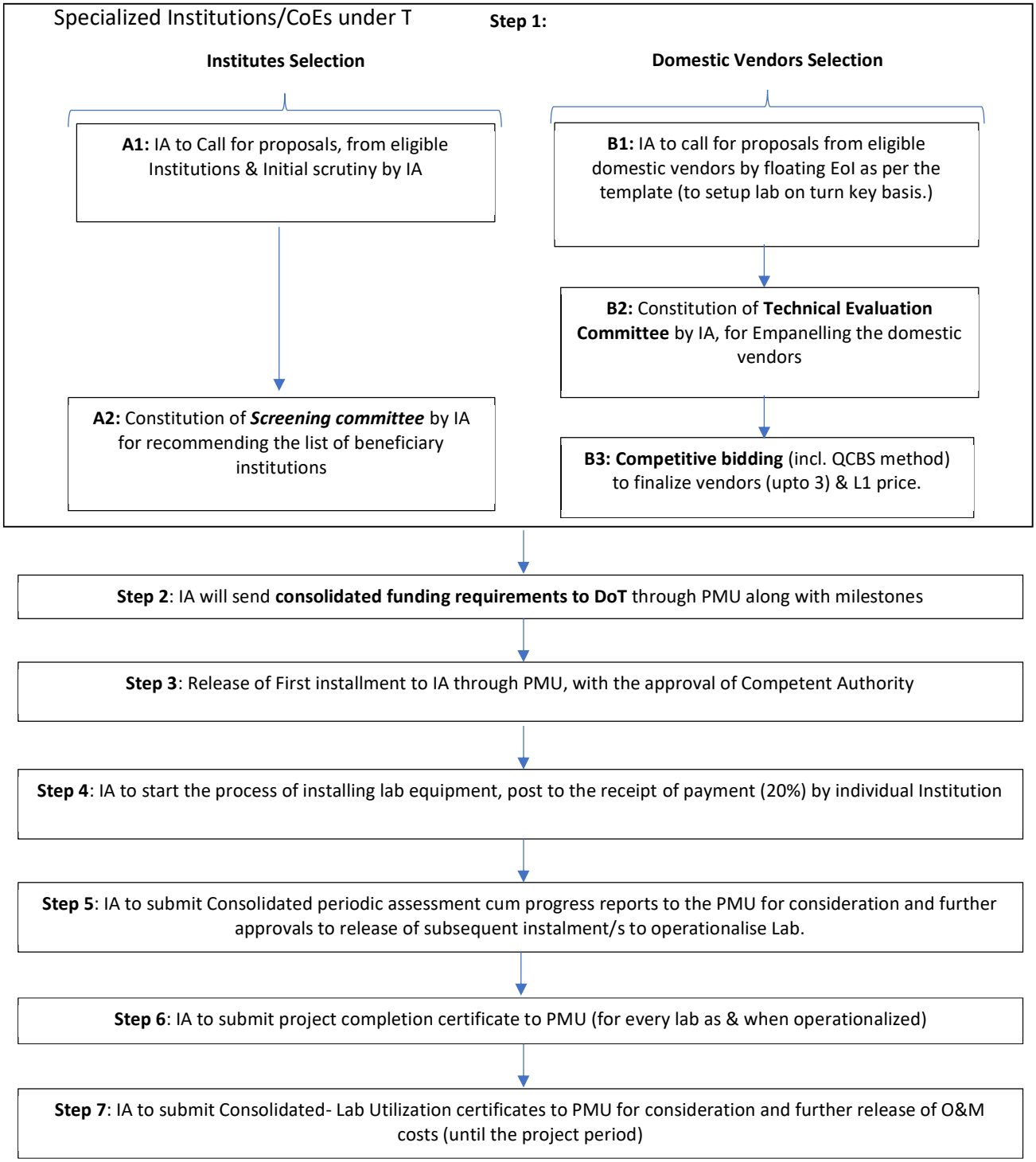
**5G Use case lab with focus on Agritech applications**

*Figure 5*

# Annexure-I: Implementation Mechanism

## 1.Implementation SOP

SOP for establishment and maintenance of 100+ 5G Use case labs at Educational & Specialized Institutions/CoEs under T




***Note**: IA shall manage a dedicated portal for managing the above scheme i.e. publishing guidelines/Terms and conditions, receiving the applications, publishing shortlisted Institutions, project Status Tracking, Instalment Releases, Monitoring Milestones, submission of periodic fund utilization certificates by beneficiary institutions, Remote O&M support, Database of institutions & industry partners, Projects / use cases developed, Lab Utilization etc .*

## 2. Implementing Agency

2.1 TCIL (here onwards called the Implementation Agency – IA) will implement the project under TDIP Scheme on behalf of DoT by following the extant guidelines.
2.2 The detailed responsibilities of IA are mentioned at Annexure-II of these guidelines

## 3. Application Format and Procedure

3.1 Online application submission by Eligible Institutions (as per the defined criterion) through a dedicated portal managed by the IA.
3.2 Online application submission by Eligible domestic companies[1] (defined) for empanelment through the portal (in response to the EoI floated by IA);
3.3 EoI format, Online Application Formats for institutions & domestic companies may be prepared in consultation with PMU/DoT.
3.4 Application format shall include the necessary terms and conditions (including 20% payment in case of academic institutions)

## 4. Project Monitoring Unit (PMU)

4.1 SRI division, DoT-HQ would function as PMU for the program
4.2 PMU will coordinate for fund disbursement, milestone monitoring, necessary approvals etc. It will also coordinate with PRSG committee for quarterly/bi-annual review of the program.
4.3 IA will provide necessary secretariat and people support to PMU to implement the program.

## 5. Selection Committee to identify beneficiary institutions

5.1 After receipt of applications for the program, IA shall constitute the Selection committee chaired by Sr DDG, NTIPRIT in consultation with PMU, to carry out detailed evaluation of applications and recommend the list of beneficiary institutions.
5.2 Members of the committee may include representatives from Ministry of Education, Training division, skill development division, WPC, Industry/Academia/Domain experts etc. with IA (as convener) ;
5.3 The committee will recommend the beneficiaries list based on the various aspects/parameters mentioned in the Selection Criterion of the document (Section 4),

[1] Domestic Company is defined as those which are owned by resident Indian citizens as defined in the FDI Circular of 2017. A company is considered as 'Owned' by resident Indian citizens if more than 50% of the capital in it is beneficially owned by resident Indian citizens and / or Indian companies, which are ultimately owned and controlled by resident Indian citizens. Further, it should be a DSIR recognized entity holding IPR ownership in India and carrying out R&D activities in India.

However the committee may also consider any other aspect/parameter as may deem fit, while finalizing the list;

5.4 The committee may also interact with the eligible institution/s to Confirm how an applicant has tied up the resources such as space, equipment, power supply, tech-manpower, advisors, Student/start up ecosystem etc. for establishing and utilization of Lab;

5.5 Due diligence will be carried out by the Committee before recommending the list i.e. reconfirming eligibility and other necessary aspects;

5.6 IA through PMU, shall submit the recommended list to DoT for information.

## 6. Empanelment Process of Vendors by IA

6.1 IA to empanel domestic companies meeting technical requirements by floating an EoI as per the template (with requirements) to setup lab on turnkey basis;

6.2 Eligible domestic companies shall apply through online on the dedicated portal

6.3 IA shall constitute a Technical Evaluation Committee chaired by Sr. DDG, TEC for technical evaluation and empanelling the domestic companies

6.4 Members of the committee may include representatives from SRI division-DoT HQ, TEC, Industry/Academia, IA (convener) etc.

6.5 IA through PMU, shall submit the empanelled list to DoT for information.

6.6 IA may continue the partnership with empanelled members to carry out similar projects (not limiting to the scope of the above project).

6.7 The technical evaluation may consider using Quality cum Cost-Based Selection (QCBS)[2]– Evaluation based on the Quality and cost factors based on technical qualification of the bidder considering the specialized nature of 5G usecase labs.

6.8 IA while onboarding the vendors will ensure that 5G and related equipment/software (procured under the program) shall be designed and made in India, wherever available, and it will be through Government e Marketplace (GeM).

## 7. Project Review & Streeting Group (PRSG):

PRSG will be constituted, to review the project progress periodically and steer it for successful completion. The composition of PRSG as follows

| | | |
|---|---|---|
| i. | Additional Secretary (Telecom), DOT | Chairperson |
| ii. | DDG (Finance), DOT | Member |
| iii. | Director, C-DOT | Member |
| iv. | DDG(IC), DoT | Member |
| v. | DDG (SRI), DoT | Member |
| vi. | DDG (NT), DGT-HQ | Member |

[2] https://www.pppinindia.gov.in/toolkit/ports/module2-leapsfp-dotpp.php?links=ctbspm1b
PPP Toolkit – Ministry of Finance

| | | |
|---|---|---|
| vii | JWA(WPC), DoT HQ | Member |
| viii. | Representative from Ministry of Education | Member |
| ix. | Representative from Ministry of Electronics and IT | Member |
| x. | Any other representative(s) from Central Government Ministry/ Department/ Organization may be co-opted with the approval of Chairperson | Member |
| xi | Director (SRI) | Member-Convener |

7.1 PRSG will meet at least once in every quarter (till labs get commissioned) to oversee and periodically suggest broad level actions for proliferation and utilization of 5G use case labs as and when convened by PMU through IA. Further to the establishment, it may meet once on 6 months to oversee the utilization.

7.2 PRSG may also recommend modification to milestones/ fund disbursement mechanisms to meet the program deliverables.

7.3 PRSG may recommend necessary modifications to guidelines, Implementation mechanism, terms and conditions of IA, Budgetary requirements, Milestones, Funding mechanism & modalities etc. from time to time.

7.4 IA shall extend necessary secretarial support to PMU in this regard

## 8. Fund Disbursement & Modalities:

8.1 Funding for this program will be done under TDIP scheme.

8.2 IA will seek commercial bids from empanelled vendors and will finalize up to three vendors and L1 price for execution of the program;

8.3 Techno-commercial Agreement will be signed between IA and the Domestic vendors to implement the project on Turnkey basis

8.4 IA will send the consolidated funding requirements (including administrative expenses) to DoT through PMU with funding milestones (Advance, supply of equipment, completion of installation, AMC etc.)

8.5 Total funds for projects shall be released to the Implementing Agency by DoT through PMU in instalments against the milestones with the approval of competent Authority.

8.6 After release of First instalment, subsequent instalments shall be released based on the achievements of the set milestones.

8.7 IA will submit Statement of Expenditure (SE) on annual basis during beginning of the Financial Year for preceding year.

8.8 PRSG will do periodic review of fund disbursement, other activities and deliverables to gauge output.

8.9 To facilitate ease of doing innovation especially by start-ups/MSMEs, IA will ensure the timely disbursal of funds to Domestic Vendors (who are part of the project) for meeting the project deliverables in a timely manner. Advances will be paid to MSMEs with an undertaking without any need for bank guarantees (as done in the case of USoF pilots).

## 9. Indicative Administrative Expenses of IA:

Indicative Administrative Expenses for the program to be paid to IA shall be Rs 4 Crores flat for the entire period (5 years) , The expenses may include co-ordination with DoT for Institutes selection and domestic vendors selection, management resources, portal management, outreach, travel & accommodation, due-diligence in scrutiny and empanelment, project evaluation & tracking, digital networking and utilization of labs, Knowledge dissemination reg successful 5G projects/pilots related to Health, Agri, intelligent Transport systems, Industry 4.0 and for other activities as defined in "Objectives & Scope" section of the guidelines.

## 10. Ownership:

10.1 The ownership of the Lab will be with the beneficiary institution, during and after the project completion

10.2 The beneficiary institution shall provide necessary facilities such as: Space, power supply, internet & intranet connectivity, other equipment, tech-manpower (for local maintenance) etc.

10.3 The beneficiary institution in co-ordination with IA, shall be taking full responsibility of safety, security, utilization of the Lab by student/faculty/start-up ecosystem etc.

## Annexure-II: Terms & Conditions for IA

1. The fund is for a specific project as approved by Department of Telecommunications (DoT) subject to the conditions listed below. The IA shall give an undertaking that they agree to be governed by these conditions.
2. The fund amount shall be i) spent for the project within the specified time; and ii) any portion of the fund, which is not ultimately required for expenditure for the approved purposes, shall be duly surrendered to DoT;
3. The IA should adhere to GFR guidelines, as applicable, while incurring expenditures out of DoT's portion of grant amount under the project.
4. Initially Scrutinized, Eligibility Check and completeness shall be carried out by IA as per the Criterion
5. The IA shall be responsible for ensuring that all procurement of equipment and services, including works, obtained through grant amount are through a transparent and competitive techno-commercial bidding process as per the applicable government rules/guidelines;
6. The IA shall render progress-cum-achievement reports at an interval of not exceeding six months/as when PMU requests- on the progress made on all aspects of the project including expenditure incurred on various approved items during the period;
7. IA to certify the completion of the project.
8. IA shall submit Consolidated periodic assessment cum progress reports to the PMU for consideration and further approvals to release of subsequent instalments.
9. The IA is not allowed to entrust the implementation of this project for which fund is received to another institution and to divert the fund received for this program for any other purpose.
10. IA shall maintain a dedicated portal for managing the above scheme i.e. publishing guidelines/Terms and conditions, receiving the applications, publishing shortlisted Institutions, project Status Tracking, Instalment Releases, Monitoring Milestones, submission of periodic fund utilization certificates by beneficiary institutions, Remote O&M support, Database of institutions & industry partners, Projects / usecases developed, Lab Utilization etc.
11. IA shall ensure timely conduction of Screening committee & Technical Evaluation Committee.
12. IA shall extend necessary support to PMU for timely conduction of PRSG meetings (Quarterly/biannually).
13. IA will extend O&M support, Remote support, Basic Training, periodic upgradations etc. to keep the Labs up & running.
14. IA will submit quarterly lab utilization certificates to PMU by the institutions.
15. IA is also encouraged to explore and execute similar projects independently to other agencies / institutions as well to promote proliferation of 5G usecase labs with different configurations and flavours (not limiting to the scope of the above project).
16. IA will maintain a separate bank account for all activities related to the program.
17. IA shall constitute the Selection committee & Technical Evaluation committee in consultation with PMU/DoT.
18. IA through PMU, shall submit the empanelled list of Domestic vendors/ recommended list of institution to DoT for information.
19. IA will provide necessary secretariat and people support to PMU to implement the program.
20. IA while onboarding the vendors will ensure that 5G and related equipment/software (procured under the program) shall be designed and made in India, wherever available, and it will be through Government e Marketplace (GeM).

## Annexure-III Template for UCL Equipment Configuration

| S.No | Category | Equipment/Devices | Specs | Remarks |
|---|---|---|---|---|
| 1.1 | 5G SA Infra (RAN+Core) | Configurations (indicative) | i) Network in a Box<br>ii)Integrated RAN(gNB) + Core<br>iii)Disintegrated RAN ( RRH, BBU/RU, DU, CU)+ Core | Other than the mentioned 3 options, Any other option to meet the required specs may also be considered |
| 1.2 | | 3GPP Release | Rel 15 Compliant & upgradable to Rel 16 (part of package) | Individual Institute may take up directly with IA/Vendor for further upgradations (Rel-17 & above) |
| 1.3 | | Data Rate (DL+UL) | >= 200Mbps | Individual Institute may take up with IA/Vendor for further enhancements |
| 1.4 | | Latency | < = 15ms | |
| 1.5 | | Max No.of active devices (That network can connect) | 30 (min) | |
| 1.6 | | Radio Power | Upto 100mW | 100mW is the limitation as per the Indoor Experimental License(indoor) |
| 1.7 | | Duplexing Mode | TDD | Individual institute may take up with IA/Vendor to procure radios in other NR Bands (incl. FDD) |
| 1.8 | | Band support | n78 | |
| 1.9 | | MIMO Configuration | 2T2R (min) | Individual Institute may take up with IA/Vendor for further enhancements to Radio ie. 4T4R/8T8R & beyond |
| | | | | |
| 2.1 | Enablers | Management Dashboard | | For configuring/controlling the Network parameters, configuring the devices |
| 2.2 | | Router with Firewall | >=500Mbps - 1(in number) | |
| 2.3 | | Application Server | 1(in number-min) | For Hosting Applications locally & implementing MEC functionality |
| 2.4 | | OT & IT Software | | Designed & Developed in India to the extent Possible |
| | | | | |

| | | | | |
|---|---|---|---|---|
| 3.1 | 5G Student/Start-up Development Kit | 5G SIMs | 10 (in number) | Individual Institution may procure additional 5G enabled smart devices directly or through IA/Vendor as per the specific requirements<br>Eg1: Health institution can opt for 5G enabled low-cost ambulance, e-ICU, smart wearables, Robotic Arm for Tele-Radiology etc.<br>Eg2: Agriculture Institution- Agri Drone, Soil/temperature/humidity sensors etc.<br>Eg3: Smart Factory- IIoT Devices, 5G Enabled AGVs etc. |
| 3.2 | | 5G Dongles | 2 (in number) | |
| 3.3 | | 5G Evaluation Board | 1 (in number) | |
| 3.4 | | IoT Gateway | 1 (in number) | |
| 3.5 | | 5G IoTs (Health/Agriculture/Industry 4.0/Automobile) etc. | 3 (in number) - Optional | |
| 3.6 | | 5G Mini Drone | 1 (in number) Optional | |
| 3.7 | | 5G XR(AR/VR/MR) headset or Device (with loaded application) | 2 (in number) Optional | |
| 3.8 | | 5G CPE | 1 (in number) | |
| 3.9 | | 5G Camera | 1 (in number) Optional | |
| 3.10 | | 5G Handsets | 2 (in number) Optional | |
| | | | | |
| 4 | Other Accessories for Installation | Antennas & Passive components such as cables, connectors, mounting brackets, etc.; Power/OFC cables, AC-DC convertors, cabinets; GPS antenna with accessories etc. | As per requirement | May be finalised during the empanelment/ before finalising commercial bid |
| | | | | |
| 5 | Support Services | Installation, Handholding, Basic Training, standard Remote support, O&M etc. | | On- premise support and Basic training (First Year/as and when upgraded), Remote support through the project period |

I/3103615/2023

Department of Telecommunications
Standards-R&D-Innovation
(**Project Monitoring Unit**)
********

No. 1-TR_5G_Startup_Kit/2022-SRI-Part(1) Dated: 24th May, 2023

**OFFICE MEMORANDUM**

**Sub:** Update on 100 5G use case labs Project (Budget Announcement 2023-24)

**I. Background**: In line with the budget announcement, DoT is setting up one hundred 5G Use case labs to facilitate the development of applications using 5G services in various socioeconomic verticals. These labs will be set up in educational institutions. The use cases could span across various socioeconomic verticals including education, agriculture, health, power, urban management, mining, logistics, resource management, tourism, sports, security, e-governance etc.

DoT would be funding 80% of CAPEX for the Lab Setup and 100% of OPEX expenses for next 4 years. The institution to contribute for the remaining 20% of the CAPEX. As per initial estimates, the institution's CAPEX contribution could be to a tune of Rs 10 lakhs or higher depending on CAPEX price discovery from financial bids for setting up the 5G lab infrastructure.

The 5G lab equipment would include 5G SA infrastructure (mid band), 5G SIMs, Dongles, IoT Gateway, Router and Application Server to meet lab needs along with management dashboard. The ownership of the Lab Assets will be with the beneficiary institution, during and after the project completion. The beneficiary institution shall provide necessary facilities such as Space, power supply, internet & intranet connectivity, other equipment, tech-manpower (for local maintenance) etc. As per the approved guidelines for 100 5G use case labs, TCIL is nominated as **implementing agency** for the project and rollout is planned for September-December 2023.

**II. Developments**:
Post to the Meeting of Member(Technology) & CMD TCIL regarding implementation mechanism and timelines, OM on implementation guidelines with identified co-ordination points from TCIL(Implementation agency), NTIPRIT(institution selection), TEC (Vendors selection), SRI division(PMU) for effective implementation was notified to the relevant stakeholders on **27th March, 2023**

I/3103615/2023

**1. Institutions Selection:**

- Call for proposals for Higher Educational Institutions was floated through TCIL website, inviting applications from on **5th April, 2023**.
- For wider visibility and ease of accessibility, application link (https://www.tcil.net.in/5glab/5glab.html) is made available at My Gov portal & DoT portal Manthan portal etc.
- LSAs(DoT field units) had reached out to the local institutions at their respective service areas and awareness sessions on 100 5G labs & 5G courses were conducted with the support of Skill development division and AICTE team.
- Last date for application submission was extended twice upto **15th May, 2023**, to encourage participation from institutions across the country's landscape and also based on the requests from various organizations
- TCIL had received 600+ applications from engineering and other streams.
- Institution selection committee chaired by DG(NTIPRIT), with a Member from the Department of Higher Education was constituted by TCIL in line with the implementation guidelines and was notified on **5th May, 2023**. Its members include JS, Dept. of Higher Education, DDG(WA), DDG(Skill Development), DDG(NICF), Sr. DWA, WPC
- First meeting of the Institution selection committee under the chairmanship of DG(NTIPRIT) was held on **12th May 2023**, where in which criteria for selection and other aspects were discussed. TCIL team under the guidance of the chair is preparing a refined list for the committee's consumption.
- Efforts are ongoing to notify the selected institutions by the end of **June 2023.**

A summary with indicative statewide analysis of the applications received is enclosed for kind reference.

**2. Vendors Selection:**

- TCIL floated the Expression of Interest (EoI) on **5th April, 2023** for the empanelment of 5G vendors to supply equipment for the proposed 100 5G labs. The EoI was uploaded in TCIL website and CPP portal as well.
- Pre-bid conference was also held to bring more clarity to the vendors and the corrigendum also issued with Technical and generic clarifications.
- To provide ample opportunity and time, Last date for submission was extended till **5th May, 2023.** System integrators in partnership with Indian OEMs, Domestic OEMs individually as well as in consortium mode are also allowed to bid with an indigenous equipment clause (as per the implementation guidelines)
- TCIL had received 20 bids for Technical evaluation, Also to encourage indigenous players bids were accepted through CPP portal and TCIL nodal mail (as notified in RFP)
- Post to the Technical evaluation, PoC demo, and Financial bids will be taken up.
- For Vendors empanelment Technical Evaluation Committee chaired by Sr. DDG(TEC) was formed and notified on **5th May 2023.** Committee members include experts from C-DOT, IISc Bangalore, Industry experts (Capgemini), DoT-HQ

I/3103615/2023

- First meeting of the Technical Evaluation was held on **12th May 2023** in which criterion for the selection of vendors was discussed in length, Under the guidance of TEC, TCIL team is preparing the detailed checklist for the committee's consumption.
- Efforts are ongoing to shortlist the vendors for PoC by **June 2023 1st week.**
- List of the technical bids received is attached for kind reference.
- RFP for a dedicated portal is also being finalised

*Post to the successful PoC and Financial bidding, the First 100 5G labs Project Review & Steering Group (PRSG) meeting under the chairmanship of the Additional Secretary(Telecom) will be convened in July- 2023. The Installation of 5G lab Equipment is expected to begin from September-October 2023*

*Submitted for Kind information please*

**Enclosures:**

i) OMs regarding the Institution selection committee and Technical Evaluation Committee
ii) Summary of the institution applications received with indicative statewide analysis.
iii) List of technical bids received.

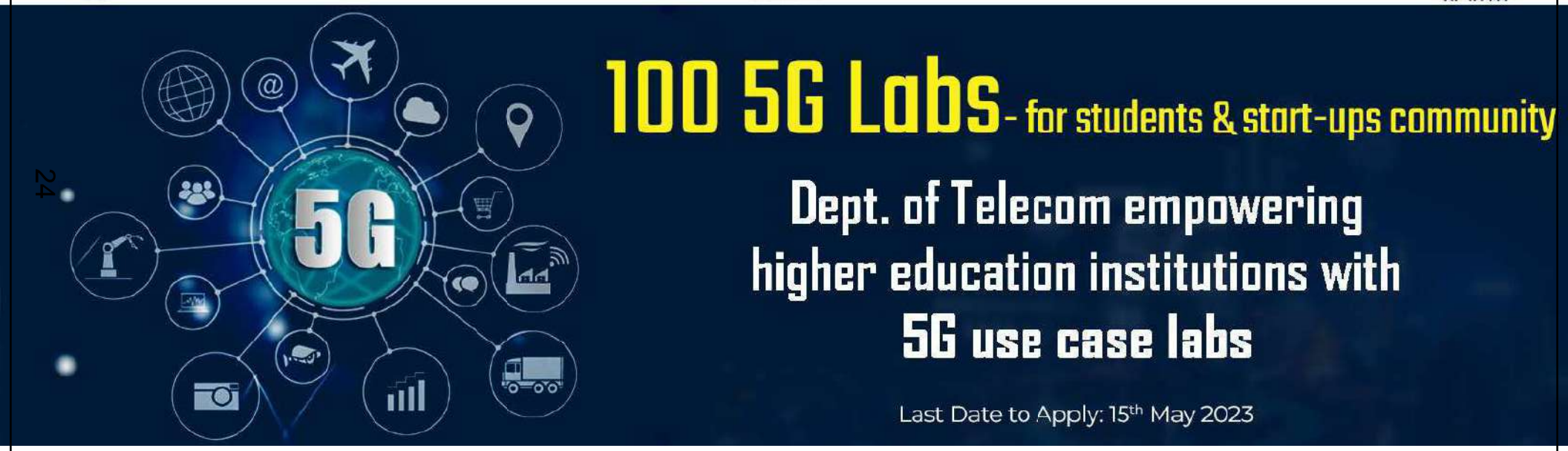
Signed by A Alex Vikas
Date: 24-05-2023 16:08:47
Reason: Approved

(A. Alex Vikas)
PMU Co-ordinator/ Asst. Director General
Mail: alex.vikas17@gov.in
Tel No: 011- 2337207

To:

1. PSO to Secretary (Telecom) & Chairman DCC - for kind information
2. Member(F)/Member(T)/Member(S)/Additional secretary (Telecom)- for kind information
3. DDG(SRI) /DDG(SU)/DDG(Budget), DoT HQ
4. Director(Budget)/Director(P)/Director(SRI), DoT HQ

I/3103615/2023



I/3103615/2023

# Telecommunications Consultants India Ltd

(A Government of India Enterprise)
TCIL Bhawan, Greater Kailash – I, New Delhi – 110048

IS/ISO 9001

CIN: U74999DL1978GO1008911

TCIL/ITTII/123/2023 Date: 04/05/2023

## OFFICE MEMORANDUM

Subject : **Institute Selection Committee**

This is in reference to DoT office memorandum No. 1-TR_5G_Startup_Kit/2022-SRI dated 27th March 2023 in regard to setting up of 100 5G use cases labs in academic institutions along with Implementation guidelines. As per the OM, one hundred 5G use case labs will be deployed by DoT through M/s TCIL as the implementation agency (IA).

Further, as per the Implementation guidelines issued by Department of Communications, committee is being constituted for Selection of Institute headed by DG(NTIPRIT) and ED(IT&T-II), TCIL as Convener. Technical Evaluation Committee consists of following officials.

1. Sh. Raju Sinha, DG(NTIPRIT) – Chairperson
2. Ms. Alka Astana, ED(IT&T-II), TCIL – Convener
3. Smt. Saumya Gupta, Joint Secretary (TE), Department of Higher Education
4. Sh. Ashok Kumar, DDG(WA), NTIPRIT
5. Sh. Alok Bharati, DDG (Skill development), DoT HQ
6. DDG(NICF), DoT
7. Sr. DWA, WPC

Sanjay Kumar
General Manager(Digital Transformation)
kumars@tcil.net.in ; +91-9811111425

**To:**

1. Sh. Raju Sinha, DG(NTIPRIT) – Chairperson
2. Ms. Alka Astana, ED(IT&T-II), TCIL – Convener
3. Smt. Saumya Gupta, Joint Secretary (TE), Department of Higher Education
4. Sh. Ashok Kumar, DDG(WA), NTIPRIT
5. Sh. Alok Bharati, DDG (Skill development), DoT HQ
6. DDG(NICF), DoT
7. Sr. DWA, WPC

**Copy to:**

1. PSO to Secretary (Telecom) & Chairman DCC - for kind information
2. Member(T)/Member(S)/AS(T)
3. Member(F)/DG(NICF)/WA WPC for kind nomination of suitable officer please
4. Director(P) / CMD, TCIL
5. DDG(SRI) / DDG(SU), DoT HQ

**Enclosure** : DoT office memorandum No. 1-TR_5G_Startup_Kit/2022-SRI dated 27th March 2023

I/3103615/2023

# Telecommunications Consultants India Ltd

(A Government of India Enterprise)
TCIL Bhawan, Greater Kailash – I, New Delhi – 110048

IS/ISO 9001

CIN: U74999DL1978GO1008911

TCIL/ITTII/123/2023 Date: 05/05/2023

## OFFICE MEMORANDUM

Subject : **Technical Evaluation Committee.**

This is in reference to office memorandum No. 1-TR_5G_Startup_Kit/2022-SRI dated 27th March 2023 in regard to setting up of 100 5G use cases labs in academic institutions along with Implementation guidelines. As per the OM, one hundred 5G use case labs will be deployed by DoT through M/s TCIL as the implementation agency (IA).

Further, as per the Implementation guidelines issued by Department of Communications, committee is being constituted for technical evaluation headed by Sr. DDG(TEC) and ED(IT&T-II), TCIL as Convener. Technical Evaluation Committee consists of following officials.

1. Sr. DDG, TEC - Chairperson
2. ED(IT&T-II), TCIL - Convener
3. Director (SRI), DoT
4. Expert, C-DOT
5. Prof KVS Hari, IISC Bangalore
6. Ms. Monica Gupta, VP, Capgemini (Industry expert)

Sanjay Kumar
General Manager(Digital Transformation)
kumars@tcil.net.in ; +91-9811111425

**To:**

1. Sr. DDG, TEC - Chairperson
2. ED(IT&T-II), TCIL - Convener
3. Director (SRI), DoT
4. Expert, C-DOT
5. Prof KVS Hari, IISC Bangalore
6. Ms. Monica Gupta, VP, Capgemini (Industry expert)

**Copy to:**

1. PSO to Secretary (Telecom) & Chairman DCC - for kind information
2. Member(T)/Member(S)/Member(F)/AS(T)
3. Director(P) / CMD, TCIL
4. DDG(SU), DoT HQ
5. DDG(SRI)/CEO C-DOT to nominate suitable officer.

**Enclosure** : DoT office memorandum No. 1-TR_5G_Startup_Kit/2022-SRI dated 27th March 2023

I/3103615/2023

**Annexure**

High-level view of the applications received from Higher educational Institutions for 100 5G labs

| **Summary (indicative analysis as on 15.05.2023 )** | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **State & Institute NIRF Categories wise** | | | | | | | | | | |
| | **Overall** | **Universities** | **Colleges** | **Research Institutes** | **Engineering** | **Medical** | **Others** | **Total** | **NIRF** | **Non-NIRF** |
| | | | | | | | | | | |
| **Andhra Pradesh** | 1 | 6 | 6 | 0 | 30 | 0 | 0 | **43** | **15** | **28** |
| **Andaman and Nicobar Islands** | 0 | 0 | 0 | 0 | 0 | 0 | 0 | **0** | **0** | **0** |
| **Arunachal Pradesh** | 0 | 2 | 1 | 0 | 1 | 0 | 0 | **4** | **1** | **3** |
| **Assam** | 1 | 3 | 0 | 0 | 4 | 0 | 0 | **8** | **5** | **3** |
| **Bihar** | 0 | 1 | 0 | 0 | 2 | 0 | 1 | **4** | **2** | **2** |
| **Chandigarh** | 0 | 1 | 0 | 0 | 1 | 0 | 1 | **3** | **2** | **1** |
| **Chhattisgarh** | 1 | 4 | 0 | 0 | 6 | 0 | 0 | **11** | **6** | **5** |
| **Dadar and Nagar Haveli** | 0 | 0 | 0 | 0 | 0 | 0 | 0 | **0** | **0** | **0** |
| **Daman and Diu** | 0 | 0 | 0 | 0 | 0 | 0 | 0 | **0** | **0** | **0** |
| **Delhi** | 2 | 4 | 2 | 1 | 7 | 0 | 0 | **16** | **11** | **5** |
| **Lakshadweep** | 0 | 0 | 0 | 0 | 0 | 0 | 0 | **0** | **0** | **0** |
| **Puducherry** | 0 | 2 | 1 | 0 | 4 | 0 | 0 | **7** | **5** | **2** |
| **Goa** | 1 | 0 | 0 | 0 | 3 | 0 | 0 | **4** | **2** | **2** |
| **Gujarat** | 2 | 7 | 8 | 0 | 11 | 0 | 3 | **31** | **8** | **23** |
| **Haryana** | 0 | 4 | 0 | 0 | 3 | 0 | 2 | **9** | **4** | **5** |
| **Himachal Pradesh** | 0 | 3 | 0 | 0 | 3 | 0 | 0 | **6** | **4** | **2** |
| **Jammu and Kashmir** | 2 | 3 | 0 | 0 | 3 | 0 | 2 | **10** | **7** | **3** |

I/3103615/2023

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Jharkhand** | 0 | 0 | 0 | 0 | 4 | 0 | 0 | **4** | **4** | **0** |
| **Karnataka** | 2 | 14 | 5 | 0 | 48 | 0 | 2 | **71** | **39** | **32** |
| **Kerala** | 6 | 4 | 9 | 0 | 17 | 1 | 2 | **39** | **17** | **22** |
| **Madhya Pradesh** | 2 | 2 | 0 | 0 | 15 | 0 | 1 | **20** | **10** | **10** |
| **Maharashtra** | 3 | 10 | 6 | 1 | 52 | 0 | 2 | **74** | **24** | **50** |
| **Manipur** | 0 | 0 | 0 | 0 | 1 | 0 | 1 | **2** | **1** | **1** |
| **Meghalaya** | 0 | 0 | 0 | 1 | 2 | 0 | 0 | **3** | **2** | **1** |
| **Mizoram** | 0 | 0 | 0 | 0 | 1 | 0 | 0 | **1** | **0** | **1** |
| **Nagaland** | 0 | 0 | 0 | 0 | 1 | 1 | 0 | **2** | **1** | **1** |
| **Odisha** | 0 | 5 | 1 | 0 | 11 | 0 | 0 | **17** | **15** | **2** |
| **Punjab** | 2 | 3 | 2 | 0 | 4 | 0 | 3 | **14** | **9** | **5** |
| **Rajasthan** | 1 | 11 | 1 | 0 | 14 | 0 | 0 | **27** | **11** | **16** |
| **Sikkim** | 0 | 1 | 0 | 0 | 0 | 0 | 0 | **1** | **1** | **0** |
| **Tamil Nadu** | 5 | 19 | 10 | 3 | 75 | 0 | 2 | **114** | **59** | **55** |
| **Telangana** | 0 | 4 | 6 | 4 | 28 | 1 | 0 | **43** | **23** | **20** |
| **Tripura** | 1 | 0 | 0 | 0 | 2 | 0 | 0 | **3** | **2** | **1** |
| **Uttar Pradesh** | 2 | 8 | 3 | 0 | 21 | 0 | 1 | **35** | **21** | **14** |
| **Uttarakhand** | 0 | 1 | 0 | 0 | 1 | 0 | 1 | **3** | **2** | **1** |
| **West Bengal** | 0 | 0 | 0 | 0 | 2 | 0 | 0 | **2** | **1** | **1** |
| | | | | | | | | | | |
| **Total** | **34** | **122** | **61** | **10** | **377** | **3** | **24** | **631** | **314** | **317** |
| | | | | | | | | | | |

I/3103615/2023

**List of Bidders who participated in - 100 5G labs vendor empanelment**

| Sl.No. | Bidder's name |
|---|---|
| 1 | Amantya Technologies Private Limited |
| 2 | Design and Manufacturing Vista Electronics Pvt. Ltd. |
| 3 | Galore Networks Private Limited |
| 4 | L&T Technology Services Limited |
| 5 | Nipun Net Solutions Pvt Ltd |
| 6 | NTT India Pvt. Ltd. |
| 7 | NuRe Edgetech Private Limited |
| 8 | Protean eGov Technologies Limited |
| 9 | Sphere Com Services Pvt. Ltd |
| 10 | TELECOM SECTOR SKILL COUNCIL |
| 11 | WIRELESS 4 SCALE LABORATORY PRIVATE LIMITED |
| 12 | Nokia Solutions & Networks India Pvt Ltd |
| 13 | Meridian Infotech Ltd |
| 14 | Josh Tech Services Pvt Limited |
| 15 | Coral Telecom Ltd. |
| 16 | PRECISION ELECTRONICS LTD. |
| 17 | Signaltron Systems Pvt Ltd |
| 18 | Mashmari, India |
| 19 | VVDN Technologies Pvt Ltd. |
| 20 | Sooktha Consulting Pvt. Ltd |

I/3115714/2023

**Department of Telecommunications**
Standards-R&D-Innovation
(*Project Monitoring Unit*)
****

Sanchar Bhavan, 20 Ashoka Road,
New Delhi-110001

**No.** 1-TR_5G_Startup_Kit/2022-SRI **Dated:** 23rd July, 2023

**OFFICE MEMORANDUM**

**Sub:** Modified SOP for setting up of 100+ 5G Use case labs under TDIP Scheme

**Ref:** (i) OM on implementation guidelines for setting up of 100 5G labs dt. 27th March, 2023
(ii) Approved Record of the Discussion chaired by Secretary(T) "No. 1-TR_5G_startup_ Kit/2022-SRI", dt 17th July, 2023

DoT vide OM under reference issued implementation guidelines for setting up of 100+ 5G labs at Higher educational institutions which elaborates on the roles & responsibilities of TCIL(Implementation Agency), NTIPRIT & TEC along with implementation SOP (as Annexure).

Subsequently Secretary (Telecom) took a review meeting regarding the progress of 100+ 5G use case labs project for higher educational institutions on 19th June 2023. The meeting was attended by Additional Secretary(T), Member(T), Member(F), DG(NTIPRIT), Sr. DDG(TEC), CMD TCIL & other senior officials from DoT. Key decisions taken during the meeting w.r.t. i) Funding modalities ii) institutions Selection iii) Indigenous Vendors selection are captured as part of the approved record of discussion under reference, accordingly SOP is suitably updated and approved by the competent authority.

*Modified SOP is enclosed for necessary compliance*

Signed by A Alex Vikas
Date: 23-07-2023 23:20:07
Reason: Approved

(A. Alex Vikas)
Mission Co-ordinator(TSUM)/ Asst. Director General
Mail: alex.vikas17@gov.in;
Tel No: 011- 2337207

**To:**

CMD (TCIL) / DG(NTIPRIT) / Sr. DDG(TEC)

**Copy to:**

1. PSO to Secretary (Telecom) & Chairman DCC - for kind information
2. Member(T)/Member(S)/Member(F)/AS(T)
3. ED (IT&T-2), TCIL/Director(P), TCIL
4. DDG(SRI), DoT HQ / DDG(MT), TEC/ DDG(WA), NTIPRIT

I/3115714/2023

# Annexure-I: Implementation Mechanism

## SOP for setting up of 100+ 5G Use case labs at Higher educational institutions.

**Step 1:**

**Institutes Selection**

**A1:** IA to Call for proposals, from eligible Institutions & Initial scrutiny by IA

**A2:** Constitution of ***Institution selection committee*** by IA for recommending the list of beneficiary institutions

**Domestic Vendors Selection**

**B1:** IA to call for proposals from eligible domestic vendors by floating EoI as per the template (to setup lab on turn key basis.)

**B2:** Constitution of **Technical Evaluation Committee** by IA, for Empanelling the domestic vendors

**B3: Competitive bidding** (incl. QCBS method) to finalize vendors (up to 4 vendors with different Flavours of OEM partners) & L1 price

**Step 2**: IA will send **consolidated funding requirements to DoT** as per the tender discovered rates through PMU along with milestones.

**Step 3**: Release of the first instalment (50% CAPEX) to IA through PMU based on PO/Agreement with Vendors & IA to start the process of installing lab equipment.

**Step 4**: IA to submit consolidated equipment installation/delivery status for release of subsequent instalment (25% CAPEX) to operationalize Labs.

**Step 5**: IA to submit the project completion certificate (Fully functional lab) to PMU for release of final instalment (25% CAPEX)

**Step 6**: IA to submit Consolidated Lab Utilization certificates post the adjustment of Institution contribution (20% CAPEX) to PMU for consideration & further release of adjusted O&M costs annually in advance till project period

***Note**: IA shall manage a dedicated portal for managing the above scheme i.e. publishing necessary guidelines/Terms and conditions, publishing shortlisted Institutions, project Status Tracking, Instalment Releases, Monitoring Milestones, submission of periodic fund utilization certificates by beneficiary institutions, Remote O&M support, Database of institutions & industry partners, Connecting all the Labs, Knowledge dissemination-Projects / use cases developed, Lab Utilization etc.*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I/3129640/2023

**Department of Telecommunications**
**Standards-R&D-Innovation**
(*Project Monitoring Unit- 100 5G Labs*)

No. 1-5GLabsPMU/2023-SRI Dated: 30.09.2023

**SUB: MoM of Technical Evaluation Committee chaired by Sr.DDG TEC regarding 100 5G labs RFP floated by TCIL**

In line with the budget 2023 announcement, DoT is setting up the 100 5G Labs for Higher educational institutions across India. TCIL was nominated as the Implementation Agency for the project and had floated Technical-Expression of Interest for selecting the vendors on 5th April 2023. 100 Institutions spanning 28 states & 4 Union Territories have already been shortlisted by the institution selection committee where in 5G labs will be installed. Based on the response of Technical-Expression of Interest 15 vendors have been shortlisted for PoC by the Technical Evaluation Committee and the Vendors will be finalized by October 1st week. (Outcome of the Financial RFP). PoCs have been conducted for the shortlisted vendors from July to Sep. TCIL had published RFP (with final specs) on GeM portal, No. GEM/2023/B/3990770 dt 21.09.2023 inviting price bids from the shortlisted vendors after taking due inputs from Vendors(Technical inputs), Committee members and chair. Clarifications were invited from bidders and Pre-bid consultation was held on 27.09.2023.

Technical Evaluation committee meeting under the chairmanship of Sr.DDG (TEC) was held on 29.09.2023 to finalize the clarifications on Tender clauses and also deliberate on Vendor queries. The meeting was attended by following

1. *Sh. Tripti Saxena Sr. DDG, TEC – Chair*
2. Prof. K.V.S. Hari, IISc Bangalore
3. Ms. Leena Ambwani, Team Leader, C-DOT Delhi
4. Sh. Gokulraj, Group Leader, C-DOT Bangalore
5. Ms. Monika Gupta, VP, Capgemini (Industry expert)
6. Sh. Sanjay Kumar, GM (Digital Transformation), TCIL
7. Ms. Chanchal Pathak, JGM (Digital Transformation), TCIL
8. Sh. A. Alex Vikas ADG (SRI), DoT-HQ, SRI (PMU)

The committee had discussed in detail w.r.t Tender clauses and queries raised by Vendors and the following decisions were taken

i. **5G Equipment/devices:** Some equipment' were kept optional as part of the Initial Technical RFP (Annexure-I), However to bring in more relevance for use cases and optimal utilization of 5G features by students & startups all the relevant use case equipment such as 5G Drone, AR/VR equipment, 5G camera, IoTs etc. (Equipment List at Annexure-II) are made part of the RFP package (non-optional). Also, as a part of the package these

I/3129640/2023

institutions will get best 5G devices technically/commercially without going through the integration efforts (needed for latching into private 5G network setup).

ii. **Amendments**: Other decisions that were agreed upon by the members related to Technical & Generic clauses along with observations by the committee members is placed at Annexure-III

It is to be noted that Standard clause w.r.t variations (increase or decrease) in Taxes & Duties are made part of the Tender (details at Annexure-IV) and considering the strict timelines-Clarifications were already uploaded by the TCIL on 29.09.2023 in line with the suggested changes by the committee members.

This issued with the concurrence of chair, Technical Evaluation committee, 100 5G Labs

Signed by A Alex Vikas
Date: 30-09-2023 19:28:21
Reason: Approved

(A. Alex Vikas)
PMU Co-ordinator/ Asst. Director General
Mail: alex.vikas17@gov.in
Tel No: 011- 2337207

To

Chair & all the Members of the Committee

Copy to

1. Member(T), DCC, DoT HQ for kind information please
2. CMD, TCIL /Director (P), TCIL
3. DDG(SRI), DoT HQ

I/3129640/2023

**Annexure -I**

Technical Expression of Interest NO. TCIL/IT&T-II/5G Labs/Vendor Empanelment, Dt : 05.04.2023 to identify Indigenous Vendors

**Section 4.3.3 5G Student/Start-up Development Kit**

| S.No | Equipment/Devices | Specifications |
|---|---|---|
| i. | 5G SIMs | Provide details |
| ii. | 5G Dongles | Provide details |
| iii. | 5G Evaluation Board | Provide details |
| iv. | IoT Gateway | Provide details |
| v. | 5G IoTs (Health/Agriculture/Industry 4.0/Automobile) etc. (Optional) | Provide details |
| vi. | 5G Mini Drone (optional) | Provide details |
| vii. | 5G XR(AR/VR/MR) headset or Device (with loaded application) (Optional) | Provide details |
| viii. | 5G CPE | Provide details |
| ix. | 5G Camera (Optional) | Provide details |
| x. | 5G Handsets (Optional) | Provide details |

I/3129640/2023

**Annexure -II**

Financial RFP GeM Portal No. GEM/2023/B/3990770 dt 21.09.2023 to finalise among shortlisted Vendors for award of contract (Extract)

**Section 4.1**

| S.No | Equipment/Devices | Qty | Remarks |
|---|---|---|---|
| 9 | IoT Gateway | 1 | |
| 10 | IoT sensors with analytics software (Loaded in Application server) | | |
| | Temperature & Humidity sensor | 1 | |
| | Light sensor | 1 | |
| | Soil sensor (NPK) | 1 | |
| | Water (TDS/chlorine) sensor | 1 | |
| 11 | 5G Mini Drone | 1 | |
| 12 | 5G XR(AR/VR/MR) headset or Device (with loaded application such as remote maintenance / training/skill development/ education etc. ) | 1 | |
| 13 | 5G Indoor CPE | 1 | |
| 14 | 5G Camera with AI enabled Video Analytics (Face/Object/Motion detection, people counting etc.) | 1 | Analytics to be loaded in Application server, Support can be taken from cloud (VPN) |
| 15 | 5G Handsets | 2 | Support testing and testing tools |
| 16 | Adjustable Tripod Pole (3m) | 1 | Radio/Antenna mounting |
| 17 | 24 U Rack | 1 | |
| 18 | UPS 5KVA (1hr backup) | 1 | |
| 19 | 32 inch FHD Display | 1 | |
| 20 | L2 Managed Switch (24 port) | 1 | |
| 21 | Testing & Tracing Tools | 1 | For end-to-end 5G network viz.<br>UE- Radio,<br>Radio-Core, etc. (Open source includes proprietary/third party tools) |

I/3129640/2023

**Annexure-III:**

Corrigendum to RFP

| Clause | **Original RFP Clause** | **Modified Clause** | **Observations by the Committee** |
|---|---|---|---|
| 1.16 b) | The work shall be awarded to L1, L2, L3 and L4 bidder at L1 price as discovered in the financial bid evaluation. | The work shall be awarded to 4 bidders with different combination of Core+RAN offering. The work shall be awarded to L1, L2, L3 and L4 bidder (with different offering) at L1 price as discovered in the financial bid evaluation.<br><br>For additional work beyond 100 labs, work shall be distributed to bidder L1:L2:L3:L4 in ratio of 40:30:20:10. | In line with the decisions captured in Record of Discussion chaired by Secretary(T) No. 1-TR_5G_Startup_Kit/2022-SRI, dt. 17.07.2023. 4 bidders with different flavours of 5G equipment will be chosen.<br>Also there needs to be some incentive for L1 (opportunities for Additional work allocation) |
| 4.10.7 a) | Minimum 8 x PCIe 16x 4.0 slots for I/O Connectivity or higher as required. | Minimum 4 x PCIe 16x 4.0 slots for I/O Connectivity or higher"<br>as required. | Cost Optimization & availability |
| 4.1.1 | 4th Generation Intel® Xeon Gold 2.4 GHz or equivalent 24 Core (minimum). Each servers with minimum 1 Processors and 2 Socket or higher as required. | 3rd Gen Intel Xeon Gold 2.1GHz , 24Cor or equivalent. Each servers with minimum 1 Processors and 2 Socket or higher as required. | Cost Optimization & availability |
| 4.1.2 | DDR5-4800 or above 64 GB scalable upto 256 GB or above | DDR4 or above 64 GB scalable upto 256 GB or above. | Cost Optimization & availability |
| 4.2 | 4th Generation Intel® Xeon Gold 2.4 GHz or equivalent, 24 Core or higher. Each servers with minimum 1 Processors and 2 | 3rd Gen Intel Xeon Gold 2.1GHz , 24Cor or equivalent. Each servers with minimum 1 Processors and 2 Socket or higher as required | Cost Optimization & availability |

| | | | |
|---|---|---|---|
| | Socket or higher as required. | | |
| | DDR5-4800 or higher, 128 GB scalable upto 512 GB | DDR4 or above 64 GB scalable upto 256 GB or above. | Cost Optimization & availability |
| 3.5 | Additional clause | 10% of CAPEX PBG shall be returned on completion of warranty and upon submission of 10% of OPEX PBG for O&M period. PBG during O&M shall be submitted each year for the remaining value of OPEX. | More Clarification w.r.t PBG to facilitate milestone based cashflow. |
| 4.8.1 b) | WAN Interface<br>2 x 10 Gigabit Ethernet SFP+ ports | 2 x 1G Gigabit Ethernet SFP+ ports. | Cost Optimization & internet backhaul limitation |
| 4.7 | Network Management System (Single interface for RAN, Core, MEC, connected devices etc.) | Network Management System (Single interface for RAN, Core, and MEC). | Optimization of the Link |
| 4.15.2 e) | NavIC or GPS location support | NavIC or GPS location support (optional) | Non- Availability of the features |
| 4.15.2 f) | Additional clause | In-built 5G Modem (optional) | Non-Availability of the features |

I/3129640/2023

**Annexure-IV**

## 3.1 TAXES AND DUTIES

3.6.1 All rates shall be inclusive of applicable taxes and duties.

3.6.2 Where no specific mention of the taxes chargeable is made by the bidder, it shall be understood that these taxes are either not applicable or being paid at source by the bidder, or it shall be borne by him. Subsequent claim from the bidder/contractor for payment of these taxes shall not be entertained.

3.6.3 Any change in the rate quoted on account of imposition of new tax/duty or increase in the existing tax/duty where percentage of applicable tax/duty is specifically mentioned in the financial bid, the same shall be reimbursed only on production of valid documents proving increase in such duties/taxes by Competent Authority. However, in case there is decrease in rate of duties/taxes it shall be obligatory upon the successful bidder to inform the same to the Purchaser and reduce the rate accordingly failing which it shall be treated as breach of conditions of the contract.

3.6.4 Any reference regarding variation in Govt., taxes, duties, octroi or any other levies shall not be entertained subsequently in case break up is not clearly mentioned. Any form seeking exemption directly by supplier shall not be provided.

I/3142609/2023

**Department of Telecommunications**
Standards-R&D-Innovation
(*Project Monitoring Unit*)
****

Sanchar Bhavan, 20 Ashoka Road,
New Delhi-110001

**No.** 1-TR_5G_Startup_Kit/2022-SRI **Dated: 4th** Dec, 2023

**OFFICE MEMORANDUM**

**Sub: 100 5G Labs Project by DoT through TCIL under TDIP scheme - Budgetary outlay, payment structure, needful compliance by Implementation Agency(TCIL) for timely installation and optimal utilization of 5G Labs-reg**

**Ref:**

i. OM on Implementation guidelines No. 1-TR_5G_Startup_Kit/2022-SRI, dt 27.03.2023 & Modified SOP No. 1-TR_5G_Startup_Kit/2022-SRI, dt 23.07.2023
ii. Approved Record of the Discussion chaired by Secretary(T) "No. 1-TR_5G_startup_ Kit/2022-SRI", dt 17th July, 2023
iii. TCIL submissions and other MoMs reg price discovery, technicalities, and budget requirements, RFPs floated by TCIL dt. 05.04.2023 and dt. 21.09.2023, MoM of Technical Evaluation Committee dt 30.09.2023 finalizing the Lab equipment items. RFPs floated by TCIL RFPs floated by TCIL dt. 05.04.2023 and dt. 21.09.2023.
iv. MoM chaired by Secretary(T) regarding 100 5G labs portal update held on 07.11.2023
v. DO letter to DG NTIPRIT by Member(T) regarding 5G Awareness workshops.
vi. F.No: 1-TR_5G_Startup_Kit/2022-SRI approved by competent authority

**Background**: In line with the budget 2023 announcement, to build competencies and engagement in 5G technologies for students & startup communities, as per the approved implementation guidelines (under ref.), DoT through TCIL (Implementation Agency) is setting up 100 5G Use case labs to facilitate the development of applications using 5G services in various socio-economic verticals. TCIL was nominated as the Implementation Agency(IA) and SRI division has been designated as PMU. 100 Institutions spanning across all 28 states were shortlisted by the Inter-ministerial selection committee chaired by DG NTIPRIT and the 5G labs were awarded by **Hon'ble PM** during the inaugural ceremony of Indian Mobile Congress on 27th October, 2023. TCIL had shortlisted indigenous vendors through a bidding process under the guidance of the Technical Evaluation Expert Committee headed by Sr.DDG(TEC).

**2. Revised Budget:** The Competent authority had given approval of Rs. 82.77 Cr. (**97.67Cr** inclusive of GST) for the Revised Budgetary outlay excluding the institutions contributions Cr for the 100 5G Lab's project in line with tender discovered rates (L1 cost submitted by TCIL & accepted by 4 Vendors) and optimal utilization of 5G labs by Institutions, faculty, students, Start-ups. The revised budget & cost components for the 100 5G Labs project are attached as Annexure-I and details of the tender discovered rate (L1 price) & shortlisted vendors are attached as Annexure-II. All the expenditure towards the project will be made available to TCIL in advance through the TDIP fund for

each milestone on completion of the previous milestone as per the procedure defined in the Implementation Guidelines.

**3. 100 Shortlisted Institutions**: Upon communication by TCIL, 99 institutions among 100 shortlisted and 1 among the waitlisted have submitted their acceptance letters/undertakings from the Head/Director with the necessary commitments. The final 100 Institutions List is attached as Annexure-III.

In line with the GoI **"Act Fast for Northeast"** policy, taken into consideration the difficult terrain and accessibility to the institutions, an additional O&M cost of Rs. 50K per Lab per annum will be paid to the allotted vendors (zone) over and above the standard O&M price. This will encourage the vendors to provide continuous operational support to 5G labs of Institutions in North Eastern(NE) region. The institutions of Jammu & Kashmir will also be given similar additional cost of Rs. 50K per lab per annum. In case new labs are adopted/established in NE and J&K, the same additional cost will be made applicable to them. . The 13 institutions which will be provided this additional support is attached as Annexure-IV

**4. Institutions Contributions**: To meet the Project milestones in a timebound manner without any unwarranted delays, initially DoT will be funding 100% Capital Expenditure (CAPEX) (instead of 80% CAPEX) span across FY 2023-2024 & FY 2024-25.

Further, the institutions will pay 20% of CAPEX (Institution contribution) to TCIL and this amount will be used for the OPEX/O&M cost of the Labs starting from FY 2025-26. The institutions will be given considerable time (up to 9 months from the date of the agreement with TCIL) to pay their contribution (20% of the CAPEX) to TCIL. [As per Approved Record of the Discussion chaired by Secretary(T) dt 17.07.2023 and Modified SOP dt 23.07.2023 under reference]

Provisioning of the revised budget is made by considering the 20% CAPEX i.e Rs. 12.08 Cr (14.25 Cr inclusive of 18% GST) from institutions. In case TCIL is unable to receive the amount by institution/s due to any unforeseen circumstances, TCIL will utilize the earmarked OPEX/Misc funds to keep the Labs operational and suitable provisions will be made through revised budget estimates for FY 2025-26 with the approval of competent authority.

**5. 100 5G Labs Portal (Digital Network of 100 Labs)**: Domain names for the 100 5G labs portal are Bharat5glabs.gov.in (serving the current needs), Bharat5g.gov.in & Bharat6g.gov.in considering the broader scope of the project. The Bharat5glabs portal is envisaged to be a one-stop solution for all the 5G/6G/Adv. Comm.-related works capturing Academic, R&D developments, Standards, Industry, OEMs, Startups/MSMEs, Students, subject matter experts, etc. Also, other Digital services-related State portals like the TN industry 4.0 portal will be published and linked through this portal. As the portal grows in the due course of time, data from the Bharat Digicom portal (https://www.bharatdigicom.in/) managed by TCOE will be integrated into the 5G Labs portal and the Bharat Digicom portal shall be discontinued from March'2024. Also, other upcoming facilitating portals such as IPR facilitation for Startups & MSMEs (C-DOT) and Telecom PoC/Pilots for Startups & MSMEs (BSNL) will also be suitably made available in the Bharat5G portal. [As per Scope of the portal outlined in implementation guidelines & MoM chaired by Secretary(T) under ref]

**6. Zone-wise & State-wise Hackathons**: Hackathons are an effective way to ensure optimal utilization of labs and maximize the students/start-ups/Faculty participation to address real on-ground issues leveraging 5G & beyond technologies. Therefore, a provision of Rs. 2Cr per year i.e. 50L per zone (Rs. 2.36 Cr inclusive of GST) has been made for conducting Hackathons (including motivational incentives for winners etc.) starting from FY 2024-25. The 100 Institutions will be divided into 4 zones

i.e 25 Institutions per zone and a detailed SOP for conducting hackathons will be prepared by TCIL in consultation with the concerned stakeholders

**7. 5G workshops/Training & adoption of Labs for mentoring:** TCIL may extend necessary support to DoT/NTIPRIT/LSAs/NICF etc.

(i) for conducting 5G workshops/training,

(ii) adoption of Labs by TSPs /OEMs,

(iii) for mentoring & ranking of institutions to encourage the utilization of labs,

(iv) periodic updating of Bharat5glabs portal and

(v) adaption of best practices by institutions for developing innovative use cases.

**8. Payment structure to Implementation Agency (TCIL) in line with Allotted Budget for 100 5G Labs Project:**

**A) FY 2023-24:** As per the procedure outlined in modified SOP dt 23.07.2023 and Record of Discussion chaired by Secretary(T) under reference, the first instalment of Rs 30.5 Cr (Rs. **36** Cr inclusive of GST**)** under TDIP fund will be released to TCIL (Approx. 50 % of the CAPEX) in advance to start the process of procurement and installation of labs by selected 4 Vendors (refer Annexure). Subsequent to the release of Purchase Orders (POs) by TCIL to 4 vendors, amount of **Rs 18.14 Cr** ( Rs **21.4Cr** inclusive of GST) i.e approx. 25% of the CAPEX + 3Cr Administrative charges will be released to TCIL in the Last Quarter of FY 2023-24 (upon submission of Lab equipment delivery/installation status). TCIL shall ensure timely payments to vendors (as per the payment norms mentioned in the RFP including PBG) without any additional BG/financial burden on the vendors so that vendors can expedite the procurement of necessary equipment/line items of 5G Lab to meet the project deliverables in the timebound manner. , This is in line with the RFP objectives to encourage Start-ups/MSMEs (Vendors) in supplying various competitive Indigenous 5G technologies for Lab setup.

**B) FY 2024-25 to FY 2028-29**

i. For the subsequent period of project duration (FY 2024-25 to FY 2028-29), funds will be released as per the allotted budget in advance to TCIL based on the fund requirements submitted by TCIL. This submission should include the previous year's fund utilization certifications/Statement of Expenditure, consolidated Lab utilization certificate (once operational) and Hackathon impact assessment (if any) for consideration by the competent authority. [in line with SOP and implementation guidelines under reference]
ii. In FY 2024-25, an amount of Rs 2Cr (Rs 2.36Cr inclusive of GST) for O&M/Miscellaneous cost component will be paid in advance during Q1 of FY 2024-25. This advance payment is intended to support oversight and coordination activities, as defined in the attached Annexure-I, necessary for the operationalization & functional testing of the Labs. However, the remaining amount of Rs 16.13 Cr (Rs. 19.03 Cr inclusive of GST) -comprising 25% of CAPEX and 1Cr administrative charges will be disbursed to TCIL upon submission of the project completion certificate (training and completion of Stability period as per the tender)
iii. Furthermore, the Hackathon cost component of Rs 2 Cr (Rs. 2.36 Cr inclusive of GST) will be paid annually in advance to TCIL upon submission of the roadmap/procedure for conducting hackathons with an objective for optimal utilization of Labs by students and start-ups/MSMEs, among others in line with objectives & deliverables of the program.
iv. During the submission of fund requirements for FY 2025-29, TCIL shall include

20% CAPEX (14.25 Cr inclusive of 18% GST) received from institutions in FY 2024-2025.

v. Suitable provisions for additional funds under the Miscellaneous component (details at Annexure) were already made in the budget, However, in case TCIL requires additional funds beyond the allotted budget due to any unforeseen circumstances (including cost overruns, delays in institution payments to TCIL, additional Labs establishment, adaptation of new labs, or overrun in coordination activities etc.), the same will be considered and suitably provisioned through Revised Estimates from FY 2025-26 with the approval of Competent Authority.

vi. Based on the availability of funds corresponding to different components of the Budget, TCIL as per the requirement of the project through internal administrative approval can utilise funds allocated to various components interchangeably (especially for oversight & co-ordination perse) without exceeding the overall & individual limit to meet the milestones about timely installation, testing, operationalization, utilization of 5G Labs (Components of the Budget are outlined in the Annexure-I) the same may be recorded in the periodic fund utilization certifications/Statement of Expenditure certificates submitted by TCIL.

vii. Periodic fund utilization certifications/Statement of Expenditure certificates submitted by TCIL need to be duly vetted by TCIL internal committee with finance member in it.

viii. For the Budget allocation standard 18% GST is considered, however claim will be on actual rates as applicable from time to time.

**9.** The following will be ensured by TCIL for timely installation, operationalization and utilization of 5G Labs:

i. Adherence to the following rollout Timelines:

a. Installation & operationalization of 2 Labs (Up & running) by December 2023 per each shortlisted vendor (to enable pre-testing of features by TCIL/DoT team & further guide the vendors)
b. Delivery and installation of the Equipment (remaining) by February 2024
c. Operationalization of all the Labs by March 2024.

In line with the tender timelines, TCIL will make its best efforts to ensure that the four Vendors will meet the deliverables in a time-bound manner.

ii. Award of Work/Issuance of PO to the shortlisted (4) vendors as per the approved L1 cost (discovered through GeM price bid) without any further delay to meet the timelines of the project.

iii. MoU/Agreement needs to be signed by TCIL with DoT with all the necessary modalities clearly defining the roles and responsibilities of all the stakeholders (incl. Institutions, vendors, NTIPRIT, TEC etc.), Terms & Conditions, Milestones, rollout timelines (Delivery, Installation, Operationalization of Labs), dedicated manpower, allotted revised budget payment modalities, corresponding milestones etc. Draft MoU/Agreement between DoT and TCIL shall be submitted by TCIL at the earliest for consideration and approval by the competent authority.

iv. As per the RFP, Zone wise allocation (25 Labs each) needs to be allotted to 4 Vendors (based on preferences of L1, L2, L3, and L4). It shall be ensured that while allocating labs to the 4 vendors based on their preferences, 4 institutions in Delhi region may be allotted among 4 vendors (to enable ease of demo visits by Senior officers of DoT-HQ/other Ministries). Later these labs may be transformed into Model Labs.

v. The Rollout Roadmap ( Zone-wise) needs to be submitted by TCIL in coordination with 4 vendors with the commitment to timelines (as mentioned in Para 9-i) .and the same shall be part of the MoU/Agreement.

vi. Also, agreements between TCIL and shortlisted Institutions (100) shall include all necessary clauses regarding roles & responsibilities of the institution including 20% CAPEX contribution, Space, Ownership, Safety & security of equipment, tech-manpower for local maintenance, support to DoT/TCIL/NTIPRIT/LSAs etc. for

I/3142609/2023

organizing Hackathons, 5G workshops/Trainings and other required commitments for ensuring successful installation and utilization of Lab equipment by faculty, students & Startups to meet the objectives & deliverables of the 100 5G Lab's program in line with the Budget Announcement 2023-24. [as per Implementation guidelines, DO letter by M(T) to NTIPRIT under reference]

vii. Empanelment of PoC-Qualified vendors/OEMs/Solution-providers for exploring opportunities for 5G & Beyond.

viii. Timely Allocation of dedicated human resources preferably zone wise & centrally regarding needful oversight, monitoring and coordination activities (field visits for checking the installation status, functional testing of features & roadmap Sep 2024 compliance, acceptance testing, Travel, Logistics, on-ground support to NTIPRIT/DoT field units (LSAs/CCAs/WMOs) for holding Hackathons, workshops, recording of best practices & use cases and updating the same in portal, expertise support, Periodic inspections, extended support to NE, J&K institutions etc. as defined in Annexure-I) for ensuring timely installation, testing, operationalization, optimal utilization of Labs.

ix. Launch of Bharat5glabs (100 5G labs) portal-

a. with all the information, activities related to 100 5G Lab's portal connecting all the stakeholders as per the defined scope in guidelines & SOP (DoT-monitoring, Institutions- updating the info, common public, etc.),
b. the broader scope given by Secretary(T)- building portal as a one-stop platform for all the 5G/6G/Adv. Comm.-related works capturing- Academia/Standards /OEMs/ Startups-MSMEs, students, subject matter experts, and
c. to host DoT facilitation initiatives such as IPR facilitation for Startups & MSMEs (C-DOT) and Telecom PoC/Pilots for Startups & MSMEs (BSNL), Telecom Startups & MSMEs facilitation cell etc.

**10.** This is issued with the approval of the Competent Authority

Encl:

- Annexure-I: Revised budget outlay & cost components for the 100 5G Labs project
- Annexure-II: Tender discovered rate (L1 price) & shortlisted vendors
- Annexure-III: Final 100 Institutions List
- Annexure-IV: 13 institutions list from North East and J&K attached

Signed by A Alex Vikas
Date: 04-12-2023 19:23:45
Reason: Approved

(A. Alex Vikas)
Asst. Director General(S-R-I)
Mail: alex.vikas17@gov.in
Tel No: 011- 2337207

**To:**
CMD (TCIL)

**Copy to:**

1. PSO to Secretary (Telecom) & Chairman DCC - for kind information
2. Member(T)/Member(S)/Member(F)/AS(T)
3. DDG(SRI), DoT HQ
4. Director(P)/ED, TCIL
5. GM(Digital Transformation), TCIL for needful compliance please

I/3142609/2023

**Annexure-I**

**Revised Budget & Cost components for 100 5G Labs project:**

| S.No | Type/Year | Y1 | Y2 | Y3 | Y4 | Y5 | Y6 |
|---|---|---|---|---|---|---|---|
| | | 2023-24 | 2024-25 | 2025-26 | 2026-27 | 2027-28 | 2028-29 |
| 1 | Capital Cost(CAPEX) in Rs Cr | 45.65 | 15.12 | - | - | - | - |
| 2 | O&M Cost (OPEX)/ Miscellaneous in Rs Cr | - | 2 | 2 | 2 | 2 | 2 |
| 3 | Administrative expenses Rs. Cr | 3 | 1 | - | - | - | - |
| 4 | Hackathon/ Motivational incentives Rs. Cr | - | 2 | 2 | 2 | 2 | 0 |
| 5 | Total (Excl. GST) Rs. Cr | 48.65 | 20.12 | 4 | 4 | 4 | 2 |
| 6 | Total (incl. 18% GST) Rs. Cr | 57.40 | 23.75 | 4.72 | 4.72 | 4.72 | 2.36 |
| Total Budgetary outlay (excluding GST): **82. 77 Cr** (60.77 Cr for CAPEX+ 10 Cr for 4 years CAPEX/Miscellaneous + 8 Cr for Hackathons + 4 Cr Administrative charges) | | | | | | | |
| Total Budgetary requirements (including 18% GST): **97.67** Cr | | | | | | | |

The components of the funding are as below:

1. **CAPEX cost** includes the following:

- 5G Lab Infrastructure & services (CAPEX) discovered through tender by TCIL (60.4 Cr exclusive of Taxes)
- CAPEX costs towards dedicated portal development & hosting by TCIL-In line with the scope as defined in the implementation guidelines under reference & MoM chaired by Secretary(T) regarding 100 5G labs portal update held on 07.11.2023. (0.37Cr exclusive of Taxes)

2. **OPEX/Misc. costs** for 5 years includes the following :

- O&M of Lab Infrastructure discovered through tender by TCIL (by considering 20% CAPEX to be paid by institutions to TCIL that to be utilised for O&M)
- Portal O&M, needful upgradations (online applications for Hackathons, workshops, best practices, DoT initiatives supporting academia, Start-ups & MSMEs etc.), Management
- Additional funding of Rs.6Cr (IA Miscellaneous) is for the following

a. Adjustment of O&M costs if there is a delay in Institution payment (20% CAPEX – to the tune of 12.08 Cr exclusive of Taxes )

b. Any additional Labs need to be established (refer to para II of Approved Record of the Discussion chaired by Secretary(T) dt 17.07.2023)

c. Oversight and coordination for O&M and utilization by IA: Ensuring the Installation & optimal utilization of Labs - Field visits for checking the installation status, functional testing of features & roadmap (Sep 2024) compliance, Acceptance testing, Travel, Logistics, on-ground support to NTIPRIT/DoT field units (LSAs/CCAs/WMOs) for holding Hackathons, workshops, recording of best practices & use cases and updating the same in portal, additional resources (incl. expertise), Periodic inspections, Extended support to NE, J&K institutions etc.

d. Any additional OPEX requirements for adaptation of 5G labs (incl. NEC 5G Labs) if required

- For any given financial year if funds are exhausted by TCIL for the aforementioned purposes suitable provisions will be made in the Revised estimates with the approval of competent authority

**3. Administrative charges**: As per the Implementation guidelines (OM dt 27.03.2023 under reference) administrative charges of 4Cr (4.72 Cr inclusive of GST) will be paid to TCIL during the Installation & commissioning phase of the Labs therefore under the proposed budget 3cr (3.54 Cr inclusive of GST) will be paid to TCIL given in FY 2023-2024 and 1 Cr (1.18 Cr inclusive of GST) will be paid in FY 2024-2025 upon submission of Project completion certificate by TCIL i.e post to Installation and operationalization of 100 Labs.

**4. Zone-wise & State-wise Hackathons**: Provision of Rs. 2Cr per year i.e Rs. 50Lakhs per zone (Rs. 2.36 Cr inclusive of GST) has been made for conducting Hackathon (including Motivational incentives - cash awards etc.)

I/3142609/2023

**Annexure-II**

**Tender discovered cost (L1 Price) & Shortlisted Vendors:**

TCIL under the guidance of the Technical Evaluation committee headed by Sr. DDG(TEC) having members from TEC, C-DOT, Academia(IISc Bangalore), Industry, TCIL, DoT have finalized 4 Indian vendors (Expert Committee with members from TEC, C-DOT, Academia, Industry is formed with members formed as part of Implementation guidelines). As outlined in the SOP under reference, Vendor selection process includes the following

- Floating of technical-RFP with broader technical specs (RFP No: TCIL/IT&T-II/5G Labs/Vendor Empanelment Date of Issue: 05.04.2023) in TCIL website & CPP portal inviting indigenous 5G OEMs (esp. Start-ups & MSMEs)/consortium of OEMs/ System Integrator- (RFP No: TCIL/IT&T-II/5G Labs/Vendor Empanelment Date of Issue: 05.04.2023 )- **21 bids** have been received.

- Preliminary shortlisting based on the eligibility criterion especially based on the supply of Indian Core & RAN equipment etc. and PoC testing were conducted by the testing teams comprising TCIL, DoT, TEC, IITD, IISc, IITB etc. at locations Delhi, Bangalore, Mumbai & Chennai to validate the key working features of 5G **– 14 vendors** were shortlisted in this process.

- Limited Financial RFP with detailed technical specs on 5G Equipment & use case application- devices (No-optional) have been floated through GeM Portal (RFP No GEM/2023/B/ 3990770 dt 21.09.2023 )

- 4 Vendors (L1, L2, L3, L4) have been shortlisted based on cost competitiveness and these vendors are also offering different Core & RAN combinations (part of Tender conditions & SOP)

**L1 price discovered** by TCIL through EoI No. GEM/2023/B/3990770 dated 21.09.2023 for Calling of Price Bid from Shortlisted OEM/ their authorized distributors / System integrators (SIs) in the field of 5G for deployment of 5G use case labs is as follows.

| S.No | Fund aspect | INR (1 Lab) | INR (100 Labs) | INR (100 Labs with 18% GST) |
|---|---|---|---|---|
| 1 | CAPEX | 60,37,617 | 60,37,61,700 | 71,24,38,806 |
| 2 | OPEX (4 years) | 14,75,661 | 14,75,66,100 | 17,41,27,998 |
| 3 | Total | 75,13,278 | 75,13,27,800 | 88,65,66,804 |
| | | | 75.13278 Cr | 88.6566 Cr |

TCIL informed that all Four Bidders in the Price bid ranking have confirmed for supply of equipment **as per L1 rates.** The final selected vendors are as below.

| S.No | Qualified Bidder | |
|---|---|---|
| 1 | Coral Telecom | L1 |
| | | |

I/3142609/2023

| | | |
|---|---|---|
| 2 | Signaltron Systems Pvt Ltd | L2 |
| 3 | VVDN Technologies Pvt Ltd. | L3 |
| 4 | Wireless 4 Scale Laboratory Private Limited | L4 |

Waitlisted vendors are as below :

| | | |
|---|---|---|
| 5 | Meridian Infotech Limited | L5 |
| 6 | Amantya Technologies Private Limited | L6 |

I/3142609/2023

**Annexure-III**

| **List of 100 5G Lab Institutions** | | |
|---|---|---|
| **S.No.** | **Name of institute** | **State** |
| 1 | Indian Institute of Information Technology Design and Manufacturing Kurnool | Andhra Pradesh |
| 2 | Koneru Lakshmaiah Education Foundation | Andhra Pradesh |
| 3 | Vignans Foundation for Science Technology and Research | Andhra Pradesh |
| 4 | GITAM Institute of Technology and Management, Deemed to be University | Andhra Pradesh |
| 5 | Indian Institute of Technology Tirupati | Andhra Pradesh |
| 6 | National Institute of Technology Arunachal Pradesh | Arunachal Pradesh |
| 7 | Indian Institute of Technology Guwahati | Assam |
| 8 | National Institute of Technology Silchar | Assam |
| 9 | National Institute of Technology Patna | Bihar |
| 10 | Indian Institute of Technology Patna | Bihar |
| 11 | Dr Rajendra Prasad Central Agricultural University | Bihar |
| 12 | Punjab Engineering College Deemed to be University | Chandigarh |
| 13 | University Institute of Engineering and Technology Panjab University | Chandigarh |
| 14 | IIT Bhilai | Chhattisgarh |
| 15 | National Institute of Technology Raipur | Chhattisgarh |
| 16 | DELHI TECHNOLOGICAL UNIVERSITY | Delhi |
| 17 | Indraprastha Institute of Information Technology Delhi | Delhi |
| 18 | JAMIA HAMDARD | Delhi |

| | | |
|---|---|---|
| 19 | Indian Agricultural Research Institute | Delhi |
| 20 | Birla Institute of Technology and Science, K. K. Birla Goa Campus | Goa |
| 21 | National Institute of Technology Goa | Goa |
| 22 | Sardar Vallabhbhai National Institute of Technology | Gujarat |
| 23 | School of Cyber Security and Digital Forensics National Forensic Sciences University Institute of National Importance Ministry of Home Affairs GoI | Gujarat |
| 24 | Indian Institute of Technology Gandhinagar | Gujarat |
| 25 | Central University of Haryana | Haryana |
| 26 | National Power Training Institute | Haryana |
| 27 | National Institute of Technology Hamirpur | Himachal Pradesh |
| 28 | Indian Institute of Technology Mandi | Himachal Pradesh |
| 29 | Institute of Technology, University of Kashmir | Jammu and Kashmir |
| 30 | Indian Institute of Technology Jammu | Jammu and Kashmir |
| 31 | Birla Institute of Technology, Mesra | Jharkhand |
| 32 | Indian Institute of Technology, Dhanbad. | Jharkhand |
| 33 | National Institute of Technology Jamshedpur | Jharkhand |
| 34 | Siddaganga Institute of Technology | Karnataka |
| 35 | Indian Institute of Information Technology Dharwad | Karnataka |
| 36 | BMS College of Engineering | Karnataka |
| 37 | Indian Institute of Science | Karnataka |
| 38 | VTUS Centre for Post Graduate Studies ,Kalaburgi | Karnataka |
| 39 | Manipal Institute of Technology | Karnataka |
| 40 | M. S Ramaiah Institute of Technology | Karnataka |

| | | |
|---|---|---|
| 41 | National Institute of Technology Karnataka, Surathkal | Karnataka |
| 42 | Department of Electronics, Cochin University of Science and Technology | Kerala |
| 43 | National Institute of Technology Calicut | Kerala |
| 44 | Indian Institute of Space science and Technology | Kerala |
| 45 | Indian Institute of Technology Palakkad | Kerala |
| 46 | PDPM Indian Institute of Information Technology, Desing and Manufacturing Jabalpur | Madhya Pradesh |
| 47 | Indian Institute of Technology Indore | Madhya Pradesh |
| 48 | Atal Bihari Vajpayee Indian Institute of Information Technology and Management, Gwalior, Madhya Pradesh | Madhya Pradesh |
| 49 | Maulana Azad National Institute of Technology, Bhopal, M.P. | Madhya Pradesh |
| 50 | DPU, Dr D Y Patil School of Science and Technology | Maharashtra |
| 51 | Symbiosis Institute of Technology, Pune | Maharashtra |
| 52 | Dr P M Padole<br>Director, VNIT Nagpur, Mob: 9822220713 | Maharashtra |
| 53 | Dr. Babasaheb Ambedkar Technological University, Lonere | Maharashtra |
| 54 | COEP Technological University | Maharashtra |
| 55 | Government College of Pharmacy Amravati | Maharashtra |
| 56 | DKTEs Textile and Engineering Institute, Ichalkaranji | Maharashtra |
| 57 | National Institute of Technology Manipur | Manipur |
| 58 | Indian Institute of Information Technology Senapati, Manipur | Manipur |
| 59 | National Institute of Technology Meghalaya | Meghalaya |
| 60 | National Institute of Technology Mizoram | Mizoram |
| 61 | National Instiutue of Technology Nagaland | Nagaland |
| 62 | Christian Institute of Health Sciences and Research | Nagaland |

| | | |
|---|---|---|
| 63 | Indian Institute of Technology Bhubaneswar | Odisha |
| 64 | Siksha O Anusandhan Deemed to be University | Odisha |
| 65 | Kalinga Institute of Industrial Technology Deemed to be University | Odisha |
| 66 | National Institute of Technology Rourkela | Odisha |
| 67 | National Institute of Technology Puducherry | Puducherry |
| 68 | Pondicherry University | Puducherry |
| 69 | Dr B R Ambedkar NIT Jalandhar | Punjab |
| 70 | Thapar Institute of Engineering and Technology | Punjab |
| 71 | Guru Nanak Dev University, Amritsar | Punjab |
| 72 | Indian Institute of Technology Ropar | Punjab |
| 73 | Banasthali Vidyapith | Rajasthan |
| 74 | Birla Institute of Technology and Science Pilani | Rajasthan |
| 75 | Indian Institute of Technology Jodhpur | Rajasthan |
| 76 | Malaviya National Institute of Technology Jaipur | Rajasthan |
| 77 | MBM University, Jodhpur | Rajasthan |
| 78 | Sikkim Manipal Institute of Technology | Sikkim |
| 79 | National Institute of Technology Tiruchirappalli | Tamil Nadu |
| 80 | Vellore Institute of Technology | Tamil Nadu |
| 81 | College of Engineering Guindy | Tamil Nadu |
| 82 | SASTRA Deemed University | Tamil Nadu |
| 83 | Sri Sivasubramaniya Nadar College of Engineering | Tamil Nadu |
| 84 | Saveetha Institute of Medical and Technical Sciences | Tamil Nadu |

I/3142609/2023

| | | |
|---|---|---|
| 85 | SRM Institute of Science and Technology | Tamil Nadu |
| 86 | IIIT Hyderabad | Telangana |
| 87 | National Institute of Technology Warangal | Telangana |
| 88 | University of Hyderabad | Telangana |
| 89 | Tripura Institute of Technology | Tripura |
| 90 | Jaypee Institute of Information Technology, Noida | Uttar Pradesh West |
| 91 | Motilal Nehru National Institute of Technology Allahabad | Uttar Pradesh East |
| 92 | Indian Institute of Information Technology Allahabad | Uttar Pradesh East |
| 93 | Aligarh Muslim University | Uttar Pradesh West |
| 94 | Indian Institute of Technology Kanpur | Uttar Pradesh East |
| 95 | Indian Institute of Technology BHU | Uttar Pradesh East |
| 96 | College of Technology, Govind Ballabh Pant University of Agriculture and Technology | Uttarakhand |
| 97 | THDC Institute of Hydropower Engineering and Technology | Uttarakhand |
| 98 | Indian Institute of Technology Roorkee | Uttarakhand |
| 99 | National Institute of Technology Durgapur | West Bengal |
| 100 | Indian Institute of Information Technology, Kalyani | West Bengal |

I/3142609/2023

**Annexure-IV**

| **5G Use Case Labs : J&K & NE INSTITUTES** | | |
|---|---|---|
| **S.No.** | **Name of institute** | **State** |
| 1 | National Institute of Technology Arunachal Pradesh | Arunachal Pradesh |
| 2 | Indian Institute of Technology Guwahati | Assam |
| 3 | National Institute of Technology Silchar | Assam |
| 4 | Institute of Technology, University of Kashmir | Jammu and Kashmir |
| 5 | Indian Institute of Technology Jammu | Jammu and Kashmir |
| 6 | National Institute of Technology Manipur | Manipur |
| 7 | Indian Institute of Information Technology Senapati, Manipur | Manipur |
| 8 | National Institute of Technology Meghalaya | Meghalaya |
| 9 | National Institute of Technology Mizoram | Mizoram |
| 10 | National Instiutue of Technology Nagaland | Nagaland |
| 11 | Christian Institute of Health Sciences and Research | Nagaland |
| 12 | Sikkim Manipal Institute of Technology | Sikkim |
| 13 | Tripura Institute of Technology | Tripura |

**Ministry of Communication**
**Department of Telecommunications**
Standards-R&D-Innovation Division
(*PMU- 100 5G Labs project)*
****

1405, Sanchar Bhavan, 20 Ashoka Road,
New Delhi-110001

**No.** 1-Role of LSAs(100 5G labs)/2024-SRI **Dated: 15th Feb, 2024**

## OFFICE MEMORANDUM

**Sub: Role of DoT field units(LSAs) reg 100 5G Labs project & Beyond**

**Ref:**

i. OM on Assignment of roles to LSA field Units No. 1-1/2020-DGT/1 dt 28.02.2020 reg engaging with DCT development ecosystem
ii. Meeting chaired by Member(T) & DG(Telecom) with LSA heads dt. 13.12.2023
iii. TSUM facilitation cell OM- No. 1_TSUM/2022-SRI dt. 19.08.2023 and Sanchar Mitra Scheme OM- No. MEDIA-SANCHAR SAATHIAWARENESS-2023 dt 27.12.2023
iv. DO letters by Member(T) to NTIPRIT for organizing workshops
v. DO letters by Member(T) to TSPs/OEMs for adaption of 5G Lab institutions

1. **Background**: *Hon'ble Prime Minister*, during the inaugural ceremony of Indian Mobile Congress on October 27, 2023, awarded 100 "5G Use Case Labs" to educational institutions across the countries with an objective to build competencies and engagement in 5G technologies for students & start-up communities. Setting up of labs is an endeavour to realise the opportunities associated with 5G technology and to facilitate the development of applications using 5G services in various socio-economic verticals.
2. SRI Division, DoT-HQ has been designated as Project Management Unit (PMU) and TCIL as the Implementing Agency (IA). TCIL has awarded the supply and installation work to indigenous vendors through a competitive bidding process. Efforts are underway to operationalise the Labs by March 2024. In addition, Zonal/State- Level 5G workshops are also being organised to facilitate capacity building.
3. All the 100+ Labs/institutions are connected through a dedicated portal (https://bharat5glabs.gov.in/), The portal also acts as a knowledge dissemination platform for institutions/students/start-ups. Bharat 5G Labs platform is envisaged to be a one-stop solution for all the 5G/6G/IPR/ Quantum-related works capturing academic R&D developments/Standards

Industry ecosystem etc. It also hosts experts registration, 6G R&D proposals, IPR facilitation, PoCs/Pilots etc.

4. In this regard, the undersigned has been directed to intimate that roles & responsibilities of DoT Field units(LSAs) have been identified reg 100 5G Labs & Beyond and the same has been attached as Annexure-I.
5. DDGs(Technologies) from each LSAs shall be the focal point to carry out the activities (Annexure-I). DDG (Technology), DGT HQ shall be nodal officer for overall co-ordination with LSAs and S-R-I, Division DoT HQ [ as per OM dt 28.02.2020- under reference]
6. Further, the DoT fields units (Annexure-II) are hereby requested to take necessary action at their end for fulfilling the roles and responsibilities assigned to them.

This is issued with the approval of Member(Technology) & DG(Telecom)

Encl: As Above

**Signed by A Alex Vikas**
**Date: 15-02-2024 20:28:37**

(A. Alex Vikas)
Mission Co-ordinator(TSUM)/ Asst. Director General(SRI)
Mail: alex.vikas17@gov.in
Tel No: 011- 2337207

To:

1. All Heads of LSAs, DoT.
2. DDGs(Technology) of All LSAs, DoT.

Copy to:

1. PSO to Secretary(Telecom) & Chairman DCC- for information please
2. Member(Finance)Member(Technology)/Member(Services)- for information please.
3. DG (Telecom)/DG(NTIPRIT)/DG(NICF)/CGCA
4. Sr.DDG(TEC) for information please
5. Wireless Advisor, WPC/ Director WMO/ JWAs of respective RLOs
6. CMD (TCIL) for needful support to LSAs
7. DDG(Technology), DGT HQ for needful co-ordination & uploadin
8. DDG (Administration & HR), DGT HQ
9. DDG(Media) for information please.
10. DoT Nodal for uploading OM in the DoT website

**Annexure-I**

| S.No | Activity | Role of LSA |
|---|---|---|
| 1 | Agreement with Institutions (by TCIL) | Facilitation. |
| 2 | Installation/Commissioning – 100 5G Labs | Facilitation, due observation and necessary intervention. |
| 3 | User Acceptance Testing – 1005G Labs | Part of testing team |
| 4 | Hackathon (LSA Wise/State Wise)- 100 5G Labs | Coordination with stakeholders and extending necessary support including publicity & logistics |
| 5 | Workshops/Trainings for Institutions- 100 5G Labs | Organizing the workshops at LSA/State level and coordination with NTIPRIT for zonal level workshops including logistics support |
| 6 | Capacity Building sessions for LSAs by SRI | Proactive participation |
| 7 | Mentoring/Adoption of Institutes by OEMs/TSPs/CoEs | Coordinate with Institution, TCOE and OEMs/TSPs/CoEs |
| 8 | ITU Academia Membership | Encouraging India's academia membership in ITU |
| 9 | Sanchar Mitra Scheme/Sanchar Saathi volunteers | i. Selection of students in Co-ordination with Institutions.<br>ii. Guidance for activities of Sanchar Mitra volunteers in co-ordination with Institutions, LEAs, State-gov departments etc. |
| 10 | SPoC for Institutions | Updating the List of Nodal officers |
| 11 | Updating the repositories in Bharat5G/6G portal | Co-ordination and validating/ updating repositories wherever required like-<br>i. R&D activities, Projects, Infra, thesis/abstract from Academia<br>ii. Use cases, products from Start-ups/MSMEs<br>iii. IPRs and standardization from Academia/Start-ups/MSME<br>iv. Schemes/Programs from State Agencies, Incubation Centres/ Centre of Excellence. |
| 12. | Telecom Start-ups & MSMEs | i. State/LSA level TSUM facilitation |

| | | |
|---|---|---|
| | (TSUM) Facilitation Cell (State gov departments, customs etc. ) | cell in co-ordination with DoT HQ for IPR facilitation, PoC testing & certification, establishing sandbox environments, etc.<br>ii. Creating awareness regarding DoT initiatives/funding Schemes/ Programs- TTDF, DCIS, 5G Hackathons, 100 5G labs, 6G R&D proposals, Experts registration etc.<br>iii. Periodic meetings with representatives of state government, State-PSUs, VCs and other key bodies. |
| 13. | Support for 5G/4G/IoT Projects (including CNPN) | Co-ordination for the following<br>i. Rural zones- 5G Model Village etc.<br>ii. Urban/Industrial zones- CNPN networks (Power, Mining, Ports & Airports, Factories etc.) |
| 14. | 5G Test Beds Utilisation | Facilitation & awareness for utilization of 5G test beds (https://user.cewit.org.in/5gtb/index.jsp ) at CEWIT/IITM / IISC/IITD/ IITK. |
| 15. | Monthly Report | Periodic submission of progress cum action taken reports through portal |

In order to perform the aforementioned activities in an effective way, LSAs are encouraged to co-ordinate with CCAs/RLOs/WMOs.

**Annexure-II**

| 100 5G Lab Institutions mapped to LSA units | | | |
|---|---|---|---|
| **S.No** | **LSA (Focal Point)** | **Mapped States** | **No. of Institutions** |
| 1 | Andhra Pradesh LSA | Andhra Pradesh,Telangana | 8 |
| 2 | Assam LSA | Assam | 2 |
| 3 | Bihar LSA | Jharkhand,Bihar | 6 |
| 4 | Delhi LSA | Delhi | 4 |
| 5 | Gujarat LSA | Gujarat | 3 |
| 6 | Himachal Pradesh LSA | Himachal Pradesh | 2 |
| 7 | Haryana LSA | Haryana | 2 |
| 8 | Jammu and Kashmir LSA | Jammu and Kashmir | 2 |
| 9 | Karnataka LSA | Karnataka | 8 |
| 10 | Kerala LSA | Kerala | 4 |
| 11 | Kolkata LSA | Sikkim | 1 |
| 12 | West Bengal LSA | West Bengal | 2 |
| 13 | Madhya Pradesh LSA | Madhya Pradesh,Chattishgarh | 6 |
| 14 | Mumbai LSA | Goa | 2 |
| 15 | Maharashtra LSA | Maharashtra | 7 |
| 16 | North East LSA | Mizoram, Nagaland,Arunachal Pradesh,Manipur,Meghalaya,Tripura | 8 |
| 17 | Odisha LSA | Odisha | 4 |
| 18 | Punjab LSA | Punjab,Chandigarh | 6 |
| 19 | Rajasthan LSA | Rajasthan | 5 |
| 20 | Tamil Nadu LSA | Tamil Nadu,Puducherry | 9 |
| 21 | Uttar Pradesh-East LSA | Uttar Pradesh (East) | 4 |
| 22 | Uttar Pradesh-West LSA | Uttarakhand, Uttar Pradesh(west) | 5 |
| | | **Total Institutions** | **100** |

100 Institutions list can be accessed at https://www.bharat5glabs.gov.in/institutes

Government of India
Ministry of Communications
Department of Telecommunications
Wireless Planning and Coordination (WPC) Wing
Sanchar Bhawan, Ashoka Road, New Delhi-110001

File No. P-11014/02/2020-PP Date: 04.04.2024

**OFFICE MEMORANDUM**

**Subject: Simplification in the procedure for obtaining Experimental Licence for 100 5G Use Case Labs -Reg**

The Department of Telecommunications has awarded '100 5G Use Case Labs' to educational institutions across the country. These labs will be using 5G frequency bands for carrying out various experiments and testing use cases and need to obtain Experimental (Non-Radiating) category of license from WPC Wing to ensure interference free operations to licensed Telecom Service Providers.

2. To simplify the grant of experimental licence for these labs, a specific Approval Type, known as 'Experimental Licence for 100 5G Use Case Labs,' has been introduced on the National Single Window System (NSWS) Portal. The NSWS portal redirects the applicant to the Saralsanchar Portal. In this the application process is simplified wherein technical specifications and schematic details are auto-filled. Applicants need to fill only the details of institute address, authorised person and pay the requisite fees of Rs 5000 for one year. The licence can be downloaded instantaneously, making the entire process very simple.

3. The detailed user manual is available on https://saralsanchar.gov.in. This initiative is poised to significantly support innovation and cultivate competencies and engagement in 5G technologies among students and start-up communities.

RADHIKA KOLLU
Digitally signed by RADHIKA KOLLU
Date: 2024.04.04 18:11:51 +05'30'

(K. Radhika)
AWA/WPC Wing

To:
1. All the Academic Institutions under 100 5G Use Case Labs
2. All Sr. DWA of WPC Wing
3. DDG (SRI) for coordinating with the institutes
4. Director (IT) for uploading on the Saralsanchar portal

Copy To:
1. Sr. PPS to WA; PPS to JWA (WPC)
2. Director (WMO)
3. JWA- RLO (ER/ NR/ WR/ SR), Sr. DWA, RLO-NE

# User Manual

(For Applicant)

## Experimental License for 100 5G Labs

## Department of Telecommunications

**Government of India**

01 April, 2024

| Sl. No | Contents | Page No. |
|---|---|---|
| 1 | ***Section-1*** | 3 |
| 1.1 | General Introduction | 3 |
| 1.2 | About Licenses | 3 |
| 2 | ***Section-2 Applicant Part*** | 4 |
| 2.1 | First Login | 4 |
| 2.2 | Creation of Application in NSWS | 5 |
| 2.3 | Application Form | 11 |
| 2.4 | Online Payment | 16 |
| 2.5 | Upload Signed Application | 18 |
| 2.6 | Download License | 19 |

# Section-1

## 1.1 General Introduction

**SARALSANCHAR'( Simplified Application For Registration and Licenses)** a Web based Portal, for Issuing of various types of Licenses and Registration certificates is part of various Digital initiatives being taken by Department of Telecommunications. It is a unified portal to issue various types of Licenses and registrations in a digitized manner which will not only ensure transparency but also make the process more efficient. This will pave the way for a paperless, secure and hassle-free platform for various applicants.

### Acronyms and Abbreviations

List of the acronyms and abbreviations used in this document and the meaning of each.

- **DoT** : Department of Telecommunications.
- **NSWS** : National Single Window System.

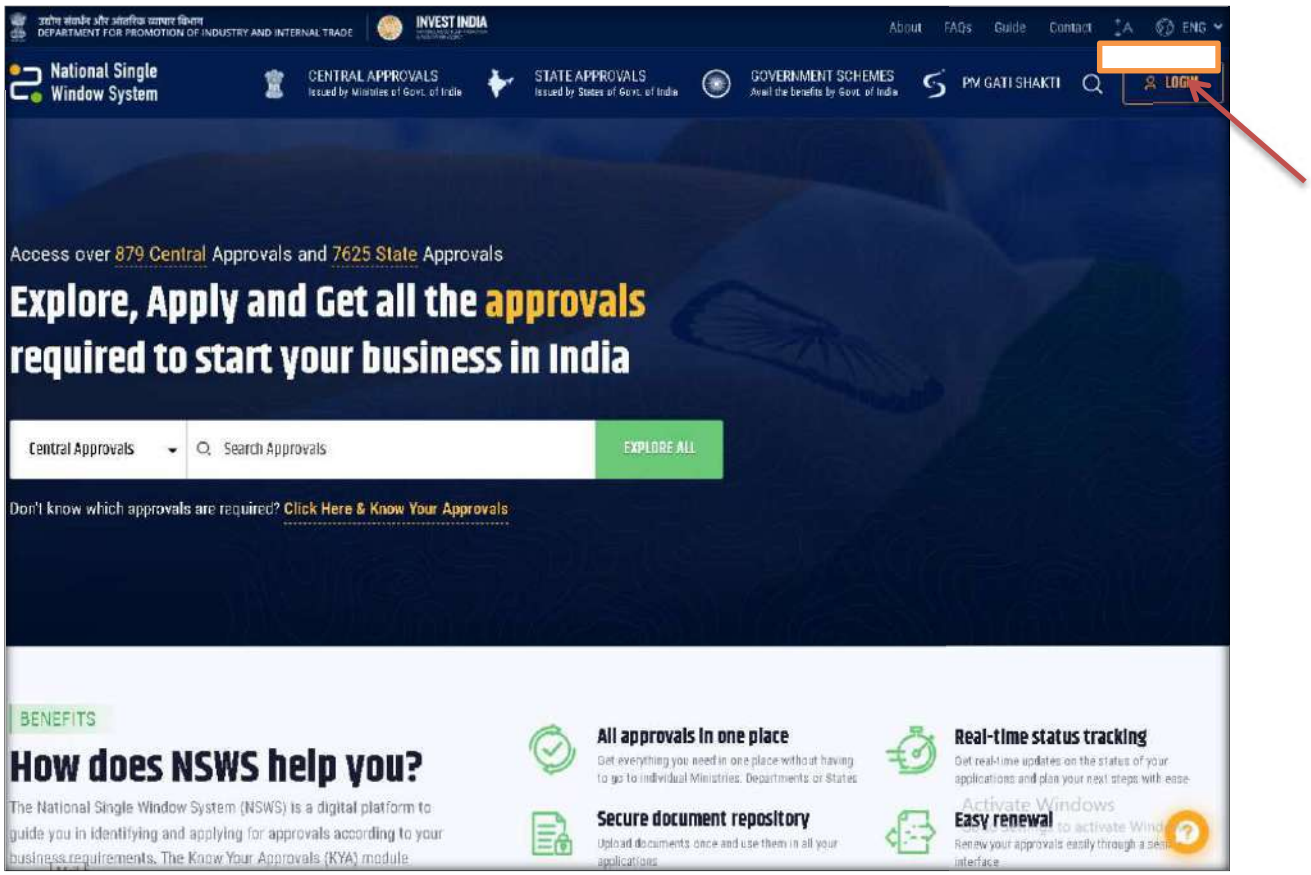
## 1.2 About Licenses

The Department of Telecommunications has awarded '100 5G Use Case Labs' to educational institutions across the country. The primary objective behind this initiative is to cultivate competencies and engagement in 5G technologies among students and start-up communities.

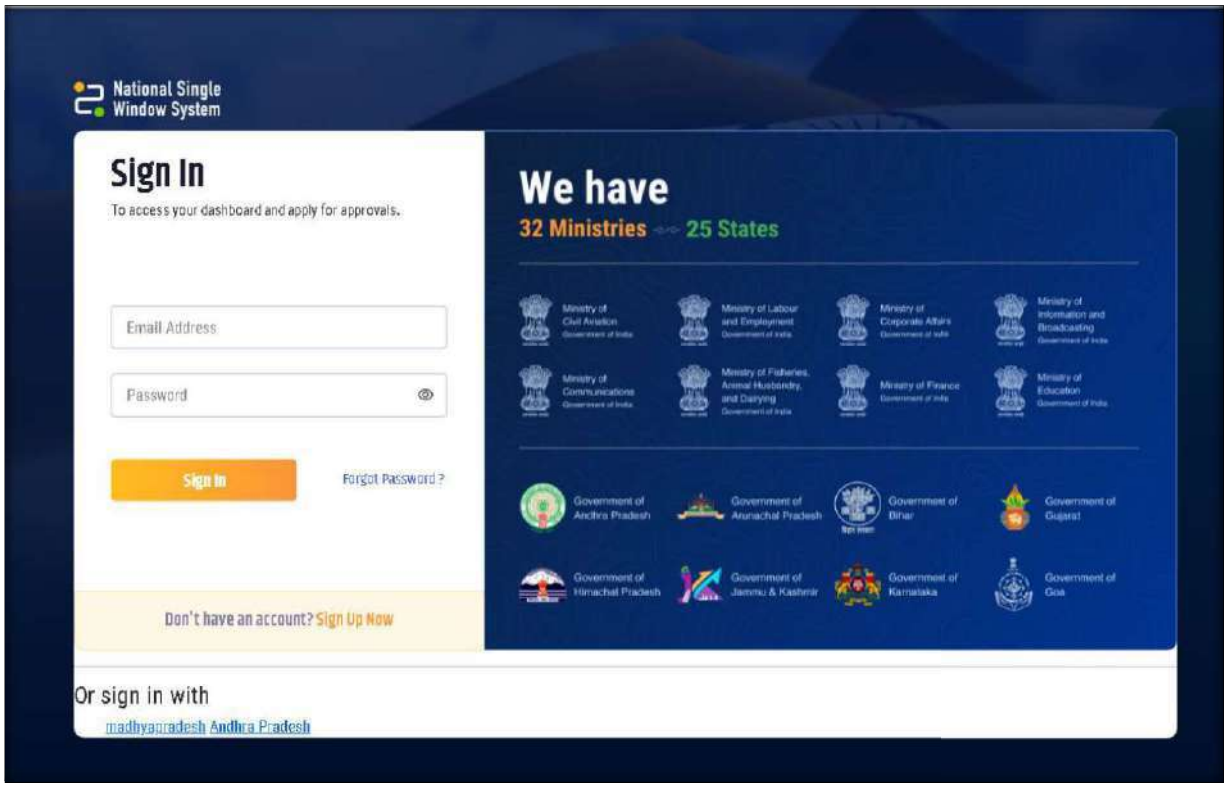
# Section-2 Applicant Part

## 2.1 Login

Go to National Single Window System using the URL https://www.nsws.gov.in/.



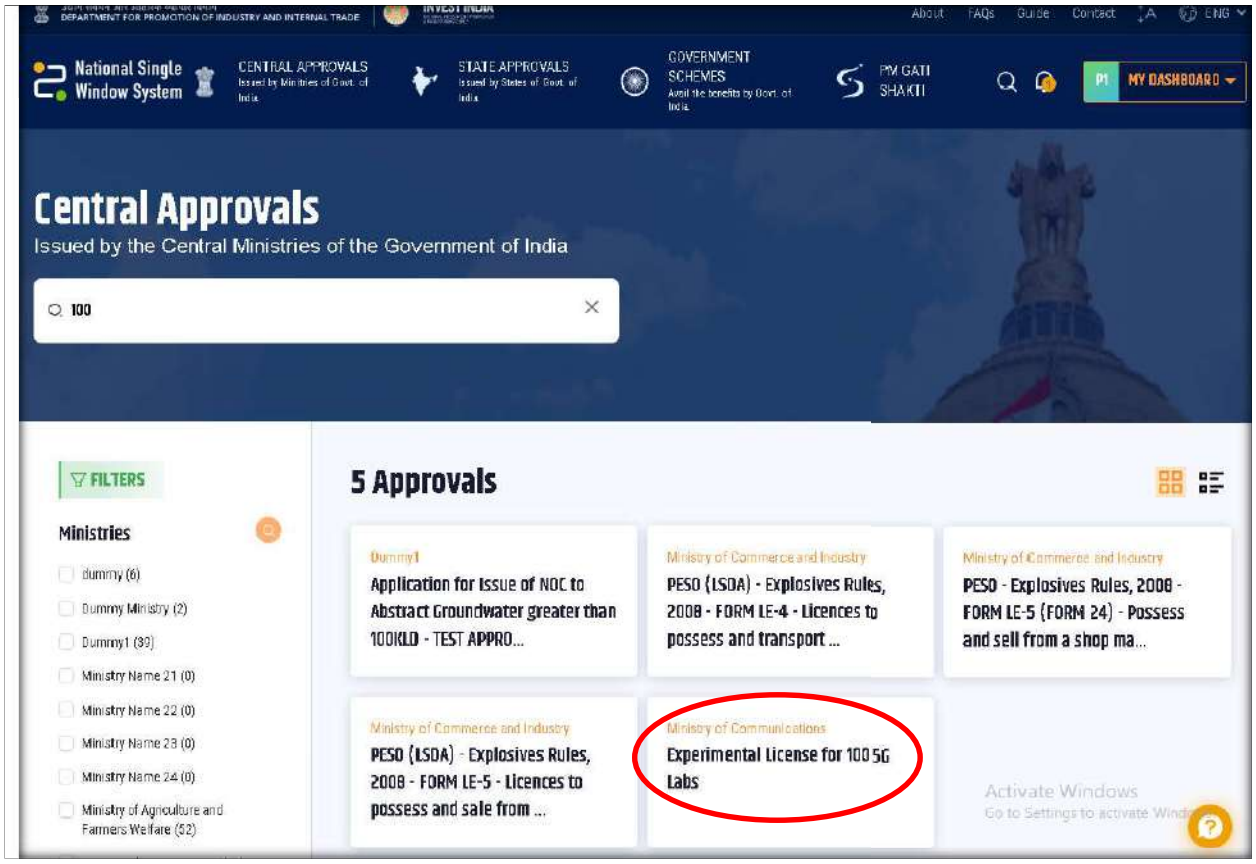

Click on Login. Following login form will open.



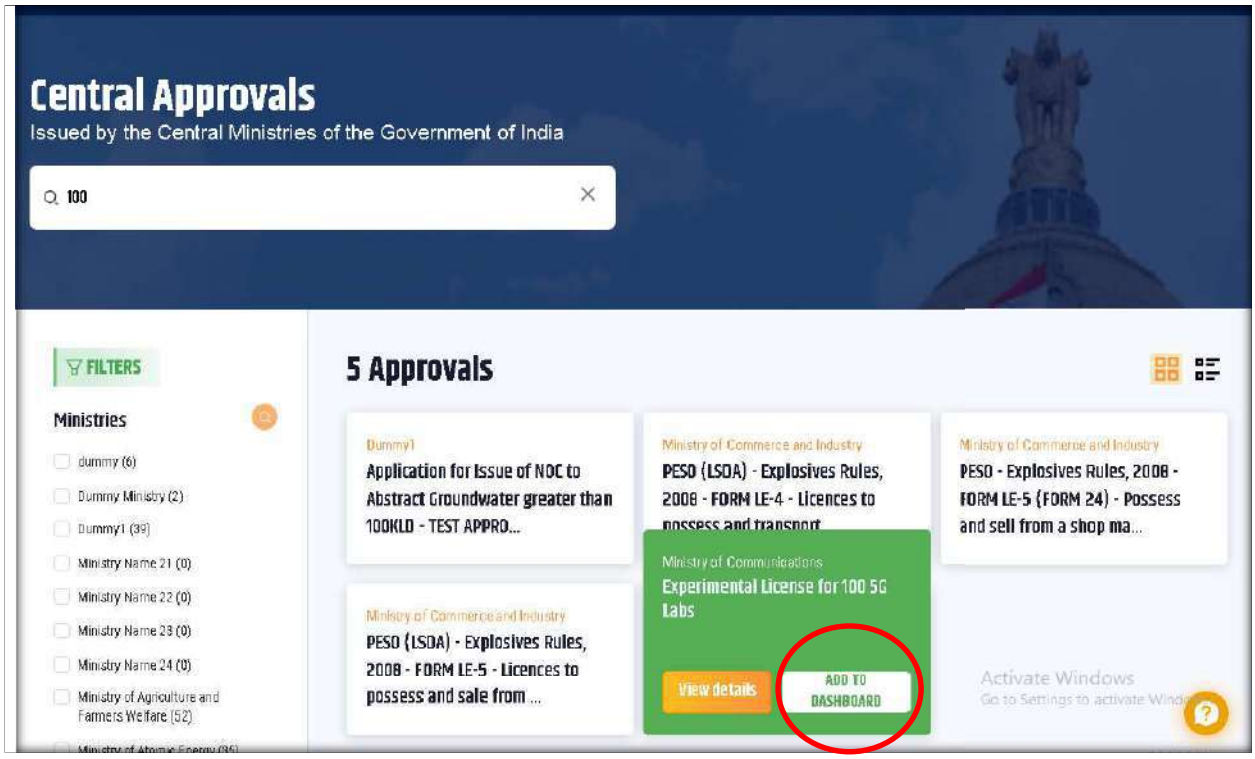

If you have account in the NSWS portal ,enter the credentials and sign in. If don not have account , click on Sign Up Now to create the account.
After successful Sign in you will be landed to Dashboard as shown in the following screenshot.

## 2.2 Creation of application in NSWS

Click on Central Approvals, following option will come.



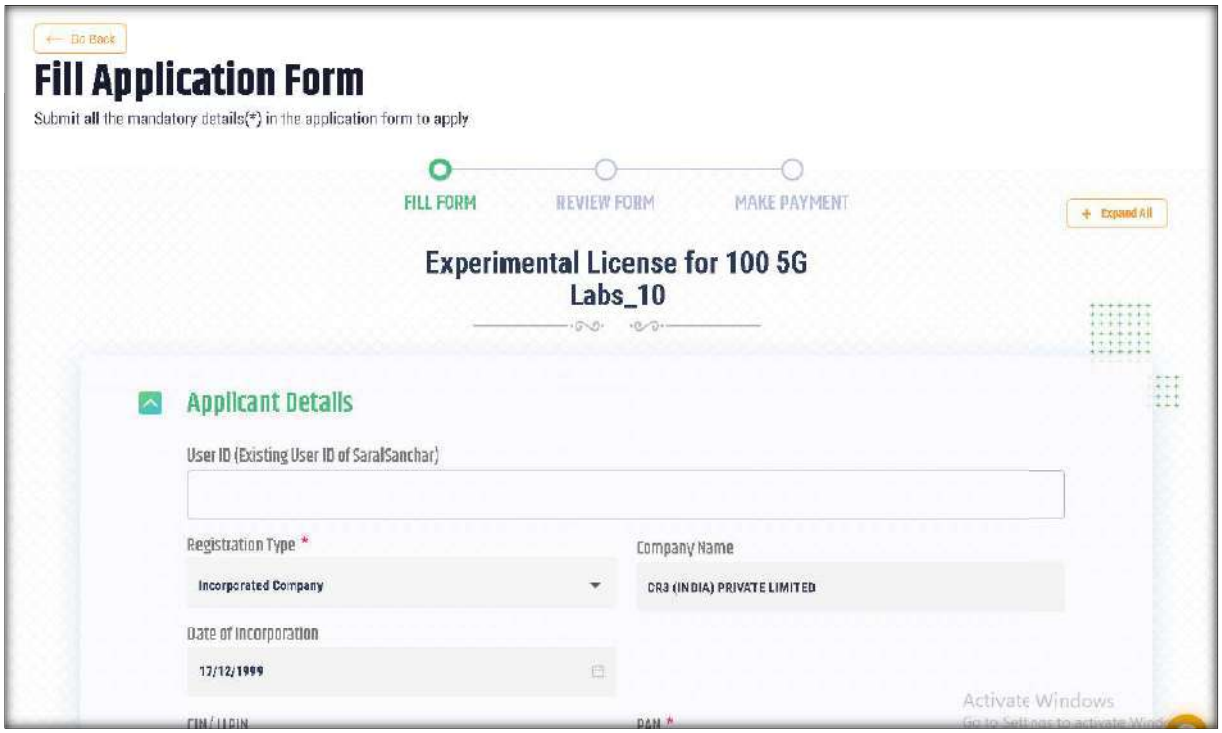

Search for Experimental License for 100 5G Labs. Click on "Experimental License for 100 5G Labs". Then a popup will come to view details of license and add to Dashboard as shown in the following image.



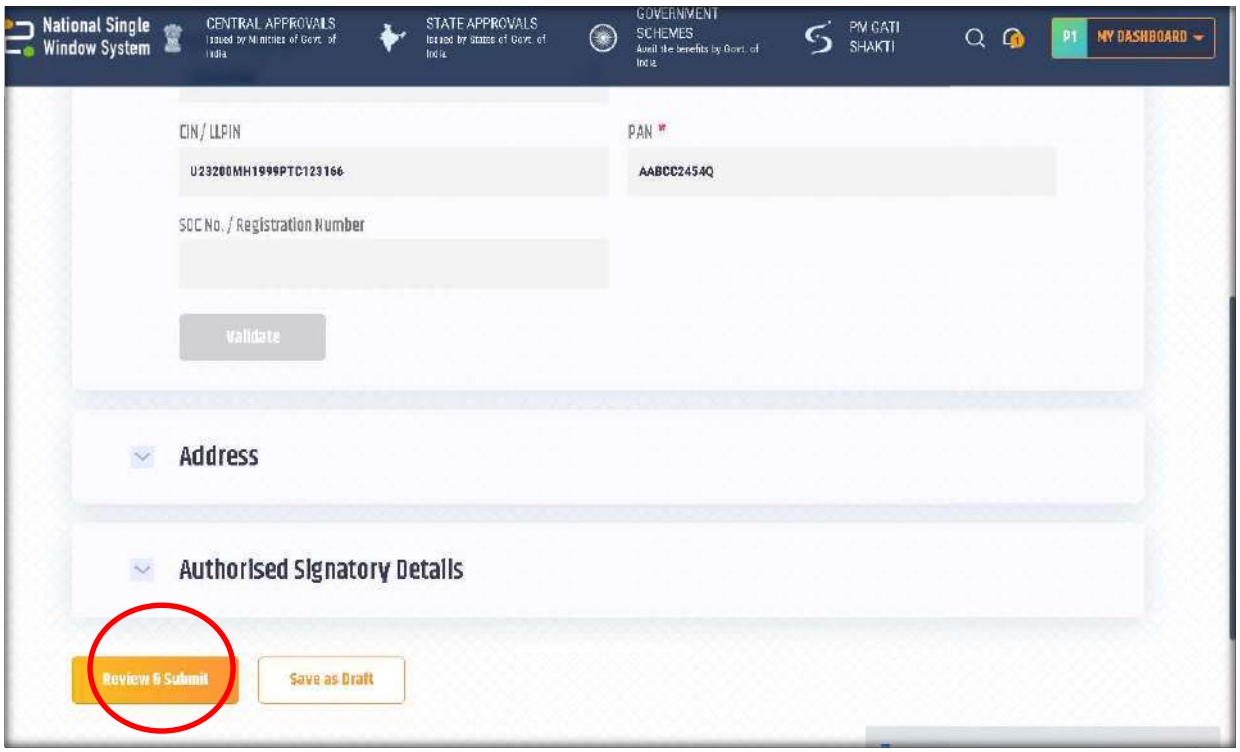

Click on "ADD TO DASHBOARD" this approval will be added in the dashboard.

Then go to the dashboard

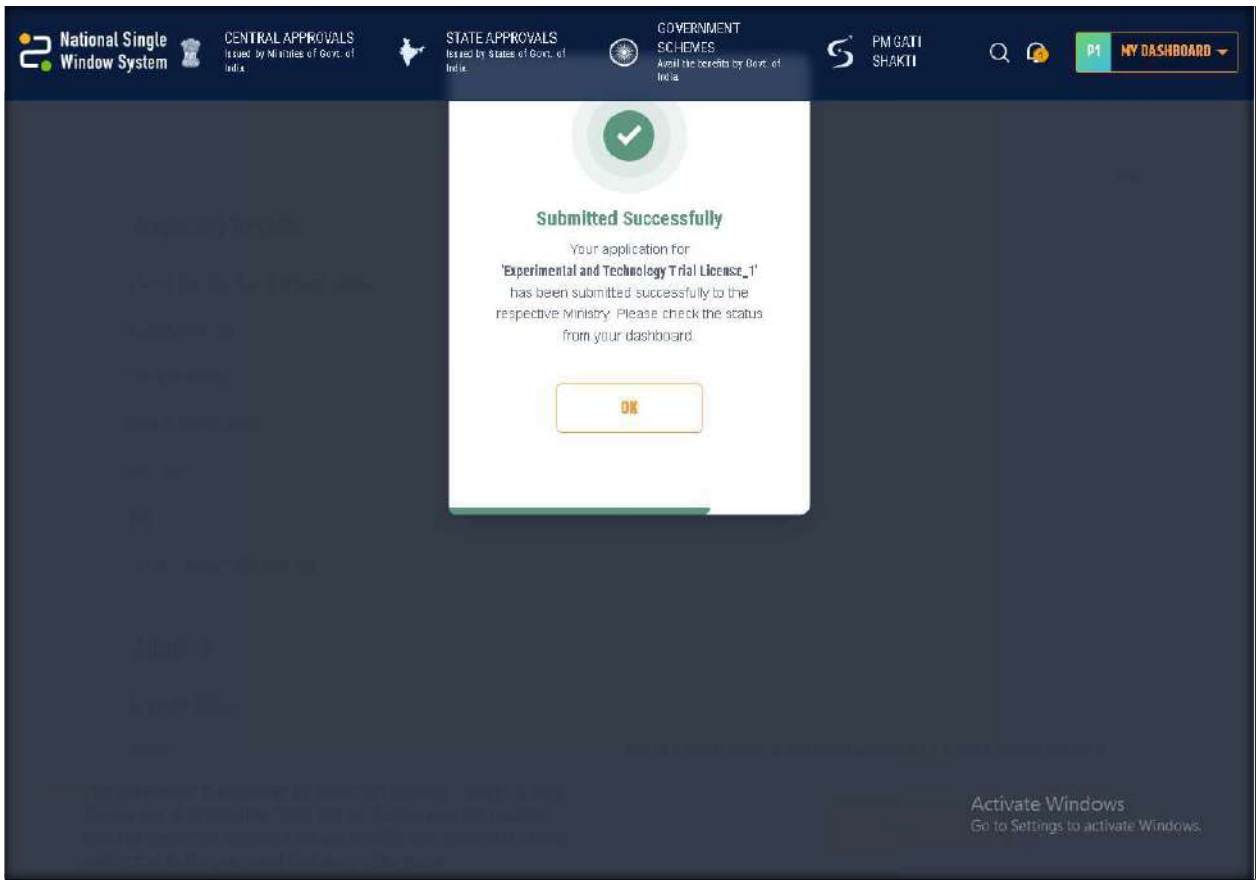
Click on "Apply Now" as shown in the above image, following form will open




Enter existing User ID of Saralsanchar portal and click on validate button .



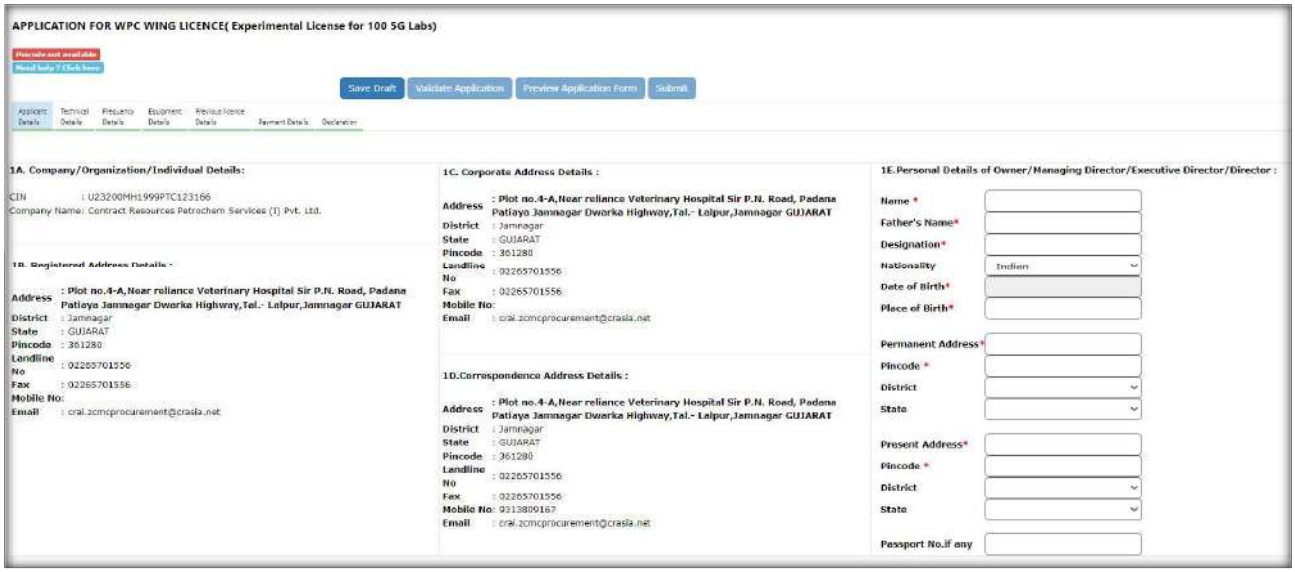

After successful validation click on "Review & Submit" button.




Application is submitted and a link(Go to Ministry Portal) will be available to get the application form as shown in the following image.

Click on Go to Ministry Portal button, you will be redirected to following page.

Click on “Go” button , the application form will open .



## 2.3 Application Form:

Below is a sample demo for Application of an Experimental License For 100 5G Labs.

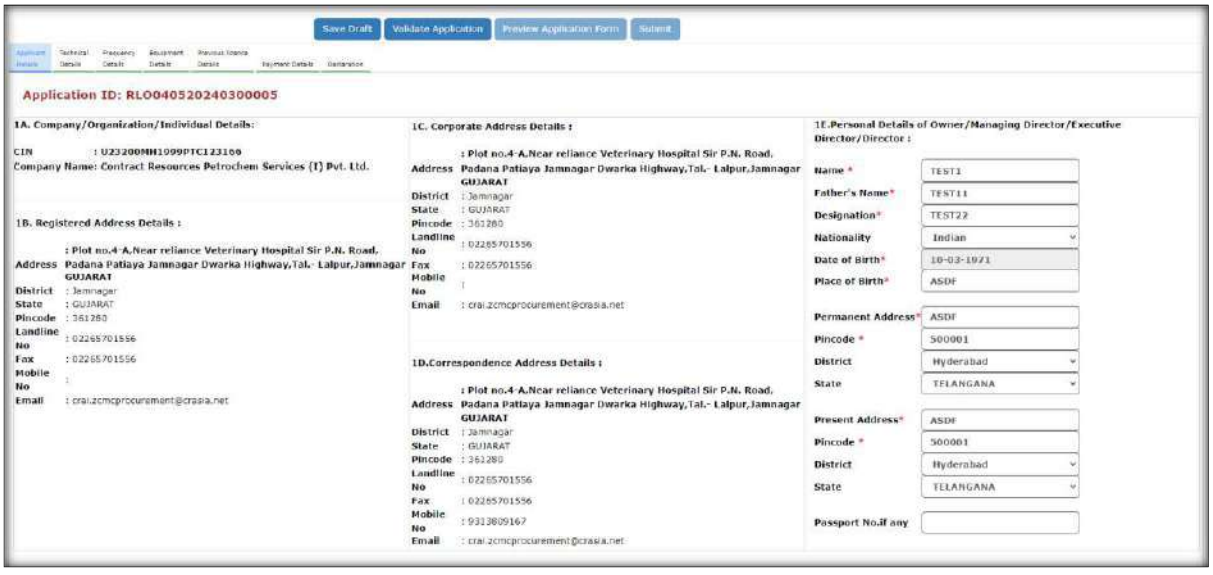
Application form will appear as follows. Form comprises of 7 steps viz. Applicant Details, Technical Details, Frequency Details, Equipment Details, Payment details and Declaration.

## IMPORTANT

**Digitally Signed Document :** All the document to be uploaded hereafter must be in **"pdf format"** and **"digitally Signed"** . The portal will automatically reject any document which is not **digitally signed** or in **pdf format**.

### 2.3.1 Applicant Details

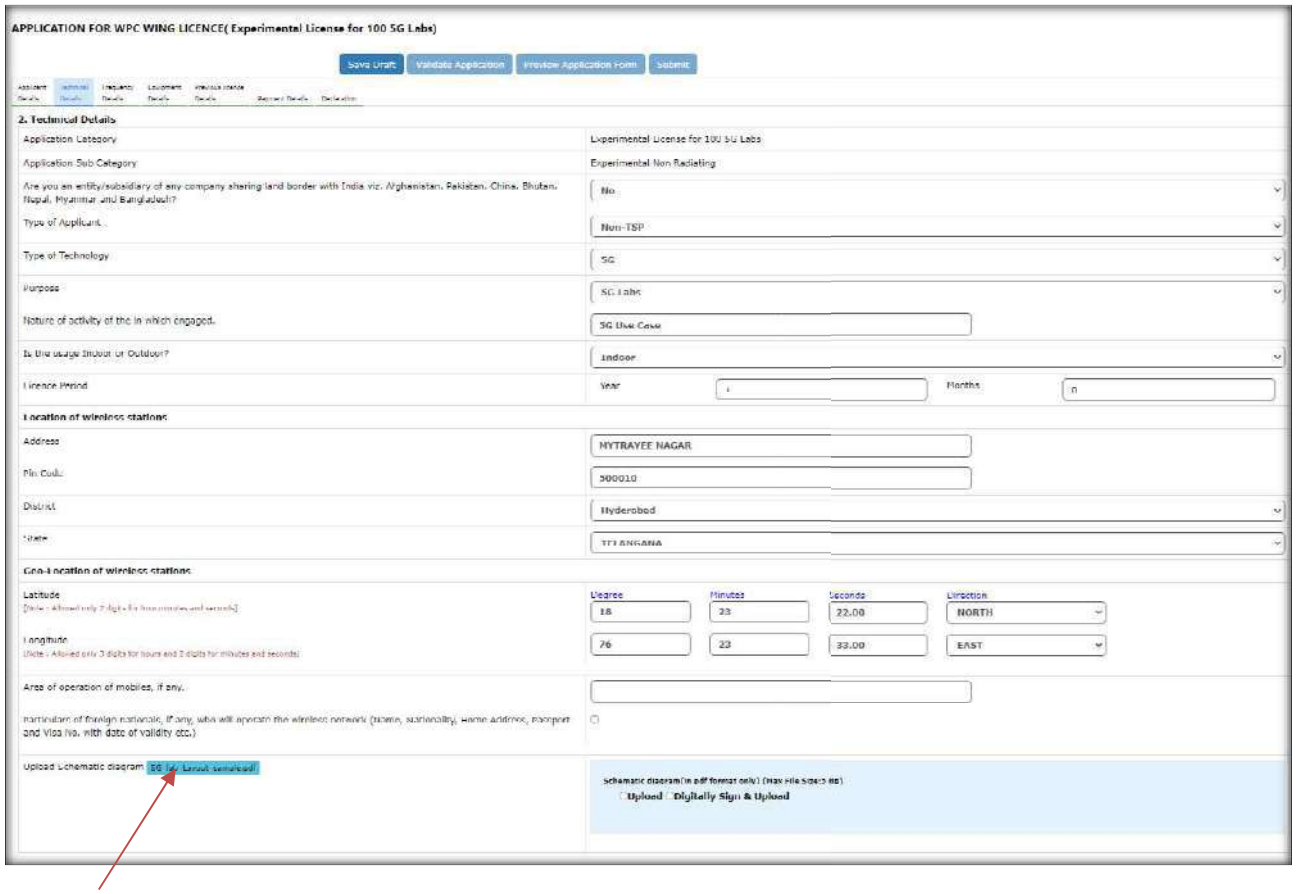
***Applicant Details***:
In this step the information about Applicant.1A,1B,1C and 1D will be auto populated as per user profile entered during registration or update profile.




1E Personal Details of Owner/Managing Director/Executive Director/Director details have to be entered.

## 2.3.2 Technical Details

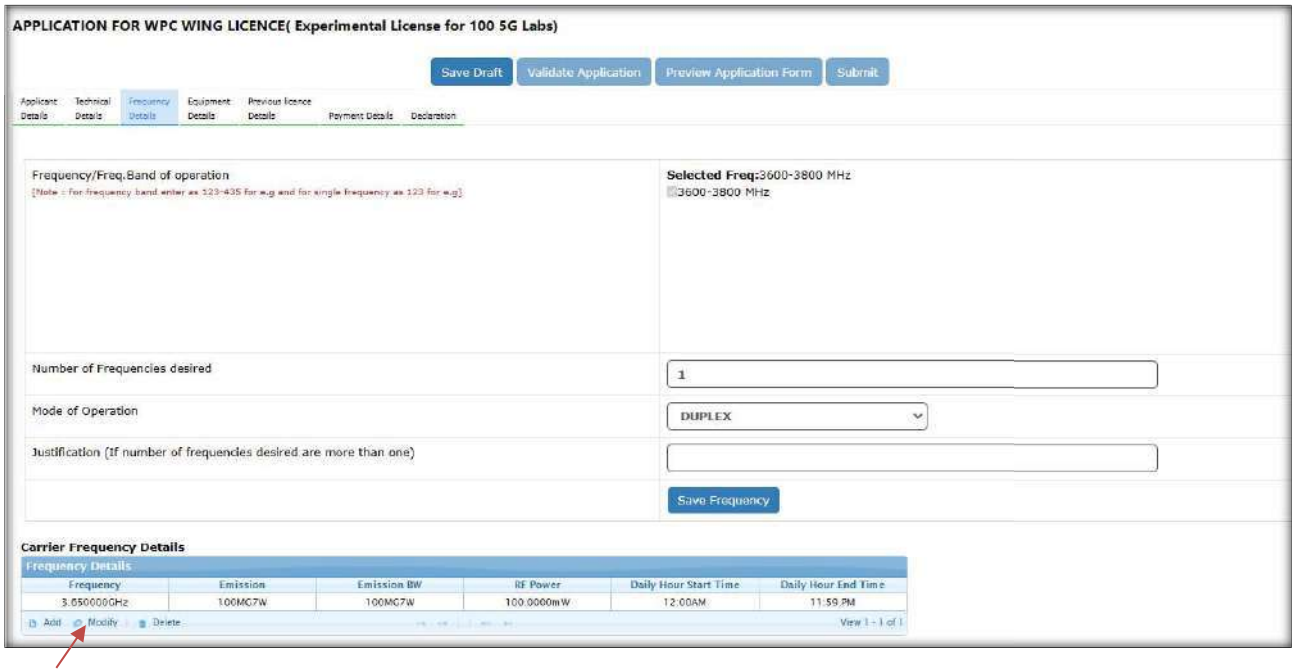
***Technical Details***: In this step the user has to fill up location of wireless station and go location of wireless station and click on "Save Draft" then application will created. After that by clicking "**choose File**" button against document and then selecting **file** and **upload** using "**upload button**"**.** Sample copy of schematic diagram may be downloaded and signed it digitally and upload it.



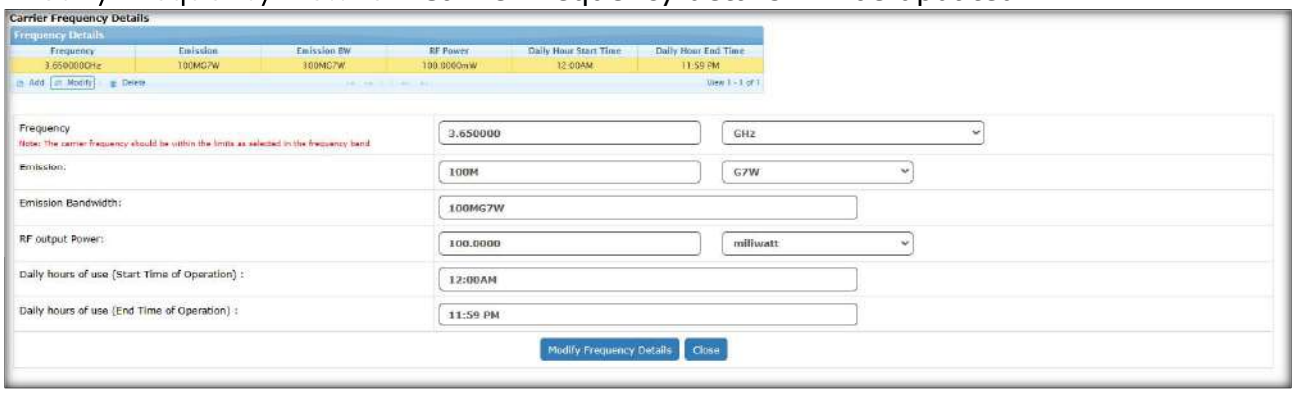

The form can be saved at any intermediate stage by clicking "save draft" button. After completion of this step, next step will be active using "next" button.

## 2.3.3 Frequency details

Frequency/Frequency Band of operation, number of frequency and mode of operation and carrier frequency details will be auto populated.




If you want to modify the carrier frequency details by clicking on "Modify" button as shown in the above image. On clicking Modify button , carrier frequency details form will open as shown in the below image. Modify as per requirement and click on "Modify Frequency Details". Carrier frequency details will be updated.



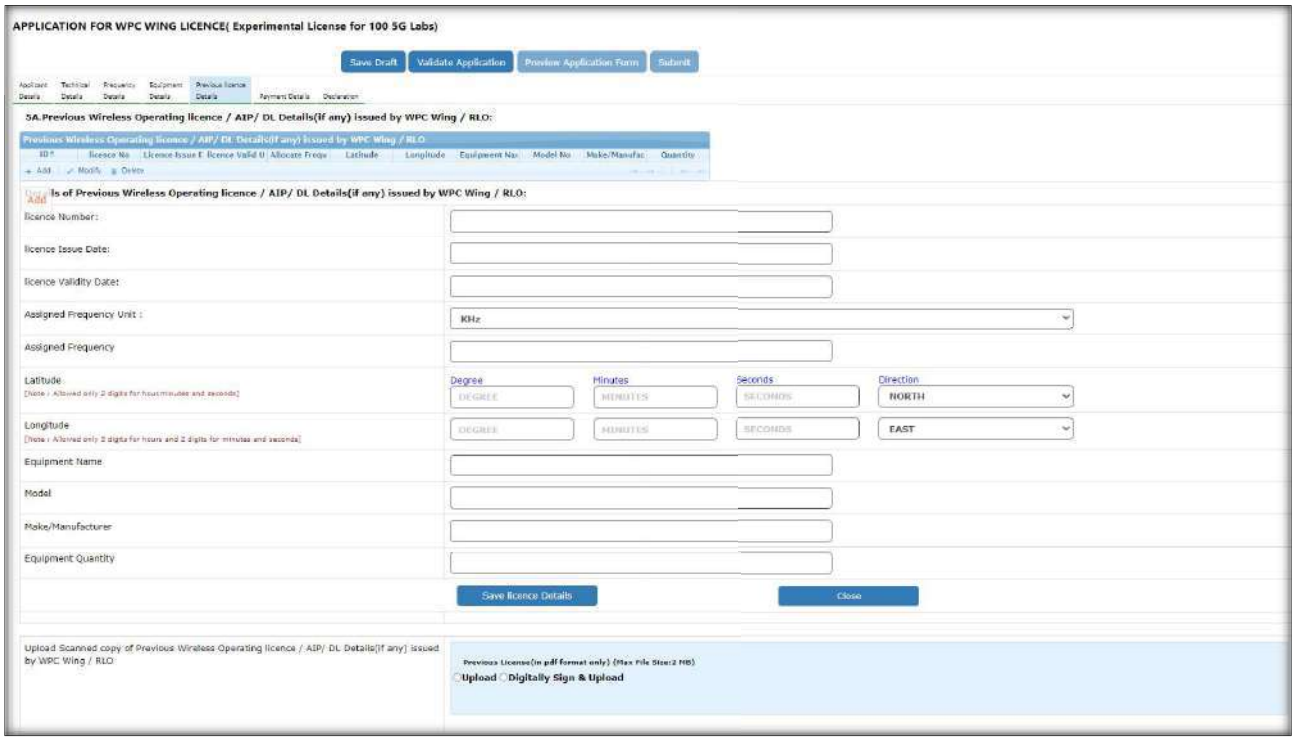
## 2.3.4 Equipment Details

Equipment details will be auto populated as shown in the following image.

## 2.3.5 Previous License Details

This is optional.

If you want to add details click on Add button below form will open. Fill up all fields and click on save license details.




### 2.3.6 Payment Details

Payment option will be available after submitting this application.

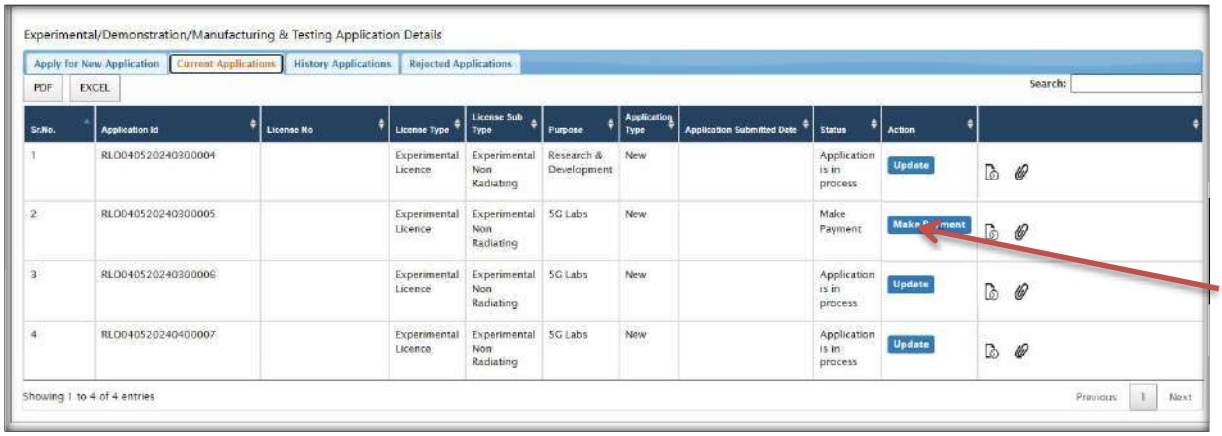
### 2.3.7 Declaration

Accept the declaration.

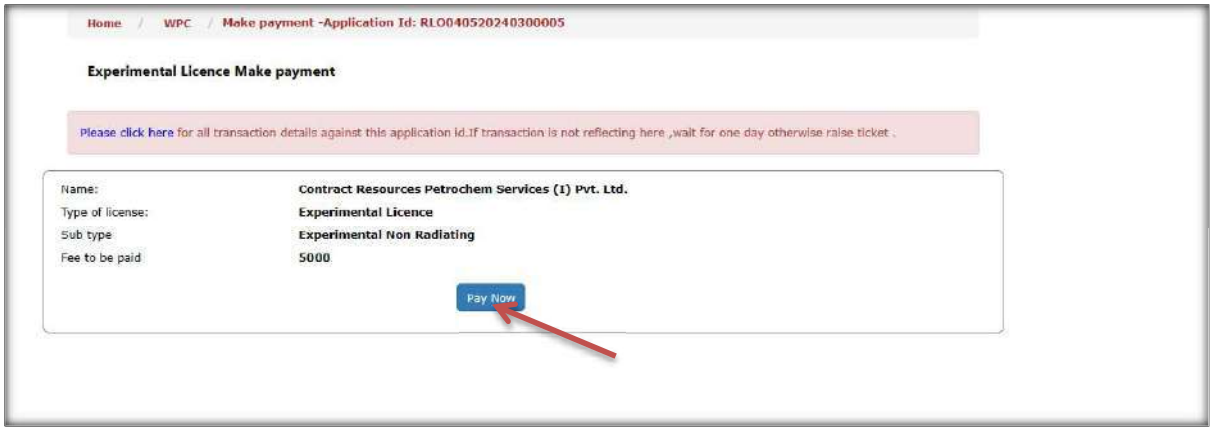
After filling up all details, click on save draft button. Then validate application, once application is validated "Preview Application Form" and "Submit" buttons will be enabled. Check the application pdf by clicking on "Preview Application Form" button. If all are correct, click on "Submit" button to submit the application. Once application is successfully submitted, the payment option will come and you will not be able to modify the application form.

## 2.4 Online Payment

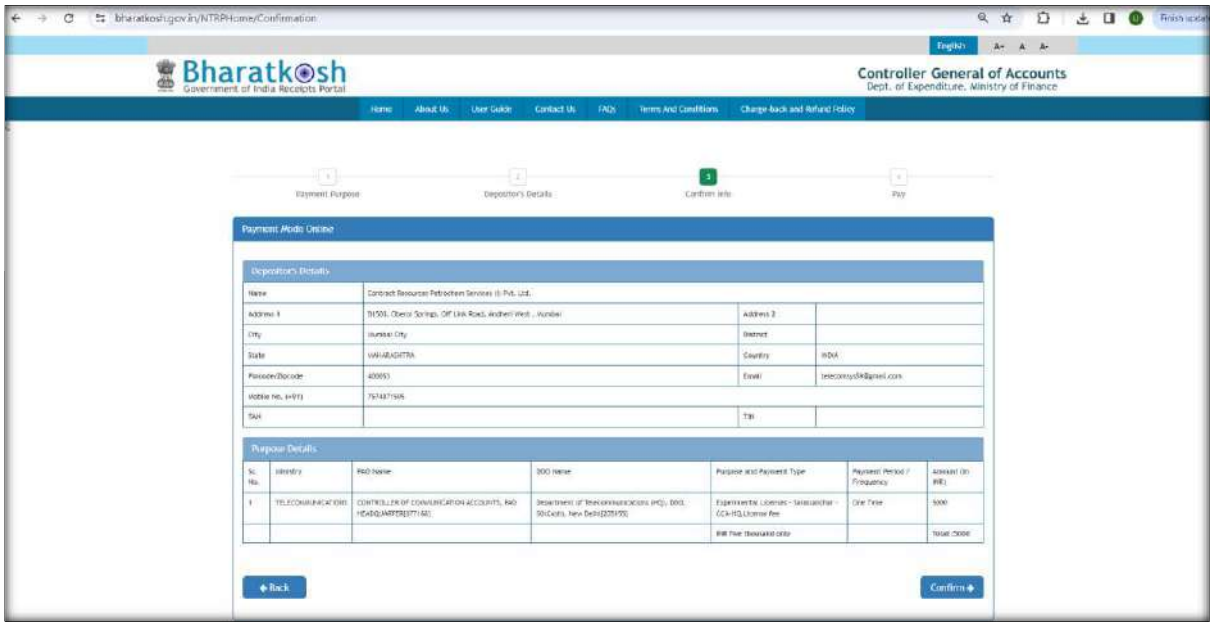
After submitting the application ,payment option will be available as shown in the below image.



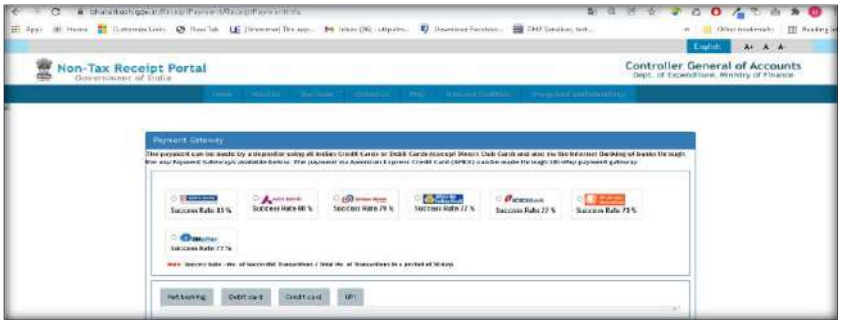

Click on "Make Payment" following payment link page will come.

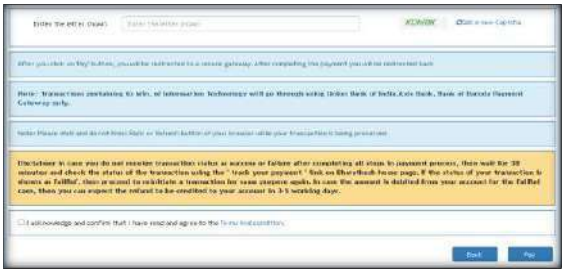



On click of "Pay Now" button, you will be directed to the Saralsanchar integrated Bharatkosh portal.



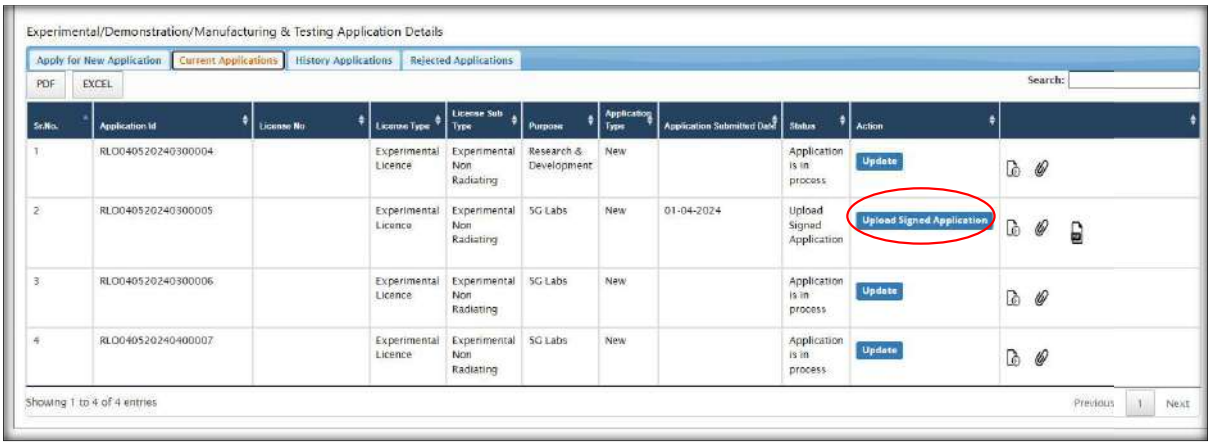

Click on Confirm button , following page will come for payment mode



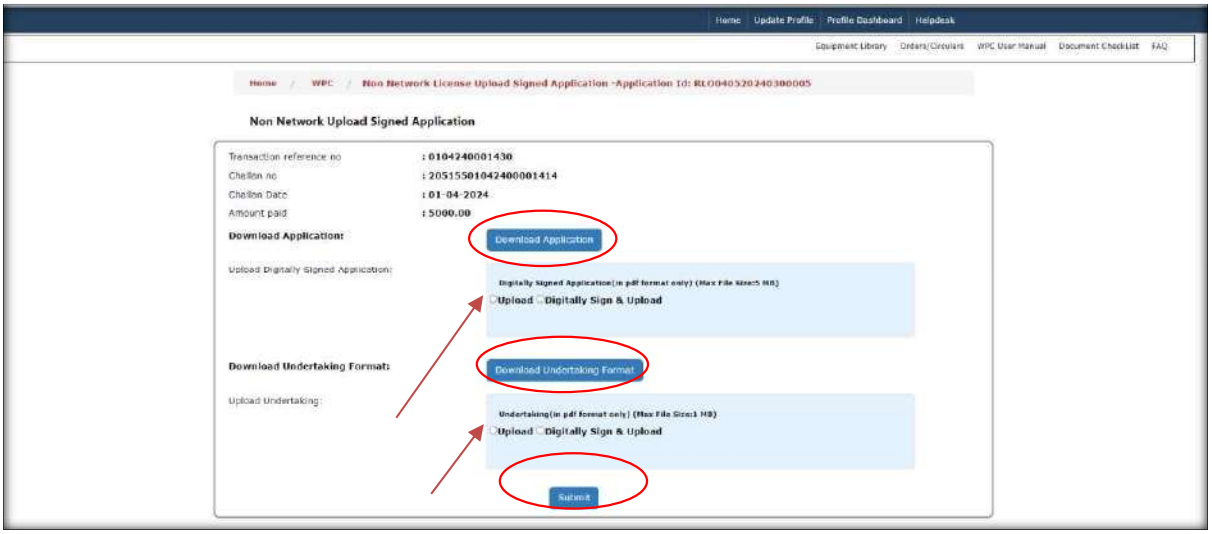



Click pay button to get redirected to enter details of Credit card/bank details. On completion of successful payment, you will be redirected to Saralsanchar portal with the transaction ref no , challan number.
After payment application upload signed application will be available.

## 2.5 Upload Signed Application




Click on Upload Signed Application, following form will open.




Click on "Download Application", Application pdf will be downloaded. Signed this application pdf and upload. Download Undertaking format and upload digitally signed undertaking pdf file.

After uploaded these two files, click on "Submit" button application will be submitted and License will be issued.

## 2.6 Download License

You can download the License from your dashboard by clicking the **Download License** button which is shown below.

सत्यमेव जयते

**Government of India**
**Department of Telecommunications**
**Wireless Planning and Coordination Wing**

**Application ID:RLO040520240300005** **Date: 02/04/2024**

**Licence No:EXP-SD-RLO_SR_EXP-2024000003** **Valid up to: 01/04/2025**

**Sub: Grant of Experimental (Non-Radiating ) Licence**

M/s Contract Resources Petrochem Services (I) Pvt. Ltd.,Plot no.4-A,Near reliance Veterinary Hospital Sir P.N. Road, Padana Pattaya Jamnagar Dwarka Highway,Tal.- Lalpur,Jamnagar GUJARAT,Jamnagar,GUJARAT,361280, with represented by Gunavel Rathinam is hereby issued Experimental Licence(Non-Radiating) for indoor usage on 02/04/2024 - based on the details submitted and respective undertaking under Section 4 of the Indian Telegraph Act, 1885 and the scope of the licence described under 1.A. of the OM No. R-14016/01/2019-NT (Pt.) dated 23rd July, 2019 .

A. Purpose: 5G Labs

B. Spectrum Usage: Restricted to indoor environment in non-radiating or very low power radiating mode (Up to 100 mW).

C. Licence fees: Rs. 5000 paid vide Transaction Ref. No.0104240001430 Dated: 01/04/2024.

D. Date of Issue: 02/04/2024

E. Valid Up to: 01/04/2025

F. Frequency Band: 3600-3800 MHz

G. Frequency Parameters:

Operating Frequency Range:

| Sr.NO | Carrier(s)/ Frequency Spot(s) | Bandwidth & Emission | RF output power | Mode | Operating Hours |
|---|---|---|---|---|---|
| 1 | 3.650000GHz | 100MG7W | 100.0000mW | DUPLEX | H24 |

H. Location of the Setup/Wireless Station:

| | |
|---|---|
| Address | MYTRAYEE NAGAR |
| Pincode | 500010 |
| District | Hyderabad |
| State | TELANGANA |
| Latitude | 18°23'22.00"N |
| Longitude | 78°23'33.00"E |

1/3

I. Equipment Details of the Experimental setup:

Section A: Non- Imported:

| Sr.NO | Name of Equipment | Model | Name of Manufacturer | Manufacturer Address | Quantity |
|---|---|---|---|---|---|
| 1 | gNodeB | STGNB2214-020W-1D | Signaltron Systems Pvt Ltd | 4C-116, 3rd Floor, 4th Cross, OMBR Layout, Banaswadi, Bengaluru, Karnataka,560043India | 1 |
| 2 | 5G Drone | Suparna | Menthosa | H.No 240/2 Purani India, Mandi, Himachal Pradesh 175001India | 1 |

No.: 1-20/2022-NTI.WA
Government of India
Ministry of Communications
National Telecommunications Institute for Policy Research, Innovation &Training
Govt. of India Enclave, Near Rajnagar, Ghaziabad, UP- 201002

Dated: 12.07.2024

Subject: Collection and sending of regular information about Innovations and new Use Cases on 5G technology developed by the select Institutions/Startups to NTIPRIT_regarding

The 5G Use Case Labs are set up/being setup in 100 select Institutions across the country for which DDG(Technology) in LSA Units are the nodal points regarding a number of activities such as coordination and validating/updating repositories about the R&D activities, Use Cases, IPRs from Startups etc besides facilitation in Installation and commissioning, user acceptance testing, LSA wise/State wise Hackathon , conducting workshops at LSA/State level etc(reference DoT letter 1-Role of LSAs(100 5G Labs)/2024-SRI dated 15th February, 2024 enclosed)

2. A 5G Use Case Lab has been set-up in NTIPRIT Ghaziabad which has been inaugurated by Secretary (Telecom) DoT on 27th June 2024. A Workshop on 'Role of LSAs in 5G Lab Project" was also conducted for all LSA field unit officers during 27-28 June 2024 with sessions on Installation, configuration and User Acceptance Testing of the equipment besides other deliberations.
3. A proficiency level-2 training was also conducted at IIT Kanpur on '5G Technology' from 18 June to 20 June 2024 with exposure to minor practical and standards development work on 5G/6G being taken up by IIT Kanpur
4. The regional workshops were also held by NTIPRIT at IIT Roorkee (29-30 January 2024), IIT Guwahati (11-12 February), IIT Gandhinagar (18-19 March 2024) and IIT Madras (1-2 April 2024) with participation from all select Institutions and LSAs.
5. During the inauguration of the 5G Use Case Lab and workshop for DoT LSA officers on 27th June 2024, **Secretary(T) DoT has desired that NTIPRIT should regularly interact and gather the information about the innovations and IPR generation activities being done in the select Institutions across the country** and disseminate the information further.
6. It is therefore requested that LSA units may regularly gather information about the innovative works and IPR generation related activities and **send the same on a quarterly basis** as per the attached format.
7. The first quarterly information about such works may please be sent for q.e. June 2024 by 31st July 2024 on email id dir.wa2-nti@gov.in

Enclosure: As above

(Atul Wakhle)
Director (WA)
National Telecommunications Institute for Policy Research, Innovation & Training (NTIPRIT)
Department of Telecom, Government of India
Mail:dir.wa2-nti@gov.in
Mob: +91 9422085858

To:
LSA Heads (Special DGT/Addl DGT)

Copy to:

i. PSO to Secretary(T) DoT
ii. Member (Services), DoT HQ
iii. DGT, DoT HQ with a kind request to further sensitize the LSAs in this regard
iv. DDG(SRI), DoT HQ
v. DoT Website

**ANNEXURE**

| S No. | Name of LSA | Name of Institution | Type of activity (5G Use Case/IPR/Other) | Name of the Startup/Enterprise/Researcher with contact email/mobile number | Details of the innovation/work carried out | Remarks |
|---|---|---|---|---|---|---|
| | | | | | | |
| | | | | | | |
| | | | | | | |
| | | | | | | |
| | | | | | | |
| | | | | | | |

**100 5G LABS FOR HIGHER EDUCATIONAL INSTITUTIONS**
**USER ACCEPTANCE TEST**

Date:

Name of the Institutions:
Implementation Agency : Telecommunications Consultants India Limited
Equipment Supplier: Wireless 4 Scale
UAT Committee Member:

1. DoT LSA :
2. TCIL :
3. Institute :
4. W4S

### 4.1 Bill of Material Verification

| S.No | Equipment/Devices | Qty | Remarks | Make/Model | Serial Nos |
|---|---|---|---|---|---|
| 1 | 5G Core | 1 | Installed on COTS Standard server | | |
| 1.1 | 5G Core Server | 1 | Rack mountable | | |
| 2 | 5G Radio | | | | |
| | • Integrated RAN | 1 | Pole Mountable | | |
| | or | | | | |
| | • Disintegrated RAN | | | | |
| | • RU | 1 | Pole Mountable | | |
| | • CU+DU | 1 | Installed on COTS Standard server | | |
| | • CU+DU Server | | Appropriately dimensioned and included as a separate server. (Rack mountable), | | |
| 3 | IMS solution | 1 | Part of 5G core – COTS standard Server | | |
| 4 | MEC & Application server | 1 | Installed on COTS Standard Server. | | |
| | MEC Server | | Rack Mountable. | | |
| 5 | NMS (with Dashboard) | 1 | Part of 5G core – COTS standard Server | | |
| 6 | Router with Firewall | 1 | (5 years subscription) | | |
| 7 | 5G SIMs | 50 | | | |
| 8 | 5G Evaluation Board / Hardware and Software Development | 1 | | | |

| | | | | | |
|---|---|---|---|---|---|
| | Kit. | | | | |
| 9 | IoT Gateway | 1 | | | |
| 10 | IoT sensors with analytics software (Loaded in Application server) | | | | |
| | Temperature & Humidity sensor | 1 | | | |
| | Light sensor | 1 | | | |
| | Soil sensor (NPK) | 1 | | | |
| | Water (TDS/chlorine) sensor | 1 | | | |
| 11 | 5G Mini Drone | 1 | | | |
| 12 | 5G XR(AR/VR/MR) headset or Device (with loaded application such as remote maintenance / training/skill development/ education etc. ) | 1 | | | |
| 13 | 5G Indoor CPE | 1 | | | |
| 14 | 5G Camera with AI enabled Video Analytics (Face/Object/Motion detection, people counting etc.) | 1 | Analytics to be loaded in Application server, Support can be taken from cloud (VPN) | | |
| 15 | 5G Handsets | 2 | Support testing and testing tools | | |
| 16 | Adjustable Tripod Pole (3m) | 1 | Radio/Antenna mounting | | |
| 17 | 24 U Rack | 1 | | | |
| 18 | UPS 5KVA (1hr backup) | 1 | | | |
| 19 | 32 inch FHD Display | 1 | | | |
| 20 | L2 Managed Switch (24 port) | 1 | | | |
| 21 | Testing & Tracing Tools | 1 | For end-to-end 5G network viz.<br>UE- Radio,<br>Radio-Core, etc. (Open source includes proprietary/third party tools) | | |

## 4.2 5G Network specs (Core+RAN)

| S.No | Technical Specifications | Configuration | Pass/Fail | Remarks |
|---|---|---|---|---|
| | **5G Architecture** | **Standalone** | | |
| i. | Radio Configurations | 1. Integrated RAN (gNB)<br><br>OR<br><br>2. Disintegrated RAN (RU, DU, CU) | | |
| ii. | 3GPP Release | Rel 15 Compliant & upgradable to Rel 16 (by Sep 2024) | | |
| iii. | Data Rate (DL+UL) | >= 200Mbps | | |
| iv. | Latency | < = 15ms. Upgrade to Rel 16 should meet the latency as per the 3GPP standards. | | |
| v. | Max No. of active devices (That network can connect) | 30 (min) | | |
| vi. | Radio Power | Upto 100mW | | |
| vii. | Duplexing Mode | TDD | | |
| viii. | Channel Bandwidth | 20MHz, 40MHz, 60MHz, 80MHz, 100MHz | | |
| ix. | Band support | n78<br>(Must be Tunable to centre frequency with desired channel bandwidth-100MHz in the whole n78 band i.e 3300MHz to 3800MHz) | | |
| x. | MIMO Configuration | 2T2R (min) | | |
| xi. | Synchronization | IEEE 1588 | | |
| xii. | Antenna | External or Internal | | |
| xiii. | Mechanical/Environmental | IP54/QM333 where ever applicable. (Environmental Testing). Temperature and Humidity as per the Indian Geographical conditions | | |
| xiv. | MIMO Configuration | 2T2R (min) | | |
| xv. | Network Slicing | Support for slicing, | | |

| | | | | |
|---|---|---|---|---|
| | | Preconfigured 3 different slices (min). Support for configuring slices by Institutions. | | |
| xvi. | 5G Use case scenarios | eMBB (R15),<br>URLLC(R16),<br>MMTC (R16) | | |
| xvii. | Uptime | 99% | | |
| xviii. | MEC | Should support integration with edge compute and cloud platforms. | | |
| xix. | VoNR | Should support VoNR. | | |
| xx. | APIs | Should support APIs – for example any NMS, MEC platform integration, OT/IT integrations. | | |
| xxi. | NMS with dashboards | Integrated with Core / Radio / other IT devices. | | |
| xxii. | Security | Features to ensure system security (security features for Core, RAN and devices, solution for internal and external attack vectors, proactive security monitoring, reporting and alarms, are the security alerts stored including supporting data & logs)<br>Security considerations in deployment of solution – both interface and application level to be considered | | |
| xxiii. | 5G Devices | All 5G devices to support private PLMN IDs | | |
| xxiv. | IP communication | IP V4 and IP V6 should be supported | | |

| **4.3** | **Functional Specifications : 5G SA Core Networks** | | |
|---|---|---|---|
| **SN** | **Functional Description** | **Pass/Fail** | **Remarks** |
| 4.3.1 | The 5G System architecture is defined to support data connectivity and services enabling deployments to use techniques such as e.g. Network Function Virtualization and Software Defined Networking. The 5G System architecture shall leverage service-based interactions between Control Plane (CP) Network Functions where identified. Some key principles and concept are to: | | |

| | | | |
|---|---|---|---|
| a) | Separate the User Plane (UP) functions from the Control Plane (CP) functions, allowing independent scalability, | | |
| b) | evolution and flexible deployments e.g. centralized location or distributed (remote) location. | | |
| c) | Modularize the function design, e.g. to enable flexible and efficient network slicing. | | |
| d) | Wherever applicable, define procedures (i.e. the set of interactions between network functions) as services, so that their re-use is possible. | | |
| e) | Enable each Network Function to interact with other NF directly if required. The architecture does not preclude the use of an intermediate function to help route Control Plane messages (e.g. like a DRA). | | |
| f) | Minimize dependencies between the Access Network (AN) and the Core Network (CN). The architecture is defined with a converged core network with a common AN – CN interface which integrates different Access Types i.e.3GPP access and non-3GPP access (in Roadmap). | | |
| g) | Support a unified authentication framework. | | |
| h) | Support "stateless" NFs, where the "compute" resource is decoupled from the "storage" resource. | | |
| i) | Support capability exposure. | | |
| j) | Support concurrent access to local and centralized services. To support low latency services and access to local data networks, UP functions can be deployed close to the Access Network. | | |
| | | | |
| 4.3.2 | **The 5G System architecture consists of the following network functions (NF).** | | |
| a) | Authentication Server Function (AUSF) | | |
| b) | Access and Mobility Management Function (AMF) | | |
| c) | Data Network (DN), e.g. operator services, Internet access or 3rd party services | | |
| d) | Network Exposure Function (NEF) | | |

| | | | |
|---|---|---|---|
| | Network Repository Function (NRF) | | |
| f) | Network Slice Selection Function (NSSF) | | |
| g) | Policy Control Function (PCF) | | |
| h) | Session Management Function (SMF) | | |
| i) | Unified Data Management (UDM) | | |
| j) | Unified Data Repository (UDR) | | |
| k) | User Plane Function (UPF) | | |
| l) | Application Function (AF) | | |
| m) | User Equipment (UE) | | |
| n) | (Radio) Access Network (IAN) | | |
| 4.3.3 | Network functions should comply to and support all functionalities as per 3GPP SA Rel 15 architecture upgradable to Rel 16. | | |
| 4.3.4 | Should support Network segmentation/slicing for custom QoS for data connectivity. | | |
| 4.3.5 | Should support integration with edge compute and cloud platforms | | |
| 4.3.6 | Should support VoNR (mention timeline if in roadmap) | | |

## 4.4 5G NR Radio

4.4.1 The gNB should host the following functions: -

| SN | gNodeB Functions | Pass/Fail | Remarks |
|---|---|---|---|
| a) | Functions for Radio Resource Management: Radio Bearer Control, Radio Admission Control, Connection Mobility Control, Dynamic allocation of resources to UEs in both uplink and downlink (scheduling); | | |
| b) | IP header compression, encryption and integrity protection of data; - Selection of an AMF at UE attachment when no routing to an AMF can be determined from the information provided by the UE; | | |
| c) | Routing of User Plane data towards UPF(s); | | |
| d) | Routing of Control Plane information towards AMF; | | |
| e) | Connection setup and release; | | |
| f) | Scheduling and transmission of paging messages; | | |

| | | | |
|---|---|---|---|
| | Scheduling and transmission of system broadcast information (originated from the AMF or OAM); | | |
| h) | Measurement and measurement reporting configuration for mobility and scheduling; | | |
| i) | Transport level packet marking in the uplink; | | |
| j) | Session Management; | | |
| k) | Support of Network Slicing; | | |
| l) | QoS Flow management and mapping to data radio bearers; | | |
| m) | Support of UEs in RRC_INACTIVE state; | | |
| n) | Distribution function for NAS messages; | | |
| 4.4.2 | **Network Interfaces** | **Pass/Fail** | **Remarks** |
| 4.4.3 | **NG User Plane :** | | |
| a) | The NG user plane interface (NG-U) is defined between the NG-RAN node and the UPF. | | |
| b) | The transport network layer is built on IP transport and GTP-U is used on top of UDP/IP to carry the user plane PDUs between the NG-RAN node and the UPF. | | |
| c) | NG-U provides non-guaranteed delivery of user plane PDUs between the NG-RAN node and the UPF. | | |
| d) | Further details of NG-U compliance as per 3GPP standards | | |
| 4.4.4 | **NG Control Plane :** | Pass/Fail | Remarks |
| a) | NG-C provides the following functions: | | |
| b) | NG interface management; | | |
| c) | UE context management; | | |
| d) | UE mobility management; | | |
| e) | Transport of NAS messages; | | |
| f) | Paging; | | |
| g) | PDU Session Management; | | |
| h) | Configuration Transfer; | | |
| i) | Further details of NG-C compliance as per3GPP standards | | |

| | **Radio Protocol Architecture** | **Pass/Fail** | **Remarks** |
|---|---|---|---|
| a) | Protocol stack for the user plane, i.e. SDAP, PDCP, RLC and MAC sublayers (terminated in gNB on the network side) should perform the functions as per 3GPP Release 15. | | |
| b) | The protocol stack for the control plane should perform the function as detailed in 3GPP Release 15 : | | |
| c) | PDCP, RLC and MAC sublayers (terminated in gNB on the network side) should perform the functions listed in 3GPP Standards. | | |
| d) | RRC (terminated in gNB on the network side) performs the functions listed in 3GPP Standards. | | |
| e) | NAS control protocol (terminated in AMF on the network side) performs the functions listed in 3GPP Standards.for instance: authentication, mobility management, security control. | | |
| **4.5** | **Multi-access Edge Computing framework** | | |
| **SN** | **Features & Functionalities** | **Compliance** | |
| 4.5.1 | Multi-access Edge Computing enables the implementation of MEC applications as software-only entities that run on top of a Virtualisation infrastructure, which is located in or close to the network edge. | | |
| 4.5.2 | The multi-access edge system shall consists of the MEC hosts and the MEC management necessary to run MEC applications. | | |
| 4.5.3 | The MEC host shall contains a MEC platform and a Virtualisation infrastructure which provides compute, storage, and network resources, for the purpose of running MEC applications. | | |
| 4.5.4 | The MEC platform shall be the collection of essential functionality required to run MEC applications on a particular Virtualisation infrastructure and enable them to provide and consume MEC services. | | |
| 4.5.5 | The MEC platform may be accompanied by API gateway functionality for MEC applications to access the MEC service APIs. | | |
| 4.5.6 | A MEC service shall be provided and consumed either by the MEC platform or a MEC application. | | |
| 4.5.7 | The Radio Network Information service, when available, shall provide authorized applications with radio network related information. | | |

| | **Firewall with Router / SD-WAN Functional / Technical Specification**<br><br>(Functionalities can be combined or separate) | **Pass/Fail** | **Remarks** |
|---|---|---|---|
| 4.6.1 | **Networking Interfaces** | | |
| a) | 1 Gigabit Base-T (Cu) = 6 ports or higher(as required) | | |
| | 1 Gs Fiber (SFP) = 4 ports | | |
| b) | WAN Interface | | |
| | 2 x 1 Gigabit Ethernet SFP+ ports | | |
| c) | Management Port | | |
| | 10/100 Mbps RJ 45 Management port | | |
| 4.6.2 | Firewall Features | | |
| 4.6.3 | Performance requirements | | |
| a) | Firewall throughput of 1 Gbps | | |
| b) | Threat Protection – 500 Mbps | | |
| 4.6.4 | Web Filtering | | |
| a) | Control access to websites and website categories (e.g. business, news, sports, social media) | | |
| b) | Integrate into Active Directory, applying policies to specific users or user groups. | | |
| c) | Enforce Safe Search | | |
| d) | Block or control access to Cloud services. | | |
| e) | Detailed web access reports (by user, group, website) | | |
| f) | Full SSL Inspection support | | |
| g) | Transparent solution (or explicit proxy support) | | |
| 4.6.5 | Optional: Intrusion Prevention System | | |
| a) | Detect and block network borne attacks | | |
| b) | Protect mail, web and remote-access servers from attacks (IIS, Exchange, Citrix) | | |
| c) | Protect staff and internal systems from application level attacks (e.g. Office, Adobe Acrobat) | | |

| | | | |
|---|---|---|---|
| d) | Common web platform protection (e.g. .NET, PHP, WordPress, CRM) | | |
| e) | Customizable rule-set and signatures | | |
| 4.6.6 | Secure Remote Access | | |
| a) | SSL and IPSec VPN support | | |
| b) | Web-Portal and full VPN client application | | |
| c) | Site-to-Site IPSec VPN (with Dynamic Routing) | | |
| d) | Optional: RADIUS/Active-Directory integration | | |
| e) | Optional: Granular network access policies | | |
| f) | Two-Factor authentication support (e.g. Secure Token) | | |
| g) | High performance VPN concentrator solution | | |
| 4.6.7 | Application Control | | |
| a) | Control and monitor cloud application usage (e.g. Dropbox, Box, Google Apps) | | |
| b) | Full network visibility of HTTP and HTTPS traffic | | |
| c) | Control access to specific applications (e.g. Web Games, Video Streaming) | | |
| d) | Block access to malicious content or websites | | |
| e) | Detailed reporting and network visibility | | |
| f) | Optional: Deep-inspection of traffic for granular reporting | | |
| 4.6.8 | Intelligent Networking | | |
| a) | Dynamic routing (OSPF/BGP/RIP) | | |
| b) | Load-Balancing and ECMP | | |
| c) | 802.1Q VLAN and Layer2 switching support | | |
| d) | Rich traffic/shaping QOS support | | |
| e) | Dynamic routing over VPN (e.g. OSPF) | | |

| **4.7** | **Server 1 with 5G Core, IMS and NMS software** | **Pass/Fail** | **Remarks** |
|---|---|---|---|

| | | | |
|---|---|---|---|
| | Processors | | |
| a) | 3rd Generation Intel® Xeon Gold 2.1 GHz 24 Core or equivalent. Each servers with minimum 1 Processors and 2 Socket or higher as required. | | |
| 4.7.2 | Memory | | |
| a) | DDR4 or above 64 GB scalable upto 256 GB or above | | |
| b) | Support for advanced memory redundant technologies like Advanced error-correcting code (ECC) and memory mirroring | | |
| 4.7.3 | **Storage** | | |
| a) | SAS SSD Storage = 1TB Usable expandable to 5 TB | | |
| b) | The server should have front-accessible, hot-swappable, Enterprise class SSD drives. | | |
| 4.7.4 | Operating System And Virtualization Software | | |
| a) | Virtualization and OS for all environment. | | |
| 4.7.5 | **Network Interface** | | |
| a) | Dual port 1G SFP+ card, 4 x 1G BaseT | | |
| 4.7.6 | PCIe Slots | | |
| a) | Minimum 4 x PCIe 16x 4.0 slots for I/O Connectivity or higher as required. | | |
| 4.7.7 | **Management** | | |
| a) | Support out of band upgrades, Agentless out-of-band management, integrated diagnostics and Power monitoring, Virtual Media, HTML5 remote control. | | |
| b) | The server should support industry standard management protocols i.e. IPMI, Red Fish API etc. | | |
| c) | One 1-Gbps RJ-45 management port | | |
| d) | Supports a stateless environment where server identity is created by the administrator who defines the server BIOS version, MAC ID, NIC firmware version, WWPN FC-HBA firmware version, Adapter QoS, Management module firmware version, UUIDs, Server Boot Policies, KVM IP etc. | | |
| e) | Auto generated technical support logs and forwarding to technical support for faster resolution. | | |
| f) | Automated hardware configuration and Operating System deployment to multiple servers | | |

| | | | |
|---|---|---|---|
| g) | The proposed management solution should provide proactive security & software advisory alerts and should outline the fixes required to address the issues | | |
| h) | The proposed management solution should analyze current configurations & identify potential issues due to driver & firmware incompatibility | | |
| i) | The management system should report validation based on compatibility of server model, processor, firmware, adapters, operating system and drivers & share compliance status | | |
| j) | The server should support multiple management interfaces including web user interface and command line interface. | | |
| 4.7.8 | **Security** | | |
| a) | The server should provide cryptographic firmware updates, (Every SW upgrade will be FIPS 104-2 validated) or equivalent. | | |
| b) | Server management system should provide anticounterfeit | | |
| c) | The server should provide hardware policy based security, Detection of any Configuration drifts | | |
| d) | The proposed server OEM should certify malicious code free design (undertaking to be submitted on OEM letterhead) | | |
| e) | The server should provide Hardware root of trust for Secure Boot Image Failsafe as per NIST SP800-147(B) or equivalent. | | |
| 4.7.9 | Modules/Licenses | | |
| a) | All the modules/licenses (perpetual) relevant to the above functionalities should be included as part of the solution. | | |
| 4.7.10 | Power and Others | | |
| a) | To be fully populated with fans & redundant power supplies. | | |
| b) | Rail Kit and cable management arm to be provided along with the server | | |
| c) | All allied components such as power cable, FC, ethernet cables and sockets/connectors as required for provisioning and integration of the solution should be provided by Contractor. Should be housed in suitable rack. | | |
| | | | |

| 4.8 | **Server 2 with 5G MEC Platform and Application Server** | **Pass/Fail** | **Remarks** |
|---|---|---|---|
| 4.8.1 | **Processors** | | |
| a) | 3rd Generation Intel® Xeon Gold 2.1 GHz 24 Core or equivalent. Each servers with minimum 1 Processors and 2 Socket or higher as required. | | |
| 4.8.2 | Memory | | |
| a) | DDR4 or higher, 64 GB scalable upto 256 GB | | |
| 4.8.3 | **Storage** | | |
| 4.8.4 | SAS SSD Storage = 4TB Usable expandable to 8 TB or above<br>**HDD Storage =** 4TB Usable expandable to 8 TB or above | | |
| 4.8.5 | Operating System and Virtualisation | | |
| a) | Virtualisation and OS for all environment. | | |
| 4.8.6 | Network | | |
| a) | Single port 1G SFP+ card, 2 x 1G BaseT | | |
| 4.8.7 | PCIe Slots | | |
| a) | Minimum 4 x PCIe 16x 4.0 slots for I/O Connectivity or higher as required. | | |
| 4.8.8 | **Management** | | |
| a) | Support out of band upgrades, Agentless out-of-band management, integrated diagnostics and Power monitoring, Virtual Media, HTML5 remote control. | | |
| b) | The server should support industry standard management protocols i.e. IPMI etc. | | |
| c) | One 1-Gbps RJ-45 management port | | |
| d) | Supports a stateless environment where server identity is created by the administrator who defines the server BIOS version, MAC ID, NIC firmware version, WWPN FC-HBA firmware version, Adapter QoS, Management module firmware version, UUIDs, Server Boot Policies, KVM IP etc. | | |
| e) | Auto generated technical support logs and forwarding to technical support for faster resolution. | | |
| f) | Automated hardware configuration and Operating | | |

| | | | |
|---|---|---|---|
| | System deployment to multiple servers | | |
| g) | The proposed management solution should provide proactive security & software advisory alerts and should outline the fixes required to address the issues | | |
| h) | The proposed management solution should analyze current configurations & identify potential issues due to driver & firmware incompatibility | | |
| i) | The management system should report validation based on compatibility of server model, processor, firmware, adapters, operating system and drivers & share compliance status | | |
| j) | The server should support multiple management interfaces including web user interface and command line interface. | | |
| 4.8.9 | **Security** | | |
| a) | The server should provide cryptographic firmware updates, (Every SW upgrade will be FIPS 104-2 validated) or equivalent | | |
| b) | Server management system should provide anticounterfeit | | |
| c) | The server should provide hardware policy based security, Detection of any Configuration drifts | | |
| d) | The proposed server OEM should certify malicious code free design (undertaking to be submitted on OEM letterhead) | | |
| e) | The server should provide Hardware root of trust for Secure Boot Image Failsafe as per NIST SP800-147(B) or equivalent | | |
| 4.8.10 | Modules/Licenses | | |
| a) | All the modules/licenses (perpetual) relevant to the above functionalities should be included as part of the solution. | | |
| 4.8.11 | Power and Others | | |
| a) | To be fully populated with fans & redundant power supplies. | | |
| b) | Rail Kit and cable management arm to be provided along with the server | | |
| c) | All allied components such as power cable, FC, ethernet cables and sockets/connectors as required for provisioning and integration of the solution should be provided by Contractor. Should be housed in suitable rack. | | |

| | **5G rewriteable SIM (programable)** | **Pass/Fail** | **Remarks** |
|---|---|---|---|
| a) | Network Supported : 5G SA | | |
| b) | File Structure : 3GPP Compliant | | |
| c) | Data Security : Secure channel | | |
| d) | Authentication : 3GPP Compliant | | |
| e) | Should be Vibration, shock, impact and chemical resistance | | |
| f) | Material : PVC | | |
| g) | Card Characteristics : as per ISO/IEC 7810 | | |
| h) | Any future upgrade should be supported through SIM upgrade or replacement or through OTA | | |
| i) | SIM card programmer/writer or support to provision SIMs in agreed response time throughout the project period | | |
| j) | SIM cards to support all sizes. | | |
| | | | |
| **4.10** | **5G Evaluation Board** | **Pass/Fail** | **Remarks** |
| a) | 5G Qualcomm Snapdragon™ X62/X65 or higher 5G Modem-RF system Supported | | |
| b) | **Cellular Modem-RF Specs:** 5x20 MHz carrier aggregation (DL) and above, 100/200 MHz bandwidth (sub-6 GHz),<br><br>Optional: 8 carriers (mmWave), 3x20 MHz carrier aggregation (UL), 800 MHz bandwidth (mmWave). | | |
| c) | **Performance Enhancement Technologies:** 100 MHz envelope tracking, Adaptive antenna tuning, Uplink Data Compression (UDC), 5G envelope tracking. | | |
| d) | Physical and Wi-Fi Interface Modules. | | |
| e) | Peripheral connectors on the platform support UART, SPI, I2C and GPIO. | | |
| f) | Sensor interfaces for medical devices, smart-home appliances and smart-city applications. | | |

| | | | |
|---|---|---|---|
| | SD Card Connector. | | |
| h) | PCI to USB Connector. | | |
| i) | Dual (U)SIM card supported: 1.8 V and 2.95 V | | |
| j) | 1 digital audio codec board interface | | |
| k) | COM1 and COM2. Also support Main URAT, Debug URAT | | |
| l) | USB Type C Interface, USB 3.0 & USD 2.0 supported | | |
| m) | LEDs indication | | |
| n) | Power key, Reset key, USB Boot, Power Switch, etc. | | |
| o) | 12 Antenna Interfaces | | |
| p) | should include minimum Cables, Antenna, Audio earphone, USB Driver, Required modules of Wi-Fi, Ethernet, Audio, PCI TO USB, Power Adaptors | | |
| q) | AT command (external MCU integration) and SDK support (within the module) | | |
| r) | Application MCU Development Board for Applications like Smart Home, Smart City, Medical Devices, Agritech etc.<br><br>32bit 400MHz Dual Core or higher<br><br>Interface. I2C, SPI, UART, GPIO, ADC, Timer, EEPROM, Flash/NVS, etc<br><br>RTOS Support for Real-Time Application<br><br>MCU Development Board Should able to integrate with the 5G Development Board via Wi-Fi / Ethernet / USB. | | |
| | | | |
| **4.11** | **IOT Gateway** | Pass/Fail | Remarks |
| a) | CPU : Qualcomm or equivalent, Dual core or Higher | | |
| b) | RAM: 256 MB | | |
| c) | Module: 5G Sub-6Ghz SA 1/2 Gbps DL (4x4 MIMO) or higher , 900/600 Mbps UL (2x2); | | |
| d) | Status: Signal strength (RSSI), SINR, RSRP, RSRQ, EC/IO, | | |

| | | | |
|---|---|---|---|
| | RSCP Bytes sent/received, connected band, IMSI, ICCID. | | |
| e) | SMS/Call status | | |
| f) | Supports sending and reading Unstructured Supplementary Service Data messages | | |
| g) | APN | | |
| h) | Possibility to use different PDNs for multiple network access and services | | |
| 4.11.2 | Access Points | | |
| a) | Supports 802.11b/g/n/ac | | |
| b) | WiFi Security : WPA X /WEP/ TKIP variants as per the latest version. | | |
| c) | Support minimum 30 simultaneous WiFi users connections | | |
| d) | Optional: Support LoRa, Sigfox any of the LPWAN | | |
| 4.11.3 | Networks | | |
| a) | Supports all required network protocols. | | |
| b) | Supports Static and dynamic routing | | |
| c) | Supports static and dynamic IP allocation | | |
| 4.11.4 | VLAN | | |
| a) | Multiple clients and a server can run simultaneously | | |
| b) | WireGuard VPN client and server support | | |
| c) | Port and tag-based VLAN separation | | |
| d) | Supports up to 5 x VPn IPsec tunnels with different AES/DES encryption methods. | | |
| 4.11.5 | Security | | |
| a) | Pre-configured firewall rules can be enabled via the WebUI, unlimited firewall configuration via CLI; NAT; NAT-T | | |
| b) | DDOS prevention (port scan prevention | | |
| c) | Sets up custom data limits for SIM card | | |

| | | | |
|---|---|---|---|
| | Blacklist for blocking out unwanted websites, Whitelist for specifying allowed sites only | | |
| 4.11.6 | Management | | |
| a) | Status, configuration, FW update, CLI, troubleshoot, event log, system log. | | |
| b) | Allows monitoring of: Device data, Mobile data, Network info, Availability | | |
| c) | Allows monitoring of: Device Model, Revision and Serial Number, Mobile Cell ID, ICCID, IMEI, Connection Type, Operator, Signal Strength, WAN Type and IP | | |
| d) | Allows monitoring of: WAN Type, WAN IP, Mobile Operator Name, Mobile Signal Strength, Mobile Network Type. | | |
| e) | Firmware update from server, automatic notification | | |
| f) | Management API over HTTP/HTTPS | | |
| 4.11.7 | Others | | |
| a) | Supports 1 or more SIMs | | |
| b) | Supports location tracking | | |
| c) | Supports USB. | | |
| d) | Supports digital input/output | | |
| e) | Supports minimum 4 nos 10/100/1000 Ethernet ports. | | |
| f) | Supports 1G WAN port. | | |
| g) | Supports status LED | | |
| h) | Supports PoE and DC power socket. | | |
| | | | |
| **4.12** | **5G Mini Drones** | **Pass/Fail** | **Remarks** |
| a) | 5G enabled and controlled drone | | |
| b) | Supports 5G nano SIM | | |
| c) | Connectivity Options: Cellular ( 5G ) and Wi-Fi for ubiquitous and long-range connectivity | | |

| | | | |
|---|---|---|---|
| | Qualcomm or equivalent onboard processing engine for AI/ML analytics<br><br>Should support MEC applications | | |
| e) | Autonomy: Visual Inertial Odometry (VIO), path planning, PX4, GPS-denied navigation, BVLOS, follow me. | | |
| f) | 360 obstacle avoidance: Depth estimation, obstacle detection, mapping and VOA | | |
| g) | Payload Capacity – 500g/1 Kg or above | | |
| h) | GPS or NAVIC enabled | | |
| i) | Geo Fencing | | |
| j) | Processing Specs | | |
| k) | Tracking sensor | | |
| l) | Image Sensor | | |
| m) | 5G Modem (n78 band) | | |
| n) | Ethernet / management port | | |
| o) | Wi-Fi (2.4/5 GHz) | | |
| | | | |
| **4.13** | **XR (AR/VR ) Device** | **Pass/Fail** | **Remarks** |
| 4.13.1 | AR/VR Device | | |
| a) | Chipset : 2.0 GHz 8-core Qualcomm | | |
| b) | Memory : 64GB internal storage / 4GB RAM / MicroSD slot | | |
| c) | Ruggedized against moisture, dust and free fall. | | |
| d) | Port : USB type C and Audio | | |
| e) | Inbuilt Application : Skill development/remote maintenance / training/skill development/ education. | | |
| f) | Qualcomm Trusted Execution Environment, Crypto Engine, Secure Boot | | |
| 4.13.2 | Connectivity / Sensors | | |
| a) | Bluetooth 5.0 | | |

| | | | |
|---|---|---|---|
| b) | 802.11 a/b/g/n/ac - 2.4 GHz and 5GHz | | |
| c) | Qualcomm® Sensor-Assisted Positioning | | |
| d) | 3 IMUs (including 3-axis accelerometers, gyroscopes, eCompass) | | |
| e) | NavIC or GPS location support (optional) | | |
| f) | In-built 5G Modem (optional) | | |
| 4.13.3 | Input/Output | | |
| a) | Display: 24° field-of-view, 24-bit color LCD, outdoor visible, HD resolution | | |
| b) | Microphone | | |
| c) | Speaker : Internal 94 dBA loudspeaker | | |
| | | | |
| **4.14** | **5G Indoor CPE** | **Pass/Fail** | **Remarks** |
| 4.14.1 | **Processor – Qualcomm or equivalent** | | |
| a) | CPU Clock Speed 1.0GHz or higher | | |
| b) | CPU Cores Dual-core Cortex-A53 Arm or equivalent | | |
| 4.14.2 | Memory : 512 MB RAM or higher | | |
| 4.14.3 | **5G** | | |
| a) | 3GPP Release 15 or above support | | |
| b) | 5G Sub 6 Band: n78 TDD | | |
| c) | 5G Modem : Snapdragon X62/X65 or equivalent | | |
| d) | Bandwidth Support : 20/40/60/80/100 MHz | | |
| e) | MIMO : 4 x 4 or higher | | |
| f) | **Download Speed:** Up to 1000 mbps | | |
| 4.14.4 | **GPS Supported.** | | |
| 4.14.5 | **Wireless** | | |
| a) | AP type: Indoor, dual radio, 2.4, 5-GHz | | |
| b) | Dynamic frequency selection (DFS] | | |

| | | | |
|---|---|---|---|
| c) | Standards: IEEE 802.11 a/b/g/n/ac wave 2 | | |
| d) | 2x2 MU-MIMO with 2 spatial streams | | |
| e) | PHY rate: 500 Mbps (2.4 GHz) 2048 Mbps (5GHz) | | |
| f) | IPv4 and IPv6 dual stack support | | |
| 4.14.6 | **RF** | | |
| a) | Integrated Omni-directional Antenna | | |
| b) | Transmit Power :<br>• 25 dBm on 2.4 GHz;<br>• 25 dBm on 5 GHz | | |
| c) | Supports -95 dBm or better Receiver Sensitivity | | |
| d) | Supports RF management. | | |
| e) | Intelligent RF control plane for self-healing and self-optimization | | |
| 4.14.7 | **Security** | | |
| a) | Portal and guest accounts | | |
| b) | Flexible guest access with device isolation Captive. | | |
| c) | All version and Latest WPA support. | | |
| d) | Rogue access point detection and prevention. | | |
| e) | IP Security (IPSec), PPTP, IP-Filtering | | |
| f) | *Optional: Wireless intrusion detection systems (**WIDS**) and wireless intrusion protection systems (**WIPS**)* | | |
| g) | MAC address authentication | | |
| h) | Optional: X.509 digital certificates | | |
| i) | Optional: Support for locally-significant certificates using Public Key Infrastructure (PKI) | | |
| 4.14.8 | **Interface** | | |
| a) | 2 X 1000 Base-T Ethernet | | |
| b) | 1 X SIM card slot (Nano-SIM) | | |
| c) | Power-over-Ethernet or DC Power | | |
| 4.14.9 | **Management** | | |

| | | | |
|---|---|---|---|
| a) | Management: Standalone (via GUI) or through on premise based solution or cloud-based. | | |
| b) | Radio Resource Management for power and channel. | | |
| c) | Compliant with IEEE 802.11e Standard. | | |
| d) | Intelligent RF control plane for self-healing and self-optimization. | | |
| | | | |
| **4.15** | **5G Camera** | **Pass/Fail** | **Remarks** |
| 4.15.1 | The camera shall supports advanced AI/ML video analytics<br><br>Applications on MEC platform: Face Detection/ Recognition, People Counting, LPR(License Plate Recognition), Object Detection/Classification, and object pattern recognition models Etc. | | |
| 4.15.2 | 5G enabled for Private Network.<br>• Supports Sub-6G bands:<br>• FR-1, N78 | | |
| 4.15.3 | SIM Card Supported : Nano SIM | | |
| 4.15.4 | Interface<br>• 10/100/1000 Ethernet port – 1 nos<br>• Audio<br>• Alarm | | |
| 4.15.5 | **Camera** | | |
| a) | Sensor :<br>• 5M / 8M effective resolution<br>• | | |
| b) | Wide Dynamic Range (WDR) = 120dB<br><br>• to improves a cam'ra's image quality under high-contrast lighting conditions where both dimly and brightly lit areas are present in the field of view. | | |
| c) | IR distance supported = 30 meter | | |
| d) | Memory : Micro SD & SDXC slot up to 256GB; | | |
| 4.15.6 | **Lens** | | |

| | | | |
|---|---|---|---|
| | Lens: Motorized, vary focal (2.7mm-13.5mm) | | |
| b) | Field of View(FOV) :<br>• Horizontal – 98 Deg wide.<br>• Vertical – 50 Deg wide. | | |
| c) | Focus and IRIS Control : Auto | | |
| 4.15.7 | **Video** | | |
| a) | Video Compression : H.265 / H.264 / MJPEG or equivalent | | |
| b) | High speed live Streaming Capability. | | |
| c) | Frame Rate = 1 – 30/60 fps. | | |
| 4.15.8 | Power : PoE (802.3at) | | |

**4.16** Other Accessories for Installation

| S.No | Equipment/Devices | Compliance |
|---|---|---|
| i. | Antennas & Passive components such as cables, connectors, mounting brackets, etc.; Power/OFC cables, AC-DC convertors, GPS antenna with accessories etc. | |

**4.17 Technical Specification for Rack and other accessories**

| SN | Rack Technical Specifications | Compliance |
|---|---|---|
| 4.17.1 | Standard Intelligent racks to host 5G SA equipment. | |
| 4.17.2 | Each cabinet should include 2 nos of Rack PDUs come with up to 16 outlets per strip or more. | |
| 4.17.3 | Shall consist of 24U heights of Steel material to accommodate supplied Servers (3 or more), Routers, UPS, Battery, Monitor etc. | |
| 4.17.4 | should have open perforation area (more than 75%) on the door to help better cool servers. | |
| 4.17.5 | The cabinet shall be comprised of modular cable management panels or finger sections. | |
| 4.17.6 | 32 inch FHD Display screen.(Wall/Tabletop) | |
| 4.17.7 | UPS 5KVA with Battery for 1hour backup on full load ( whole Lab setup) | |
| 4.17.8 | L2 Managed Switch 24x 1G ports +2 x 10G (fiber) with standard security, management, protocol, VLAN and IPV6 support | |
| 4.17.9 | Optional: 1 U sliding rack optimized keyboard, mouse with KVM switch | |

| | | |
|---|---|---|
| | to monitor 3 severs and networking routers/switches. | |

# gNB User Manual

# Table of Contents

## Contents

1. Introduction ..... 3
2. Product description ..... 4
3. Package contents ..... 5
4. CU-DU server ..... 5
4.1 Access the CU-DU server ..... 5
4.2 checklist covering all services ..... 6
*4.2.1 Check ts2phc service is active or not by using the following commands ..... 6*
*4.2.2 Check PTP service is active or not by using the following commands ..... 6*
*4.3.3 Check PHC service is active or not by using the following commands ..... 6*
5. Steps to Bring up gNB process ..... 7
5.1 Start Docker application steps ..... 7
5.2 Run the gNB ..... 8
*5.2.1. SMO GUI ..... 8*
*5.2.2 To run and execute gNB O1 Handler ..... 9*
*5.3.3. To start the gNB process ..... 10*
*5.3.4. To start or stop the gNB process ..... 11*
*5.3.5 Camp the UE once all status are running on proprietary-system options ..... 12*
*5.3.6. Manual Steps to Add gNB Node ..... 12*
6. To check the CU-DU logs ..... 14
6.1. To check the L1 logs follow the below steps ..... 14
6.2. To check the CU logs follow the below steps ..... 15
6.3 To check the PDCP logs follow the below steps ..... 16
6.4. To check the DU logs follow the below steps ..... 17
7. Following are the steps to check the L1 config files ..... 18
8. Following are the steps to check the DU config files ..... 19
9. Following are the steps to check CU config files ..... 20
10. Diagnose ..... 21

WiSig Networks - Confidential

# 1. Introduction

This document provides comprehensive instructions for Accessing, bringing up, and diagnose the CU-DU server. Please read this manual carefully before using the product to ensure optimal performance and safety.

# 2. Product description

The CU-DU is an essential part of a mobile telecommunications system that connects individual mobile devices to the core network. It serves as the interface between mobile devices, such as smartphones or tablets, and the mobile network infrastructure.

# 3. Package contents

- CU-DU server -1 Qty
- Power cord -2 Qty
- CAT6 LAN cable -1 Qty
- 5dBm GNSS antenna -1 Qty
- VGA cable -1 Qty
- SFP+ SR Modules -2Qty
- SFT to Ethernet – 1Qty

# 4. CU-DU server

## 4.1 Access the CU-DU server

Connect the FH and BH ports according to the following diagram. The image contains the Server IP/BH , BMC IP, and FH IP

*Figure 1:CU-DU server*

Once the Server is deployed in Network , Access the CU-DU server using the following credentials

ssh username@**172.16.0.4**
password: **wisig321**

Note: All the usernames, BMC IP, DU Virtual MAC MAC addresses are mentioned in below link with corresponding states and addresses.
https://docs.google.com/spreadsheets/d/15bCbiDfybbH7DRxhriKg7CEX3X0KI8WMuOoCne74_YY/edit#gid=0

## 4.2 checklist covering all services

Verify the status of the following services to ensure they are active. If any service is found inactive, execute the corresponding commands to start the services

### 4.2.1 Check ts2phc service is active or not by using the following commands

$ **sudo systemctl status ts2phc.service**

Review the service status information displayed. If the service is active and running, no further action is required. If the service is inactive type the following command to activate the service

$ **sudo systemctl restart ts2phc.service**

### 4.2.2 Check PTP service is active or not by using the following commands

$ **sudo systemctl status ptp4l.service**

Review the service status information displayed. If the service is active and running, no further action is required. If the service is inactive type the following command to activate the service

**$ sudo systemctl restart ptp4l.service**

### 4.3.3 Check PHC service is active or not by using the following commands

**$ sudo systemctl status phc2sys.service**

Review the service status information displayed. If the service is active and running, no further action is required. If the service is inactive type the following command to activate the service

**$ sudo systemctl restart phc2sys.service**

# 5. Steps to Bringup gNB process

## 5.1 Start Docker application steps

Open the terminal and execute the following script from the "/home/username/SMO/docker-compose"

**$ sudo su**
**$ ./start_sdnr.sh**

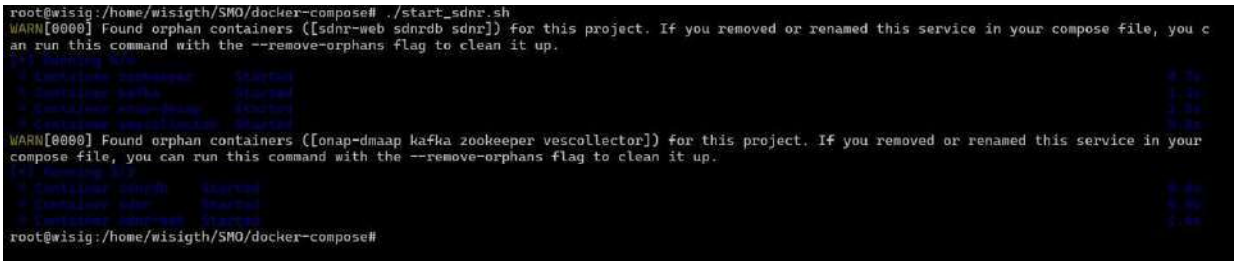



*Figure 2:sdnr_logs*

Check status

***$ docker ps***

Note: It brings up all containers

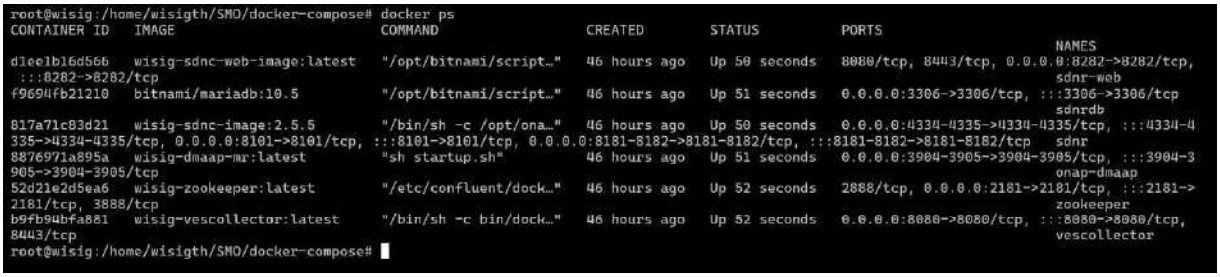



*Figure 3: Docker ps status*

Looking into the SDNR logs will wait until certificate installation script has finished executing.

***$ docker logs -f sdnr***

*Figure 4: Certification status*

## 5.2 Run the gNB

### 5.2.1. SMO GUI

Open the SMO GUI URL in Chrome: http://172.16.0.4:8282/odlux/index.html in the PC

-Username: admin
-Password: admin

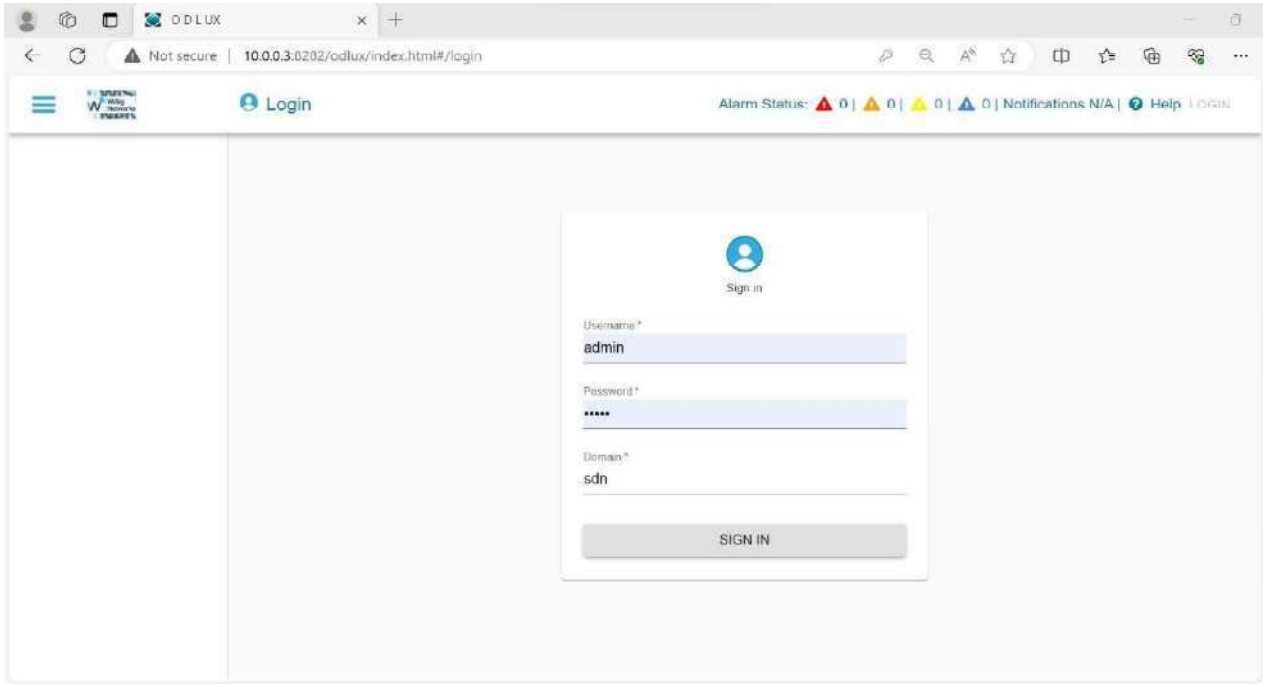



*Figure 5: Admin page*

### 5.2.2 To run and execute gNB O1 Handler

Go to the terminal and execute the gNB O1Handler, run the following script from the "/home/username/SMO/gNB_O1_hdl/bin" directory using the sudo command:

**$ sudo ./o1_handler_startup.sh**

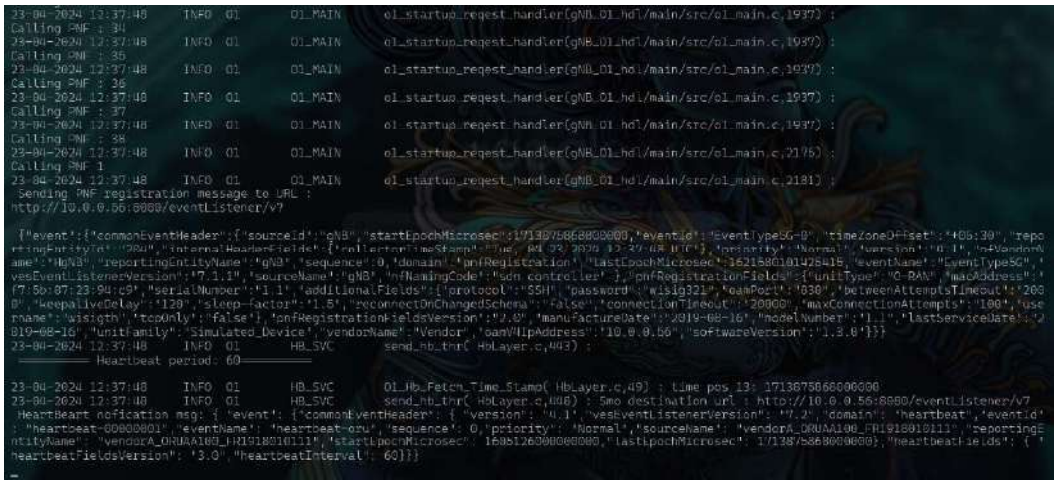


*Figure 6: O1_handler_logs*

## 5.3.3. To start the gNB process

To start the gNB process, verify that the gNB in a connected state. If it is not automatically connected, follow the manual steps provided in **section 5.3.6.**

Once connected, navigate to the Configuration page and click on the gNB. Within the gNB, locate the "proprietary-system."

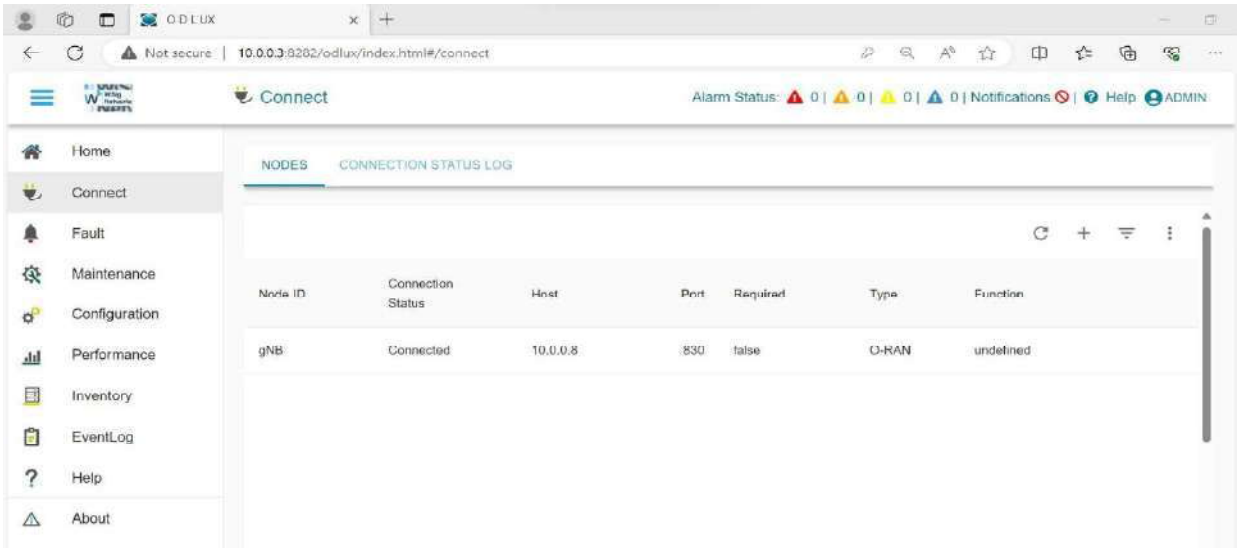



*Figure 7: gNB connection_status*

Note: Ensure that the connection status mode is set to CONNECTED. If it is not connected, please follow manual steps.

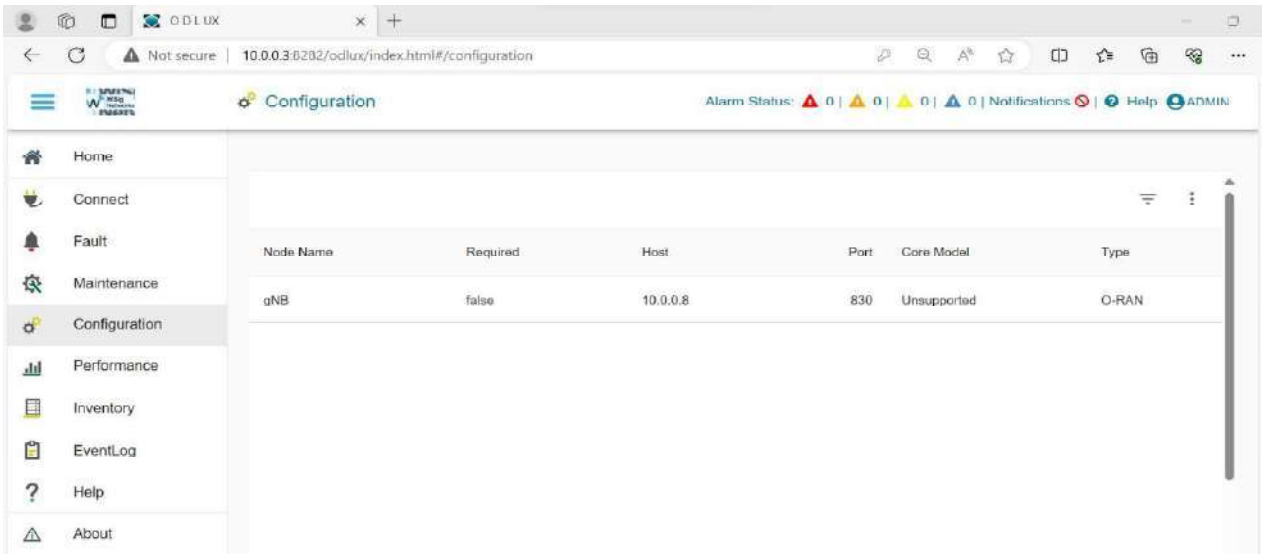



*Figure 8: gNB connected*

## 5.3.4. To start or stop the gNB process

Under "proprietary-system," click on "system-start" to initiate the gNB process, including CU, DU, and L1 or click on "system-stop" to halt the gNB process.

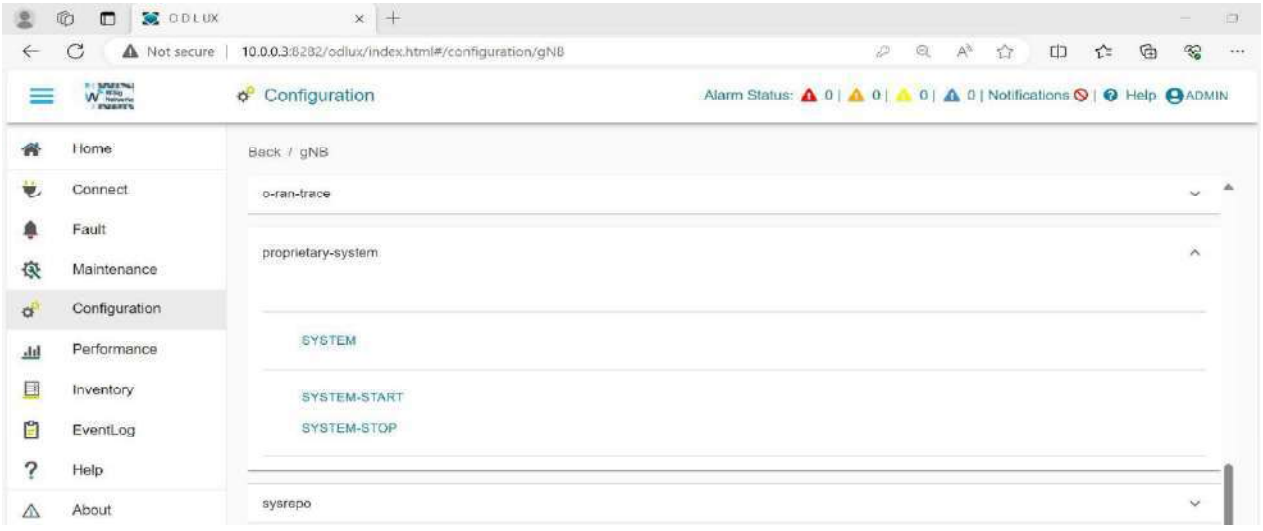



***Figure 9: System start/stop***

Note: Don't click the "system-start" multiple times, it will stall the complete execution

**To stop the entire RAN execution:**

1. Go to chrome browser, click on "**proprietary system** => **System STOP => EXEC**"
2. Go to **new** terminal and Kill manually if any gNB ( **l1app, gNB_du, kill.sh** ) process is running using below command
   **$ kill -9 <pid>**
3. Go to terminal and give "**ctrl+c**" in **o1_handler_start.sh** terminal
4. <**cd /home/username/SMO/docker-compose/**>
   - **$ ./stop_sdnr.sh**
   - $ **docker system prune -y**
   - Check if any process is running with below command
     **$ ps -al**

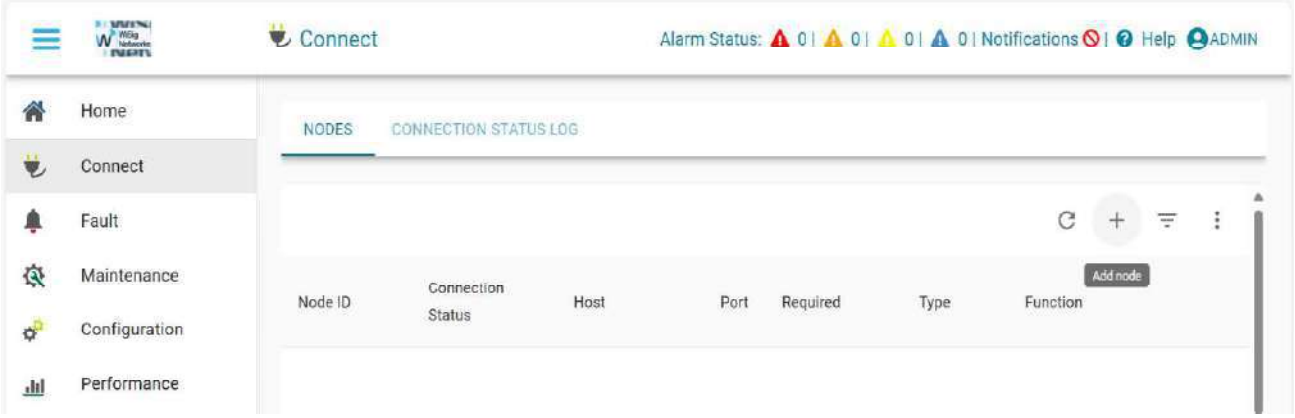
### 5.3.5 Camp the UE once all status are running on proprietary-system options

To check the status navigate to below path
Go to the configuration tab ->gNB-> proprietary-system->system-monitoring

### 5.3.6. Manual Steps to Add gNB Node

To add a gNB node manually, enter the following fields to match the system credentials.

Node ID: gNB
Host/IP address: [Enter the system's IP address]
Netconf Port: [Enter the Netconf port number]
Username: [Enter the username]
Password: [Enter the password]



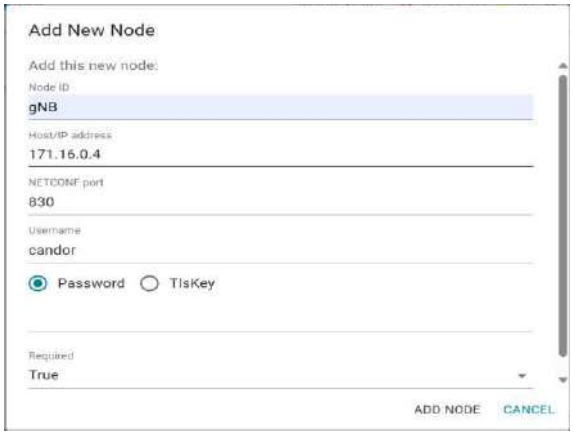


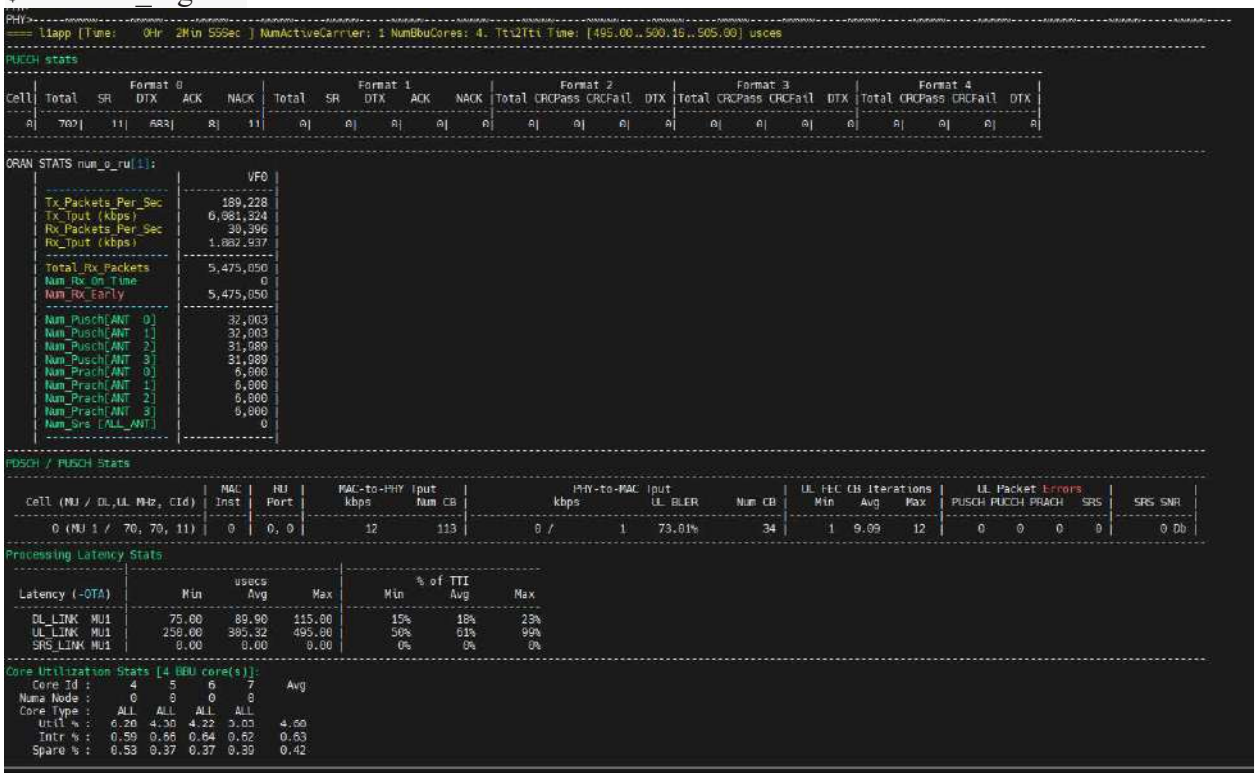

*Figure 10:Adding gNB node*

Note: Ensure that the provided information is accurate to successfully add the gNB node to the system.

# 6. To check the CU-DU logs

## 6.1. To check the L1 logs follow the below steps

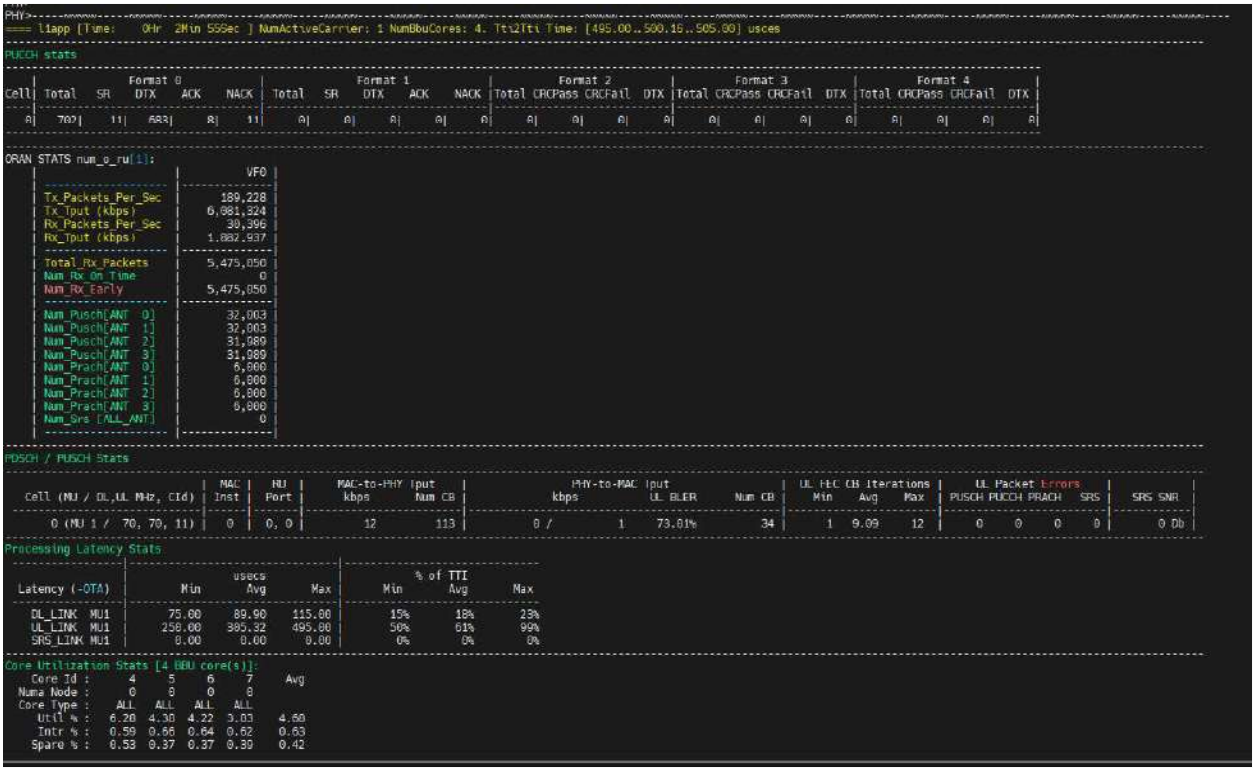
```
$ cd /home/candor/SMO/gNB_O1_hdl/O-RAN/Slot0/runBharatRan/logs
$ tail –f l1_logs.txt
```



## 6.2. To check the CU logs follow the below steps

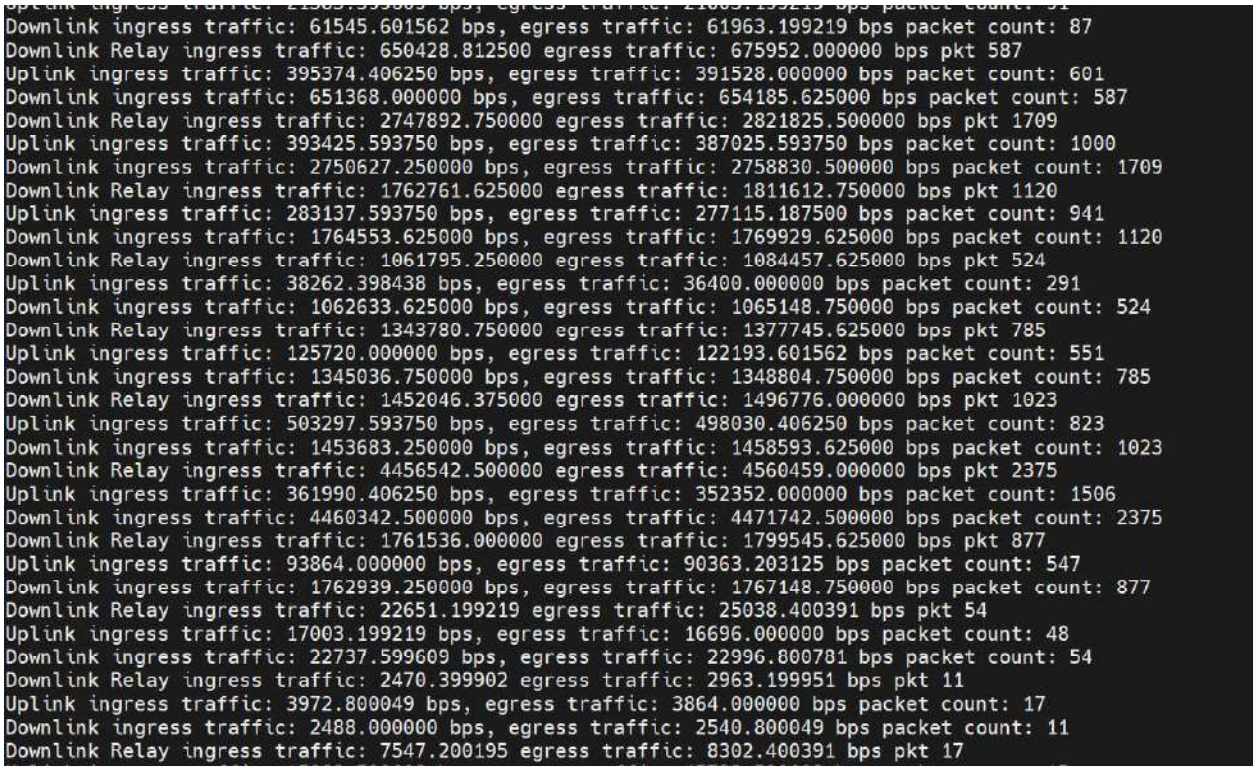
$ cd /home/candor/SMO/gNB_O1_hdl/O-RAN/Slot0/runBharatRan/logs

$ tail –f cu_logs.txt



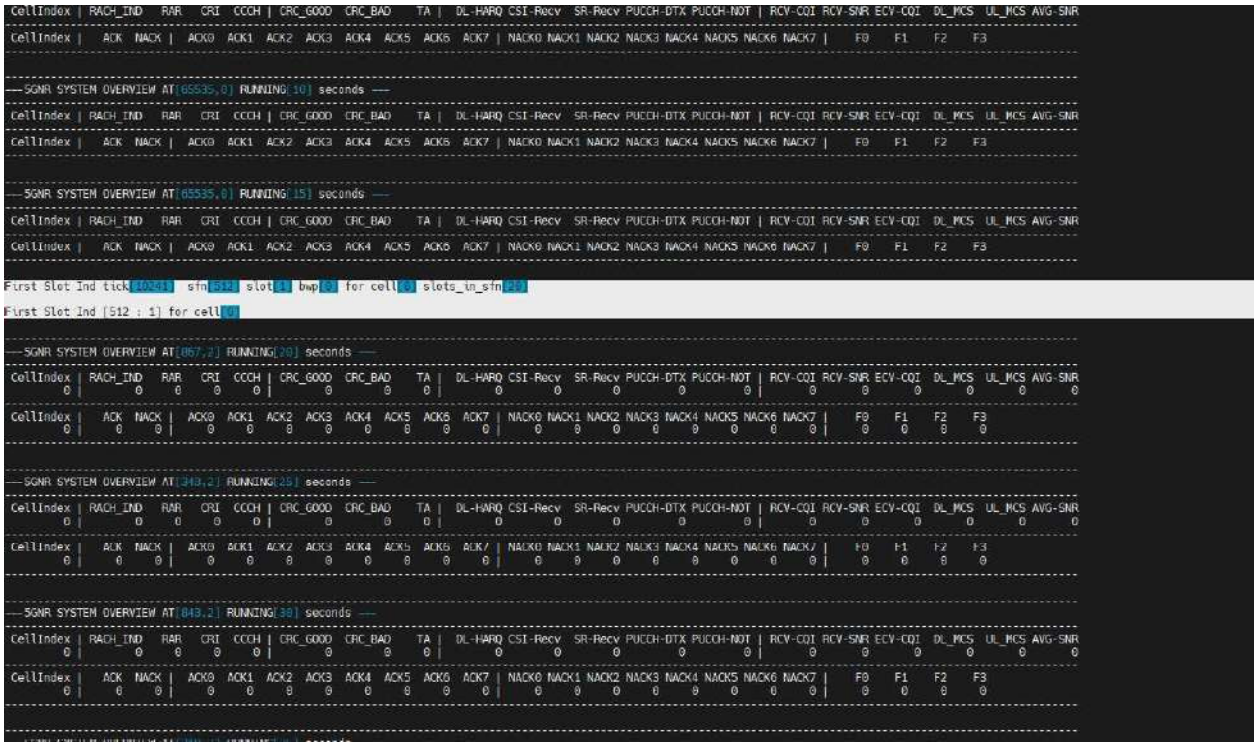
## 6.3 To check the PDCP logs follow the below steps

$ cd /home/candor/SMO/gNB_O1_hdl/O-RAN/Slot0/runBharatRan/logs
$ tail –f pdcp_logs.txt



## 6.4. To check the DU logs follow the below steps

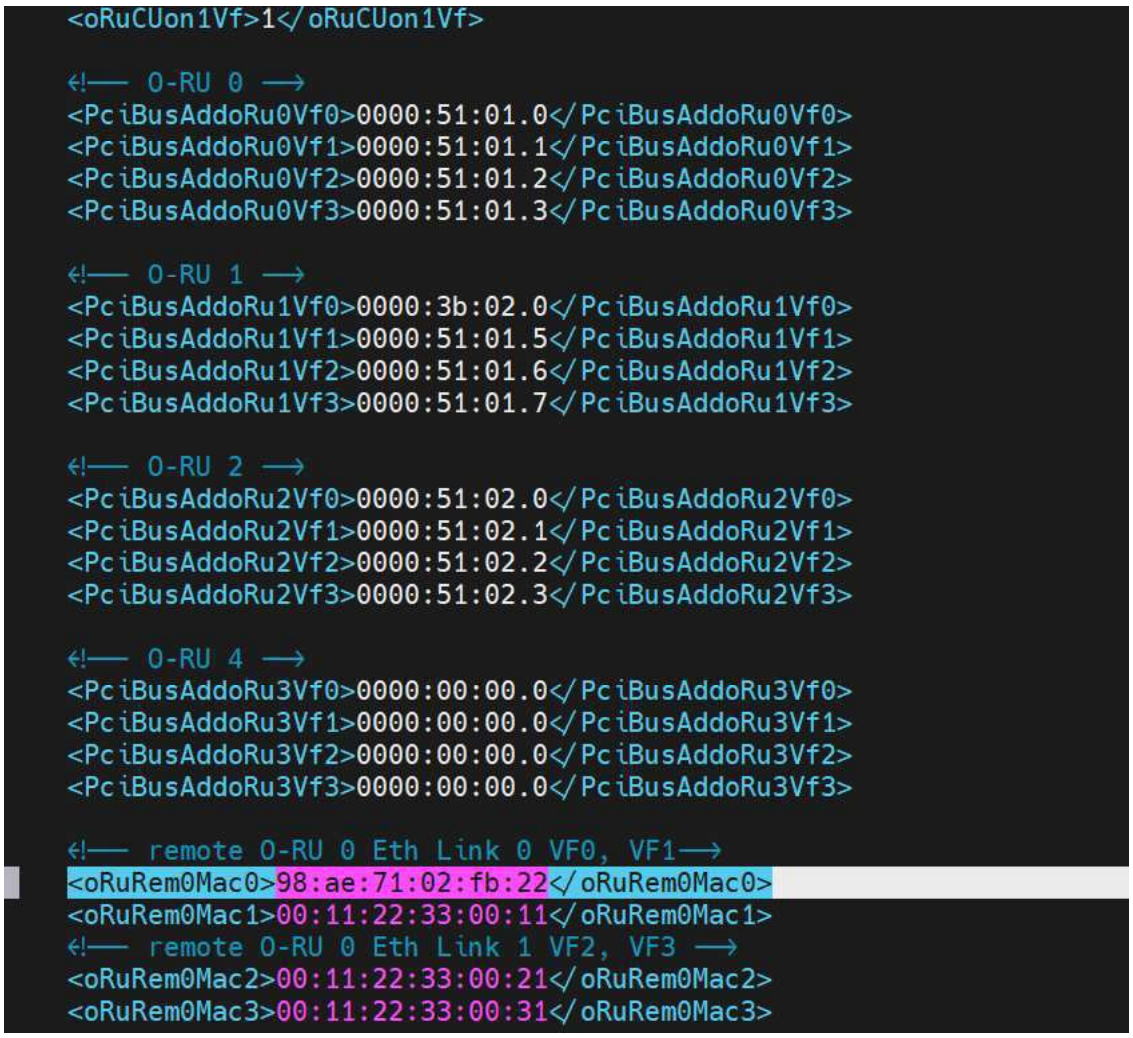
$ cd /home/candor/SMO/gNB_O1_hdl/O-RAN/Slot0/runBharatRan/logs
$ tail –f du_logs.txt



# 7. Following are the steps to check the L1 config files

1. To check the L1 config files follow the below steps
2. To check the RU MAC address in L1 config file go to below path
3. $ cd /home/candor/smo/l1/bin/nr5g/gnb/l1
4. Open the following file using vim

**vim xrancfg_sub6.xml**



# 8. Following are the steps to check the DU config files

1. $ cd /home/candor/smo/l2l3/gNB_DU/cfg
2. Open the file TR196_gNodeB_DU_Data_Model.xml & Proprietary_gNodeB_DU_Data_Model.xml file to see the the DU config using below command

**vim Proprietary_gNodeB_DU_Data_Model.xml**
**vim TR196_gNodeB_DU_Data_Model.xml**

```
gNB_DU_Configuration.cfg  Proprietary_gNodeB_DU_Data_Model.xml         TR196_gNodeB_DU_Data_Model.xsd
mu_mimo_config_info.txt   Proprietary_gNodeB_DU_Data_Model.xml_backup
pmi_precoder_info.txt     TR196_gNodeB_DU_Data_Model.xml
```

# 9. Following are the steps to check CU config files

1. $ cd /home/candor/smo/l2l3/gNB_CU/cfg
2. Open the file TR196_gNodeB_CU_Data_Model.xml & Proprietary_gNodeB_CU_Data_Model.xml file to see the the CU config using below commands

**vim Proprietary_gNodeB_CU_Data_Model.xml**
**vim TR196_gNodeB_CU_Data_Model.xml**

```
gNodeB_CU_Configuration.cfg           TR196_gNodeB_CU_Data_Model.xml
Proprietary_gNodeB_CU_Data_Model.xml  TR196_gNodeB_CU_Data_Model.xsd
```

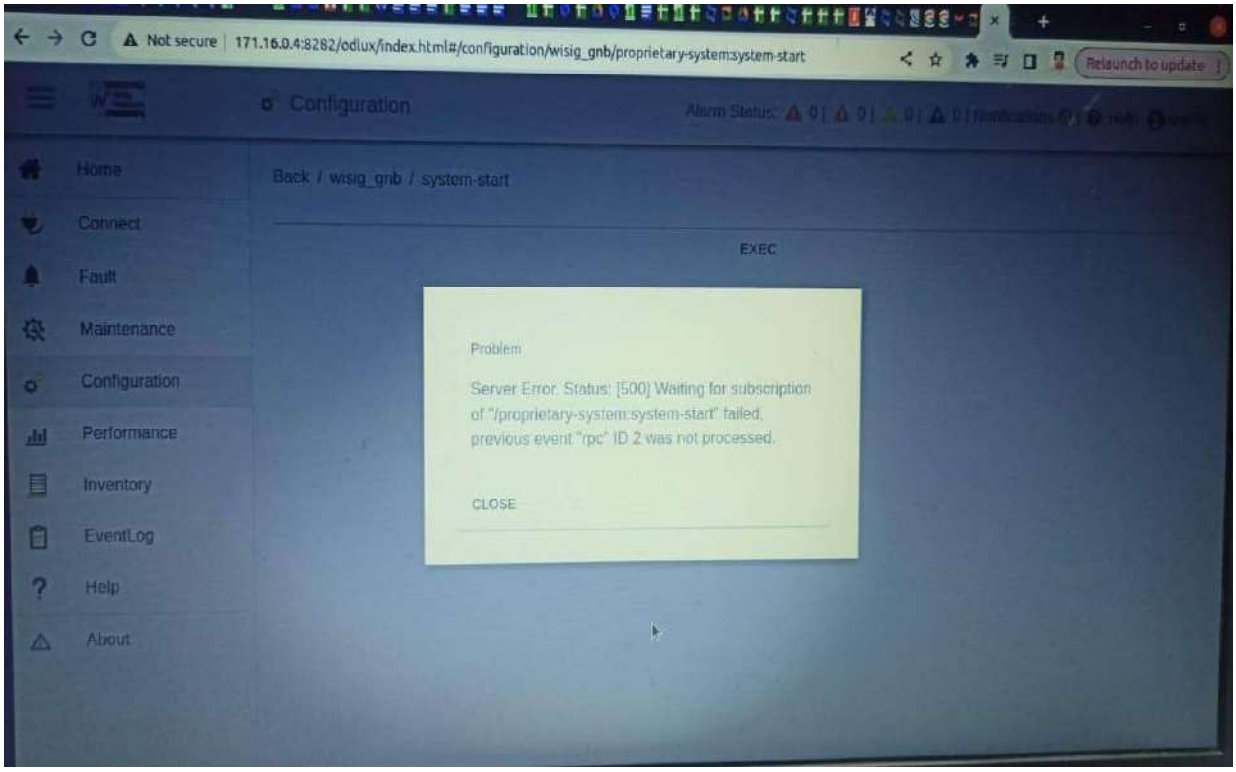
# 10. Diagnose

## Follow the below steps if you encounter any issues on GUI.

- Stop and start the docker. Refer to **Section 5.1** for instructions
  $ ./stop_sdnr.sh

- If any Server Error occurs as shown in the below image, type the following command on any of the terminals on the CU-DU server and follow from **Section 5.3.**

  $ sudo rm -rf /dev/shm/sr_*
  $ sudo rm -rf /dev/shm/srsub_*



NIRAL
NETWORKS
NiralOS™
5G
CORE
USER MANUAL

# CONTENTS


INTRODUCTION 03

Web UI

- Log-In into GUI 04
- Log-out from GUI 05
- Credentials 06

Functionalities

- Overview 07
- Topology 08
- Device information 09
- Device statistics 10
- Radio statistics 11
- Alarm Manager 12
- User and Subscriber Management 13

Operations

- Start NiralOS 5G Core 14
- Stop NiralOS 5G Core 15
- Restart NiralOS 5G Core 16

# INTRODUCTION

The NiralOS 5G Core is a cloud-native software designed specifically for Private 5G deployment within Enterprises. This software contains various 5G network functions such as Mobility Management, Authentication, Session and Policy Management along with Fast path User plane. The software is hosted on commercial off-the-shelf (COTS) servers and can run as bare-metal, virtual network functions (VNF), or cloud-native functions (CNF). Niral's Private 5G Specification is release-16 compliant and provides support for various interfaces along with a web-based dashboard for subscriber provisioning and management. The Niral Private 5G UPF can easily interoperate with TSP's centralized 5G Core Control Plane using the PFCP interface and provide in-premise edge applications with privacy, security, and low latency.

**NiralOS Private 5G Specification**

• NiralOS 5G core is compliant with the latest standards (Release-16) for private 5G networks.

• NiralOS 5G Core network includes various functions to handle data and communication, such as UPF, AMF, SMF, AUSF, UDM, and NRF.

• NiralOS relies on containerized and cloud-native Infra solutions for its operations.

NiralOS™

# Log-In into GUI

**Requirements:**

- Any modern web browser supporting JavaScript
- Standard Screen Resolution 1920x1080

**Procedure**

- Connect the Desktop Laptop to a the 5G Core Lab Firewall/Router and waiting for attaining a IP in subnet *171.16.0.X/24*
- Once connected open Web address > *http://171.16.0.8:3000*
- Enter username and password then press Log-In button.

*For username and password please refer to the credentials section.*

Web-GUI

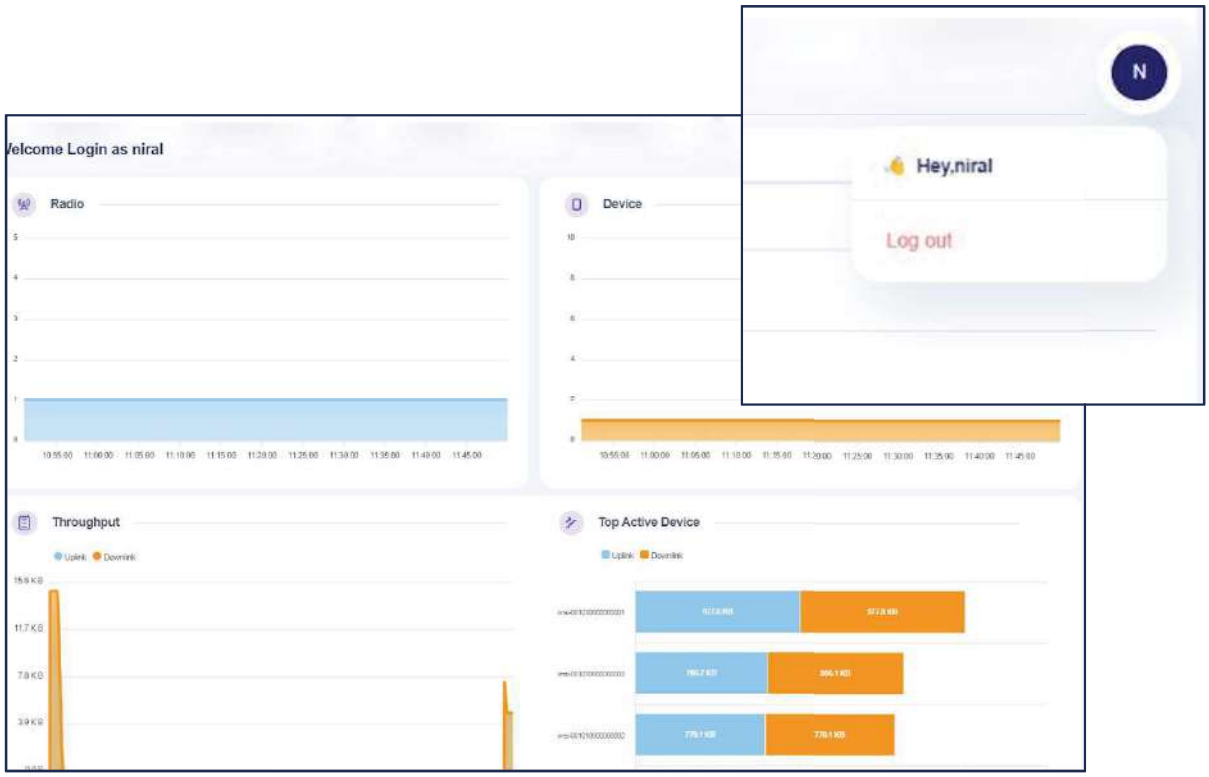
# Log-Out from GUI

**Requirements:**

- Any modern web browser supporting JavaScript
- Standard Screen Resolution 1920x1080

**Procedure**

- Click on the Profile button situated on the right hand side corner.
- Press the Logout button.
- You have successfully logged out.



Web-GUI

# Credentials

**Default username:** 5GLab

**Default password:** vvdn@123

Functionalities

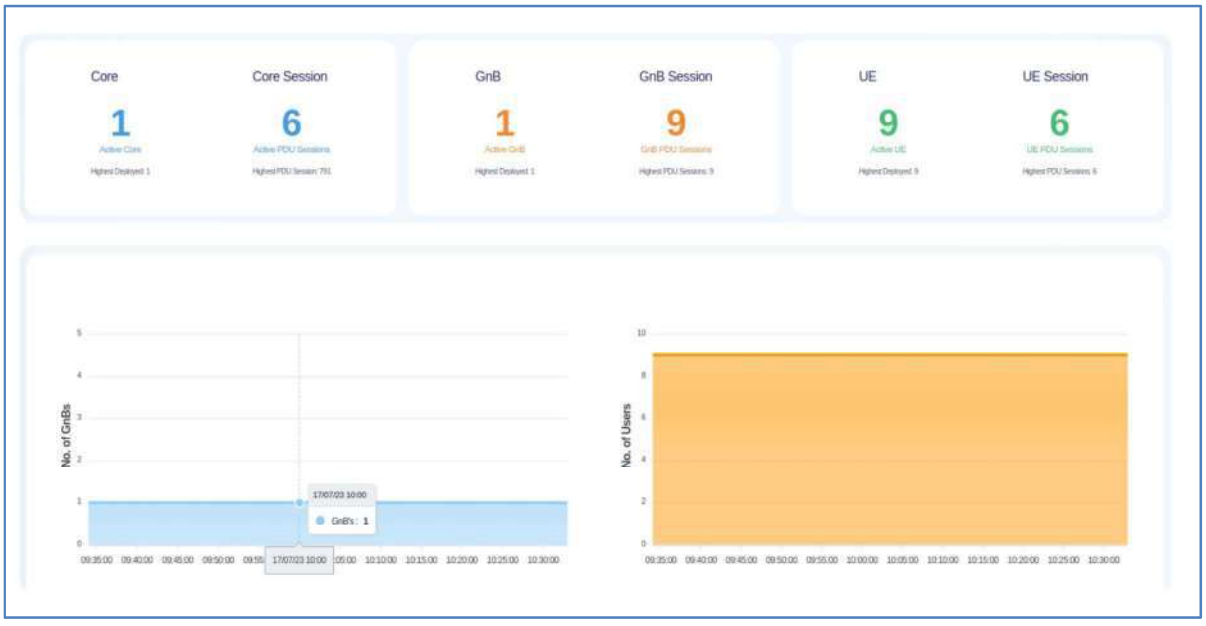
# Overview

This page provides a comprehensive overview of the User Interface (UI) and 5G core, with a particular focus on the core, GnB, and UE sections. The below block offers a detailed breakdown of these sections, furnishing information on various key metrics.

The core section presents an overview of the current state of affairs, including the number of active cores and the number of active PDU sessions. Additionally, the section provides historical data by highlighting the highest deployed core and the highest PDU session, thereby offering a basis for comparison between the current and previous states.

The GnB section offers insights into the GnB's current state, while the UE section focuses on the User Equipment, furnishing details on its status and performance.

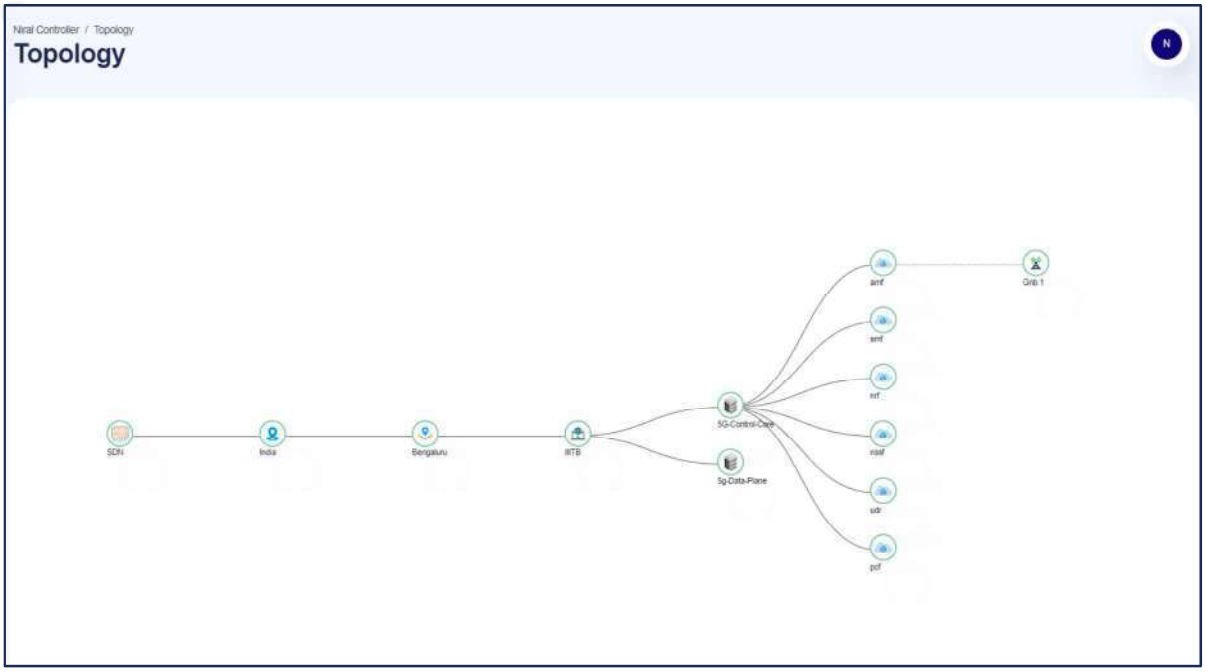
Overall, this page serves as a valuable resource for professionals seeking to quickly grasp the essentials of the UI and 5G core, as well as those looking to monitor and analyze the performance of these critical components.



Functionalities

# Topology

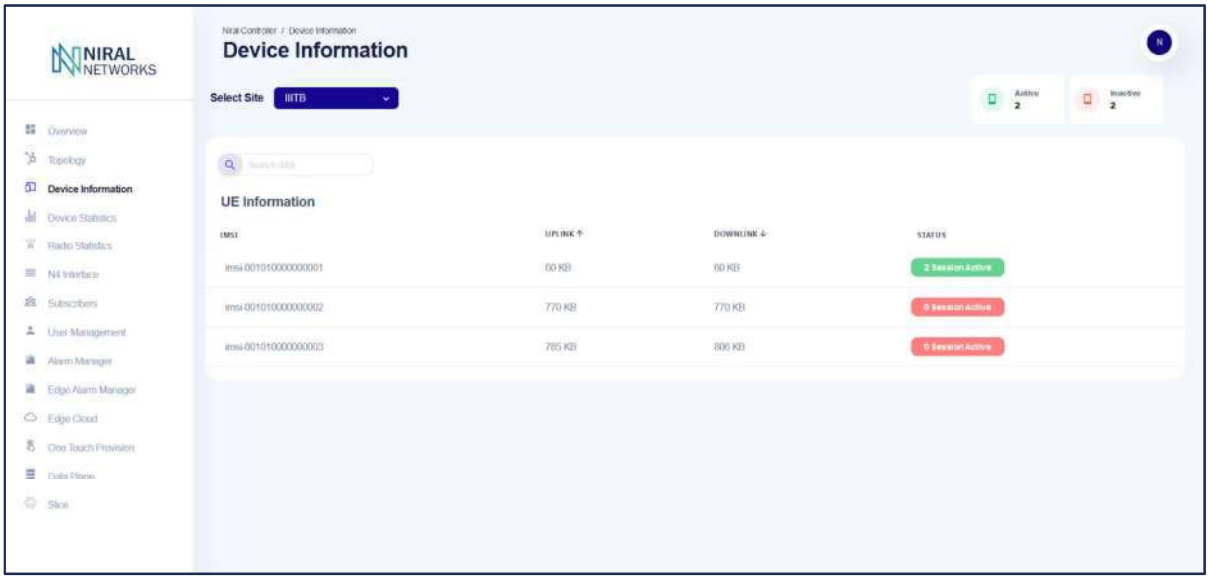
This page offers a visual representation of our deployment topology, which encompasses the arrangement of nodes and links within the deployment environment. The diagram incorporates information pertaining to zones, subzones, deployments, devices, network functions, and GNB details. To facilitate rapid comprehension, a standard color-coding scheme is employed, whereby a red outline surrounding an icon indicates that the component is offline, while a green outline signifies that it is operational or running. By clicking on the GNB icon, you can determine which User Equipment (UE) is connected to which GNB. This color scheme enables swift identification of component statuses within the deployment, thereby facilitating efficient troubleshooting and management.



Functionalities

# Device Information

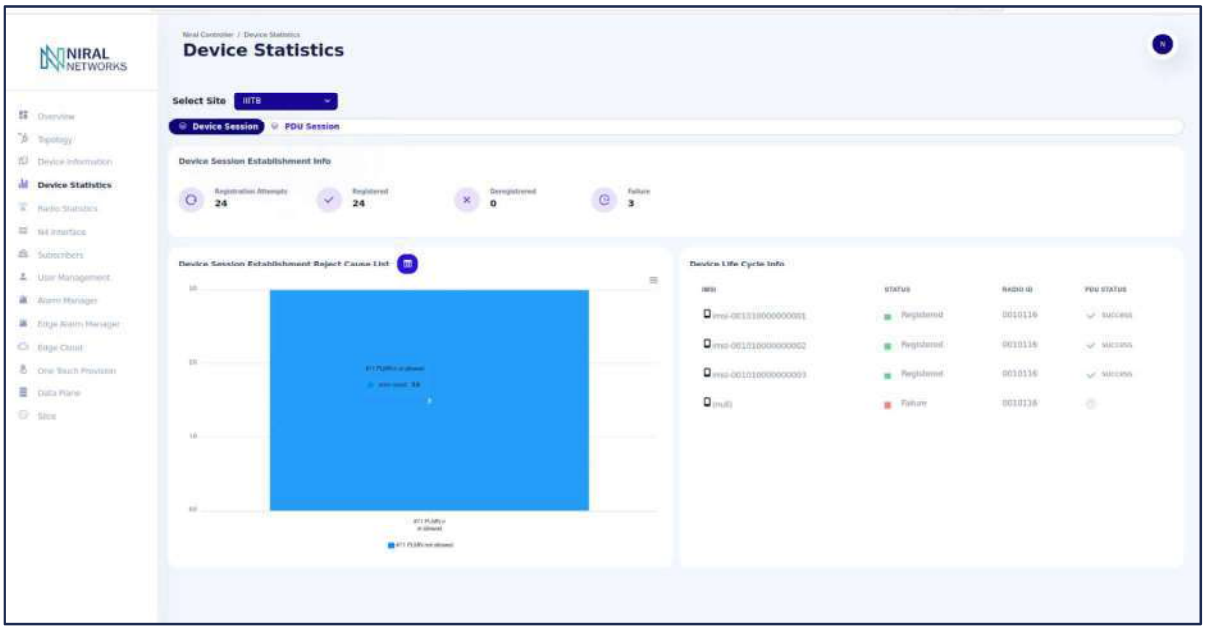
The Device Information tab provides real-time monitoring capabilities for the User Equipment (UE), including aggregated uplink and downlink data and the number of active PDU sessions. For more detailed statistics, users can click on a specific item to be redirected to a dedicated page that offers comprehensive information about the selected UE. This includes information such as Data Network Name (DNN), Packet Session Identifier (PSI), assigned IP address for that session, slice information (SST and SD), current uplink and downlink data, session duration, and status for each session. This enables network operators to efficiently monitor and manage the performance of their network.



Functionalities

# Device Statistics

The Device Statistics page provides a comprehensive overview of device performance and session data, with a focus on two key aspects: Device Session and PDU Session. The Device Session section offers insights into various metrics related to registration attempts, successful registrations, and failures. In contrast, the PDU Session section primarily concentrates on displaying establishment data and failures associated with PDU Sessions, while also providing information on PDU Session release. This layout allows for easy monitoring and analysis of device performance and session data, enabling efficient troubleshooting and optimization of network operations.



Functionalities

# Radio Statistics

This section provides a comprehensive overview of gNB (Next-Generation NodeB) registration activities:

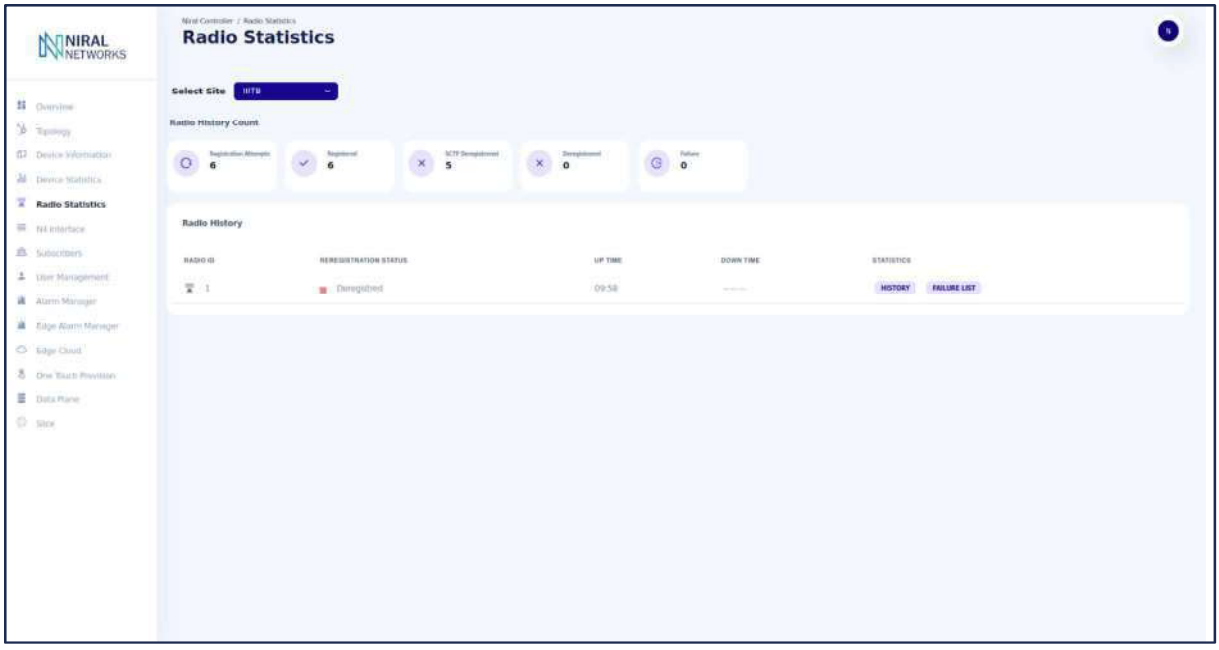
1.Registration Attempt:
- Displays the total number of attempts made for gNB registration.

2.Registered:
- Indicates the current total count of successfully registered gNBs.

3.SCTP Deregister:
- Presents the total number of gNBs that were deregistered due to SCTP layer breakdown.

4.Deregister:
- Shows the overall count of manually deregistered gNBs.

5.Failure:
- Highlights the total number of gNBs that failed to register.



Functionalities

# Alarm Manager

In the event of a network function failure, an alert is generated, as per the 5G architecture and the network connectivity flow. This is in accordance with the standard operating procedures and protocols established for ensuring seamless network performance and reliability.

Alarm Severity Options:

- Minor (MN)
- Major (MJ)
- Critical (CR)
- Not Reported (NR)
- Not Alarmed (NA)

# User & Subscriber Management

**CAUTION:**

This section is **NOT** supposed to be altered or changed until and unless suggested by qualified technician as this may result to malfunction of the NiralOS 5G Core and also may lead to data loss.

**Procedure:**

- Press the Power Button Physically on the server
- Wait for 10 minutes so that the setup can boot up and everything should be in normal running state.

# STOP NiralOS 5G Core

**Requirements:**

- SSH Capable device ( Desktop/Laptop) installed with OpenSSH Client

**Procedure:**

- Connect the SSH Capable device to the Firewall/Router situated in the 5G Lab Premises
- Run command > *ssh user@171.16.0.2*
- Accept by typing (*Yes*) for RSA Fingerprint Check Message
- Enter Host OS Password ( Default Password: *vvdn@123* )
- Run command > *sudo shutdown now*
- Wait for 2 minutes the NiralOS 5G Core should gracefully shutdown.

# RESTART NiralOS 5G Core

**Requirements:**

- SSH Capable device ( Desktop/Laptop) installed with OpenSSH Client

**Procedure:**

- Connect the SSH Capable device to the Firewall/Router situated in the 5G Lab Premises
- Run command > *ssh user@171.16.0.2*
- Accept by typing (*Yes*) for RSA Fingerprint Check Message
- Enter Host OS Password ( Default Password: *vvdn@123* )
- Run command > *sudo reboot now*
- Wait for 10 minutes so that the setup can boot up and everything should be back to normal running state.

# DEVICE ONBOARDING PROCESS

## USER GUIDE

Version: 1.01

Prepared by: Nishant Malik

## REVISION HISTORY

| Version | Author | Status | Date | Reason for change |
|---|---|---|---|---|
| 1.01 | Nishant Malik | Initial Draft | 01/07/24 | Document created |

TABLE OF CONTENTS


**INTRODUCTION** **4**

User Guide 4
Audience 4

**DEFINE** **5**

INFINXT SDWAN ARCHITECTURE 5
CONTROL PLANE 6
DATA PLANE 6
WAN EDGE ONBOARDING PROCESS 6

**DESIGN** **7**

ONBOARDING OPTIONS 7
InfiNxt WAN EDGE DEVICES 8

**DEPLOY** **9**

WAN INTERAFCE CONFIGURATION 10
PRE-REQUISITES 13
MANUAL ONBOARDING 16
AUTOMATED 25

**OPERATION** **26**

ISSUES 26
TROUBLESHOOTING 26

# INTRODUCTION

## USER GUIDE

This guide is intended to provide guidance to onboard InfiNxt SD-WAN Edge devices into the enterprise SD-WAN Infrastructure. The guide focuses on the step-by-step procedures to configure each of the onboarding options available, along with the use cases specific to WAN Edge deployment. The physical or virtual WAN Edge onboard options include manual or the automated deployment process for InfiNxt edge devices on centralized controller.

This document focuses on how to deploy an InfiNxt SDWAN Edge device in enterprise customer network. In this guide, WAN Edge routers are deployed either at remote sites or at the datacentre and are connected to single or two WAN transports, either of the Internet or MPLS or both.

Although this deployment guide is about onboarding InfiNxt SD-WAN Edge devices. It is presumed that

- InfiNxt SD-WAN Controllers are already deployed
- InfiNxt SDWAN Edge has reachability to the controllers which are reachable via across the WAN transport either Internet or MPLS (on-prem)

This document contains four major sections:

**Define**: This section provides a high-level overview of the SD-WAN architecture and components, WAN Edge devices and options available to onboard for a physical or virtual WAN Edge router.

**Design:** This section provides detailed discussion on the design considerations and prerequisites needed for each of the onboarding options to build a secure SD-WAN enterprise infrastructure.

**Deploy:** This section discusses step-by-step procedures to onboard an InfiNxt SD-WAN Edge device in the customer network.

**Operation:** This section briefly discusses how to monitor and troubleshoot the onboarding issues, if necessary, in the SD-WAN environment.

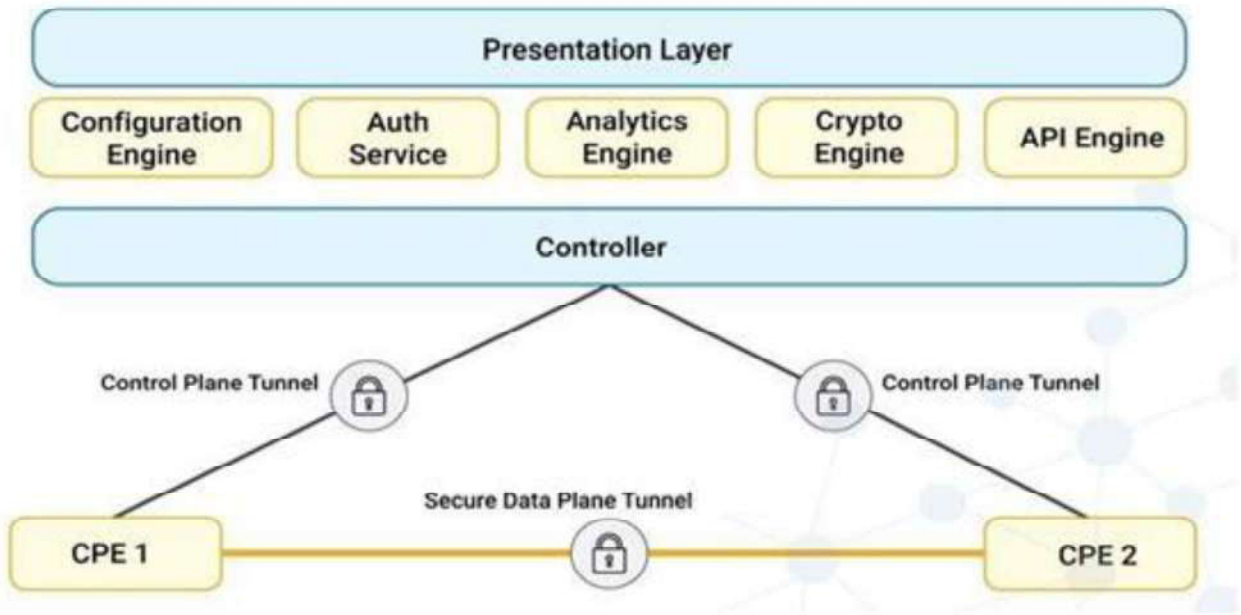
## AUDIENCE

The audience for this document includes network design engineers and network operations personnel who have deployed the InfiNxt SD-WAN controllers and are looking for the best viable option to onboard the WAN Edge devices in their respective network environment.

## DEFINE

The InfiNxt SD-WAN solution is a robust enterprise-grade overlay architecture that facilitates digital and cloud transformation within enterprises. This comprehensive solution seamlessly integrates routing, security, centralized policy management, and orchestration across large-scale networks. It effectively tackles the typical challenges and issues encountered in traditional WAN deployments.

## INFINXT SDWAN ARCHITECTURE




The InfiNxt SD-WAN solution consists of distinct components:

Configuration Orchestrator: This component oversees device configurations, facilitates initial activation, and manages device authentication. It communicates with the device/CPE through a control plane tunnel dedicated to management and configuration traffic. The Orchestrator handles the entire configuration lifecycle for the CPE and plays a crucial role in Zero Touch Provisioning (ZTP) and Zero Touch Deployment (ZTD) processes.

Crypto Engine: This component is accountable for establishing control tunnels from the CPE to the controller. It determines the encryption and tunnel protocols used. Additionally, it manages tunnel identifiers for each connected CPE, ensuring logical segregation between organizations. The cryptographic engine also handles Security Associations (SA) for dataplane devices, managing policies such as key rotation and exchanges.

API Engine: This component provides controller functionalities to CPE devices in the southbound direction and to other OSS systems in the northbound direction. It exposes numerous API endpoints for consumption by various services. The controller's user interface interacts with the API engine to execute commands.

Analytic Engine: This component facilitates the monitoring and management of health Key Performance Indicators (KPIs) for various client-facing elements of the solution, such as CPEs and WAN links. It employs monitoring mechanisms to track device and link KPIs, storing them in a time-series database for subsequent analytics. Real-time streaming telemetry is utilized to offer immediate insights and analytics.

Authentication Engine: This component handles authentication within the SD-WAN solution, serving two key functions. Firstly, it supports Zero Touch Provisioning (ZTP), authenticating CPE devices with the ZTP server using methods such as PKI or device serial number whitelisting. Secondly, it forms the foundation for the controller's logical separation and multi-tenancy by verifying identities through authentication and granting necessary permissions via authorization mechanisms.

## CONTROL PLANE

This is the secure management tunnel established from CPE edge devices to the controller. In Infinxt, this tunnel employs TLS v1.2 to ensure data security. The primary traffic passing through this tunnel includes configuration commands from the controller to the device, network monitoring data, KPI monitoring, and API communications between the CPE and controller. The control plane tunnel is automatically created when a CPE device is powered on and connected to the network.

## DATA PLANE

This tunnel is established directly between two CPE devices, utilizing the WireGuard VPN protocol for enhanced security. It primarily carries user traffic between these endpoints. Because it operates solely between user CPE devices, no business information or user data passes through the controller; instead, communication occurs exclusively between user locations.

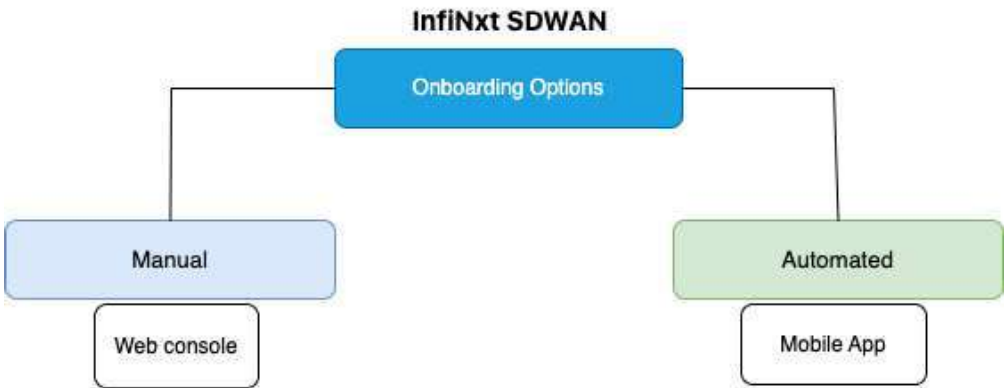
## WAN EDGE ONBOARDING PROCESS

In this solution, our emphasis is on establishing secure data plane connections by integrating physical or virtual WAN Edge devices and ensuring secure control connections across all SD-WAN components within the network environment. With InfiNxt SD-WAN solution, a whitelist model is employed, requiring that WAN Edge devices intending to join the SD-WAN overlay network must be pre-registered with all SD-WAN controllers in advance. This process involves adding the WAN Edge device's serial number to the inventory database stored on the controller, associating it with the respective tenant account. Admin must authorize the device and assign it to a tenant to enable device activation and onboarding.

Upon boot up with active WAN connection, The CPE starts by initiating a TLS connection with the controller at the configured URL location, which defaults to https://app.infinxt.co.in. Upon successful TLS handshake, the CPE establishes a bi-directional communication channel with the controller using the socket.io protocol. Once this connection is set up, the CPE proceeds to initiate the device registration process over the secure TLS connection using JSON RPC. This sequence ensures secure and efficient communication between the CPE and the controller, facilitating seamless integration and operation within the network environment.

Once approved by the admin, the device is ready for activation in the tenant's account under the "Devices" tab. Upon activation by the tenant, the controller initiates the Zero Touch Provisioning (ZTP) process by sending default configurations to the device, facilitating its onboarding to the controller. Once activated, the controller retrieves and updates the controller database with the device's public WAN IPs, VPN keys, and current configuration. Following this, the tenant admin proceeds to configure additional features and services as needed.

## ONBOARDING OPTIONS




### Manual Onboarding (Web console)

InfiNxt WAN edge SDWAN can be manually onboarded either through the management or console port on the hardware platform, or via the KVM / ESXI console for virtual edge devices. When using this option, configure the device with a bare minimum configuration that is needed for the device to reach the InfiNxt SD-WAN controller.

With this option, the edge device requires an active internet connection, either through a broadband link with a dynamic WAN IP obtained via DHCP or an internet lease line with a static IP. Each InfiNxt SDWAN model comes pre-configured with a default configuration boot file. This file includes a management IP configured on one port, intended for managing the device and establishing initial configurations. These configurations are minimal and are aimed at enabling connectivity to the controller IP hosted on the internet.

Once the device is configured with an internet connection, the user should log in to their customer tenant account where the device will be visible for manual activation and onboarding. Once successfully onboarded, the device is ready to be further configured with the necessary features available on the controller's web console.

### Automated (Mobile App)

Mobile app based automated onboarding ensures the true day-zero experience of securely onboarding and deploying the WAN Edge device, with default-shipped factory settings, into the SD-WAN network.

To get started with the onboarding process, users simply need to install the InfiNxt app available on the play store and log in using the username and password created for their customer tenant account. Once the broadband internet connection is active and linked to the edge device, users can scan the code on the device and click "activate". This action will onboard the device onto the tenant account, allowing further feature configuration through the controller GUI.

WAN Edge devices intending to join the SD-WAN overlay network must be pre-registered with all SD-WAN controllers in advance. This process involves adding the WAN Edge device's serial number to the inventory database stored on the controller, associating it with the respective tenant account.

## INFINXT WAN EDGE DEVICES

The table below lists the supported edge devices for InfiNxt SDWAN models. All models support both onboarding options, depending on the type of WAN links and the deployment of the controller.

- Inhouse Infinity Labs controller instance hosted on AWS cloud
- On-prem controller instances hosted on customer DC or network infra running on either public Internet or private MPLS cloud
- Customer owned cloud-based controller instance

| InfiNxt Models | Onboarding Options | |
|---|---|---|
| | Manual | Automated |
| iEdge1000M | Yes | Yes |
| iEdge1000 | Yes | Yes |
| iEdge100F | Yes | Yes |
| iEdge100L | Yes | Yes |
| iEdge100 | Yes | Yes |
| iEdge50 | Yes | Yes |

The table below enumerates the supported virtual edge devices for InfiNxt SDWAN models. Virtual CPEs exclusively support the manual onboarding process, as these instances are integrated and installed on VMs managed either by the customer or through cloud-based solutions, with no code available for scanning via a mobile app.

| InfiNxt Models | Onboarding Options | |
|---|---|---|
| | Manual | Automated |
| vCPE iEdge1000M | Yes | NA |
| vCPE iEdge1000 | Yes | NA |
| vCPE iEdge100F | Yes | NA |
| vCPE iEdge100L | Yes | NA |
| vCPE iEdge100 | Yes | NA |
| vCPE iEdge50 | Yes | NA |

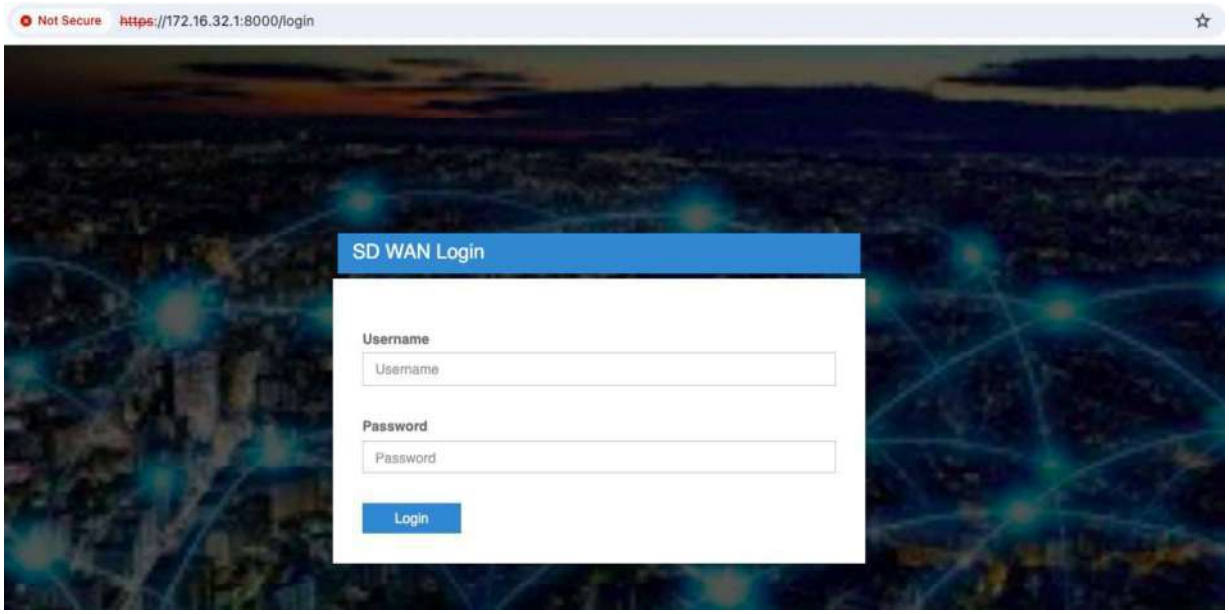
Our Infinity OS for SDWAN solution is designed to support all models seamlessly. Upon installation of the OS image on edge devices, it automatically loads the default config file. This file is pre-configured with the management IP 172.16.32.1 on one interface of SDWAN appliance, intended for initial setup and management purposes.

Furthermore, all SDWAN iEdge hardware appliances are accessible via a local user interface at http://172.16.32.1:8000. Moreover, the OS installation process incorporates updating the devices' location file, which defaults to https://app.infinxt.co.in which is essential for the device authentication process required during onboarding.

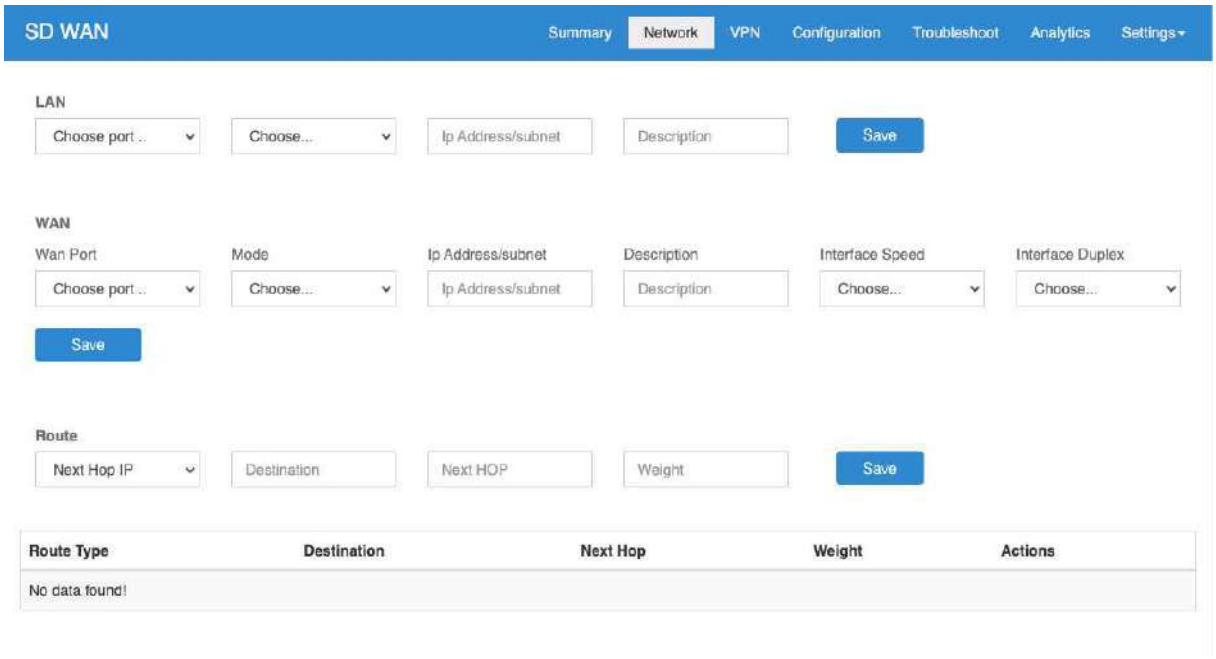
## DEPLOY

The deployment section is organized to cover the configuration of WAN interface, prerequisites, device onboarding procedure and verification. User can access the device via local UI: htts://172.16.32.1:8000 or via CLI connected to management port or console port depending on the options available with respective models.

Local UI:




Username: infinity | Password: Infi@Labs




After logging into the device using the local user interface, users can manage the device based on the available WAN link to configure for onboarding process.

Login to CLI via management port:

Username: infinity | Password: @Infi@Labs

## WAN INTERAFCE CONFIGURATION

This section includes the step to follow to configure the WAN interface for device on-barording with different WAN link types.

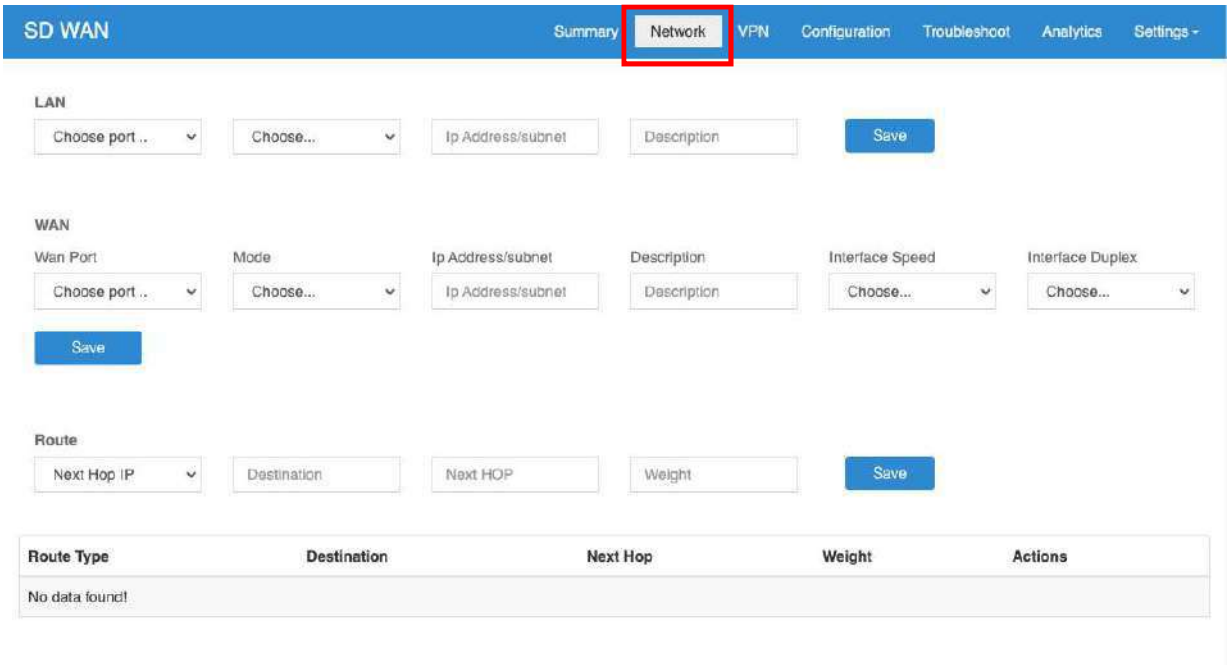
- WAN IP via DHCP (Broadband)
- STATIC (Internet Lease Line | MPLS)

WAN link type – Broadband: As part of the solution design, the first two ports on each device model are initially configured with the DHCP command with default config.boot file. Upon termination of the broadband link onto the device, it automatically receives an IP address via DHCP and sets up a default route in the routing table, directing traffic to the valid next-hop IP address of the service provider's DHCP server for internet access.

The screenshot below displays the IP address assigned to the eth0 WAN interface by DHCP and the status of Controller and internet reachability, highlighted in red.

WAN link type – STATIC: Below are the steps to activate the link with WAN IP as STATIC

- Access the local UI of the device and navigate to the network tab.




- Navigate to the WAN settings and select the WAN interface as eth0. Set the mode to static, then enter the IP address with its CIDR subnet for configuring the WAN interface along with a description. Finally, click on save to apply the settings.

Click on save, it will delete the DHCP command from wan interface and configure the static IP as required and the below success message will pop up

- You can verify via CLI and GUI under summary tab

```
Interface        IP Address                     S/L  Description
---------        ----------                     ---  -----------
eth0             192.168.100.254/24             u/D  WAN-ILL
```

**Interfaces Details :**

| Interface | Status | IP | Description |
|---|---|---|---|
| eth0 | Down | 192.168.100.254/24 | WAN-ILL |

- Go to the route tab in the local UI under the network tab. Choose the next-hop IP option to set up the static route. Enter the destination IP as either the default (0.0.0.0/0) or a specific destination according to your needs or the link type, along with a valid next-hop IP. Value for weight is equivalent to distance and can be configured as per network requirement. By default, it will be considered as 1. Finally, click on save to apply the changes.

Post success, below route entry will appear on the summary tab on local UI.

| Route Type | Destination | Next Hop | Weight | Actions |
|---|---|---|---|---|
| static-route | 0.0.0.0/0 | 192.168.1.1 | | |

```
infinity@l2iv804c:~$ sh config commands | match static
set protocols static route 0.0.0.0/0 next-hop 192.168.1.1
```

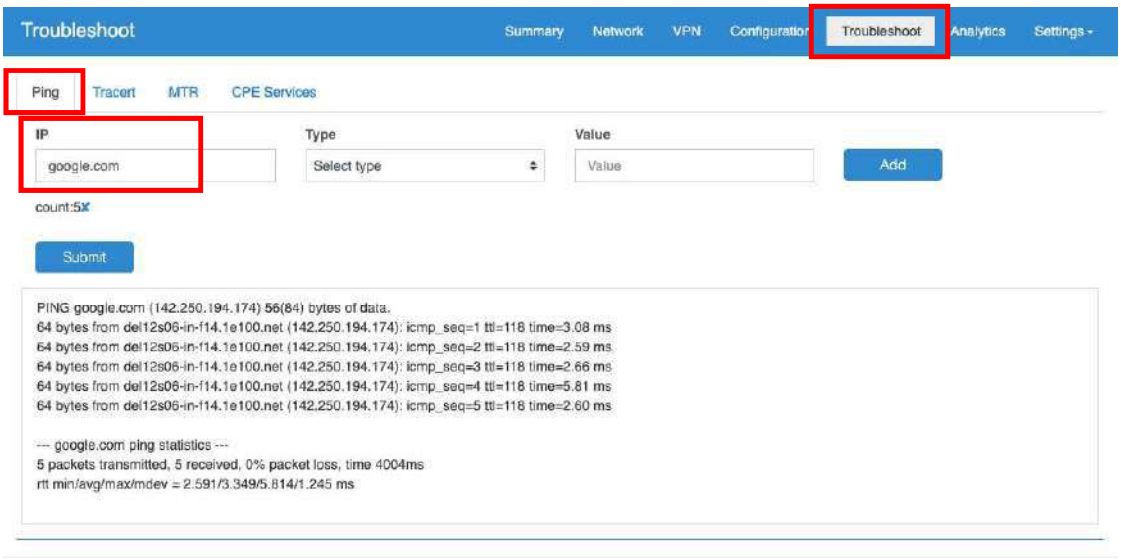
## PRE-REQUISITES

The below checklist showcases the prerequisites that are needed before proceeding with the WAN Edge onboarding process.

- Ensure that the device is connected to an active internet connection

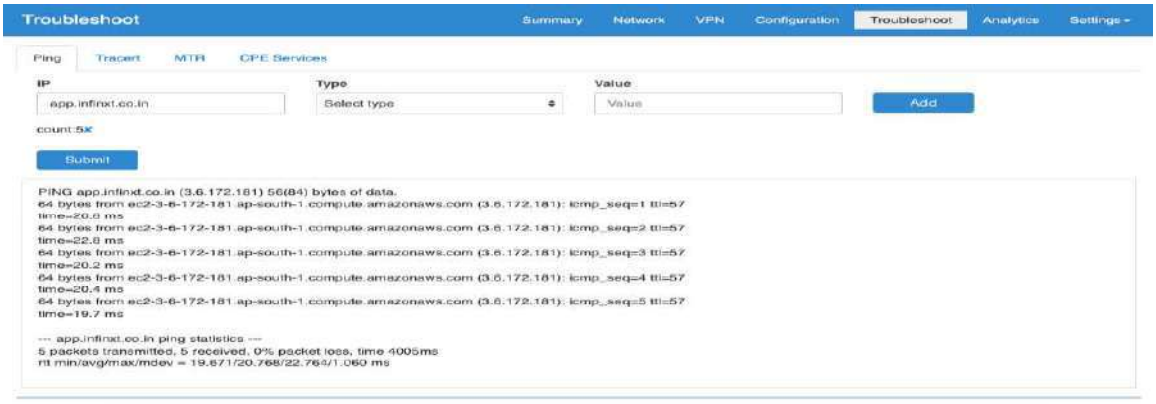
```
PING google.com (142.250.194.46) 56(84) bytes of data.
64 bytes from del12s02-in-f14.1e100.net (142.250.194.46): icmp_seq=2 ttl=117 time=12.8 ms
64 bytes from del12s02-in-f14.1e100.net (142.250.194.46): icmp_seq=4 ttl=117 time=11.8 ms
64 bytes from del12s02-in-f14.1e100.net (142.250.194.46): icmp_seq=5 ttl=117 time=12.3 ms
64 bytes from del12s02-in-f14.1e100.net (142.250.194.46): icmp_seq=6 ttl=117 time=16.8 ms
64 bytes from del12s02-in-f14.1e100.net (142.250.194.46): icmp_seq=7 ttl=117 time=11.9 ms
64 bytes from del12s02-in-f14.1e100.net (142.250.194.46): icmp_seq=9 ttl=117 time=12.2 ms
```

- Goto Troubleshoot tab on local UI → select ping and input he destination → click on add and submit



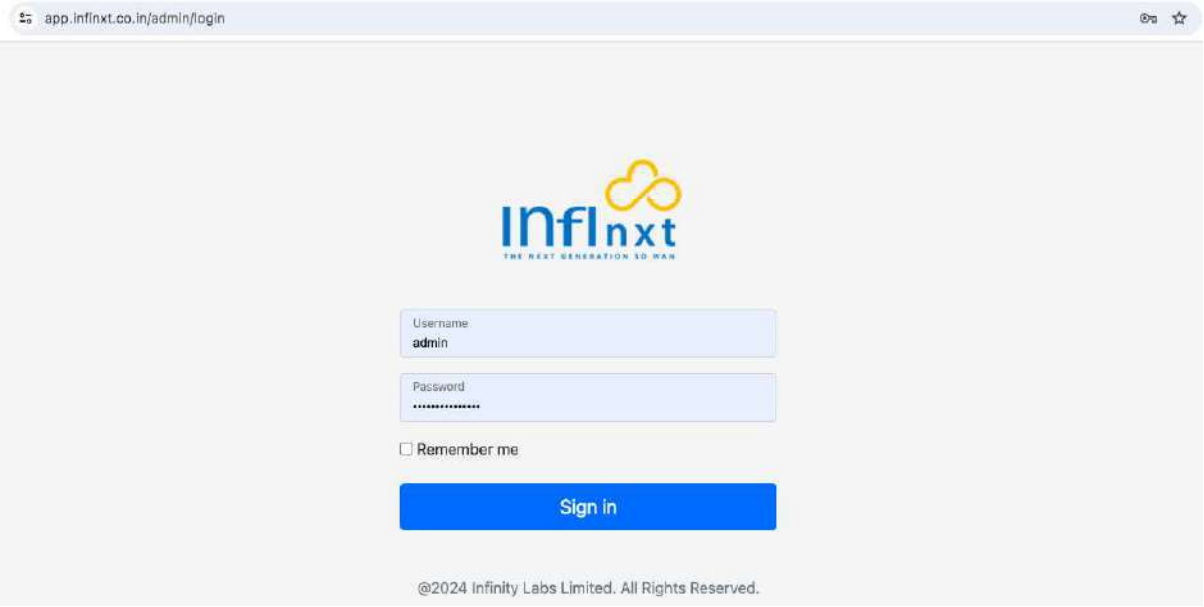

- Make sure the WAN Edge device has reachability to the InfiNxt SDWAN controller (default) or the controller IP if using an on-premises solution via a private MPLS network

```
PING app.infinxt.co.in (3.6.172.181) 56(84) bytes of data.
64 bytes from ec2-3-6-172-181.ap-south-1.compute.amazonaws.com (3.6.172.181): icmp_seq=2 ttl=56 time=30.2 ms
64 bytes from ec2-3-6-172-181.ap-south-1.compute.amazonaws.com (3.6.172.181): icmp_seq=3 ttl=56 time=27.6 ms
64 bytes from ec2-3-6-172-181.ap-south-1.compute.amazonaws.com (3.6.172.181): icmp_seq=4 ttl=56 time=27.3 ms
64 bytes from ec2-3-6-172-181.ap-south-1.compute.amazonaws.com (3.6.172.181): icmp_seq=5 ttl=56 time=26.7 ms
```

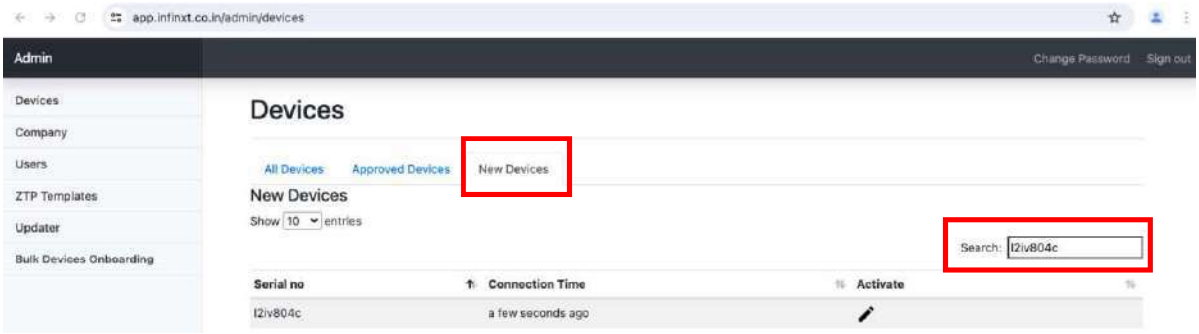


- Ensure that the device has been activated on the admin portal and is awaiting approval under new device tab of controller admin portal

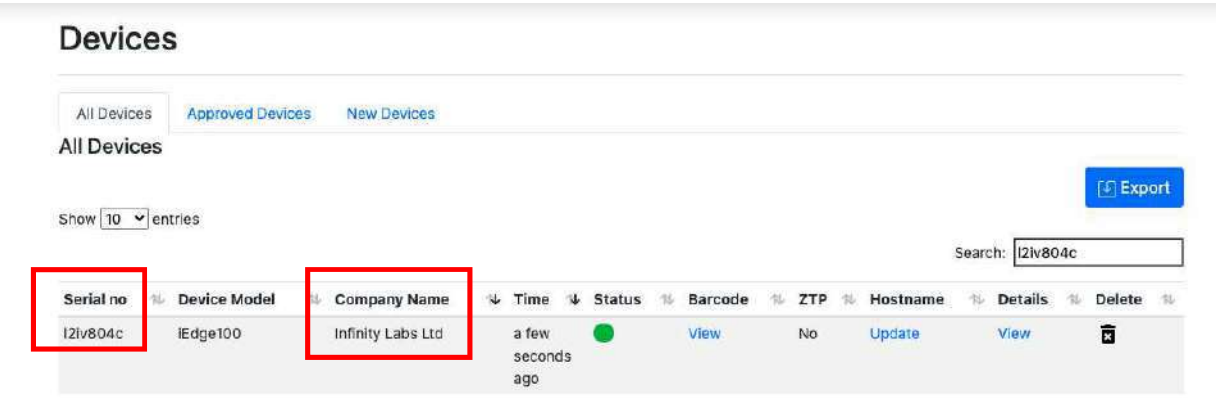
- User to login to controller admin portal – https://app.infinxt.co.in/admin



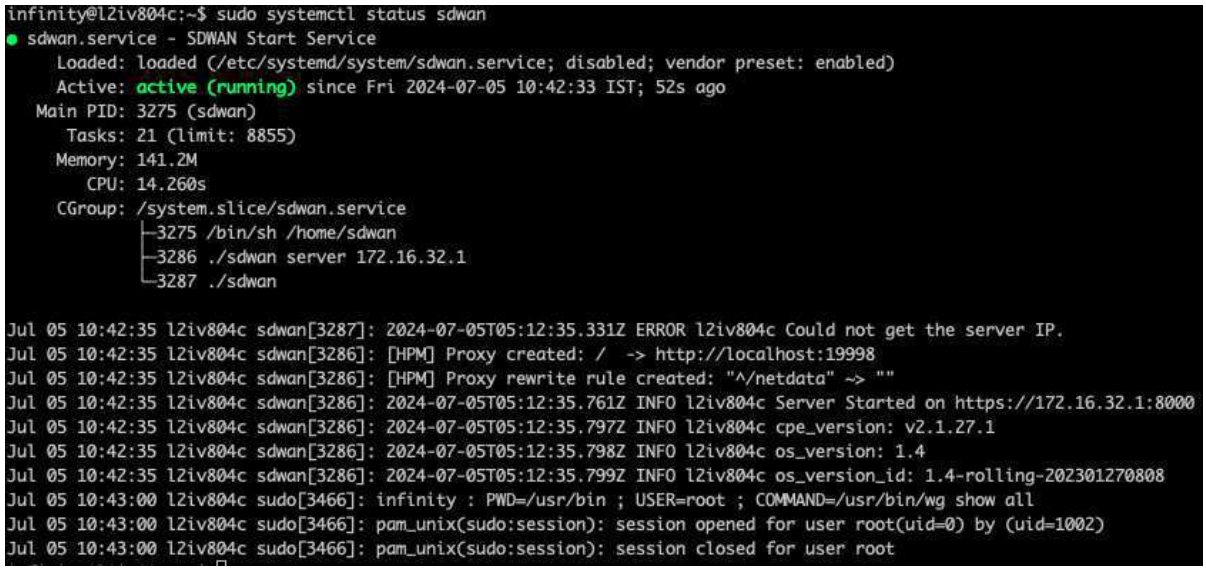

- Post successful login, navigates to new devices tab and put the device serial number in search tab and click on activate and submit




- Ensure that the administrator has created a tenant account, and the device has been approved and added to the respective customer's tenancy (Steps for tenant creation and device addition → Goto Manual Onboarding section)

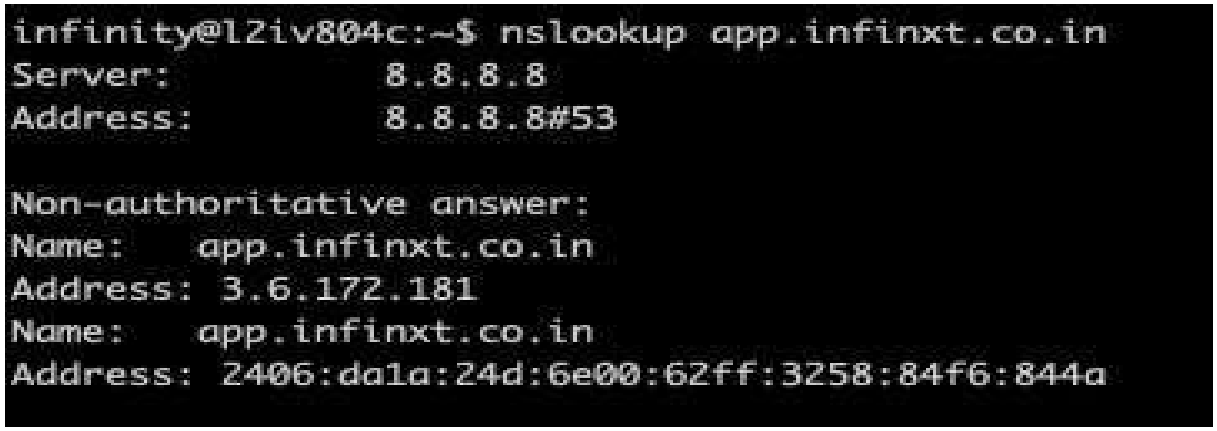


- Ensure that the SDWAN service is activated and operational on the edge device to initiate secure communication for the authentication process




- Ensure that the device's location file contains the correct controller details for on-premises or customer-owned cloud-based VM to host the controller instance

- Default iEdge router should be able to resolve the FQDN https://app.infinxt.co.in




- If the controller is behind a firewall, ensure that ports 443/TCP and 19998/TCP are open on the firewall

## MANUAL ONBOARDING

This section explains briefly the steps to follow for onboarding process.

The following table lists the management supported interfaces that can be leveraged to onboard devices with manual onboarding process.

| InfiNxt Models | Management Port | |
|---|---|---|
| | Ethernet | Console Port |
| vCPE iEdge1000M | eth3 | Yes |
| vCPE iEdge1000 | eth3 | Yes |
| vCPE iEdge100F | eth3 | Yes |
| vCPE iEdge100L | eth5 | NA |
| vCPE iEdge100 | eth3 | NA |
| vCPE iEdge50 | eth1 | NA |

The default config.boot file of each device includes a management interface configured with an IP address 172.16.32.1/24. DHCP is also configured to allow laptops/PCs to receive IP addresses automatically for the management console.

The first two interfaces of every edge SDWAN device are designated as WAN ports. The default config.boot file includes commands on both WAN interfaces to automatically obtain IP addresses to support broadband links as well as links assigning the IP address via DHCP as WAN.

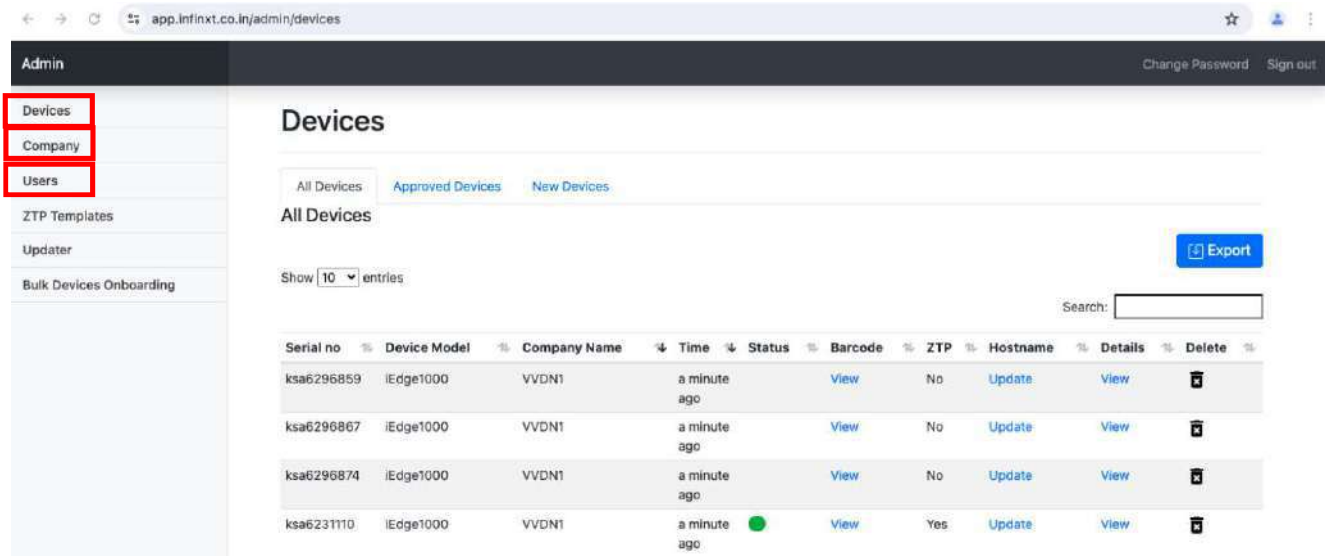
$**set interface ethernet <eth0> address dhcp**

Let's begin by creating a tenant on the controller admin portal, then proceed with setting up a user admin account for the newly created tenant, and finally add devices to their respective tenant.

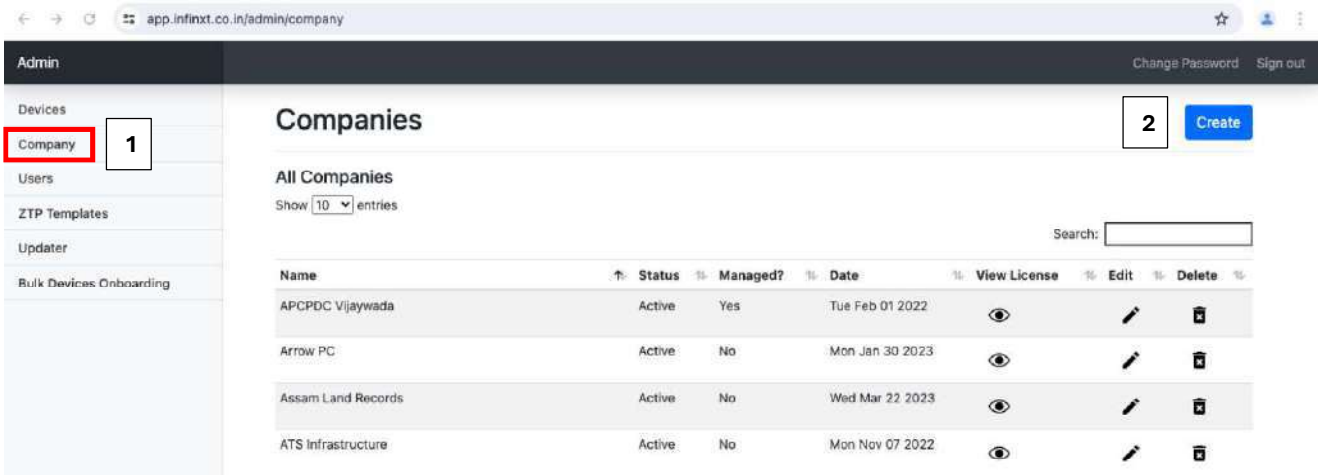
### Tenant Account and user credentials creation Process:-

**Step1: -** Access the controller's admin portal by enter the following URL into the web browser: https://app.infinxt.co.in/admin. Username | Password to be created while installing/integration of controller instance and should be available with administrator

After a successful login, the home screen will display the status of all available devices, along with options to create a company, manage user credentials for the tenant account, and activate devices.



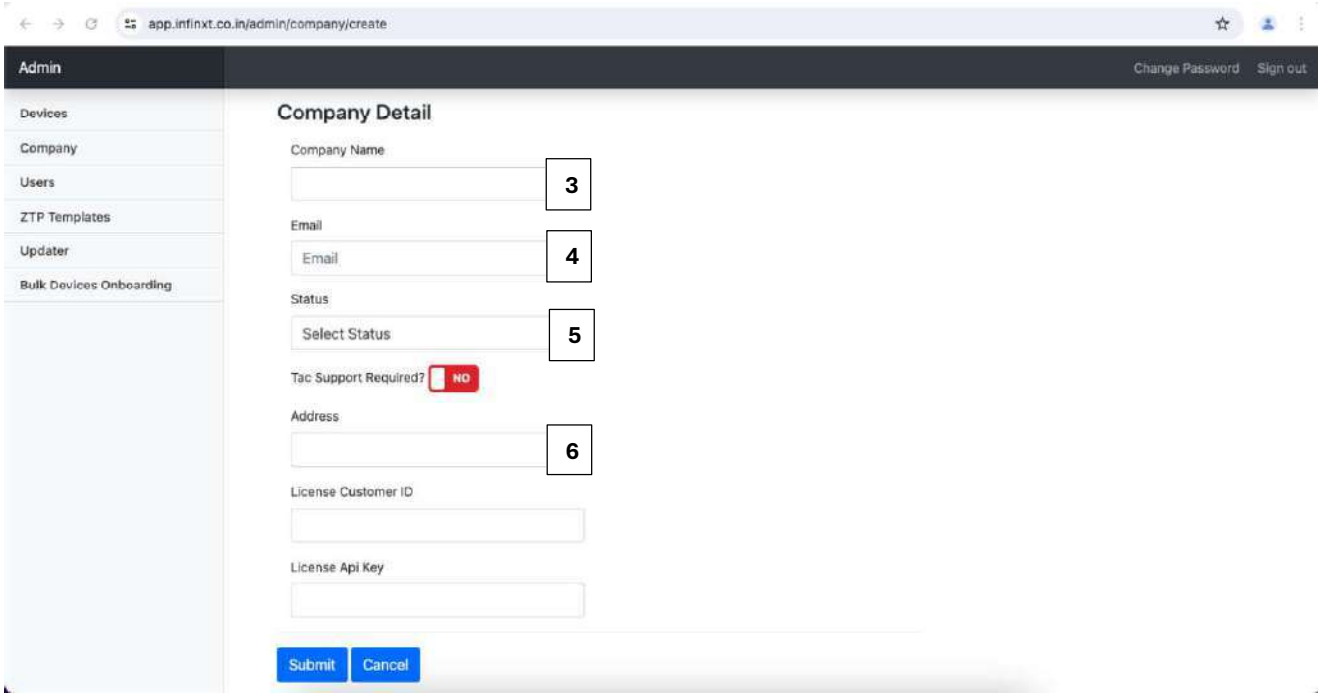

**Step2: -** To create a tenant account, click on the company tab followed by create button and move to step 3



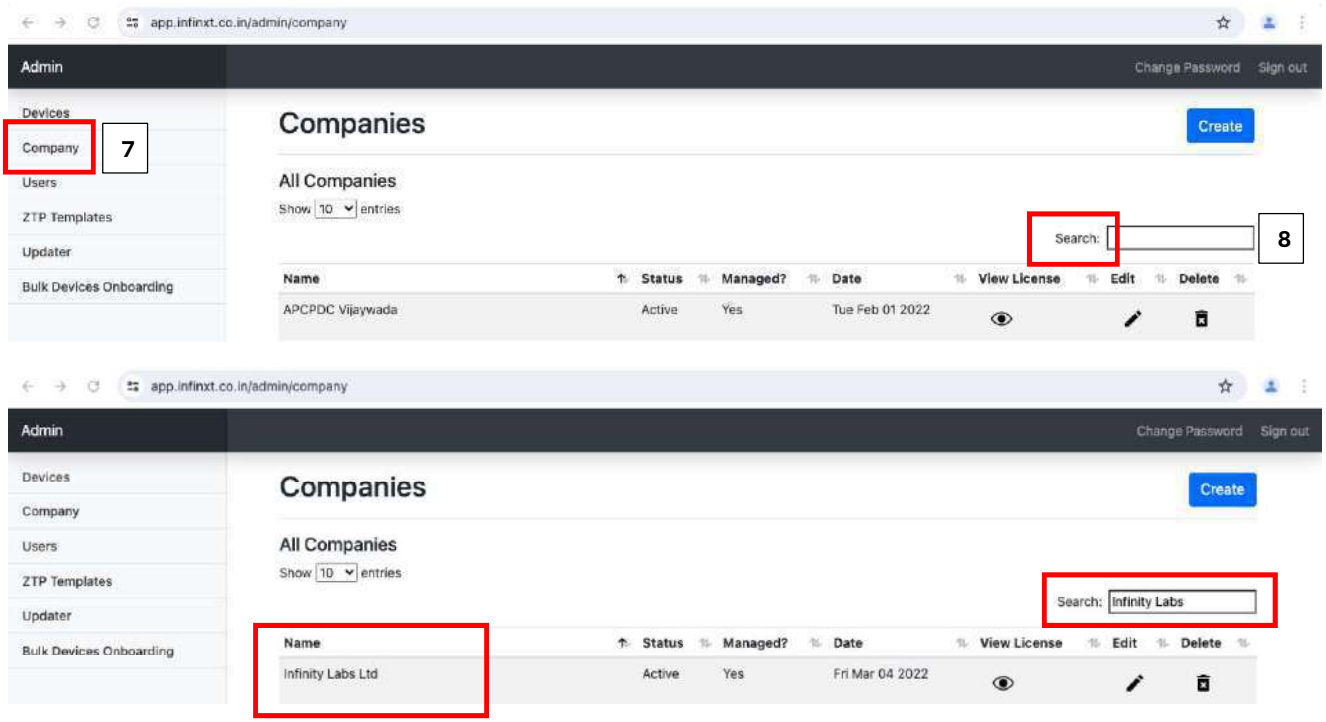

**Step3: -** To create a tenant, the following details are required for input during company creation and click on submit

- Company name (Eg.: Infinity Labs Limited)
- Valid email ID (Eg.: support@infinitylabs.in)
- Status to input as active
- Valid address (Eg: Fifth Floor, KR signature, Plot No. 3, Sector 135, Noida, Uttar Pradesh 201305)
- License Customer ID (non-mandatory and can be updated later)
- License API Key (non-mandatory and can be updated later)

Note: We have a dedicated portal for generating licenses based on the solution purchased by the customer. This includes the API Key and customer ID, incorporating details such as tenure, device count, and security features activation.




**Step4: -** Post successful submission, again go back to company tab to verify the company details. Click on the company and go to the search tab and input the company name created (eg: Infinity Labs Limited)

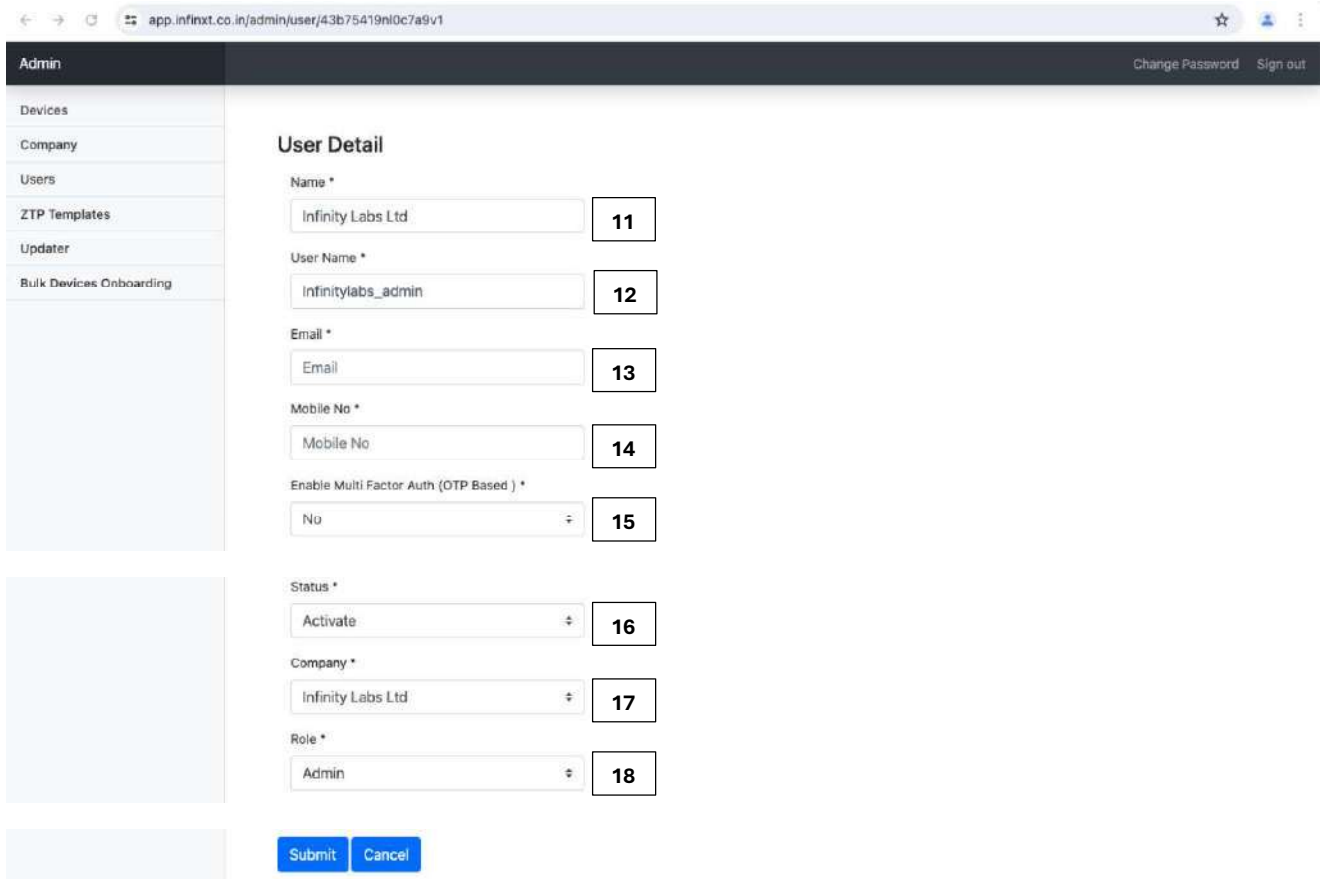



This facilitates the immediate activation of the tenant account, allowing for the creation of login credentials to that specific account.

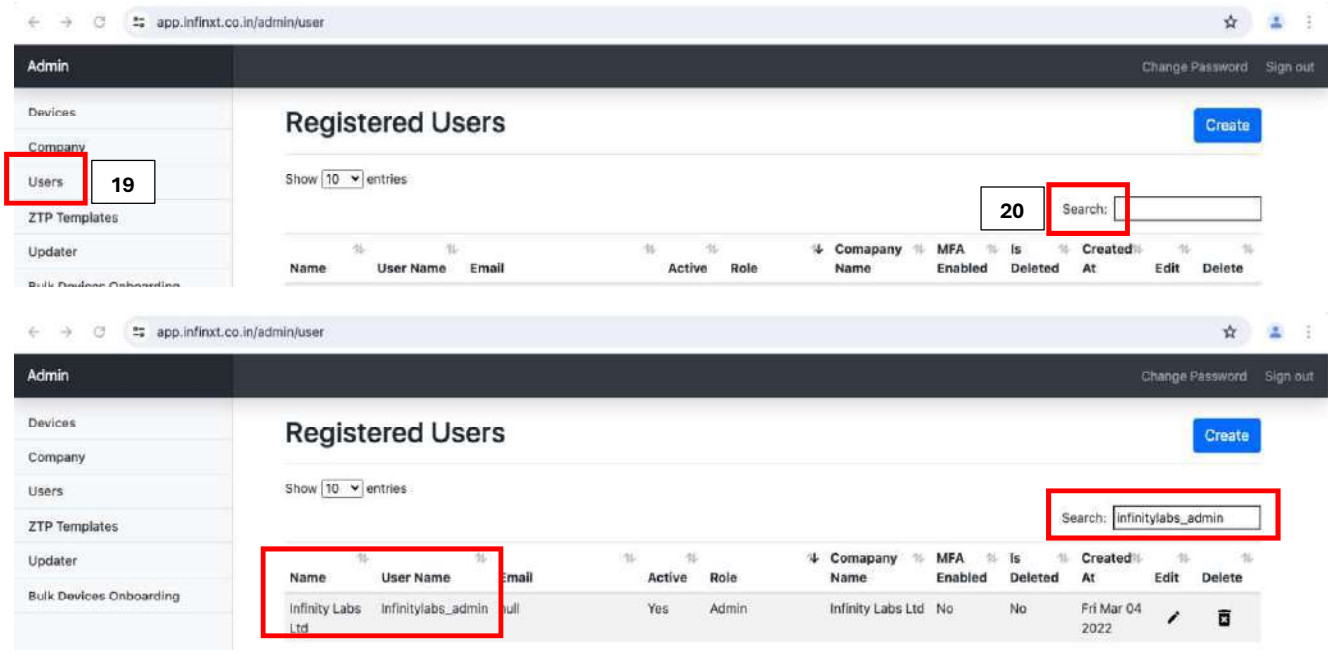
**Step5: -** To create a user login credentials for the respective tenancy, click on the user tab followed by create button and move to step 6

**Step6: -** To create a user account, the following details are required for input during user account creation for respective tenancy and click on submit

- Name – Full Name of the user
- Username – username which will be used as login credentials and must be unique
- Email – Valid email ID (Eg: support@infinitylabs.in)
- Enable Multifactor Auth: Optional field (Yes/No) – If yes, then admin will get the message on updated mobile number with an OTP to authenticate for login or other activities
- Mobile no. – Valid mobile no.
- Status – Status to input as active
- Company – Select the tenant/company name for which the admin is creating the user account
- Role – Select the role as admin. Admin user can create further user from tenant account based on the role (RBAC)

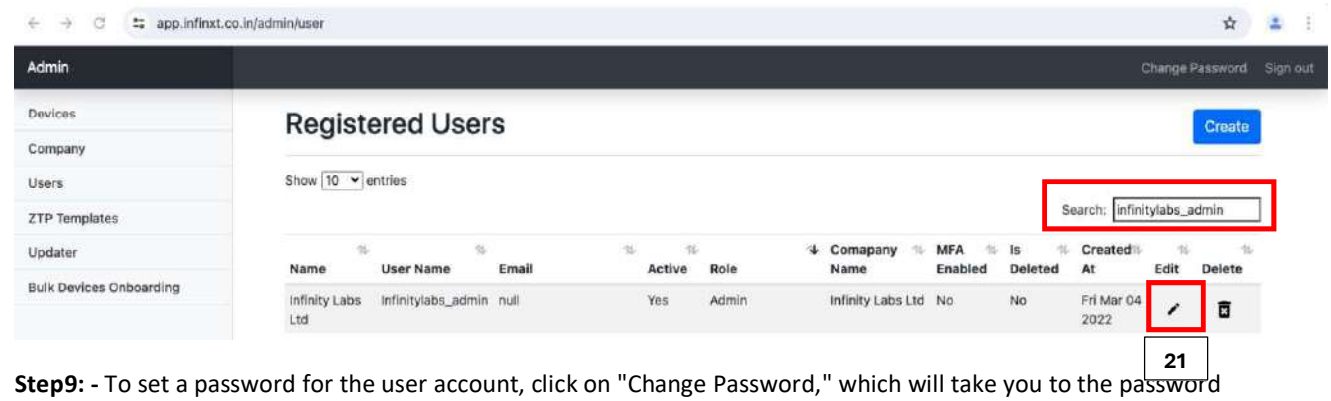


**Step7: -** Post successful submission, again go back to user tab to verify the details. Click on the user and go to the search tab and input the username created (eg: Infinitylabs_admin)



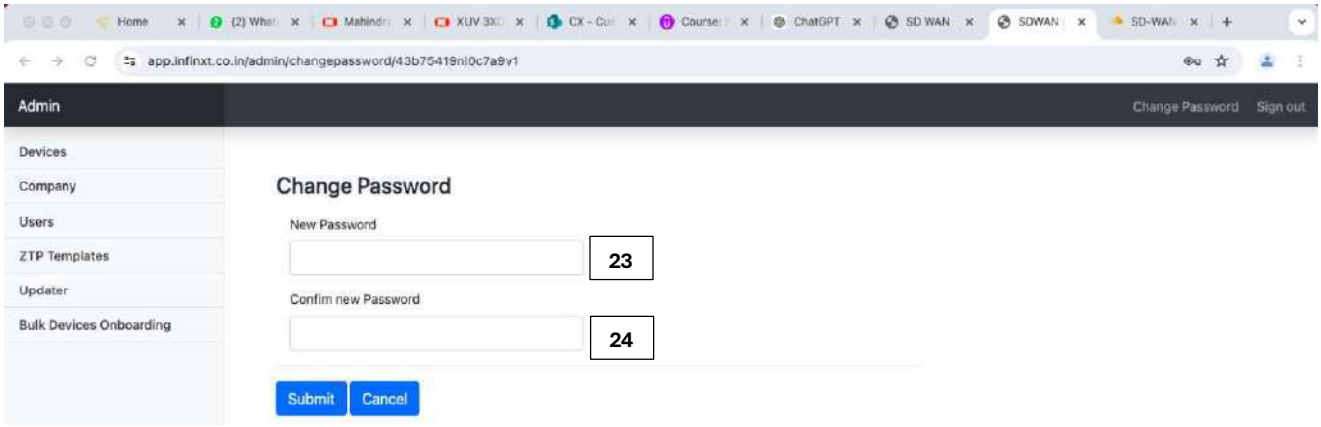

This facilitates the immediate activation of the user account for user login for respective tenancy, allowing for setting the password to that specific username.

**Step8: -** To set a password for the user account, search the username created for new tenant same as step 7 and click on edit (display as pencil icon)



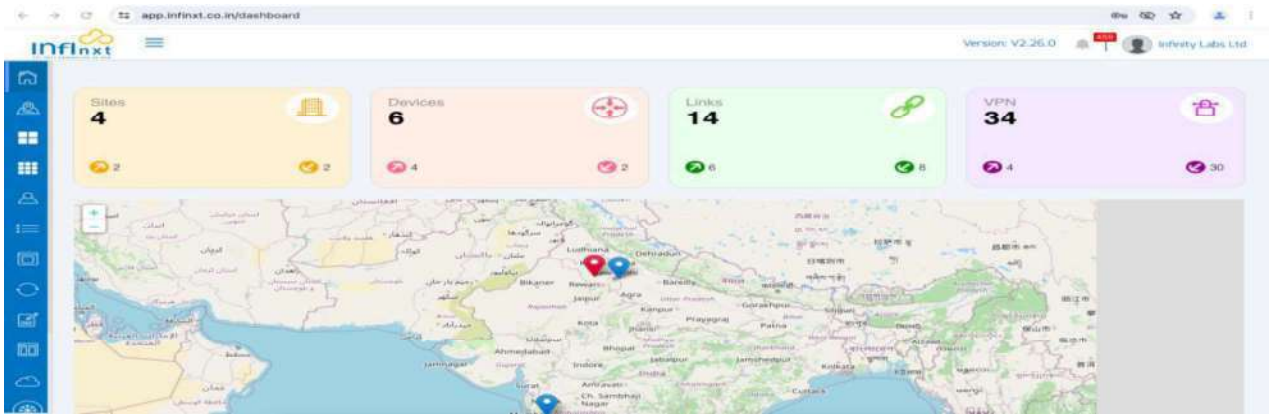

**Step9: -** To set a password for the user account, click on "Change Password," which will take you to the password modification page.

**Step10: -** Input the password for the user account, click on submit and the credentials are ready for user account login to start with device on-boarding




**Step11: -** Verify the credentials by login to user account with URL: https://app.infinxt.co.in

And after successful authentication to user account, it navigates to dashboard of user account to onboard and manage the edge devices.



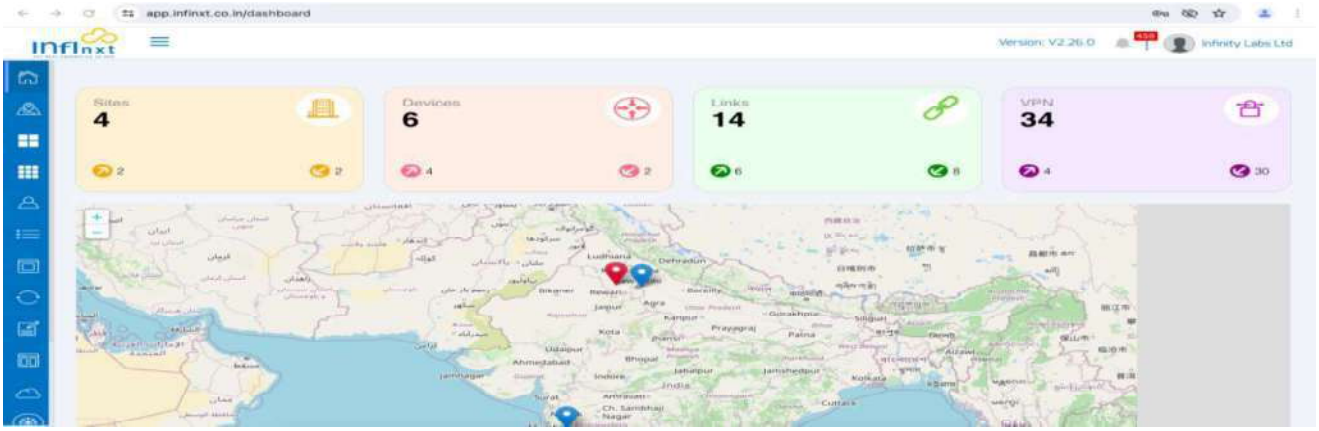

## Device Onboarding Process:-

Assuming the tenant has already been created and device is added to the respective account.

Option 1: WAN link connected to edge router is of Broadband with Dynamic IP assigned via DHCP.

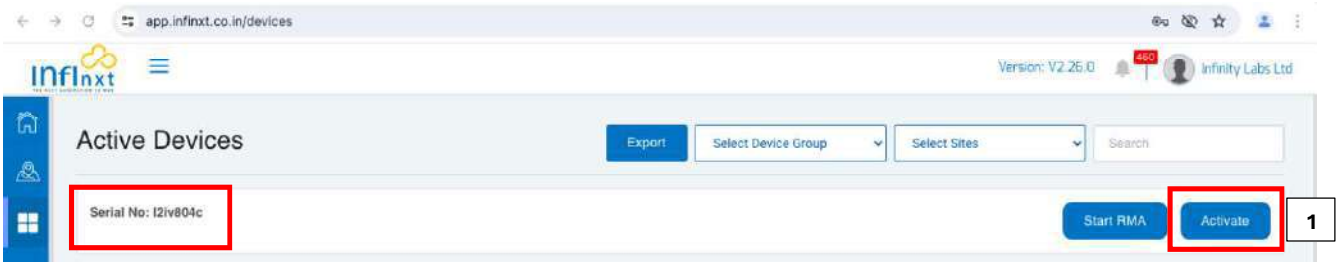
Configuration steps to connect and configure the active internet connection has already been explained and captured under deploy section (initial).

**Step1: -** Login to tenant account to initiate an onboarding process and connect an active internet link.



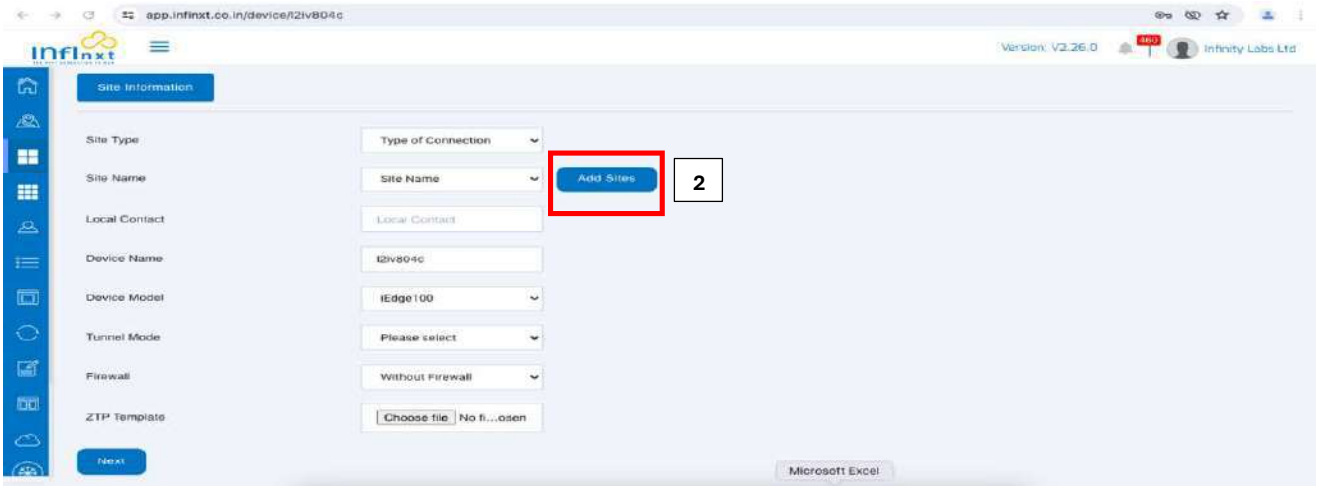

**Step2: -** Once the device boot-up and sense the active internet link, it initiates a TLS connection with controller for authentication and establish a bi-directional communication channel using the socket.io protocol and post successful setup device initiates a registration process.

Post registration device is visible under tenant account ready for activation process. Click on activate button




**Step3: -** It navigates to next page to enter the details of site and device to start onboarding. Click on add site button

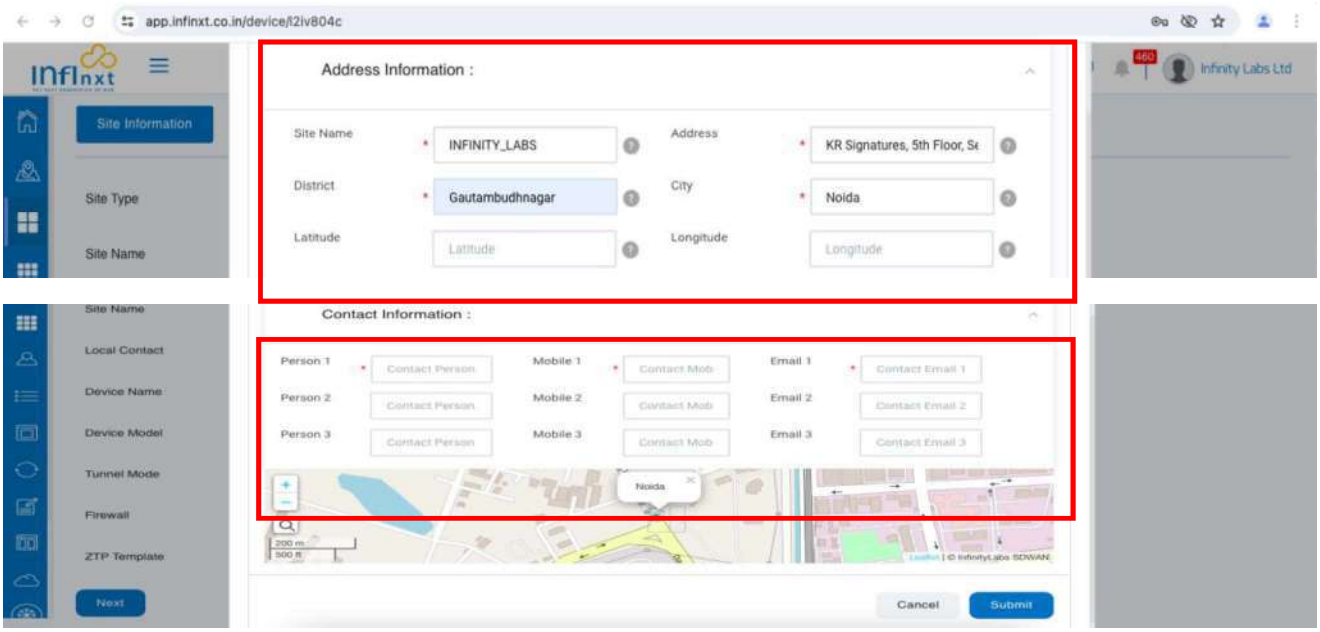
- **Site Type:** This refers to the functional mode of the device in customer network Hub | Branch | MESH
- **Site Name:** Site address details along with Lat/Long where the device belongs. Details can be added either in advance from site details tab or from the activation page explain in step 4.
- **Local contact:** Site SPOC
- **Device Name:** User defined name of the site through which the device can be identified
- **Device Model:** Can be selected from dropdown. Currently it is iEdge100 and other options available - iEdge50|iEdge100L|iEdge100F|iEdge1K|iEdge1000M



**Step3: -** Add site button navigates to page where user need to put the site address details along with site SPOC/ADMIN details required for on-boarding.

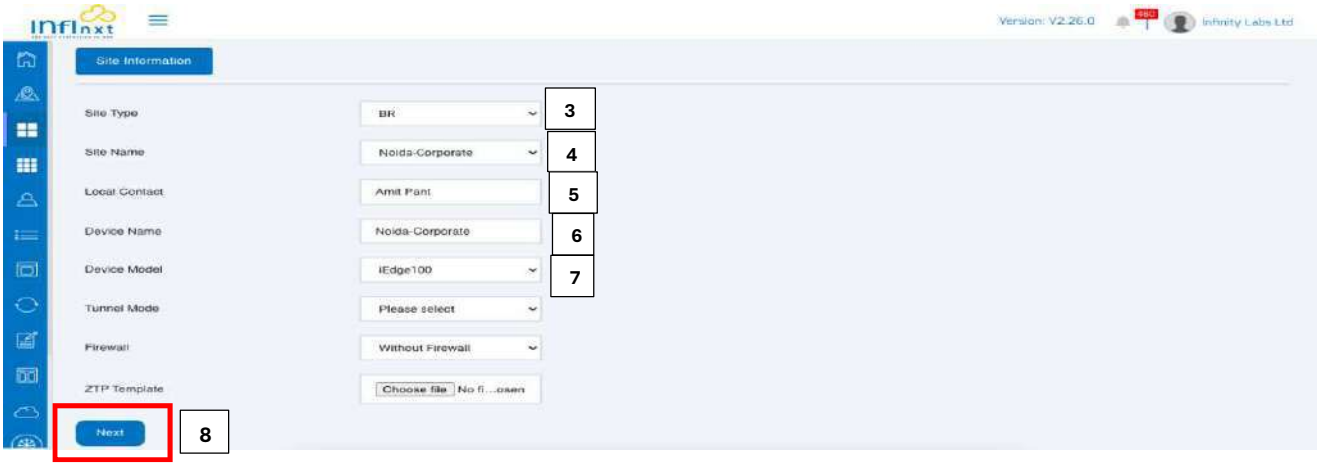
- **Site Name:** Site name should be relevant to site address for easy identification on dashboard for tracking and management
- **Address:** Address of the site here the device is installed
- **District|City:** Where the site belongs for identification
- **Lat/Long:** Optional field which refers to actual location
- **Contact Information:** Site admin/SPOC details with site contact person (admin) and valid email ID|Mobile no.

Although the user can put the address into the search of google map to know the actual device location. Once detail populated, click on submit to add the site details, which then available to select in site name drop down option on activation tab (Step 2).




**Step4: -** Once the site details added, it navigates back to activation page.

Select the site type as required, Eg: BR refer to branch, site name as the user added based on site location, local contact (admin), device name recommended to use same as site name, device model and then click on next button

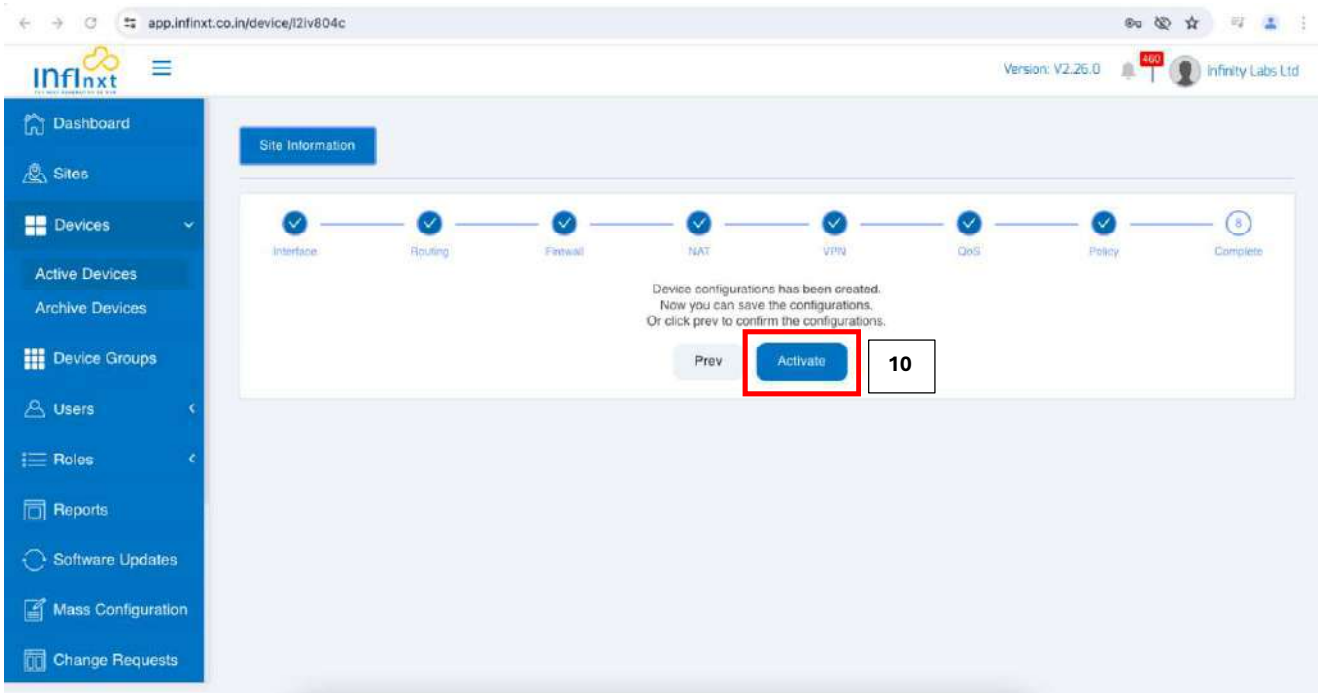


**Step5: -** Next button navigates to option to select the device terminal

- **Auto:** In this mode, the device activation process registers a new device in the managed devices database and pre-configures it with default ZTP settings tailored to its model. This includes interface configurations, automatically assigning LAN interface IP addresses, setting up DHCP for LAN, enabling Auto VPN with automatic Infinxt VPN tunnel configuration for Hub or Spoke site types, enabling auto MESH VPN for MESH sites, and adding the device to existing MESH VPN groups or networks.
-
- **Manual:** In manual mode, the device activation process registers a new device in the managed devices database and configures it based on the settings manually provided by the user during the activation process.

Click "Save" to proceed to the final activation step, where the device will be onboarded to the tenant user account for further feature configuration management.

**Step6: -** Click the "Activate" button to activate and onboard the device to the tenant account.




The device is now ready for further feature enablement and management with.

**Status:** Green indicates that the device status is up, while red indicates that it is down.

**Controller:** Green indicates that the controller's reachability status of the device is up, while red indicates that it is down.

**VPN:** Green indicates that the tunnel to the Hub is up, while red indicates that it is down (as indicated by the below status, tunnel has not yet been created).

AUTOMATED

## OPERATION

This section will address common issues encountered during the onboarding process and provide troubleshooting steps to ensure successful integration of edge devices.

Below are some common issues users may encounter:

## ISSUES

Here are some common issues that may occur:

- The controller IP cannot be reached via the WAN link
- The location file is not updating on edge devices to the respective controller's FQDN
- Ports 443 and 80 are not enabled in the network
- Socket connections between the edge device and controller are not being established

## TROUBLESHOOTING

1. The controller IP cannot be reached via the WAN Internet link – User can
- test ping to the controller's FQDN/IP address using either the local UI or CLI
- If the ping fails, the user should verify the WAN link status or contact their service provider for assistance.

2. The location file is not updating on edge devices to the respective controller's FQDN – User to check if

- The location file is updated with correct controller details. Use the following command to verify the location from the CLI:

```
infinity@l2iv804c:~$ location
{'domain': 'https://app.infinxt.co.in', 'backup_hosts': []}
```

Here are the steps to update the location with the correct FQDN:

**Step1:** Navigate to the directory using the command `$cd /config/.sdwan/` and press Enter

```
infinity@l2iv804c:~$ cd /config/.sdwan
infinity@l2iv804c:/config/.sdwan$
```

**Step 2:** Execute the command `sudo nano _location` and press Enter

```
infinity@l2iv804c:/config/.sdwan$ sudo nano _location
```

**Step 3:** The user will enter edit mode and use the arrow keys to modify the domain to the correct FQDN or IP address

```
  GNU nano 5.4
{"domain":"https://app.infinxt.co.in","backup_hosts":[]}
```

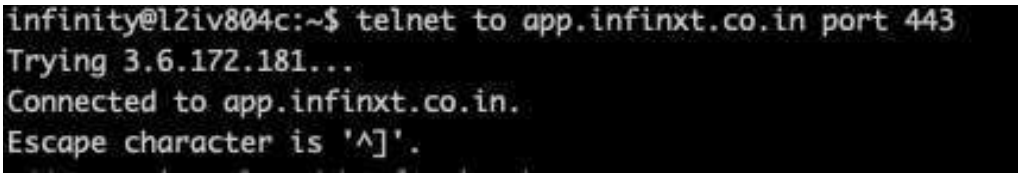
**Step 4:** After editing, press `Ctrl + X`, followed by `Y`, and then `Enter` to save the changes.

**Step 5:** To verify the location, execute the command `location`.

**Step 6:** Execute the command to change the directory to default mode by entering `$cd` and press enter

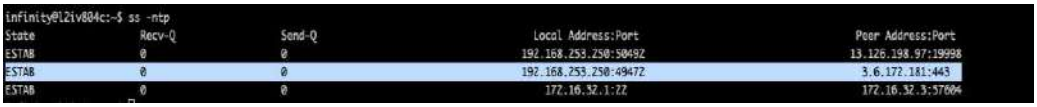
3. Ports 443 and 80 are not enabled in the network – User can check if

- ***telnet to app.infinxt.co.in port 443*** | 80 is getting connected



- If it is not connected, then the user should check the firewall policies, network settings, or coordinate with their service provider.

4. Socket connections between the edge device and controller are not being established – user can check if




The connection to the controller IP should be in an established state. If not, the user should verify network reachability

# RUTX50 Manual



This page contains the **user manual** for RUTX50 devices. Here you will find information summaries on all the features and functions available in RUTX50.

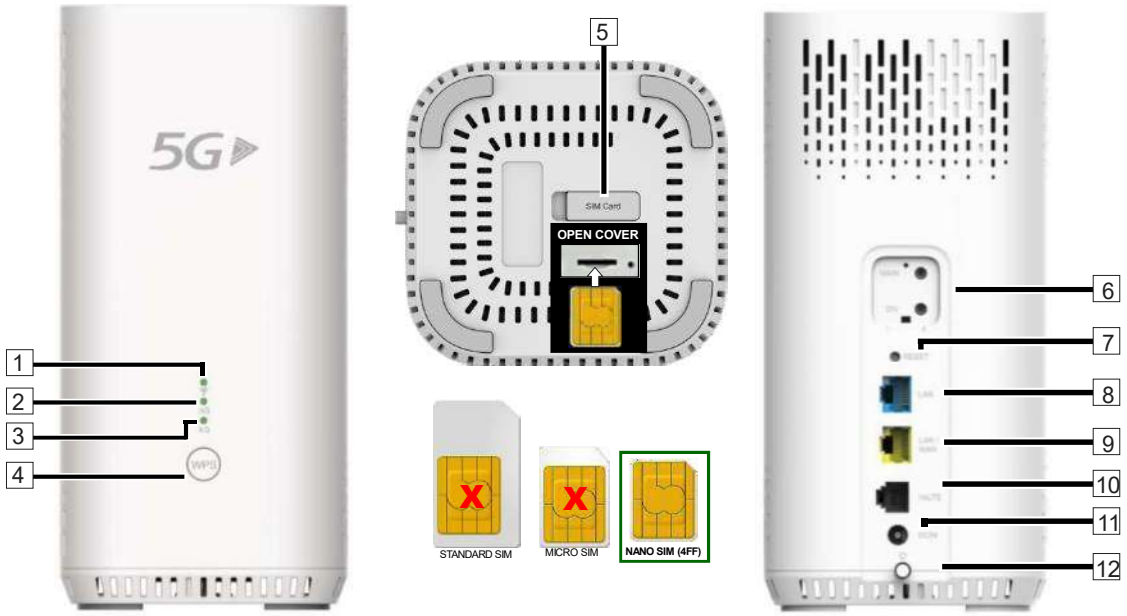

- WebUI
  - Search bar
  - Status
  - Network
  - Services
  - System
- Setting up your device
  - Front & Back Panels
  - LEDs
  - Spatial Measurements & Weight
- Remote Monitoring & Administration
- Powering Options
- Power Consumption
- Device Recovery Options
- Product Labels


For more more in-depth configuration guides, visit the **RUTX50 Configuration Examples** page.

# 5G NR Wi-Fi 6 CPE Router

# User Manual

# Product overview




### Front - System LEDs

| No. | Function | Off | Green | Blue | Red |
|---|---|---|---|---|---|
| 1 | Wi-Fi | No Wi-Fi | 2.4 & 5GHz | 5GHz only | 2.4GHz only |
| 2 | 5G NR | No 5G NR signal | Strong | Moderate | Weak |
| 3 | 4G LTE | No 4G LTE signal | Strong | Moderate | Weak |

### Front - WPS button

| | |
|---|---|
| 4 | Push to easily connect to a secure network (disabled after 2 mins) |

### Bottom - Nano SIM slot

| | |
|---|---|
| 5 | Install a 5G/4G nano SIM card for wireless connectivity to your mobile network provider if you wish to use this feature. |

### Rear - Buttons and Ports

| No. | Name | Function |
|---|---|---|
| 6 | 5G/4G antenna ports | [Optional] For 2 (MMCX) external antennas |
| 7 | Reset button | Router (software) reboot: Press & hold 3 secs<br>Reset to factory defaults: Press & hold 10 secs |
| 8 | LAN (RJ-45) | Connect this to your wired PC, network device, or a network hub; for devices not using Wi-Fi |
| 9 | LAN/WAN (RJ-45) | Connect this to your internet network provider if you subscribed to one (may be with TV service) |
| 10 | VoLTE (RJ-11) | [Optional] Connect this to an analog telephone |
| 11 | 12V DC input | Connect the supplied DC power adapter here |
| 12 | Power On/Off button | Press in to turn on, Press again to turn off |

## Setting up the Router

Obstacles such as concrete or metal walls/doors may affect the transmission of WiFi signals. Therefore, install the Router in an open space and keep the Router away from electronic devices that produce strong electrical or magnetic fields, such as microwave ovens, satellite dishes, cordless phones, or large appliances.

**Setting up the Router procedure and notes:**

1. Unbox the Router and DC power adaptor.
2. If you wish to use 5G/4G connectivity to your mobile network provider, first install a nano SIM card on the bottom of the Router. If not, you can continue using a typical Ethernet cable to your network provider's device. Alternatively both a nano SIM card and wired Ethernet can be used at the same time.
3. Plug the DC power adapter into an electrical outlet, then connect the adapter to the power jack on the back of the Router.
4. Press the power button to turn on the Router. During initialization, the WiFi LED on the front of the Router will:
   - **Blink green:about 1 min**
   - **Glow red: about 1 min**
   - **Glow green:about 17 sec**
5. See "Connecting to the Router" on the next page.

► The computer should not be setup to use a static IP address (this is very rare). Change the settings back to its default using Dynamic Host Configuration Protocol (DHCP) when connecting to the Router's web interface.

► The computer should not be setup to use a proxy server when connecting to the Router's web interface.

# Connecting to the Router

You can connect to the Router's network through a wired (using an Ethernet cable) or through a wireless (WiFi) connection.

**Connect to the Router's network using a wired (Ethernet) connection**

You can connect your computer to the Router using an Ethernet cable and join the Router's local area network (LAN).

1. Make sure that the Router is turned on, and about 3 mins has passed.
2. Connect an Ethernet cable to an Ethernet port on your computer.
3. Connect the other end of the Ethernet cable to the LAN port on the Router, and make sure the Ethernet ports' LEDs are lighted to show a successful connection.
4. Your computer will connect to the Router's local area network (LAN).

**Connect to the Router's network using a wireless (WiFi) connection**

You can connect your computers, mobile phones, or other devices to the Router using wireless (WiFi) and join the Router's local area network (LAN).

1. Make sure that the router is turned on, and about 3 mins has passed.
2. On your computer or WiFi device, find and select the WiFi network and enter your WiFi key (password). If you have not set a password or changed the SSID, the factory default names and passwords are:

   >SSID: **Raptor** / password: (leave empty)

   >SSID: **Raptor-5G** / password: (leave empty)
3. Your WiFi device will connect to the Router's WiFi network.

**WiFi connection using WPS**

You can connect your WPS-enabled device to the Router's WiFi network with Wi-Fi Protected Setup (WPS). This is an alternative to finding and selecting the Router's WiFi network manually. To use WPS to connect to the WiFi network:

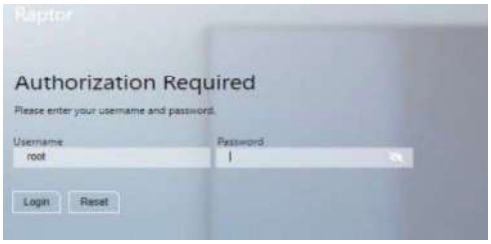
1. Make sure that the router is turned on, and about 3 mins has passed.
2. Check the WPS instructions for your WPS-enabled device.
3. Press the WPS button on the router, the front LED will start blinking.
4. Within two minutes, on your WPS-enabled device, press its WPS button or follow its instructions for WPS connections. (If the Router's front LED is no longer blinking, it has timed out. You can press the WPS button on the Router again if needed.)
5. Your WPS-enabled device will connect to the Router's WiFi network.

# Quick setup for 5G network or wired Ethernet network

Setting up, configuring, and managing the Router can be done through the Router's internal utility accessed using any Internet browser.

1. Connect to the Router using the provided Ethernet cable or via WiFi.

2. Enter "**192.168.1.1**" in the address bar of your web browser to enter the Router's login page. Click **Login** without changing the Username and leave Password blank.



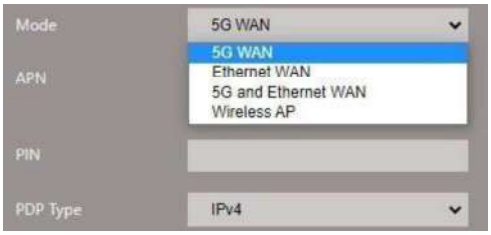

► If you do not see this Login page, switch to the other LAN port on the Router.

3. The first time you login, you will be directed to the **System Setup Wizard**, set your own login password and click **Next**.

► You must set a Password or else you will not be allowed to continue configuration.

► Passwords must be at least 5 characters long and case sensitive (0–9, a–z, A–Z); symbols are also supported.

4. Select the **Operation Mode** as wireless **5G WAN** (SIM card), wired **Ethernet WAN** (LAN cable), or both; and click **Next**.



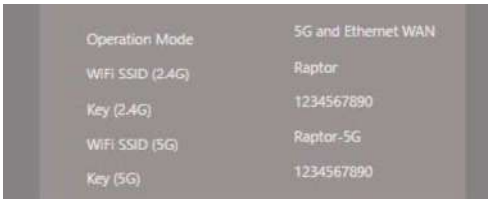

► If you select either of the **5G** modes, enter the SIM **PIN** code. If **PIN** was never set, leave this empty. Note that your mobile carrier may have a set a default **PIN** code.

5. **Wi-Fi 2G**: Edit the SSID (name of your hotspot), for example you can add your name. Select an encryption method and click **Next,** then repeat for **5G** network.

► It is highly recommended that you enable Encryption to secure your Wi-Fi network.

6. Click the yellow key icon to reveal the default key (password). You can change this to your own (at least 8 characters long and case sensitive (0–9, a–z, A–Z)). This is the last screen, click **Apply** to save settings.

7. Your settings will be displayed. Note your WiFi key (password) or write down in a safe location and click **Finish**.




► Your Router connection will reset. Wait a few minutes if you wish to re-login.

## Setup the Router mobile App

Search and install **PegaWrt** from your mobile Apps store. Launch **PegaWrt** App on your mobile phone.

Connect your phone to the Router 's Wi-Fi using settings from your web setup or these factory defaults:

SSID: "**Raptor**" Password: (blank)

SSID: "**Raptor-5G**" Password: (blank)

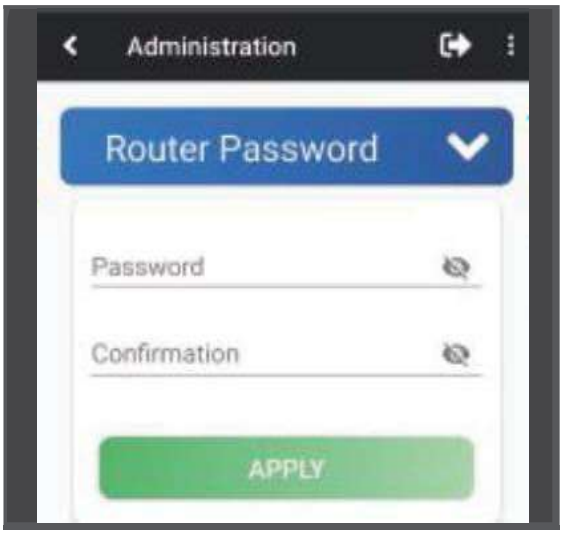
**LOGIN -** Enter your Password (from your web setup) to login or leave blank if you did not previously setup.

**Home -** Available settings will be shown in the mobile App.

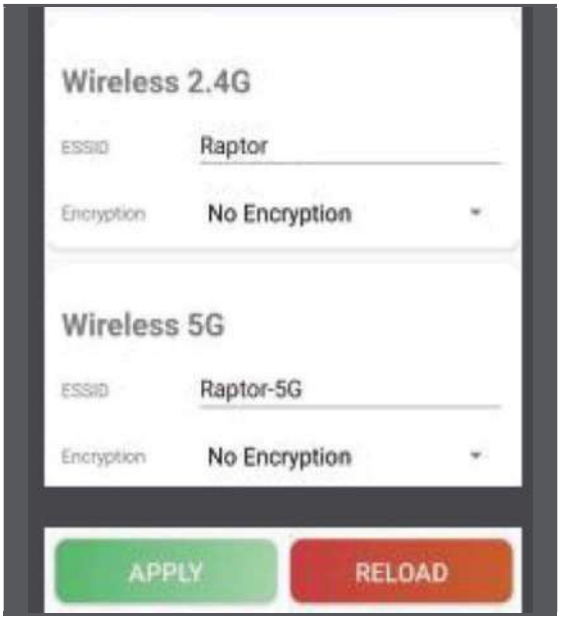
## Setup the Router mobile App

**Operation Mode -** Select wireless **5G WAN** (SIM card), wired **Ethernet WAN** (LAN cable), or both; and click **APPLY**.

► If you select **5G WAN**, enter the SIM **PIN**. Leave **PIN** blank if never set. Note that your mobile carrier may have a set a default **PIN** code.




**Administration -** Change or set your password here if you have not done so in web setup.




**WiFi -** Change or set Wi-Fi SSID name or encryption if you have not in web setup.

For example, set Wi-Fi 2.4GHz/5GHz Encryption to WPA2-PSK, the WiFi key (password) will be set to **1234567890,** and then click **APPLY.**

Note: After you press APPLY, the App will bring you back to the LOGIN page because your connection will use the new encryption method.

## Safety precautions

- Do not use the Router where flammables or explosives are stored, for example, in a gas station, oil depot, or chemical plant. Otherwise, explosions or fires may occur. In addition, follow the instructions indicated in text or symbols in area used.
- Place the Router on a stable surface.
- Keep the Router far from electronic devices that generate strong magnetic or electric fields, such as microwave ovens, satellite dish antennas, or large appliances.
- Place the Router in a cool and well-ventilated indoor area. Do not expose the device to direct sunlight.
- Do not use the Router in an area under or over temperatures ranging from 0°C to 40°C.
- Do not cover the Router. Reserve a minimum space of 10 cm around the Router for heat dissipation.
- Do not use any other power adapter except the one is provided with this Router. Use of other adapters could damage the Router.
- Before connecting and disconnecting cables, ensure that your hands are dry.
- To avoid electric shock, damage, and fire resulting from short-circuiting, do not use the product if wet, with wet hands, or while holding or drinking any liquids.
- Only insert relevant cables or devices into the ports, do not insert metal or other foreign objects.
- Do not place naked flame sources, such as lighted candles, on the Router.
- Keep the Router far from sources of heat and fire, such as a heater or a candle.
- Do not attempt to open the Router, there are no user serviceable parts.
- Do not perform or attempt any customizing, adjustments, repair or maintenance on the Router.
- Damage caused by user misuse or not following instructions are not covered by warranty.
- **Rated Voltage: 12V (DC Voltage)**
- **Rated Current: 3.0A**
- **Max Operating Temps: 0ºC to 40ºC**
- **Operating humidity: 5% to 95%, (non-condensing)**

# Network Camera Manual

# Contents


1. NOTE ..... 6
2. IMPORTANT SAFETY INSTRUCTION ..... 7

Warning: ..... 8

Attention: ..... 8

3. INSTALLATION ..... 9
4. WEB CLIENT USER MANUAL ..... 11

Install Active-X ..... 11

WEB CONFIGURATION ..... 16

Preview interface : ..... 16

Local Config: ..... 17

Encode: ..... 17

1. **Encode setting:** ..... 17
2. **Audio:** ..... 21

Prev Mode: ..... 22

1. **Output model:** ..... 22
2. **Display config:** ..... 23
3. **Profile:** ..... 24
4. **Privacy Masking** ..... 25
5. **Overlay** ..... 26

Network: ..... 27

**TCP/IP:** ..... 27

Net Service: ..... 28

1. **Email:** ..... 28
2. **DDNS:** ..... 29
3. **RTSP:** ..... 29
4. **FTP:** ..... 30
5. **UPnP:** ..... 30
6. **P2P:** ..... 31
7. **IP filter:** ..... 31

Record: ..... 32

1. **Record Plan:** ..... 33

2. **Snapshot:** ..... 33
3. **Storage:** ..... 34
Alarm: ..... 35
1. **Motion Detection:** ..... 35
2. **Video Blind:** ..... 35
3. **Alarm Input:** ..... 36
4. **Alarm Out:** ..... 36
5. **Abnormity:** ..... 36
6. **Insight:** ..... 36
7. **Human:** ..... 37
System Config: ..... 37
1. **System Config:** ..... 37
2. **General Config:** ..... 38
3. **Time Setting:** ..... 39
Security: ..... 39
System Tool: ..... 40
System Info: ..... 42
ALARM ..... 43
PLAYBACK ..... 44

# Welcome

Thank you for choosing SPARSH Network camera.

We hope this user manual is intended to serve as a better reference tool for your system.

## Copyright Statement!

This manual may not be reproduced in any form or by any means to create any derivative such as translation, transformation.

# 1. NOTE

Please prevail in kind, the instructions are for reference only.

This product may contain technical inaccuracies or typographical errors.

The products described in this manual may be updated at any time without notice.

This product manual shot may not from the same product, it is just for illustration.

If any doubt, or to obtain the latest procedures and additional documentation, please contact the company service department.

# 2. IMPORTANT SAFETY INSTRUCTION

Read these instructions carefully and retain them for future use. If this product is passed to a third party, then these instructions must be included.

- Read these instructions.
- Keep these instructions.
- Heed all warnings.
- Follow all instructions.
- Clean only with dry cloth

When using electrical products, basic safety precautions should always be followed to reduce the risk of fire, electric shock, and/or injury to persons.

The safety measures are categorized into two parts "**Warning**" & "**Attention.**

**Warning :** **Without Warning may lead to death or serious injury.**

**Attention :** **Without attention may lead to injury or loss of property.**

| | |
|---|---|
| **Warning** To remind users to keep away.<br>from potential death or danger of serious injury. | **Attention** To remind users to keep away from<br>potential injury or danger of property from loss |

## Warning:

This product is not intended for use by children.

Please use the required power meets SELV (safety extra-low voltage circuit) to supply. In the accordance of

IEC60950-1, the power should comply with the rated voltage of DC 12V and AC 24V (Depends on the specific model) of Limited Power Source.

If the equipment works abnormal, please contact your purchasing store or the nearest service center. Do not disassemble or modify the equipment in any methods. (If do without permission, user will be responsible for the result.)

To reduce the risk of fire or electric shock, please prevent the product from raining and moisture.

The installation should be done by the professionals and comply with local regulations.

Please install the easy-used electric failure equipment into wiring of the building.

The instruction of equipment installation on the ceiling: After installation, please to make sure the connection can withstand 50Newton(N) downward pull.

## Attention:

Before the camera running, please make sure whether the supply power is right.

Please do not drop off or heavily strike the product. Do not open the product housing. Do not insert any foreign objects inside of the housing.

Please do not touch the image sensor optical element. If it is necessary to clean, please use the clean cloth slightly moist with alcohol to wipe dust; When not in use, please cover the lens to protect the image sensors.

Please avoid focusing on glare (e.g., lighting, sunlight etc.), or it will be resulting in too bright or colorful vertical stripes on the screen (which is not camera's error) and affect life of image sensors.

The laser beam may burn down image sensors. When the installation of laser beam used, please make sure that the surface of image sensors does not expose under the laser beam.

Please avoid followed places: moisture, dusty, hottest, coldest (Normal work temperature range: -14°F～+ 140°F) and strong electromagnetic radiation etc.

Please do not accumulate more heat and maintain ventilation flow around the camera.

When using the camera, please do not make water and any liquid flow into it.

When delivering the camera, please packing as shipping out or with the same quality material of the factory.

Parts change regularly: Some parts of the products (e.g., Electrolytic Capacitors) should change regularly according to their average lifespan. Their lifespan would be different for the using environment and using time, so please check them regularly. For more information, please consult with your purchasing distributors.

## 3. INSTALLATION

Ensure that the device in the package is in good condition and all the provided accessories is in box.

Choose an installation location and height where you can get the desired view without any obstacle.

Don't touch glass of the camera directly & make sure that al the related equipment is power -off

during the installation.

### Tools required for CCTV Installation (Not included)

**Drilling Machine** **he** **Pencil** **ps** **Screw Drivers** **Wall Plugs (included)** **Screw (included)**

**camera.**

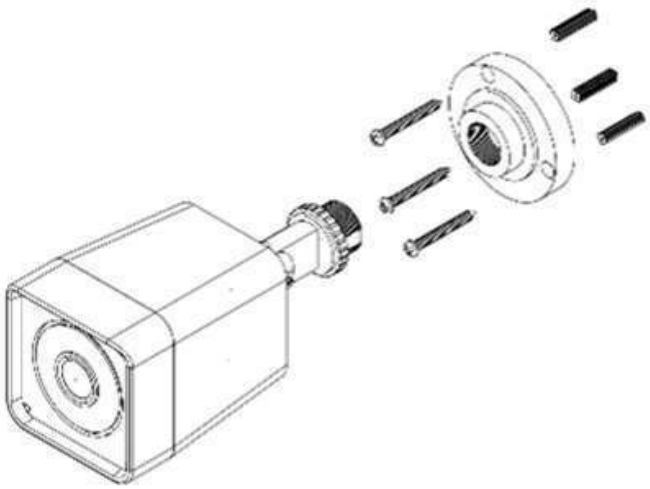
**Step 1.** Place the wall bracket onto the surface of the chosen installation. Location after then mark the drilling holes with the help of pencil. (Dome camera bracket can be unscrewed with the three screws on the back)

**Step 2.** Drill the 3 holes as per provided PVC Wall plug and then insert the wall plugs onto the drilled holes.

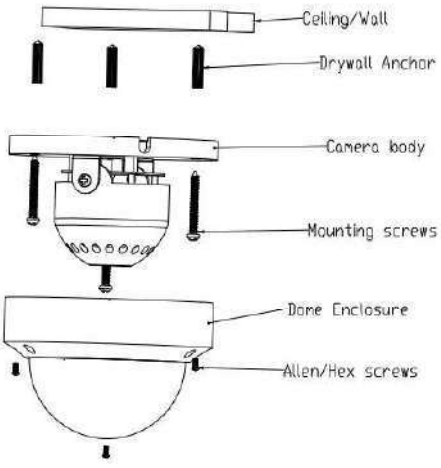
**Step 3.** Also drill one big hole for cabling. (Skip this step if not required)

**Step 4.** Now secure the wall bracket using the screws.

**Step 5.** Insert the screws and tighten it to the base and install the camera.



Bullet Camera

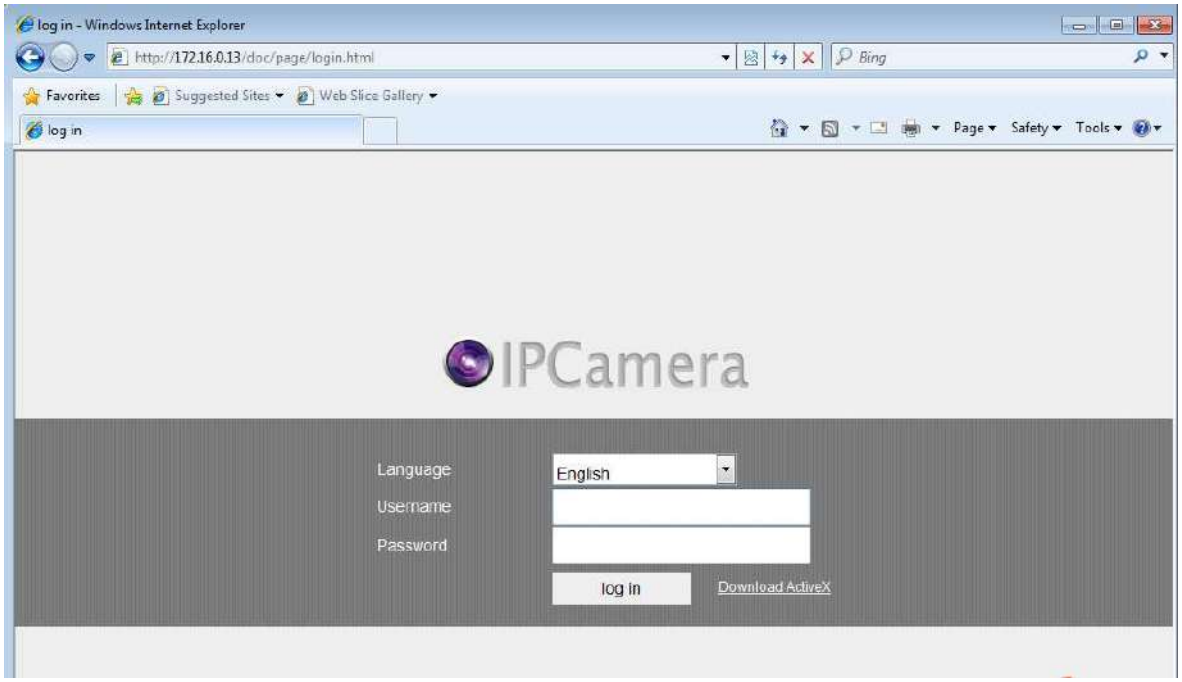



Dome Camera

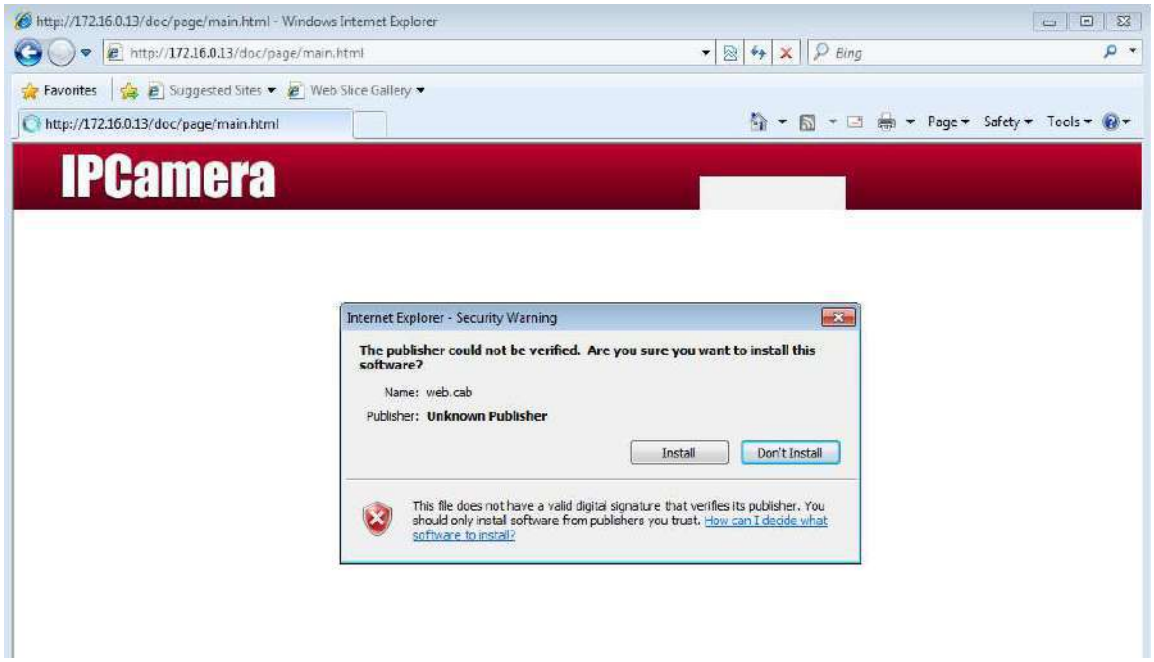
# 4. WEB CLIENT USER MANUAL

**Install Active-X**

1. Input username and password to log in (When first access, need to download the Active-X) :

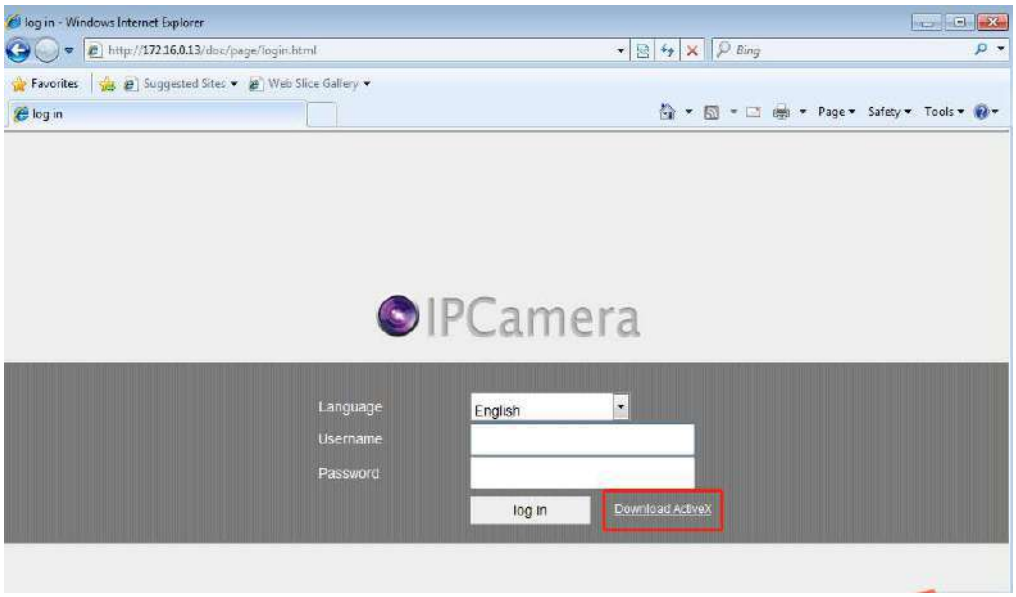


2. Choose a language Input username and password to log in, automatically prompt to install Active-x, Click “Install” to complete the Active-x installation.




When installation finished, it will appear a file folder named Web client in C:\Program Files \IPC Web. It is the downloaded file. If delete the file, will result in unable to log in the Web client. You should download again.

3. If you are not prompted to install the control, click " Download Active X"



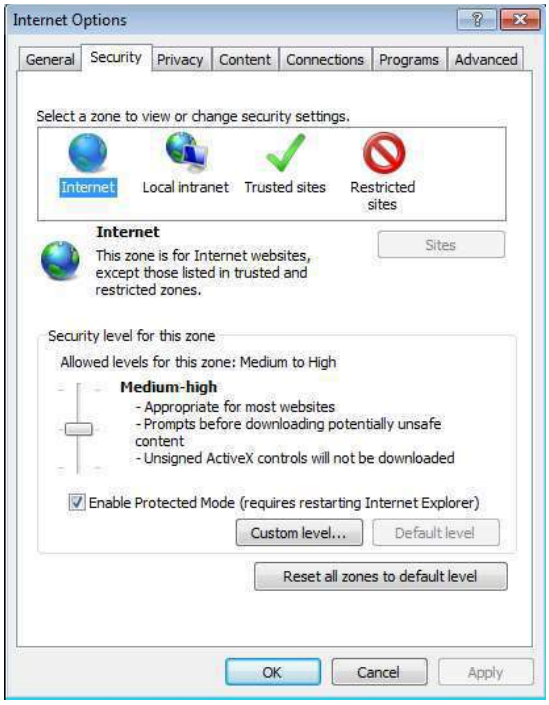

The file will be downloaded in C:\Users\Administrator\Downloads \TSIPCWebClient_EN_SetUp.exe, Double clicks to run this program.

**FAQ:**

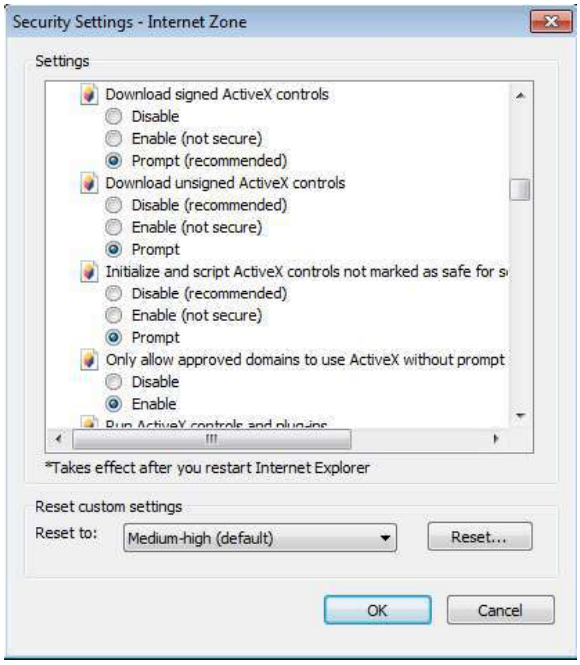
Question：Why I cannot install the control?

Answer：Your IE browser intercepts the installation; you need to change the settings in IE. Choose the security option.

Choose “Security”



Choose “Custom Level” , Enable “Download unsigned Active X controls.”




Then Re-enter the IP address of device, choose “install Active-X controls”.

**Note:** enter the username and password of device click log in to access web client.

Question: I have already downloaded the controls, why I still cannot log in Web client?

Answer: Please check if the network is normal or if the IP conflict with another device.

# WEB CONFIGURATION

**Preview interface：**

| | | |
|---|---|---|
| Preview interface | Main stream Sub stream Sub stream2 | Select Mainstream or Sub stream or Sub stream 2 to play video |
| | | Input on the device, turn on the switch, you can hear the sound |
| | | Turn on the switch, the device and the web can talk |
| | | Record： Record local recordings and save them in C:\Record |
| | | Capture the preview screen and save it in C:\Picture |
| | | Continuously capture three pictures and save them in C:\Picture |
| | 1X | Original resolution display |
| | W:H | Suitable for display resolution |
| | | Full-screen display |

| | | |
|---|---|---|
| | 3 | fluency |

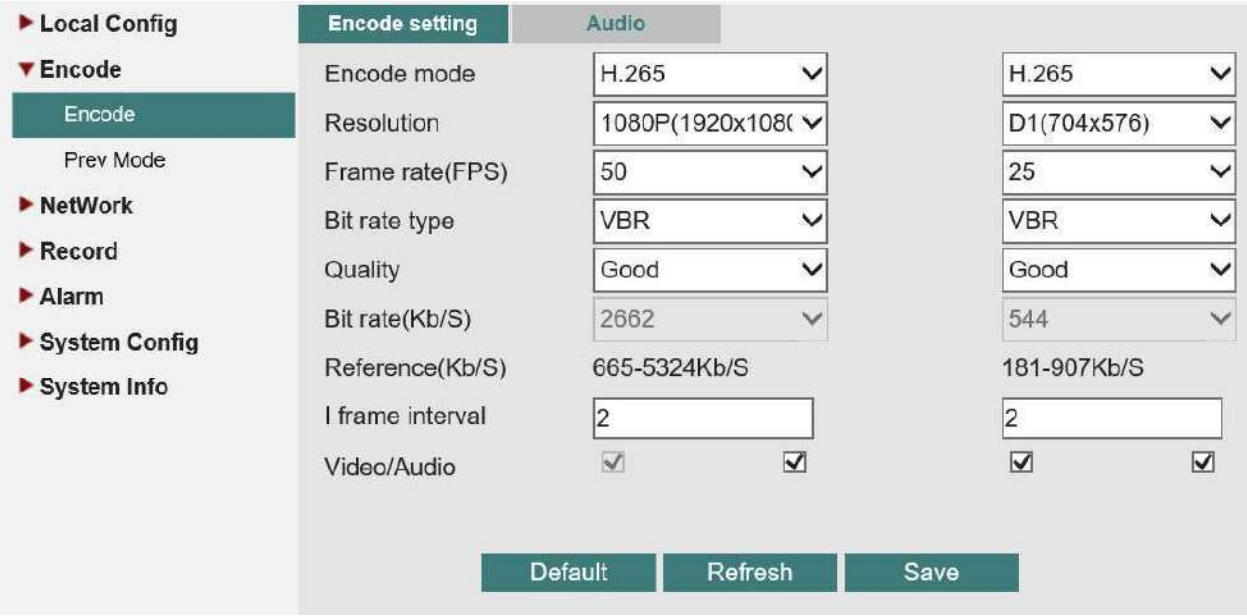
**Local Config:**

In this configuration of the camera can be done.

**Encode:**

Encoding of video and audio are as follow:

1. **Encode setting:**




Encode mode:

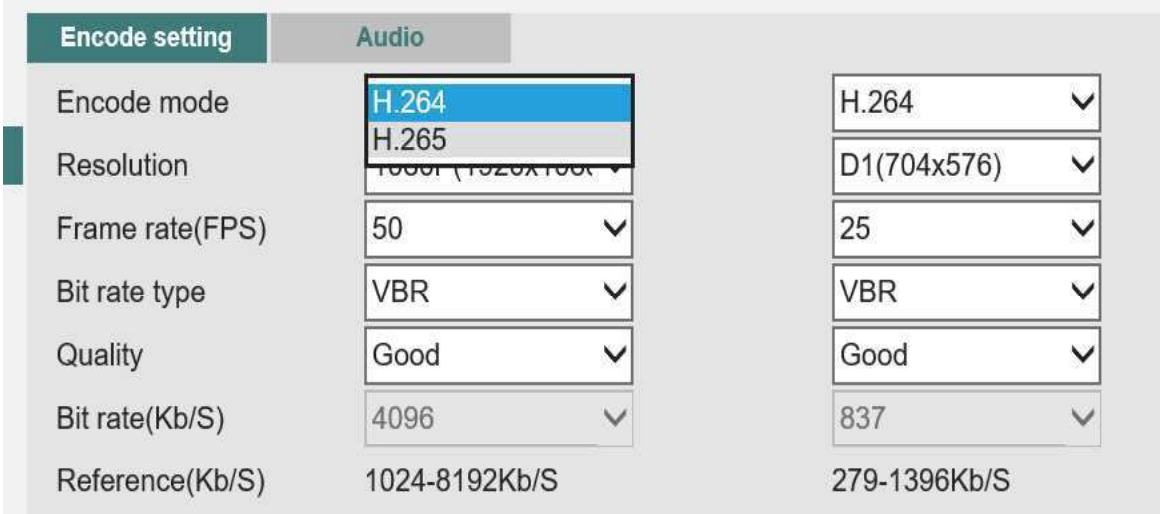
Camera is running on three streams:

Mainstream

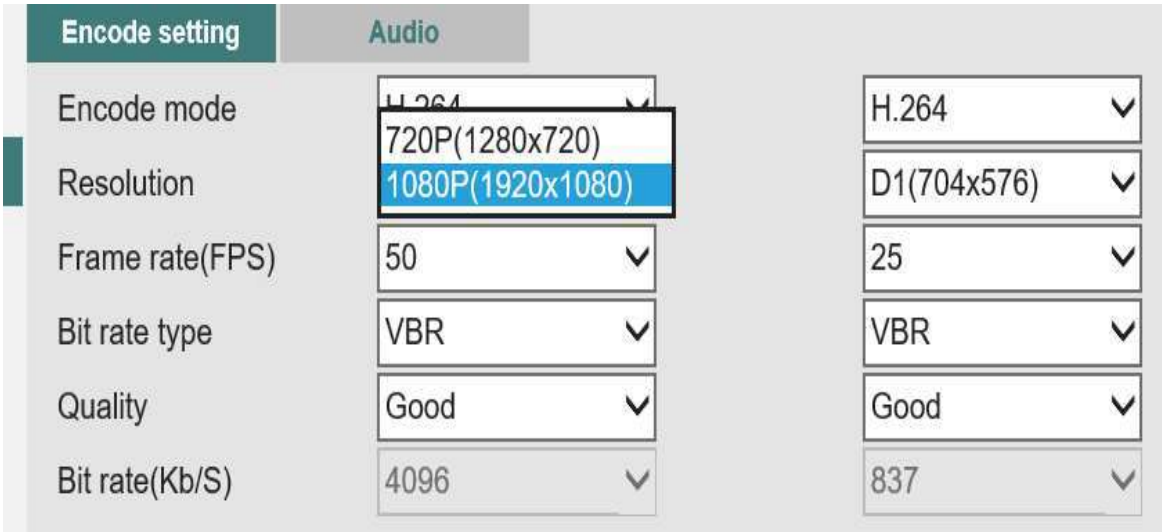
Sub stream

Third Stream

Choose video compression standard maintain on the 2 MP camera.




Resolution: We can set the resolution according video quality need.

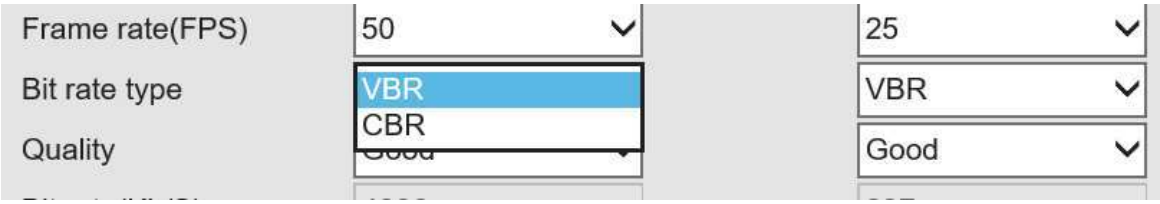


Frame rate: In mainstream we can only set the fps upto 50 but in sub stream upto 25 only.

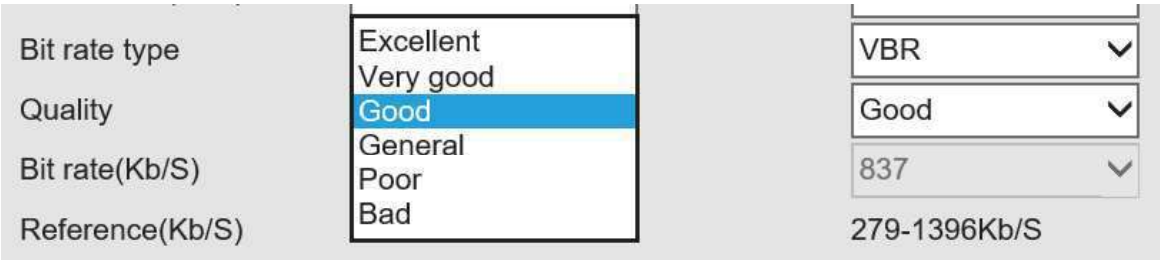
Bit rate type: There are two types which are follow as

(i)VBR

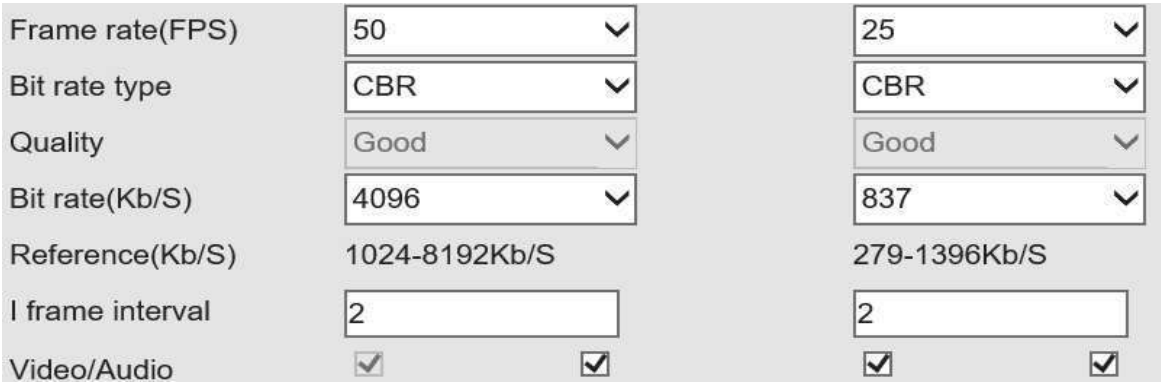
(ii)CBR



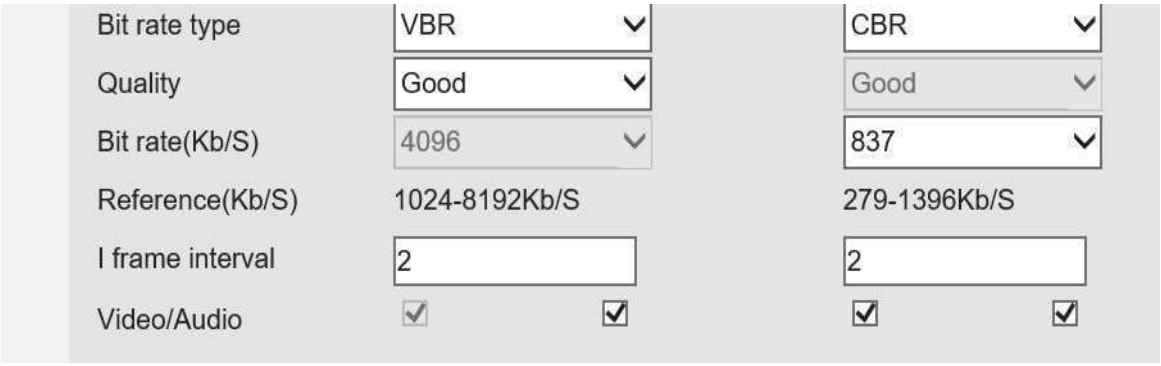

Quality: Quality can be set according the video requirement which are as:




Bit rate: When we set the video encode on CBR bit rate type then we manually config the bit rate of the video encode.



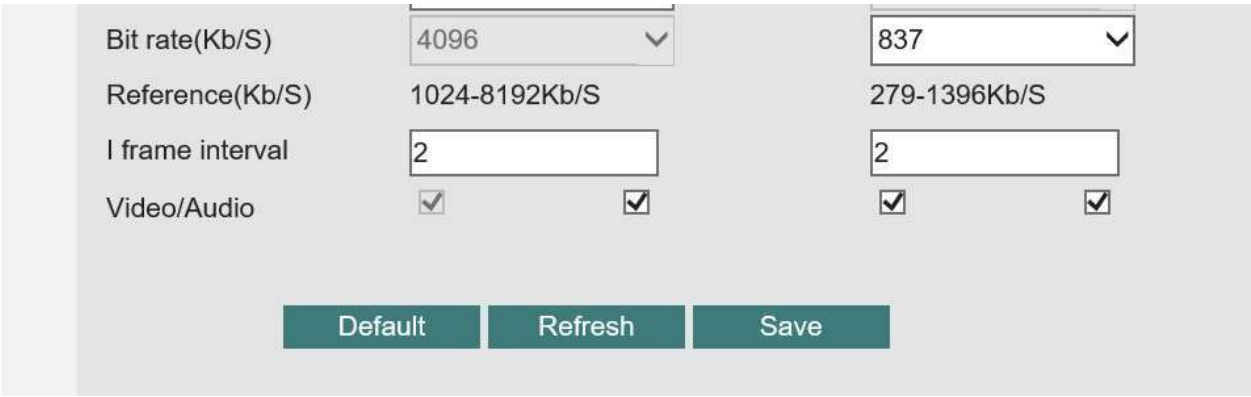
I frame interval: For 2mp bullet camera by default value is 2.




Default: It is used for setting the value as default.

Refresh: It is used for updation from previous.

Save: It is used for saving the configuration.



2. **Audio:**

In this we have done the audio encoding setting.

Encode mode: We have used the audio format “G711A” and “G711U” for audio compression standard.

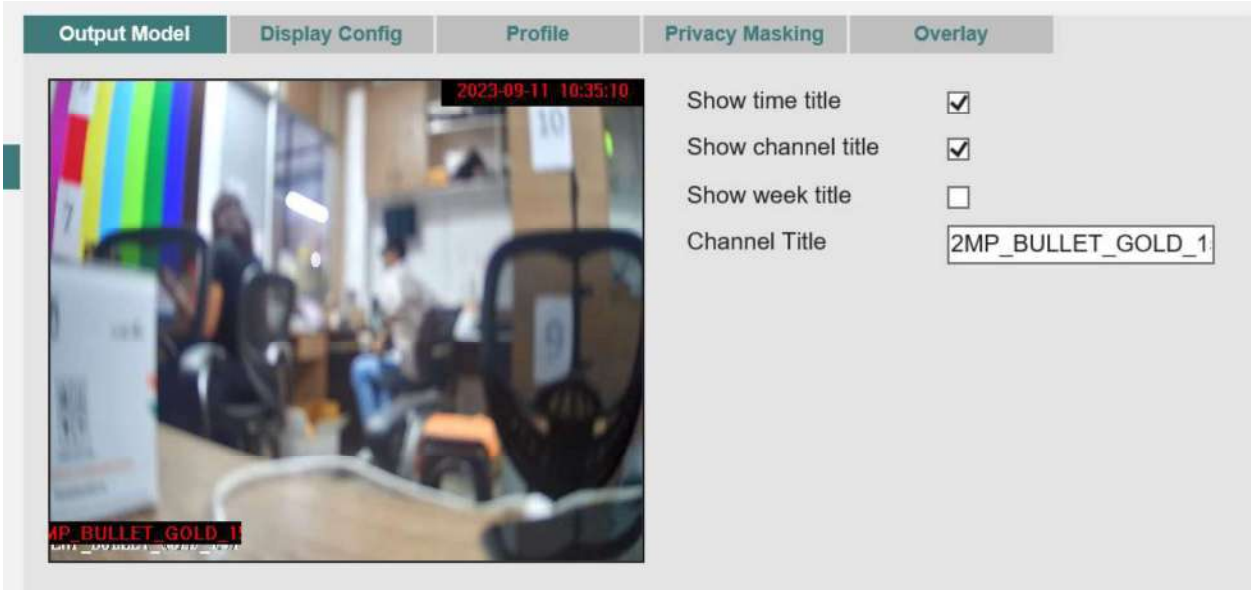
Sampling rate: the sampling rate should be 8000 Hz or more than it due to the smoothness of audio.

**Prev Mode**:

**1. Output model:**

Output model is used to set the on screen changes as

(i) Shoe time title
(ii) Shoe channel title
(iii) Show week title
(iv) Channel title



2. **Display config:**

Image Color: In this we have change the color combination of the image as

(i) Profile: If we want to change the profile as day/night.
(ii) Brightness: Brightness can be change as the outside intensity of light require to do changes.
(iii) Contrast
(iv) Saturation
(v) Tone
(vi) Sharpness

**Camera:**

Day/night mode: In this manual and auto mode provided according to the requirement of the camera

DNC threshold: the threshold is upto 1-50.

IR_CUT: it is synchronous with the amount of light sensed by LDC.

Flip: It is used to set for real image should be seen on screen if camera is set inverted or alternate.

Mirror: It is used to set the camera alternate side view.

IR reverse:

ISP:

Exposure mode: Its value is up to 1/100000 for BLC and HLC of the video encoding in the camera.

According to the amount of light the value getting changed.

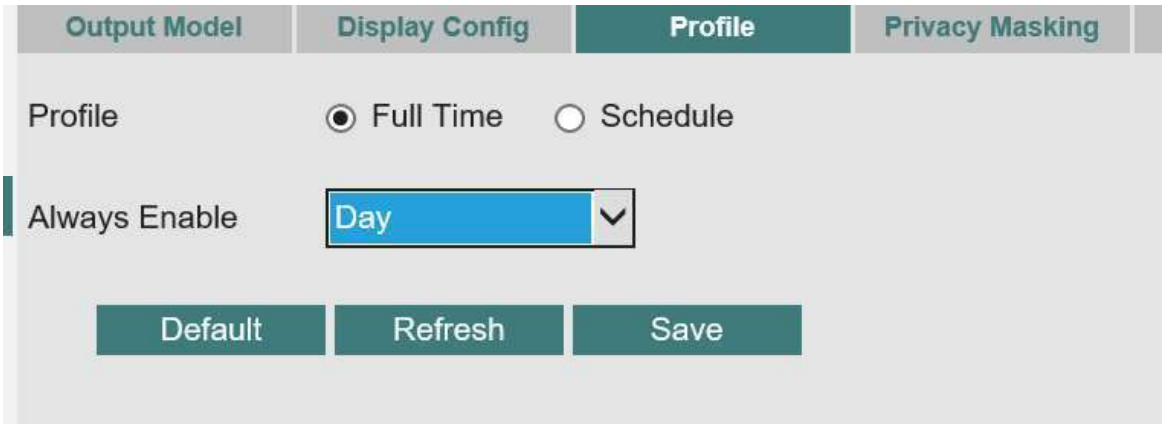
Backlight: It can be on BLC OR ON HLC.

BLC: Backlight compensation optimized the camera to increase light exposure for darkened areas.

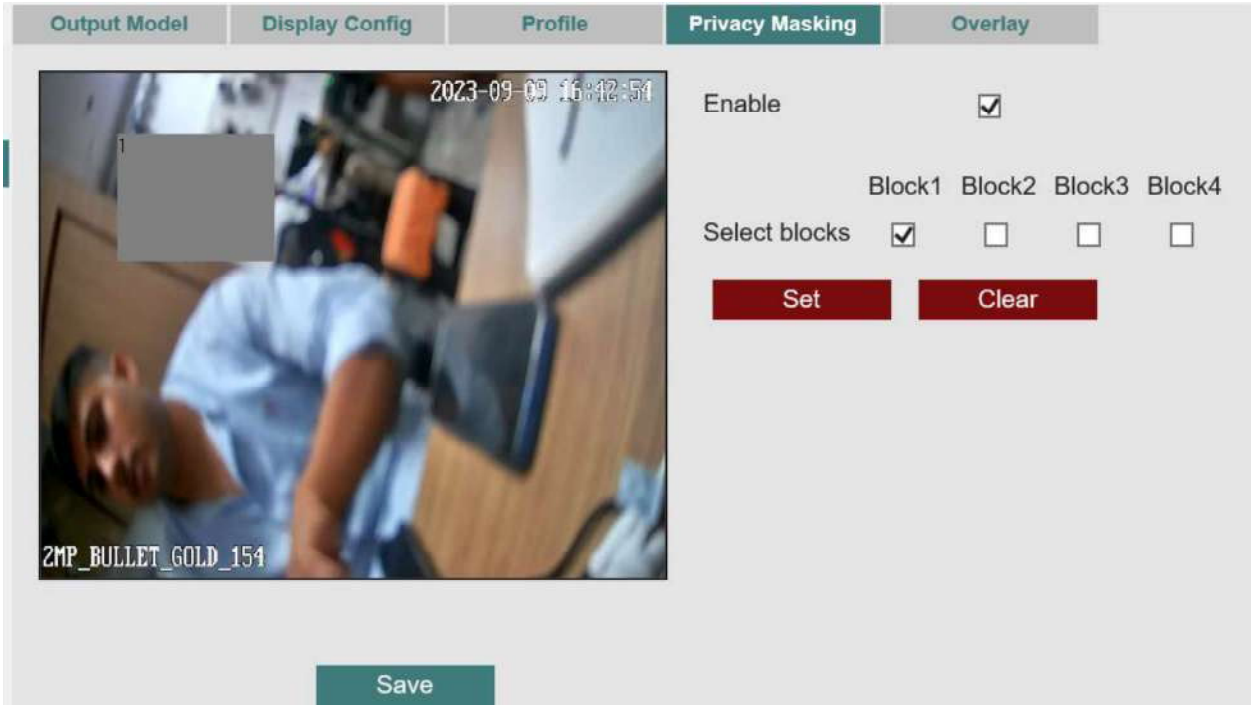
HLC: Highlight compensation automatically detects strong light sources and light exposure, greatly improving the clarity of bright areas.

Fill Light:

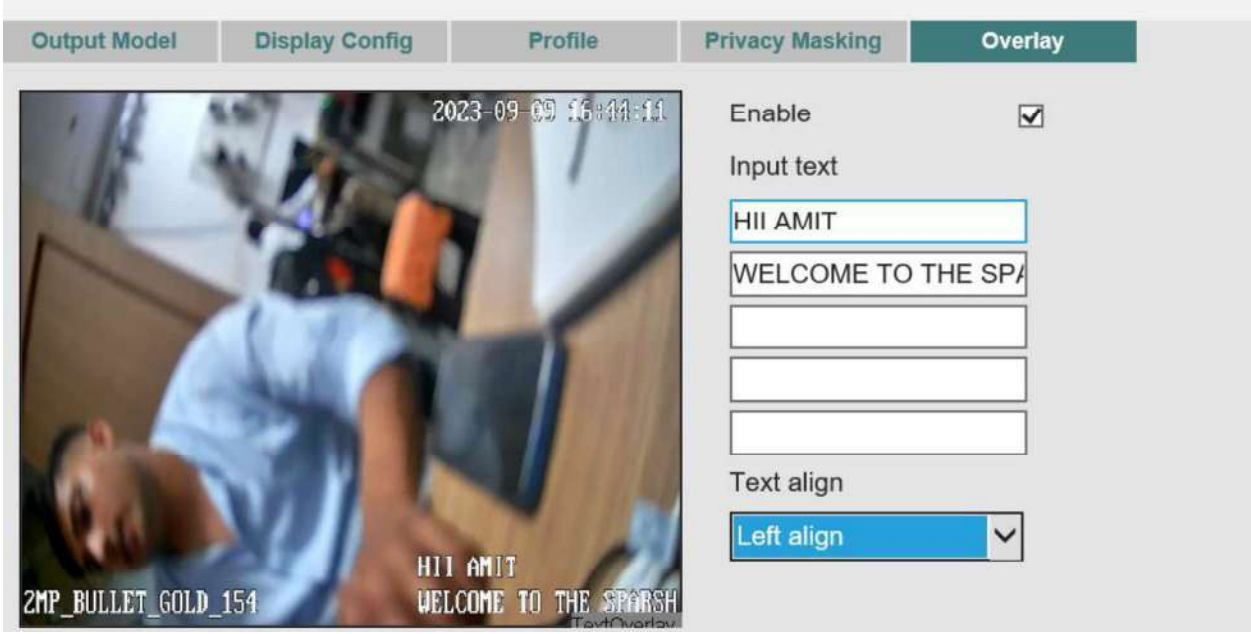
### 3. Profile:



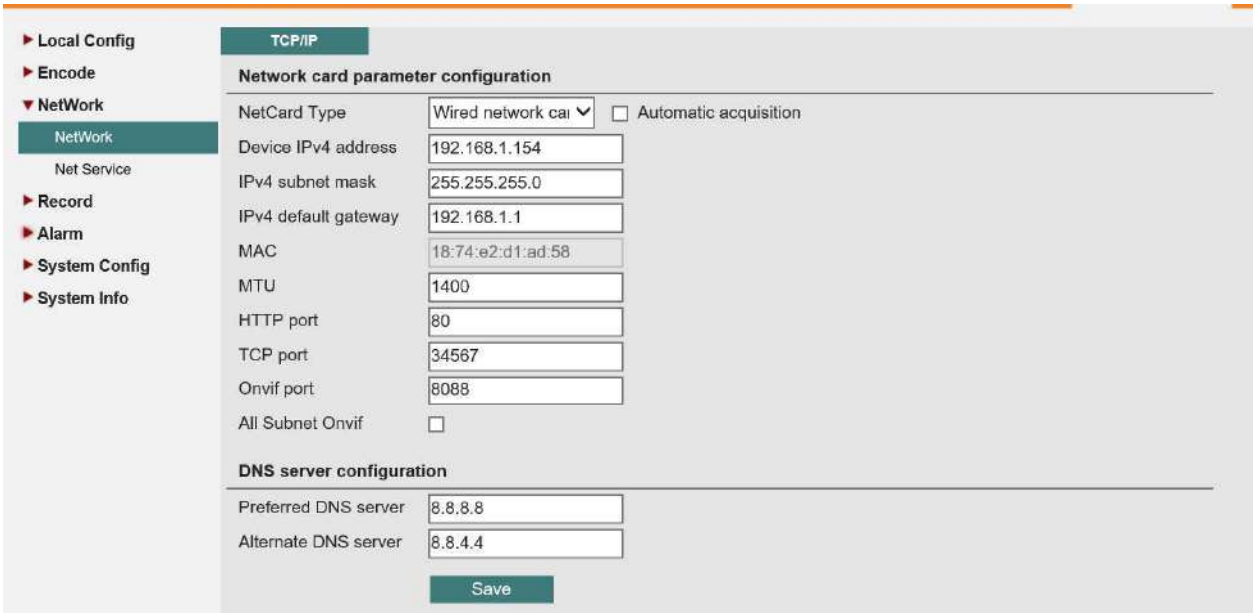
4. **Privacy Masking:** It is used to hide the part of the image.



5. **Overlay**: It is used to overwrite the text on display screen. Alignment of the text such as left align or right align can be done by text align.



**Network:**




**TCP/IP:**

We use here TCP/IP.

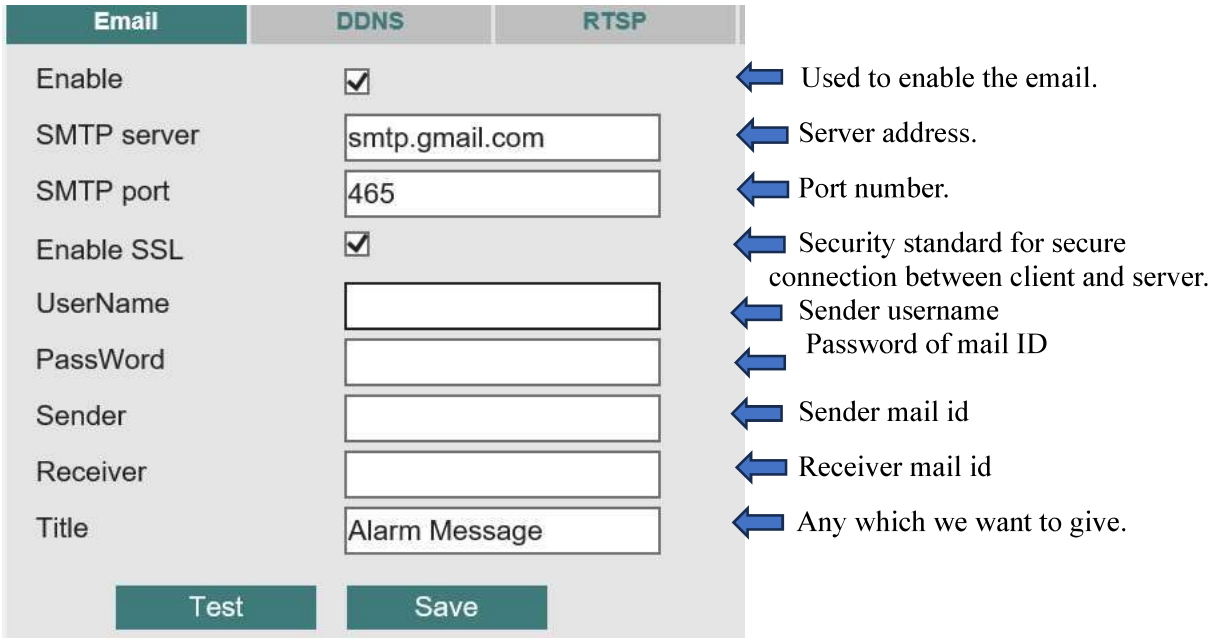
Network card parameter config:

(i) Net card type: we used wired network card.
(ii) Device IPv4 address: we manually provide the IP to the device.

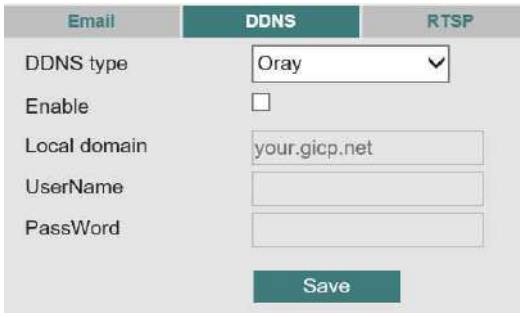
(iii) All the port no and mask are by default set.

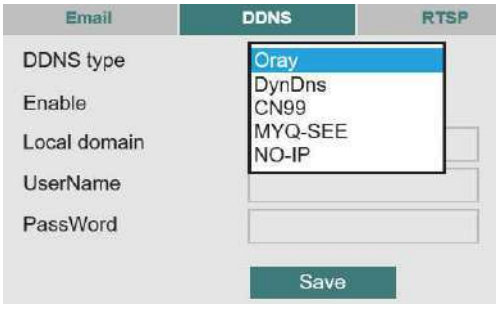
**Net Service:**

**1. Email:**



## 2. DDNS:



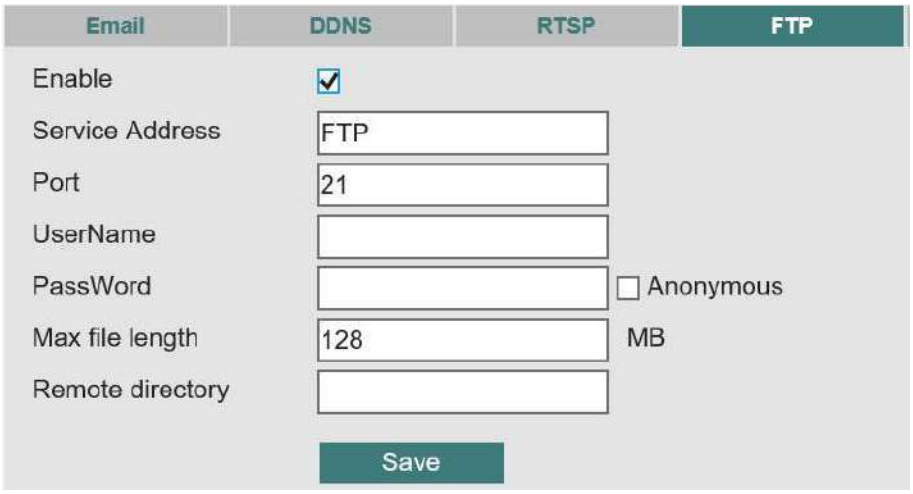




In DDNS we select the DDNS type and provide user at which we want the DDNS should come.

## 3. RTSP:

In RTSP, real time data will be transferred from multimedia to a server. We can access it on VLC by copying the URL.

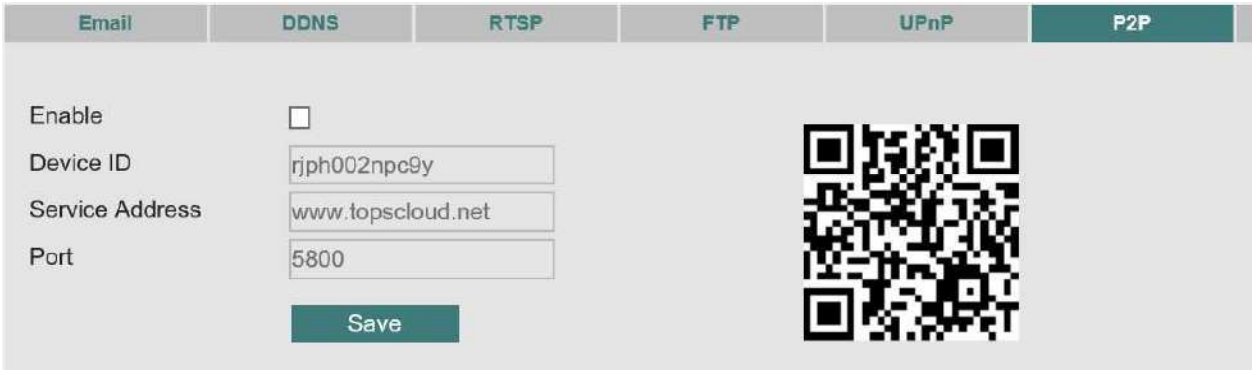
**4. FTP:**


Email
DDNS
RTSP
FTP
Enable
Service Address
FTP
Port
21
UserName
PassWord
Anonymous
Max file length
128
MB
Remote directory
Save

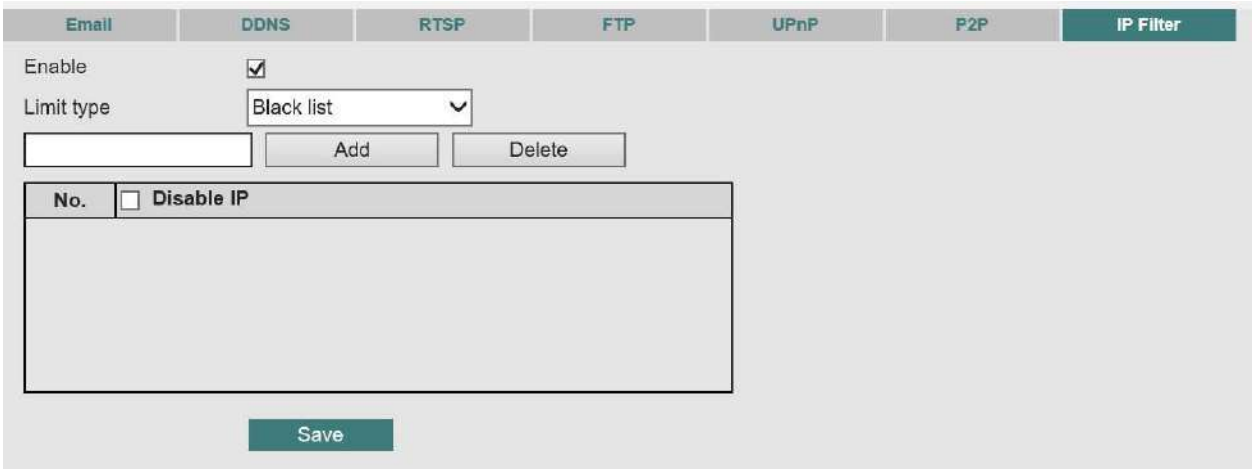
**5. UPnP:**

Email
DDNS
RTSP
FTP
UPnP
Enable
HTTP port
0
TCP port
0
Phone port
0
Save

**6. P2P:**

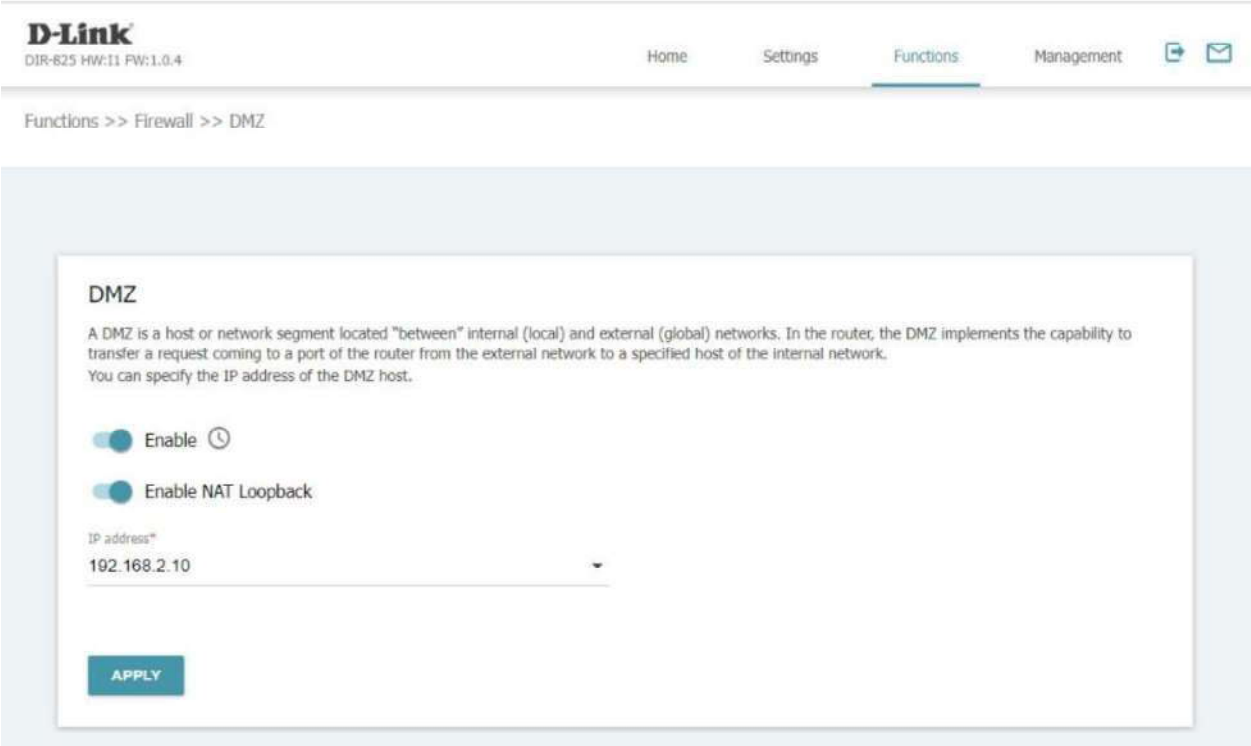



**7. IP filter:**



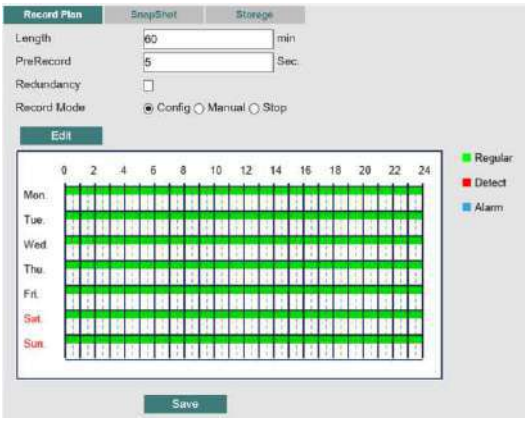
**5G:**

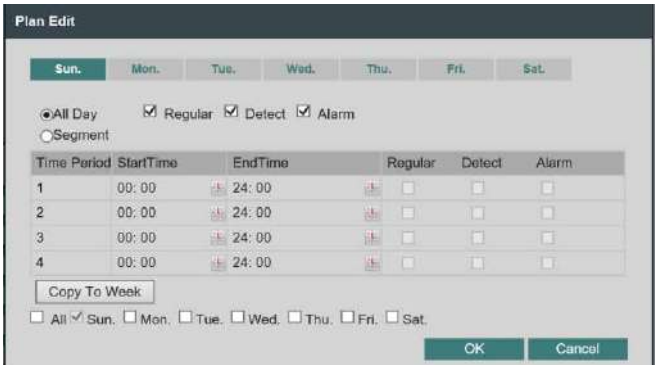
To enable 5G service of camera, enable DMZ service from router web GUI as below. Enter camera IP address and apply.




**Record:**

## 1. Record Plan:



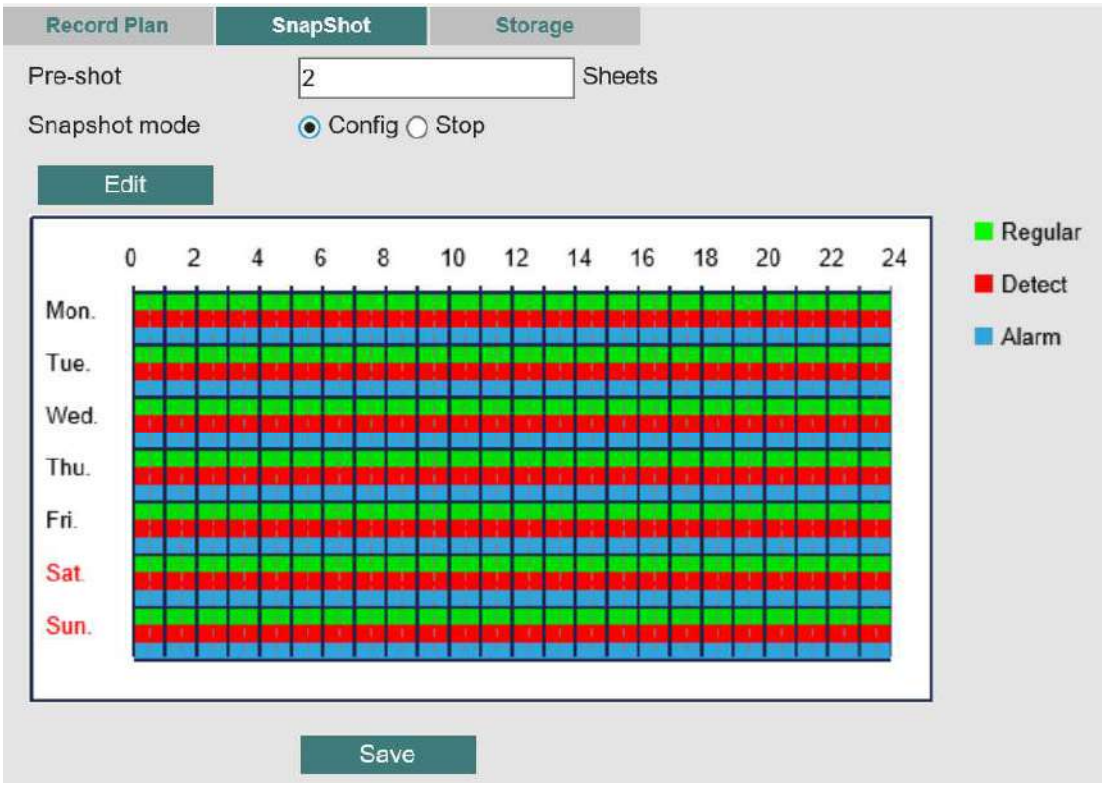

(i) Length is provided for what time the record should be require.

(ii) Pre record

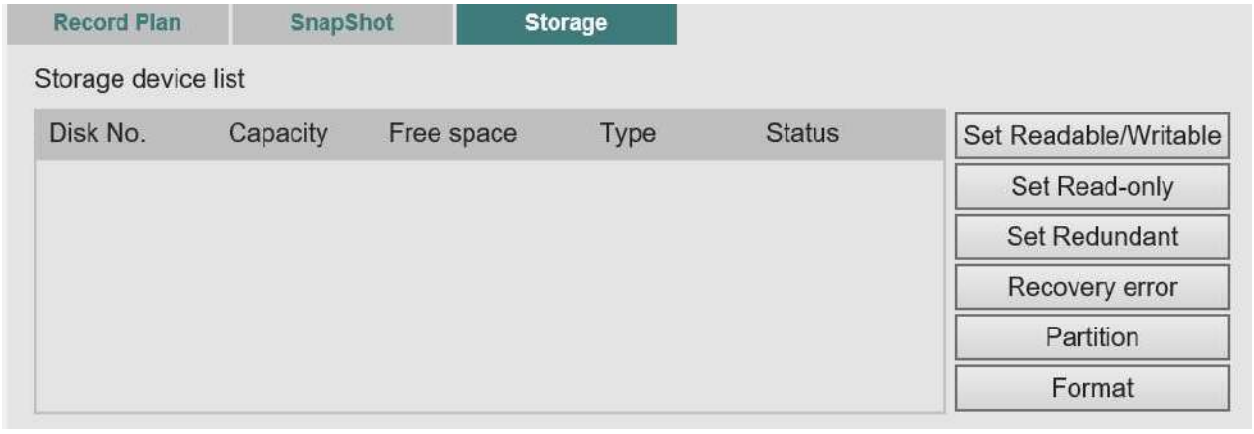
(iii)Redundancy is enable if we require unsual images for being record and increase to take more storage for recording.

(iv)Record mode: if we want to config it then it can, if manually then it can be set, if stop to record then it can be.

## 2. Snapshot:

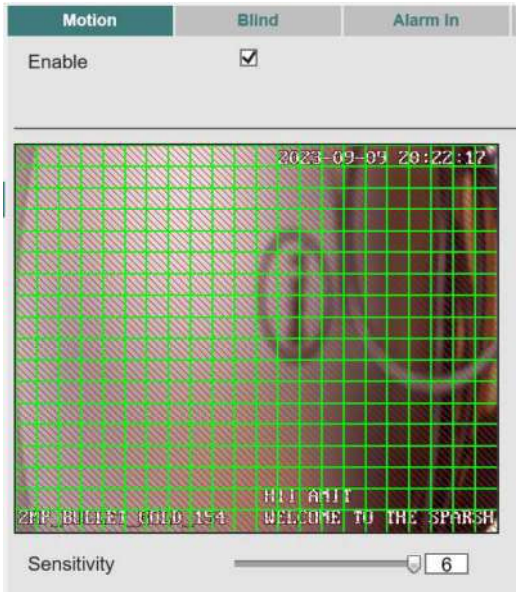

Record Plan
SnapShot
Storage
Pre-shot
2
Sheets
Snapshot mode
Config
Stop
Edit
0 2 4 6 8 10 12 14 16 18 20 22 24
Mon.
Tue.
Wed.
Thu.
Fri.
Sat.
Sun.
Regular
Detect
Alarm
Save

**3. Storage:**


Record Plan
SnapShot
Storage
Storage device list
Disk No.
Capacity
Free space
Type
Status
Set Readable/Writable
Set Read-only
Set Redundant
Recovery error
Partition
Format

**Alarm:**

## 1. Motion Detection:



In Motion Detection, the selected area is affected by any motion then the alert is come and buzzer beep.

Sensitivity tends to increase the power of senser to identify the motion.

Enable is used to enable the alarm.

This type of alert come when motion is happen .

## 2. Video Blind:

If the camera video will be tampered by any way then it give alert and play buzzer to remind the user of any threat.

| | | | |
|---|---|---|---|
| 5 | Video Tamper | 2023-09-11 16:25:07 | 1 |

This type of alert is come when it certain threat will happen.

3. **Alarm Input:**

same operation held in alarm input.

4. **Alarm Out:**

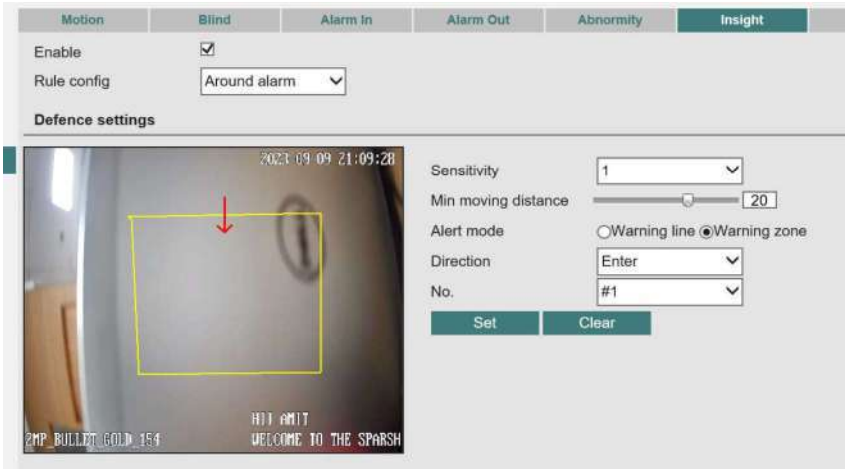
If we want to config the alarm then from here it will be done.

5. **Abnormity:**

If any abnormity happen in video of camera then it ll create a alert to the user.

6. **Insight:**

If we want to select an area by which any will through it then it ll give alert .



This type alert will be shown.

| | | | |
|---|---|---|---|
| 1 | Motion Detect | 2023-09-11 16:22:10 | 1 |
| 8 | Intelligent Alert | 2023-09-11 16:53:58 | 1 |

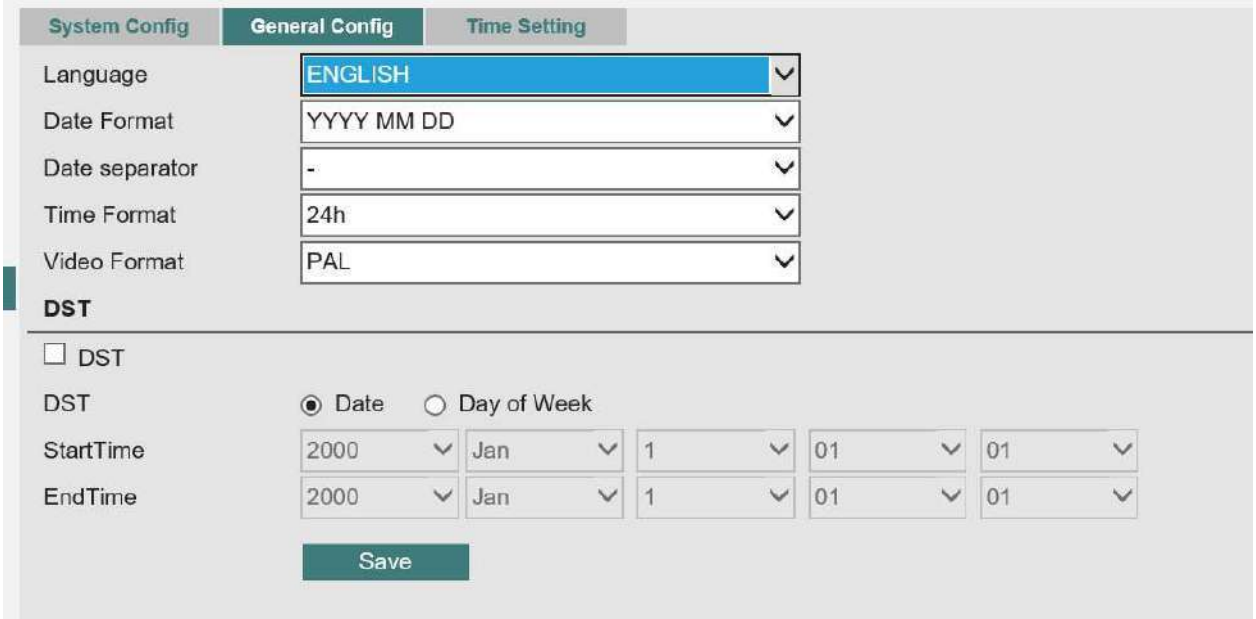
7. **Human:**

When the human motion is observed by the camera then it will also give alert to the user.

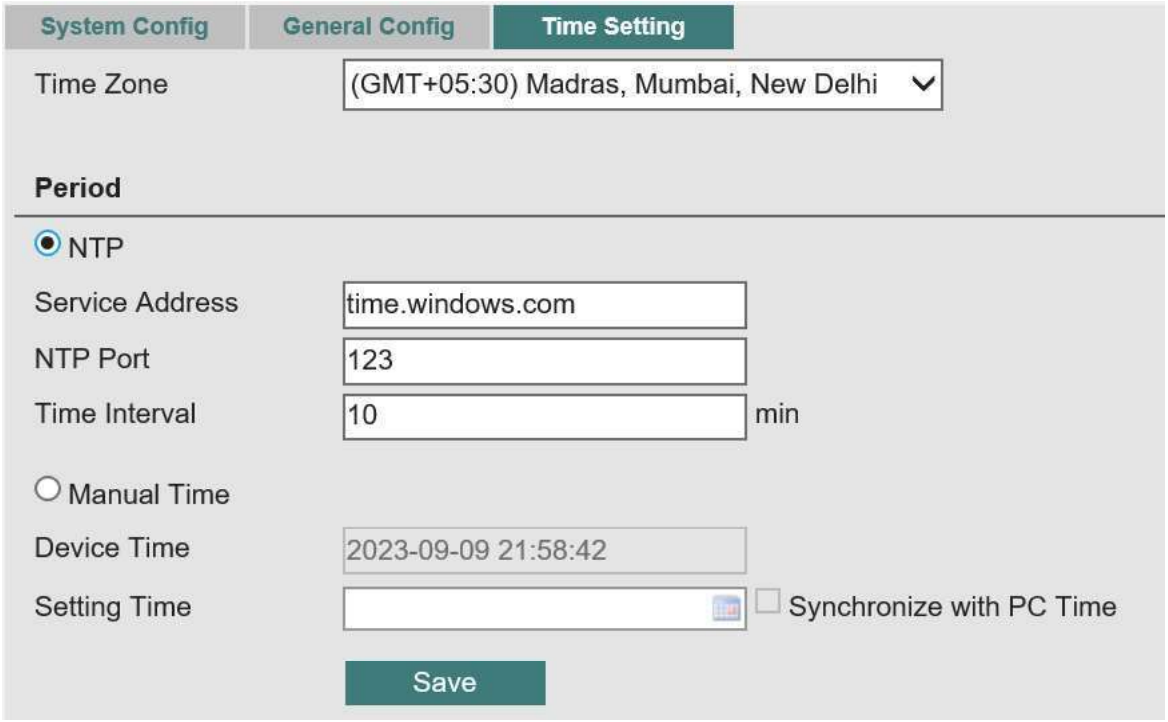
**System Config:**

1. **System Config:**

## 2. General Config:



### 3. Time Setting:




NTP is used for time management of the different time for different places.

Manual time is used to set time manually and if sync time with PC then time will be same as with PC.

**Security:**

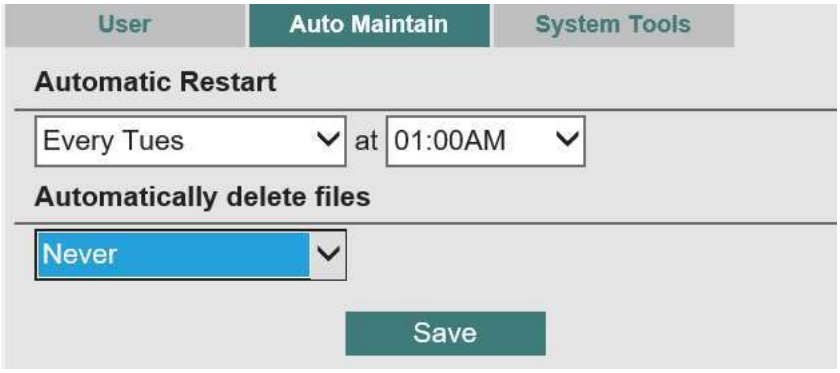
If we keep

**System Tool:**

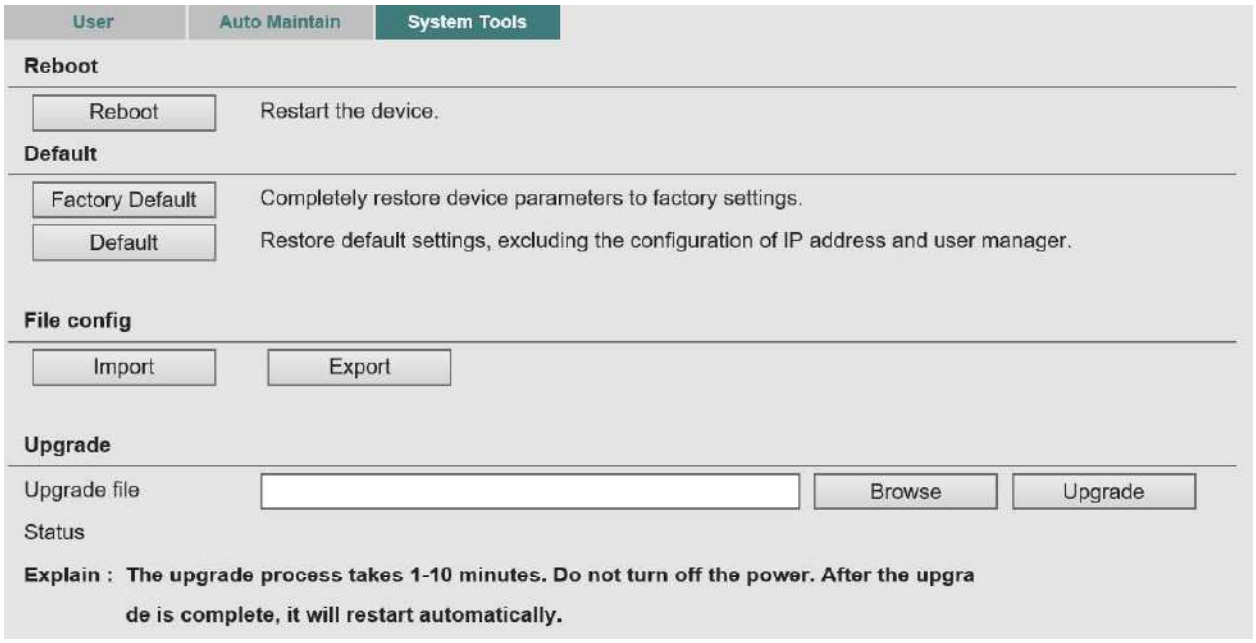
(i)If we want to assign any user for the access of camera then we can do it from here.

(ii)We can also add group from here for access of camera (as a user/admin).

(ii)Automatic restart from here for maintaining of camera and set the mode for deletion of files and it is customizable also.




System tool:




(i) Import: This function is used to import the file of camera from device.

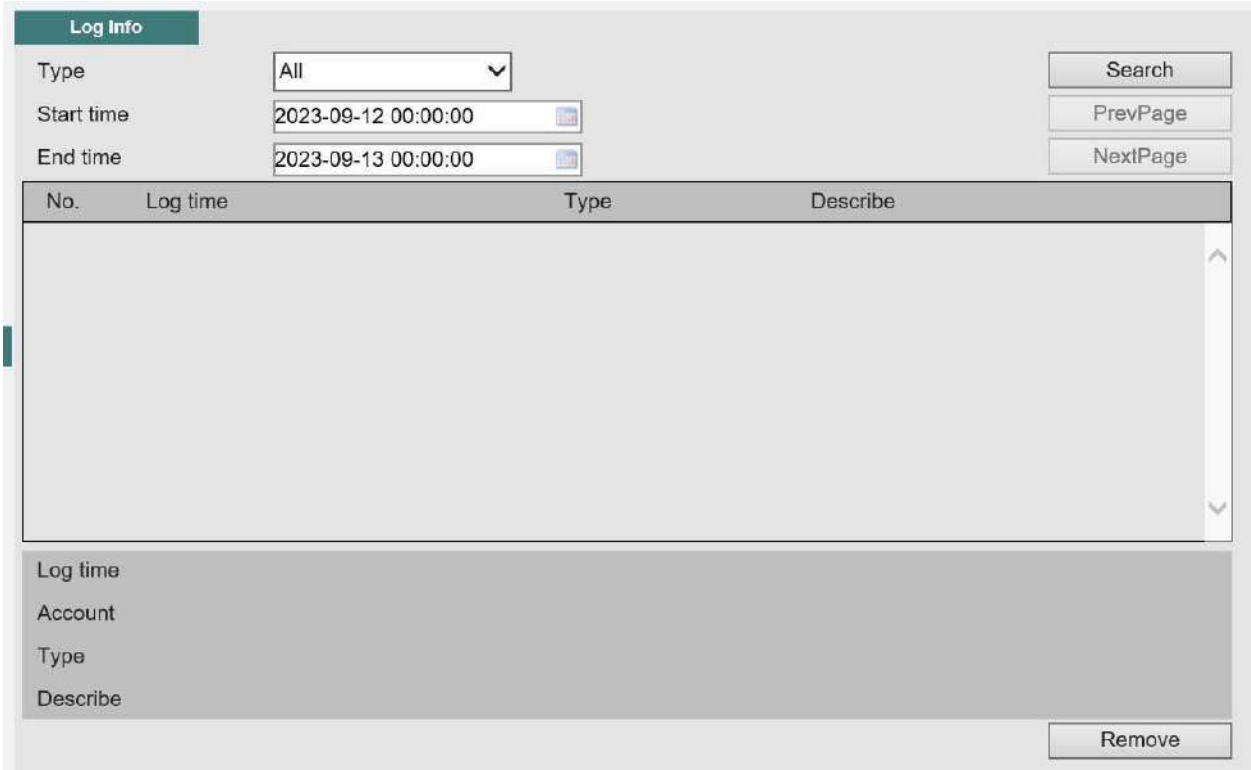
(ii) Export: This function is used to export file to a certain location (save a file of data for further use).

**System Info:**

VERSION INFO

**Log Info:**

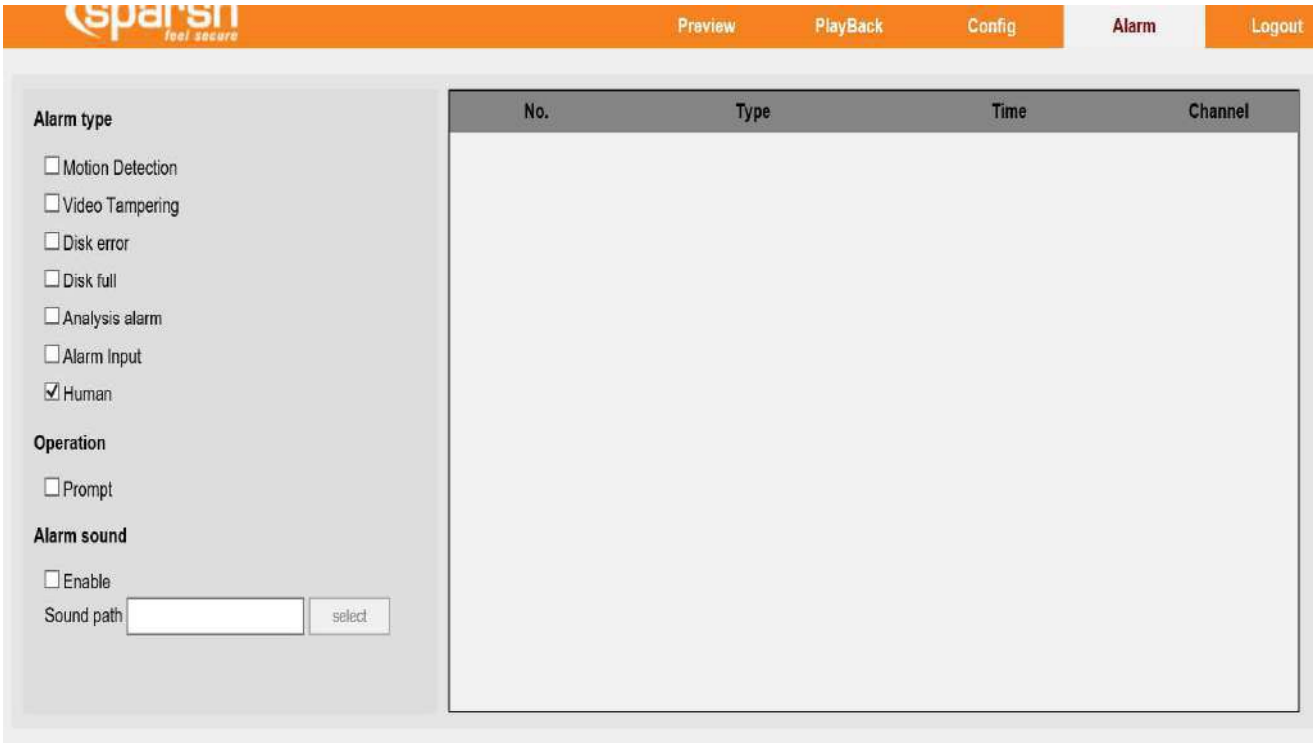
In this all the log info notify here.



# ALARM

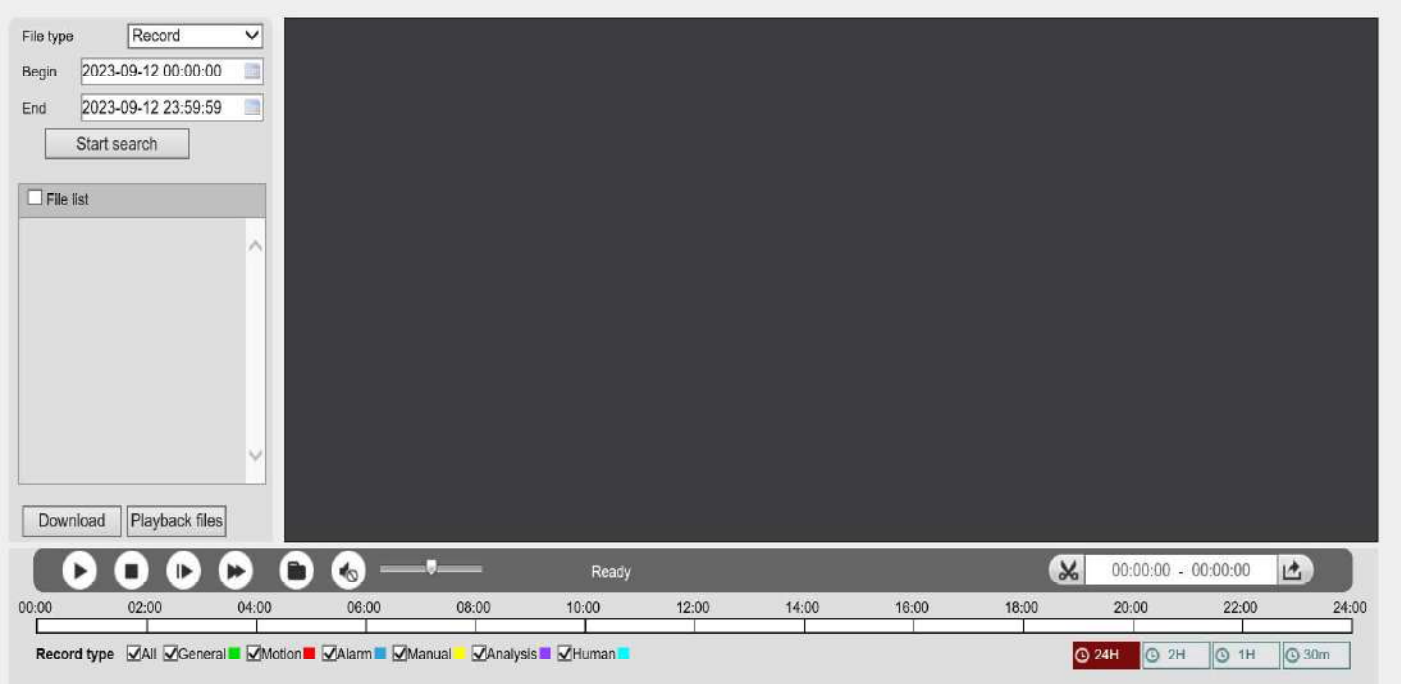
(i) All the notification of alert come here of all alarm type.

(ii)We can give the sound by apply sound path of sound file.



# PLAYBACK

Playback is used to watch recorded files.




| | | |
|---|---|---|
| Playback interface | | Select the type of playback file |
| | | Select the playback date, The default is the current date |

| | | |
|---|---|---|
| | | Display videos or pictures saved in the TF card |
| | | Download the video file to a local folder |
| | | Play the video files |
| | | Start playback |
| | | Stop playback |
| | | Slow playback |
| | | Fast playback |
| | | Play local video files |
| | | Sound Volume Up>Down and Mute |
| | | Show different recording categories |

# Evolution Board User Guide

# Overview

5G EVB can be used to develop applications based on Quectel 5G modules.
This EVB can be used to test basic functionalities of these modules.

## Key Features

The following table describes the detailed features of 5G EVB.

**Table : Key Features**

| | |
|---|---|
| Power Supply | DC power supply: 4.5–5.5 V<br>Typical: +5 V/ 3 A |
| Module TE-A Interface | Quectel 5G modules supported |
| PHY TE-A Interface | PHY AR8035 supported |
| Wi-Fi TE-A Interface | Reserved |
| AP TE-A Interface | Reserved |
| SD Card Interface | SD card connector |
| (U)SIM Card Interface | Dual (U)SIM card supported: 1.8 V and 2.95 V |
| Audio Interfaces | 1 digital audio codec board interface:<br>Supporting ALC5616-TE-A and TLV320AIC3104-TE-A codec boards<br>2 analog audio interfaces:<br>Used for loudspeaker and earphone |
| UART Interface | **COM1 (J2002):**<br>Main UART<br>For data communication<br>Default baud rate: 115200 bps<br>**COM2 (J2003):**<br>Debug UART<br>For debugging |

| | |
|---|---|
| | • Default baud rate: 115200 bps |
| USB Interface | • USB Type C interface<br>• USB 3.0 and USB 2.0 supported |
| PCIe to USB Interface | Reserved |
| Signal Indication | 5 LEDs are available for signal indication |
| Button and Switches | • PWRKEY (S0202)<br>• RESET (S0201)<br>• USB_BOOT (S0203)<br>• PCIe Configuration Switch (S1501)<br>• Power Switch (S0301)<br>• SDIO Configuration Switch (S2501)<br>• RGMII Configuration Switch (S1801/S1802)<br>• Codec Configuration Switch (S2801) |
| Antenna | 12 antenna interfaces |

# Interface Application

This chapter describes the hardware interfaces of the 5G EVB, as listed below:

- Power supply
- Module TE-A interface
- PHY TE-A interface
- USB interface
- Audio interfaces
  - Digital Audio Codec Board Connector
  - Analog Audio Interfaces
    - Loudspeaker Interface
    - Earphone Interface
- (U)SIM interfaces
- UART interfaces
- SD card interface
- PCIe to USB interface
- Switches and buttons
- Status indicators
- Wi-Fi interface
- Antenna Interfaces

It also provides information about the buttons, switches, status indication LEDs and test points to help developers use the EVB.

## 1. Power Supply (J0303)

The 5G EVB can be powered by an external power adapter through the power jack (J0303).

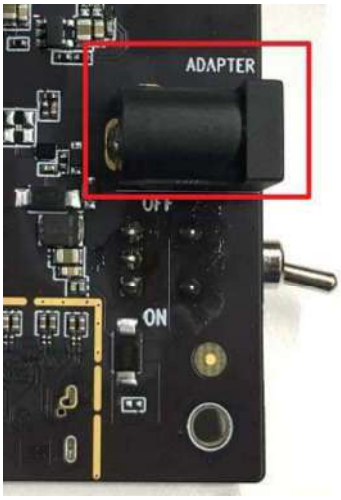



**Figure : EVB Power Supply Interface**

## 2. Module TE-A Interface (J0101/J0102)

Module TE-A interface is designed to accommodate the TE-A of the modules. The TE-A is mounted onto and connected to the EVB via BTB connectors J0101 and J0102. The developer will be able to test the functionalities of the modules easily (insert as indicated by the arrow to prevent reverse insertion).

The following figure displays the connection between the module TE-A and the EVB.

**Figure : Connection Between the Module TE-A and the EVB**

## 3. USB Interface (J1101)

A USB Type C connector, which complies with USB 3.0/3.1 and USB 2.0 standard, is provided. This USB interface is used for AT command communication, data transmission and firmware upgrade.

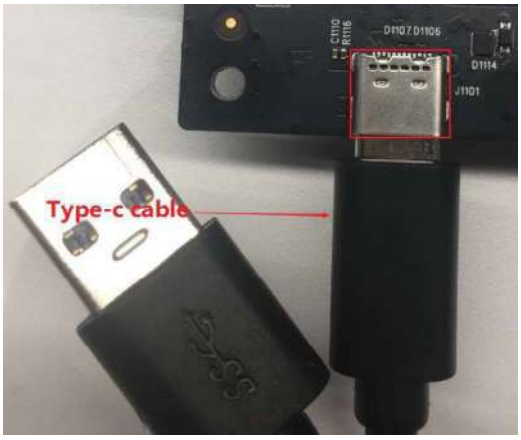



**Figure : USB Interface Connection**

## 4. Audio Interface (J0802/J0901/J0801)

EVB provides one digital audio codec board interface (I2S) J0802 and two analog audio interfaces J0901 and J0801.

### 4.1. Digital Audio Codec Board Connector (J0802)

The 5G EVB supports two different kinds of external digital audio codec TE-As named ALC5616 and TLV320AIC3104. The codec circuit is assembled on an independent small board which can be interconnected with the EVB by the BTB connector J0802.

Codecs can be selected according to specific application demands, the following figures show the connection between digital audio codec TE-A and the EVB.

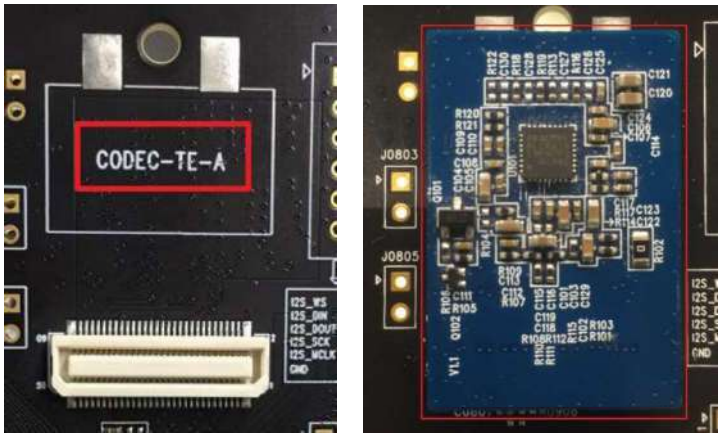



**Figure : Connection Between Codec TE-A and the EVB**

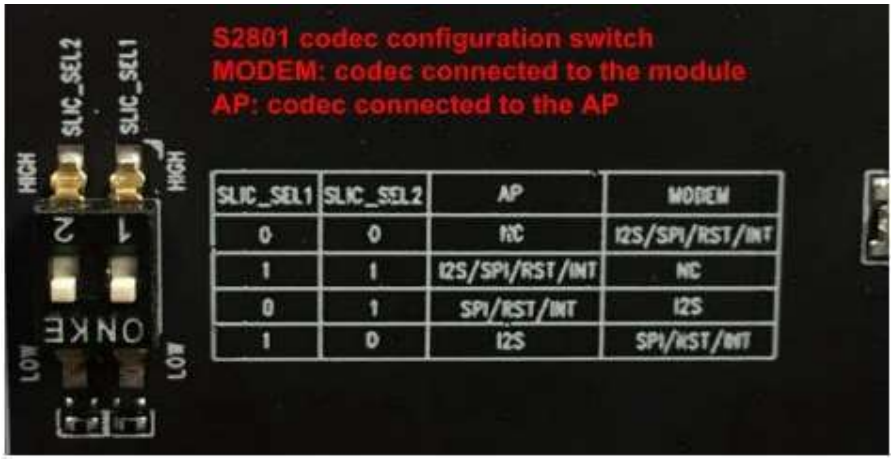



**Figure : S2801 Switch**

**Table : Codec TE-A Configuration Switch**

| | | | |
|---|---|---|---|
| Low | Low | NC | I2S/SPI/RST/INT |
| High | High | I2S/SPI/RST/INT | NC |
| Low | High | SPI/RST/INT | I2S |
| High | Low | I2S | SPI/RST/INT |

The figure and table below illustrates the pin assignment and pin definition of earphone connector J0901.

**Figure : Pin Assignment of J0901**

**Table : Pin Definition of J0901**

| | | |
|---|---|---|
| 1 | MIC | Microphone input |
| 2 | AGND | Dedicated GND for audio |
| 3 | SPK_R | Right channel of stereo audio output |
| 4 | SPK_L | Left channel of stereo audio output |
| 5, 6 | NC | Not connected |

The following figure shows a schematic of audio plug which suits the audio jack on 5G EVB.

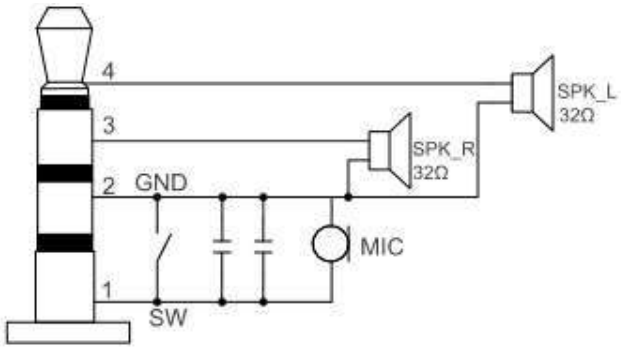



**Figure : Schematic of Audio Plug**

## 5. (U)SIM Card Interfaces (J1401/J1402)

The 5G EVB has two 8-pin push-push type (U)SIM card (1.8/2.95 V) connectors J1401 and J1402. The following figure shows a simplified connector schematic for J1401 and J1402.

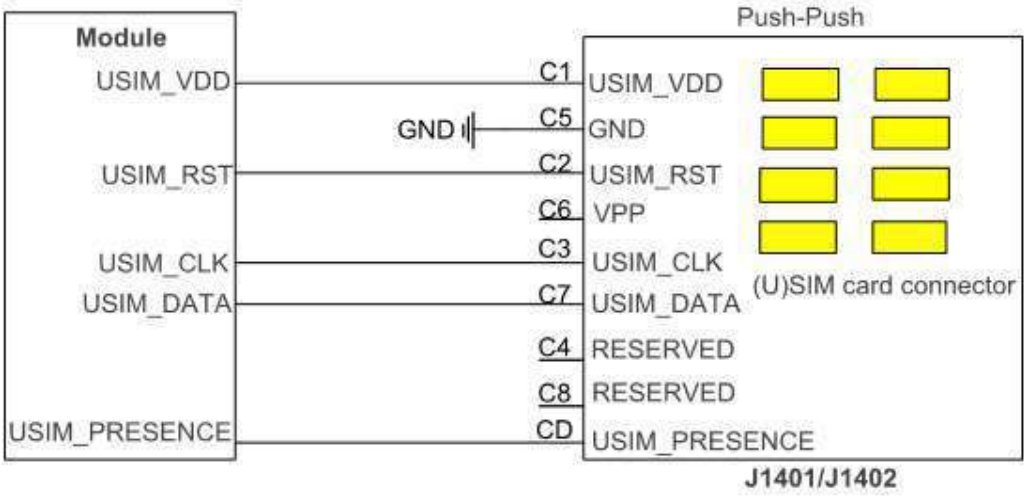



**Figure : Simplified Connector Schematic for (U)SIM Card Connectors**

The figure and table below illustrate the pin assignment and definition of (U)SIM card connector J1401. J1402 is similar to J1401.

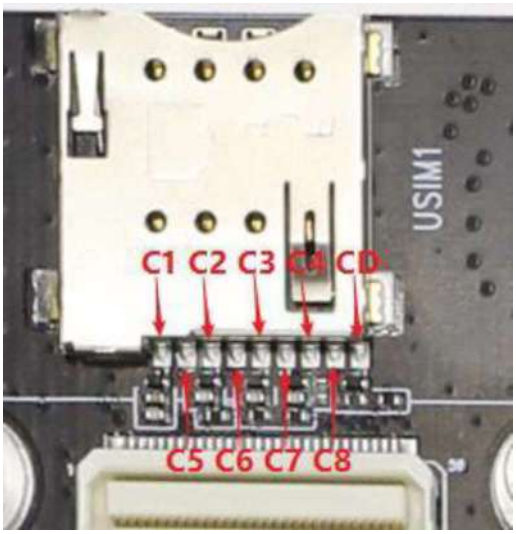



**Figure : Pin Assignment of (U)SIM Card Connector J1401**

**Table : Pin Definition of J1401**

| | | | |
|---|---|---|---|
| C1 | USIM_VDD | PO | U(SIM) card power supply, provided by module |
| C2 | USIM_RST | DO | U(SIM) card reset |
| C3 | USIM_CLK | DO | U(SIM) card clock |
| C4 | RESERVED | - | Not connected |
| C5 | GND | - | Ground |
| C6 | VPP | - | Not connected |
| C7 | USIM_DATA | I/O | Data line, bi-directional |
| C8 | RESERVED | - | Not connected |
| CD | USIM_PRESENCE | DI | U(SIM) card insertion detection. |

## 6. SD Card Interface (J1301)

The 5G EVB provides an SDIO interface, which can be used for connecting SD card. The following figure shows the simplified interface schematic for J1301.

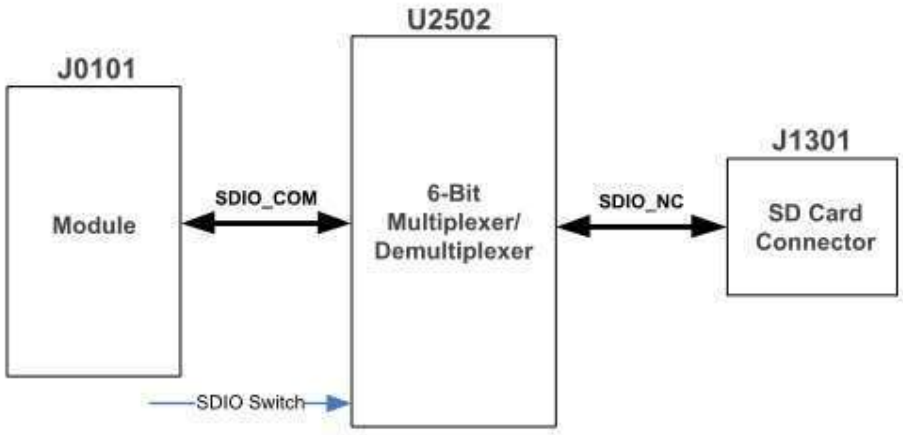



**Figure : Simplified Interface Schematic for J1301**

If SD card function is intended to be used, please switch the SDIO Switch to low level illustrated in the figure and table below, a standard SD card can be inserted into J1301. Which supports micro SD card

of maximal 32 GB. With the SD card interface, customers can easily enhance the memory capacity of modules.

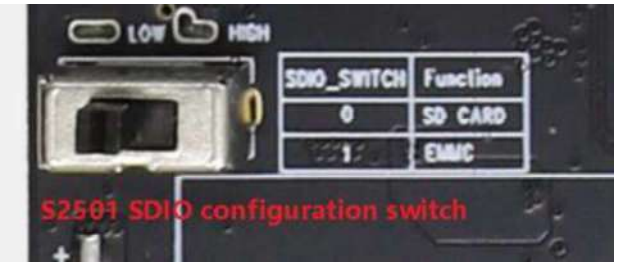



**Figure : S2501 Switch**

**Table : SDIO Switch Function**

| SDIO Switch | Function |
|---|---|
| Low | Enable SD card function |
| High | Enable eMMC function [1)] |

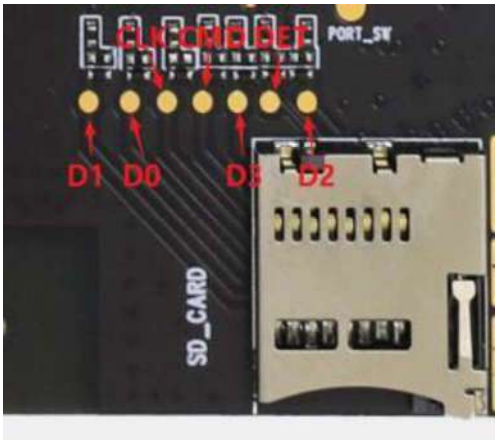



**Figure : Pin Assignment of SD Card Connector J1301**

## 7. UART Interfaces (J2002/J2003)

The 5G EVB supports two UART interfaces: main UART J2002 and debug UART J2003, supporting baud rate of 115200 bps by default.

The main UART interface is used for communication between the module and the host application.

The debug UART interface is used for Linux console and log output.

The following figure shows a block diagram of UART interfaces of the EVB.

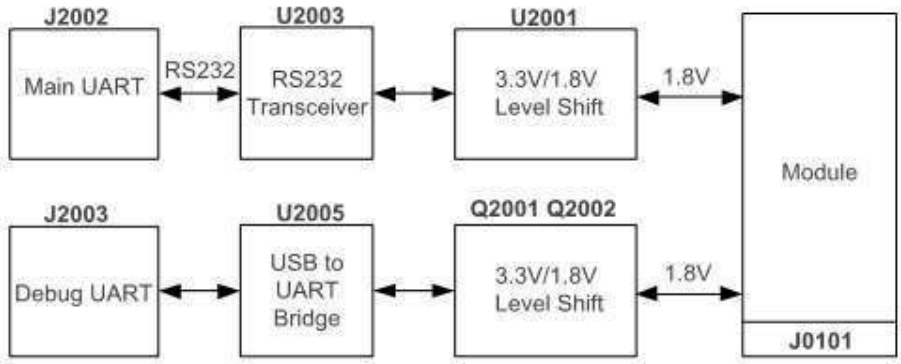



**Figure : UART Block Diagram**

## 8. PCIe to USB Interface (J1601)

The 5G EVB reserves a PCIe 3.0 signal over USB interface for developers' testing, and this function is not enabled by default. Please refer to the following block diagram.

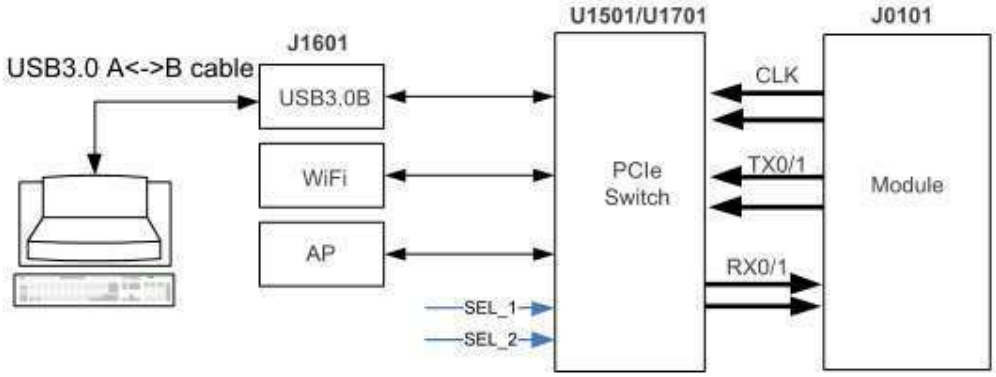



**Figure : PCIe Block Diagram**

**Figure : S1501 Switch**

**Table : PCIe Connection Truth Table**

| | | |
|---|---|---|
| 0 | 1 | Module → PC |
| 1 | 0 | Module → AP |
| 1 | 1 | Module → Wi-Fi |

## 9. Switches and Buttons

The 5G EVB includes six switches (S0301, S1501, S1801, S1802, S2501 and S2801) and three buttons (S0201, S0202 and S0203), as shown in the following figures.

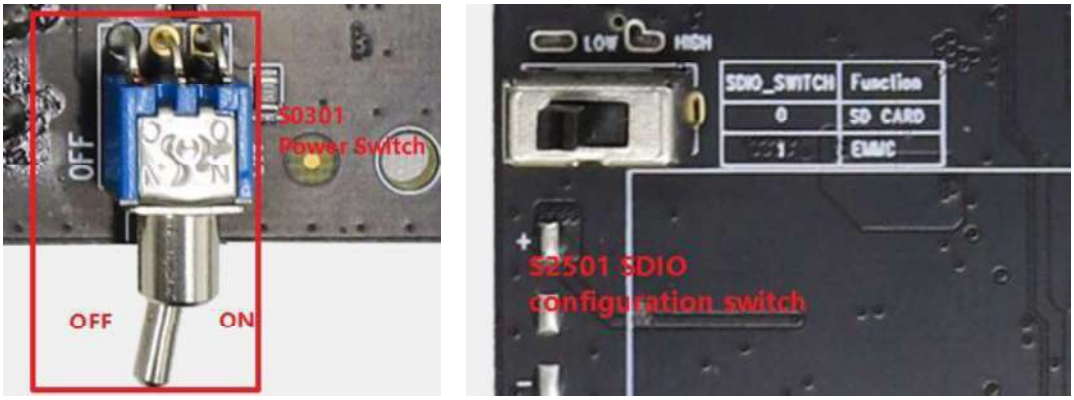



**Figure : Power Switch and S2501 Switch**

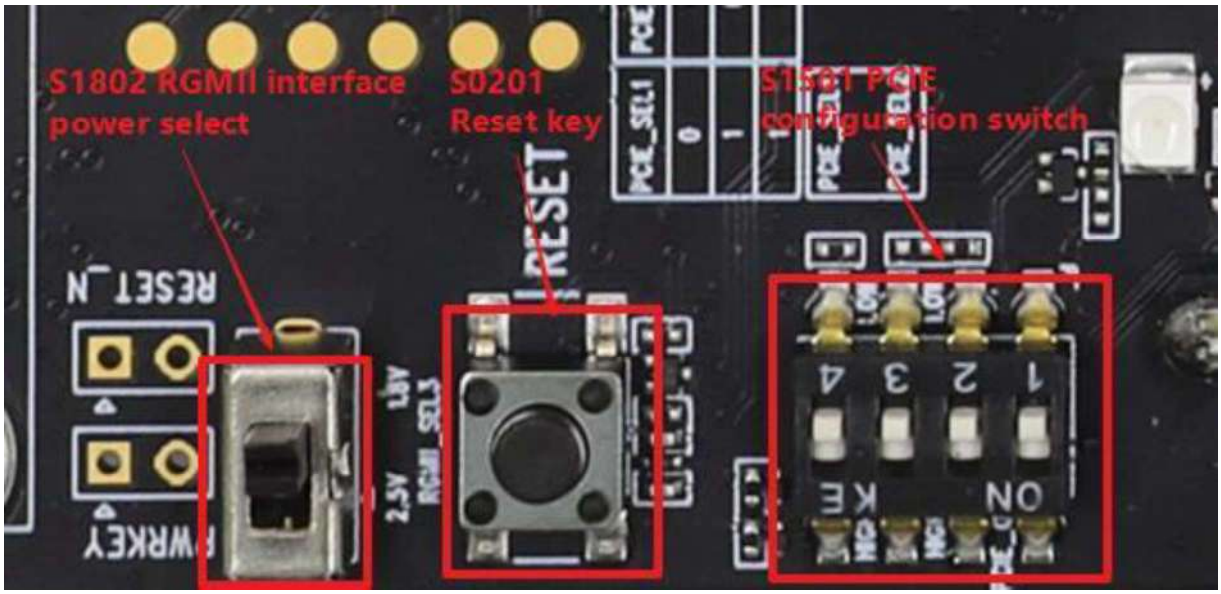



Figure : S1802/S1501 Switch and S0201 Button

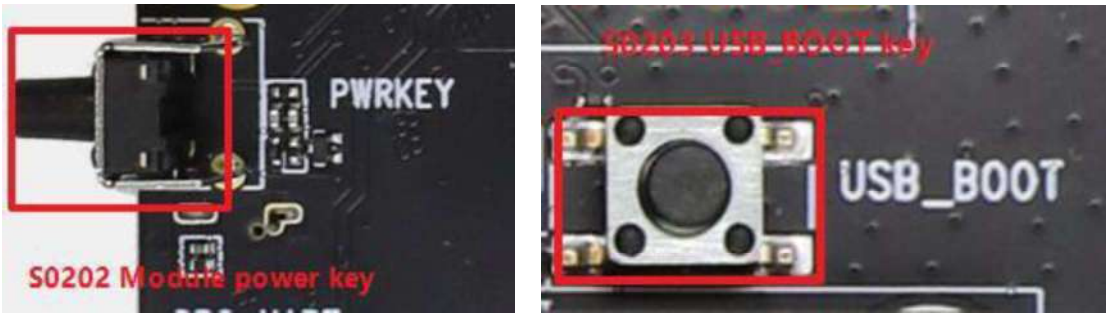



Figure : S0202/S0203 Button

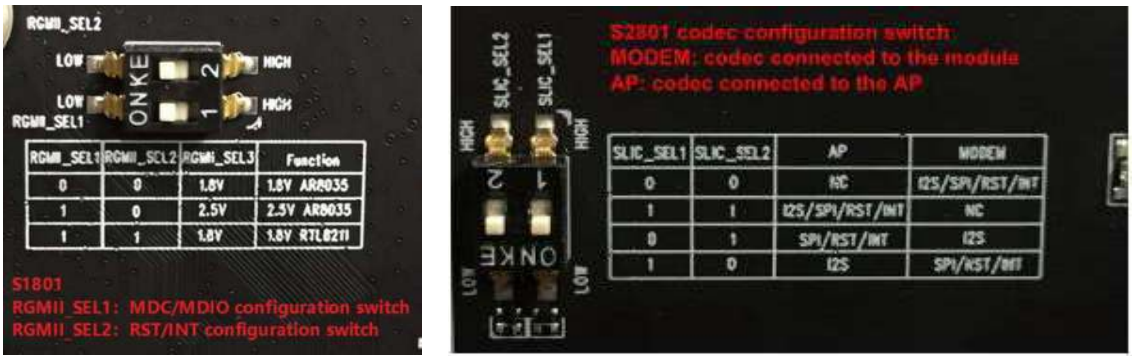



Figure : S1801/S2801 Switch

**Table : Description of Switches and Buttons**

| | | |
|---|---|---|
| Power Switch | S0301 | VBAT ON/OFF control |
| PWRKEY | S0202 | Power key used to turn on/off the module. |
| PCIe Configuration | S1501 | PCIe configuration switch. Refer to ***Chapter 3.9*** |
| RESET | S0201 | Reset button used to reset the module. |
| USB_BOOT | S0203 | Emergency download control. |
| RGMII Configuration | S1801, S1802 | RGMII configuration switch. Refer to ***Chapter 3.3*** |
| SDIO Configuration | S2501 | SDIO configuration switch. Refer to ***Chapter 3.7*** |
| Codec Configuration | S2801 | Codec configuration switch. Refer to ***Chapter 3.5*** |

## 10. Status Indicators (D0201/D0202/D0203/D0204/D0205)

There are five status indication LEDs on the EVB. The following figure manifests the positions of these LED indicators.

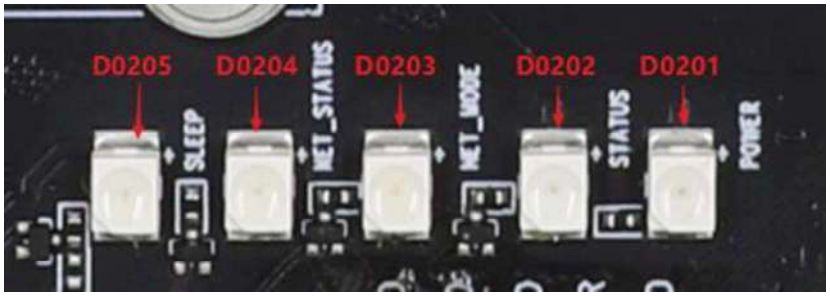



**Figure : Status Indicator**

**Table : Description of Status Indication LEDs**

| | |
|---|---|
| D0201 | Indicates whether the power supply for module is ready.<br>On: VBAT ON<br>Off: VBAT OFF |

| | |
|---|---|
| D0202 | Indicates the operation status of the module.<br>On: the module is powered on<br>Off: the module is powered off |
| D0203 | Indicate the module's NET_MODE status. |
| D0204 | Indicate the module's NET_STATUS status. |
| D0205 | Indicate the module's SLEEP status. |

## 11. Wi-Fi Interfaces (J0701/J0702)

The Wi-Fi TE-A interface is designed to accommodate the TE-A of Wi-Fi modules (paired with FG50V). The TE-A is connected to the EVB via BTB connectors J0701 and J0702. The interface allows customers to test the Wi-Fi function of the module or to develop applications with Wi-Fi function easily.

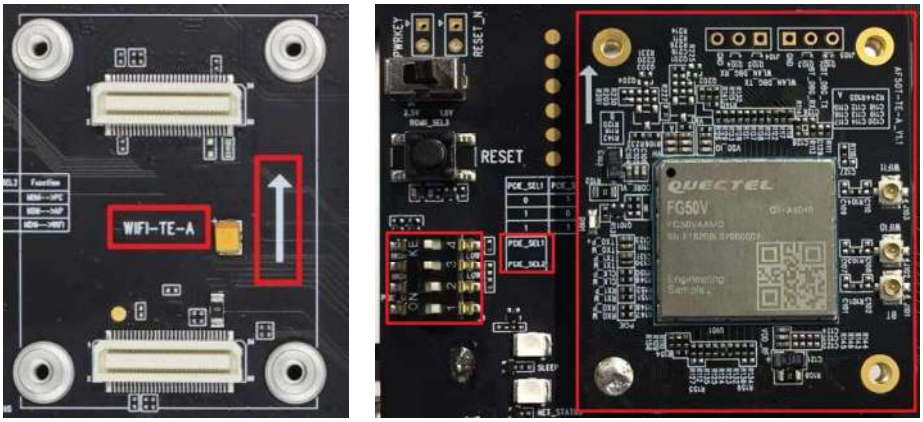



**Figure : Connection Between FG50V-TE-A and the EVB**

## 12. Antenna Interfaces

The 5G EVB includes twelve antenna interfaces. The following figure shows the assembly of these antenna interfaces.

**Figure : Antenna Interfaces**

# EVB Operation Procedures

This chapter introduces how to use the 5G EVB for testing and evaluation.

## 1. Turn on the Module

1. Connect the module TE-A to the EVB via connectors J0101 and J0102.
2. Insert a (U)SIM card into the USIM1 card connector on EVB.
3. Use RF cable to connect the module TE-A to the EVB, and connect antennas to the EVB.
4. Connect the EVB to a 5 V/ 3 A power, then switch S0301 to ON. Then D0201 (ON/OFF indicator of the module's power supply) will light up.
5. Press the S0202 (PWRKEY) for at least 500ms, then the module will be powered on and D0202 (operation indicator of the module) will light up.

## 2. Communication via USB

1. Power on the module according to the procedure in ***Chapter 1***.
2. Connect the EVB and a PC with USB cable through USB Type-C interface, and then run the driver disk on the PC to install the USB driver. The USB port numbers can be viewed in Device Manager of the PC when the USB
driver is installed, as shown below.

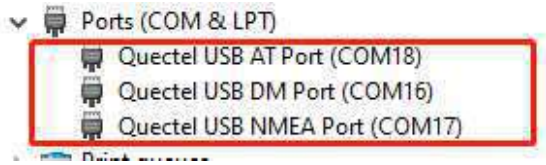



**Figure : USB Ports**

# TECHNICAL MANUAL

## IOT SOLUTION

| Details | |
|---|---|
| Client Details | VVDN Technologies |
| Contact Person | Mr. Sonu |
| Prepared By | ETE Solutions |

# QUESTIONS? CONTACT US.

+91-9717702842

support@etesolutionsindia.com

# TECHNICAL DOCUMENT

## ABOUT PROJECT

In this project, we're leveraging the power of 5G connectivity alongside a variety of **IoT sensors** to revolutionize environmental testing. With the help of our cutting-edge technology we have enhanced the accuracy and efficiency of our testing processes. By integrating IoT sensors into the labs and utilizing the high-speed, low-latency capabilities of 5G networks, we aim to elevate the quality of environmental testing across air, water, and soil domains. This innovative approach not only ensures compliance with industrial standards but also facilitates real-time monitoring and analysis for proactive environmental management.

## TECHNOLOGY USED

### MQTT (Message Queuing Telemetry Transport)

- It facilitates efficient communication between IoT devices and the central monitoring system.
- It enables real-time data exchange, critical for accurate environmental sensing.

### IoT Cloud Broker:

- An IoT Cloud Broker acts as an intermediary between IoT devices and cloud services.
- It enables centralized management and monitoring of IoT devices deployed across different locations.
- It provides seamless integration of IoT data streams with cloud-based analytics and storage platforms.

## SENSOR USED IN THIS PROJECT

- **NPK Soil Sensor**
- **TDS Sensor**
- **Temperature & Humidity Sensor**
- **Light Intensity Sensor**

# SENSOR SPECIFICATION

## LIGHT INTENSITY SENSOR

- Chip: BH1750FVI
- Power Supply: 3.3V - 5V
- Light Range：0 - 65535 lx(Lux)
- Sensor Built-in: 16 bit AD converter
- Direct digital output, bypassing the complex calculation, bypassing the calibration
- Close to the spectral characteristics of visual
- Widely used to 1-lux high precision measurement
- Standard NXP IIC communication

## TDS SENSOR

| | |
|---|---|
| Input Voltage | 3.3 ~ 5.5V |
| Output Voltage | 0 ~ 2.3V |
| Working Current | 3 ~ 6mA |
| TDS Measurement Range | 0 ~ 1000ppm |
| TDS Measurement Accuracy | ± 10% F.S. (25 °C) |
| Module Interface | PH2.0-3P |
| Electrode Interface | XH2.54-2P |

## NPK SOIL SENSOR

- Power supply: 5V Only
- Maximum power consumption: ≤0.15W
- Operating temperature: -40~80°C
- NPK parameters: Range: 0-1999 mg/kg(mg/L)
- Resolution: 1 mg/kg(mg/L)
- Precision: ±2%FS
- Response time: ≤1S
- Protection grade: IP68
- Probe material: 316 stainless steel

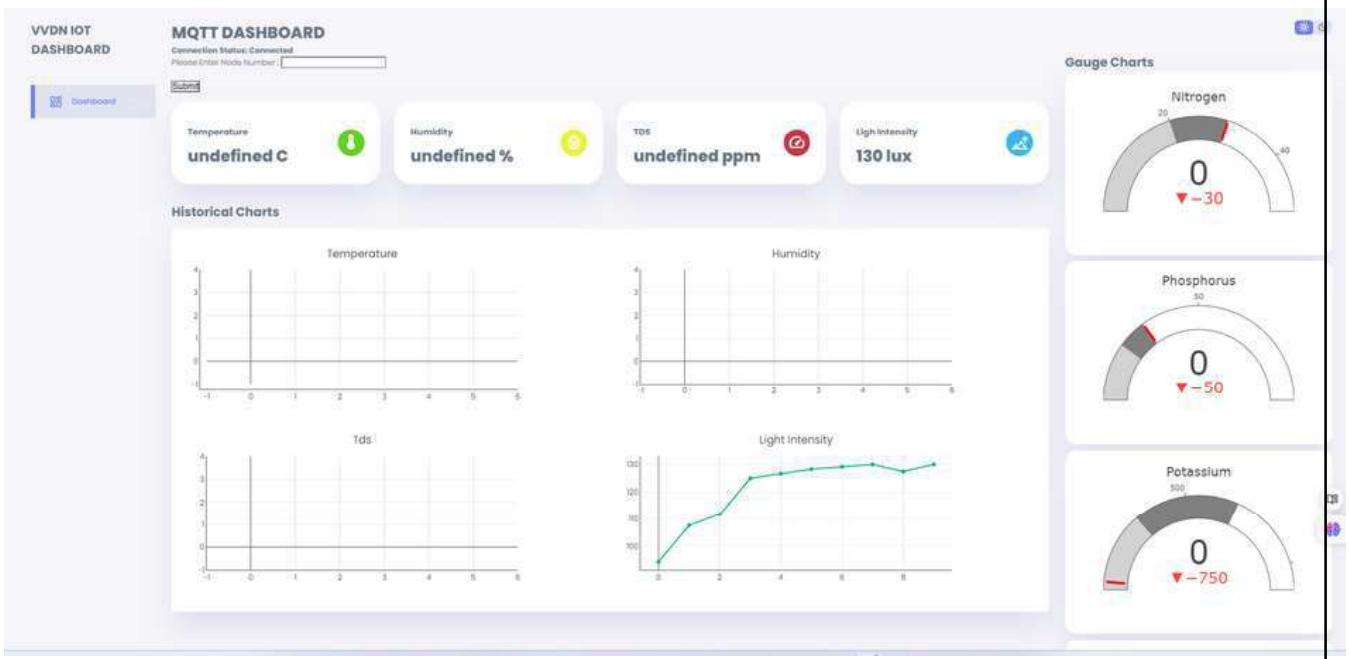
## TEMPERATURE & HUMIDITY SENSOR

- Operating Voltage: 5V
- Temperature Range: -40 to 125 C
- Humidity Range: 0~100%rh
- Probe Material: Plastic + Metal
- Typ. Temperature Accuracy: 0.3C
- Typ. Relative Humidity Accuracy:3 %RH

# DASHBOARD

## STEP 1

- Open url: **https://vvdn-dashboard.onrender.com**
- Enter the node of the sensor mentioned on sensors
- Keep same node at same site example: All sensor should have same node mentioned on there box



## 5G CPE SETUP:

- First, set the Wi-Fi credentials of TELTONIKA 5g Cpe to default
- SSID: RUT_D8B8_2G
- Password: Ec72Qsp4
- As soon as sensor find the wifi it automatically connect and send data to cloud MQTT broker

# DEBUG

- Connect sensor to pc and open tera term(window) or minicom (linux) and connect on serial (port. baudrate: 9600 [115200 for tds]) and check the log.
- If not able to connect to internet it will show connecting to wifi else sesnor reading will shown.
- Connect our team at support@etesolutionsindia.com for resolution.

TELTONIKA | Networks

# RUTX50

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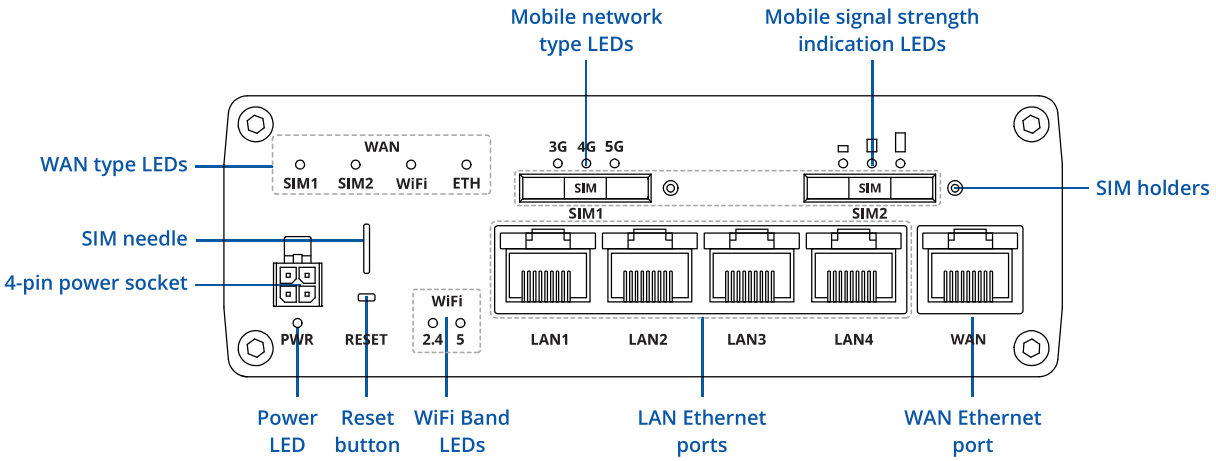
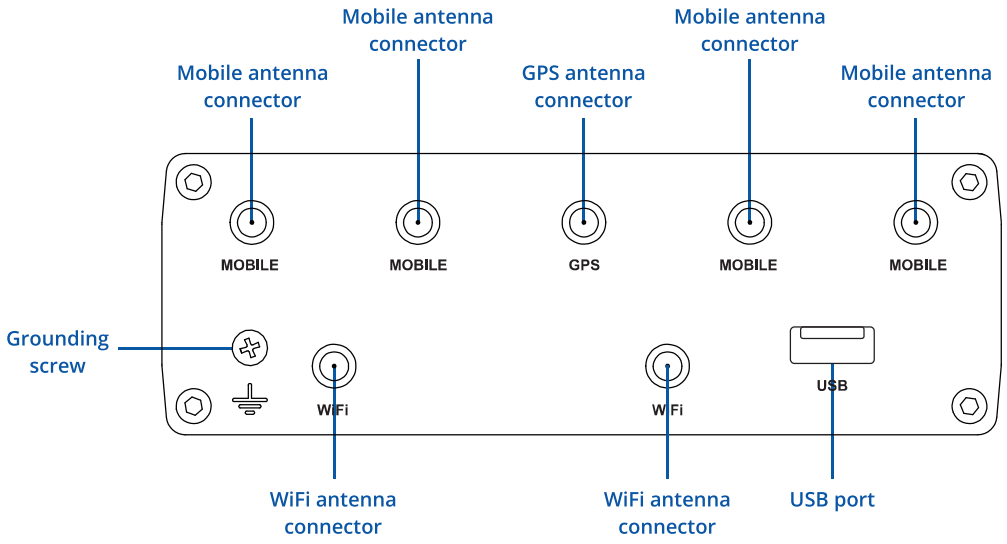
# HARDWARE

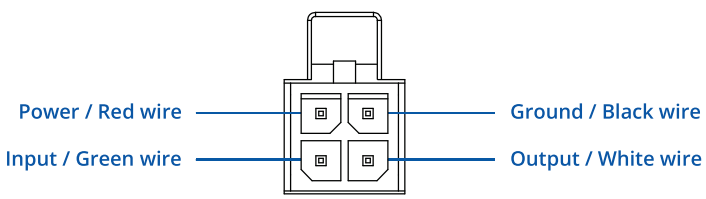
## FRONT VIEW




## BACK VIEW




## POWER SOCKET PINOUT



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# FEATURES

## MOBILE

| | |
|---|---|
| Mobile module | 5G Sub-6Ghz SA/NSA 2.1/3.3Gbps DL (4x4 MIMO), 900/600 Mbps UL (2x2); 4G (LTE) – LTE Cat 20 2.0Gbps DL, 200Mbps UL; 3G – 42 Mbps DL, 5.76Mbps UL |
| Status | Signal strength (RSSI), SINR, RSRP, RSRQ, EC/IO, RSCP Bytes sent/received, connected band, IMSI, ICCID. |
| SMS/Call | SMS status, SMS configuration, send/read SMS via HTTP POST/GET, EMAIL to SMS, SMS to EMAIL, SMS to HTTP, SMS to SMS, scheduled SMS, SMS autoreply, Call utilities |
| USSD | Supports sending and reading Unstructured Supplementary Service Data messages |
| Black/White list | Operator black/white list |
| Band management | Band lock, Used band status display |
| APN | Auto APN |
| Bridge | Direct connection (bridge) between mobile ISP and device on LAN |
| Passthrough | Gateway assigns its mobile WAN IP address to another device on LAN |
| Multiple PDN | Possibility to use different PDNs for multiple network access and services |

## WIRELESS

| | |
|---|---|
| Wireless mode | 802.11b/g/n/ac Wave 2 (WiFi 5) with data transmission rates of up to 867 Mbps (Dual Band, MU-MIMO), 802.11r fast transition, Access Point (AP), Station (STA) |
| WiFi security | WPA3-EAP, WPA3-SAE, WPA2-Enterprise-PEAP, WPA2-PSK, WEP; AES-CCMP, TKIP, Auto Cipher modes, client separation |
| ESSID | ESSID stealth mode |
| WiFi users | Up to 150 simultaneous connections |
| Wireless Hotspot | Captive portal (Hotspot), internal/external Radius server, built in customizable landing page |
| Wireless mesh/ roaming | Wireless mesh (802.11s), fast roaming (802.11r) |

## ETHERNET

| | |
|---|---|
| WAN | 1 x WAN port (can be configured as LAN) 10/100/1000 Mbps, compliance with IEEE 802.3, IEEE 802.3u, 802.3az standards, supports auto MDI/MDIX crossover |
| LAN | 4 x RJ45 port, 10/100/1000 Mbps, supports auto MDI/MDIX crossover |

## NETWORK

| | |
|---|---|
| Network protocols | TCP, UDP, IPv4, IPv6, ICMP, NTP, DNS, HTTP, HTTPS, SMTP, SSL v3, TLS, PPP, PPPoE, SSH, DHCP, SNMP, MQTT, Wake on LAN (WOL) |
| Routing | Static routing, Dynamic routing (BGP, OSPF v2, RIP v1/v2, EIGRP, NHRP) |
| Connection monitoring | Ping Reboot, Wget Reboot, Periodic Reboot, LCP and ICMP for link inspection |
| Firewall | Port forwards, traffic rules, custom rules |
| DHCP | Static and dynamic IP allocation |
| DDNS | Supported >25 service providers, others can be configured manually |
| SSHFS | Possibility to mount remote file system via SSH protocol |

## SECURITY

| | |
|---|---|
| Authentication | Pre-shared key, digital certificates, X.509 certificates |
| Firewall | Pre-configured firewall rules can be enabled via the WebUI, unlimited firewall configuration via CLI; NAT; NAT-T |
| Attack prevention | DDOS prevention (SYN flood protection, SSH attack prevention, HTTP/HTTPS attack prevention), port scan prevention (SYN-FIN, SYN-RST, X-mas, NULL flags, FIN scan attacks) |
| VLAN | Port and tag-based VLAN separation |
| Mobile quota control | Set up custom data limits for SIM card |
| WEB filter | Blacklist for blocking out unwanted websites, Whitelist for specifying allowed sites only |
| Access control | Flexible access control of TCP, UDP, ICMP packets, MAC address filter |



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## VPN

| | |
|---|---|
| OpenVPN | Multiple clients and a server can run simultaneously, 12 encryption methods |
| OpenVPN Encryption | DES-CBC, RC2-CBC, DES-EDE-CBC, DES-EDE3-CBC, DESX-CBC, BF-CBC, RC2-40-CBC, CAST5-CBC, RC2-64-CBC, AES-128-CBC, AES-192-CBC, AES-256-CBC |
| IPsec | IKEv1, IKEv2, supports up to 5 x VPN IPsec tunnels (instances), with 5 encryption methods (DES, 3DES, AES128, AES192, AES256) |
| GRE | GRE tunnel |
| PPTP, L2TP | Client/Server services can run simultaneously, L2TPv3 support |
| SSTP | SSTP client instance support |
| STUNNEL | Proxy designed to add TLS encryption functionality to existing clients and servers without any changes in the program's code |
| DMVPN | Method of building scalable IPsec VPNs |
| WireGuard | WireGuard VPN client and server support |
| ZeroTier | ZeroTier VPN |

## MODBUS TCP SLAVE

| | |
|---|---|
| ID filtering | Respond to one ID in range [1;255] or any |
| Allow remote access | Allow access through WAN |
| Custom registers | MODBUS TCP custom register block, which allows to read/write to a file inside the router, and can be used to extend MODBUS TCP slave functionality |

## MODBUS TCP MASTER

| | |
|---|---|
| Supported functions | 01, 02, 03, 04, 05, 06, 15, 16 |
| Supported data formats | 8 bit: INT, UINT; 16 bit: INT, UINT (MSB or LSB first); 32 bit: float, INT, UINT (ABCD (big-endian), DCBA (little-endian), CDAB, BADC), HEX, ASCII |

## MQTT GATEWAY

| | |
|---|---|
| Gateway | Allows sending commands and receiving data from MODBUS Master through the MQTT broker |

## DNP3

| | |
|---|---|
| Supported modes | TCP Master, DNP3 Outstation |

## DATA TO SERVER

| | |
|---|---|
| Protocols | HTTP(S), MQTT, Azure MQTT, Kinesis |

## IoT PLATFORMS

| | |
|---|---|
| Clouds of things | Allows monitoring of: Device data, Mobile data, Network info, Availability |
| ThingWorx | Allows monitoring of: WAN Type, WAN IP, Mobile Operator Name, Mobile Signal Strength, Mobile Network Type |
| Cumulocity | Allows monitoring of: Device Model, Revision and Serial Number, Mobile Cell ID, ICCID, IMEI, Connection Type, Operator, Signal Strength, WAN Type and IP |
| Azure IoT Hub | Can send device IP, Number of bytes send/received, Mobile connection state, Network link state, IMEI, ICCID, Model, Manufacturer, Serial, Revision, IMSI, SIM State, PIN state, GSM signal, WCDMA RSCP, WCDMA EC/IO, LTE RSRP, LTE SINR, LTE RSRQ, CELL ID, Operator, Operator number, Connection type, Temperature, PIN count to Azure IoT Hub server |

## MONITORING & MANAGEMENT

| | |
|---|---|
| WEB UI | HTTP/HTTPS, status, configuration, FW update, CLI, troubleshoot, event log, system log, kernel log |
| FOTA | Firmware update from server, automatic notification |
| SSH | SSH (v1, v2) |
| SMS | SMS status, SMS configuration, send/read SMS via HTTP POST/GET |
| Call | Reboot, Status, Mobile data on/off, Output on/off |
| TR-069 | OpenACS, EasyCwmp, ACSLite, tGem, LibreACS, GenieACS, FreeACS, LibCWMP, Friendly tech, AVSystem |
| MQTT | MQTT Broker, MQTT publisher |
| SNMP | SNMP (v1, v2, v3), SNMP trap |
| JSON-RPC | Management API over HTTP/HTTPS |
| Modbus | Modbus TCP status/control |
| RMS | Teltonika Remote Management System (RMS) |



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## SYSTEM CHARACTERISTICS

| | |
|---|---|
| CPU | Quad-core ARM Cortex A7, 717 MHz |
| RAM | 256 MB (100 MB available for userspace) |
| FLASH storage | 256 MB (80 MB available for userspace) |

## FIRMWARE / CONFIGURATION

| | |
|---|---|
| WEB UI | Update FW from file, check FW on server, configuration profiles, configuration backup |
| FOTA | Update FW |
| RMS | Update FW/configuration for multiple devices at once |
| Keep settings | Update FW without losing current configuration |

## FIRMWARE CUSTOMIZATION

| | |
|---|---|
| Operating system | RutOS (OpenWrt based Linux OS) |
| Supported languages | Busybox shell, Lua, C, C++ |
| Development tools | SDK package with build environment provided |

## LOCATION TRACKING

| | |
|---|---|
| GNSS | GPS, GLONASS, BeiDou, Galileo |
| Coordinates | GNSS coordinates via WebUI, SMS, TAVL, RMS |
| NMEA | NMEA 0183 |
| Server software | Supports server software: TAVL, RMS |
| Geofencing | Multiple configurable geofence zones |

## USB

| | |
|---|---|
| Data rate | USB 2.0 |
| Applications | Samba share, USB-to-serial, Modbus gateway |
| External devices | Possibility to connect external HDD, flash drive, additional modem, printer |
| Storage formats | FAT, FAT32, NTFS |

## INPUT/OUTPUT

| | |
|---|---|
| Input | 1 x Digital Input, 0 - 6 V detected as logic low, 8 - 30 V detected as logic high |
| Output | 1 x Digital Output, Open collector output, max output 30 V, 300 mA |
| Events | SMS, Email, RMS |
| I/O juggler | Allows setting certain I/O conditions to initiate event |

## POWER

| | |
|---|---|
| Connector | 4-pin industrial DC power socket |
| Input voltage range | 9 – 50 VDC, reverse polarity protection, surge protection >51 VDC 10us max |
| PoE (passive) | Passive PoE. Possibility to power up through LAN1 port , not compatible with IEEE802.3af, 802.3at and 802.3bt standards |
| Power consumption | Idle: <4 W, Max: <18 W |

## PHYSICAL INTERFACES (PORTS, LEDS, ANTENNAS, BUTTONS, SIM)

| | |
|---|---|
| Ethernet | 5 x RJ45 ports, 10/100/1000 Mbps |
| I/Os | 1 x Digital Input, 1 x Digital Output on 4-pin power connector |
| Status LEDs | 3 x connection status LEDs, 3 x connection strength LEDs, 10 x Ethernet port status LEDs, 4 x WAN status LEDs, 1x Power LED, 2 x 2.4G and 5G WiFi LEDs |
| SIM | 2 x SIM slot (Mini SIM – 2FF), 1.8 V/3 V |
| Power | 1 x 4-pin DC power connector |
| Antennas | 4 x SMA for Mobile, 2 x RP-SMA for WiFi, 1 x SMA for GNSS |
| USB | 1 x USB A port for external devices |
| Reset | Reboot/User default reset/Factory reset button |
| Grounding | 1 x Grounding screw |



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## PHYSICAL SPECIFICATION

| | |
|---|---|
| Casing material | Aluminum housing |
| Dimensions (W x H x D) | 132 x 44.2 x 95.1 mm |
| Weight | 533 g |
| Mounting options | DIN rail (can be mounted on two sides), flat surface placement |

## OPERATING ENVIRONMENT

| | |
|---|---|
| Operating temperature | -40 °C to 75 °C |
| Operating humidity | 10 % to 90 % non-condensing |
| Ingress Protection Rating | IP30 |



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# WHAT'S IN THE BOX?

## STANDARD PACKAGE CONTAINS*

- RUTX50 Router
- 18 W PSU
- 4 x Mobile antennas (swivel, SMA male)
- 2 x WiFi antennas (magnetic mount, RP-SMA male, 1.5 m cable)
- 1 x GNSS antenna (adhesive, SMA male, 3 m cable)
- Ethernet cable (1.5 m)
- SIM Adapter kit
- QSG (Quick Start Guide)
- Packaging box

RUTX50 ROUTER

18 W PSU

4 X MOBILE ANTENNAS (SWIVEL, SMA MALE)

2 X WIFI ANTENNAS (MAGNETIC MOUNT, RP-SMA MALE, 1.5 M CABLE)

1 X GNSS ANTENNA (ADHESIVE, SMA MALE, 3 M CABLE)

ETHERNET CABLE (1.5 M)

SIM ADAPTER KIT

* For all standard order codes standard package contents are the same, execpt for PSU.



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# STANDARD ORDER CODES

| PRODUCT CODE | HS CODE | HTS CODE | PACKAGE CONTAINS |
|---|---|---|---|
| RUTX50 000000 | 851762 | 8517.62.00 | Standard package |

For more information on all available packaging options – please contact us directly.

# AVAILABLE VERSIONS

| PRODUCT CODE | REGION (OPERATOR) | FREQUENCY |
|---|---|---|
| RUTX50 0***** | Europe[1], the Middle East, Africa, Oceania, Brazil | • 5G NR NSA: n1, n3, n5, n7, n8, n20, n28, n38, n40, n41, n77, n78<br>• 5G NR SA: n1, n3, n5, n7, n8, n20, n28, n38, n40, n41, n77, n78<br>• 4G (LTE-FDD): B1, B3, B5, B7, B8, B20, B28<br>• 4G (LTE-TDD): B38, B40, B41, B42, B43<br>• 3G: B1, B5, B8 |

The price and lead-times for region (operator) specific versions may vary. For more information please contact us.
1 - Regional availability - excluding Russia & Belarus.



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# MOUNTING OPTIONS

## DIN RAIL KIT

| Parameter | Value |
|---|---|
| Mounting standard | 35mm DIN Rail |
| Material | Low carbon steel |
| Weight | 57g |
| Screws included | Philips Pan Head screw #6-32×3/16, 2pcs |
| Dimensions | 82 mm x 46 mm x 20 mm |
| RoHS Compliant | V |

### DIN RAIL KIT

- DIN Rail adapter
- Philips Pan Head screw #6-32×3/16, 2pcs for RUT2xx/RUT9xx

| ORDER CODE | HS CODE | HTS CODE |
|---|---|---|
| PR5MEC00 | 73269098 | 7326.90.98 |

For more information on all available packaging options – please contact us directly.

## COMPACT DIN RAIL KIT

| Parameter | Value |
|---|---|
| Mounting standard | 35mm DIN Rail |
| Material | ABS + PC plastic |
| Weight | 6.5 g |
| Screws included | Philips Pan Head screw #6-32×3/16, 2pcs |
| Dimensions | 70 mm x 25 mm x 14,5 mm |
| RoHS Compliant | V |

### DIN RAIL KIT

- Compact plastic DIN Rail adapter (70x25x14,5mm)
- Philips Pan Head screw #6-32×3/16, 2pcs

| ORDER CODE | HS CODE | HTS CODE |
|---|---|---|
| PR5MEC11 | 73269098 | 7326.90.98 |

For more information on all available packaging options – please contact us directly.

## SURFACE MOUNTING KIT

| Parameter | Value |
|---|---|
| Mounting standard | Flat surface mount |
| Material | ABS + PC plastic |
| Weight | 2x5 g |
| Screws included | Philips Pan Head screw #6-32×3/16, 2pcs |
| Dimensions | 25 mm x 48 mm x 7.5 mm |
| RoHS Compliant | V |

### DIN RAIL KIT

- Surface mounting kit
- Philips Pan Head screw #6-32×3/16, 2pcs

| ORDER CODE | HS CODE | HTS CODE |
|---|---|---|
| PR5MEC12 | 73269098 | 7326.90.98 |

For more information on all available packaging options – please contact us directly.



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# RUTX50 SPATIAL MEASUREMENTS & WEIGHT

## MAIN MEASUREMENTS

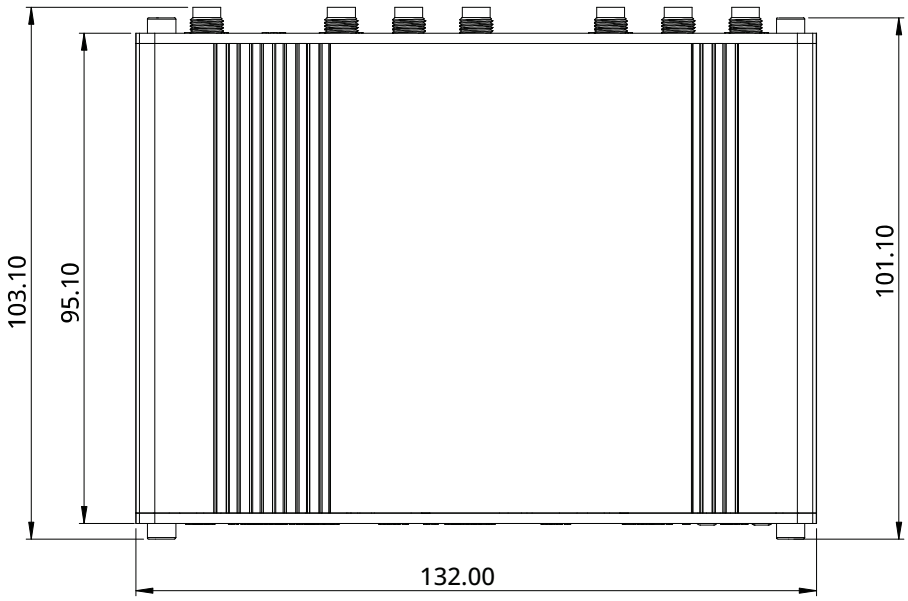
W x H x D dimensions for RUTX50:

| | |
|---|---|
| Device housing*: | 132 x 44.2 x 95.1 |
| Box: | 355 x 60 x 175 |

*Housing measurements are presented without antenna connectors and screws; for measurements of other device elements look to the sections below.

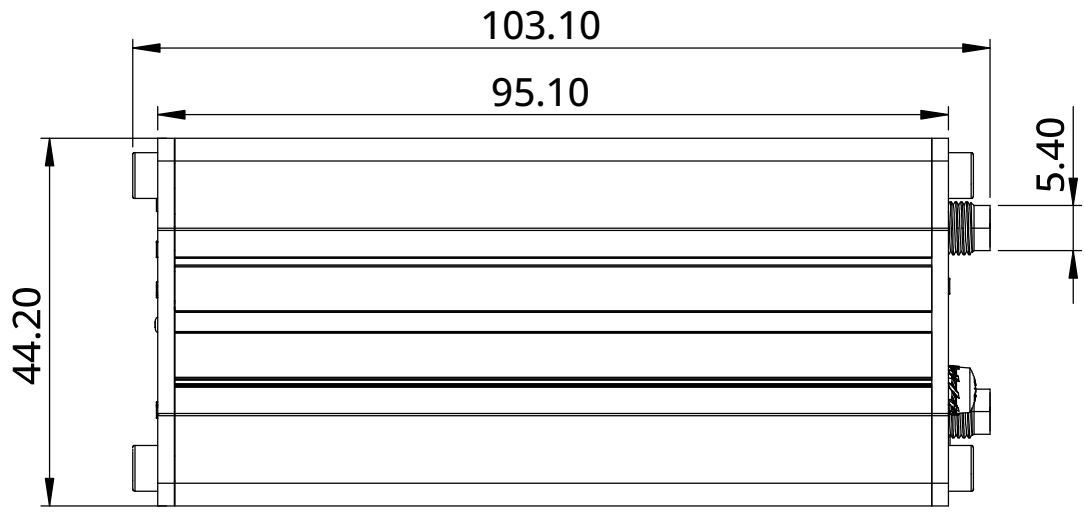
## TOP VIEW

The figure below depicts the measurements of RUTX50 and its components as seen from the top:




## RIGHT VIEW

The figure below depicts the measurements of RUTX50 and its components as seen from the right side:



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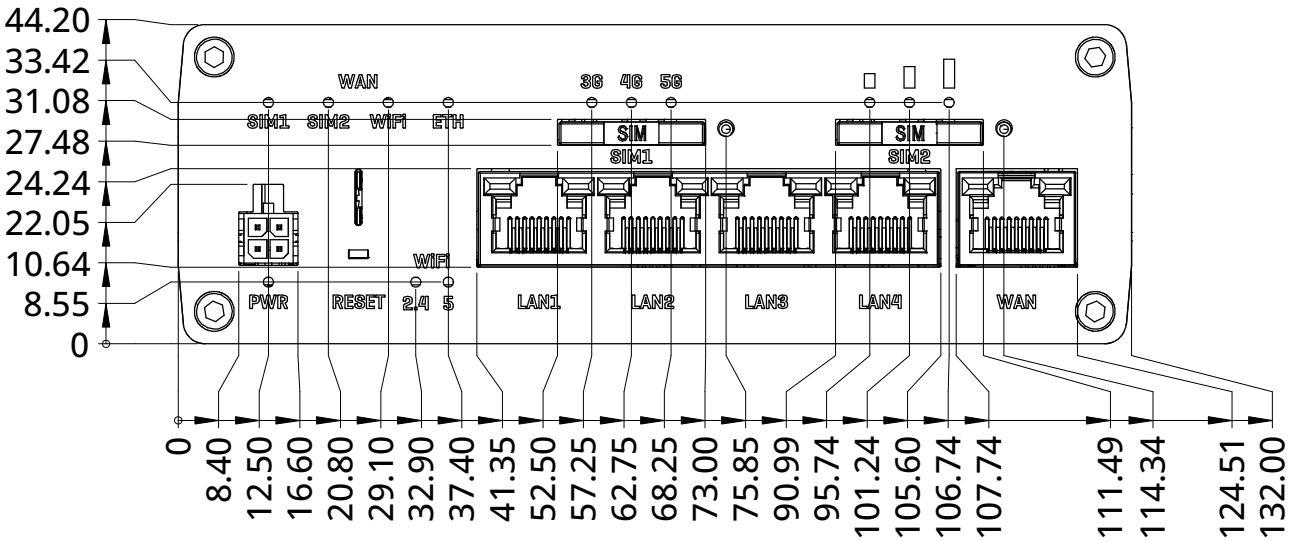
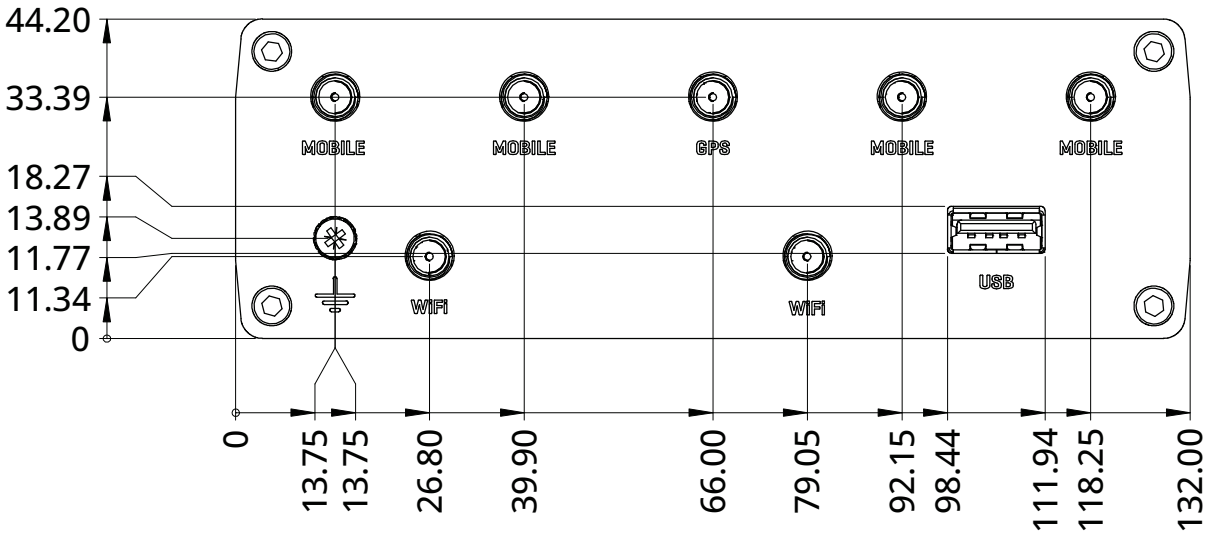
## FRONT VIEW

The figure below depicts the measurements of RUTX50 and its components as seen from the front panel side:




## REAR VIEW

The figure below depicts the measurements of RUTX50 and its components as seen from the back panel side:



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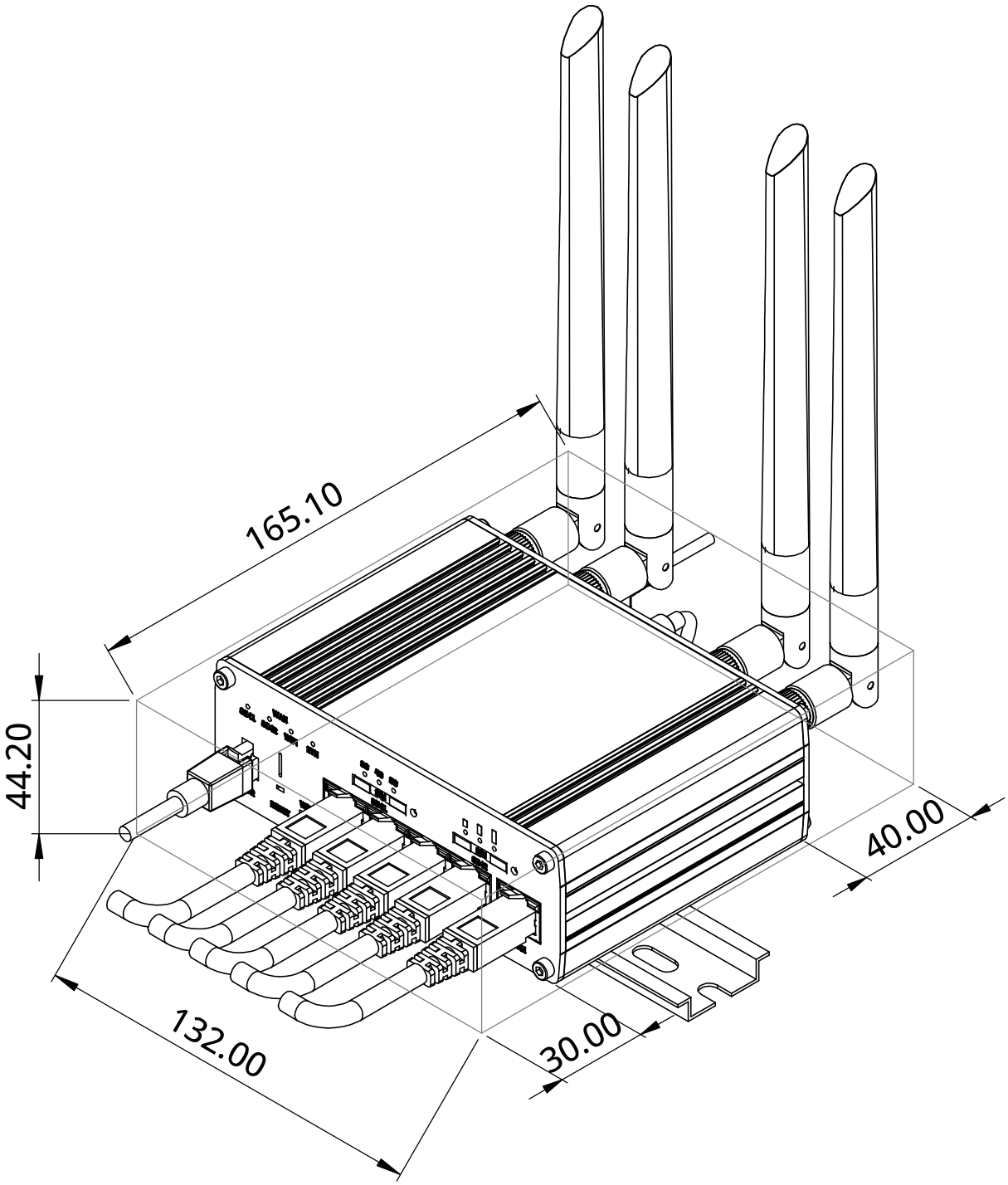
## MOUNTING SPACE REQUIREMENTS

The figure below depicts an approximation of the device's dimensions when cables and antennas are attached:



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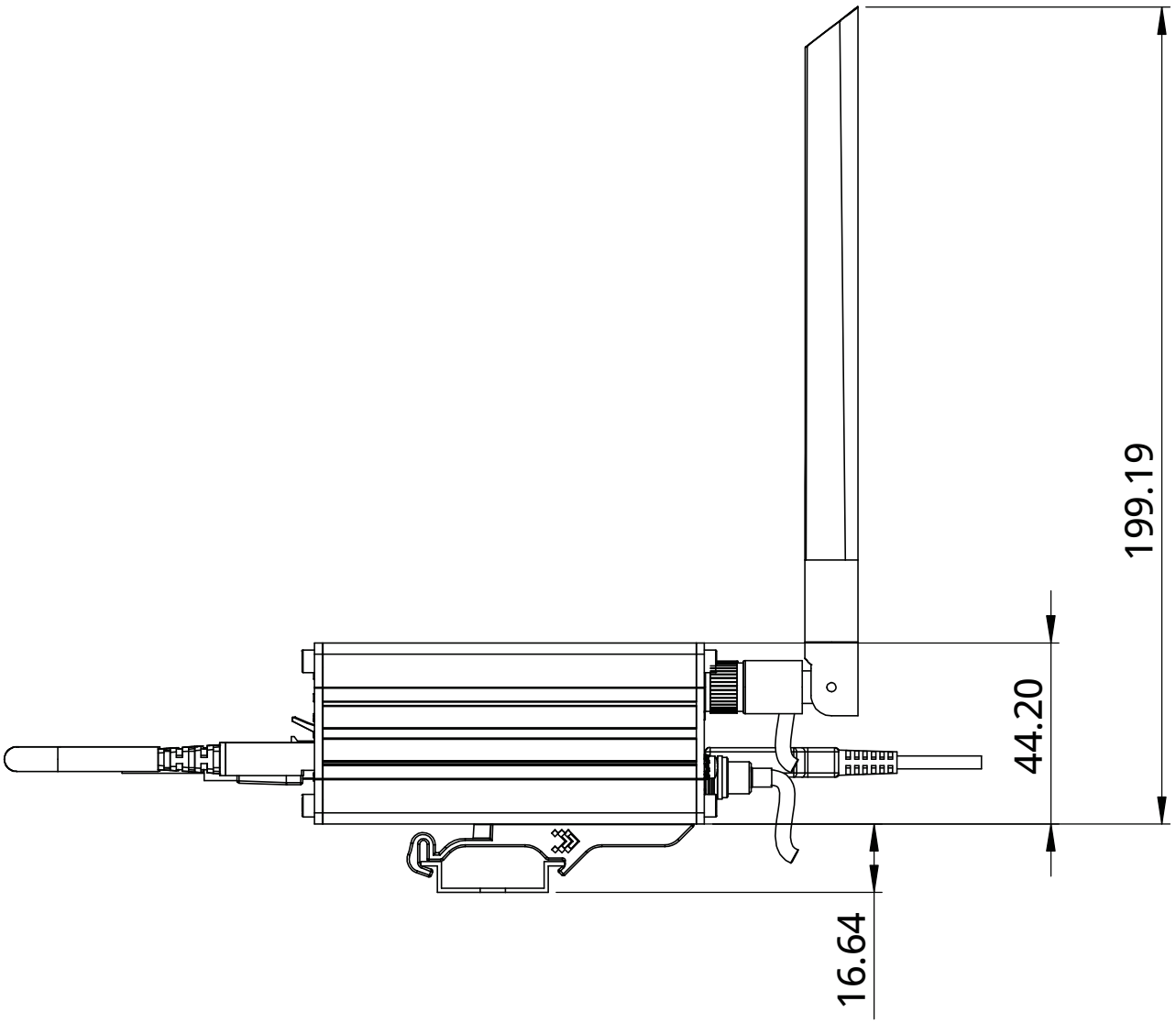
## DIN RAIL

The scheme below depicts protrusion measurements of an attached DIN Rail:



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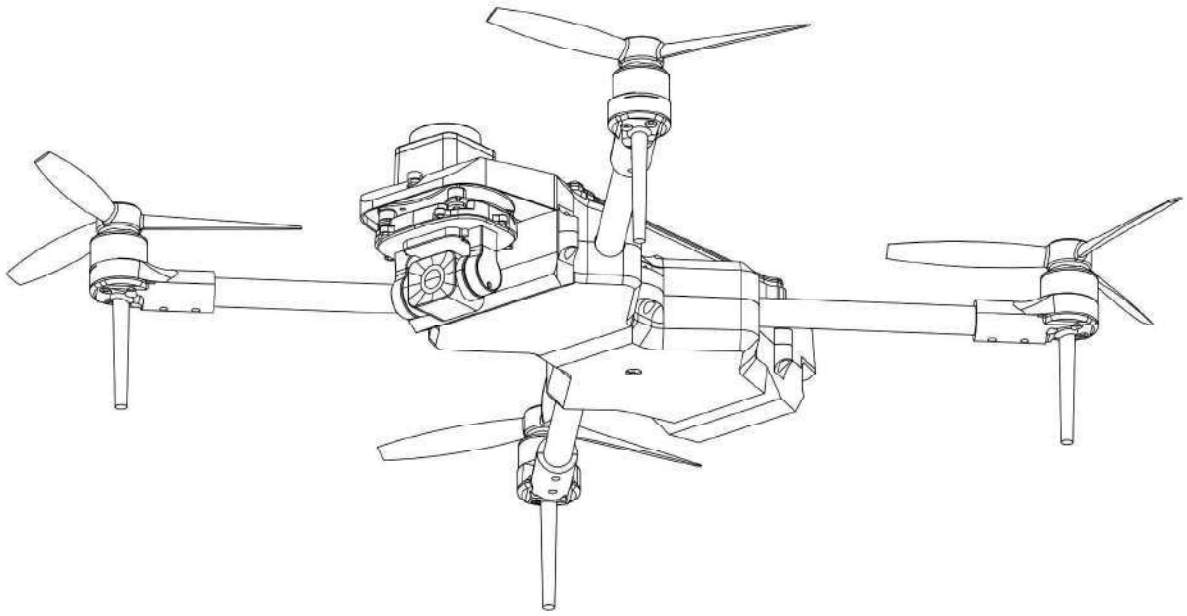
# iDronam Suparna

**User Manual** v1.2 *Jan 24, 2024*



MENTHOSA

SOLUTIONS

### 💡 Searching for Keywords

Search for keywords such as “battery” and “install” to find a topic. Press Ctrl+F on Windows or Command+F on Mac to begin a search.

### 💡 Printing this Document

This document supports high resolution printing.

| Revision Notes | | |
|---|---|---|
| **v1** | 01-Jan-2024 | Initial Version |
| **v1.1** | 14-Jan-2024 | Added Joystick Control |
| **v1.2** | 24-Jan-2024 | Added Operation Alt. Data |

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# Using this Manual

## Legend

🛑Warning ❗ Important 💡 Hints and Tips 📘Reference

## Read Before the First Flight

Read the following documents before using the **Suparna**:

1. Disclaimer and Safety Guidelines
2. Quick Start Guide
3. User Manual

❗ It is recommended to watch all tutorial videos on the official Menthosa website and read the disclaimer and safety guidelines before using for the first time. Prepare for your first flight by reviewing the quick start guide and refer to this user manual for more information

🛑 The operating temperature of this product is 0° to 40° C. It does not meet the standard operating temperature for military grade application (-55° to 125° C), which is required to endure greater environmental variability. Operate the product appropriately and only for applications that meet the operating temperature range requirements of that grade.

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# Content

**Product Profile** **6**
Introduction 6
Aircraft Highlight 6
Purpose and Target Users 8
**Aircraft** **11**
Aircraft Diagram 11
Aircraft Technical Specification 12
**Safety Precautions** **15**
Safety Procedures 16
**Battery Handling** **21**
Battery Installation 21
Handle with Care 21
**Flight** **25**
Flight Environment Requirements 25
Flight Limits 25
**Joystick** **28**
Joystick - Mode and Zoom 28
Joystick - Sticks and Control 29
Joystick - Camera Control 30
Joystick - Command and Special Modes 31
**Contact Us** **33**
Contact 33

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# Product Profile

This section introduces **Suparna** and lists the components of the aircraft.

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# Product Profile

## Introduction

**Suparna** features both an 360 degree Laser Sensing System and Downward Vision Systems, allowing for hovering and flying indoors as well as outdoors and for automatic Return to Home. The aircraft has a maximum flight speed of 54 kmph in outdoor stable wind conditions and 15 kmph indoor environment (***This has been tested under controlled environment.*** 🛑***Do not attempt to imitate***) and maximum flight time of 18 minutes.

## Aircraft Highlight

The **Suparna** represents a significant leap in unmanned aerial technology, offering a blend of high-speed connectivity, advanced navigation, and versatile functionality. This compact drone is designed for a variety of applications, from commercial and industrial use to recreational and research purposes.

- **5G Connectivity and Control**: Equipped with support for a 5G nano SIM, this drone ensures seamless operation through high-speed cellular networks. The 5G connectivity provides a robust link between the drone and its controller, ensuring real-time data transfer and remote control, even over long distances.

- **Dual Connectivity Options**: Apart from 5G, the drone also features Wi-Fi connectivity, allowing it to operate efficiently in both urban and remote areas. This dual connectivity ensures ubiquitous access and control, providing flexibility in various environments.

- **Advanced Processing Engine**: Powered by an onboard processor, the drone is adept at AI/ML analytics, making it suitable for applications that require real-time data

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processing, such as image analysis and environmental monitoring.

- **Autonomy and Navigation**: The drone boasts sophisticated autonomous capabilities like Visual Inertial Odometry (VIO) for accurate positioning, advanced path planning algorithms, and PX4 software for autonomous flight control. It also features GPS-denied navigation and Beyond Visual Line of Sight (BVLOS) capabilities, enhancing its utility in complex environments.

- **360-Degree Obstacle Avoidance** : With depth estimation and obstacle detection technology, the drone can navigate safely through challenging terrains. It employs mapping and Visual Obstacle Avoidance (VOA) systems to detect and avoid obstacles in all directions.

- **Payload Capacity**: Designed to carry payloads of 500g, this drone is suitable for a variety of tasks, including delivery, surveying, and equipment transport.

- **Outdoor Navigation System**: The inclusion of both GPS makes it reliable for critical missions.

- **Geo-Fencing**: This feature allows users to set virtual boundaries for the drone, ensuring it operates within predefined areas, enhancing safety and compliance.

- **High-Performance Sensors**: Equipped with sophisticated tracking and image sensors, the drone can capture high-quality images and videos, and efficiently track subjects or terrain features.

- **Connectivity and Ports**: The drone includes a 5G modem supporting the available 5G bands, an Ethernet/management port, and dual-band Wi-Fi-6, providing various options for data transmission and device management.

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This **Suparna**(5G-Enabled Mini Drone), with its advanced features and capabilities, is set to revolutionize the way we think about remote operations, surveillance, mapping, and a host of other applications in various sectors.

## Purpose and Target Users

The **Suparna**, with its advanced technological capabilities, is an invaluable tool tailored for a diverse group of users, including students, trainee professionals, researchers, and faculty members across various disciplines. This drone's unique features make it an ideal fit for educational, professional training, and research environments.

1. **Students**: For students pursuing studies in fields such as engineering, robotics, environmental sciences, and cinematography, this drone serves as a practical learning tool. It offers hands-on experience in cutting-edge technology, including 5G connectivity, AI/ML analytics, and advanced navigation systems. Its user-friendly interface makes it accessible for beginners, while its advanced features provide a deeper learning opportunity for more advanced students.

2. **Trainee Professionals**: Aspiring professionals in sectors like surveying, urban planning, and emergency response can greatly benefit from training with this drone. The practical skills gained in operating and managing advanced drone technology are directly transferable to their respective industries. The drone's capabilities in obstacle avoidance, payload management, and high-speed data processing provide a real-world understanding of how unmanned aerial vehicles (UAVs) can be utilized in professional scenarios.

3. **Researchers**: For researchers, particularly in fields such as environmental monitoring, agricultural technology, and geographical mapping, the drone's advanced navigational

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technologies like VIO, GPS-denied navigation, and BVLOS offer new avenues for gathering data in inaccessible or challenging terrains. Its ability to carry payloads and advanced sensors enables detailed environmental data collection and analysis, crucial for scientific advancements.

4. **Faculty Members**: In an academic setting, faculty members can integrate this drone into their curriculum to provide students with insights into the latest technological advancements. It can be used for demonstrations, practical experiments, and even as a research tool to aid in academic studies and publications. The drone's versatility makes it an excellent resource for a wide range of academic disciplines.

Overall, the **Suparna** (5G-Enabled Mini Drone) is not just a piece of technology; it's a gateway to experiential learning, professional skill development, and groundbreaking research for its diverse range of users. It offers a unique combination of practicality, innovation, and educational value, making it a significant asset in the hands of students, trainee professionals, researchers, and faculty alike.

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# Aircraft

This section introduces **Suparna** technical checklist.

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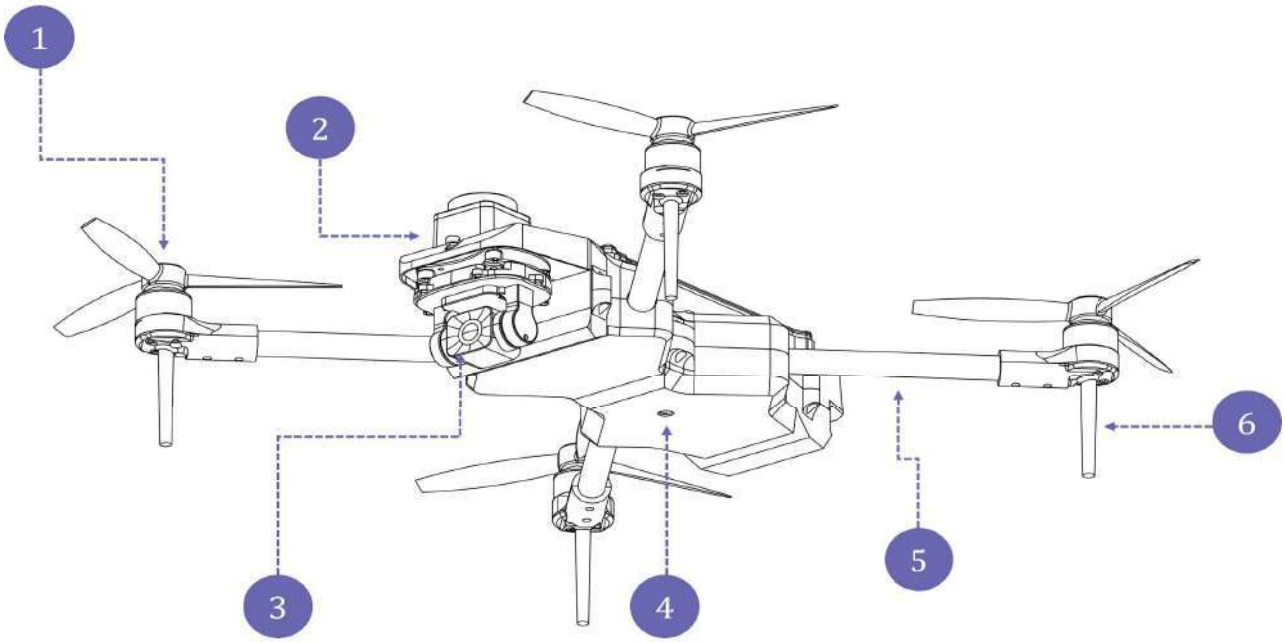
# Aircraft

## Aircraft Diagram




1: Propellers

2: 360 Lidar

3: Gimbal Camera

4: Downward Facing Vision System

5: Arms

6: Landing Gear

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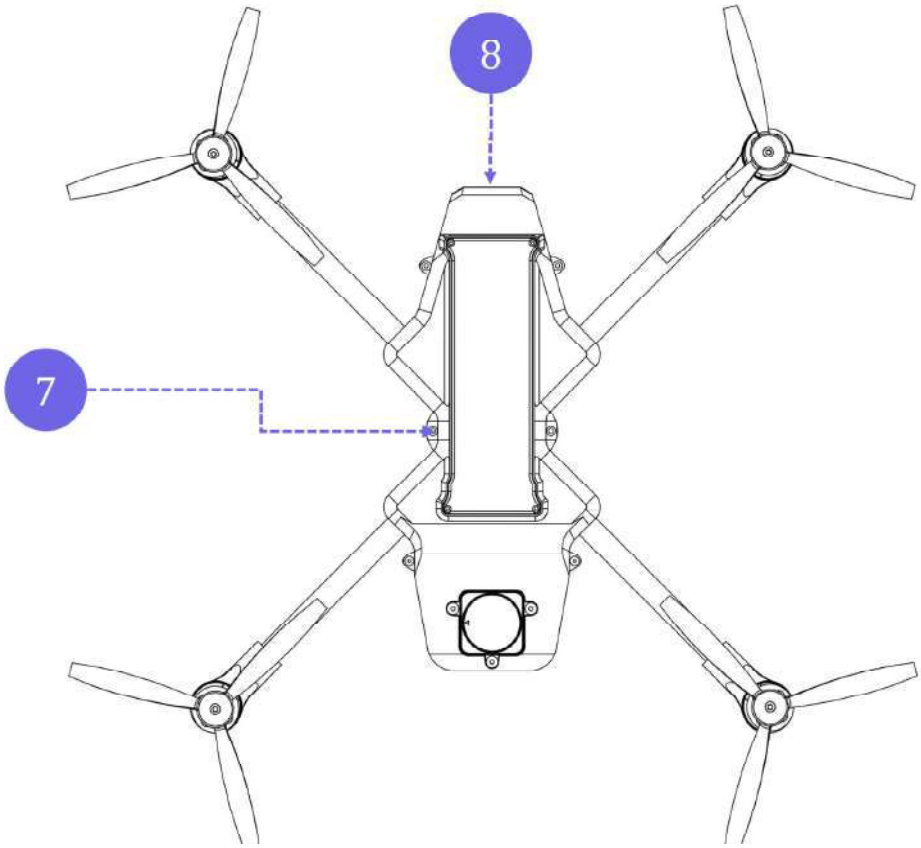


7: Removable Battery

8: GPS & External Compass

## Aircraft Technical Specification

| | |
|---|---|
| Drone Type | Quadcopter |
| Frame Material | Carbon Fiber |
| Size (Propeller to Propeller) | 470mm |
| Weight | 1300gm |

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| | |
|---|---|
| Extra Payload | 500gm |
| Max Altitude-Outdoor | 50 m AGL (🛑Check Local Rules and Regulation) |
| Max Operating Altitude | 750 m AMSL |
| Failsafe Features | 1. Low Battery<br>2. Critical Battery<br>3. Ground Station Connection Loss |
| Indoor Features | • 360 Degree Obstacle Avoidance System.<br>• Vision Based Odometry<br>• 10cm Precision for Indoor Altitude Hold.<br>• Real Time Video Streaming.<br>• Position Visualizations in 2D/3D Space.<br>• ROS Support. |
| 5G Features | • N78 Bands for Captive Networks.<br>• 5G Compatible Specialized GCS. |
| Battery Type | Li-ion 4 Cell |
| Battery Capacity | 4500mAh |
| Battery Charging/Discharging Cycle | 300 |
| Flight Time | 18 Minutes |
| Cruise Speed | 54 kmph (Outdoor)<br>15kmph(Indoor)<br><br>(🛑 **Warning!** These Data have been recorded in a professional environment. ***Do not attempt to imitate.*** Always use and fly drones with utmost caution.) |
| Communication Channel | Internet - 5G with LTE Fallback Or WiFi:6 |
| GCS | iDronam for Enterprise |
| GNSS | GPS |

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| | |
|---|---|
| Hovering Accuracy | Vertical: ±0.2 m (with Vision Positioning), ±1 m (with GNSS Positioning + Barometer)<br>Horizontal: ±0.3 m (with Vision Positioning), ±1.5 m (with GNSS Positioning) |
| Operating Temperature | 5° to 40° C |
| Vision Operating Environment | • Non-reflective, discernible surfaces with diffuse reflectivity of >20%<br>• Adequate lighting of lux >15 |
| Image Sensor (Default) | 1/2.7 Inch, 2 MP effective resolution |
| FOV | Horizontal 160° |
| Controllable Pitch Angle | -90° ~ 25° |

❗ Maximum flight time was tested in an environment with no wind while Hovering at a consistent point and the maximum flight speed was tested at sea level altitude with no wind. These values are for reference only.

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# Safety First

This section introduces **Suparna** safety Procedure

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# Safety Precautions

## Safety Procedures

Operating a **Suparna** (5G-enabled mini drone) indoors requires careful attention to safety and precautionary measures to ensure both the operator's safety and the integrity of the drone. Here are some guidelines for the safe operation of the drone in indoor environments:

- **Pre-Flight Check**:
  - Inspect the drone for any damage or loose parts.
  - Ensure all software and firmware are up-to-date.
  - Check the battery level and ensure it's adequately charged.
  - Test the drone's sensors and navigation systems to ensure they are functioning correctly.

- **Environmental Awareness**:
  - Clear the area of people, pets, and fragile objects to minimize risk.
  - Be aware of the indoor space layout, including ceilings, walls, and any obstacles like furniture.
  - Avoid flying near air vents, as the airflow can affect the drone's stability.

- **Control and Supervision:**
  - Always maintain a line of sight with the drone.
  - Do not rely solely on the drone's cameras or sensors; use visual supervision.
  - Keep the drone within a safe distance from the operator and bystanders.

- **Use of Safety Features:**
  - Utilize the drone's obstacle avoidance systems to prevent collisions.

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- Engage geo-fencing features to limit the drone's operational area.
- Use the drone's autonomy features wisely, especially in confined spaces.

- **Speed and Height Management:**
  - Operate the drone at lower speeds to maintain control.
  - Avoid flying too close to the ceiling or too low to the ground to prevent crashes.

- **Interference Considerations**:
  - Be aware of potential Wi-Fi or electronic interference in indoor environments that might affect drone control.
  - Test the 5G and Wi-Fi connectivity in the indoor area before flying.

- **Emergency Procedures:**
  - Familiarize yourself with the drone's emergency stop function
  - Have a clear plan for what to do in case of a loss of control or other emergencies.

- **Training and Skill Level:**
  - Ensure that the operator is adequately trained to handle the drone indoors.
  - Practice in a safe, controlled environment to build skill and confidence.

- **Legal and Ethical Considerations:**
  - Abide by any institutional or organizational policies regarding drone usage indoors.
  - Respect privacy if the drone is equipped with cameras or recording devices.

- **Post-Flight Check:**
  - After flying, inspect the drone for any damage.

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- Download and analyze any collected data if necessary.
- Store the drone in a safe, dry place away from potential hazards.

By adhering to these guidelines, operators can safely and effectively utilize the 5G-enabled mini drone in indoor settings for various applications, while minimizing risks and ensuring the longevity of the device.

❗ Vision Systems have limited ability to sense and avoid obstacles, and the performance may be affected by the surrounding environment. Make sure to maintain visual line of sight with the aircraft and pay attention to prompts in iDronam App.

❗ ( **Warning** )The Vision Systems cannot work properly over surfaces that do not have clear pattern variations.

The Vision Systems cannot work properly in any of the following situations. Operate the aircraft cautiously.

- Flying over monochrome surfaces (e.g., pure black, pure white, pure green).
- Flying over highly reflective surfaces.
- Flying over water or transparent surfaces.
- Flying over moving surfaces or objects.
- Flying in an area where the lighting changes frequently or drastically.
- Flying over extremely dark (< 10 lux) or bright (> 40,000 lux) surfaces.
- Flying over surfaces that strongly reflect or absorb infrared waves (e.g., mirrors).
- Flying over surfaces without clear patterns or texture.
- Flying over surfaces with repeating identical patterns or textures (e.g., tiles with the same design).
- Flying over obstacles with small surface areas (e.g., tree branches). Keep the

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sensors clean at all times.

- DO NOT tamper with the sensors. DO NOT use the aircraft in dusty or humid environments.

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# Battery Handling

This section introduces **Suparna** Battery Handling Guidelines

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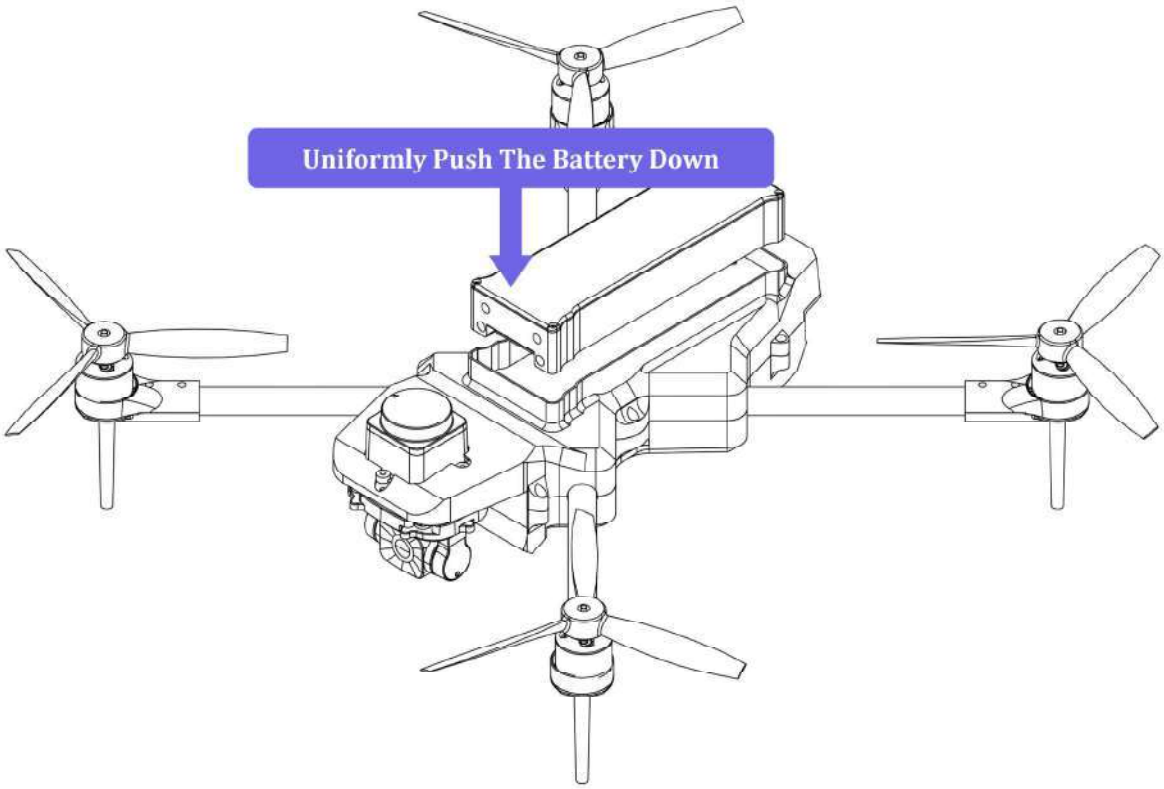
# Battery Handling

## Battery Installation




## Handle with Care

Battery handling and storage are critical for maintaining the safety and longevity of the 5G-enabled mini drone. Proper care of the drone's batteries ensures optimal performance and reduces the risk of accidents. Here are detailed instructions for handling and storing the drone's batteries:

- **Charging the Battery:**
    - Use only the charger provided with the drone or one recommended by the

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manufacturer.

- Charge the battery in a well-ventilated area away from flammable materials.
- Do not leave the battery unattended while charging.
- Avoid charging the battery immediately after flight. Allow it to cool down first.

- **Handling the Battery**:
  - Handle batteries with care. Avoid dropping them or subjecting them to impact.
  - Do not dismantle, puncture, or alter the battery in any way.
  - Keep the battery dry and away from water or moisture.
  - Avoid exposing the battery to extreme temperatures, both hot and cold.
  - If the battery appears swollen, discolored, or damaged in any way, do not use it.

- **Storage of the Battery:**
  - Store batteries in a cool, dry place, away from direct sunlight and heat sources.
  - If storing for an extended period, keep the battery at a 40-60% charge level. Avoid storing it fully charged or fully depleted.
  - Check the charge level every few months and recharge to the recommended storage level if necessary.
  - Use a fireproof bag or container designed for LiPo battery storage.

- **Transporting the Battery:**
  - When transporting, ensure the battery terminals are protected and cannot come into contact with metal objects.
  - Carry batteries in a dedicated, fireproof battery case.
  - Follow any specific regulations for transporting batteries by air or other

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means.

- **Battery Disposal:**
    - Do not dispose of batteries in regular trash. Batteries should be recycled according to local regulations.
    - If the battery is damaged, follow local guidelines for hazardous waste disposal.

- **General Precautions:**
    - Never use a battery that is not specifically designed for your drone model.
    - Keep batteries out of reach of children.
    - Regularly inspect the battery for signs of wear or damage.

By following these guidelines, users can ensure the safe operation and extended life of their drone batteries. Proper battery care is not only a matter of performance but also a critical aspect of operational safety.

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# Flight

This section introduces **Suparna** safe Flight guidelines.

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# Flight

## Flight Environment Requirements

- Do not use the aircraft in severe weather conditions including wind speeds exceeding 8.0 m/s, snow, rain, and fog.
- Tall structures and large metal structures may affect the accuracy of the onboard compass and GNSS system.
- Flying indoors flying do not force aircrafts to go through sharp turns. Have at least 4m*4m clearance from all sides.
- It is recommended to keep the aircraft at least 8 m away from structures.
- Avoid obstacles, crowds, high voltage power lines, trees, and bodies of water.
- It is recommended to keep the aircraft at least 10 m above water.
- Minimize interference by avoiding areas with high levels of electromagnetism such as locations near power lines, base stations, electrical substations, and broadcasting towers.
- Aircraft and battery performance is subject to environmental factors such as air density and temperature. Be careful when flying (5,000 m) or more above sea level, since battery and aircraft performance may be reduced.
- Aircraft cannot use GNSS within the polar regions. Use the Downward Vision System when flying in such locations.
- If taking off from a moving surface, such as a moving boat or vehicle, fly with caution.

## Flight Limits

| When GNSS is Available |
|---|

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| | |
|---|---|
| Max Altitude | 100m (Subject to Flying Permission) |
| When GNSS is not available and Downward Facing Vision System is used | |
| Max Altitude | 20m |

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# JOYSTICK

This section introduces **Suparna**'s Controlling System

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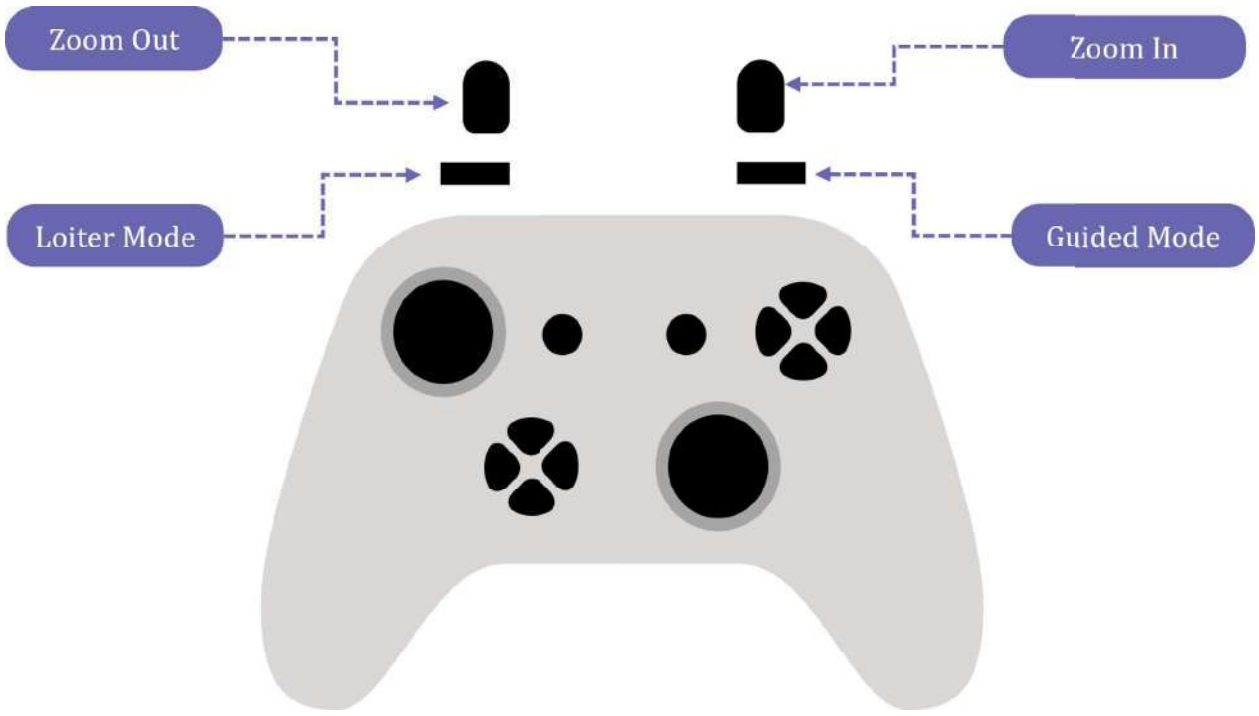
# Joystick

## Joystick - Modes and Zoom

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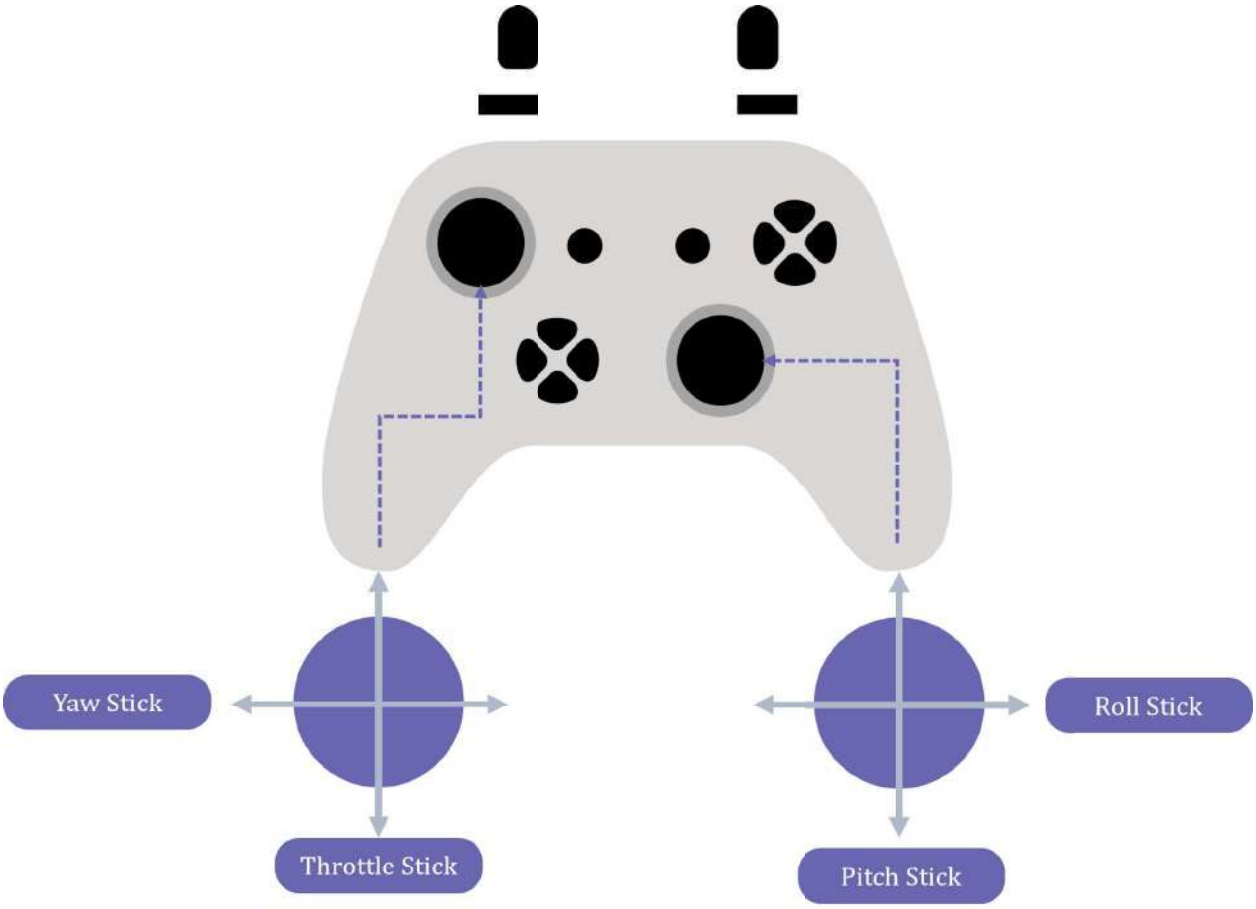
## Joystick - Sticks and Control

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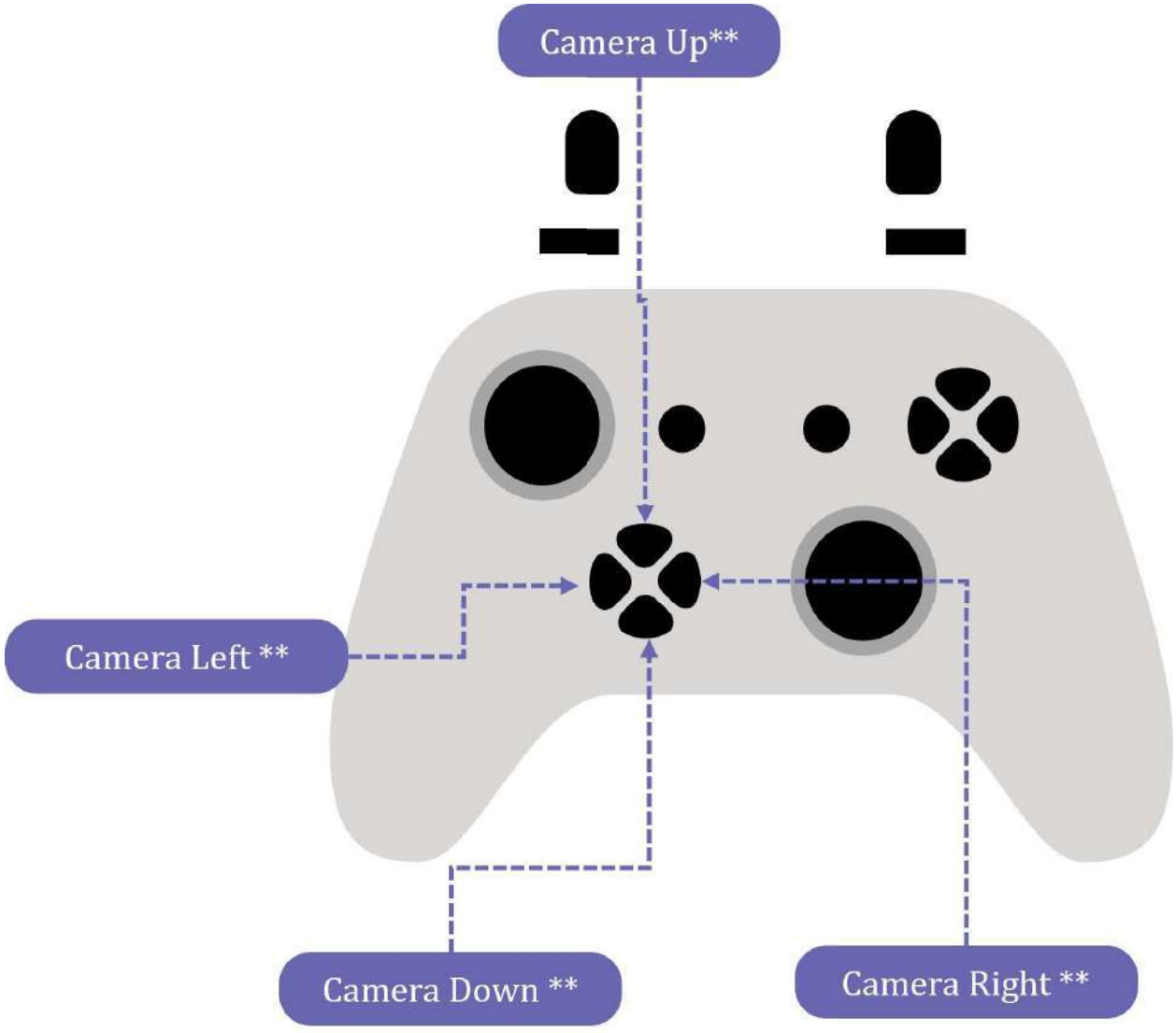
## Joystick - Camera Control

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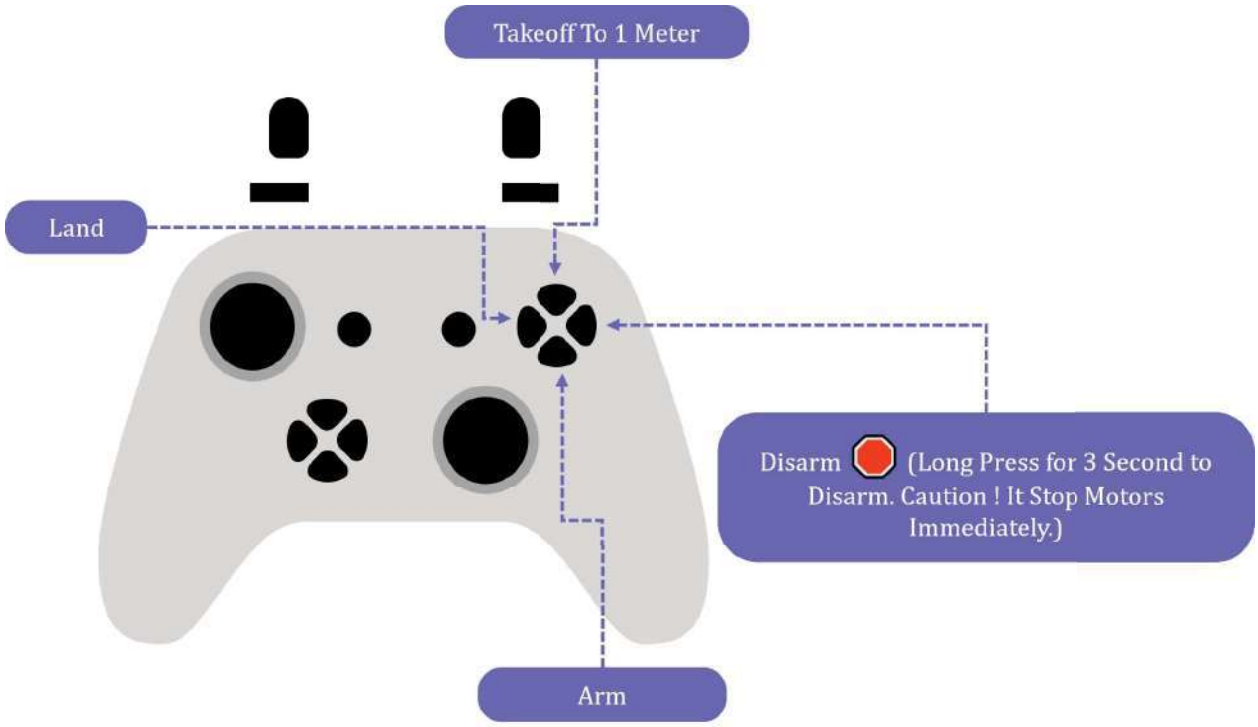
## Joystick - Command and Special Modes




** Available with only selected Model.

: Use with caution

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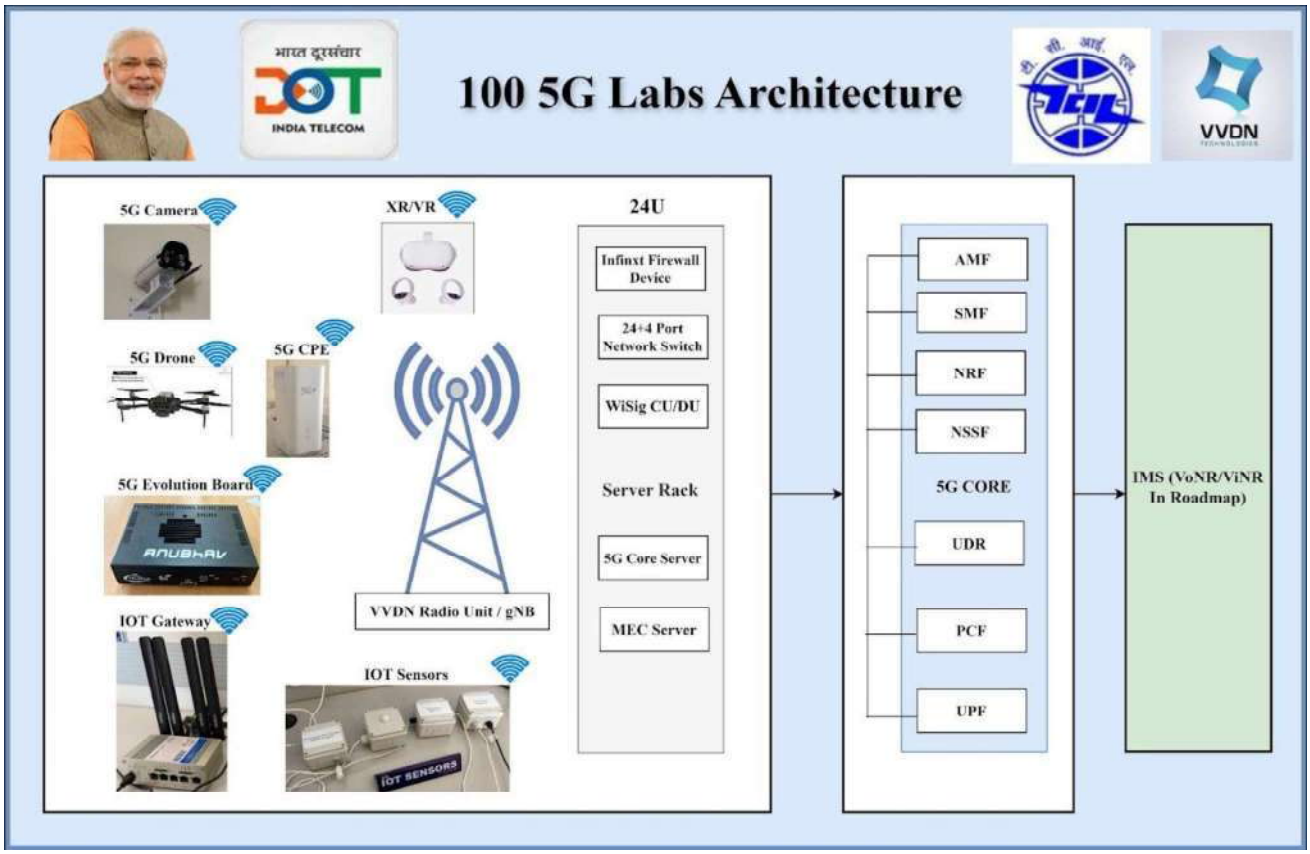
# 100 5G Use cases Lab Training Programme



## Table of Contents


1.0 5G Overview 8

1.1 Network Slicing: 8

1.2 5G Deployment Architecture 9

1.2.1 5G Standalone Architecture 10

1.2.2 Protocol Deploy and C-Plane Stack and U-Plane Stack 10

1.2.3 L2 Radio Stack Overview: 12

1.3 PHY Layer (L1): 13

1.3.1 Frequency Scan: 13

1.3.2 Time Synchronization: 13

1.3.3 Channelization: 14

1.3.4 Power Control: 14

1.3.5 Beam Management: 14

1.3.6 Frame Structure: 14

1.3.7 CSI Report: 14

1.3.8 MIMO: 14

1.4 5g Radio Frame Structure 15

1.5 Sub carrier spacing 16

1.6 SSB 17

1.7 5G Logical, Transport and Physical channels across the RAN protocol stacks 18

1.8 5G LAB ARCHITECTURE & OVERVIEW TCIL LAB 19

1.9 System In 5G LAB Setup 20

1.10 Radio access network (VVDN LPRU +CU/DU) 21

1.10.1 VVDN LPRU 21

1.10.2 Low Power Radio Unit (LPRU) 22

1.10.3 Key Features and Specifications: 22

1.10.4 Device Specification: 23

1.10.5 LPRU LED Notification 24

1.11 LPRU Implementation in 5G LAB :- 25

1.11.1 RU Configuration 25

1.11.2 U plane configuration 28

2.0 Wisig Networks CU/DU 36
2.1 INTRODUCTION 36
2.1.1 Purpose 36
2.1.2 Scope 36
2.1.3 Definitions, Acronyms, and Abbreviations Acronym Explanation 37
2.2 Reference Documents 38
2.3 High-Level System Architecture 39
2.4 System Key Parameters 41
2.5 gNB CU Architecture 42
2.5.1 Software Modules 42
2.5.2 CU OAM 42
2.5.3 NR Layer 3 42
2.5.4 NR RRM 42
2.5.5 PDCP and Packet Relay 43
2.6 gNB DU Architecture 43
2.6.1 Software Modules 43
2.6.2 DU OAM 43
2.6.3 DU MGR 43
2.6.4 DU F1U 44
2.6.5 NR MAC 44
2.6.6 NR RLC 44
2.7 Interfaces 45
2.8 CU-DU Hardware Specifications 45
2.9 Bringup Of CU-DU Hardware 46
2.9.1 Access the CU-DU server 46
2.9.2 CU/DU/L1 Installations 47
2.9.3 L1 Prerequisites 47
2.9.5 Instructions same for all service Checking 48
i. Check ts2phc service is active or not 49
ii. Check PTP service is active or not 49

iii. Check PHC service is active or not 49
2.10 CU/DU Bring-up Using SMO 49
2.11 Run the gNB 50
2.11.1 SMO GUI 50
2.11.2 To run and execute gNB O1 Handler 51
2.11.3 To start the gNB process 51
2.11.4 To start or stop the gNB process 52
2.11.5 Camp the UE once all status are running on proprietary-system options 53
2.12 To check the RAN logs 54
2.12.1 To check the L1 logs follow the below steps 54
2.12.2 To check the CU logs follow the below steps 54
2.12.3 To check the PDCP logs follow the below steps 54
2.12.4 To check the DU logs follow the below steps 55
2.12.5 WiSig Networks 5G gNodeB CUDU 55
A. Diagnose 55
2.13 Linux RAN Bring up Process 55
3.0 Niral Core MEC/NMS 60
3.1 Introduction 60
3.1.1 OVERVIEW OF NIRALOS 60
3.2 NiralOS 5G Core 61
3.3 NiralOS Private 5G Key Features 62
3.4 NiralOS 5G Core Architecture 62
3.4.1 Network function 63
3.5 Network interface - NiralOS 5G core 64
3.5.1 NiralOS Controller 65
3.5.2 NiralOS EDGE 66
3.5.3 Scope 67
3.6 Setup diagram of 5G Lab 67
3.6.1 Setup diagram 68
3.7 5G Core Controller API: 68

**3.7.1 Access Token API** 68
**3.7.2 Overview** 69
3.7.3 Alarms Count 70
**3.7.4 Topology** 70
**3.7.5 Control Plane Stats** 75
**3.7.6 5G Radio Stats** 76
**3.7.7 PFCP Stats** 77
**3.7.8 Add APN** 78
**3.7.9 View APN List** 79
**3.7.10 Slice** 79
**3.7.11 Add Slice** 84
**3.7.12 Slice list** 85
**3.7.13 Add Subscriber** 89
**3.7.14 View All Subscribers** 90
**3.7.15 Trace imsi** 95
**3.7.16 Radio list** 95
**3.7.17 list all alarms** 96
**3.7.18 Critical alarm list** 97
**3.7.19 UE History count** 97
**3.7.20 5G Core Site Info** 99
**3.7.21 Login** 100
**3.7.22 Node** 101
**3.7.23 Stats** 102
**3.7.24 Vm details** 105
**3.7.25 Create Vm** 107
**3.8 MEC** 108
**3.8.1 Overview – functionalities and description.** 108
**3.9 Implementation of MEC Platform.** 108
**3.9.1 Explanation of Hardware proposed for the lab.** 108
**3.9.2 Provisioning of server for installation** 108

A. 1OS and Virtualization software installation. 108
B. Virtual machine for each network functions 109
3.10 Procedure for Application Installation 109
3.11 Procedure to Route the traffic to MEC server. 111
3.12 Monitor analysis and management of application life cycle. 111
3.13 Network Management System (NMS) 111
3.14 Unified management layer across the 5G RAN and the 5G-Core xNFs 111
3.14.1 Resource Management (RM) - 111
3.14.2 Fault Management (FM) 112
3.14.3 Performance Management (PM) 113
3.14.4 Configuration Management (CM) 114
3.14.5 Configuring gNodeB 115
3.15 Configuring of 5G Core 116
3.16 Monitoring of 5G Network 119
3.17 NiralOS MEC – Add and Delete VM 121
3.18 Container Interface Details 122
3.19 Basic Configuration 126
3.19.1 Login ID Details 126
3.19.2 Network Interface configuration 127
3.19.3 Subscriber configuration 128
3.20 Drone Live Streaming on MEC- 129
3.21 Camera Live Streaming on MEC- 131
4.0 Infinity Firewall Secure SDWAN 133
4.1 Introduction 133
4.2 Architecture 134
4.3 Components 135
4.4 Control Plane 135
4.5 Data Plane 135
4.6 DHCP (Dynamic Host Configuration Protocol) 135
4.7 Routing 136

**4.8 Firewall** 136
**4.9 Intrusion Detection System/Intrusion Prevention System (IDS/IPS)** 136
**4.10 Web Filtering** 136
**4.11 Deep Packet Inspection (DPI)** 137
**4.12 Virus/Malware Detection Engine** 137
**4.13 Virtual Private Network (VPN)** 137
**4.14 WAN Load Balancing** 137
**4.15 Quality of Service (QoS)** 138
**4.15.1 Login(IP-72.16.0.X)** 138
**4.15.2 Dashboard** 138
**4.15.3 Sites** 138

## 1.0 5G Overview

5G, or fifth-generation wireless technology, is the latest standard for mobile networks, succeeding 4G (LTE). It promises to revolutionize the way we connect and communicate by providing faster speeds, more reliable connections, and lower latency. Here's an overview of what 5G brings to the table:

1. Higher Speeds: 5G can deliver download speeds up to 100 times faster than 4G, with speeds potentially exceeding 10 gigabits per second. This makes it ideal for high-bandwidth applications like HD video streaming, virtual reality (VR), and large-scale data transfers.

2. Lower Latency: Latency refers to the delay between sending and receiving data. 5G reduces latency to as low as 1 millisecond, making real-time communication (such as gaming, remote surgery, and autonomous vehicles) smoother and more reliable.

3. Massive Connectivity: 5G can support a much higher number of devices connected to the network simultaneously. This is crucial for the growing Internet of Things (IoT), where billions of devices need to communicate with each other, from smart home appliances to industrial sensors.

4. Improved Reliability: With better signal integrity and higher bandwidth availability, 5G can provide more consistent and reliable connections, even in crowded urban areas or rural environments.

### 1.1 Network Slicing:

5G allows for the creation of virtual networks, or "slices," which can be tailored to specific needs (e.g., autonomous vehicles, medical services, or industrial applications). This means operators can optimize the network based on different requirements.

1. **eMBB (**enhanced Mobile Broadband): There are multiple deployment options for 5G and if we talk about Non-Standalone deployment which focus on eMBB and provide more bandwidth, low latency.

2. **URLLC** (Ultra Reliable Low Latency Communications): Mission critical applications are good examples for URLLC where latency can not be compromised. It can be achieved with Standalone 5G core deployment to reduce latency.

3. **mMTC** (massive Machine Type Communications): NB-IoT which is a Low Power Wide Area Network (LPWAN) radio technology defined by 3GPP and MMTC covers most of its requirements.

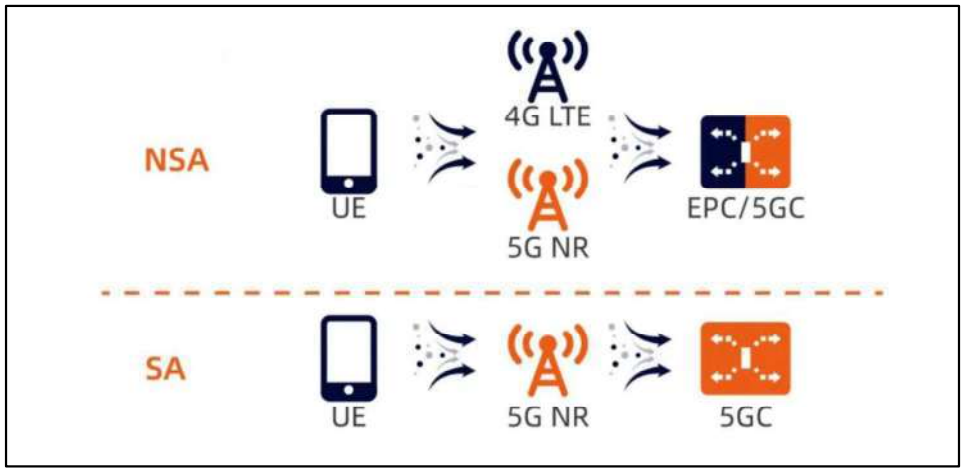
4. Enhanced Efficiency: 5G is designed to be more energy-efficient and spectrum-efficient than previous generations, helping to reduce power consumption and make more efficient use of available frequencies.

In summary, 5G is not just about faster internet speeds but a significant upgrade in how wireless networks operate, enabling new technologies and services across industries, from healthcare to transportation and beyond

**1.2 5G Deployment Architecture**

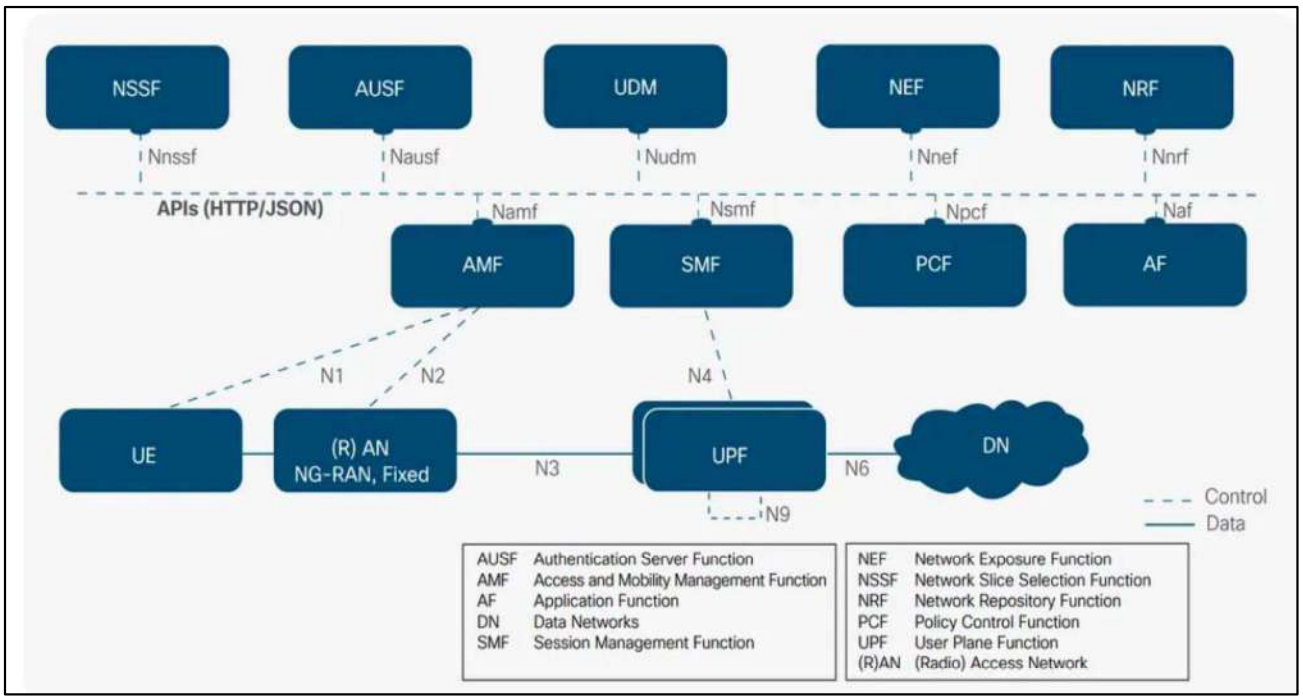
- SA- Standalone Architecture
- NSA - Non Standalone Architecture

5G Standalone (SA) networks are built on a dedicated 5G core network, while 5G Non-Standalone (NSA) networks use the existing 4G core network. SA networks are considered the long-term solution for 5G, while NSA networks are a transitional technology.




We deployed in all Labs 5G SA O-RAN Solution with RU-VVDN, CU/DU -Wisig networks and Core-Niral Networks.

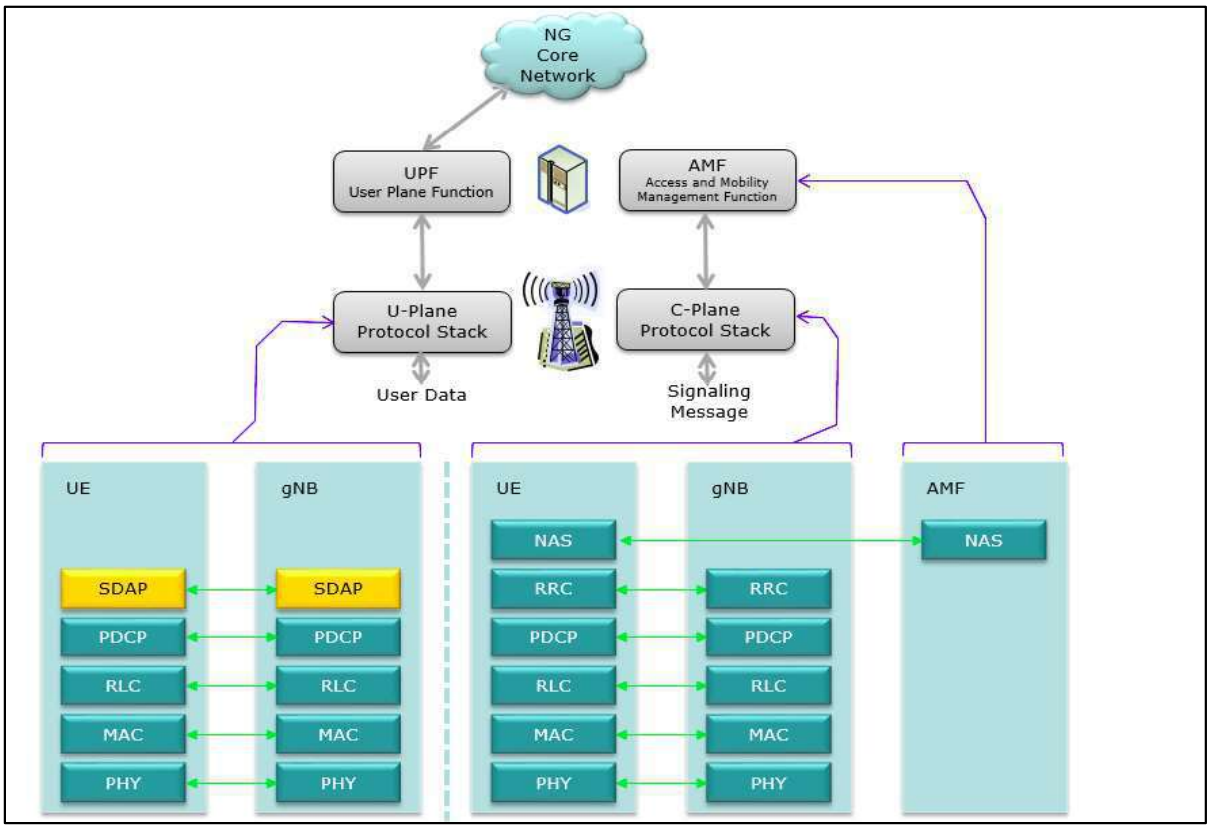
### 1.2.1 5G Standalone Architecture




### 1.2.2 Protocol Deploy and C-Plane Stack and U-Plane Stack

NR radio protocol stack has two different stack depending on the type of data that is processed by the stack. If the data is Signaling message, it goes through the C-plane stack and if it is user data, it goes through U-Plane stack. Both U-Plane and C-Plane is made up of a common structure : PHY <-> MAC <-> RLC <-> PDCP, but the components sitting on top of PHY/MAC/RLC/PDCP gets different between C-Plane and U-Plane. In case of U-Plane, a layer called SDAP is sitting at the top of the radio stack and the SDP is connected to UPF (User Plane Function). In case of C-Plane, the two layers RRC and NAS are sitting at the top of the stack. NAS layer gets connected to AMF (Access and Mobility Management Function).

What We've mentioned can be described in a block diagram as shown below.



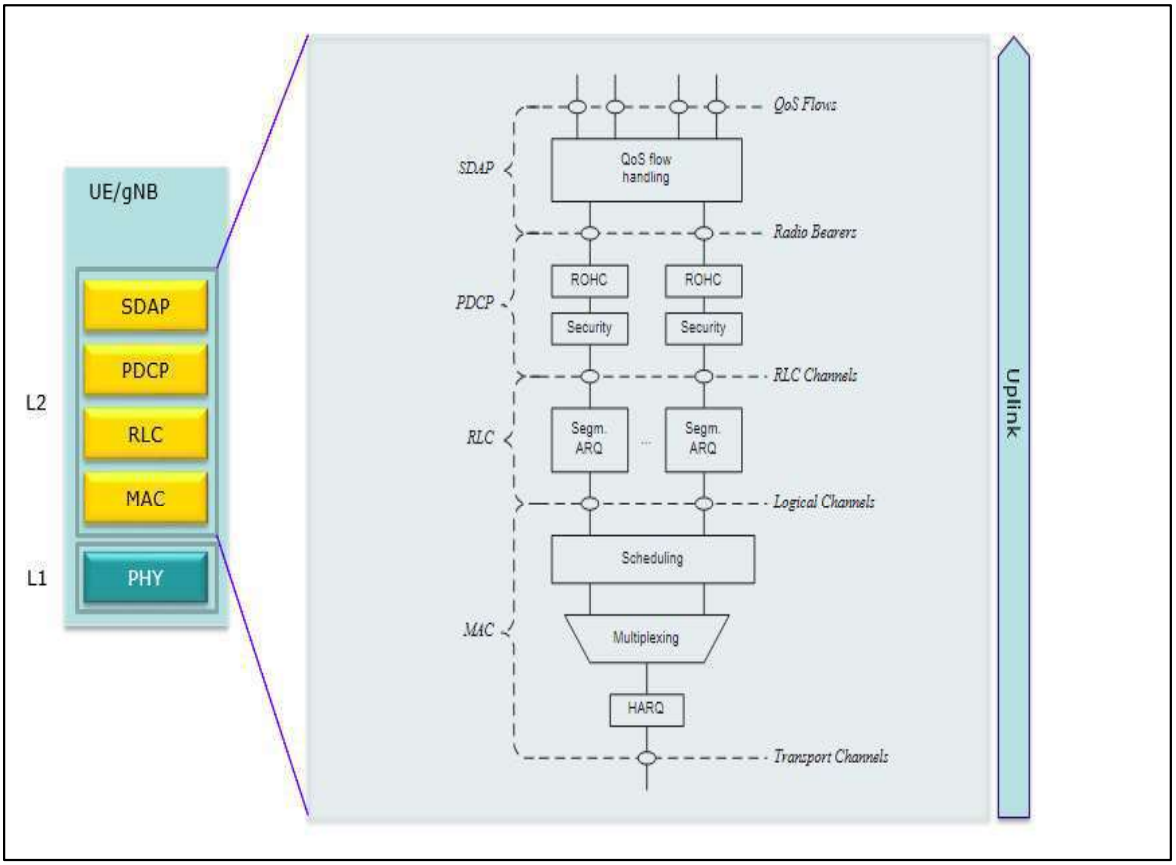

**<User Plane Protocol Stack >** **<Control Plane Protocol Stack>**

System Level Specification for Access Network

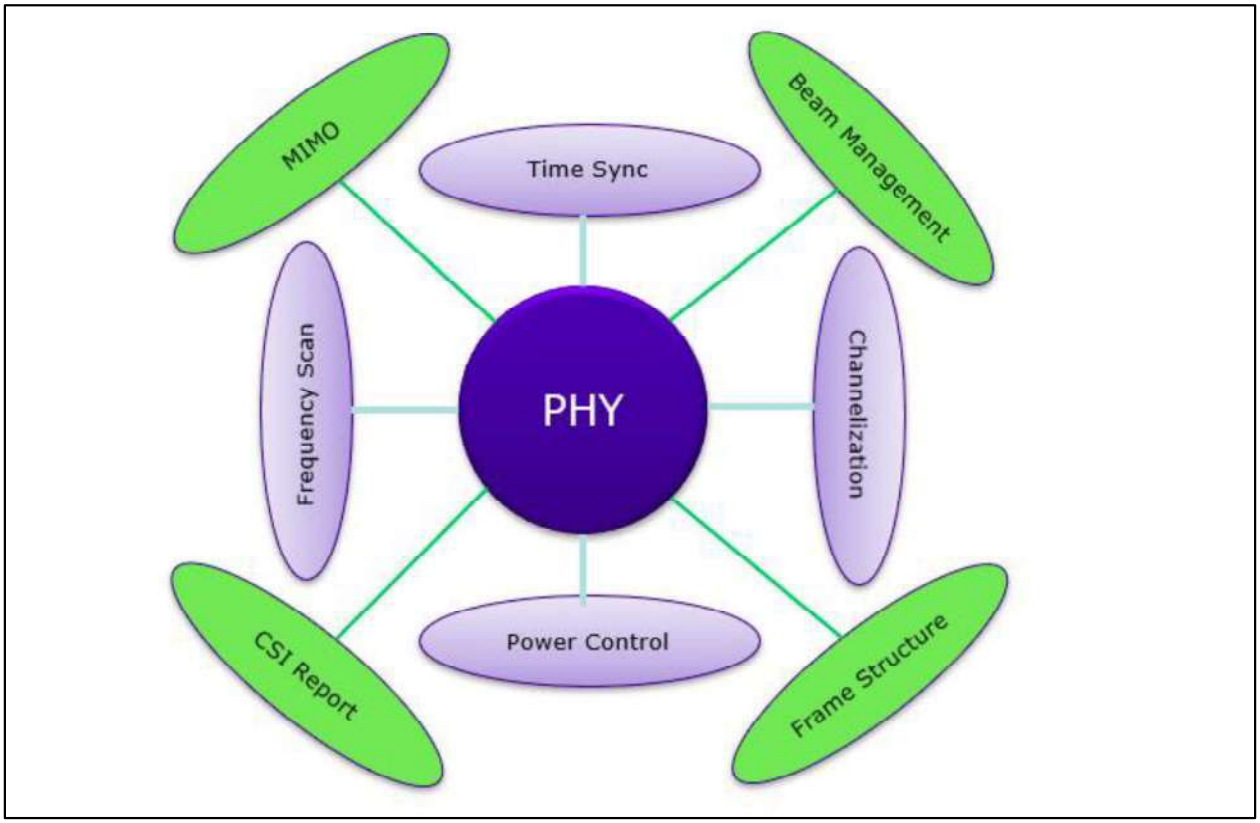
- NAS (Non-Access Stratum)
- RRC (Radio Resource Control)
- SDAP (Service Data Adaptation Protocol)
- PDCP (Packet Data Convergence Protocol)
- RLC (Radio Link Control)
- MAC (Medium Access Control)
- PHY (Physical Layer)

### 1.2.3 L2 Radio Stack Overview:

NR U-Plane Uplink radio protocol. Basic structure is same as downlink structure



### 1.3 PHY Layer (L1):




**1.3.1 Frequency Scan**:

- Scanning different frequencies to identify the best ones for use in communication.
- GCSN

**1.3.2 Time Synchronization:**

- Ensuring that the transmission and reception of signals are precisely timed across the network.
- SSB (SS Block)

### 1.3.3 Channelization:

The process of defining multiple different physical resource properties and transmitting the data (e.g, user traffic, control data, sync signal etc) via a different profile called 'channel'.

- PBCH
- PDCCH
- PDSCH
- PUSCH
- PUCCH

### 1.3.4 Power Control:

Managing the power levels used for signal transmission to optimize performance and reduce interference.

- Power Control

### 1.3.5 Beam Management:

- In the context of 5G, this would involve the management of beamforming processes for directional signal transmission and reception.
- Beam Forming
- Beam Management
- CSI Framework

### 1.3.6 Frame Structure:

- Defining the format of data frames for efficient communication and data delivery.
- Frame Structure

### 1.3.7 CSI Report:

- Channel State Information reporting, which involves the mobile device reporting back to the network on the quality of the received signal.
- CSI Framework

### 1.3.8 MIMO:

- Stands for Multiple Input Multiple Output, a method for multiplying the capacity of a radio link using multiple transmission and receiving antennas to exploit multipath propagation.
- Massive MIMO
- MIMO Configuration for Downlink

- MIMO Configuration for Uplink

### 1.4 5g Radio Frame Structure

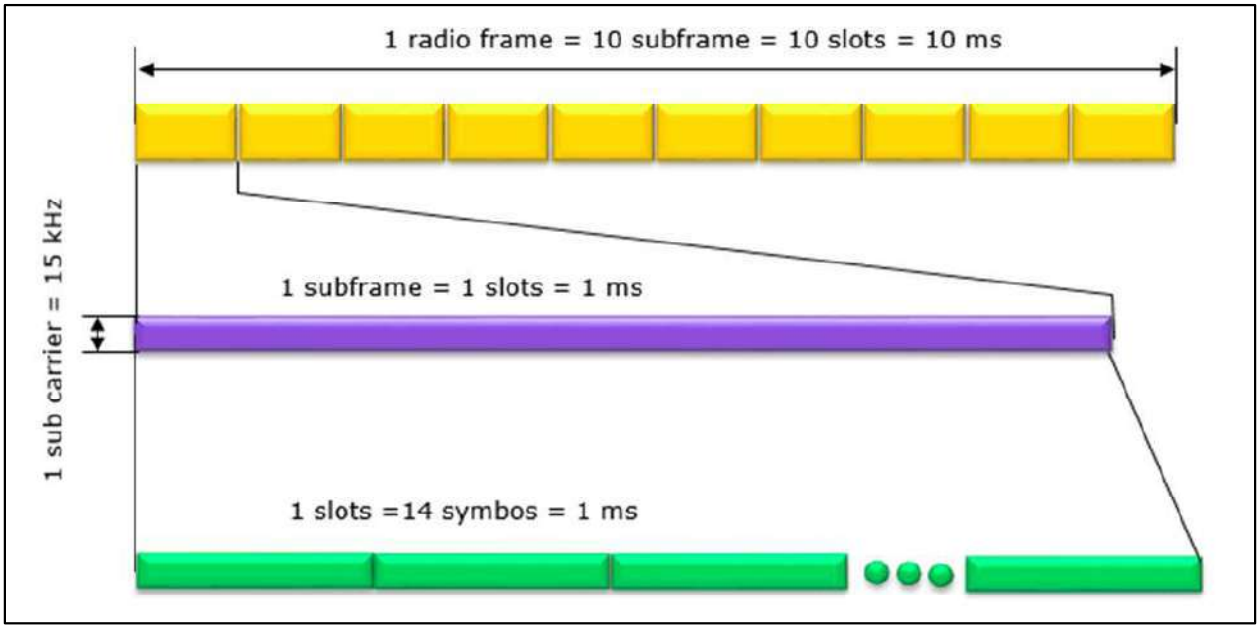
Downlink and uplink transmissions are organized into frames with 10ms duration, each consisting of ten subframes of 1ms

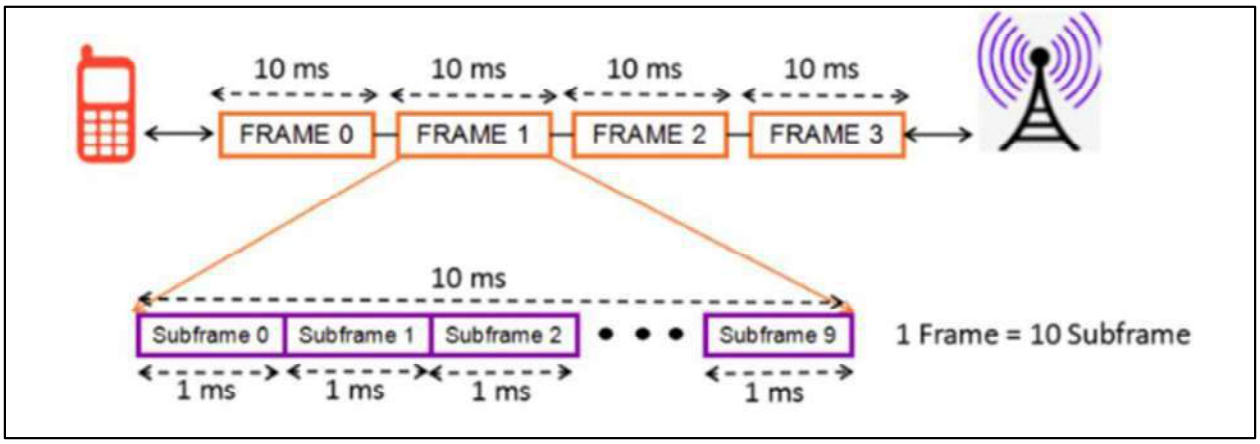
Each frame is divided into two equally-sized half-frames of five subframes each with half-frame 0 consisting of subframes 0 to 4 and half-frame 1 consisting of subframes 5 to 9.

In Total, there are 10 subframes in one frame.

The number of symbols for a slot is always 14 and for slot with extended CP, the number of symbols for a slot is always 12.

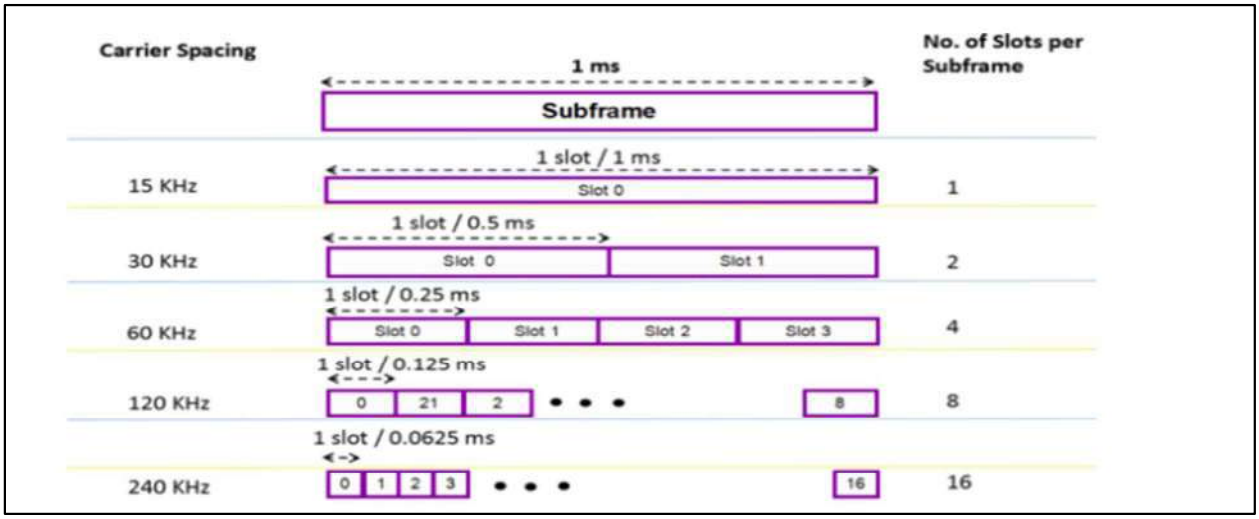
Slot length gets different depending on different subcarrier spacing. There can be 1, 2, 4, 8, or 16 slots per subframe






### 1.5 Sub carrier spacing

| $\mu$ | $N^{slot}_{symb}$ | $N^{frame,\mu}_{slot}$ | $N^{subframe,\mu}_{slot}$ |
|---|---|---|---|
| 0 | 14 | 10 | 1 |
| 1 | 14 | 20 | 2 |
| 2 | 14 | 40 | 4 |
| 3 | 14 | 80 | 8 |
| 4 | 14 | 160 | 16 |



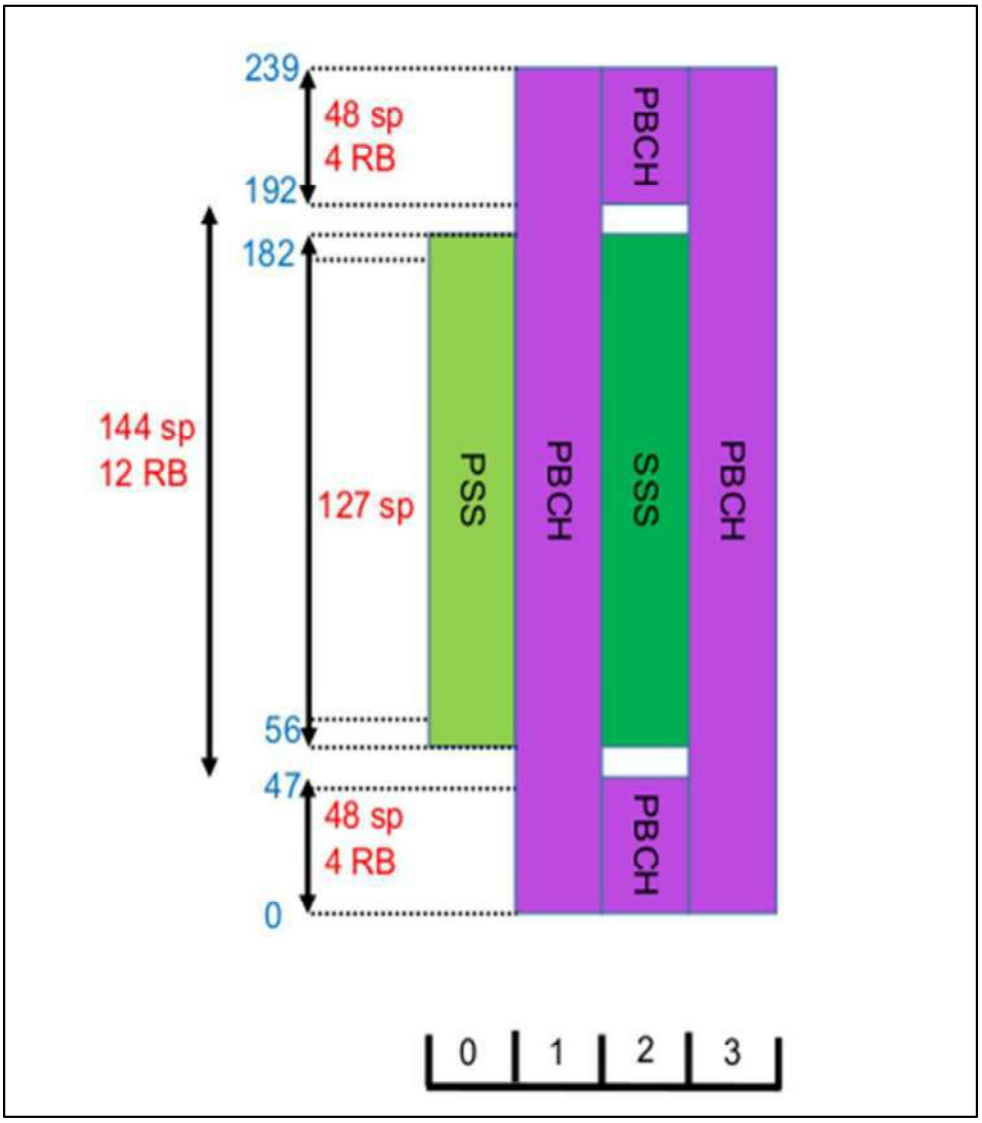
**1.6 SSB**

The SSB block consists of 4 OFDM symbols in the time domain and 240 subcarriers in the frequency domain.

The PSS signal is located on the symbol 0 and is spread over 127 subcarriers.

The SSS signal is located on symbol 2 and is spread over 127 subcarriers.

The PBCH signal is mapped on 240 subcarriers at symbol 1 and symbol 3 and split on two frequency blocks at symbol.



### 1.7 5G Logical, Transport and Physical channels across the RAN protocol stacks

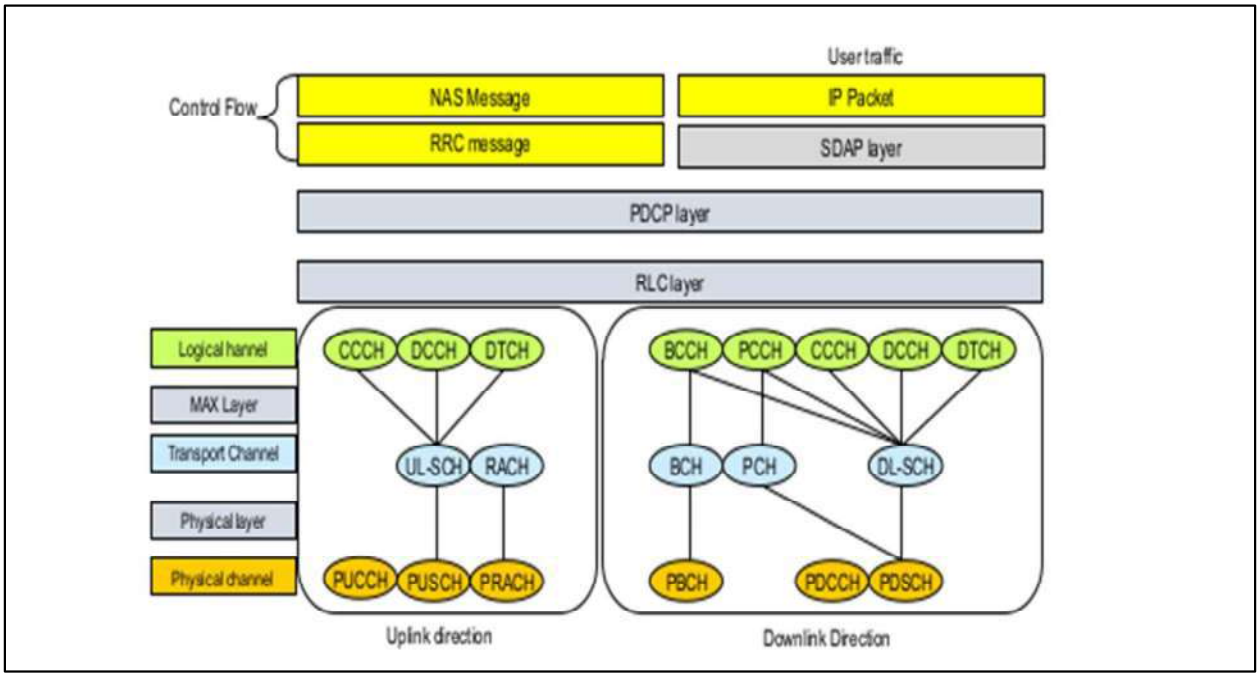
Three types of channels are defined

1-logical channel : What kind of information will be transmitted(C/T) B/WRLC and MAC

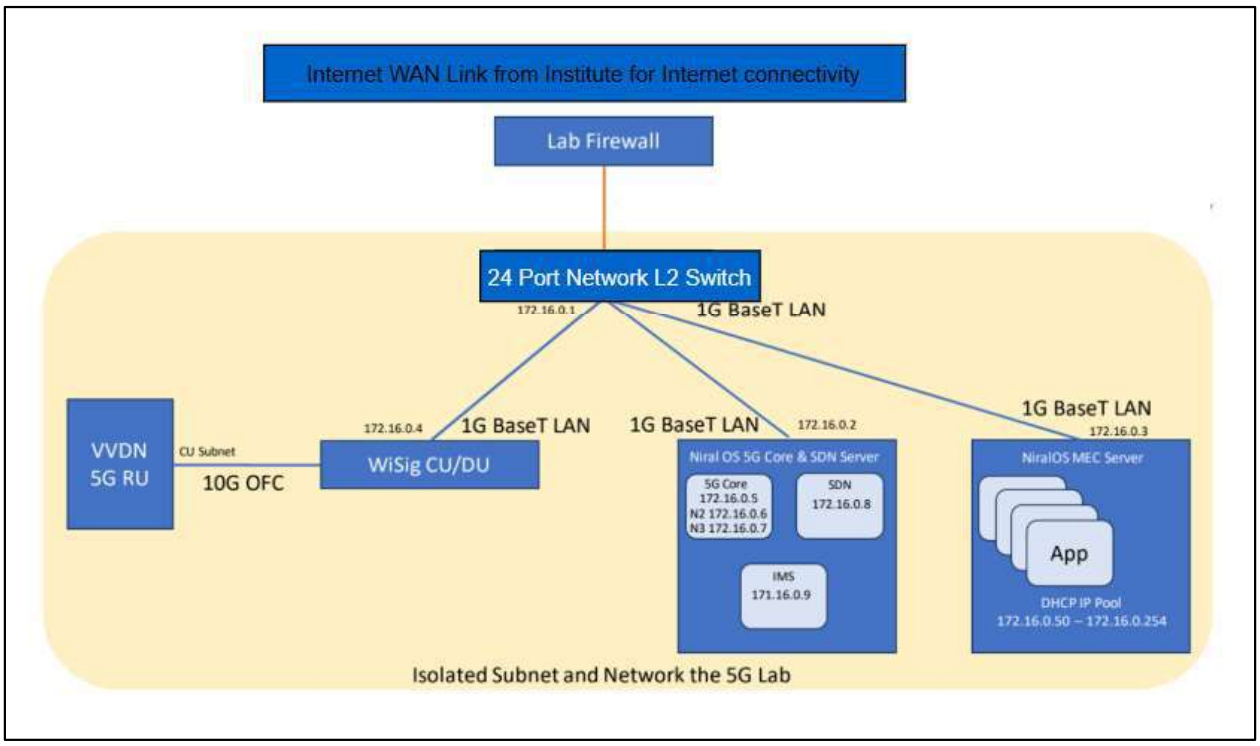
2-Transport channel : Defines How this information will be transmitted B/W the MAC and physical layer.

3-Physical channel is carrying all higher layer information over the Air-interface, Higher layer information will be transmitted within Radio -Frame in the form of PHY Channel.

- Mapping between Logical, Transport and Physical channels



## 1.8 5G LAB ARCHITECTURE & OVERVIEW TCIL LAB



### 1.9 System In 5G LAB Setup

### 1.10 Radio access network (VVDN LPRU +CU/DU)

#### 1.10.1 VVDN LPRU

### 1.10.2 Low Power Radio Unit (LPRU)

The Low Power Radio Unit (LPRU) is a single-board Optical to Radio interface solution for 5G low-power RU application, The RU has a digital and RF section realized in a single board, and the system can be used for improving 5G network coverage and can be cascaded into multiple stages.

### 1.10.3 Key Features and Specifications:

- Deployment Type: Indoor RU
- Radio Configuration: 1 CC, 100 MHz, 4Tx4R,
- IBW :200MHz
- OBW : 100MHz
- Supported Frequency bands 3.6-3.8GHz, 3.4-3.6GHz, 3.55-3.7GHz, 3.8-4.2GHz, 4.7-4.9GHz
- Mode: Single Numerology, SCS 30KHz
- Antenna: Semi-directional / External
- Front haul: 10G over SFP+
- Powered using PoE++ Type 3 (IEEE PoE standard, 802.3bt) or 12V/5A DC Adaptor
- RF Power: 4x250mW
- O-RAN Split: 7-2a

### 1.10.4 Device Specification:

| Radio Specification | |
|---|---|
| Band | 3.6-3.8GHz, 3.4-3.6GHz, 3.55-3.7GHz, 3.8-4.2GHz, 4.7-4.9GHz |
| Bandwidth | 100MHz |
| Duplex Mode | TDD |
| Modulation | QAM 256/64/16/QPSK |
| Carriers | 1 |
| Sub Carrier Spacing | 30KHz |
| RF Output power | 24dBm/ 250mW |
| **Connectivity Specification** | |
| Data Link | 10G SFP+ |
| Interface Protocol O-RAN | FH Split 7.2 |
| **Power Specification** | |
| Powering | PoE++ Type 3, IEEE802.3bt or 12V/5A DC Adaptor |
| Max Power Consumption | 51W (max) |
| **Mechanical Specification** | |
| Design Shape | Square |
| Product Dimension | 250x213.5x92 mm |
| Weight | 5.3 Kg (With mounting bracket) |
| **Antenna Specification** | |
| Antenna | External / Internal Antenna |
| Type | Semi Directional |
| Gain | 6-9dBi |
| MIMO | 4x4 (4T4R) |
| **Environment Specification** | |
| Ambient Temperature | 0°C to +45°C |
| Storage Temperature | -40°C to +55°C |
| Cold Start Temperature | -20°C to +55°C |
| Humidity | 5% to 95% |
| Environmental | IP31/ IP65 complaint |
| Operating Environment | Indoor/Outdoor equipment |
| Thermal dissipation | Convection |
| Installation Type | Ceiling or Wall mount |

### 1.10.5 LPRU LED Notification

For all good conditions, all the Green LED will be in a stable state.

| LED Blinking Status | Power LED Indications | Colour |
|---|---|---|
| Slow | Temperature Warning | RED |
| Fast | Factory Reset | |
| Stable | Initialisation Fault | |
| Slow | Software Upgrade Process | Green |
| Stable | RU Booted Successfully | |

| Colour | Network LED | | Status LED | |
|---|---|---|---|---|
| RED | Slow | DHCP Failure | Slow | PTP Sync Fail |
| | Fast | Factory Reset | Fast | Factory Reset |
| | Stable | Link Down | Stable | Software Upgrade Fail |
| Green | Slow | Software Upgrade process | Slow | Software Upgrade process |
| | Fast | Link UP, Trying DHCP | Fast | Reserved for Future |
| | Stable | Link UP + DHCP Ready | Stable | PTP UP |
| Blue | Stable | Power failure | | |
| | stable | Power on | | |
| | Stable | Thermal shutdown | Stable | Thermal shutdown |

### 1.11 LPRU Implementation in 5G LAB :-

#### 1.11.1 RU Configuration

**First login the CU/DU via - ssh username@172.16.0.4**
User - username
Pwd - password
Then type :- sudo su and type password :- wisig321

**Then login the RU as following steps :-**

RU Login :- ssh root@192.168.4.50
Password :- vvdn

```
root@candor:/home/candor# ssh root@192.168.4.50
root@192.168.4.50's password:
root@vvdn lpru:~$
```

**Carrier Active state :**

After synch run the ./active.sh script for active carrier_state (Only reboot case).

Most importantly the Carrier has to be ACTIVATED before sending the

Data :-

■ Carrier state will be INACTIVE by default . After boot up the

operator/user has to activate the carrier once the board synchronizes

with the master and the sync-state is changed to LOCKED from

FREERUN. Please refer to 3.1 for activating the carrier.In the

FREERUN state the RU only accepts INACTIVE as an input for the

leaf <active>. If the operator tries to push ACTIVE in this scenario

an error message indicating the board is in FREERUN will be given

by the netopeer-server of the RU.

■ The <gain> parameter of the tx-array-carriers has to be adjusted

according to the power at which the DU is sending data to the RU.

```
<tx-array-carriers>
  <name>txarraycarrier0</name>
  <absolute-frequency-center>643452</absolute-frequency-center>
  <center-of-channel-bandwidth>3651780000</center-of-channel-bandwidth>
  <channel-bandwidth>100000000</channel-bandwidth>
  <active>ACTIVE</active>
  <gain>0.0</gain>
  <downlink-radio-frame-offset>0</downlink-radio-frame-offset>
  <downlink-sfn-offset>0</downlink-sfn-offset>
</tx-array-carriers>
<rx-array-carriers>
  <name>rxarraycarrier0</name>
  <absolute-frequency-center>643452</absolute-frequency-center>
  <center-of-channel-bandwidth>3651780000</center-of-channel-bandwidth>
  <channel-bandwidth>100000000</channel-bandwidth>
  <active>ACTIVE</active>
  <downlink-radio-frame-offset>0</downlink-radio-frame-offset>
  <downlink-sfn-offset>0</downlink-sfn-offset>
  <gain-correction>0.0</gain-correction>
  <n-ta-offset>25600</n-ta-offset>
</rx-array-carriers>
```

command for Synchronization:

**tail -f /var/log/synctimingptp2.log**

```
[2024-10-04_07:00:09.874488] LOOPCONTROL(101): Update clock drift: -474.07 ns/s, Average: -475.26 ns/s
[2024-10-04_07:00:09.999290] LOOPCONTROL(101): Sync: minimum timestamp difference: 326.0 (ns), variation 0.0, 2 values, 0 outliers
[2024-10-04_07:00:09.999439] LOOPCONTROL(101): Delay: average timestamp difference: 324.0 (ns), variation 1.0, 2 values, 0 outliers
[2024-10-04_07:00:10.000023] LOOPCONTROL(101): error 1 ns, e_variation 1 ns, slope 31.256, s_variation 0, delay 324, synchronized
[2024-10-04_07:00:10.000053] LOOPCONTROL(101): Update clock drift: -475.75 ns/s, Average: -475.24 ns/s
[2024-10-04_07:00:10.124091] LOOPCONTROL(101): Sync: minimum timestamp difference: 321.0 (ns), variation 0.0, 2 values, 0 outliers
[2024-10-04_07:00:10.124233] LOOPCONTROL(101): Delay: average timestamp difference: 323.0 (ns), variation 0.0, 1 values, 0 outliers
[2024-10-04_07:00:10.124833] LOOPCONTROL(101): error -1 ns, e_variation 0 ns, slope -56.088, s_variation 0, delay 322, synchronized
[2024-10-04_07:00:10.124863] LOOPCONTROL(101): Update clock drift: -474.63 ns/s, Average: -475.25 ns/s
[2024-10-04_07:00:10.249690] LOOPCONTROL(101): Sync: minimum timestamp difference: 328.0 (ns), variation 0.0, 2 values, 0 outliers
[2024-10-04_07:00:10.249851] LOOPCONTROL(101): Delay: average timestamp difference: 325.0 (ns), variation 1.0, 2 values, 0 outliers
[2024-10-04_07:00:10.250454] LOOPCONTROL(101): error 1.5 ns, e_variation 1 ns, slope -12.502, s_variation 0, delay 326, synchronized
[2024-10-04_07:00:10.250485] LOOPCONTROL(101): Update clock drift: -476.04 ns/s, Average: -475.24 ns/s
```

**start-up command :**

**sysrepocfg - -edit=vi -d startup (To config the Freq, ARFCN,Bandwidth,PRB,ExacID for MIMO, MAC Address of DU)**

```
<tx-array-carriers>
  <name>txarraycarrier0</name>
  <absolute-frequency-center>643452</absolute-frequency-center>
  <center-of-channel-bandwidth>3651780000</center-of-channel-bandwidth>
  <channel-bandwidth>100000000</channel-bandwidth>
  <active>ACTIVE</active>
  <gain>0.0</gain>
  <downlink-radio-frame-offset>0</downlink-radio-frame-offset>
  <downlink-sfn-offset>0</downlink-sfn-offset>
</tx-array-carriers>
<rx-array-carriers>
  <name>rxarraycarrier0</name>
  <absolute-frequency-center>643452</absolute-frequency-center>
  <center-of-channel-bandwidth>3651780000</center-of-channel-bandwidth>
  <channel-bandwidth>100000000</channel-bandwidth>
  <active>ACTIVE</active>
  <downlink-radio-frame-offset>0</downlink-radio-frame-offset>
  <downlink-sfn-offset>0</downlink-sfn-offset>
  <gain-correction>0.0</gain-correction>
  <n-ta-offset>25600</n-ta-offset>
</rx-array-carriers>
```

After sync, run the ./active.sh script for active carrier_state (Only reboot case).

**To check the counters :**

root@lpru: **cd /etc/scripts**
**./stat_ru.sh**

```
-------------------------RECIEVE COUNTERS DL ----------------------------
                        LAYER-0
------ TOTAL_RX Packets      :  50705
------ RX_DUPLICATE         :  0
------ RX_CORRUPT           :  0
------ RX_LATE              :  0
------ RX_EARLY             :  0
------ RX_ON-TIME           :  51583
                        LAYER-1
------ TOTAL_RX Packets      :  51746
------ RX_DUPLICATE         :  0
------ RX_CORRUPT           :  0
------ RX_LATE              :  0
------ RX_EARLY             :  0
------ RX_ON-TIME           :  52617
                        LAYER-2
------ TOTAL_RX Packets      :  52789
------ RX_DUPLICATE         :  0
------ RX_CORRUPT           :  0
------ RX_LATE              :  0
------ RX_EARLY             :  0
------ RX_ON-TIME           :  53670
                        LAYER-3
------ TOTAL_RX Packets      :  53862
------ RX_DUPLICATE         :  0
------ RX_CORRUPT           :  0
------ RX_LATE              :  0
------ RX_EARLY             :  0
------ RX_ON-TIME           :  54717
-------------------------Receive counter DL C Plane----------------------------
                        LAYER-0
------ TOTAL_RX Packets      :  268333
------ RX_DUPLICATE         :  0
------ RX_CORRUPT           :  0
------ RX_LATE              :  0
------ RX_EARLY             :  0
------ RX_ON-TIME           :  268350
                        LAYER-1
------ TOTAL_RX Packets      :  268353
------ RX_DUPLICATE         :  0
------ RX_CORRUPT           :  0
------ RX_LATE              :  0
------ RX_EARLY             :  0
------ RX_ON-TIME           :  268370
                        LAYER-2
------ TOTAL_RX Packets      :  268374
------ RX_DUPLICATE         :  0
------ RX_CORRUPT           :  0
------ RX_LATE              :  0
------ RX_EARLY             :  0
------ RX_ON-TIME           :  268391
                        LAYER-3
------ TOTAL_RX Packets      :  1878750
------ RX_DUPLICATE         :  0
------ RX_CORRUPT           :  0
------ RX_LATE              :  0
------ RX_EARLY             :  0
------ RX_ON-TIME           :  1878867
```

### 1.11.2 U plane configuration

```
<user-plane-configuration xmlns="urn:o-ran:uplane-conf:1.0">
<low-level-tx-links>
<name>lltxlink0</name>
<processing-element>element1</processing-element>
<tx-array-carrier>txarraycarrier0</tx-array-carrier>
<low-level-tx-endpoint>slltx-endpoint0</low-level-tx-endpoint>
</low-level-tx-links>
<low-level-tx-links>
<name>lltxlink1</name>
<processing-element>element1</processing-element>
<tx-array-carrier>txarraycarrier0</tx-array-carrier>
<low-level-tx-endpoint>slltx-endpoint1</low-level-tx-endpoint>
</low-level-tx-links>
<low-level-tx-links>
<name>lltxlink2</name>
<processing-element>element1</processing-element>
<tx-array-carrier>txarraycarrier0</tx-array-carrier>
<low-level-tx-endpoint>slltx-endpoint2</low-level-tx-endpoint>
</low-level-tx-links>
<low-level-tx-links>
<name>lltxlink3</name>
<processing-element>element1</processing-element>
<tx-array-carrier>txarraycarrier0</tx-array-carrier>

<low-level-tx-endpoint>slltx-endpoint3</low-level-tx-endpoint>
</low-level-tx-links>

<low-level-rx-links>
<name>llrxlink0</name>
<processing-element>element1</processing-element>
<rx-array-carrier>rxarraycarrier0</rx-array-carrier>
<low-level-rx-endpoint>sllrx-endpoint0</low-level-rx-endpoint>
</low-level-rx-links>
<low-level-rx-links>
<name>llrxlink1</name>
<processing-element>element1</processing-element>
```

<rx-array-carrier>rxarraycarrier0</rx-array-carrier>
<low-level-rx-endpoint>sllrx-endpoint1</low-level-rx-endpoint>
</low-level-rx-links>
<low-level-rx-links>
<name>llrxlink2</name>
<processing-element>element1</processing-element>
<rx-array-carrier>rxarraycarrier0</rx-array-carrier>
<low-level-rx-endpoint>sllrx-endpoint2</low-level-rx-endpoint>
</low-level-rx-links>
<low-level-rx-links>
<name>llrxlink3</name>
<processing-element>element1</processing-element>
<rx-array-carrier>rxarraycarrier0</rx-array-carrier>
<low-level-rx-endpoint>sllrx-endpoint3</low-level-rx-endpoint>
</low-level-rx-links>

<low-level-tx-endpoints>
<name>slltx-endpoint0</name>
<compression>
<compression-type>STATIC</compression-type>
<iq-bitwidth>16</iq-bitwidth>
</compression>
<cp-length>352</cp-length>
<cp-length-other>288</cp-length-other>
<offset-to-absolute-frequency-center>0</offset-to-absolute-frequency-center>
<number-of-prb-per-scs>
<scs>KHZ_30</scs>
<number-of-prb>273</number-of-prb>
</number-of-prb-per-scs>
<e-axcid>
<o-du-port-bitmask>65024</o-du-port-bitmask>
<band-sector-bitmask>448</band-sector-bitmask>
<ccid-bitmask>56</ccid-bitmask>

<ru-port-bitmask>7</ru-port-bitmask>
<eaxc-id>1</eaxc-id>
</e-axcid>

```
</low-level-tx-endpoints>
<low-level-tx-endpoints>
<name>slltx-endpoint1</name>
<compression>
<compression-type>STATIC</compression-type>
<iq-bitwidth>16</iq-bitwidth>
</compression>
<cp-length>352</cp-length>
<cp-length-other>288</cp-length-other>
<offset-to-absolute-frequency-center>0</offset-to-absolute-frequency-center>
<number-of-prb-per-scs>
<scs>KHZ_30</scs>
<number-of-prb>273</number-of-prb>
</number-of-prb-per-scs>
<e-axcid>
<o-du-port-bitmask>65024</o-du-port-bitmask>
<band-sector-bitmask>448</band-sector-bitmask>
<ccid-bitmask>56</ccid-bitmask>
<ru-port-bitmask>7</ru-port-bitmask>
<eaxc-id>2</eaxc-id>
</e-axcid>
</low-level-tx-endpoints>
<low-level-tx-endpoints>
<name>slltx-endpoint2</name>
<compression>
<compression-type>STATIC</compression-type>
<iq-bitwidth>16</iq-bitwidth>
</compression>
<cp-length>352</cp-length>
<cp-length-other>288</cp-length-other>
<offset-to-absolute-frequency-center>0</offset-to-absolute-frequency-center>
<number-of-prb-per-scs>
<scs>KHZ_30</scs>
<number-of-prb>273</number-of-prb>
</number-of-prb-per-scs>
<e-axcid>
<o-du-port-bitmask>65024</o-du-port-bitmask>
```

<band-sector-bitmask>448</band-sector-bitmask>
<ccid-bitmask>56</ccid-bitmask>
<ru-port-bitmask>7</ru-port-bitmask>
<eaxc-id>3</eaxc-id>
</e-axcid>

</low-level-tx-endpoints>
<low-level-tx-endpoints>
<name>slltx-endpoint3</name>
<compression>
<compression-type>STATIC</compression-type>
<iq-bitwidth>16</iq-bitwidth>
</compression>
<cp-length>352</cp-length>
<cp-length-other>288</cp-length-other>
<offset-to-absolute-frequency-center>0</offset-to-absolute-frequency-center>
<number-of-prb-per-scs>
<scs>KHZ_30</scs>
<number-of-prb>273</number-of-prb>
</number-of-prb-per-scs>
<e-axcid>
<o-du-port-bitmask>65024</o-du-port-bitmask>
<band-sector-bitmask>448</band-sector-bitmask>
<ccid-bitmask>56</ccid-bitmask>
<ru-port-bitmask>7</ru-port-bitmask>
<eaxc-id>4</eaxc-id>
</e-axcid>
</low-level-tx-endpoints>

<low-level-rx-endpoints>
<name>sllrx-endpoint0</name>
<compression>
<compression-type>STATIC</compression-type>
<iq-bitwidth>16</iq-bitwidth>
</compression>
<cp-length>352</cp-length>
<cp-length-other>288</cp-length-other>

<offset-to-absolute-frequency-center>0</offset-to-absolute-frequency-center>
<number-of-prb-per-scs>
<scs>KHZ_30</scs>
<number-of-prb>273</number-of-prb>
</number-of-prb-per-scs>
<ul-fft-sampling-offsets>
<scs>KHZ_30</scs>
</ul-fft-sampling-offsets>
<e-axcid>
<o-du-port-bitmask>65024</o-du-port-bitmask>
<band-sector-bitmask>448</band-sector-bitmask>
<ccid-bitmask>56</ccid-bitmask>
<ru-port-bitmask>7</ru-port-bitmask>
<eaxc-id>1</eaxc-id>

</e-axcid>
<non-time-managed-delay-enabled>true</non-time-managed-delay-enabled>
</low-level-rx-endpoints>
<low-level-rx-endpoints>
<name>sllrx-endpoint1</name>
<compression>
<compression-type>STATIC</compression-type>
<iq-bitwidth>16</iq-bitwidth>
</compression>
<cp-length>352</cp-length>
<cp-length-other>288</cp-length-other>
<offset-to-absolute-frequency-center>0</offset-to-absolute-frequency-center>
<number-of-prb-per-scs>
<scs>KHZ_30</scs>
<number-of-prb>273</number-of-prb>
</number-of-prb-per-scs>
<ul-fft-sampling-offsets>
<scs>KHZ_30</scs>
</ul-fft-sampling-offsets>
<e-axcid>
<o-du-port-bitmask>65024</o-du-port-bitmask>
<band-sector-bitmask>448</band-sector-bitmask>

<ccid-bitmask>56</ccid-bitmask>
<ru-port-bitmask>7</ru-port-bitmask>
<eaxc-id>2</eaxc-id>
</e-axcid>
<non-time-managed-delay-enabled>true</non-time-managed-delay-enabled>
</low-level-rx-endpoints>
<low-level-rx-endpoints>
<name>sllrx-endpoint2</name>
<compression>
<compression-type>STATIC</compression-type>
<iq-bitwidth>16</iq-bitwidth>
</compression>
<cp-length>352</cp-length>
<cp-length-other>288</cp-length-other>
<offset-to-absolute-frequency-center>0</offset-to-absolute-frequency-center>
<number-of-prb-per-scs>
<scs>KHZ_30</scs>
<number-of-prb>273</number-of-prb>
</number-of-prb-per-scs>
<ul-fft-sampling-offsets>
<scs>KHZ_30</scs>
</ul-fft-sampling-offsets>
<e-axcid>

<o-du-port-bitmask>65024</o-du-port-bitmask>
<band-sector-bitmask>448</band-sector-bitmask>
<ccid-bitmask>56</ccid-bitmask>
<ru-port-bitmask>7</ru-port-bitmask>
<eaxc-id>3</eaxc-id>
</e-axcid>
<non-time-managed-delay-enabled>true</non-time-managed-delay-enabled>
</low-level-rx-endpoints>
<low-level-rx-endpoints>
<name>sllrx-endpoint3</name>
<compression>
<compression-type>STATIC</compression-type>
<iq-bitwidth>16</iq-bitwidth>

```
</compression>
<cp-length>352</cp-length>
<cp-length-other>288</cp-length-other>
<offset-to-absolute-frequency-center>0</offset-to-absolute-frequency-center>
<number-of-prb-per-scs>
<scs>KHZ_30</scs>
<number-of-prb>273</number-of-prb>
</number-of-prb-per-scs>
<ul-fft-sampling-offsets>
<scs>KHZ_30</scs>
</ul-fft-sampling-offsets>
<e-axcid>
<o-du-port-bitmask>65024</o-du-port-bitmask>
<band-sector-bitmask>448</band-sector-bitmask>
<ccid-bitmask>56</ccid-bitmask>
<ru-port-bitmask>7</ru-port-bitmask>
<eaxc-id>4</eaxc-id>
</e-axcid>
<non-time-managed-delay-enabled>true</non-time-managed-delay-enabled>
</low-level-rx-endpoints>
<tx-array-carriers>
<name>txarraycarrier0</name>
<absolute-frequency-center>[ARFCN]</absolute-frequency-center>
<center-of-channel-bandwidth>[CENTER FREQUENCY]</center-of-channel-bandwidth>
<channel-bandwidth>[BANDWIDTH]</channel-bandwidth>
<active>ACTIVE</active>
<gain>0</gain>
<downlink-radio-frame-offset>0</downlink-radio-frame-offset>
<downlink-sfn-offset>0</downlink-sfn-offset>
<rw-duplex-scheme>TDD</rw-duplex-scheme>
<rw-type>NR</rw-type>
</tx-array-carriers>

<rx-array-carriers>
<name>rxarraycarrier0</name>
<absolute-frequency-center>[ARFCN]</absolute-frequency-center>
<center-of-channel-bandwidth>[CENTER FREQUENCY]</center-of-channel-bandwidth>
```

<channel-bandwidth>[BANDWIDTH]</channel-bandwidth>
<active>ACTIVE</active>
<downlink-radio-frame-offset>0</downlink-radio-frame-offset>
<downlink-sfn-offset>0</downlink-sfn-offset>
<gain-correction>0</gain-correction>
<n-ta-offset>25600</n-ta-offset>
</rx-array-carriers>
</user-plane-configuration>

***Note: For 4T4R the eAxC ID mapping is PDSCH- 0,1,2,3***
***PUSCH-0,1,2,3***
***PRACH-4,5,6,7***

***ARFCN of center frequency chosen within the band eg : ARFCN : 646667, Freq : 3700.005 Mhz Bandwidth within ibw/obw can be used.***

***Note: We need to push the ACTIVE leaf in the U-plane each time after the RU gets synchronized.***

**Gen3 :- v3.0.5**

**test.xml --->**

```
<lpru_config>
   <bandwidth>100</bandwidth>
   <du_mac_address>00:11:22:33:44:66</du_mac_address>
   <cu_plane_vlan>100</cu_plane_vlan>
   <ud_compression>ON</ud_compression>
   <dl_compression_method>1</dl_compression_method>
   <dl_compression_value>9</dl_compression_value>
   <ul_compression_method>1</ul_compression_method>
   <ul_compression_value>9</ul_compression_value>
   <num_prb>273</num_prb>
   <prach_layer0_PCID>4</prach_layer0_PCID>
   <prach_layer1_PCID>5</prach_layer1_PCID>
   <prach_layer2_PCID>6</prach_layer2_PCID>
   <prach_layer3_PCID>7</prach_layer3_PCID>
   <pxsch_layer0_PCID>0</pxsch_layer0_PCID>
   <pxsch_layer1_PCID>1</pxsch_layer1_PCID>
```

<pxsch_layer2_PCID>2</pxsch_layer2_PCID>
<pxsch_layer3_PCID>3</pxsch_layer3_PCID>
</lpru_config>

## 2.0 Wisig Networks CU/DU

### 2.1 INTRODUCTION

#### 2.1.1 Purpose

The purpose of this document is to submit to the TCIL team as part of the CU/DU Training for the project of 100 5G Use case Labs for Higher Educational institutions and this document provides high level design of the CU-DU software entity which is the part of 5G New radio (NR) network.

#### 2.1.2 Scope

This document describes the design and functionality of the control unit and distributed unit in the gNB protocol Stack for 5G NR (valid for SA mode) developed by WiSig. For simplicity, the control unit will be referred to as CU and the distribution unit will be referred to as DU. This document describes external interfaces and high-level design of the CU-DU Software Entity. This document also elaborates the interconnection between the CU-DU and other parts of NR stack and WiSig NR Common Framework

### 2.1.3 Definitions, Acronyms, and Abbreviations Acronym Explanation

| Acronym | Explanation |
|---|---|
| CU | Control Unit |
| DU | Distributed Unit |
| GTP | GPRS Tunnelling Protocol |
| GTP-U | GTP User Plane |
| IP | Internet Protocol |
| IPv4 | Internet Protocol version 4 |
| IPv6 | Internet Protocol version 6 |
| MeNB | Master eNodeB |
| NR | New radio |
| PDCP | Packet Data Convergence Protocol |
| PDN | Packet Data Network |
| PDU | Protocol Data Unit |
| QFI | Qos flow ID |
| QoS | Quality of Service |
| RAB | Radio Access Bearer |
| RDI | Reflective QoS flow to DRB mapping Indication |
| RQI | Reflective QoS Indication |
| SDAP | Service Data Adaptation Protocol |
| SGW | Serving Gateway |
| UDP | User Datagram Protocol |
| UE | User Equipment |
| RAN | Radio Access Network |
| RU | Radio Unit |
| SMO | Service Management & Orchestration |
| RIC | RAN intelligent controller |
| MAC | Media Access Control |
| QOS | Quality of Service |
| PHY | Physical Layer |
| FAPI | Femto Application Platform Interface |
| NR | New Radio |

## 2.2 Reference Documents

3GPP TS 29.281 V15.6.0 (2019-09) 3rd Generation Partnership Project; Technical Specification Group Core Network and Terminals; General Packet Radio System (GPRS) Tunnelling Protocol User Plane (GTPv1-U)

3GPP TS 37.324 V15.1.0 (2018-09): 3rd Generation Partnership Project; Technical Specification Group Radio Access Network; E-UTRA and NR; Service Data Adaptation Protocol (SDAP) specification (Release 15)

3GPP TS 38.413 V15.5.0 (2019-09): 3rd Generation Partnership Project; Technical Specification Group Radio Access Network; NG-RAN; NG Application Protocol (NGAP) (Release 15)

3GPP TS 38.300 V15.7.0 (2019-09): 3rd Generation Partnership Project; Technical Specification Group Radio Access Network; NR; NR and NG-RAN Overall Description; Stage 2 (Release 15)

3GPP TS 38.415 V15.2.0 (2018-12): 3rd Generation Partnership Project; Technical Specification Group Radio Access Network; NG-RAN; PDU Session User Plane Protocol (Release 15)

3GPP TS 33.501 V 15.0.0 (2018-03): “Security architecture and procedures for 5G system”

3GPP TS 38.300 Version 15.4.0 (2018-03): “NR; NR and NG-RAN Overall Description" Stage 2

3GPP TS 38.322: RLC Protocol Specification, version 15.4.0

3GPP TS 38.323 V15.4.0. (2018-03) “Packet Data Convergence Protocol (PDCP) specification”

3GPP TR 21.905 Version 9.4.0 (2009-12): "Vocabulary for 3GPP Specifications" (Release 9).

GPP TS 38.425 V15.4.0. (2018-06): “NR User Plane Protocol”.

3GPP TS 38.322 Version 15.5.0: NR Radio Link Protocol RLC Protocol Specification

3GPP TS 38.331 Version 15.5.0: NR Radio Resource Control (RRC) Protocol Specification

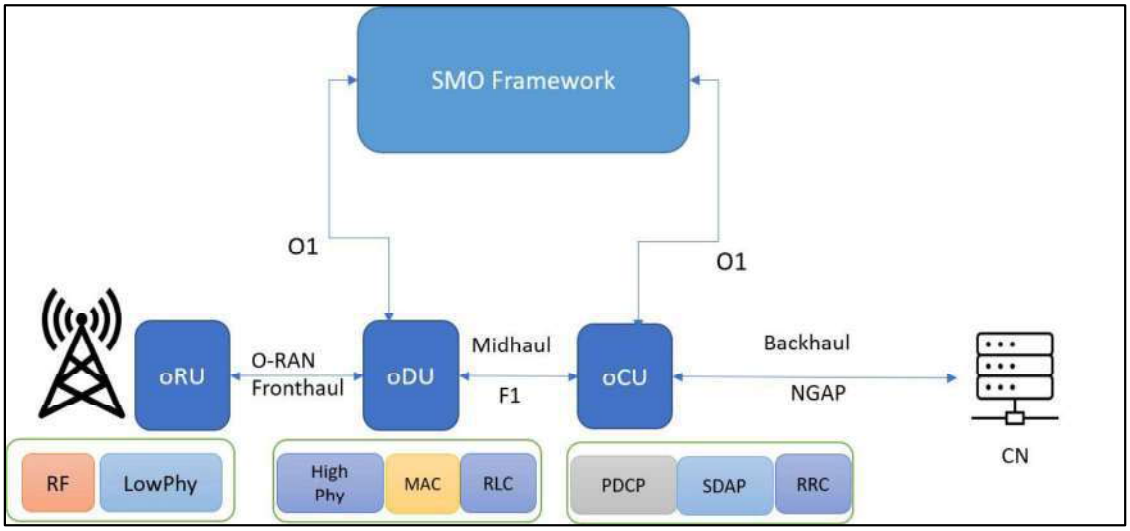
3GPP TS 38.321 Version 15.4.0: NR MAC Protocol Specification

3GPP TS 38.323 Version 15.5.0: NR PDCP Protocol specification

3GPP TS 38.473 Version 15.5.0: NR F1AP Protocol Specification

WiSig Networks 5G gNodeB CU/DU

### 2.3 High-Level System Architecture




The above block diagram illustrates a high level architecture of the 5G Cellular Network with CU, DU, RU, Core Network & SMO components. Following are the key system level specifications of this system.

| | | |
|---|---|---|
| 5G NR Radio, 3GPP Rel-15 | Supporting Access Modes: SA | 7.2x Category A Split |
| Frequency Bands: n78 | MIMO Modes: 4T4R | Upto 250mW/port output power |
| SCS, 30KHz | Bandwidth: upto 100MHz | Use Cases: eMBB |

**5**G NR Radio, 3GPP Rel-15

Supporting Access Modes: SA

7.2x Category A Split Frequency Bands: n78

MIMO Modes: 4T4R

Upto 250mW/port output power SCS, 30KHz

Bandwidth: upto 100MHz

Use Cases: eMBB

## 2.4 System Key Parameters

| System Paramenter | Values |
|---|---|
| Network Architecture | SA |
| Duplexing | TDD |
| 3GPP Compliance | Release 15 |
| Maturity | CDoT TRL-6 |
| Service Management & Orchestration (SMO) | Yes |
| Bandwidth | Upto 100 MHz BW |
| Frequency Band | n78 |
| Numerology | 30KHz |
| MIMO | DL: SU-MIMO upto 4 layers UL: upto 2 layers |
| Modulation UL | UL: QPSK to 64 QAM [CP-OFDM] |
| Modulation DL | DL: QPSK to 256 QAM [CP-OFDM] |
| FH Split | 7.2a ORAN Compliant |
| Synchronization | G 8275.1 (PTP) |
| Transmit Power | 250Milli Watt/Channel |
| ORAN Support | C/U/S Planes |
| Peak Data Rate per Cell per CC | Up to 400-500Mbps (DL) |
| | Up to 20Mbps (UL) |
| # Active Users / Cell | 512 |
| Multi Cell /DU Support | Up to 3 Cells / DU |
| UE Admission Control | Based on ARP |
| Physical Fronthaul | ORAN 7.2x, Category A Two optical 25GbE ports |
| Mobility | Yes |
| ARQ/HARQ | Supported |
| Power Control | Yes |
| Link Adaptation | Yes |
| VoNR | Yes |
| ViNR | Yes |

## 2.5 gNB CU Architecture

**This chapter provides an overview of gNB CU software architecture for gNB framework.**

### 2.5.1 Software Modules

gNB CU Framework components include the following:

- CU OAM
- NR Layer 3
- NR RRM

### 2.5.2 CU OAM

CU OAM interacts with NR RRM, NR SON, NR Layer3, NR Packet-Relay and NR PDCP layers of gNB/SgNB and provides configuration related data to the modules by reading xml files.

### 2.5.3 NR Layer 3

NR Layer3 from here onwards will be referred to as L3 in this document. L3 comprises of RRC, F1AP-CU layer, NGAP layer, XNAP layer and ASN Encoder/Decoder
The Radio Resource Control protocol layer of gNB is used to configure and control the radio resource between the gNB and UE. The main function of L3 layer is to establish, release and manage the RRC connection between UE and gNB.
F1AP Protocol layer provides the signalling service between CU and DU
SA Mode: NGAP Protocol layer provides the signalling service between gNB and AMF.
SA Mode: XNAP Protocol layer provides the signalling service between two gNBs. XNAP services are divided into two groups:
o XNAP UE associated procedures.
o XNAP global procedure.

### 2.5.4 NR RRM

NR RRM shall be referred to as RRM from here onwards in this document. RRM (Radio Resource Management), consists of three elements.
RRM management Interface Handler,
Cell Manager,
UE Manager
RRM Module provides Radio Resource Management functions for

Cell setup,
UE admission control,
PDU Session setup

### 2.5.5 PDCP and Packet Relay

CU L2 receives downlink data from serving gateway, process the data and sends to lower layers. Packet relay acts as mapping entity or interceptor between eGTPu and PDCP. Major function and services of PDCP layer include:
Control Plane signalling
RLC AM/UM Mode
Ciphering/Deciphering for SRB and DRB
UL/DL Data transfer procedures
Integrity Protection for SRB and DRB
SDAP Functionality (for SA mode)

## 2.6 gNB DU Architecture

**This chapter provides an overview of gNB DU software architecture for gNB framework.**

### 2.6.1 Software Modules

gNB ESW Framework components include the following:
DU OAM
DU MGR
DU F1U
NR RLC
NR MAC

### 2.6.2 DU OAM

DU OAM spawns other DU Process and performs initialization / provisioning. It reads the parameters from XML configuration files which are required for initialization and provisioning of different layers in 5G gNB DU stack (DUMGR, RLC, MAC and Packet Relay).

### 2.6.3 DU MGR

DU MGR consists of sub-modules performing different tasks related to Cell management, UE management and resource allocation at the cell level. Following are the key tasks.
It interacts with DU OAM for configuration and provisioning related tasks.
Functionalities related to Cell Configuration / Cell Reconfiguration / cell start / cell stops are handled by DU MGR

It handles the procedures related to a UE for e.g., gNB/SgNB Addition procedure for a UE.
It interacts with the CU (Control Unit) to exchange the F1AP messages over the SCTP (Linux kernel provided lksctp) interface.

### 2.6.4 DU F1U

DU F1U module is a user plane module which interacts with CU to exchange DRB data packets over the F1U interface. It takes the data from RLC and provides that to CU and vice-versa.

### 2.6.5 NR MAC

The major functions and services of MAC layer include.
Mapping between logical channels and transport channels.
Multiplexing/demultiplexing of MAC SDUs belonging to one or different logical channels into/from transport blocks (TB) delivered to/from the physical layer on transport channels.
Error correction through HARQ
Priority handling between UEs by means of dynamic scheduling.
Priority handling between logical channels of one UE
Transport format selection.

### 2.6.6 NR RLC

The major functions and services of MAC layer include.
Creation and deletion of AM/UM Entities
UM Data Transfers
AM Data transfer, including indication of successful delivery of upper layers SDU(s).

## 2.7 Interfaces

| Interface | Components Connected | Layers Involved | Protocols | Functions |
|---|---|---|---|---|
| **NG** | gNB ↔ 5G Core Network (5GC) | Control Plane (NG-C), User Plane (NG-U) | NGAP (control), GTP-U (user | Control signaling, user data transport |
| **F1** | CU ↔ DU | Control Plane (F1-C), User Plane (F1-U) | F1AP (control), GTP-U (user) | RAN control, user data transport |
| **E1** | CU-CP ↔ CU-UP | Control Plane | E1AP | Session management, control-user plane split |
| **Front haul** | DU ↔ RU | PHY Layer | eCPRI, ORAN Fronthaul | Real-time signal processing, beamforming, RF control |
| **Xn** | gNB ↔ gNB | Control Plane (Xn-C), User Plane (Xn-U) | XnAP (control), GTP-U (user) | Inter-gNB handovers, user data forwarding |

## 2.8 CU-DU Hardware Specifications

- **Make:** Intel Icelake Supported Server
- **CPU:** Intel ICX 6338N 1P 32C/64T 2.2GHz 185W
- **DDR :** 128GB ECC DDR4 3200 MHz
- **ACC100:**Intel MT Bryce JJACC100 S RKQD
- **SSD:**M.2 SSD SATA 1000GB
- **GPS:**GPS Module
- **NIC:** Intel CAM1 E-810
- **Dimensions H * W * D :**44.45 mm x 438 mm x 380 mm

**(Figure:CU-DU server)**

### 2.9 Bring-up of CU-DU Hardware

#### 2.9.1 Access the CU-DU server

● Connect the FH and BH ports according to the following diagram. The image contains the Server IP/BH, BMC IP, and FH IP.
WiSig Networks 5G gNodeB CU/DU

● Once the Server is deployed in Network, Access the CU-DU server using the following credentials

**ssh username@172.16.0.4**
**password: wisig321**

● Note: All the usernames, BMC IP, DU Virtual MAC MAC addresses are mentioned in the link below with corresponding states and addresses.
● URL List of deployed servers

### 2.9.2 CU/DU/L1 Installations

**Prerequisites**

- Below is the software version of the Prerequisites to install CU, DU and L1.
- Download and Install the following Ubuntu version on the server
- UbuntuOS version: Ubuntu 22.04.x LTS ( Jammy Jellyfish)
- Kernel version: 5.15.0-x-realtime

### 2.9.3 L1 Prerequisites

1. Install the Linux kernel tools package corresponding to a specific version of the real-time (RT) Linux kernel.
2. This package provides various utilities and tools that can be useful for performance analysis, debugging, and monitoring of the real-time kernel.

**$ apt install linux-tools-5.15.0-1030 realtime**

- Install the "build-essential" package on a Linux system. The "build-essential" package includes a collection of essential software tools and libraries required for compiling and building software from source code.

**$ apt install build-essential**

- Install the "libnuma-dev" package on a Linux system. This package provides development files and headers for the libnuma library, which is used to manage Non-Uniform Memory Access (NUMA) systems.

**$ apt install libnuma-dev**

- Install the DPDK and OneAPI by using the following procedure

- DPDK_VERSION=22.11.1

**$wget http://fast.dpdk.org/rel/dpdk-${DPDK_VERSION}.tar.xz && mkdir dpdk-${DPDK_VERSION} && **

**$tar xvf dpdk-${DPDK_VERSION}.tar.xz**

- OneAPI

**wget -O-** https://apt.repos.intel.com/intel-gpg-keys/GPG-PUB-KEY-INTEL-SW-PRODUCTS.PUB \
| gpg --dearmor | sudo tee /usr/share/keyrings/oneapi-archive-keyring.gpg > /dev/null && \
echo "deb [signed-by=/usr/share/keyrings/oneapi-archive-keyring.gpg] https://apt.repos.intel.com/oneapi all main" \
| sudo tee /etc/apt/sources.list.d/oneAPI.list && \
apt update -y && \
apt install -y intel-basekit intel-hpckit

### 2.9.4 CU-DU Installation process

i.apt update && apt install -y libncurses5-dev libxml2-dev git libyang2-dev libyang2- tools libsysrepo-dev libnetconf2-dev \

cmake lsb-core ninja-build pkg-config

wget meson python3-pyelftools libsctp-dev curl libcurl4-openssl-dev && \

ln -s /usr/include/libxml2/libxml /usr/include/libxml && \

ldconfig

ii. DPDK_VERSION=22.11.1
$wget http://fast.dpdk.org/rel/dpdk-${DPDK_VERSION}.tar.xz && \ rm -rf dpdk-${DPDK_VERSION} && mkdir -p dpdk- ${DPDK_VERSION} && \ tar xvf dpdk-${DPDK_VERSION}.tar.xz -C dpdk-${DPDK_VERSION} --strip-components 1 && \
rm dpdk-${DPDK_VERSION}.tar.xz

Check all service active or not

### 2.9.5 Instructions same for all service Checking

i. Review the service, status information displayed. If the service is active and running, no further action is required.

If the service is inactive type the following command to activate the service

**A. checklist covering all services**

Verify the status of the following services to ensure they are active. If any service is found inactive,
execute the corresponding commands to start the services

**i. Check ts2phc service is active or not**

$ sudo systemctl status ts2phc.service
Review the service status information displayed. If the service is active and running, no further action is required. If the service is inactive type the following command to activate the service
$ sudo systemctl restart ts2phc.service

**ii. Check PTP service is active or not**

$ sudo systemctl status ptp4l.service
Review the service status information displayed. If the service is active and running, no further action is required. If the service is inactive type the following command to activate the service
$ sudo systemctl restart ptp4l.service

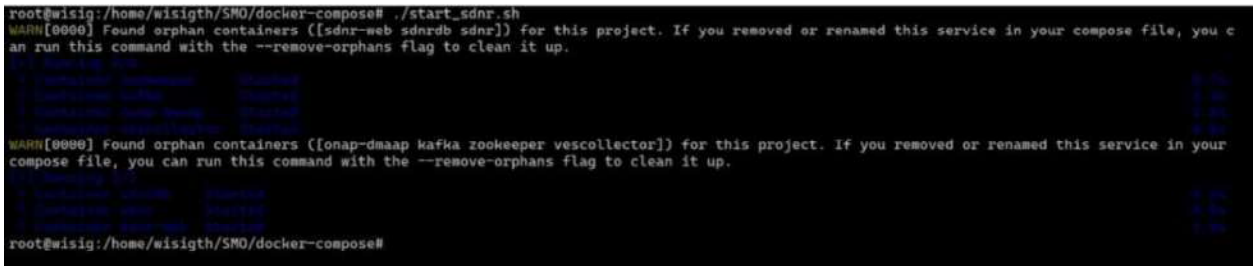
**iii. Check PHC service is active or not**

$ sudo systemctl status phc2sys.service
Review the service status information displayed. If the service is active and running, no further action is required. If the service is inactive type the following command to activate the service
$ sudo systemctl restart phc2sys.service

## 2.10 CU/DU Bring-up Using SMO

**Start Docker application steps**

i. Open the terminal and execute the following script from the "/home/username/SMO/docker-compose"
$ sudo su
$ ./start_sdnr.sh




**(Figure 2:sdnr_logs)**

ii. Check status

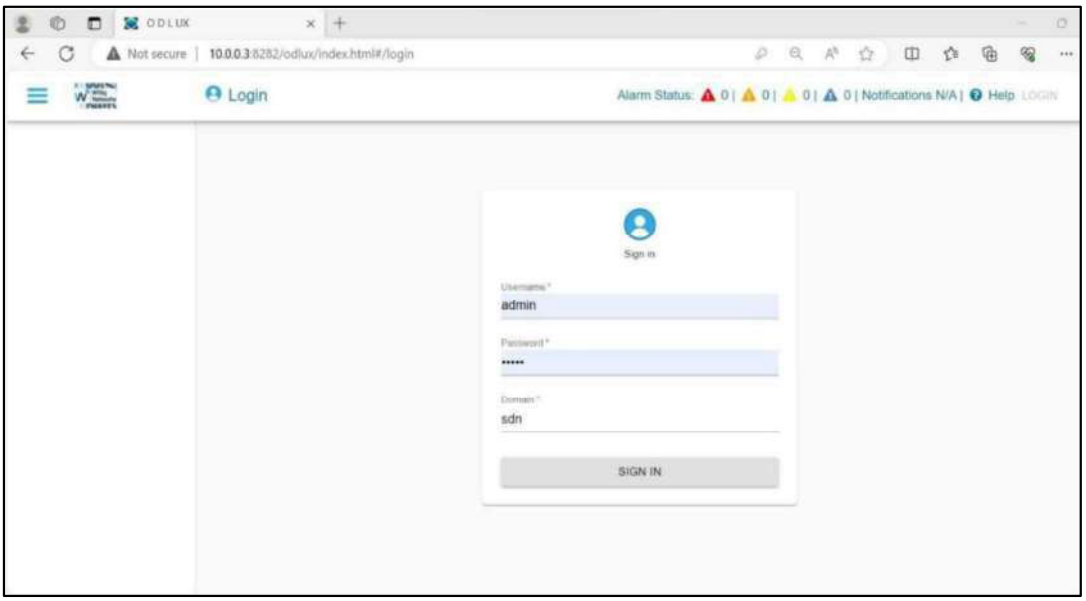
$ docker ps

Note: It brings up all containers

Looking into the SDNR logs will wait until the certificate installation script has finished executing.

$ docker logs -f sdnr

```
root@wisig:/home/wisigth/SMO/docker-compose# docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED        STATUS          PORTS                                                
```




**(Figure 5: Admin page)**

### 2.11.2 To run and execute gNB O1 Handler

● Go to the terminal and execute the gNB O1Handler, run the following script from the "/home/username/SMO/gNB_O1_hdl/bin" directory using the sudo command:

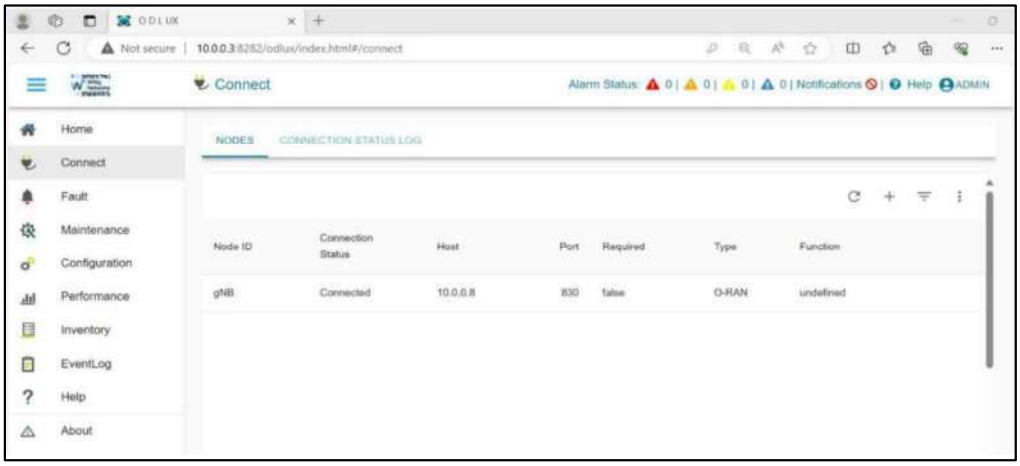
$ sudo ./o1_handler_startup.sh

**(Figure 6: O1_handler_logs)**

### 2.11.3 To start the gNB process

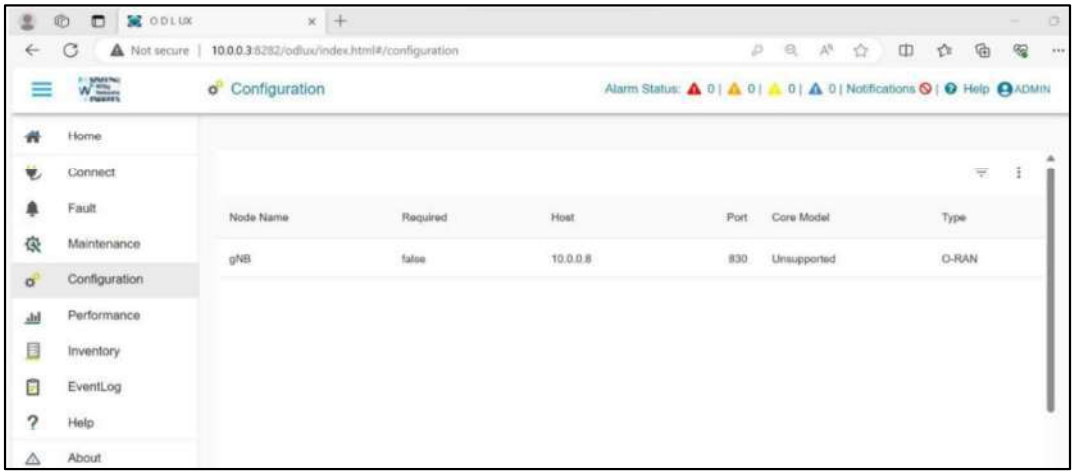
● To start the gNB process, verify that the gNB is in a connected state. If it is not automatically connected, follow the manual steps provided in section 5.3.6.

● Once connected, navigate to the Configuration page and click on the gNB. Within the gNB, locate the "proprietary-system."




**(Figure 7: gNB connection_status)**

**Note:** Ensure that the connection status mode is set to CONNECTED. If it is not connected, please follow manual steps.

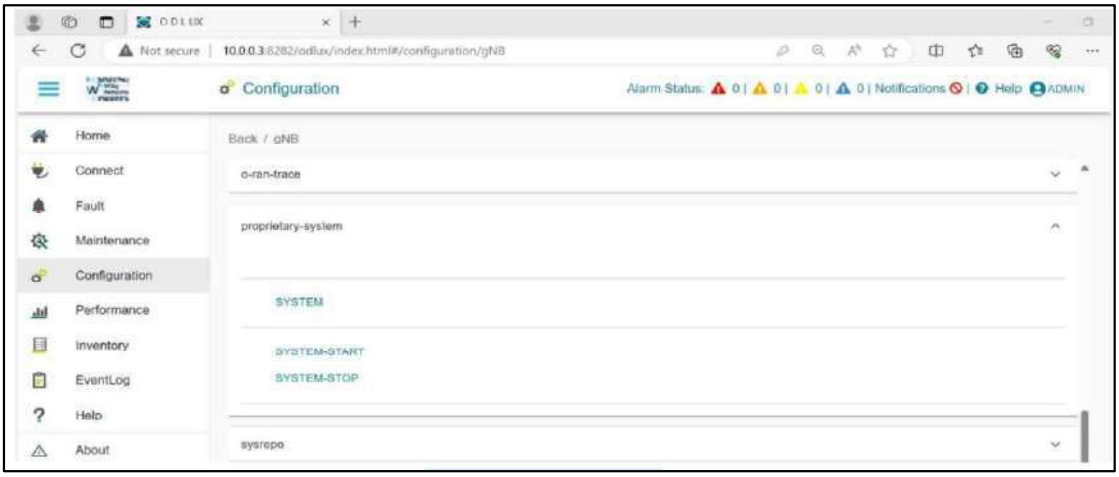



**(Figure 8: gNB connected)**

### 2.11.4 To start or stop the gNB process

Under "proprietary-system," click on "system-start" to initiate the gNB process, including CU, DU,

and L1 or click on "system-stop" to halt the gNB process.




**(Figure 9: System start/stop)**

Note: Don't click the "system-start" multiple times, it will stall the complete execution

### 2.11.5 Camp the UE once all status are running on proprietary-system options

- To check the status navigate to below path
- Go to the configuration tab ->gNB-> proprietary-system->system-monitoring

8.2.6

8.2.6 Manual Steps to Add gNB Node

- To add a gNB node manually, enter the following fields to match the system credentials.

Node ID: gNB

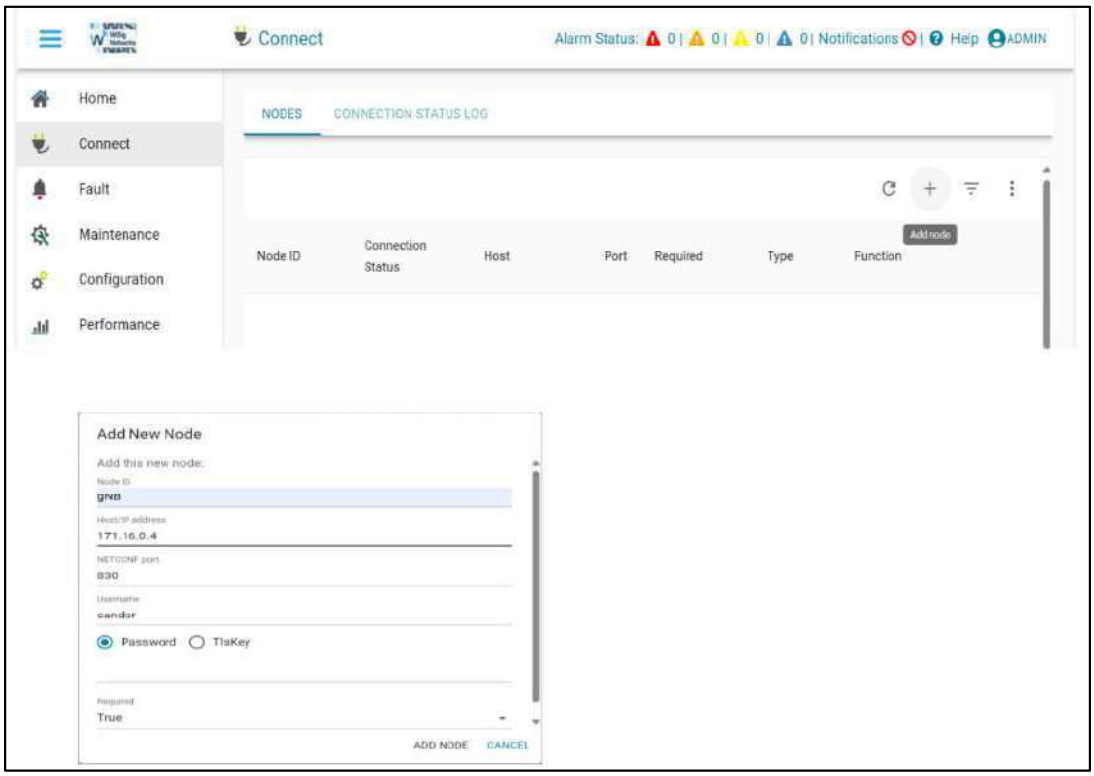
Host/IP address: [Enter the system's IP address]

Netconf Port: [Enter the Netconf port number]

Username: [Enter the username]

WiSig Networks 5G gNodeB CU/DU

Password: [Enter the password]




**(Figure 10:Adding gNB node)**

Note: Ensure that the provided information is accurate to successfully add the gNB node to the system.

### 2.12 To check the RAN logs

#### 2.12.1 To check the L1 logs follow the below steps

$ cd /home/candor/SMO/gNB_O1_hdl/O-RAN/Slot0/runBharatRan/logs
$ tail –f l1_logs.txt

#### 2.12.2 To check the CU logs follow the below steps

$ cd /home/candor/SMO/gNB_O1_hdl/O-RAN/Slot0/runBharatRan/logs
$ tail –f cu_logs.txt

#### 2.12.3 To check the PDCP logs follow the below steps

$ cd /home/candor/SMO/gNB_O1_hdl/O-RAN/Slot0/runBharatRan/logs
$ tail –f pdcp_logs.txt

### 2.12.4 To check the DU logs follow the below steps

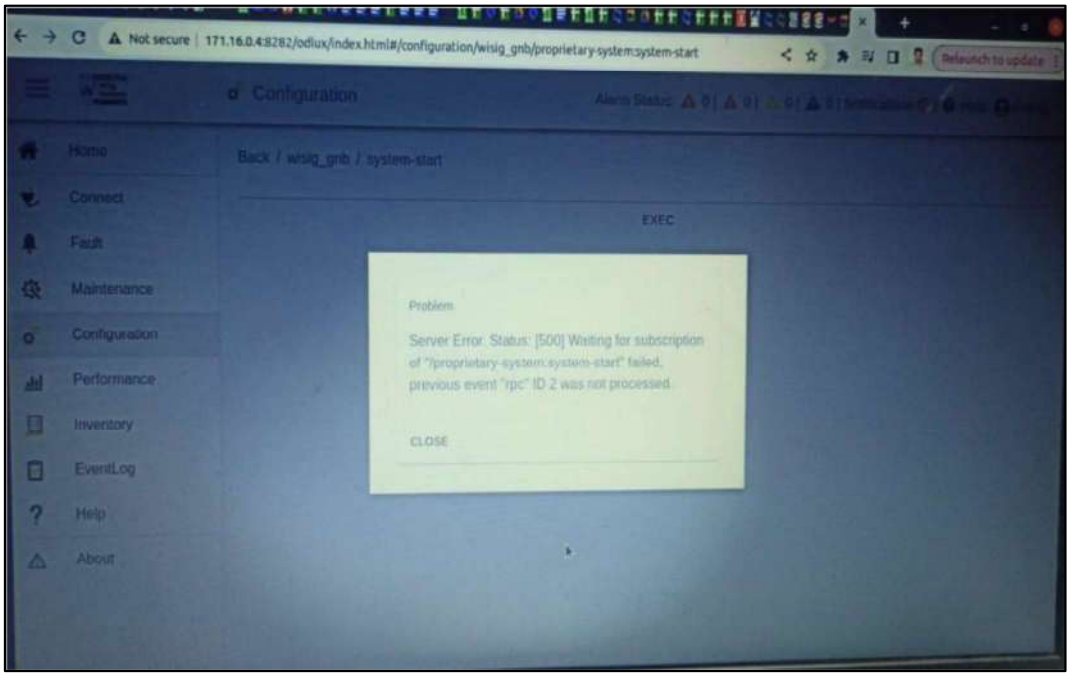
$ cd /home/candor/SMO/gNB_O1_hdl/O-RAN/Slot0/runBharatRan/logs
$ tail –f du_logs.txt

### 2.12.5 WiSig Networks 5G gNodeB CUDU

**A. Diagnose**

- Follow the below steps if you encounter any issues on GUI.
- Stop and start the docker. Refer to Section 5.1 for instructions - $ ./stop_sdnr.sh
- If any Server Error occurs as shown in the below image, type the following command on any of the terminals on the CU-DU server and follow from Section 5.3.

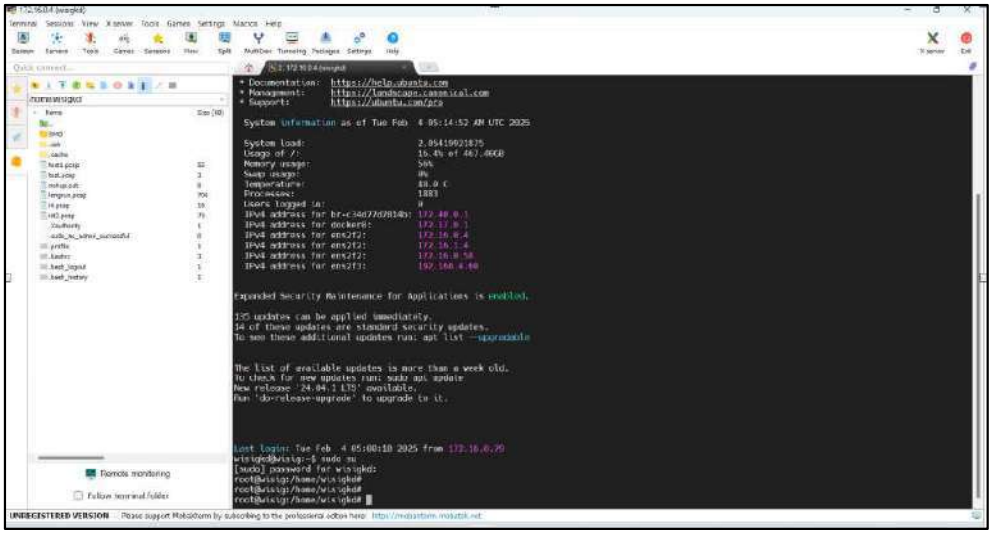
$ sudo rm -rf /dev/shm/sr_*
$ sudo rm -rf /dev/shm/srsub_*




### 2.13 Linux RAN Bring up Process

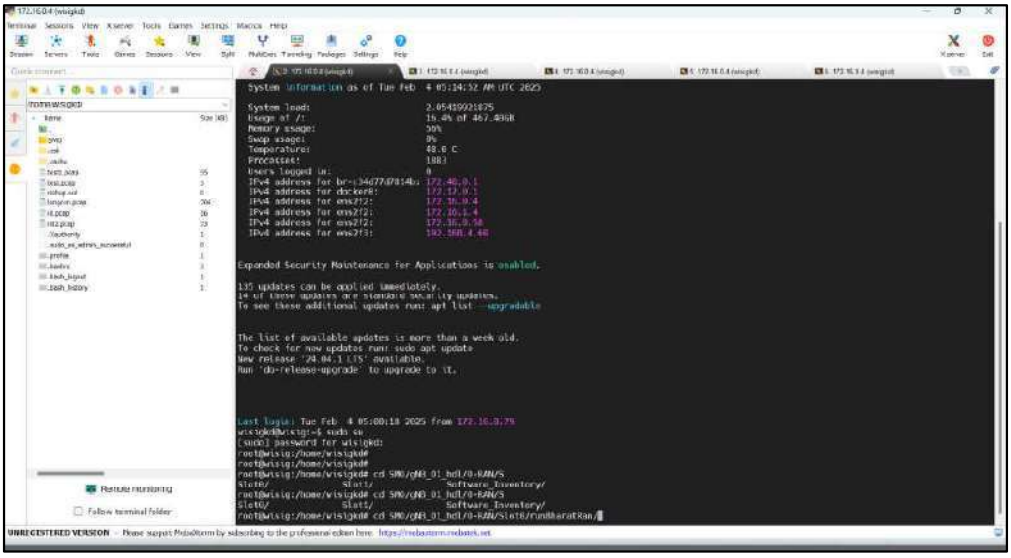
Open 4 Terminal in Linux machine or open Mobaxtem in window machine and login CU/DU Server.

**Username :Given for every institute at time of installation**
**Password:wisig321**

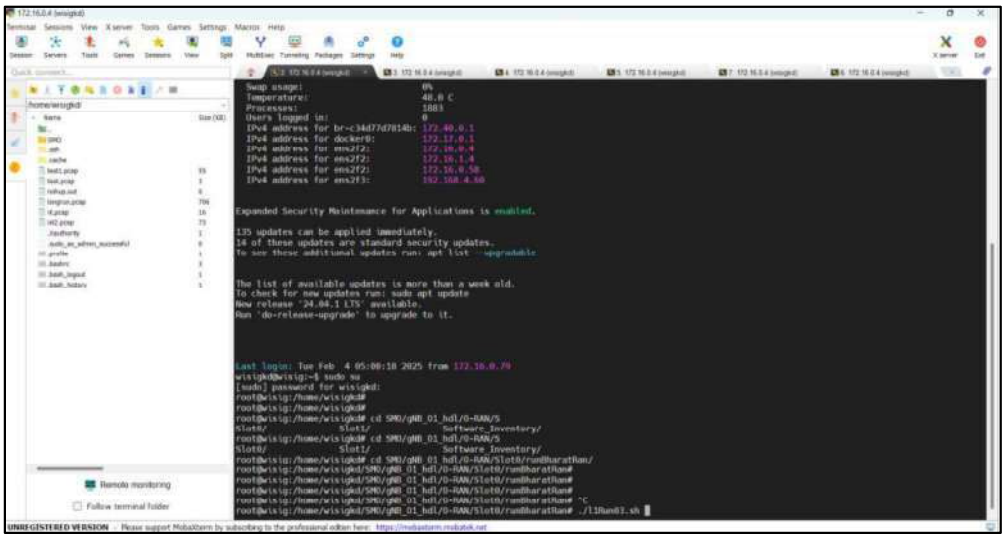


Run mentioned command in all 4Terminal as below

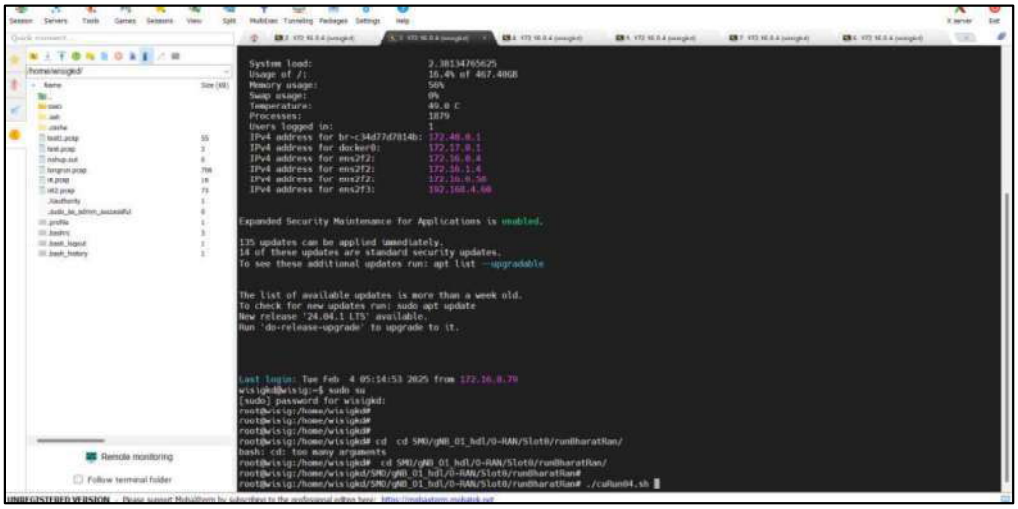
cd /SMO/gNB_O1_hdl/O-RAN/Slot0/runBharatRan



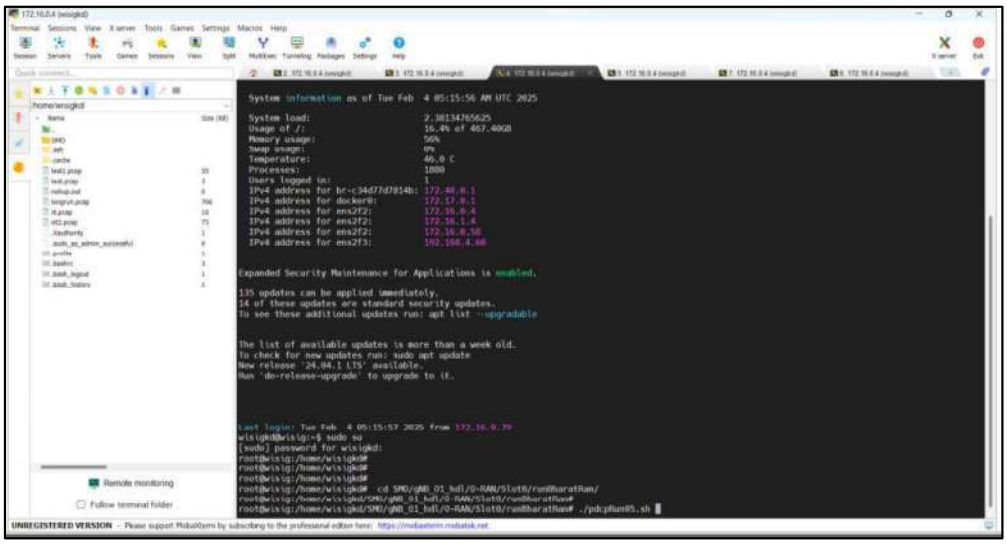
**Run l1 in first Terminal ./l1Run03.sh**



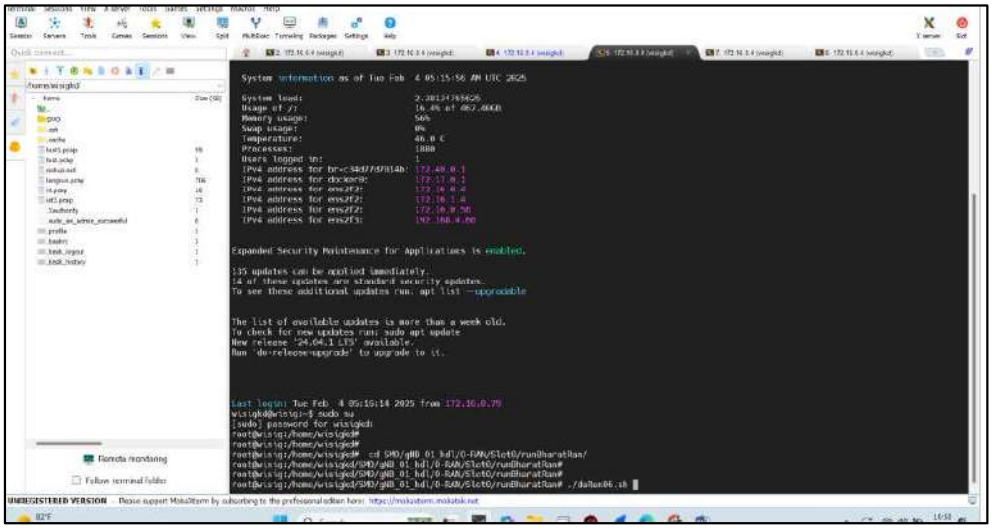
**Run ./cuRun04.sh in 2nd Terminal**



**Run *./pdcpRun05.sh* in 3rd Terminal**



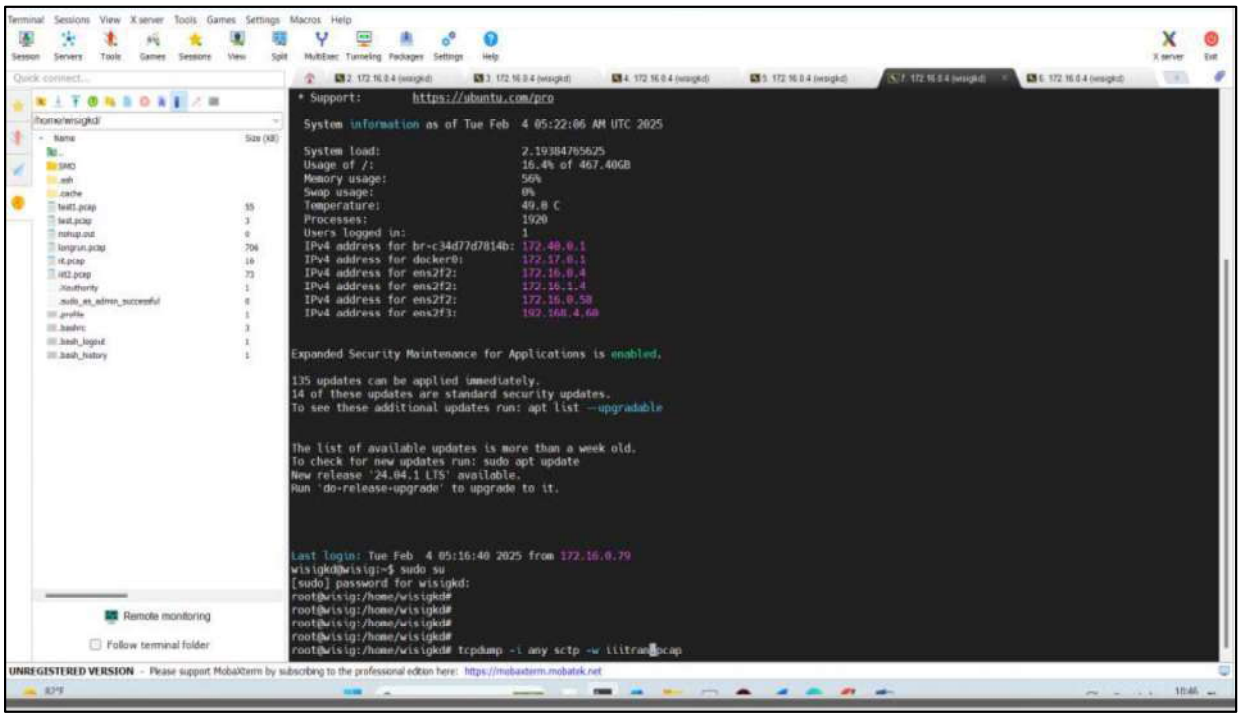
**Finally run Du ./duRun06.sh in 4th terminal**

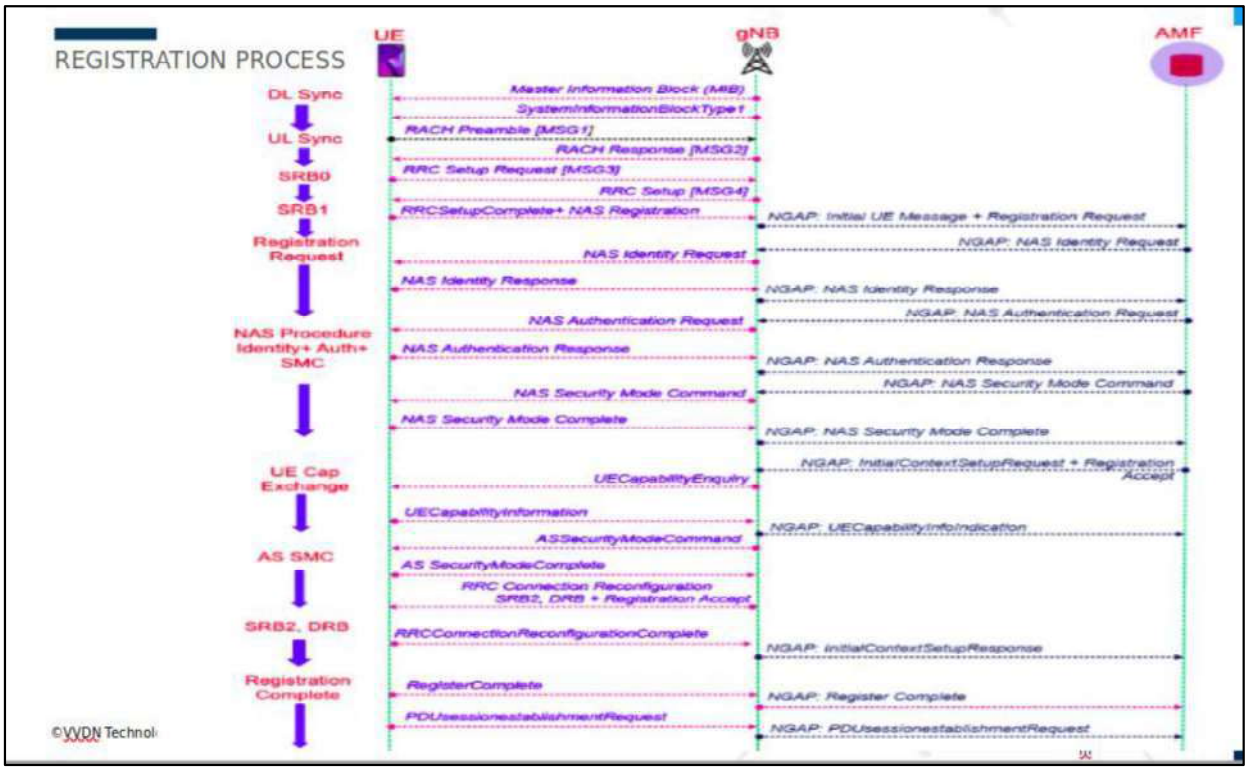


**Registration Process and Capturing Pcap log.**

Run mentioned command to capture Pcap log in CU/DU

*tcpdump -i any sctp -w (file name).pcap*


* Support: https://ubuntu.com/pro
System information as of Tue Feb 4 05:22:06 AM UTC 2025
System load: 2.19384765625
Usage of /: 16.4% of 467.40GB
Memory usage: 56%
Swap usage: 0%
Temperature: 49.0 C
Processes: 1920
Users logged in: 1
IPv4 address for br-c34d77d7814b: 172.40.0.1
IPv4 address for docker0: 172.17.0.1
IPv4 address for ens2f2: 172.16.0.4
IPv4 address for ens2f2: 172.16.1.4
IPv4 address for ens2f2: 172.16.0.58
IPv4 address for ens2f3: 192.168.4.60
Expanded Security Maintenance for Applications is enabled.
135 updates can be applied immediately.
14 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.
Last login: Tue Feb 4 05:16:40 2025 from 172.16.0.79
wisigkd@wisig:~$ sudo su
[sudo] password for wisigkd:
root@wisig:/home/wisigkd#
root@wisig:/home/wisigkd#
root@wisig:/home/wisigkd#
root@wisig:/home/wisigkd#
root@wisig:/home/wisigkd# tcpdump -i any sctp -w iitrae.pcap
Remote monitoring
Follow terminal folder
UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net


REGISTRATION PROCESS
UE
gNB
AMF
DL Sync
UL Sync
SRB0
SRB1
Registration Request
NAS Procedure Identity+ Auth+ SMC
UE Cap Exchange
AS SMC
SRB2, DRB
Registration Complete
Master Information Block (MIB)
SystemInformationBlockType1
RACH Preamble [MSG1]
RACH Response [MSG2]
RRC Setup Request [MSG3]
RRC Setup [MSG4]
RRCSetupComplete+ NAS Registration
NGAP: Initial UE Message + Registration Request
NGAP: NAS Identity Request
NAS Identity Request
NAS Identity Response
NGAP: NAS Identity Response
NGAP: NAS Authentication Request
NAS Authentication Request
NAS Authentication Response
NGAP: NAS Authentication Response
NGAP: NAS Security Mode Command
NAS Security Mode Command
NAS Security Mode Complete
NGAP: NAS Security Mode Complete
NGAP: InitialContextSetupRequest + Registration Accept
UECapabilityEnquiry
UECapabilityInformation
NGAP: UECapabilityInfoIndication
ASSecurityModeCommand
AS SecurityModeComplete
RRC Connection Reconfiguration SRB2, DRB + Registration Accept
RRCConnectionReconfigurationComplete
NGAP: InitialContextSetupResponse
RegisterComplete
NGAP: Register Complete
PDUsessionestablishmentRequest
NGAP: PDUsessionestablishmentRequest
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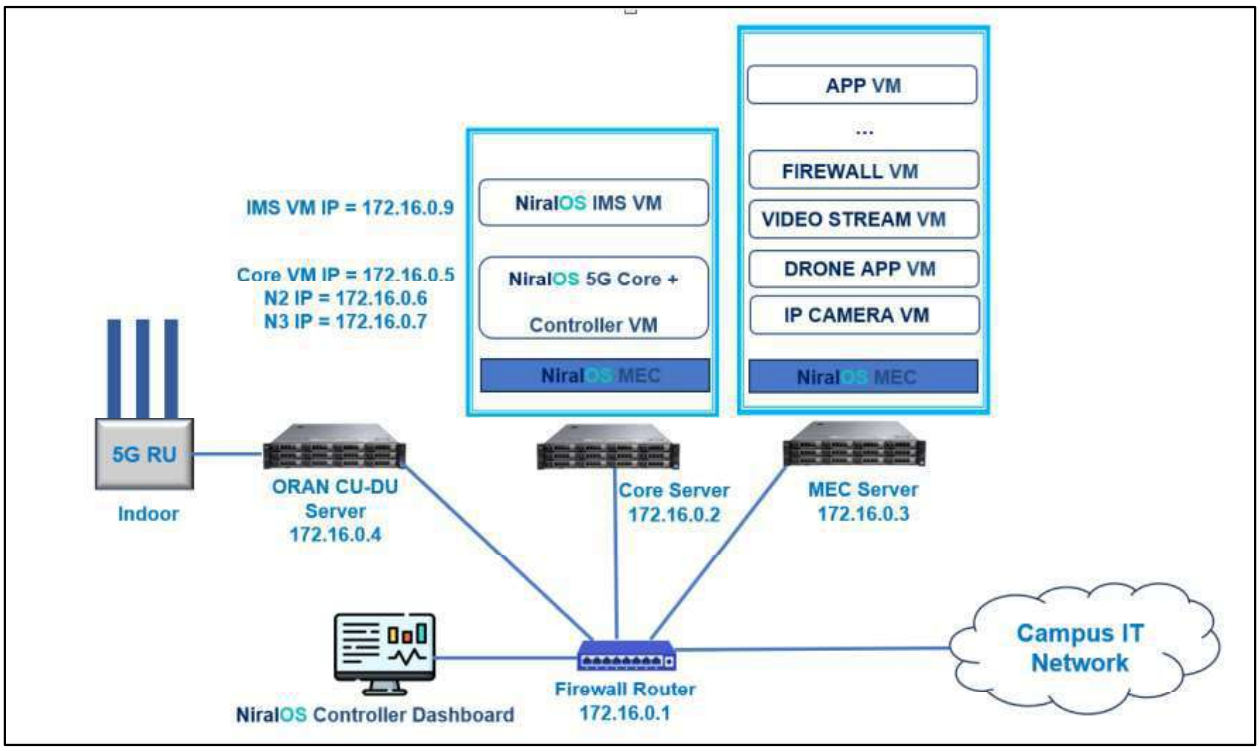
3.0 Niral Core MEC/NMS

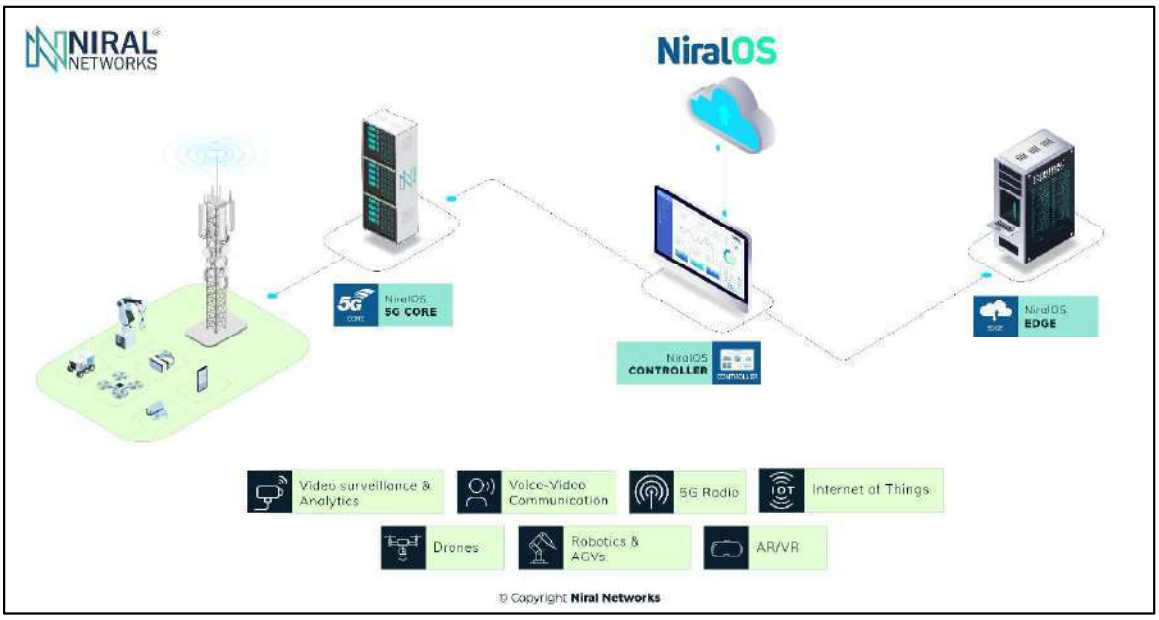



### 3.1 Introduction

#### 3.1.1 OVERVIEW OF NIRALOS

Niral Network's cutting-edge Product Suite, featuring the robust NiralOS 5G Core, Multisite Controller, and Multi-Access Edge Compute Platform, is revolutionizing the landscape of private 5G networks. With scalability and reliability at its core, our solution is unlocking the full potential of private 5G networks, paving the way for unparalleled connectivity and innovation.

At Niral Networks, we are focused on rolling out our private 5G solution across labs and enterprises, we're empowering users with transformative features like VoNR, Network APIs, Analytics, and Edge Computing. This technical user guide delves into detail on architecture overview, product installation, and monitoring features.



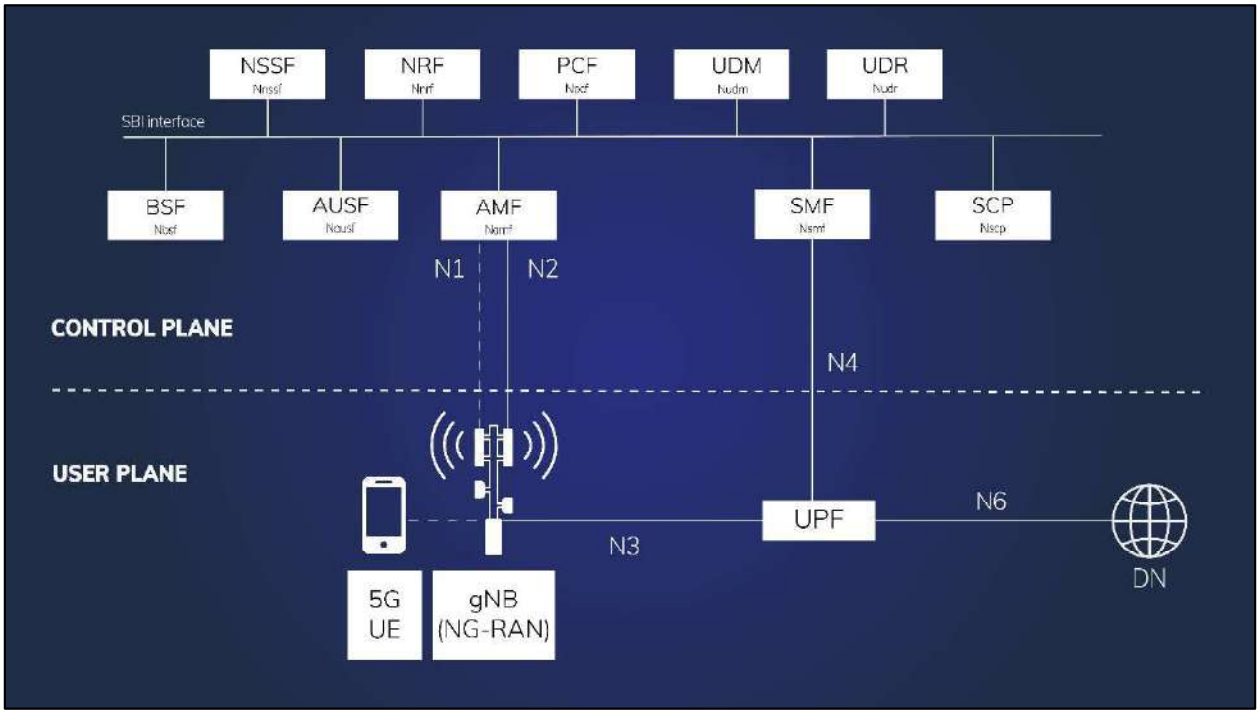

### 3.2 NiralOS 5G Core

The NiralOS 5G Core is a cloud-native software designed specifically for private 5G deployment within enterprises. This software contains various 5G network functions such as Mobility Management, Authentication, Session, and Policy Management along with a Fast path User Plane. The software is hosted on commercial off-the-shelf (COTS) servers and can run bare-metal, virtual network functions (VNF), or cloud-native functions (CNF). Niral's Private 5G Specification is Release-16 compliant and provides support for various interfaces along with a web-based dashboard for subscriber provisioning and management. It also has open APIs for the integration of third-party applications and multi-tenant controller functionality for ease of management. NiralOS Private 5G UPF can easily interoperate with centralized 5G Core Control Plane using the PFCP interface and provide on-premises edge applications with privacy, security, and low latency. It's a powerful tool for businesses looking to deploy their private 5G network while keeping costs in check.

### 3.3 NiralOS Private 5G Key Features

| |
|---|
| 3GPP Rel 16 Compliant Control Plane Network Functions: AMF, SMF, NRF, NSSF, PCF, AUSF, UDM, NEF, NWDAF |
| 3GPP Rel 16 Compliant distributed 5G User Plane Network Function UPF with local breakout |
| Service-Based Architecture and Interfaces. Support Container, VNFs and Bare Metal |
| 3GPP Rel 16 Compliant N1, N2, N3, N4, N6 Interface and PCFP with CUPS Architecture |
| Integrated gNB and O-RAN (CU-DU-RU) based Radios with 15 different 5G Radio vendors |
| Supports Static Network Slicing, Network Scaling, MEC and SDN Integration |
| Supports web-based dashboard for Alarm, SIM Provisioning and Management, Tracing and Tracking |
| Support web-based dashboard Network Topology, Statistics, Telemetry, Top User and Network Analytics |
| Support use-case integration in Mining, Manufacturing, Oil & Gas, Ports, Campuses |
| Finally, our infrastructure can support multiple businesses or organizations, each with its own separate network and management tools |

### 3.4 NiralOS 5G Core Architecture



### 3.4.1 Network function

| | |
|---|---|
| **Access and Mobility Management Function (AMF)** | AMF manages the mobility of devices within the network, handling tasks such as registration, authentication, and connection management. |
| **Session Management Function (SMF)** | SMF establishes and maintains data sessions between user devices and the network, ensuring efficient data transfer. |
| **User Plane Function (UPF)** | UPF processes and forwards user data packets, optimizing data flow to achieve low latency and high throughput. |
| **Network Slice Selection Function (NSSF)** | NSSF selects and configures network slices tailored to the specific requirements of users and applications, optimizing resource allocation and network efficiency. |
| **Binding Support Function (BSF)** | BSF enables policy/charging (PCF/CHF) scaling of the 5G network while ensuring session correlation for HTTP/2. |
| **Policy Control Function (PCF)** | PCF governs the enforcement of network policies, including quality of service (QoS) management and traffic prioritization, to meet the diverse needs of applications and users. |
| **Authentication Server Function (AUSF)** | AUSF handles user authentication and authorization, verifying the identity of devices and users before granting access to network resources. |
| **Unified Data Management (UDM)** | UDM stores and manages user-related data, such as subscriber profiles and service subscriptions, ensuring accurate and secure access control. |
| **User Data Repository (UDR)** | UDR stores user-related data, including session and subscriber data, for access and reference by various network functions. |
| **Service Communication Proxy (SCP)** | SCP provides a centralized interface for managing service-related communication, enabling efficient interaction between network functions and external services. |
| **Network Repository Function (NRF)** | NRF maintains a repository of network function instance (NF instance) information, facilitating service discovery and routing within the network. |

### 3.5 Network interface - NiralOS 5G core

The NiralOS 5G Core supports a variety of 5G interfaces, each playing a crucial role in enabling communication between different network functions and external systems. Here is an overview of the key interfaces: N1, N2, N3, N4, and N6.

| | |
|---|---|
| N1 Interface | Description: Connects the User Equipment (UE) with the Access and Mobility Management Function (AMF).<br>Function:<br>- Facilitates the initial registration and connection management of the UE.<br>- Handles UE authentication and authorization.<br>- Manages mobility procedures and session establishment.<br>Key Features:<br>- NAS (Non-Access Stratum) signaling<br>- Secure communication via encryption and integrity protection |
| N2 Interface | Description: Connects the Next Generation Node B (gNB) with the AMF<br>Function:<br>- Manages control signaling for the Radio Access Network (RAN).<br>- Facilitates handover procedures between different gNBs.<br>- Supports the setup, modification, and release of radio bearers.<br>Key Features:<br>- Uses NG-AP (Next Generation Application Protocol signaling<br>- Ensures seamless mobility and connectivity management |
| N3 Interface | Description: Connects the gNB with the User Plane Function (UPF).<br>Function:<br>- Carries user data packets between the RAN and the core network.<br>- Ensures efficient routing and forwarding of user data.<br>Key Features:<br>- Uses GTP-U (GPRS Tunneling Protocol - User Plane)<br>- Supports QoS (Quality of Service) enforcement and traffic management |

| | |
|---|---|
| N4 Interface | **Description:**<br>Connects the Session Management Function (SMF) with the UPF.<br>**Function:**<br>• Manages the control and configuration of user plane resources.<br>• Handles session establishment, modification, and release.<br>• Enforces policy and QoS rules on the user plane.<br>Key Features:<br>• Uses PFCP (Packet Forwarding Control Protocol)<br>• Enables dynamic control of user plane behavior |
| N6 Interface | **Description:**<br>Connects the UPF with external Data Networks (DN), such as the Interne or private enterprise networks.<br>**Function:**<br>• Facilitates data transfer between the 5G core network and externa networks.<br>• Supports integration with various application services and externa data sources.<br>**Key Features:**<br>• Ensures secure and efficient data routing and forwarding<br>• Supports different types of data services and network protocols |

The NiralOS 5G Core utilizes these key interfaces (N1, N2, N3, N4, and N6) to enable seamless communication between different network functions and external

systems. Each interface plays a specific role in managing control signaling, user data transfer, session management, and external network integration. By supporting these interfaces, the NiralOS 5G Core ensures efficient, secure, and reliable operation of private 5G networks, catering to the diverse needs of modern enterprises.

### 3.5.1 NiralOS Controller

Niral Networks has developed the NiralOS Controller, a ground-breaking product that provides centralized orchestration, policy control, zero-touch provisioning, and network slicing for private networks. This cloud-based platform has visibility of all Edge Gateways and private networks and can be deployed in a variety of ways to meet clients' needs. The controller is available as CNFs or

VNFs within the cloud and can be installed on-premises, on a cloud, or in a hybrid model. Additionally, the controller can be integrated with the client's Edge Application to streamline dashboard management.

NiralOS Controller can be installed in different ways depending on the client's needs. It can be installed on the client's premises on a physical machine or on a virtual machine called a hypervisor. NiralOS controllers can also be installed on a cloud service provider or data center as required by the client themselves. Each installation method has its own advantages and limitations.

The NiralOS Controller can be installed in multiple ways based on requirements:

- On-Premises Installation: Provides greater control and security over the network but requires more hardware and maintenance efforts.
- Cloud Installation: More flexible and scalable but may come with additional costs and dependence on internet connectivity.
- Hybrid Installation: Provides a balance between the two approaches and can cater to different use cases with varying network requirements. In this type of installation, the controller is installed both on-premises and in the cloud. The Edge

Application connects directly to the on-premises installation, while control and monitoring of the network are done through the cloud dashboard.

### 3.5.2 NiralOS EDGE

The NiralOS EDGE is a unique platform that offers a virtualized edge infrastructure, enabling businesses to create their private cloud infrastructure on a variety of commodity hardware. With a focus on privacy, security, and low-latency connectivity, the platform is ideal for hosting third-party applications that support emerging technologies like AR/VR, robotics, drones, and AI/ML within an enterprise.

At the core of NiralOS EDGE is the Niral Controller, which seamlessly provisions and centrally manages the local EDGE at various sites. The platform is designed to enable open APIs, allowing businesses to easily manage third-party applications from the Niral Controller. Furthermore, the platform offers the ability to scale in or scale out the third-party application resources based on the compute requirement in real time.

NiralOS EDGE is ideal for businesses that require modern cloud solutions that are versatile and scalable enough to support various emerging technologies like drones, robotics, IoT, AR/VR, AGV, and AI. Companies operating in manufacturing, power, mining, oil & gas, railways, shipping, and ports will particularly find NiralOS EDGE useful.

### 3.5.3 Scope

NiralOS private 5G solutions consist of NiralOS 5G Core, NiralOS Controller, and NiralOS EDGE and this Training Guide provides a complete functional Overview, and Application Programming Interface from the perspective of different functionality such as configuration, statistics of control and user plane and Alarms. The guide has a prerequisite of NiralOS User Guide, NiralOS Implementation Guide and latest NiralOS Release Notes.

| Date | Version | Author/Editor | Description |
|---|---|---|---|
| | Draft | Niral Networks Customer Success Team | Initial Release |

### 3.6 Setup diagram of 5G Lab

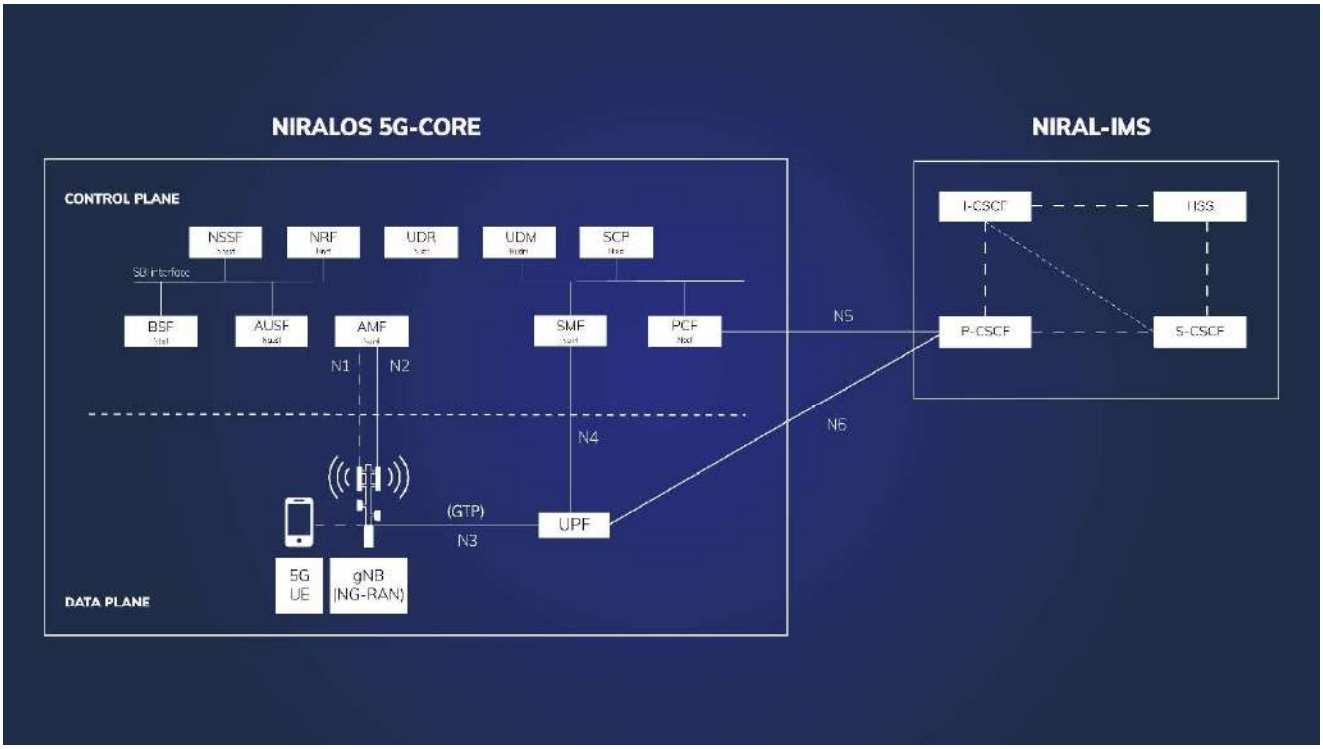



**Figure 1: NiralOS 5G Core & IMS**

### 3.6.1 Setup diagram

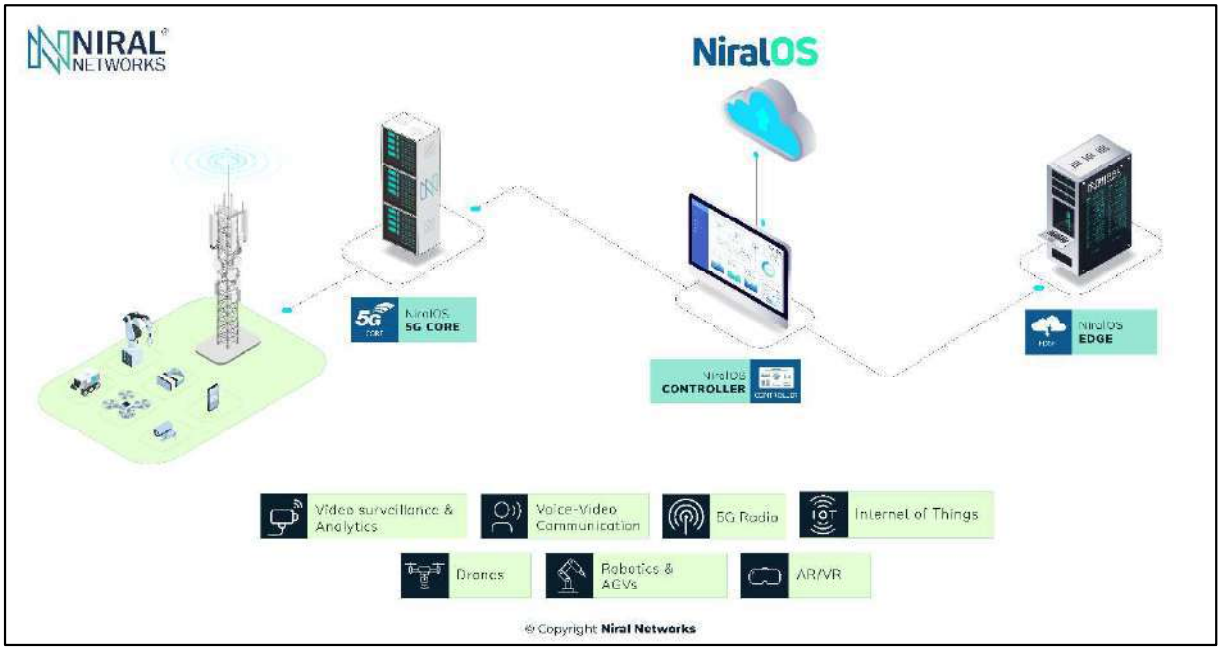



Figure 1 : 5G Lab Setup Diagram

### 3.7 5G Core Controller API:

### 3.7.1 Access Token API

Login API for NiralOS Controller

**Description:** This API endpoint allows you to authenticate a user by providing a username and password, and it responds with an access token and refresh token for further authorization. The tokens are used for secure access to protected resources.

| | |
|---|---|
| *API URL:* | *http://192.168.0.34:8082/token* |
| *METHOD:* | *POST* |
| *REQUEST BODY TYPE:* | *JSON* |
| *REQUEST BODY:* | |
| { "username": "admin", "password": "admin@1234" } | |

| | |
|---|---|
| *RESPONSE BODY TYPE:* | *JSON* |
| *RESPONSE KEY* | |
| { "accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA", "refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"} | |

**3.7.2 Overview**

| | |
|---|---|
| *API URL:* | *http://192.168.0.34:8082 /v2/5gcore/overview/cards_core_gnb_ue_graph/range=hour* |
| *METHOD:* | *GET* |
| *REQUEST BODY TYPE:* | *JSON* |
| *REQUEST BODY:* | |
| *RESPONSE BODY TYPE:* | *JSON* |
| *RESPONSE* | |
| { "accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA", "refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"} | |

3.7.3 Alarms Count

| *API URL* | http://192.168.0.34:8082 /v2/alarm_manager/widget/alarms |
|---|---|
| *METHOD:* | *GET* |
| *REQUEST BODY TYPE:* | *JSON* |
| *REQUEST BODY:* | |
| {<br>"listofCriticalAlarms": [],<br>"countLevelsofAlarms": {<br>"critical": "0",<br>"major": "0",<br>"minor": "0",<br>"notAlarmed": "0",<br>"notReported": "0"<br>}} | |
| *RESPONSE BODY TYPE:* | *JSON* |
| *RESPONSE KEY* | |
| {<br>"accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",<br>"refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"} | |

**3.7.4 Topology**

| *API URL:* | http://192.168.0.34:8082 */v2/5gcore/topology* |
|---|---|
| *METHOD:* | *GET* |
| *REQUEST BODY TYPE:* | *JSON* |
| *REQUEST BODY:* | |
| {<br>"name": "Controller",<br>"level": "Main",<br>"status": 1,<br>"children": [ | |

```
{
   "name": "5G-Core-User-Plane",
   "level": "device",
   "status": 1,
   "children": [
      {
         "nfName": null,
         "nfType": "upf-1",
         "nfIp": "niralos-upf-1",
         "nfStatus": "1",
         "device": null,
         "children": null
      }
   ]
},
{
   "name": "5G-Core-Control-Plane",
   "level": "device",
   "status": 1,
   "children": [
      {
         "nfName": null,
         "nfType": "nrf",
         "nfIp": "niralos-nrf-1",
         "nfStatus": "1",
         "device": null,
         "children": null
      },
      {
         "nfName": null,
         "nfType": "ausf",
         "nfIp": "niralos-ausf-1",
         "nfStatus": "1",
         "device": null,
         "children": null
      },
      {
```

```
        "nfName": null,
        "nfType": "nwdaf",
        "nfIp": "niralos-nwdaf-1",
        "nfStatus": "1",
        "device": null,
        "children": null
    },
    {
        "nfName": null,
        "nfType": "scp",
        "nfIp": "niralos-scp-1",
        "nfStatus": "1",
        "device": null,
        "children": null
    },
    {
        "nfName": null,
        "nfType": "nef",
        "nfIp": "niralos-nef-1",
        "nfStatus": "1",
        "device": null,
        "children": null
    },
    {
        "nfName": null,
        "nfType": "bsf",
        "nfIp": "niralos-bsf-1",
        "nfStatus": "1",
        "device": null,
        "children": null
    },
    {
        "nfName": null,
        "nfType": "smf",
        "nfIp": "niralos-smf-1",
        "nfStatus": "1",
        "device": null,
```

```
        "children": null
      },
      {
        "nfName": null,
        "nfType": "udr",
        "nfIp": "niralos-udr-1",
        "nfStatus": "1",
        "device": null,
        "children": null
      },
      {
        "nfName": null,
        "nfType": "amf",
        "nfIp": "niralos-amf-1",
        "nfStatus": "1",
        "device": null,
        "children": []
      },
      {
        "nfName": null,
        "nfType": "nssf",
        "nfIp": "niralos-nssf-1",
        "nfStatus": "1",
        "device": null,
        "children": null
      },
      {
        "nfName": null,
        "nfType": "pcf",
        "nfIp": "niralos-pcf-1",
        "nfStatus": "1",
        "device": null,
        "children": null
      },
      {
        "nfName": null,
        "nfType": "pcscf",
```

```
            "nfIp": "niralos-pcscf-1",
            "nfStatus": "1",
            "device": null,
            "children": null
        },
        {
            "nfName": null,
            "nfType": "icscf",
            "nfIp": "niralos-icscf-1",
            "nfStatus": "1",
            "device": null,
            "children": null
        },
        {
            "nfName": null,
            "nfType": "scscf",
            "nfIp": "niralos-scscf-1",
            "nfStatus": "1",
            "device": null,
            "children": null
        }
      ]
    }
  ]
}
```


| *RESPONSE BODY TYPE:* | *JSON* |
|---|---|

*RESPONSE KEY :*


```
{
  "accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",
  "refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"
}
```

### 3.7.5 Control Plane Stats

| *API URL:* | http://192.168.0.34:8082 */v2/5gcore/ue/history/count* |
|---|---|
| *METHOD:* | *GET* |
| *REQUEST BODY TYPE:* | *JSON* |
| *REQUEST BODY:* | |


```
{
  "ueInfo": [
    {
      "totalNumberOfRegAttempts": 0,
      "noOfRegistered": 0,
      "noOfDeregistered": 0,
      "noOfFailure": 0,
      "causeList": []
    }
  ],
  "pduEstInfo": [
    {
      "noOfpduEstReq": 0,
      "noOfpduEstSuccess": 0,
      "noOfpduEstFailure": 0,
      "pduCauseList": []
    }
  ],
  "pduReleaseInfo": [
    {
      "noOfpduReleaseReq": 0,
      "noOfpduReleaseSuccess": 0
    }
  ]}
```

| | |
|---|---|
| *RESPONSE BODY TYPE:* | *JSON* |
| *RESPONSE KEY* | |
| {<br>"accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",<br>"refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"<br>} | |

**3.7.6 5G Radio Stats**

| | |
|---|---|
| *API URL* | http://192.168.0.34:8082 /v2/5gcore/gnb/Info |
| *METHOD* | *GET* |
| *REQUEST BODY TYPE* | *JSON* |
| *REQUEST BODY* | |
| [<br>{<br>"id": 1,<br>"totalNumberOfAttempts": "0",<br>"totalnoOfRegistered": "0",<br>"noOfSCTP_Deregistered": "0",<br>"noOfDeregistered": "0",<br>"totalNoOfFailure": "0",<br>"nfType": "amf",<br>"nfName": "niralos-amf-1"<br>}<br>] | |
| *RESPONSE BODY TYPE* | *JSON* |
| *RESPONSE KEY* | |

```
{
  "accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",
  "refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"
}
```


### 3.7.7 PFCP Stats

| | |
|---|---|
| *API URL* | http://192.168.0.34:8082 /v2/5gcore/pfcpInfo |
| *METHOD* | *GET* |
| *REQUEST BODY TYPE* | *JSON* |
| *REQUEST BODY* | |


```
{
  "total_cp_assoc_req": 1,
  "total_cp_assoc_resp": 1,
  "totalEstReq": 0,
  "sucessEstReq": 0,
  "totalEstRsp": 0,
  "sucessEstRsp": 0,
  "failEstRsp": 0,
  "totalModReq": 0,
  "sucessModReq": 0,
  "totalModRsp": 0,
  "sucessModRsp": 0,
  "failModRsp": 0,
  "totalDelReq": 0,
  "sucessDelReq": 0,
```

| |
|---|
| "totalDelRsp": 0,<br>"sucessDelRsp": 0,<br>"failDelRsp": 0<br>} |
| *RESPONSE BODY TYPE* \| *JSON* |
| *RESPONSE KEY* \| |
| {<br>"accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",<br>"refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"<br>} |

**3.7.8 Add APN**

| | |
|---|---|
| *API URL* | http://192.168.0.34:8082 /v2/5gcore/subscriber/add/apn |
| *METHOD* | *POST* |
| *REQUEST BODY TYPE* | *JSON* |
| *REQUEST BODY* | |
| { "apn": "ims"} | |
| *RESPONSE BODY TYPE* | *JSON* |
| *RESPONSE KEY* | |
| {<br>"accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",<br>"refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQy | |

| |
|---|
| MSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"} |

**3.7.9 View APN List**

| | |
|---|---|
| *API URL* | http://192.168.0.34:8082 /v2/5gcore/subscriber/get/apn_lis |
| *METHOD* | *GET* |
| *REQUEST BODY TYPE* | *JSON* |
| *REQUEST BODY* | |


```
[  {
      "apnId": 3,
      "apnValue": "ims"
   },
   {
      "apnId": 4,
      "apnValue": "ims"
   }]
```


| | |
|---|---|
| *RESPONSE BODY TYPE* | *JSON* |
| *RESPONSE KEY* | |


```
{
   "accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",
   "refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"
}
```


**3.7.10 Slice**

| | |
|---|---|
| *API URL* | http://192.168.0.34:8082 /v2/5gcore/subscriber/slice_list/with_mongoId |
| *METHOD* | *GET* |

| *REQUEST BODY TYPE* | *JSON* |
|---|---|
| *REQUEST BODY* | |


```
[
  {
    "_id": "67bc196418417451c339a19c",
    "sliceId": "jeoo3TQOMAJmfXzhX8YFj7",
    "sst": 1,
    "sd": "000007",
    "default_indicator": true,
    "session": [
      {
        "name": "3",
        "apn_id": "3",
        "type": 1,
        "qos": {
          "index": 9,
          "arp": {
            "priority_level": 1,
            "pre_emption_capability": 2,
            "pre_emption_vulnerability": 2
          }
        },
        "ambr": {
          "downlink": {
            "value": 1,
            "unit": 3
          },
          "uplink": {
            "value": 1,
            "unit": 3
```

```
      }
    },
    "ue": {
      "ipv4": null,
      "ipv6": null
    },
    "smf": {
      "addr": null,
      "addr6": null
    },
    "pcc_rule": [
      {
        "flow": [],
        "qos": {
          "index": 1,
          "arp": {
            "priority_level": 2,
            "pre_emption_capability": 2,
            "pre_emption_vulnerability": 2
          },
          "mbr": {
            "downlink": {
              "value": 1,
              "unit": 3
            },
            "uplink": {
              "value": 1,
              "unit": 3
            }
          },
```

```
                "gbr": {
                  "downlink": {
                    "value": 1,
                    "unit": 3
                  },
                  "uplink": {
                    "value": 1,
                    "unit": 3
                  }
                }
              }
            }
          ]
        }
    ]
},
{
    "_id": "67bc1b2e18417451c339a19e",
    "sliceId": "hWDwExF6FR5u9bsNntzdxX",
    "sst": 2,
    "sd": null,
    "default_indicator": true,
    "session": [
      {
        "name": "3",
        "apn_id": "3",
        "type": 1,
        "qos": {
          "index": 9,
          "arp": {
```

```
                "priority_level": 1,
                "pre_emption_capability": 2,
                "pre_emption_vulnerability": 1
            }
        },
        "ambr": {
            "downlink": {
                "value": 1,
                "unit": 3
            },
            "uplink": {
                "value": 1,
                "unit": 3
            }
        },
        "ue": {
            "ipv4": null,
            "ipv6": null
        },
        "smf": {
            "addr": null,
            "addr6": null
        },
        "pcc_rule": []
    }
  ]
 }
]
```

| | |
|---|---|
| *RESPONSE BODY TYPE* | *JSON* |
| *RESPONSE KEY* | |


{

"accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",

"refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"

}


### 3.7.11 Add Slice

| | |
|---|---|
| *API URL* | http://192.168.0.34:8082 /v2/5gcore/subscriber/addSlice |
| *METHOD* | *POST* |
| *REQUEST BODY TYPE* | *JSON* |
| *REQUEST BODY* | |


{

"sst":2,"default_indicator":true,"session":[{"name":"3","apn_id":"3","type":"1","qos":{"index":9,"arp":{"priority_level":1,"pre_emption_capability":2,"pre_emption_vulnerability":1}},"ambr":{"downlink":{"value":1,"unit":3},"uplink":{"value":1,"unit":3}},"ue":{},"smf":{},"pcc_rule":[]}]

}


| | |
|---|---|
| *RESPONSE BODY TYPE* | *JSON* |
| *RESPONSE KEY* | |


{

"accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",

"refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"

| |
|---|
| } |

### 3.7.12 Slice list

| | |
|---|---|
| *API URL* | http://192.168.0.34:8082 /v2/5gcore/subscriber/slice_list/with_mongoId |
| *METHOD* | *GET* |
| *REQUEST BODY TYPE* | *JSON* |
| *REQUEST BODY* | |


```
[
  {
    "_id": "67bc196418417451c339a19c",
    "sliceId": "jeoo3TQOMAJmfXzhX8YFj7",
    "sst": 1,
    "sd": "000007",
    "default_indicator": true,
    "session": [
      {
        "name": "3",
        "apn_id": "3",
        "type": 1,
        "qos": {
          "index": 9,
          "arp": {
            "priority_level": 1,
            "pre_emption_capability": 2,
            "pre_emption_vulnerability": 2
          }
        },
        "ambr": {
          "downlink": {
            "value": 1,
```

```
            "unit": 3
        },
        "uplink": {
            "value": 1,
            "unit": 3
        }
    },
    "ue": {
        "ipv4": null,
        "ipv6": null
    },
    "smf": {
        "addr": null,
        "addr6": null
    },
    "pcc_rule": [
        {
            "flow": [],
            "qos": {
                "index": 1,
                "arp": {
                    "priority_level": 2,
                    "pre_emption_capability": 2,
                    "pre_emption_vulnerability": 2
                },
                "mbr": {
                    "downlink": {
                        "value": 1,
                        "unit": 3
                    },
```

```
                "uplink": {
                  "value": 1,
                  "unit": 3
                }
              },
              "gbr": {
                "downlink": {
                  "value": 1,
                  "unit": 3
                },
                "uplink": {
                  "value": 1,
                  "unit": 3
                }
              }
            }
          ]
        }
    ]
  },
  {
    "_id": "67bc1b2e18417451c339a19e",
    "sliceId": "hWDwExF6FR5u9bsNntzdxX",
    "sst": 2,
    "sd": null,
    "default_indicator": true,
    "session": [
      {
        "name": "3",
```

```
        "apn_id": "3",
        "type": 1,
        "qos": {
          "index": 9,
          "arp": {
            "priority_level": 1,
            "pre_emption_capability": 2,
            "pre_emption_vulnerability": 1
          }
        },
        "ambr": {
          "downlink": {
            "value": 1,
            "unit": 3
          },
          "uplink": {
            "value": 1,
            "unit": 3
          }
        },
        "ue": {
          "ipv4": null,
          "ipv6": null
        },
        "smf": {
          "addr": null,
          "addr6": null
        },
        "pcc_rule": []
      }
```

```
    ]
  }
]
```


| *RESPONSE BODY TYPE* | *JSON* |
|---|---|
| *RESPONSE KEY* | |


```
{
  "accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",
  "refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"
}
```


### 3.7.13 Add Subscriber

| *API URL* | http://192.168.0.34:8082 /v2/5gcore/subscriber/addSubscriber |
|---|---|
| *METHOD* | *POST* |
| *REQUEST BODY TYPE* | *JSON* |
| *REQUEST BODY* | |


```
{
"schema_version":1,"imsi":"001010000000001","deviceType":"Iot","security":{"aliseName":"new",
"amf":"8000"},"ambr":{"downlink":{"value":1,"unit":3},"uplink":{"value":1,"unit":3}},"slice":[{"s
iceId":"jeoo3TQOMAJmfXzhX8YFj7","sst":1,"sd":"000007","default_indicator":true,"session":[{"r
ame":"3","apn_id":"3","type":1,"qos":{"index":9,"arp":{"priority_level":1,"pre_emption_capability"
:2,"pre_emption_vulnerability":2}},"ambr":{"downlink":{"value":1,"unit":3},"uplink":{"value":1,"u
nit":3}},"ue":{"ipv4":null,"ipv6":null},"smf":{"addr":null,"addr6":null},"pcc_rule":[{"flow":[],"qos
":{"index":1,"arp":{"priority_level":2,"pre_emption_capability":2,"pre_emption_vulnerability":2},"
mbr":{"downlink":{"value":1,"unit":3},"uplink":{"value":1,"unit":3}},"gbr":{"downlink":{"value":
1,"unit":3},"uplink":{"value":1,"unit":3}}}}]}]},"access_restriction_data":32,"subscriber_status":0,
"network_access_mode":0,"subscribed_rau_tau_timer":12,"__v":0,"flag":false,"activation":0,"trace_
info":{"trace_ref":null,"trace_depth":null,"collection_entity_ipv4_addr":null,"collection_entity_ipv6
_addr":null,"event_list":null,"interface_list":null,"ne_type_list":null},"msisdn":[null],"mcc":"001","
mnc":"01"}
```

| | |
|---|---|
| | |
| *RESPONSE BODY TYPE* | *JSON* |
| *RESPONSE KEY* | |
| { "accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA", "refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA" } | |

**3.7.14 View All Subscribers**

| | |
|---|---|
| *API URL* | http://192.168.0.34:8082 /v2/5gcore/subscriber/viewAllSubscribers/pageNumber=1 |
| *METHOD* | *GET* |
| *REQUEST BODY TYPE* | *JSON* |
| *REQUEST BODY* | |

```
[
{
    "_id": "67bc1f1418417451c339a1a1",
    "schema_version": 1,
    "imsi": "001010000000001",
    "deviceType": "Iot",
    "msisdn": null,
    "security": {
      "aliseName": "new",
      "amf": "8000"
    },
    "ambr": {
      "downlink": {
        "value": 1,
        "unit": 3
      },
      "uplink": {
        "value": 1,
        "unit": 3
      }
    },
    "slice": [
      {
        "sliceId": "jeoo3TQOMAJmfXzhX8YFj7",
        "sst": 1,
        "sd": "000007",
        "default_indicator": true,
        "session": [
          {
            "name": "ims",
```

```
"apn_id": "3",
"type": 1,
"qos": {
   "index": 9,
   "arp": {
      "priority_level": 1,
      "pre_emption_capability": 2,
      "pre_emption_vulnerability": 2
   }
},
"ambr": {
   "downlink": {
      "value": 1,
      "unit": 3
   },
   "uplink": {
      "value": 1,
      "unit": 3
   }
},
"ue": {
   "ipv4": null,
   "ipv6": null
},
"smf": {
   "addr": null,
   "addr6": null
},
"pcc_rule": [
   {
```

```
        "flow": [],
        "qos": {
          "index": 1,
          "arp": {
            "priority_level": 2,
            "pre_emption_capability": 2,
            "pre_emption_vulnerability": 2
          },
          "mbr": {
            "downlink": {
              "value": 1,
              "unit": 3
            },
            "uplink": {
              "value": 1,
              "unit": 3
            }
          },
          "gbr": {
            "downlink": {
              "value": 1,
              "unit": 3
            },
            "uplink": {
              "value": 1,
              "unit": 3
            }
          }
        }
      }
```

```
                ]
              }
            ]
          }
      ],
      "access_restriction_data": 32,
      "subscriber_status": 0,
      "network_access_mode": 0,
      "subscribed_rau_tau_timer": 12,
      "mnc": "01",
      "mcc": "001",
      "__v": 0,
      "flag": false,
      "activation": null,
      "trace_info": null
   }
]
```


| *RESPONSE BODY TYPE* | *JSON* |
|---|---|
| *RESPONSE KEY* | |


```
{
   "accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",
   "refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"
}
```

### 3.7.15 Trace imsi

| | |
|---|---|
| *API URL* | http://192.168.0.34:8082 */v2/5gcore/subscriber/listofdeactiveOrnewimsi* |
| *METHOD* | *GET* |
| *REQUEST BODY TYPE* | *JSON* |
| *REQUEST BODY* | |
| [<br>{<br>"imsi": "001010000000001"<br>}<br>] | |
| *RESPONSE BODY TYPE* | *JSON* |
| *RESPONSE KEY* | |
| {<br>"accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",<br>"refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"<br>} | |

### 3.7.16 Radio list

| | |
|---|---|
| *API URL* | http://192.168.0.34:8082 /v2/5gcore/gnb/uniqueList |
| *METHOD* | *GET* |
| *REQUEST BODY TYPE* | *JSON* |
| *REQUEST BODY* | |
| [] | |
| *RESPONSE BODY TYPE* | *JSON* |
| *RESPONSE KEY* | |
| { "accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA", "refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA" } | |

**3.7.17 list all alarms**

| | |
|---|---|
| *API URL* | http://192.168.0.34:8082 /v2/alarm_manager/alertfilter/level=all/status=0 |
| *METHOD* | *GET* |
| *REQUEST BODY TYPE* | *JSON* |
| *REQUEST BODY* | |
| { "filtersofAlarms": [], "listofAllAlarms": [] } | |
| *RESPONSE BODY TYPE* | *JSON* |
| *RESPONSE KEY* | |
| { | |

"accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",

"refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"

}

### 3.7.18 Critical alarm list

| | |
|---|---|
| *API URL* | http://192.168.0.34:8082 */v2/alarm_manager/countCritical* |
| *METHOD* | *GET* |
| *REQUEST BODY TYPE* | *JSON* |
| *REQUEST BODY* | |
| *{0}* | |
| *RESPONSE BODY TYPE* | *JSON* |
| *RESPONSE KEY* | |

{

"accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",

"refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"

}

### 3.7.19 UE History count

| | |
|---|---|
| *API URL* | http://192.168.0.34:8082 /v2/5gcore/ue/history/count |
| *METHOD* | *GET* |
| *REQUEST BODY TYPE* | *JSON* |
| *REQUEST BODY* | |
| { | |

```
  "ueInfo": [
    {
      "totalNumberOfRegAttempts": 0,
      "noOfRegistered": 0,
      "noOfDeregistered": 0,
      "noOfFailure": 0,
      "causeList": []
    }
  ],
  "pduEstInfo": [
    {
      "noOfpduEstReq": 0,
      "noOfpduEstSuccess": 0,
      "noOfpduEstFailure": 0,
      "pduCauseList": []
    }
  ],
  "pduReleaseInfo": [
    {
      "noOfpduReleaseReq": 0,
      "noOfpduReleaseSuccess": 0
    }
  ]
}
```


| *RESPONSE BODY TYPE* | *JSON* |
|---|---|
| *RESPONSE KEY* | |


```
{
  "accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UV
```

WsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",
"refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"
}

### 3.7.20 5G Core Site Info

| *API URL* | http://192.168.0.34:8082 /v2/5gcore/site-info |
|---|---|
| *METHOD* | *GET* |
| *REQUEST BODY TYPE* | *JSON* |
| *REQUEST BODY* | |

{
"provisionStatus": null,
"siteName": null,
"tenantName": null,
"streetName": null,
"city": null,
"pinCode": null,
"state": null,
"country": "India",
"fullName": null,
"contact": null,
"email": null,
"deployment Id": null,
"localSdnVersion": "3.0.1",
"fiveGcoreVersion": null,
"kafkaPartition": null,
"globalControllerIp": null,
"globalControllerPort": null
}

| | |
|---|---|
| *RESPONSE BODY TYPE* | *JSON* |
| *RESPONSE KEY* | |
| {<br>"accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",<br>"refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"<br>} | |

**3.7.21 Login**

| | |
|---|---|
| *API URL* | *http://192.168.0.34:8082/token* |
| *METHOD* | *POST* |
| *REQUEST BODY TYPE* | *JSON* |
| *REQUEST BODY* | |
| {<br>"username": "admin",<br>"password": "admin@1234"<br>} | |
| *RESPONSE BODY TYPE* | *JSON* |
| *RESPONSE KEY* | |
| {<br>"accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",<br>"refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"} | |

### 3.7.22 Node

| *API URL* | *http://192.168.0.34:8082/edge/node/get_internal_data* |
|---|---|
| *METHOD* | *GET* |
| *REQUEST BODY TYPE* | *JSON* |
| *REQUEST BODY* | |


```
[
  {
    "edgeClientId": "xLcwQTpLkTGm",
    "hypervisorIp": "172.16.0.2",
    "hypervisorToken": "root@pam!niral1234=78f7c257-70cf-40e1-8ef6-d5c86fe09a72",
    "hypervisorNodeName": "NiralOS-5G-Core",
    "userName": "root@pam",
    "password": "#Niral@7896#",
    "hypervisorPort": "8006",
    "bucketName": "mybucket"
  },
  {
    "edgeClientId": "OsHci3mvUog1",
    "hypervisorIp": "172.16.0.3",
    "hypervisorToken": "root@pam!niral123=8e67a2cd-1e2d-4a6d-b96c-2c99bda5576f",
    "hypervisorNodeName": "MEC-Server",
    "userName": "root@pam",
    "password": "#Niral@7896#",
    "hypervisorPort": "8006",
    "bucketName": "mybucket2"
  }
]
```


| *RESPONSE BODY TYPE* | *JSON* |
|---|---|

| *RESPONSE KEY* | |
|---|---|
| {<br>"accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",<br>"refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"<br>} | |

### 3.7.23 Stats

| *API URL* | *http://192.168.0.34:8082/edge/node/get_all_hypervisor_stats/edge_client_id=xLcwQTpLkTGm* |
|---|---|
| *METHOD* | *GET* |
| *REQUEST BODY TYPE* | *JSON* |
| *REQUEST BODY* | |

[
{
"edgeClientId": "xLcwQTpLkTGm",
"totalDiskspace": 93,
"usedDiskSpace": 6,
"freeDiskSpace": 87,
"diskPercentage": 6.45,
"totalRamSpace": 62.48,
"currentRamUsage": 25,
"ramPercentage": 40.01,
"availableRamSpace": 37,

```
"cpuModel": "Intel(R) Xeon(R) Gold 6312U CPU @ 2.40GHz",
"cpuUsage": 17.79,
"cpuSockets": 1,
"totalCpus": 48,
"cpuCores": 24,
"cpuPercentage": 50.0,
"upTime": 183422,
"nodeName": "NiralOS-5G-Core",
"nodeStatus": "Active",
"cpuGraphModel": [
  {
    "edgeClientid": "xLcwQTpLkTGm",
    "cpuValue": 17.82,
    "date": "2025-02-24T15:38:45.252653"
  },
  {
    "edgeClientid": "xLcwQTpLkTGm",
    "cpuValue": 0.0,
    "date": "2025-02-24T15:39:34.033111"
  },
  {
    "edgeClientid": "xLcwQTpLkTGm",
    "cpuValue": 17.79,
    "date": "2025-02-24T15:40:18.872261"
  }
],
"memoryGraphModel": [
  {
    "edgeClientId": "xLcwQTpLkTGm",
    "memoryValue": 27.22,
```

```
            "date": "2025-02-24T15:38:45.268739"
        },
        {
            "edgeClientId": "xLcwQTpLkTGm",
            "memoryValue": 27.23,
            "date": "2025-02-24T15:39:34.048741"
        },
        {
            "edgeClientId": "xLcwQTpLkTGm",
            "memoryValue": 27.24,
            "date": "2025-02-24T15:40:18.883862"
        }
    ]
  }
]
```


| *RESPONSE BODY TYPE* | *JSON* |
|---|---|
| *RESPONSE KEY* | |


```
{
  "accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",
  "refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"
}
```

### 3.7.24 V m details:

| | |
|---|---|
| *API URL* | *http://192.168.0.34:8082/edge/node/vm/get_edge_frontend_data/edge_client_id=xLcwQTpLkTGm* |
| *METHOD* | *GET* |
| *REQUEST BODY TYPE* | *JSON* |
| *REQUEST BODY* | |


```
[
  {
    "edgeClientId": "xLcwQTpLkTGm",
    "totalDiskspace": 93,
    "usedDiskSpace": 6,
    "freeDiskSpace": 87,
    "diskPercentage": 6.45,
    "totalRamSpace": 62.48,
    "currentRamUsage": 25,
    "ramPercentage": 40.01,
    "availableRamSpace": 37,
    "cpuModel": "Intel(R) Xeon(R) Gold 6312U CPU @ 2.40GHz",
    "cpuUsage": 17.79,
    "cpuSockets": 1,
    "totalCpus": 48,
    "cpuCores": 24,
    "cpuPercentage": 50.0,
    "upTime": 183422,
    "nodeName": "NiralOS-5G-Core",
    "nodeStatus": "Active",
    "cpuGraphModel": [
```

```
    {
      "edgeClientid": "xLcwQTpLkTGm",
      "cpuValue": 17.82,
      "date": "2025-02-24T15:38:45.252653"
    },
    {
      "edgeClientid": "xLcwQTpLkTGm",
      "cpuValue": 0.0,
      "date": "2025-02-24T15:39:34.033111"
    },
    {
      "edgeClientid": "xLcwQTpLkTGm",
      "cpuValue": 17.79,
      "date": "2025-02-24T15:40:18.872261"
    }
  ],
  "memoryGraphModel": [
    {
      "edgeClientId": "xLcwQTpLkTGm",
      "memoryValue": 27.22,
      "date": "2025-02-24T15:38:45.268739"
    },
    {
      "edgeClientId": "xLcwQTpLkTGm",
      "memoryValue": 27.23,
      "date": "2025-02-24T15:39:34.048741"
    },
    {
      "edgeClientId": "xLcwQTpLkTGm",
      "memoryValue": 27.24,
```

```
            "date": "2025-02-24T15:40:18.883862"
        }
    ]
  }
]
```


| *RESPONSE BODY TYPE* | *JSON* |
|---|---|
| *RESPONSE KEY* | |


```
{
  "accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UVWsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",
  "refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"
}
```


### 3.7.25 Create Vm

| *API URL* | *http://192.168.0.34:8082/edge/node/vm/create_vm* |
|---|---|
| *METHOD* | *POST* |
| *REQUEST BODY TYPE* | *JSON* |
| *REQUEST BODY* | |


```
{" vmid":"106","name":"demo","iso":"local:iso/lubuntu-22.04.4-desktop-amd64.iso","edgeClientId":"xLcwQTpLkTGm","sockets":"1","cores":"2","memory":"2048","storageSpace":"5","provisionStatus":"false","interfaces":[{"subnet":"vmbr0","interface_Name":"net1","firewall":1}]}
```


| *RESPONSE BODY TYPE* | *JSON* |
|---|---|
| *RESPONSE KEY* | |


```
{
  "accessToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMTEyNCwiaWF0IjoxNzQwMzc3MTI0fQ.XkcZgA33I62Bc2BVHSOrTyVv5xkYomDl9Bf5wD7fg3UV
```

WsjKvJP91yTh0aSIfXAxi7Q6Lil9UXZAtQqTktoqdA",
"refreshToken": "eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhZG1pbiIsImV4cCI6MTc0MDQwMzQyMSwiaWF0IjoxNzQwMzc3MTI0fQ.dSA9txFCHEiGpkJ9At6sY632ED8Y2Ubd3Ns5Dex8YwXzEx_0Lc6YYh78BYmo1FO4I4ErE1lrTvgX8M-VbLFvkA"
}

### 3.8 MEC

#### 3.8.1 Overview – functionalities and description.

The MEC platform provides Virtualization infrastructure as part of edge cloud which is an integral part of 5G architecture. The details of

Virtualization is creating virtual infrastructure in terms of compute platform as part of edge cloud where applications can be hosted and accessed by all devices that are connected in a private 5G network.

### 3.9 Implementation of MEC Platform.

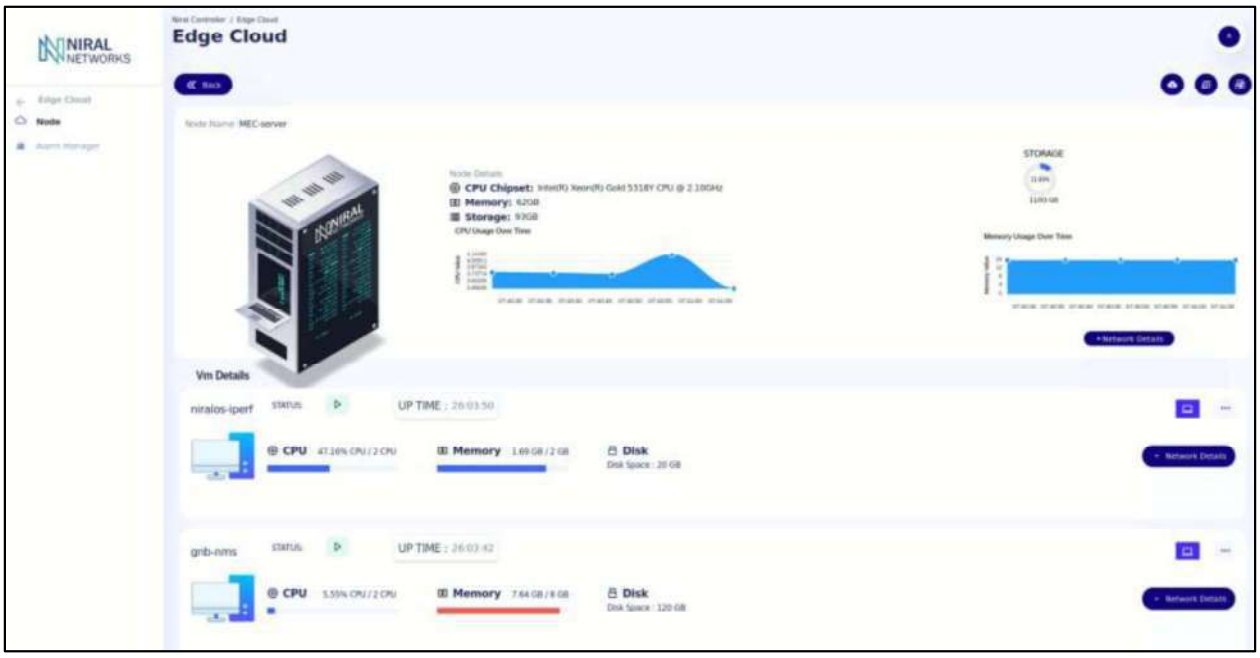
#### 3.9.1 Explanation of Hardware proposed for the lab.

MEC Edge application can be installed in any commercial server which have x86 or x64 architecture with Intel or AMD CPU.

#### 3.9.2 Provisioning of server for installation

**A. 1OS and Virtualization software installation.**

As part of the MEC platform, the base is a type1 hypervisor and it enables the creation of a virtual compute platform whereby we specify the number of storage requirements in terms of SSD, RAM requirement, number of cores required to utilize the CPU for the applications. All these can be provisioned as part of VM creation. There is also provision to delete the VMs and also know the statics of the VMs.



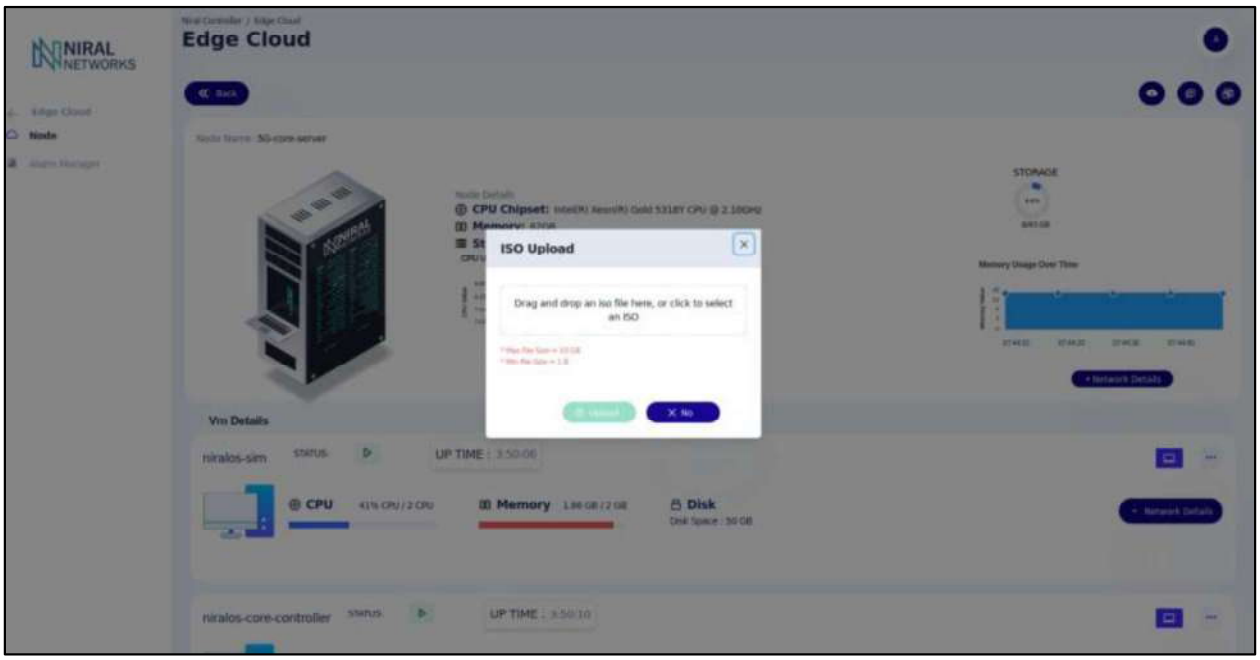

**B. Virtual machine for each network functions**

Network functions are installed as docker containers. Each network function has an individual container. We use a Macvlan concept that allows Docker containers to have their own MAC address and IP address from the host network's subnet. This enables the container to communicate directly with other devices on the physical network without NAT (Network Address Translation).

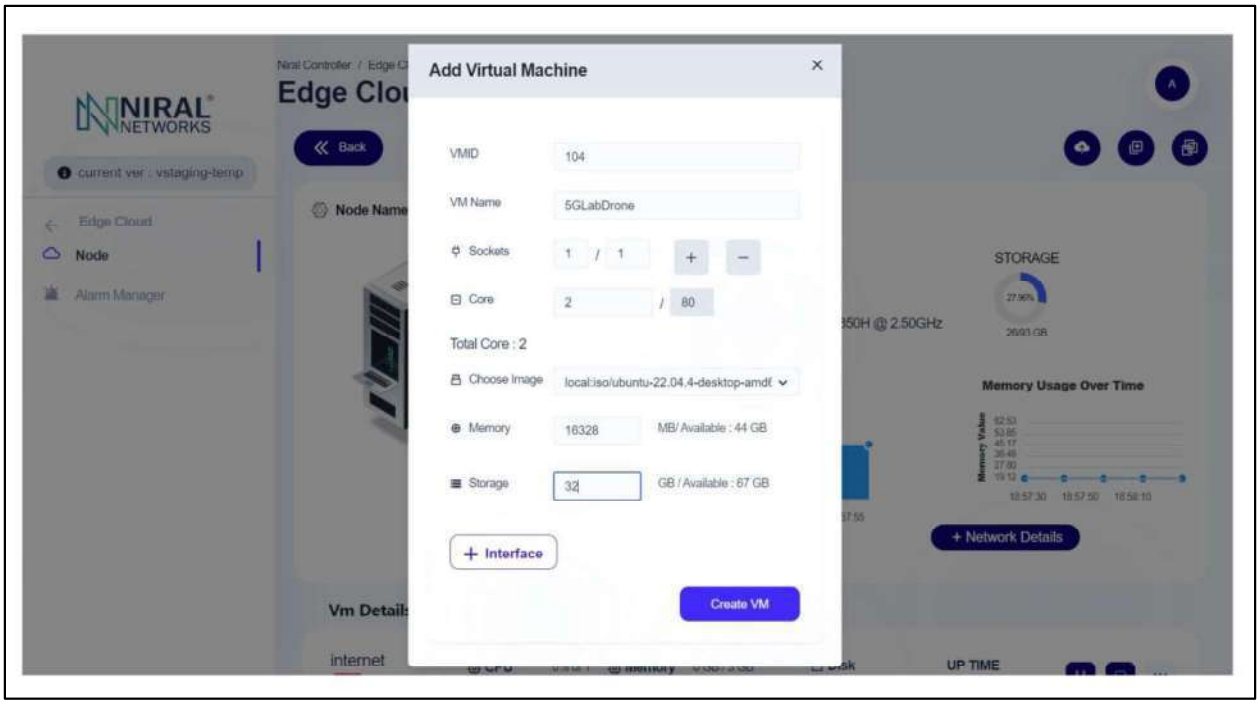
**3.10 Procedure for Application Installation**

As part of provisioning the virtualization compute platform, there is also provision to upload the operating system ISO file. So as part of vm creation OS is also installed. Once the virtualization compute platform is installed through the access of the desktop console we can install individual applications that run as part of the virtual platform. These applications are normally server applications to which devices that are connected on 5G infrastructure can communicate and successfully transfer data.

**Uploading ISO file**



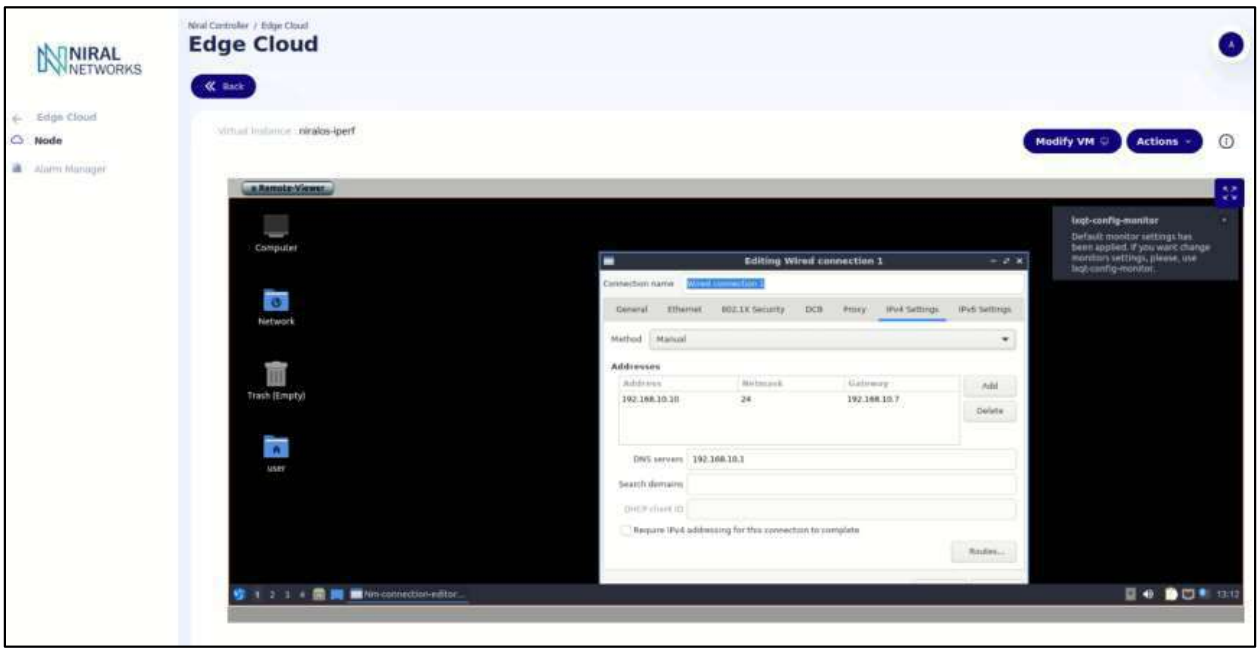

Creating Vms



### 3.11 Procedure to Route the traffic to MEC server.

When each VMs is created, the network interface UPF - N6 IP is provided as a gateway. So all traffic will be routed via gateway and we can achieve a very good latency.

Below picture shows as below.




### 3.12 Monitor analysis and management of application life cycle.

Once as part of compute virtual platform applications are installed, there are provisions in the MEC platform to monitor the working of the compute platform in terms of CPU utilization and memory utilizations. So these both can be tracked as part of functionality provided by MEC platform and apart from that various applications that are installed as part of MEC platform can also be monitored through dash board to have live monitoring of various devices that are connected in private 5G network.

### 3.13 Network Management System (NMS)

### 3.14 Unified management layer across the 5G RAN and the 5G-Core xNFs

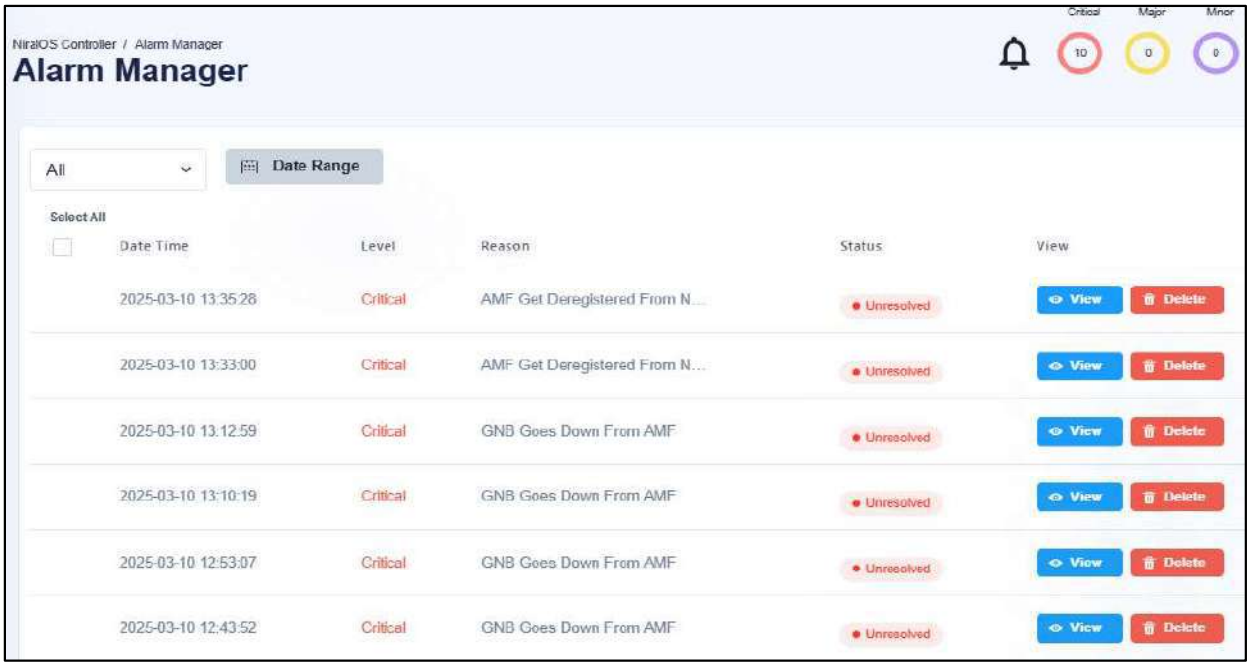
(Virtual Network Functions: VNFs & Physical Network Functions: PNFs) providing functions such as:

#### 3.14.1 Resource Management (RM) -

In 5G core, RAN and mobile resources
connected to the network are displayed.

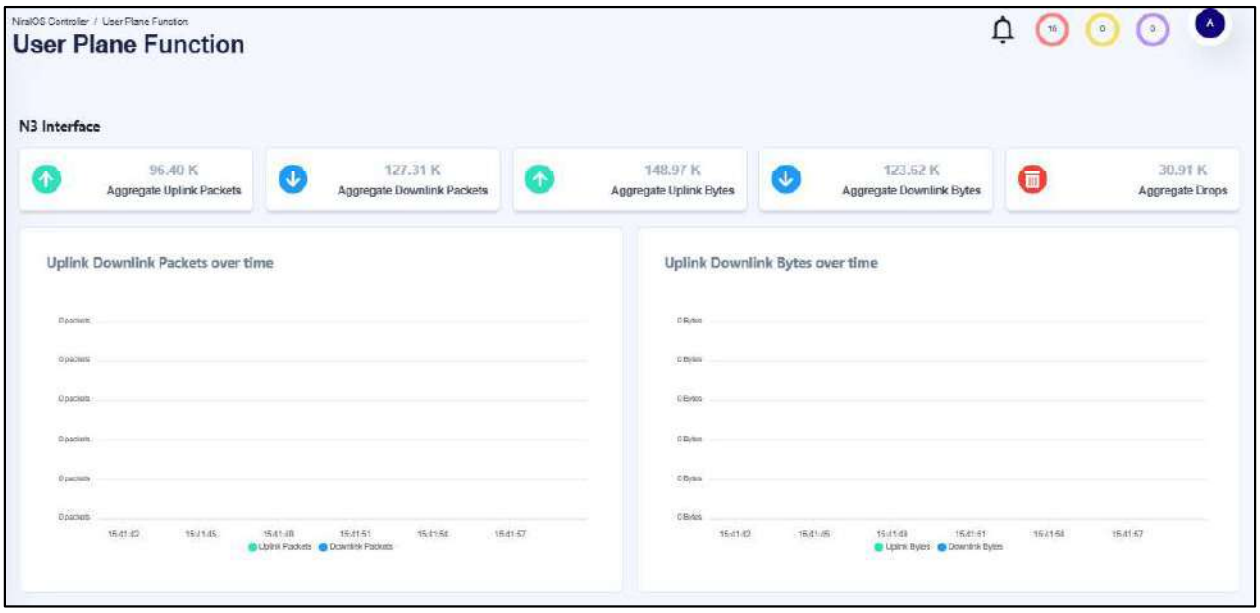
### 3.14.2 Fault Management (FM)

Alarms in 5G core provide erroneous status of the system including RAN connectivity to core.



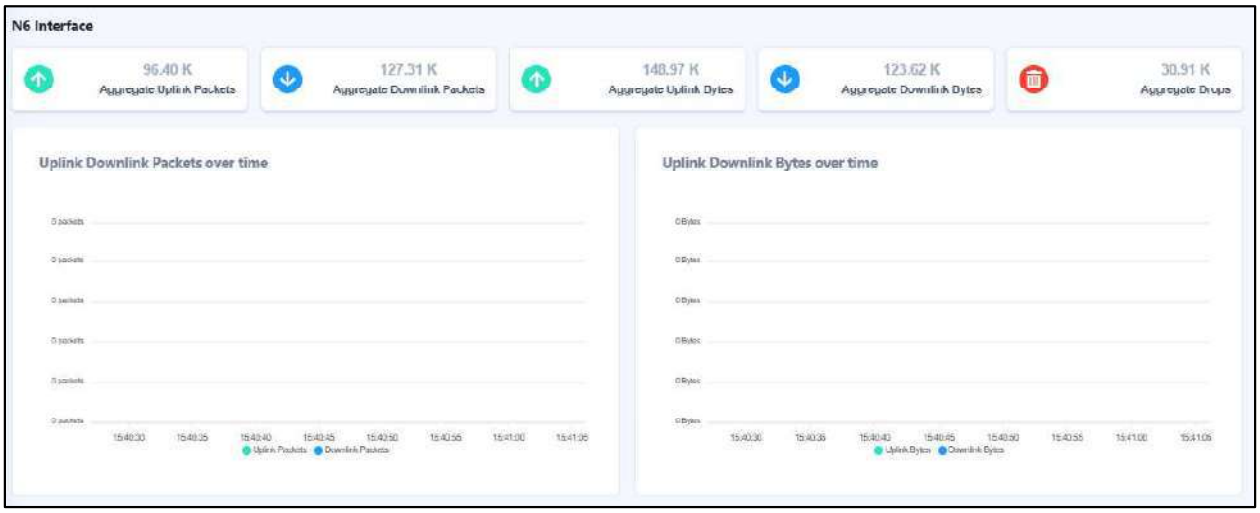
### 3.14.3 Performance Management (PM)

N3 interface traffic details in packets and graphical representation of packet flow are also sown as below.




N4 interface traffic details in packets and graphical representation of packet flow are also sown as below.

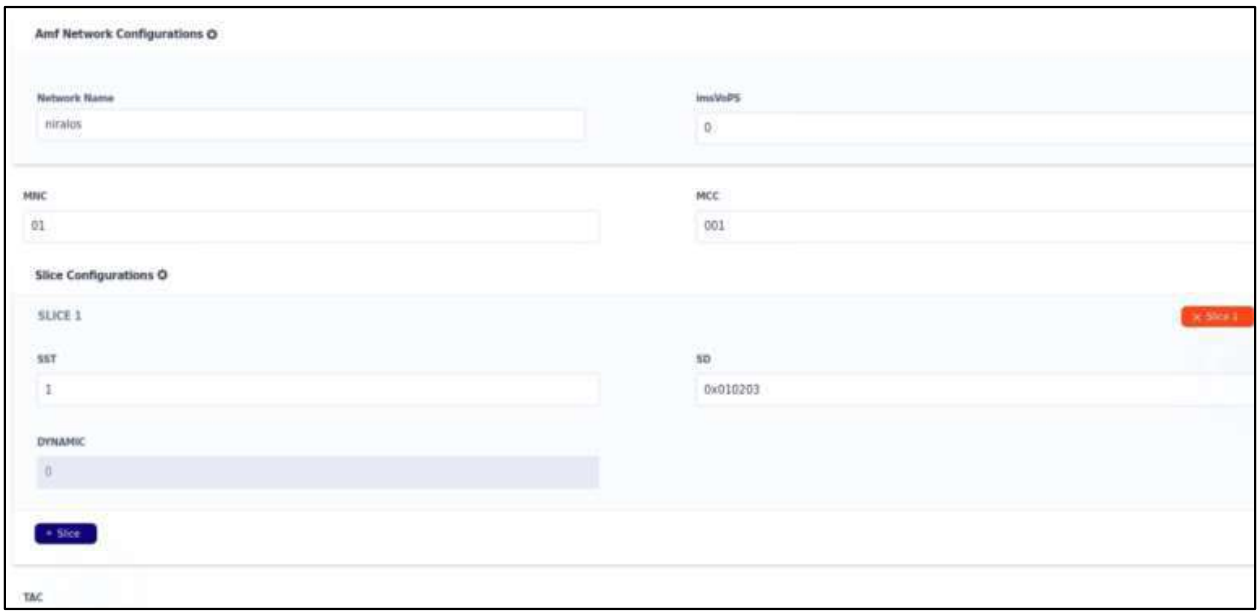
N6 interface traffic details in packets and graphical representation of packet flow are also sown as below.




### 3.14.4 Configuration Management (CM)

The tunnel ip configuration of the UPF:

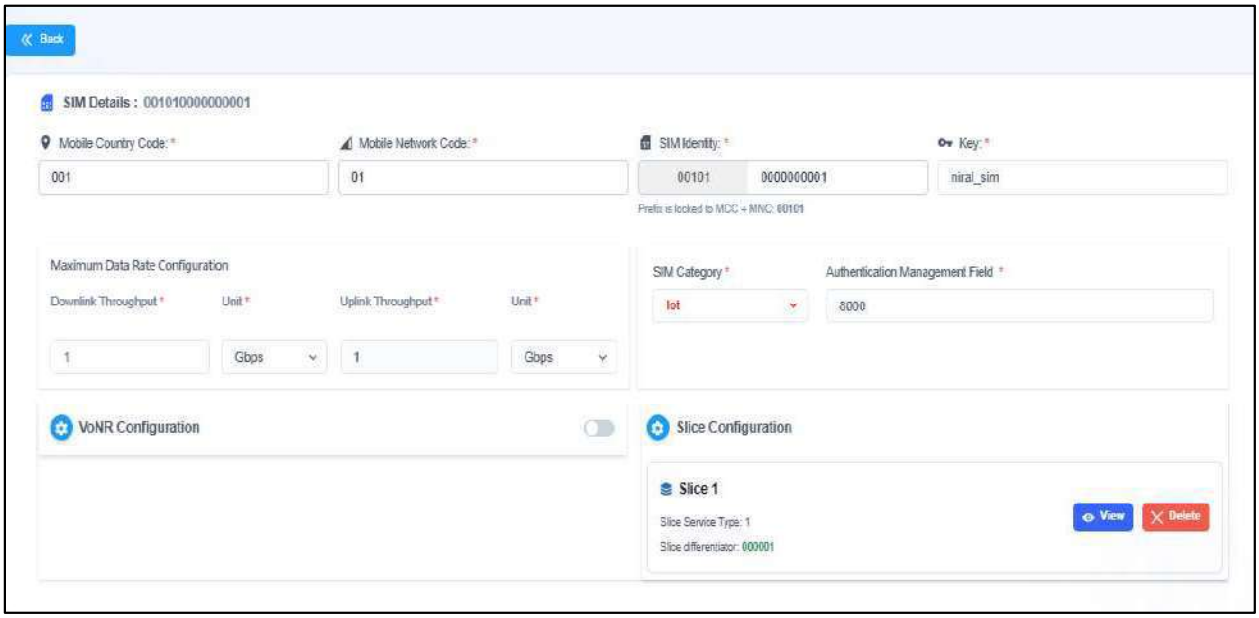
**Slice and plmn configuration:**




### 3.14.5 Configuring gNodeB

**PLMN ID**

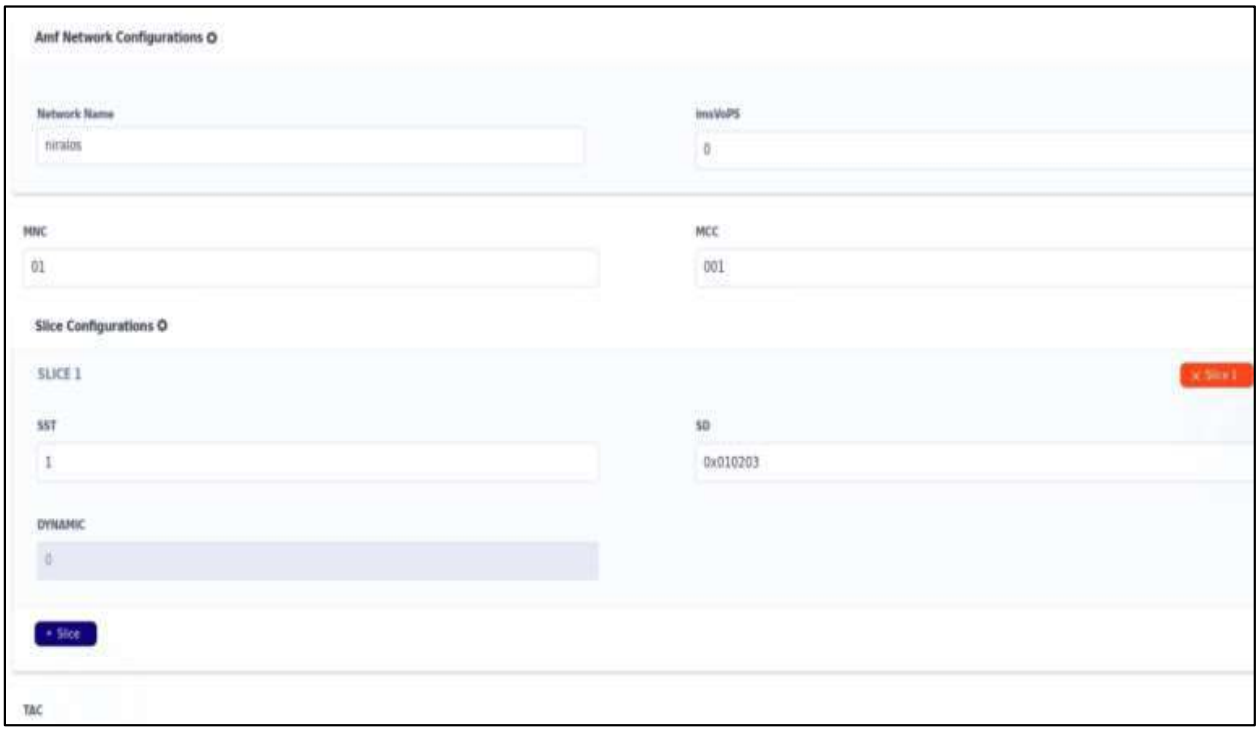
PLMN ID configuration in subscriber details. The MCC and MNC value constitute to PLMN



### 3.15 Configuring of 5G Core

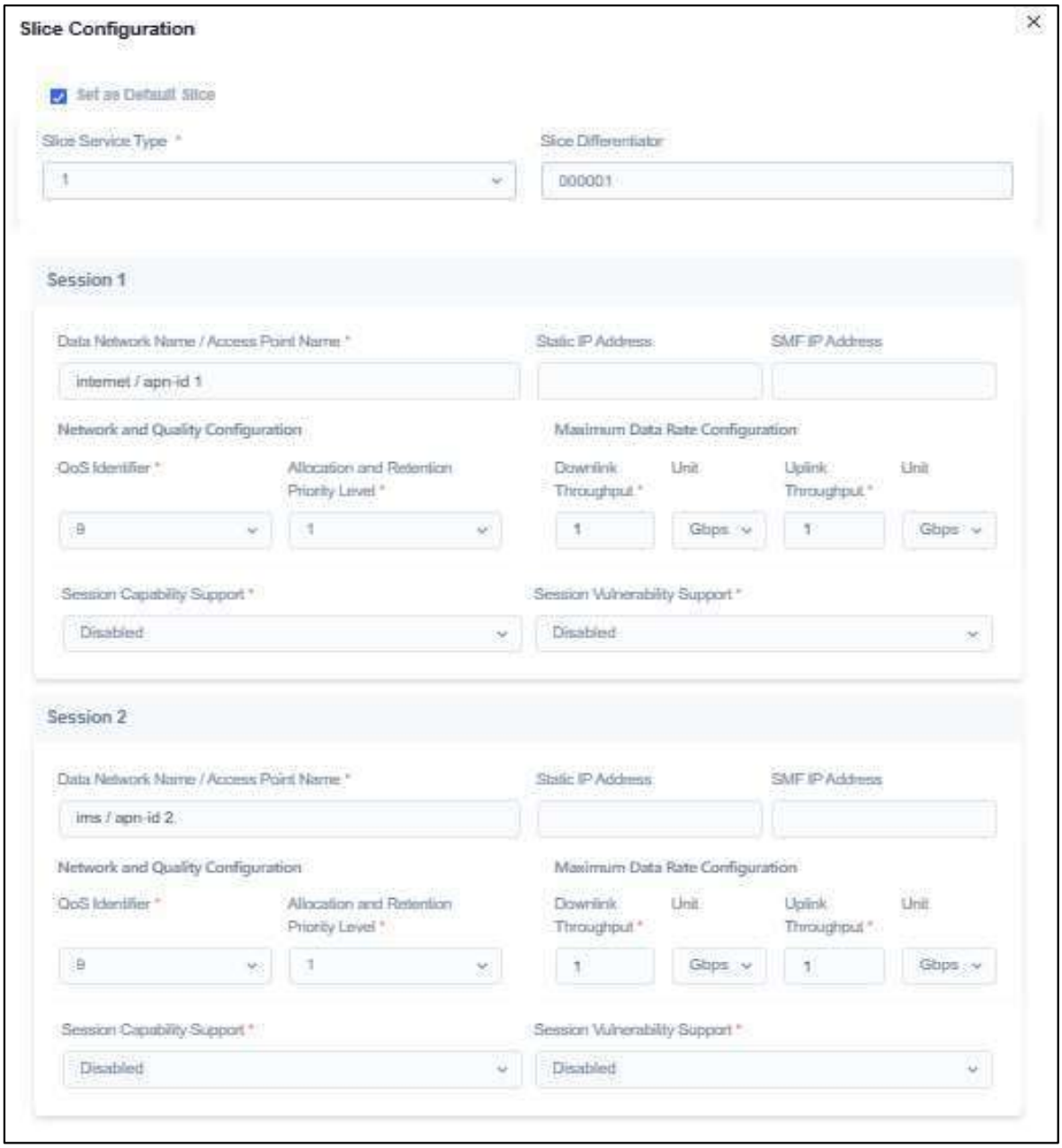
### Configuring AMF PLMN ID

AMF PLMN ID configuration can be done in core as shown below.



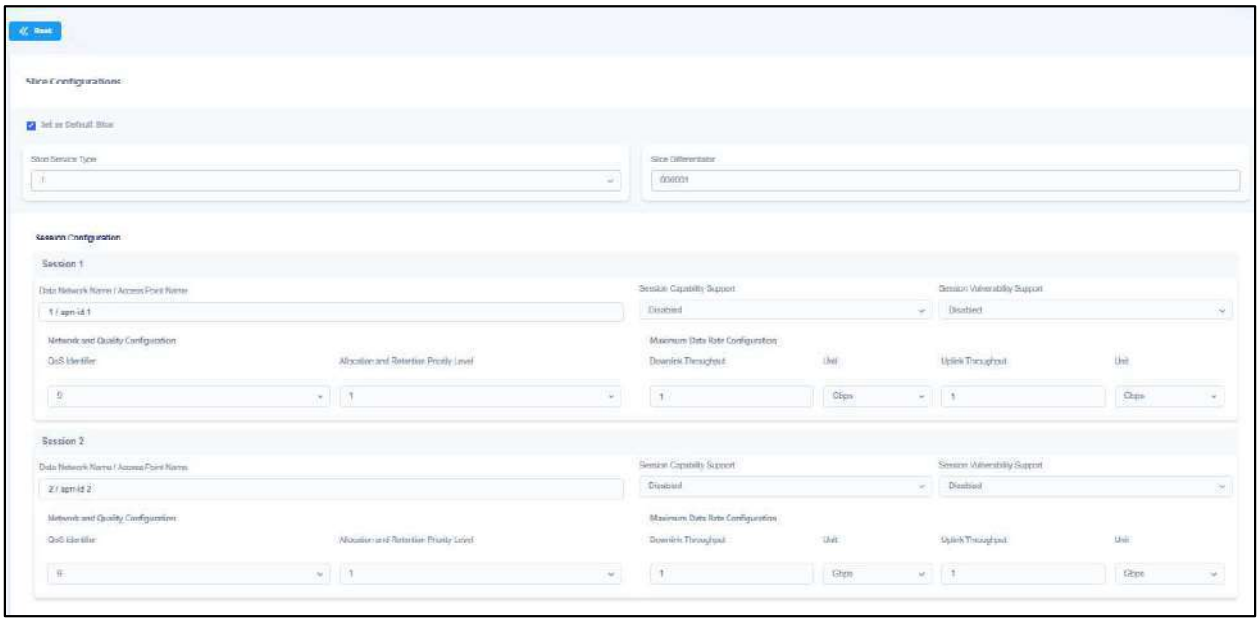
**Steps involved in assigning static IP addresses.**

IP address can be configured in slice details for each slice configuration as shown below.



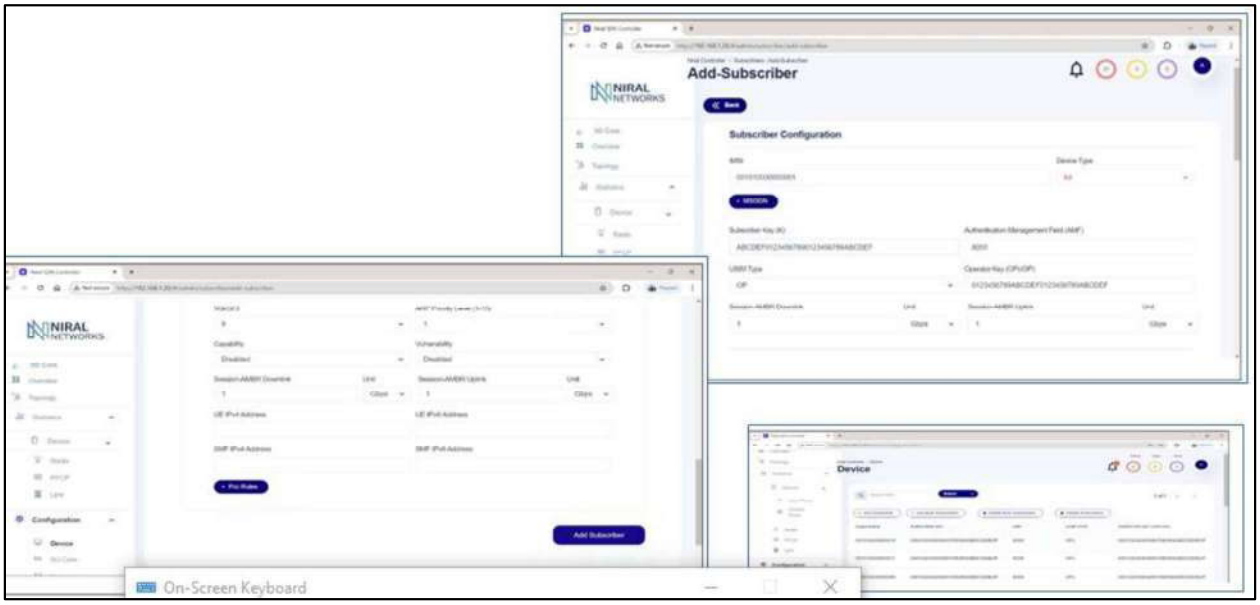
**Steps to configure other parameters**

Slice Configuration in Core




APN Configuration in Core

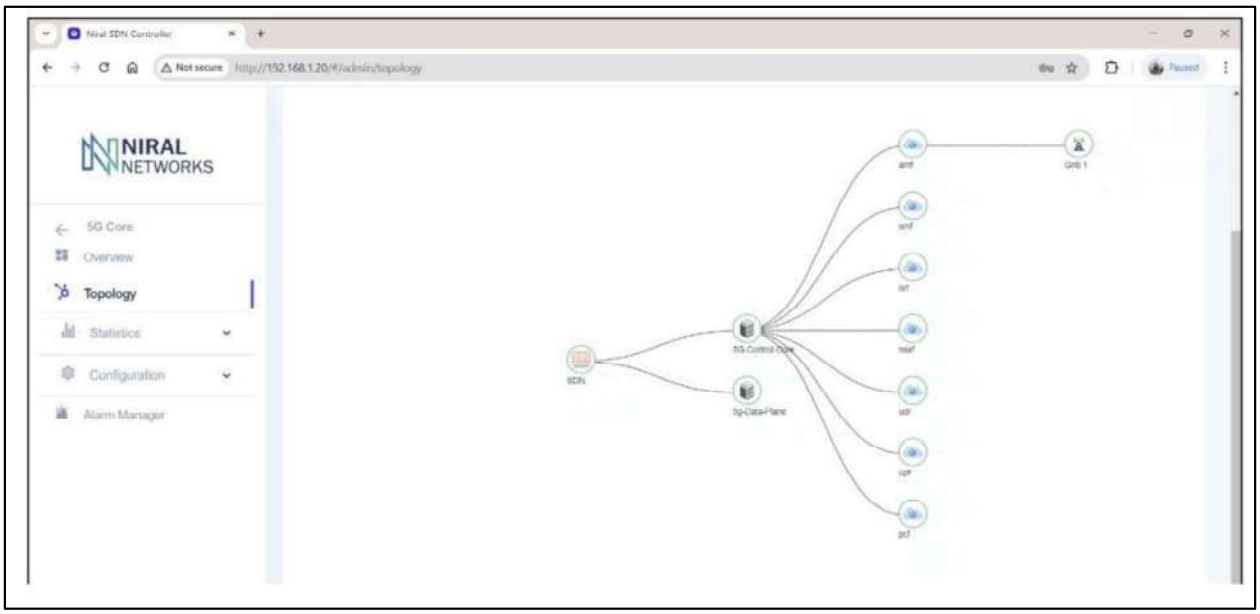
**Subscriber configuration in core as shown below:**




**3.16 Monitoring of 5G Network**

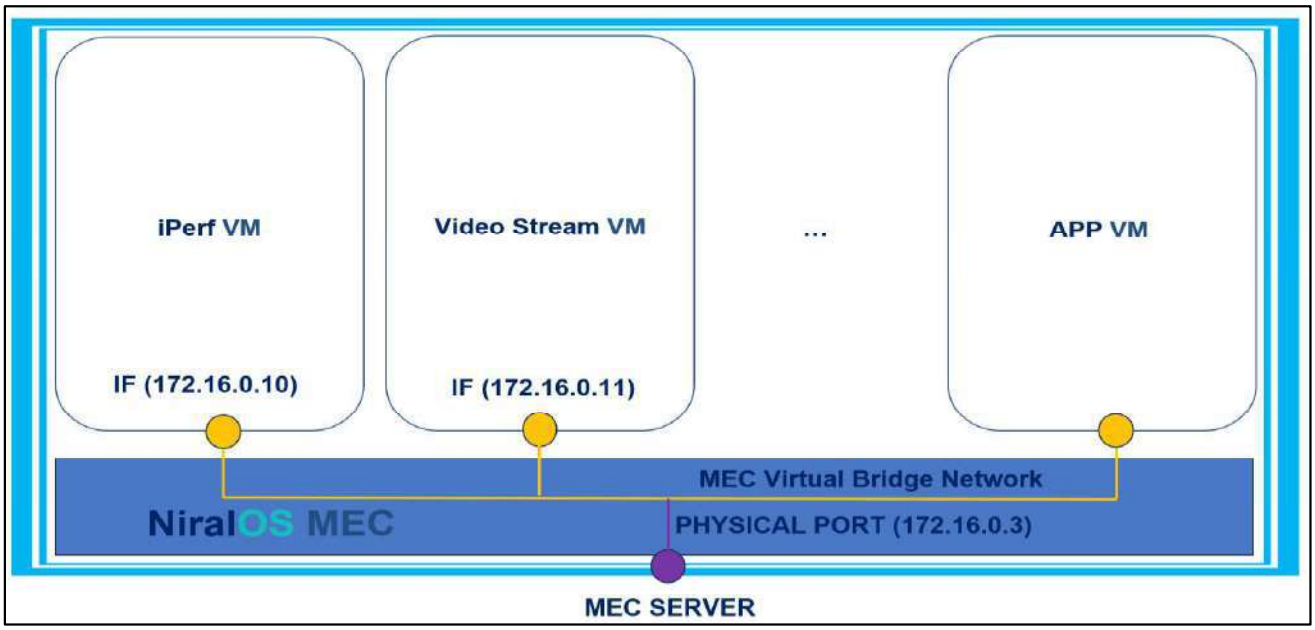
**Number of active devices**

Figure 2: Dash Board Before Running Core, Radio & Device



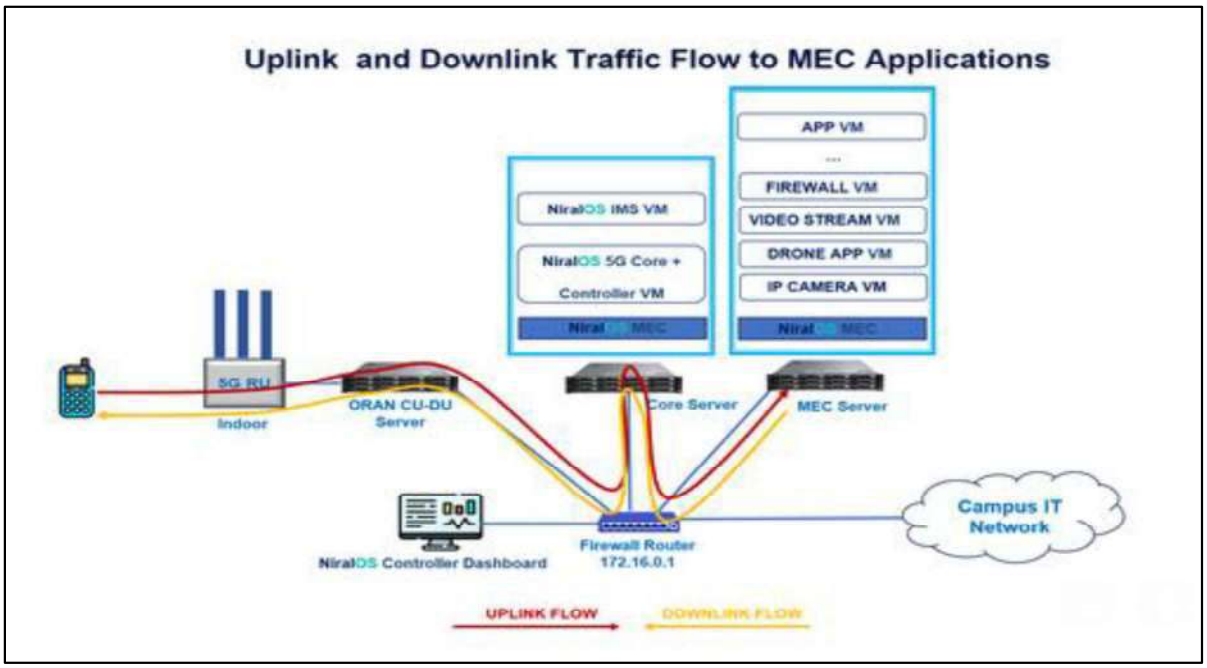

Data Usage per device

### 3.17 NiralOS MEC – Add and Delete VM

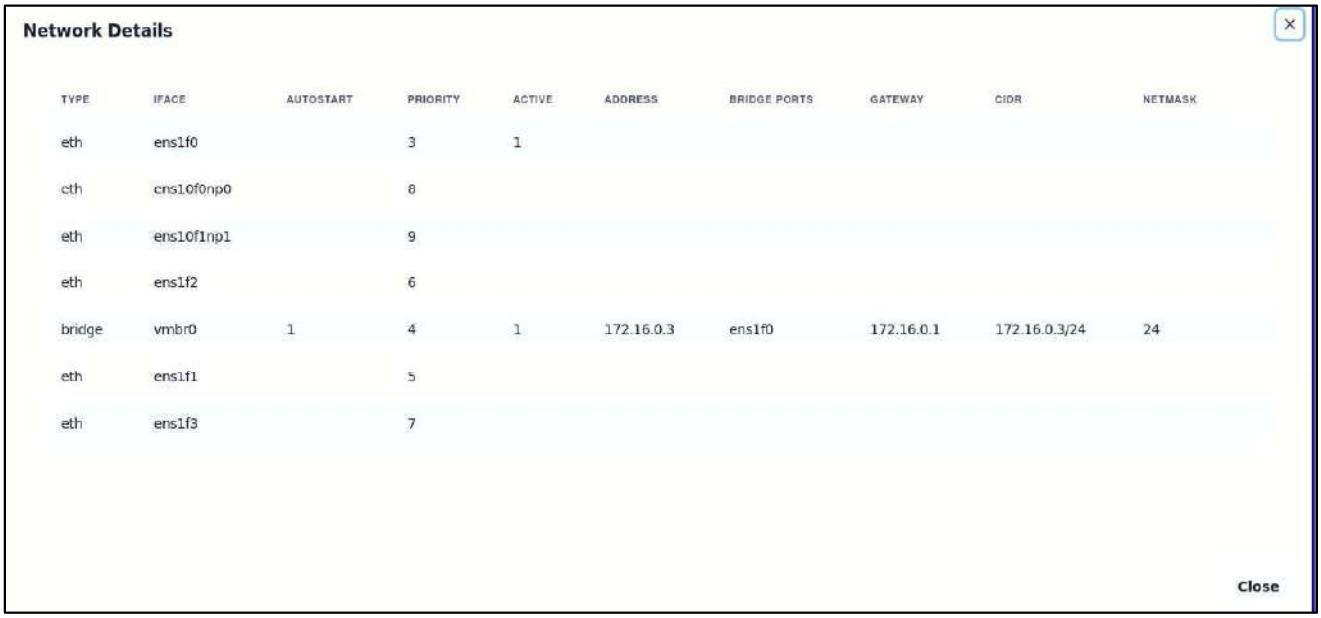



NiralOS Edge platform offers virtualized edge infrastructure to create VMs for enterprise edge applications to take advantage of privacy, security, and low latency features. The controller provides access to an edge platform through which Virtual environment instances can be created, Monitored, and also managed.

The following diagram illustrates a control flow passing through 5G Lab setup firewall, specific use case is regarding traffic from UE passing through RAN and Core to access the MEC application framework.




### 3.18 Container Interface Details:



**Network Details**

| TYPE | IFACE | AUTOSTART | PRIORITY | ACTIVE | ADDRESS | BRIDGE PORTS | GATEWAY | CIDR | NETMASK |
|---|---|---|---|---|---|---|---|---|---|
| eth | ens1f0 | | 3 | 1 | | | | | |
| eth | ens10f0np0 | | 8 | | | | | | |
| eth | ens10f1np1 | | 9 | | | | | | |
| eth | ens1f2 | | 6 | | | | | | |
| bridge | vmbr0 | 1 | 4 | 1 | 172.16.0.3 | ens1f0 | 172.16.0.1 | 172.16.0.3/24 | 24 |
| eth | ens1f1 | | 5 | | | | | | |
| eth | ens1f3 | | 7 | | | | | | |

Close

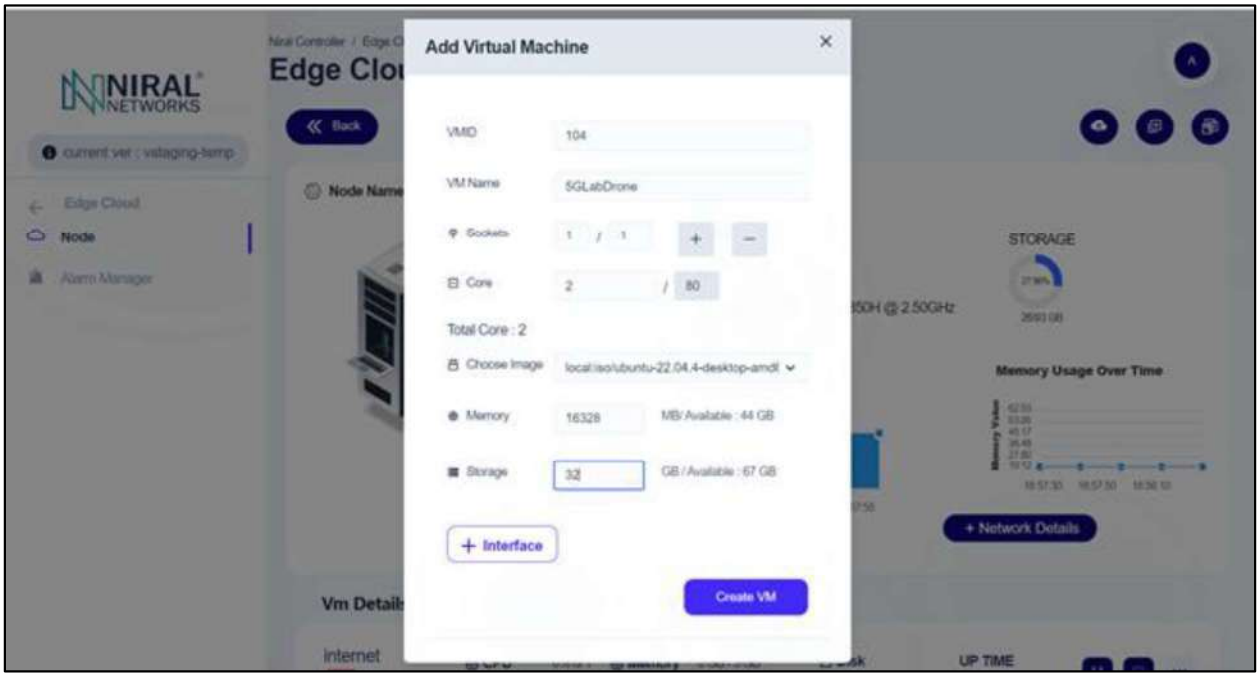
**Above image is taken from T2 Server network details section.**

**Equipment:** T2 of 1X Quad Port NIC card and 1X Dual Port NIC Card
LAN Cable from Firewall Router Connected to 1st Port of the Quad Port NIC Card.
Niral EDGE automatically creates a bridge network interface **vmbr0** using physical interface **ens1f0.**
Now all the VM Interfaces that are added from the vmbr0 subnet attains IP address in subnet **172.16.0.X/24** same as the **vmbr0** interface.
** Remaining all the In-Active Ports can be utilised for future expansions.
**Index:**
Active = 1 ( Interface is active )
AutoStart = 1 ( Interface Automatically comes up on boot of the server.

1) Create VMs with the information such as VM Name, IP address and compute resource definition to be associated with the VMs
2) Display a list of VMs created with basic information such as memory and CPU usage.
3) Desktop access to the VMs for normal operations such as maintenance, server application installation, troubleshooting, etc.

The NiralOS Edge infrastructure includes basic firewall features enabled for every VM created. The default firewall features enabled at the Hypervisor level are replicated to all VM(s) created and edit of the firewall table is not provided.

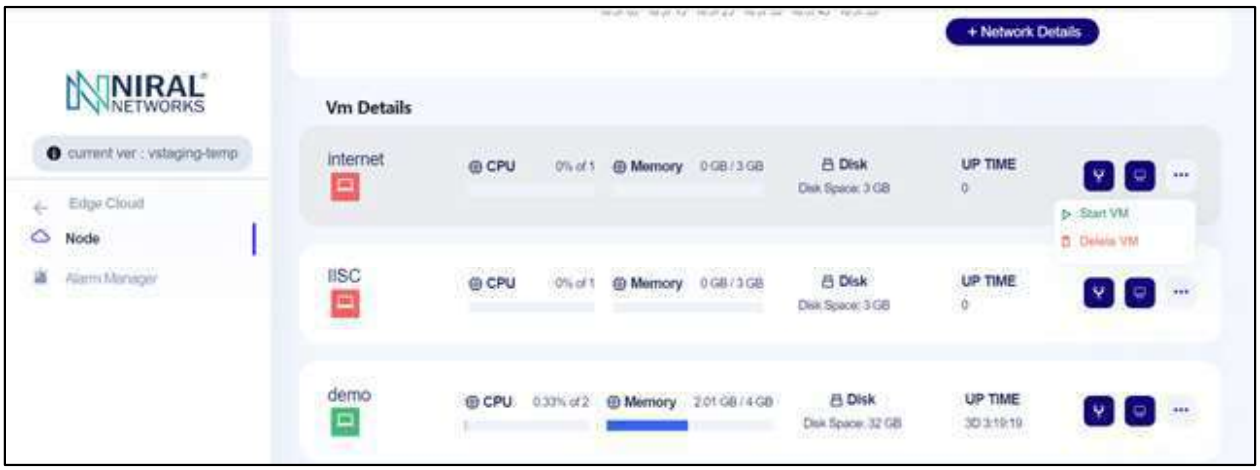
The following diagram covers the add VM scenario:




The following Table details the VM creation parameters and their description in Hypervisor framework:

| VM Creation Field Nam | VM Creation Field Description |
|---|---|
| VMID (104) | This is the unique identifier for the virtual machine. Each VM on th system has a different VMID. |
| VM Name (5GLabDrone) | The name of the virtual machine for identification. In this case, it named "5GLabDrone." |
| Sockets (1) | Refers to the number of physical CPU sockets to allocate to the VM This form has 1 socket. |
| Core (2): | The number of CPU cores per socket. This field shows that 2 core are allocated out of 80 available cores. |
| Total Core (2): | Displays the total number of CPU cores assigned to the VM. It's combination of sockets and cores (1 socket × 2 cores = 2 total cores |
| Choose Image ( local:iso/ubuntu-22.04.4-desktop-amd64.iso): | This indicates the operating system or image that will be installed o the VM. In this case, it's an Ubuntu 22.04.4 desktop image. |
| Memory (16328 MB) | The amount of RAM allocated to the VM, shown in megabyte (16,328 MB = around 16 GB). The system indicates 44 GB o available memory. |
| Storage (32 GB) | The amount of disk space allocated to the VM. In this case, it's 3 GB, with 67 GB available. |
| Interface | This button likely allows adding network interfaces to the VM configuring networking options. |
| Create VM | This button will create the VM based on the defined settings. |
| The configuration defines a VM with 2 CPU cores, 16 GB of RAM, 32 GB of disk storage, an the Ubuntu operating system. | |

The following diagram covers the delete VM scenario:




### 3.19 Basic Configuration

#### 3.19.1 Login ID Details

The NiralOS Controller dashboard is available on the IP 172.16.xx.xx

Username: admin

Password: admin@1234

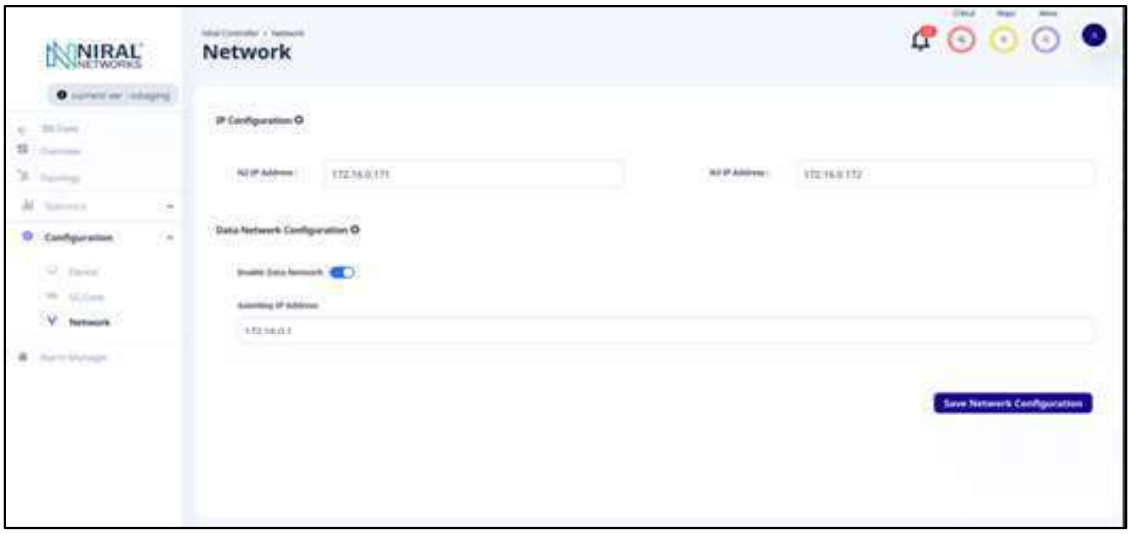
Note: Use 172.16.xx.xx IP for Controller Dashboard The NiralOS private 5G solution configuration from the perspective of the following use cases will be illustrated in detail in the user guide.

After login overview page, which contains 3 sections

a) NiralOS 5G Core: For core functionalities
b) NiralOS Controller
c) NiralOS EDGE: For EDGE Cloud-related functionalities

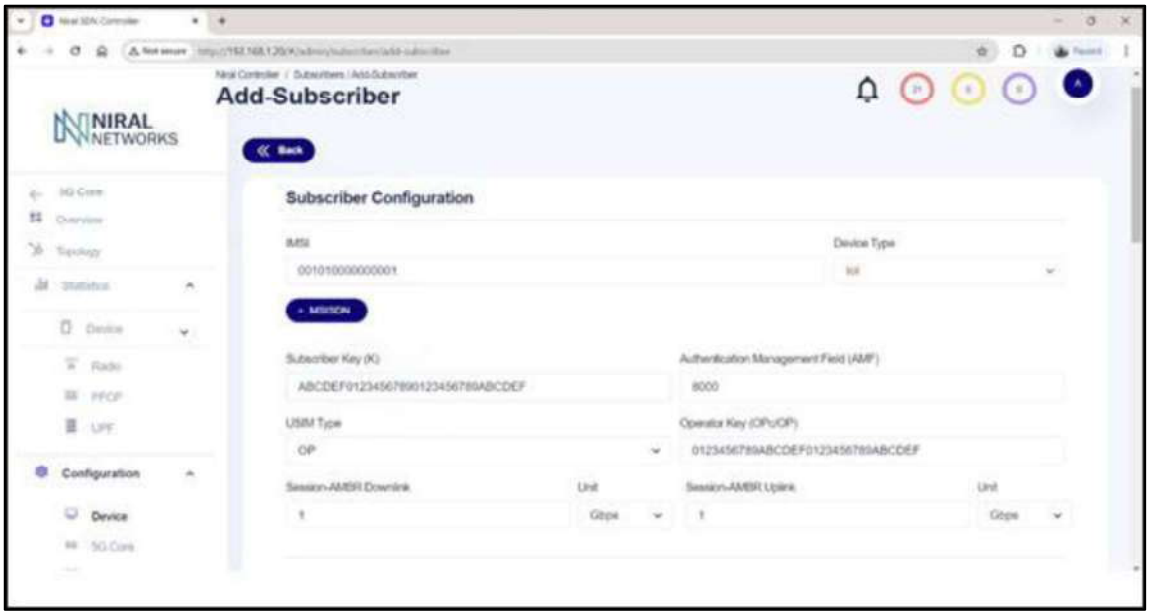
### 3.19.2 Network Interface configuration

Core provides configuration options to configure the N2, N3, and Gateway IP addresses:



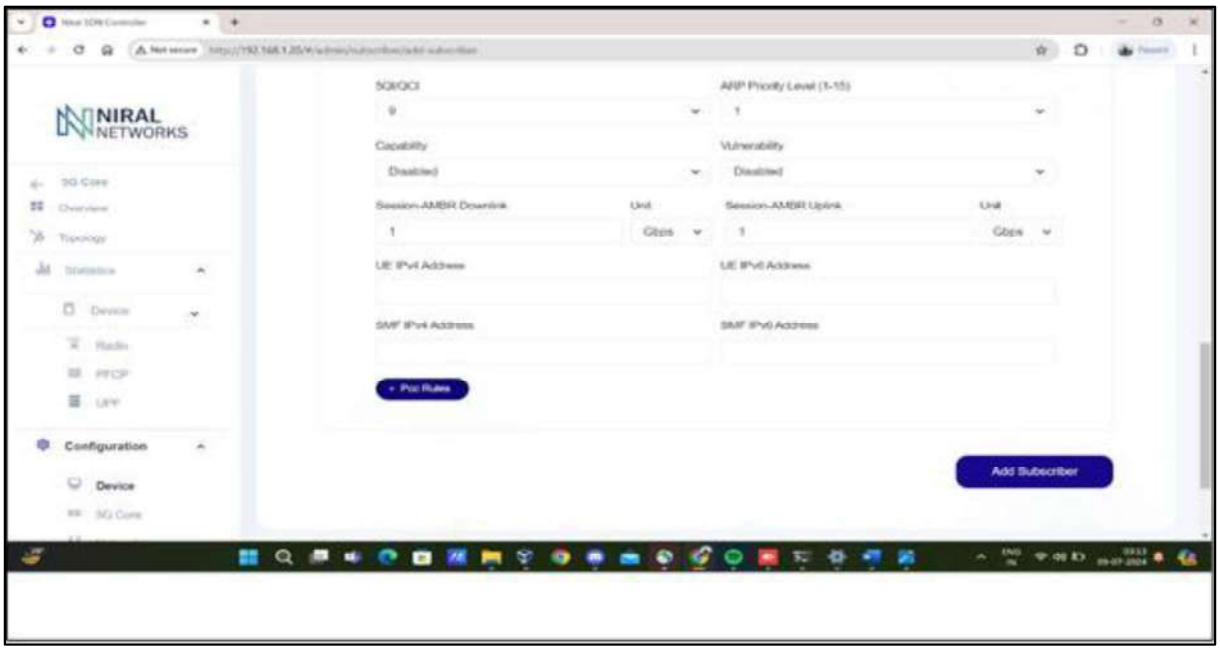
### 3.19.3 Subscriber configuration

Choose this option under the Configuration menu from the left pane for adding UE details. Once you are on this page you can add UE one by one or multiple UEs on a single go by clicking the add bulk Subscribers option. The additional deletion of the user or deletion of the bulk users can also be done by choosing the delete subscriber or delete bulk subscriber option.




The following details for Subscriber can be configured:

1. IMSI (MCC and MNC as prefix)
2. Subscriber and operator key
3. SIM Type and Device Type
4. Session Aggregate Maximum Bit Rate (UL and DL)
5. Network slice details




**3.20 Drone Live Streaming on MEC-**

Steps to open Drone MEC -

Step 1- Open Niral Network 172.16.XX.XX

**Step 2 - Go to NiralOS EDGE cloud**

Step 4 - Open Drone Vm and access the URL.

Step 5 - Enter the I.P. Address of Sim card Inserted in Drone (10.101.XX.XX)

Fig- showing Drone Camera Visuals on MEC VM

Fig- Showing API integration of Drone in MEC

**3.21 Camera Live Streaming on MEC-**

Steps to open Camera MEC -

Step 1- Open Niral Network 172.16.XX.XX

Step 2 - Go to NiralOS EDGE cloud

Step 4 - Open Camera Vm and access the URL.

Step 5 - Enter the I.P. Address of Sim card Inserted in Camera (10.101.XX.XX)

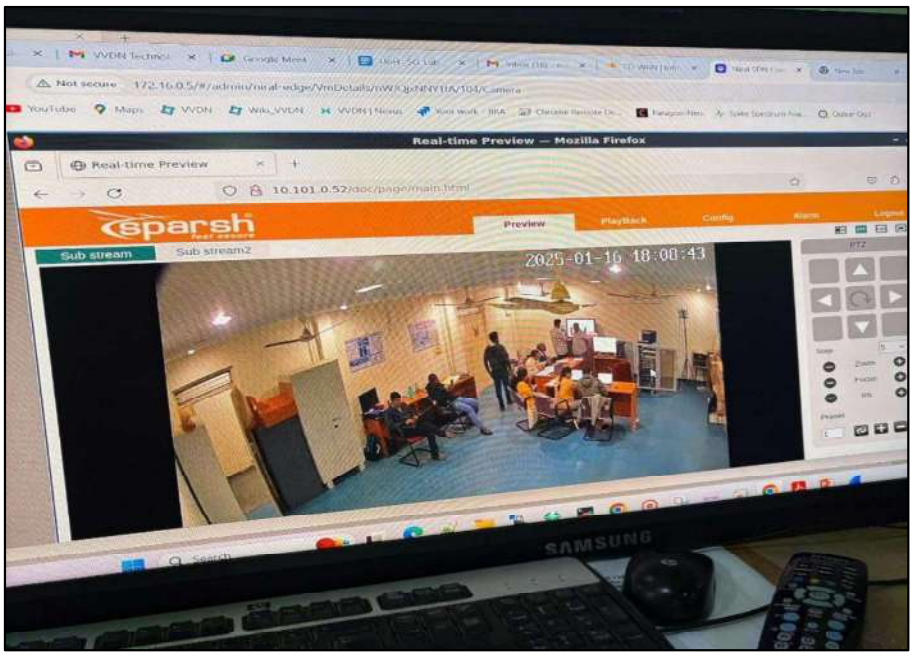


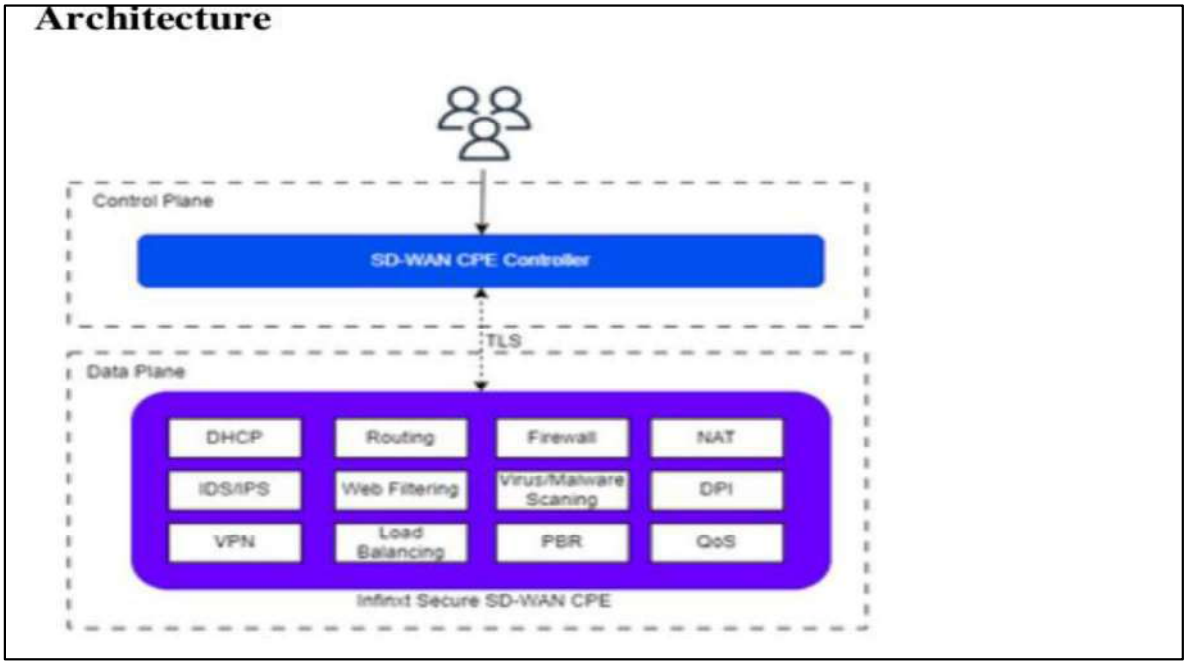

**Fig- Showing Camera footage on MEC VM**

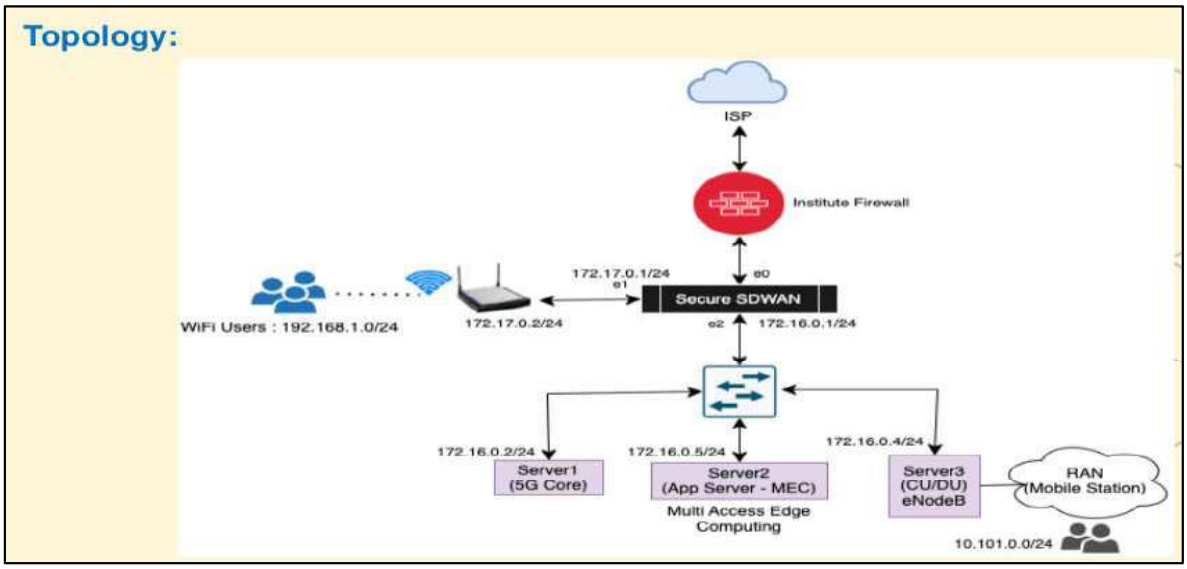
## 4.0 Infinity Firewall Secure SDWAN

### 4.1 Introduction

The Secure Next Generation SD-WAN offering from Infinity Labs provides its customers with the best of both Network connectivity and Application security. It's a unique combination to provide a secured application experience to the users.Along with SD-WAN features it also offers NGFW features such as Application layer visibility,Intrusion Detection and PreventionSystem, Virus/Malware protection and Web Filtering.

### 4.2 Architecture



**5.2 Implementation:**

• Network segmentation will separate external users and the lab setup into distinct broadcast domains, and policies will be implemented to restrict external user access to the lab setup.

• Enabling the Intrusion Prevention System (IPS) and Intrusion Detection System (IDS) features enhanced network security by monitoring and controlling traffic to prevent unauthorized access to the internal network. These systems actively detect and respond to suspicious activities or threats, ensuring that any attempts to breach the network are swiftly identified and mitigated.

**4.3 Components**

Infinxt secure SD-WAN solution is a true SD-WAN solution created on the principles of SDN. It provides clear separation between the Data plane and the Control Plane. Following SDN concepts, the Data Plane still continues to function even if the Control plane is down.

**4.4 Control Plane**

The control plane is responsible for network management, routing decisions, policy enforcement, and overall orchestration of the SD-WAN services. The control plane consists on SDWAN Controller which provides centralized management console Web UI which is responsible for managing the SD-WAN CPE configurations to control data plane services, traffic steering, managing firewall policies, etc.

**4.5 Data Plane**

The data plane is responsible for the actual forwarding of user traffic (data packets) between different locations over the WAN. It handles the movement of data based on the instructions received from the control plane. This includes Secure SD-WAN edge devices which are deployed at branch offices, data centers and cloud environments.

**4.6 DHCP (Dynamic Host Configuration Protocol)**

DHCP is a network management protocol used to automatically assign IP addresses and other network configuration parameters to devices on a network. By eliminating the need for manual IP address configuration, DHCP simplifies network management and reduces the potential for errors. When a device connects to the network, the DHCP server provides it with an IP address, subnet mask, default gateway, and DNS server information, allowing the device to communicate effectively within the network.

The DHCP server has a mac binding feature to bind specific ip address to a mac address of a device/system connected in LAN network to ensure that device always gets a fixed assigned ip

address from IP pool. This helps to identify the device in the network using its IP address and thus useful when applying specific security policies or load balancing policies based on IP address. The DHCP client is responsible for obtaining ip address from DHCP server. The interfaces which are configured in DHCP mode receive the IP address from upstream DHCP server.

**4.7 Routing**

Routing is the process of directing data packets from their source to their destination across multiple networks. Routers use routing tables and algorithms to determine the best path for data to travel through complex networks. This process involves analyzing packet headers, making forwarding decisions, and managing routes to ensure efficient and reliable data transmission Routing can be static (manually configured) or dynamic (adjusted based on network conditions). Infinxt SD-WAN supports static routing as well as dynamic routing protocols such as OSFP and BGP for large enterprise networks.

**4.8 Firewall**

Firewall is a security feature designed to monitor, filter, and control incoming and outgoing network traffic based on predetermined security rules. It acts as a barrier between a trusted internal network and untrusted external networks, such as the internet. Firewall helps to prevent unauthorized access, protect against cyber threats, and enforce security policies by allowing or blocking traffic based on criteria like IP addresses, domain names, port numbers, protocols and user/group based policies.

**4.9 Intrusion Detection System/Intrusion Prevention System (IDS/IPS)**

IDS and IPS are security solutions used to detect and respond to malicious activities on anetwork. An IDS monitors network traffic and identifies potential threats or suspicious behavior and generates alerts when malicious activity is detected. An IPS, on the other hand, not only detects threats but also takes proactive measures to block or mitigate them in real-time, providing a more active defense mechanism against attacks.

The firewall performs deep packet inspection to decode application layer information and use its anomaly detection capabilities to detect anomalies in traffic patterns. Firewall uses its signatures database to detect and mitigate various threats providing defense mechanism against attacks and exploits.

**4.10 Web Filtering**

Web filtering is a technique used to restrict or control access to websites and online content based on firewall policies. It is commonly used in both corporate and educational environments to enforce policies, enhance productivity, and protect users from inappropriate or harmful content. The L7 Firewall allows creating policies to block access to websites based on categories or website domain names in URLs.

### 4.11 Deep Packet Inspection (DPI)

DPI is a network packet analysis technique that examines the data part (payload) of packets in addition to their header information. This detailed inspection allows for more granular monitoring and control of network traffic, enabling the identification of applications, protocols, and specific content within packets. DPI can be used for purposes such as network security, performance optimization, and traffic shaping.

### 4.12 Virus/Malware Detection Engine

Virus/malware detection engine is responsible for scanning and detecting malicious files transferred over the network through the firewall. The firewall inspects the traffic flowing through its protected network interfaces and uses its signature database to examine files transferred over the network for signs of viruses, worms, trojans, ransomware and other types of malwares. By continuously scanning and monitoring network traffic the firewall can protect users and systems in private network from virus/malware infections and ensure data integrity and safety.

### 4.13 Virtual Private Network (VPN)

A VPN is a technology that creates a secure and encrypted connection over a less securenetwork, such as the internet. It allows users to access private networks remotely while ensuring that their data is protected from interception and unauthorized access. VPNs are commonly used for secure remote access, maintaining privacy, and bypassing geographic content restrictions by masking the user's IP address.

Infinxt Secure SD-WAN provides various VPN solutions depending on use-cases such as IPSec VPN, Remote Access VPN (RAVPN), Infinxt Hub and Spoke VPN, Infinxt Auto MESH VPN, etc.

### 4.14 WAN Load Balancing

WAN Load Balancing is the process of distributing network or application traffic across multiple WAN links to ensure optimal performance, reliability, and availability. By balancing the load, organizations can prevent any single link failure from becoming a bottleneck, enhance user experience, and improve overall system efficiency. If a path fails, traffic is balanced across the remaining healthy paths, a recovered path is automatically added back to the routing table and used by the load balancer. The load balancer automatically adds routes for each path to the routing table and balances traffic across the configured interfaces, determined by interface health and weight.The WAN load balancer can also be configured in Active/Standby failover mode where one interface is set to be the primary interface and other interfaces are set as secondary. In this approach, traffic is sent through only the primary interface and in case of failure, a secondary interface is selected from the pool of available interfaces. The primary interface is selected based on its weight and health.

### 4.15 Quality of Service (QoS)

QoS is a network management technique that prioritizes certain types of traffic to ensure optimal performance for critical applications and services. By allocating bandwidth, managing latency, and controlling jitter, QoS helps maintain the quality of service for applications such as VoIP, video streaming, and web applications. Implementing QoS policies ensures that high-priority traffic receives the necessary resources, even during times of network congestion.

#### 4.15.1 Login(IP-72.16.0.X)

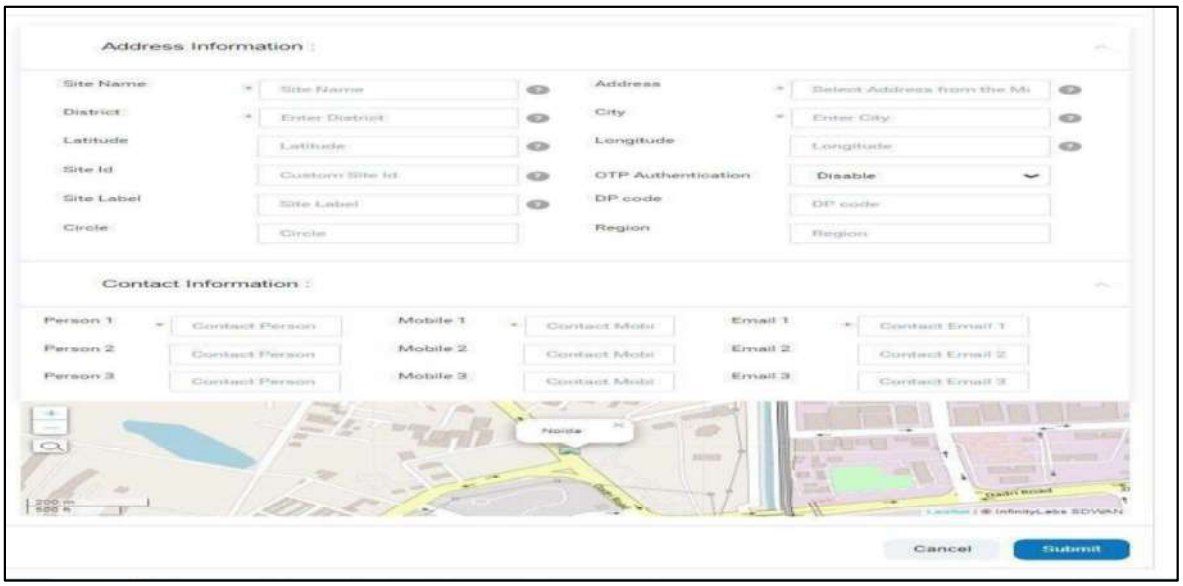
This is the initial page of the InfiNxt Controller. Please log in using the correct credentials.

#### 4.15.2 Dashboard

Upon logging into the controller with their credentials, users will land on the first page of the Controller. Here, they can access a dashboard displaying information such as the number of sites, devices, links, VPNs, and other relevant metrics.

#### 4.15.3 Sites

1. This feature displays information about existing sites. To add a new site, go to the **Sites** tab and click on “Add”
2. Complete all the required fields and then click on Submit.


Address Information :
Site Name
Site Name
Address
Select Address from the Mi
District
Enter District
City
Enter City
Latitude
Latitude
Longitude
Longitude
Site Id
Custom Site Id
OTP Authentication
Disable
Site Label
Site Label
DP code
DP code
Circle
Circle
Region
Region
Contact Information :
Person 1
Contact Person
Mobile 1
Contact Mobi
Email 1
Contact Email 1
Person 2
Contact Person
Mobile 2
Contact Mobi
Email 2
Contact Email 2
Person 3
Contact Person
Mobile 3
Contact Mobi
Email 3
Contact Email 3
Noida
Cancel
Submit

## Table of Contents

1. **Table of Contents** 1

5G CAMERA 11

1.1 Use Case Development with 5G Camera: 11

1.2 Devices Used for Development 12

1.2.1 Device Parts 12

*1.2.2 S*ensor Specifications 13

1.3 Utilities and Configuration 15

1.3.1 Camera page login & Configuration 15

1.3.2 Preview interface： 15

1.3.3 Local Config: 16

A. Encode: 16

i. Encode setting: 16

ii. Audio: 19

1.4 Prev Mode: 19

1.4.1 Output model: 19

1.4.2. Display config: 20

1.4.3 Camera: 21

1.4.4 ISP: 21

1.4.5 Profile: 22

1.4.6 Privacy Masking: 22

1.4.7 Overlay: 22

1.5 Network: 23

1.5.1 TCP/IP: 23

1.6 Net Service: 24

1.6.1 Email: 24

1.6.2 DDNS: 24

1.6.3 RTSP: 25

1.6.4 FTP: 25

1.6.5 UPnP: 25

1.6.6 P2P: 26

1.6.7 IP filter: 26

1.7 Record: **Error! Bookmark not defined.**
1.7.1 Record Plan: 27

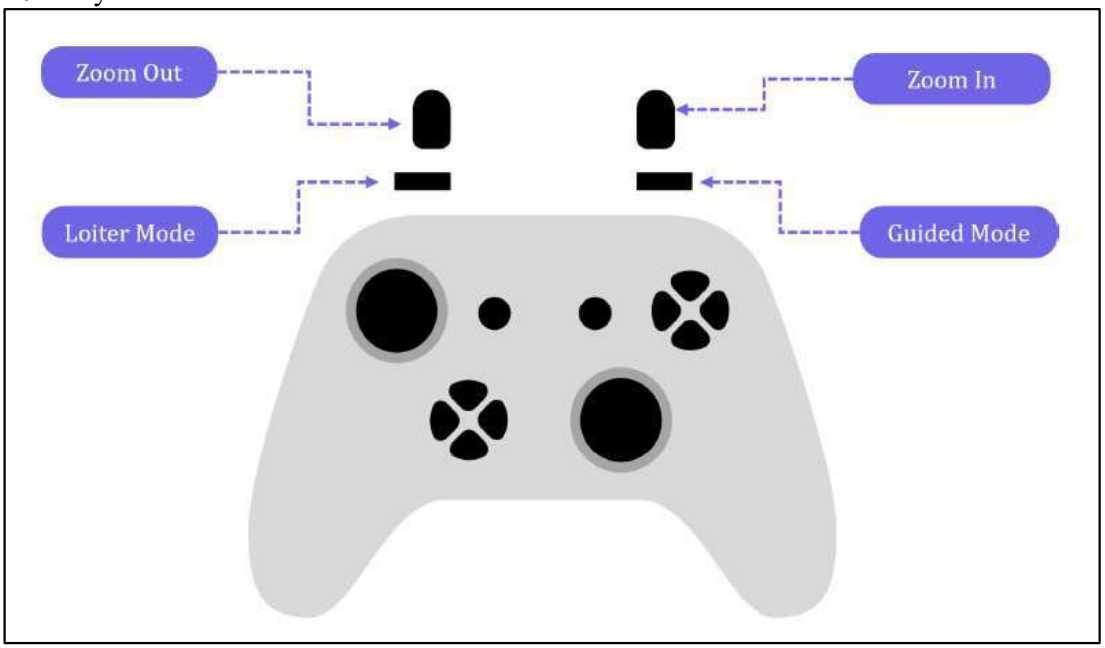
27
1.7.2 Snapshot: 27
1.7.3 Storage: 28
1.8 Seamless integration with 5G networks 28
1.8.1 Enabling 5G setting 28
1.8.2 To Access the 5G Modem info follow below steps: 29
1.9 Demonstration of Use Cases 29
1.9.1 Steps to Connect and Integrate 5G Sparsh Camera Feed with AI/ML Object & Human Detection 29
A. Connect to the Camera's LAN Network 29
B. Retrieve 5G Modem Information 29
C. Check the 5G SIM Camera's Assigned IP Address 29
D. Access the Camera Feed 29
E. Integrate Camera Feed with AI/ML Object & Human Detection 29
1.10 Troubleshooting (If Unable to Access the Feed) 30
2. IOT GATEWAY- 30
2.1 Features of IoT Gateway 30
2.1.1 Hardware 30
2.1.2 Software 31
2.2 IOT Gateway Architecture 32
2.3 Communication Protocols Used In Iot Gateways 32
2.3.1 Cellular Protocols 32

2.3.2 Wi-Fi (802.11 a/b/g/n/ac) 33
2.3.3 Ethernet (Wired LAN) 33
2.3.4 VPN (Virtual Private Network) Protocols 33
2.3.5 MQTT (Message Queuing Telemetry Transport) 33
2.3.6 Modbus (RTU/TCP) 33
2.3.7 CoAP (Constrained Application Protocol) 33
2.3.8 SNMP (Simple Network Management Protocol) 33
2.3.9 TR-069 (CPE WAN Management Protocol) 34
2.3.10 HTTP/HTTPS 34
2.3.11 NTP (Network Time Protocol) 34
2.3.12 IPv4 and IPv6 34
2.3.12 LAN/WAN Routing 34
2.3.13 PPP (Point-to-Point Protocol) 34
2.3.14 RADIUS (Remote Authentication Dial-In User Service) 34
2.3.15 DLNA (Digital Living Network Alliance) 34
2.4 VLAN 35
2.4.1. Simultaneous Server and Multiple Clients (VLAN for Server and Client Networks) 35
A. Steps to Configure VLAN for Server and Client: 35
B. VPN Client and Server Support 35
C. VPN Server Configuration (e.g., OpenVPN Server): 36
D. VPN Client Configuration (e.g., OpenVPN Client): 36
E. Port and Tag-Based VLAN 36
F. Steps to Configure Port and Tag-Based VLAN: 36
2.5 Security 37
2.5.1. Firewall Rules 37
A. How to Configure Firewall Rules on the RUTX50: 37
2.6 DDoS Prevention 38
2.6.1 DDoS Prevention Techniques on RUTX50: 38
2.7 Data Limit for SIM Card 39
2.7.1 How to Set Data Limits on the RUTX50: 39
2.8 Blocking Unwanted Websites 39
2.8.1 How to Block Websites on RUTX50: 39
2.9 Guideline for SIM insertion 40
2.9.1 Power Off the RUTX50 40
2.9.2 Locate the SIM Card Slot 40

2.9.3 Open the SIM Slot Cover 40
2.9.4 Insert the SIM Card 40
2.9.5 Close the SIM Slot Cover 41
2.9.6 Power On the Router 41
2.9.7 Check for Network Signal 41
2.9.8 Troubleshooting 41
2.10 APN Settings 41
2.10.1 Accessing the Web Interface 41
2.10.2 Navigate to the Mobile Settings 42
2.10.3 Configuring the APN Settings 42
2.10.4 Save the Settings 42
2.10.5 Reboot the Router 42
2.10.6 Verify the Connection 42
2.10.7 Common Troubleshooting Tips 43
3. 5G Indoor CPE 43
3.1 CPE system architecture 43
3.2 Wireless Access Interfaces 43
3.2.1 5G NR (New Radio) 44
3.2.2 Wi-Fi 6 (802.11ax) 44
3.2.3 Ethernet (Wired) 44
3.2.4 LTE (Fallback) 44
3.2.5 Key Wireless Access Interfaces for the MG54AX: 45
3.3 Security 45
3.3.1 Latest WPA Support 45
3.3.2 Rogue Access Point Detection and Prevention 45
3.3.3 IP Security (IPSec), PPTP, IP Filtering 45
3.3.4 MAC Address Authentication 46
3.4 5G Network settings 46
3.4.1 Network Mode 46
3.4.2 5G Band Selection 46
3.4.3 Carrier Aggregation (CA) 46
3.4.4 APN Settings 46
3.4.5 Roaming Settings 46
3.4.6 IPv6/IPv4 46
3.4.7 QoS Settings 46

3.4.8 DNS Settings 46
3.4.9 Signal Monitoring 47
3.4.10 Antenna Settings 47
3.4.11 Security 47
3.5 Wifi Settings 47
3.5.1 Wi-Fi Mode 47
3.5.2 Wi-Fi SSID 47
3.5.3 Wi-Fi Security 47
3.5.4 Channel Settings 47
3.5.5 Wi-Fi Bandwidth 47
3.5.6 Guest Network 48
3.5.7 MAC Address Filtering 48
3.5.8 Wi-Fi Power Settings 48
3.5.9 Wi-Fi Optimization Features 48
3.5.10 Wi-Fi Scheduling 48
3.6 Connecting 5G Indoor CPE to 5G and WiFi Network 48
3.6.1 Connecting to the 5G Network 48
3.6.3 Connecting to the Wi-Fi Network 49
3.6.4 Flexible guest access with device isolation Captive 49
3.7 Trouble Shooting 50
3.7.1 General Troubleshooting Steps 51
4. 5G Drone 52
4.1 Introduction - 52
4.2 Drone System Architecture 53
4.3 System Specification- 54
4.4 Aircraft Highlights- 55
4.5 Sensitivity Of A Sensor 56
4.6 Limitation Of Sensors Devices 56
4.6.1 Camera (for Visual Inertial Odometry - VIO) 56
4.6.2 LiDAR (Light Detection and Ranging) 57
4.6.3 Barometer 57
4.6.4 GPS (Global Positioning System) 57
4.7 Safety Procedure- 58
4.7.1 Pre-Flight Check: 58
4.7.2 Environmental Awareness: 58

4.7.3 Control and Supervision: 58
4.7.4 Use of Safety Features: 58
4.7.5 Speed and Height Management: 58
4.7.6 Interference Considerations: 58
4.7.7 Emergency Procedures: 59
4.7.8 Training and Skill Level: 59
4.7.9 Legal and Ethical Considerations: 59
4.7.10 Post-Flight Check: 59
4.8 Battery Handling- 59
4.8.1 Charging the Battery: 60
4.8.2 Handling the Battery: 60
4.8.3 Storage of the Battery: 60
4.8.3 Transporting the Battery: 60
4.8.4 Battery Disposal: 60
4.8.5 General Precautions: 61
4.9 Flight And Companion Controller - 61
4.9.1 Joystick - Modes and Zoom




61
4.9.2 Joystick - Sticks and Control 61
4.9.3 Joystick - Sticks and Control 62
4.9.4 Joystick - Camera Control 62
4.9.5 Joystick - Command and Special Modes 63
4.10 DATA AND CONTROL FLOW ARCHITECTURE 63

4.11 ON-BOARDING PROCESSING 65
4.12 MISSION PROTOCOL 65
4.12.1 Flight plans: 66
4.12.2 MISSION_ITEM_INT vs MISSION_ITEM 67
4.12.3 Upload a Mission to the Vehicle 69
4.12.4 Mission Upload Sequence 70
4.12.5 Sequence: Download mission 71
4.12.6 Set mission item 72
4.12.7 Clear Missions 73
4.13 GEO-FENCING 74
4.13.1 Key Features of GEO-FENCING: 74
4.14 AUTONOMY ARCHITECTURE 74
4.14.1 Visual Inertial Odometry (VIO) 74
4.14.2 Path Planning 74
4.14.3 PX4 74
4.14.4 GPS-denied navigation 75
4.14.5 BVLOS (Beyond Visual Line of Sight) 75
4.14.6 Follow Me 75
4.15 360 OBSTRACLE AVOIDANCE ARCHITECTURE 75
4.15.1 Depth estimation 75
4.15.2 Object detection 75
4.16 Mapping and Visual Odometry (VOA) 76
4.17 Wireless Access Option 76
4.17.1 Connection Modes 76
A. 5G Mode 76
B. Hotspot Mode 76
C. WiFi Mode 77
D. Network Mode Switch 79
E. Network mode switch webpage 79
i. Current Connection Type : 79
ii. WiFi / Hotspot Info : 79
a. WiFi Info - 79
b. Hotspot Info - 80
c. Add New WiFi : 80
d. Available Networks : 80

4.18 Flight Controller 81
4.19 Software Installation 82
4.20 Install iDronam For Enterprise- 82
4.20.1 Software Setup 83
A. Indoor Mode 83
B. Outdoor Mode- 86
4.21 AI/ML Use Case- 89
4.22 Sim Insertion And 5g Registration- 89
4.22.1 SIM Insert 89
A. SIM Slot: 89
B. SIM REGISTRATION 89
4.23 Installing The Drone Camera Applicatiom At Mec: 90
4.23.1 Steps for installation - 90
4.24 TROUBLESHOOTING 90
4.25 Take off and General Flight precautions- 90
4.25.1 Before taking off there a few things that pilot must observe: 91
4.25.2 Pre-Arm Checks- 92
4.26 PRE FLIGHT CHECKS 92
4.26.1 Battery Checks 92
4.26.2 Structure Checks 93
4.26.3 Landing Gears - 93
4.26.4 Propellers - Drone use the propellers to fly 93
4.27 Flight Area Checks 93
4.28 GCS Software, C2 Link and Controller Check- 94
5. META QUEST XR/VR 94
5.1 Introduction 94
5.2 What Include In The Box 95
5.3 Key Features- 95
5.3.1 Standalone VR System: 95
5.3.2 Display and Visuals: 95
5.3.3 Tracking: 95
5.3.4 Controllers: 96
5.3.5 Software and Content: 96
5.3.6 Audio: 96
5.3.7 Comfort and Design: 96

5.3.8 Storage Options: 96
5.3.9 Battery Life: 96
5.3.10 Social Features: 96
5.4 WHAT IS META QUEST 2? 96
5.3.1 Employee Training: 97
5.3.2 Medical Training: 97
5.3.3 Education: 98
5.3.4 Soft Skills Training: 98
5.3.5 Virtual Classrooms: 99
5.4 AR (Augmented Reality), VR (Virtual Reality), and MR (Mixed Reality) 99
5.4.1 Augmented Reality (AR) 99
5.4.2 Virtual Reality (VR) 100
5.4.3 Mixed Reality (MR) 100
5.5 XR Foundation Features for Design and Development 101
5.5.1 Steps to Install Required Hardware and Software 101
A. Hardware Requirements 101
B. Software Requirements 102
5.6 Install and Build application usage 102
5.7 TROUBLESHOOTING 102
5.7.1 VR Headset Not Turning On 102
5.7.2 No Display / Black Screen 103
5.7.3 Controllers Not Working 103
5.7.4 Tracking Issues 103
5.7.5 Wi-Fi Connectivity Issues 103
5.7.6 App Not Launching or Freezing 103
5.7.7 Audio Problems (No Sound or Low Sound) 104
5.7.8 Performance Lag or Low Frame Rate 104
5.7.9 Overheating 104
5.7.10 Pairing with the Meta Quest App 104
5.7.11 Factory Reset 104
5.8 Benefits of Using XR/VR : 105
5.9 WHAT YOU HAVE TO DO ? 105
6. EVALUATION BOARD 105
6.1 Overview 105
6.2 Specifications 106

6.3 Interface Application 107
6.3.1 Power Supply 108
6.3.2 Module TE-A Interface (J0101/J0102) 108
6.3.3 USB Interface (J1101) 109
6.3.4 Audio Interface (J0802/J0901/J0801) 109
6.3.5 Digital Audio Codec Board Connector (J0802) 109
6.3.6 Table : Pin Definition of J0901 111
6.3.7 (U)SIM Card Interfaces (J1401/J1402) 111
6.3.8 Table : Pin Definition of J1401 112
6.3.9 D Card Interface (J1301) 113
6.3.10 Table : SDIO Switch Function 113
6.4 UART Interfaces (J2002/J2003) 114
6.4.1 PCIe to USB Interface (J1601) 114
6.4.2 Table : PCIe Connection Truth Table 115
6.4.3 Switches and Buttons 115
6.4.4 Table : Description of Switches and Buttons 117
6.4.5 Status Indicators (D0201/D0202/D0203/D0204/D0205) 118
6.4.6 Wi-Fi Interfaces (J0701/J0702) 118
6.4.7 Antenna Interfaces 119
6.5 Evaluation Board Operation Procedures 120
6.5.1 Turn on the Module 120
6.5.2 Communication via USB 120
6.6 SETUP 120
6.6.1 ABBREVIATION:- 120
6.6.2 Steps to Run Evaluation Board 121
6.6.3 Integrating With Raspberry Pi 129
6.7 SPECIFICATION 129
7. IOT SENSOR 132
7.1 TECHNOLOGY USED 132
7.2 SENSOR USED IN THIS PROJECT 132
7.3 SENSOR SPECIFICATION 132
7.3.1 LIGHT INTENSITY SENSOR 132
7.3.2 TDS SENSOR 133
7.3.3 NPK SOIL SENSOR 133
7.3.4 TEMPERATURE & HUMIDITY SENSOR 133

7.5 DASHBOARD 134
7.6 How to Access Sensor Data 134
7.6.1 OUTPUT 136
7.6.2 5G CPE SETUP: 137
7.6.3 DEBUG 138
8. 5G Use-Case Of Evaluation Board **Error! Bookmark not defined.**
8.1 Steps to Start with Evaluation Board **Error! Bookmark not defined.**
8.2 Basic Configuration Of Evaluation Board **Error! Bookmark not defined.**
8.3 Introduction to MobaXterm **Error! Bookmark not defined.**
8.3.1 Steps To Access Coral Anubhav with MobaXterm: **Error! Bookmark not defined.**
A. Types of AT Commands **Error! Bookmark not defined.**
8.4 Configuring 5G Evaluation for 5G Registration **Error! Bookmark not defined.**
8.5 Raspberry Pi Use Case With Coral Anubhav **Error! Bookmark not defined.**
8.6 Use Cases of Automating 5G Evaluation Board (Coral Anubhav) Configuration with Bash Scripts on Raspberry Pi **Error! Bookmark not defined.**
8.6.1 Introduction **Error! Bookmark not defined.**
8.6.2 Prerequisites **Error! Bookmark not defined.**
8.6.3 Setup Instructions **Error! Bookmark not defined.**
8.6.4 Bash Script for Modem Configuration **Error! Bookmark not defined.**
8.6.5 Running the Script **Error! Bookmark not defined.**
8.6.6 Creating a System Service for Automation **Error! Bookmark not defined.**
8.7 Basic Checks **Error! Bookmark not defined.**

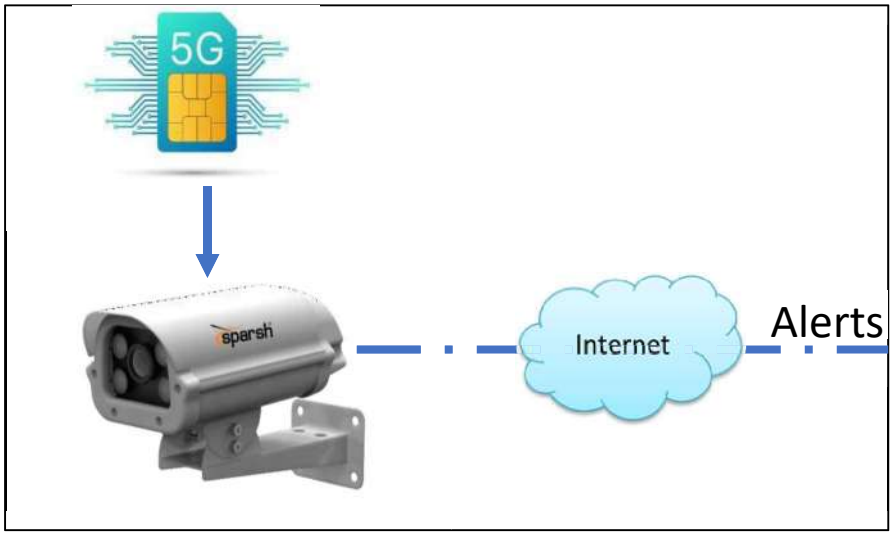
**5G Use Cases**

1. **5G CAMERA**

**1.1 Use Case Development with 5G Camera:**

**INTRODUCTION: 5G CAMERA:**

It is a Next Generation camera with a perfect blend of looks and features for any security requirement. A 100% “Made in India” product with best-in-class technology, robust mechanical, cloud support and service centers all over the country. Leveraging the high transmitted power of 5G, Sparsh 5G Camera can transmit or receive video signals over the Internet wirelessly by simply powering the camera and inserting a SIM card, eliminating the need of complicated cable and fiber connections. Compared with add-on routers, the built-in 5G module enhances the mobility of its deployment, providing a reliable security solution.



### 1.2 Devices Used for Development

#### 1.2.1 Device Parts

The Sparsh 5G Camera consists of the following essential components:

- **Camera Sensor**: High-resolution imaging sensor with night vision support.
- **Processor**: Integrated AI-capable processing unit for edge computing.
- **5G Module**: High-speed 5G connectivity for seamless data transfer.
- **Storage**: Internal and expandable storage for video retention.
- **Power Supply**: Power over Ethernet (PoE) or battery support.
- **I/O Ports**: USB, Ethernet, and GPIO for connectivity.

### 1.2.2 Sensor Specifications

| Specifications | SC-IM82NP-I (Z)(S)(H) |
|---|---|
| **Camera** | |
| Image sensor | 1/2.7" Progressive CMOS Sensor |
| Effective Pixels | 3840(H)x2160(V) |
| Min. Illumination | Color: 0.01 Lux @F1.2, B/W: 0.001 Lux@F1.2, 0 Lux at IR ON |
| S/N Ratio | >60dB |
| IR Distance | 50-60Mtr |
| Shutter Speed | 1/25sec ~ 1/100000 sec |
| IR On/Off Control | Auto/ Manual |
| IR LEDs | 4 Array LEDs |
| **Lens** | |
| Lens Type | Motorized |
| Mount Type | M14 |
| Focal Length | Motorized 2.7-13.5mm Lens |
| Max. Aperture | F1.2 |
| Angle Of View | Horizontal: 98° to 34° Vertical: 52° to 19° Diagonal: 116° to 40° |
| Focus Control | Auto/Manual |
| **Video** | |
| Compression | H.265SE/H.265/H.264 |
| Streaming Capability | 3 Streams |
| Resolution | 8MP/5MP/4MP/2MP/D1/ VGA /CIF |
| Streams | **Mainstream:** 3840x2160(4K) @15fps/ 2592X1944 (5MP)/ 2560X1440 (4MP)/ 2304X1296 (3MP) 1920X1080(2MP) @25/30fps,<br>**Sub Stream:** VGA (640x480)/CIF (352×288) @15/25fps<br>**Third Stream:** D1 (704x576)/ VGA (640x480)/CIF (352×288)/720P (1280x720) @15/25fps |
| WDR | 120dB |
| Bit Rate | 221Kbps-8Mbps |
| Bit Raye Control | CBR/VBR |
| Day/ Night | Automatic/Color/Black and white |
| Exposure Mode | Automatic/Manual |
| BLC Mode | On/Off |
| HLC Mode | On/Off |
| Profiles | Auto, Low Light Priority, High Light Priority |
| White Balance | Indoor/Outdoor/Automatic |
| Gain Control | Auto/Manual |
| Noise Filter | Auto (1-6 Levels) |
| Digital Zoom | 16X, Area's selection with Mouse |
| Anti - Flicker | On/Off |

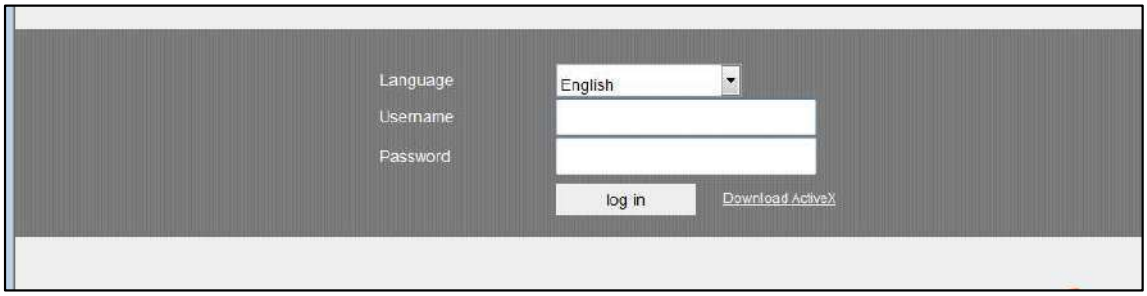
Sensor Specifications continued….

| | |
|---|---|
| AE Reference | 1-255 Level |
| AE Sensitivity | 1-10 Level |
| Flip | Off / On |
| IR Reverse | Support |
| Mirror | Off / On |
| Text Overlay | Title, Date & Time, Weekday |
| Privacy Masking | Yes (4 zones) |
| **Events** | |
| Motion Detection | 22X18 Zones |
| Video Blind | Support |
| Abnormality | No Disk, Disk Error, Disk No Space, Network Cable Disconnected, IP Address Conflict |
| Smart Functions (By Camera) | Human Detection, Intrusion Detection, Line crossing, Object Lost, Object Left, Scene Change, Audio Detection |
| Smart Functions (By Software) | License Plate Recognition, Face Detection & Recognition, People Counting, Object Detection/Classification, object pattern recognition models supported through software |
| Event Actions | Record alarm, Linkage alarm output, Sound alarm, Email link, Upload FTP, Snapshot |
| **PTZ (Manual)** | |
| Pan | 0° ~360° |
| Tilt | 0° ~90° |
| **Safety** | |
| Security | Authorized username and password, HTTPS with password, RTSP Validation, User Access Log, User Authentication, Watermark, IP address filtering, 802.1x; IP/MAC filtering; HTTPS; trusted upgrade; trusted boot; firmware encryption; generation of X.509 certification |
| ONVIF Authentication | ON, OFF |
| RTSP Authentication | Digest, None |
| **Audio** | |
| Compression | G.711A/U |
| Talk Back | Support |
| **Network** | |
| Ethernet | 10/100M adaptive Ethernet interface |
| Protocol | TCP/IP, UDP, RTP, RTSP, RTCP, HTTP, HTTPS, DNS, DDNS, DHCP, FTP, NTP, SMTP, UPNP, SMTP, IPv4, 3G/4G/5G |
| Band Support | 5G: LB: n5/8/20/28; MB: n1/3/4/66, HB: n7/38/40/41/48/77/78/79<br>4G: LB: B5(18/19/26)/8/20/28; MB: B1/3/4/66, HB: B7/34/38/39/40/41/42/43/48 |
| SIM Card | 1x Nano SIM card support |
| Interoperability | ONVIF Profile S & G |
| Max. user Access | 20 Users |
| Edge storage | Local PC for instant recording, one clicks Snapshot, one clicks triple Snapshot |
| Web Viewer | IE, Chrome, Firefox, Safari |
| Management Software | Sparsh VMS |

## 1.3 Utilities and Configuration

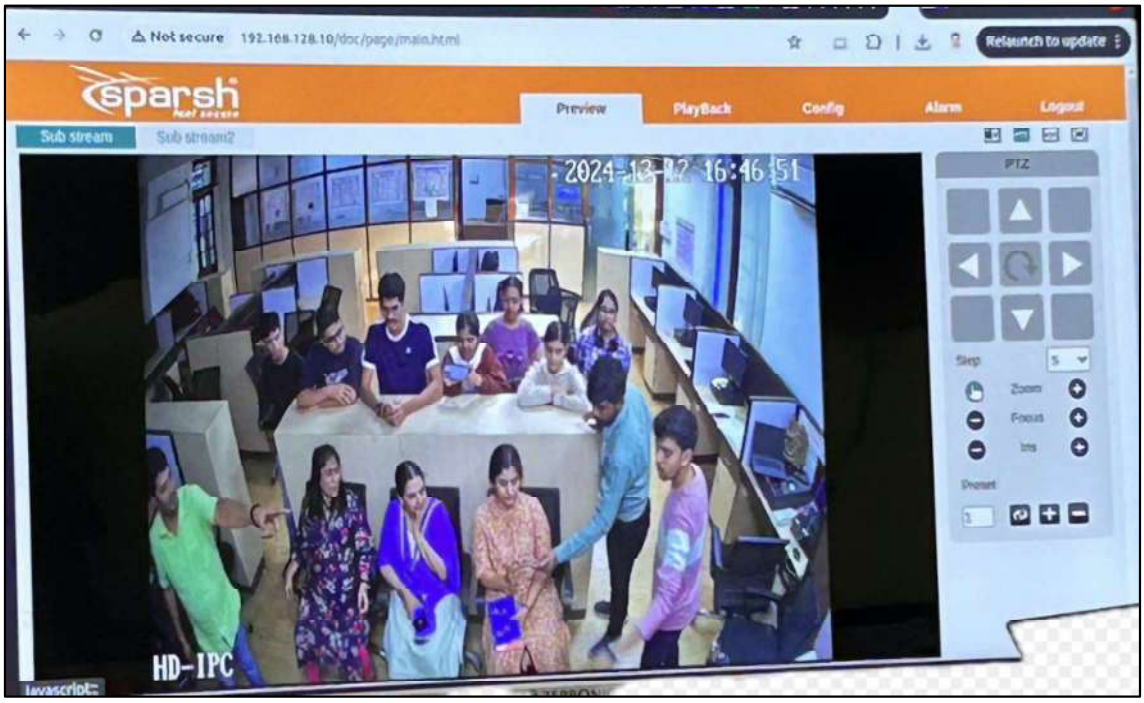
### 1.3.1 Camera page login & Configuration

Connect the lan cable to your PC open the web browser and use https://192.168.128.10 IP to open the camera login page use below credential
USER NAME : admin
PASSWORD : admin123




### 1.3.2 Preview interface：

The **camera preview interface** is the user interface (UI) that allows users to view and manage live feeds from their 5G cameras.




### 1.3.3 Local Config:

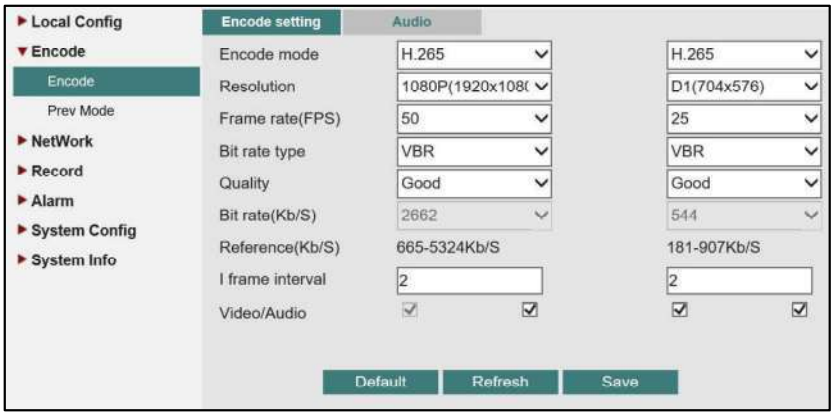
This configuration of the camera can be done.

**A. Encode:**

Encoding of video and audio are as follow:

**i. Encode setting:**

Options to adjust the video resolution (e.g., 720p, 1080p, 4K) bit rate, frame rate or to adjust the video stream quality, particularly for bandwidth management.



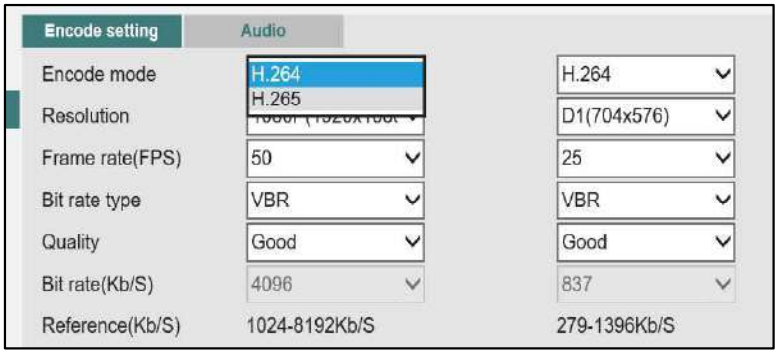

Camera is running on three streams:

a) Mainstream

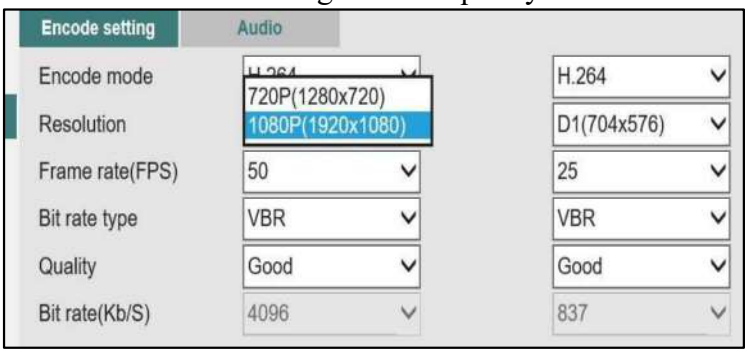
b) Sub stream

c) Third Stream

Choose the video compression standard maintained on the 2 MP camera.



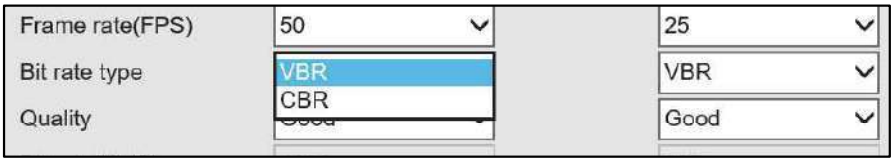

**Resolution**: We can set the resolution according to video quality need.




**Frame rate**: The number of frames shown or recorded in one second of video. A higher frame rate results in smoother motion, while a lower frame rate can cause motion to appear choppy.In mainstream we can only set the fps upto 50 but in sub stream upto 25 only. Select according to use.

**Bit rate type**: There are two types which are follow as (i)VBR

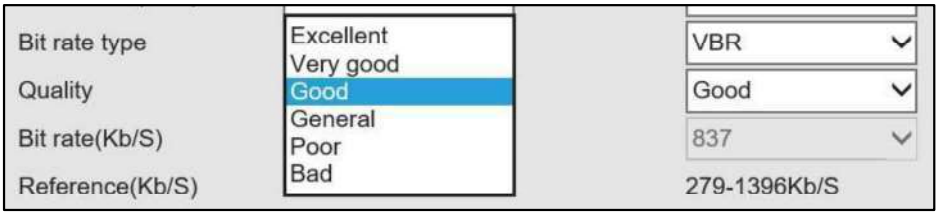
(ii)CBR



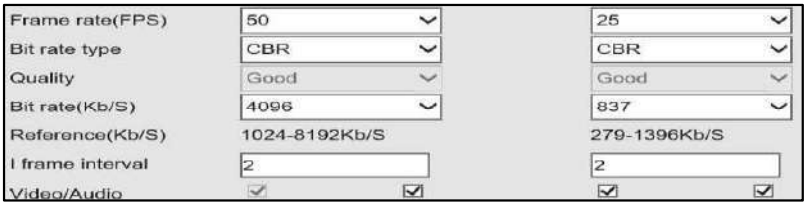

**VBR (Variable Bit Rate)** and **CBR (Constant Bit Rate)** are two common video encoding methods used in media compression. They refer to how the **bit rate** (the amount of data used

per second in video/audio encoding) is managed during the encoding process. The bit rate directly affects video quality and file size.

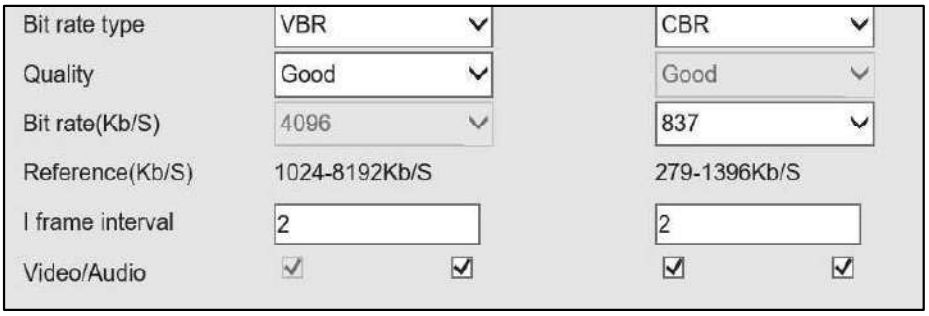
**Quality**: Quality can be set according the video requirement which will directly affect the bandwidth consumed.




**Bit rate**: When we set the video encode on CBR bit rate type then we manually config the bit rate of the video encode.



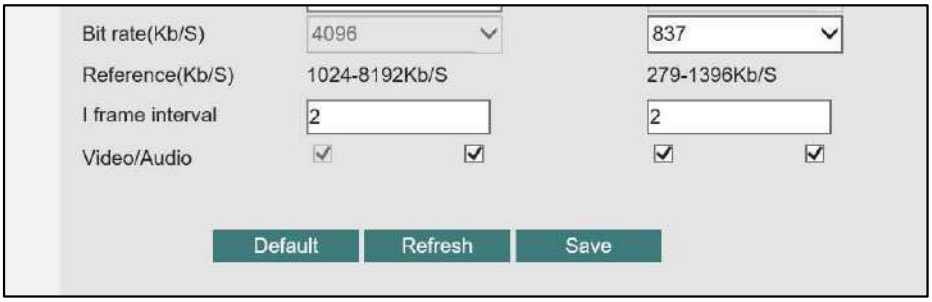

**I frame interval**: The **I-frame interval** (also known as **Keyframe interval**) is an important setting in video encoding and streaming. It refers to the frequency at which **I-frames (Intra-coded frames)** are inserted into the video
stream..For 2mp bullet camera by default value is 2.




**Default:** It is used for setting the value as default.

**Refresh:** It is used for updating from previous.

**Save**: It is used for saving the configuration.




**ii. Audio:**

In this we have done the audio encoding setting.

Encode mode: We have used the audio format “G711A” and “G711U” for audio compression standards.

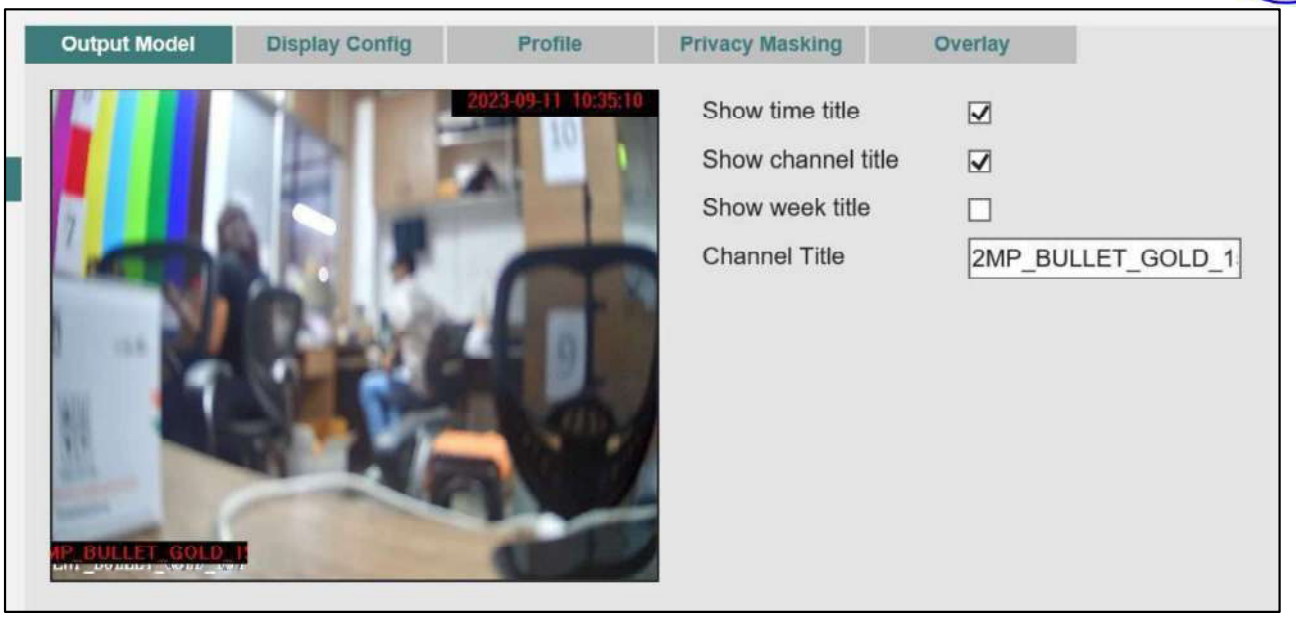
**Sampling rate**: the sampling rate should be 8000 Hz or more than it due to the smoothness of audio.

### 1.4 Prev Mode:

#### 1.4.1 Output model:

Output model is used to set the on screen changes to be displayed on the live view feed.

(i) Show time title
(ii) Show channel title
(iii) Show week title
(iv) Channel title




### 1.4.2. Display config:

Image Color: **Image color settings** control how the camera processes and displays colors in the video feed.

(v) **Profile:** If we want to change the profile as day/night.

(vi) **Brightness:** Controls the overall lightness or darkness of the image.

(vii)**Contrast**: Enhances the clarity of an image by making shadows darker and highlights brighter, making the subject stand out more clearly.

(viii) **Saturation:** Controls the intensity of colours in the image. Increasing saturation makes colors more vivid, while decreasing it results in a more washed-out or grayscale image.

(ix) **Tone:** Helpful when colors appear unnatural due to lighting (e.g., a yellowish tint under tungsten lights).

(x) **Sharpness**: Controls the clarity and crispness of the image.

### 1.4.3 Camera:

**Day/night mode: Determines how the camera adjusts to different lighting conditions (daytime or nighttime).**

**DNC threshold: By adjusting the DNC threshold**, you can manage how aggressive the camera is at removing noise, which helps maintain image clarity, especially in dark environments

IR_CUT: it is synchronous with the amount of light sensed by LDC.

Flip/mirror: The **Flip/Mirror** function is a setting that allows you to **flip or mirror** the video feed, which is particularly useful when the camera is mounted in certain orientations or positions.It is used to set for real image that should be seen on screen if camera is set inverted or alternate position.

IR reverse: The purpose of this function is to adjust the way infrared light is used or the way it impacts the camera's image.

### 1.4.4 ISP:

Exposure mode: The **exposure mode** controls how the camera adjusts the exposure time, or how long the camera's sensor is exposed to light, to capture an image. It is crucial in ensuring the camera produces well-lit images, especially in environments with changing light conditions.Its value is up to 1/100000 for BLC and HLC of the video encoding in the camera.

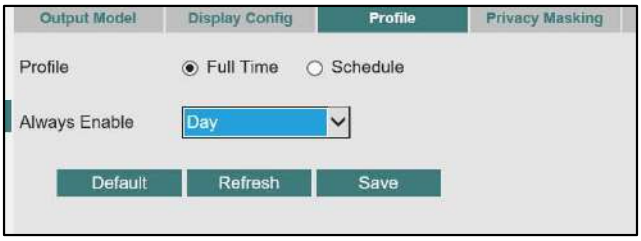
BLC: It is a feature designed to help the camera deal with situations where there is **strong light** coming from behind the subject being captured, which typically causes the subject to appear very dark.

HLC: It is a feature that helps reduce the effects of **bright light sources** (such as headlights, street lights, or spotlights) that can create glare or overexposure, making the subject difficult to see.

Fill Light :It refers to the additional light sources used to **illuminate** a scene in low-light conditions. It supplements the camera's existing light (usually infrared light at night) to provide better visibility.

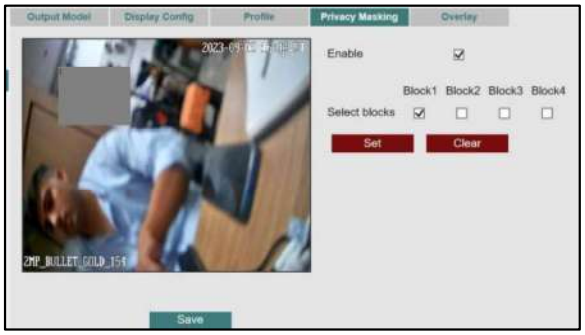
### 1.4.5 Profile:

Used to schedule camera view settings for full time day/night or can be scheduled for particular hours.




### 1.4.6 Privacy Masking:

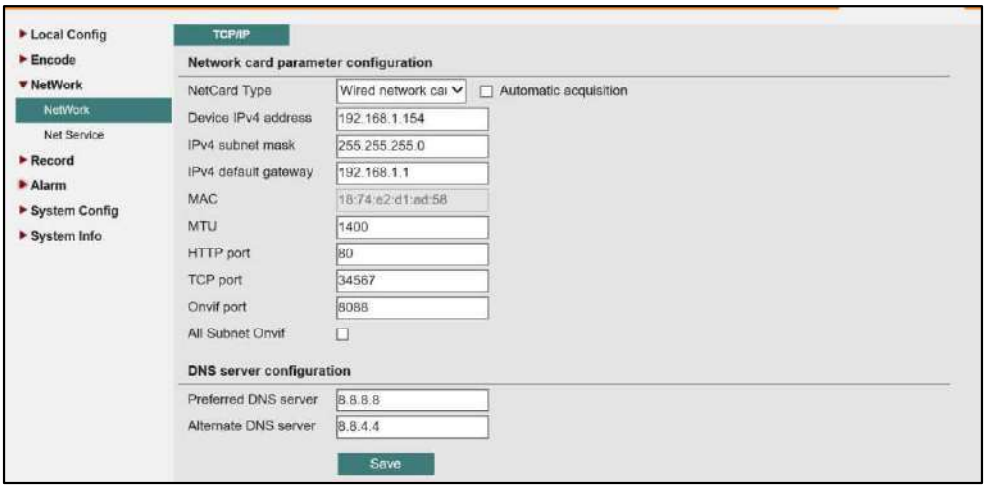
It is a feature that allows you to block out certain areas of the camera's field of view, so they are not recorded. It allows you to block out sensitive areas of the camera's view while still recording the rest of the scene.




### 1.4.7 Overlay:

It displays additional information on the video feed, such as text, logos, timestamps, or other visual markers, that are superimposed onto the live or recorded video.Alignment of the text such as left align or right align can be done by text align.

### 1.5 Network:




#### 1.5.1 TCP/IP:

We use TCP/IP.

Network card parameter config:

(xi) Net card type: we used wired network cards.

(xii) Device IPv4 address: we manually provide the IP to the device.

(xiii) All the port no and mask are by default set.

(xiv) A **MAC (Media Access Control) Address** is a unique identifier assigned to the network interface card (NIC) of a device, used for communication on a network.

(xv) MTU: **MTU (Maximum Transmission Unit)** is a network setting that defines the largest size of a data packet that can be sent over a network without needing to be fragmented. It is an important setting for ensuring efficient and reliable communication in computer networks, as it determines the maximum amount of data that can be transmitted in a single packet.

(xvi) HTTP port: The **HTTP (Hypertext Transfer Protocol)** port is primarily used to access the **web interface** of network cameras, NVRs, and other devices over the network using a **web browser**.By default is 80.

(xvii) TCP port: **TCP (Transmission Control Protocol)** ports are used for the **data transmission** between devices in a Camera system. These ports allow for reliable, ordered

delivery of video streams, remote controls, and management commands.

(xviii) Onvif port: **ONVIF ports** are essential for the proper functioning and interoperability of Camera systems, especially for **IP cameras** and **network video recorders (NVRs)**.DNS server configuration: When configuring a Camera system, setting up the DNS server correctly ensures that devices can access the network and communicate with each other.

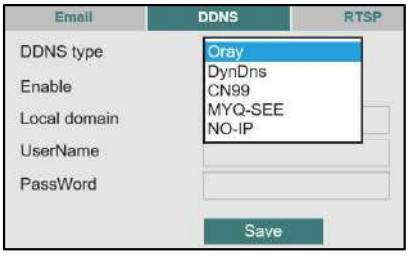
### 1.6 Net Service:

#### 1.6.1 Email:

Setting up **email alerts** allows you to receive notifications (like motion detection alerts, system errors, or triggered alarms) directly to your email address. This feature is useful for remotely monitoring your security system, ensuring that you're always notified of any events that require attention, even when you're not actively checking the system.

#### 1.6.2 DDNS:

When using DDNS with your Camera system, you can assign a **domain name** (like myCamera.dyndns.org) to your
camera changing public IP address. This allows you to access your Camera system by typing a domain name into a browser instead of a changing IP address and can be accessed from anywhere in the world.




In DDNS we select the DDNS type and provide the user at which we want the DDNS to come.

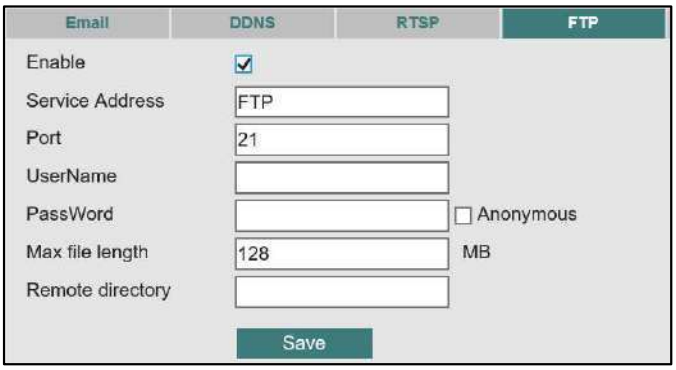
### 1.6.3 RTSP:

RTSP (Real-Time Streaming Protocol) is commonly used for transmitting video and audio from Camera cameras.

Camera cameras that support RTSP allow you to stream live footage over a network to devices like computers, mobile phones, or specialized monitoring software.Can access it on VLC by copying the URL.

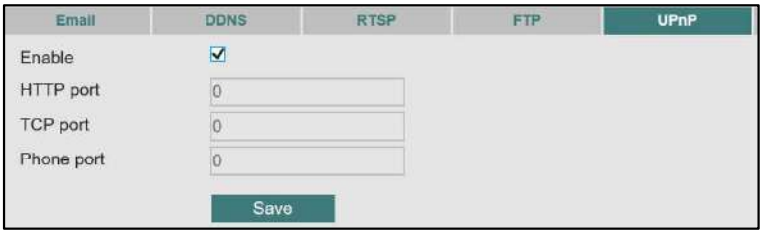
### 1.6.4 FTP:

FTP is used to upload or download video footage from cameras to a remote server or cloud storage.



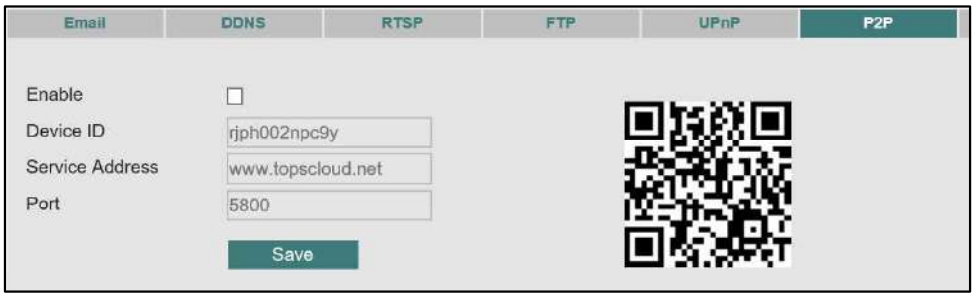

### 1.6.5 UPnP:

It can automatically configure certain network settings, such as port forwarding, without needing manual intervention. This can make setting up remote access to the camera easier, as UPnP enables the camera to automatically open the necessary ports in your router to allow access from outside your local network



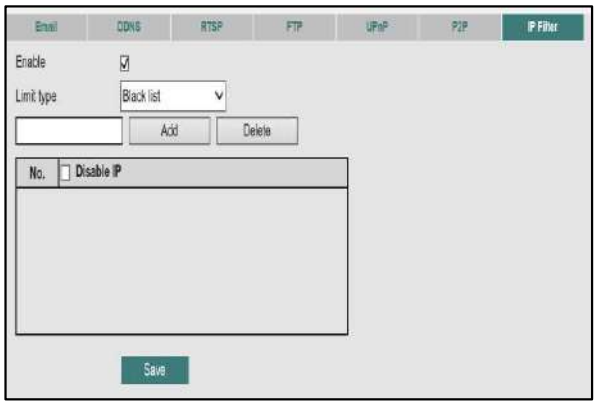

### 1.6.6 P2P:

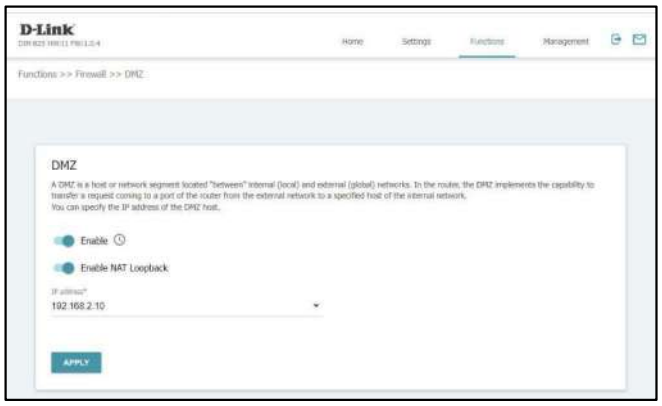
P2P (Peer-to-Peer) Camera refers to a type of surveillance camera system that uses P2P technology to establish a
direct connection between your Camera camera and a viewing device, such as a smartphone, tablet, or PC, without the need for port forwarding or a static IP address.




### 1.6.7 IP filter:

IP filtering works by specifying which IP addresses or ranges are allowed (whitelisted) or blocked



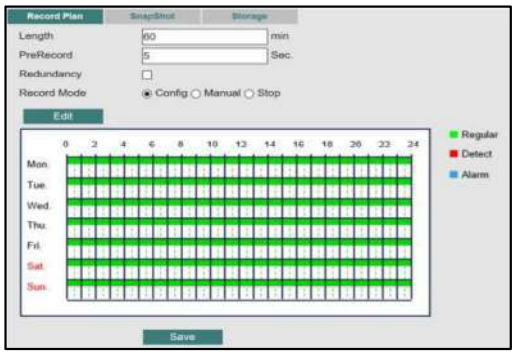



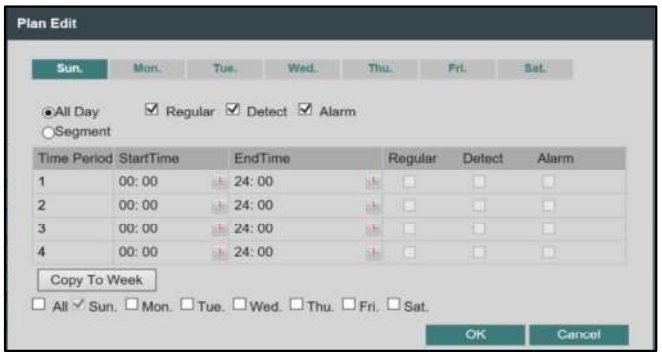

(blacklisted) from accessing the camera or Camera system. For example, you can set your Camera system to allow access only from specific IP addresses, such as your home or office IP, and block others

### 1.7 Record:

### 1.7.1 Record Plan:

A **record plan** allows you to define when and how your Camera system records footage. This feature is useful for managing storage space and ensuring cameras record during specific times, based on your needs.




(i) Length is provided for what time the record should be require.
(ii)Pre-record: the camera records a few seconds or minutes **before** an event (such as motion detection or an alarm trigger) actually occur.
(iii) Redundancy The main goal of redundancy in recording is to protect against data loss, minimize downtime, and ensure that the system continues recording even if a failure occurs in the primary storage system.
(iv) Record mode: if we want to config, it then it can, if manually then it can be set, if stop to record then it can be.

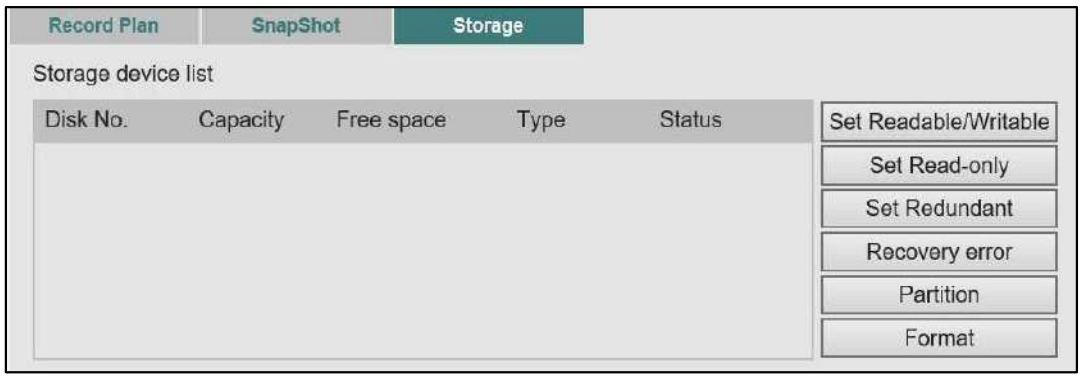
### 1.7.2 Snapshot:

The **Snapshot Function** lets you take still images from the live video feed or from recorded footage. This can be useful for capturing specific moments (like an intruder entering a room or an important event) and storing them separately from video recordings.

### 1.7.3 Storage:

Storage device shows here and the setting can be made how to use the storage device and can be formatted if required.

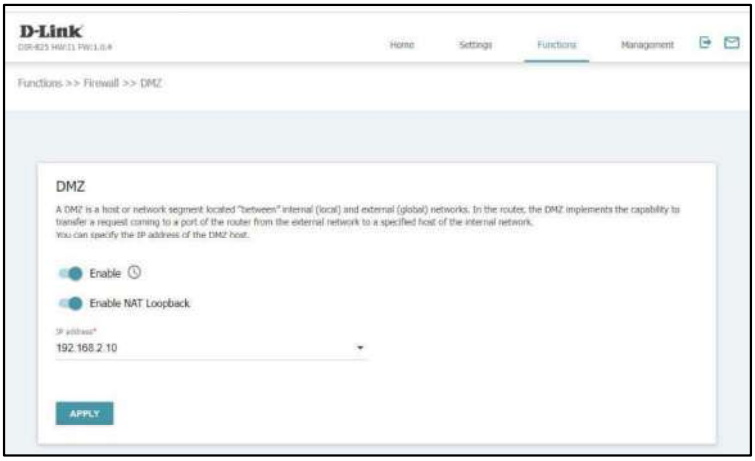
**Storage Read/Write**: Ensures efficient data transfer speeds for continuous recording and smooth playback. **Redundant Storage**: Duplicates data across multiple storage devices to prevent data loss in case of failure. **Recovery Cover**: Implements backup and failover systems to restore footage and maintain functionality after a failure.




## 1.8 Seamless integration with 5G networks

### 1.8.1 Enabling 5G setting

5G is enabled as soon as the SIM card is activated in the camera. To enable viewing of cameras over 5G, enable DMZ service from router web GUI as below. Enter camera IP address and save.



### 1.8.2 To Access the 5G Modem info follow below steps:

- Connect to the camera's LAN network by using the LAN port from camera and connect with the PC
- To get the 5G modem Info open browser(Chrome , firefox) in address bar enter the 192.168.128.1 ip address
- Check the IP address assigned to 5G SIM camera in that page
- Once you have identified the 5G SIM IP address, open Firefox in the Camera VM.
- Enter the 5G SIM IP address in the browser to access the camera feed

## 1.9 Demonstration of Use Cases

### 1.9.1 Steps to Connect and Integrate 5G Sparsh Camera Feed with AI/ML Object & Human Detection

#### A. Connect to the Camera's LAN Network

- Use an Ethernet cable to connect the **camera's LAN port** to your **PC**.
- Ensure that the PC is set to obtain an IP address automatically or configure it manually if required.

#### B. Retrieve 5G Modem Information

- Open a web browser (**Chrome or Firefox**).
- In the address bar, enter **192.168.128.1** and press **Enter**.
- This will open the 5G modem's web interface.

#### C. Check the 5G SIM Camera's Assigned IP Address

- Navigate to the **status or network settings page** in the modem interface.
- Locate the **IP address assigned to the 5G SIM camera**.
- Note down this **5G SIM IP address** for the next step.

#### D. Access the Camera Feed

- Open **Firefox** inside the **Camera VM**.
- In the address bar, enter the **5G SIM IP address** and press **Enter**.
- The camera's live feed should now be accessible.

#### E. Integrate Camera Feed with AI/ML Object & Human Detection

- To integrate the live camera feed into the **AI/ML object & human detection system**, use the **RTSP stream link**:
  rtsp://admin:admin123@<camera_IP>/avstream/channel=1/stream=0.sdp
- Where <camera_IP> is the **5G SIM IP address** retrieved earlier.
- This **RTSP stream** can be fed into AI/ML-based detection models for real-time processing and analysis.

### 1.10 Troubleshooting (If Unable to Access the Feed)

- Ensure that the camera is powered on and properly connected to the **LAN network**.
- Verify that the **5G modem** is active and providing an IP address.
- Check firewall or security settings that may block network access.
- Restart the camera, PC, or modem if necessary and repeat the steps.
- Confirm that the **AI/ML system supports RTSP streaming** and is configured correctly.

## 2. IOT GATEWAY-

### 2.1 Features of IoT Gateway

#### 2.1.1 Hardware

| | |
|---|---|
| Mobile | 5G Sub-6Ghz SA/NSA 2.1/3.3Gbps DL (4x4 MIMO), 900/600 Mbps UL (2x2); 4G (LTE) – LTE Cat 20 2.0Gbps DL, 200Mbps UL; 3G – 42 Mbps DL, 5.76Mbps UL |
| CPU | Quad-core ARM Cortex A7, 717 MHz |
| Flash storage | 256 MB |
| RAM | 256 MB |
| Powering options | 4-pin power socket, 9-50 VDC |
| SIM | 2 x SIM slot (Mini SIM – 2FF), 1.8 V/3 V |
| Antenna connectors | 4 x SMA for Mobile, 2 x RP-SMA for WiFi, 1 x SMA for GNSS |
| Ethernet | 5 x 10/100/1000 Ethernet ports: 1 x WAN (configurable as LAN), 4 x LAN |
| Wifi | 802.11b/g/n/ac Wave 2 (WiFi 5) with data transmission rates of up to 867 Mbps (Dual Band, MU-MIMO), 802.11r fast transition, Access Point (AP), Station (STA) |
| Wireless mesh/roaming | Wireless mesh (802.11s), fast roaming (802.11r) |
| GNSS | GPS, GLONASS, BeiDou, Galileo, QZSS |
| Inputs/Outputs | On 4-pin socket: 1 x Digital input, 1 x Digital open collector output |
| Other | 1 x USB host, 1 x Grounding screw |
| Status LEDs | 3 x connection status LEDs, 3 x connection strength LEDs, 10 x Ethernet port status LEDs, 4 x WAN status LEDs, 1x Power LED, 2 x 2.4G and 5G WiFi LEDs |
| Operating temperature | -40 °C to 75 °C |

| | |
|---|---|
| Housing | Aluminum housing, DIN rail (can be mounted on two sides), flat surface placement |
| Dimensions | (W x H x D) 132 x 44.2 x 95.1 mm |
| Weight | 533 g |

### 2.1.2 Software

| | |
|---|---|
| Operating system | RutOS (OpenWrt based Linux OS) |
| Mobile features | Multiple PDN, Auto APN, Band lock, SIM switch, Operator black/white list, Data/SMS limits |
| Network features | Routing, Failover, Firewall, DHCP, DDNS, Load Balancing, VoIP passthrough, Connection monitoring |
| Monitoring and Management | WEB UI, CLI, SSH, CALL, SMS, TR-069, SNMP, JSON-RPC, MQTT, MODBUS, RMS |
| VPN and tunneling | OpenVPN, IPsec, GRE, PPTP, L2TP, Stunnel, DMVPN, SSTP, WireGuard, ZeroTier |
| Cloud solutions | RMS, FOTA, Azure IoT Hub, Cloud of Things, Cumulocity, ThingWorx |
| Hotspot | External/Internal Radius, SMS OTP, MAC authentication, Walled Garden |
| GNSS | NMEA forwarding, AVL, Geofencing |

**To know more about the feature of RUTX50 you can go to the link given below:**

RUTX50 Industrial 5G Router

## 2.2 IOT Gateway Architecture

| Component | Description |
|---|---|
| **Processor (CPU)** | ARM-based processor, likely ARM Cortex-A7 for routing and data processing. |
| **Networking Interfaces** | 4G LTE WAN, Gigabit Ethernet LAN, Dual-band Wi-Fi (2.4 GHz and 5 GHz), Serial (RS232/RS485), SIM card slot. |
| **Operating System** | OpenWrt-based Linux OS for flexibility and customization. |
| **Storage** | Flash memory with optional microSD for additional storage. |
| **Power Supply** | DC input range with PoE (Power over Ethernet) support. |
| **Security Features** | VPN support (IPsec, OpenVPN, WireGuard), firewall, encryption. |
| **Management and Monitoring** | Web interface for configuration, RutOS Cloud for centralized management. |
| **Redundancy & Failover** | WAN failover for automatic connection switching between LTE, Ethernet, and Wi-Fi. |
| **Environment & Durability** | Ruggedized design, built to endure harsh conditions (temperature, dust, vibration). |
| **Expansion & Customization** | I/O ports for sensors and custom industrial applications. |

## 2.3 Communication Protocols Used In IoT Gateways

The **RUTX50** is a powerful industrial IoT gateway by **Teltonika Networks**, designed for various industrial applications. It supports multiple communication protocols, ensuring reliable and flexible communication across different IoT devices and networks.

Here's a breakdown of the key **communication protocols** supported by the **RUTX50**:

### 2.3.1 Cellular Protocols

**2G (GSM/GPRS)**: For basic connectivity in areas where higher-speed networks aren't available.
**3G (UMTS/HSPA)**: Offers improved data speeds compared to 2G, typically used in medium-level data communication.
**4G LTE**: High-speed cellular network for modern IoT applications requiring fast data transfer and reliable connectivity.
**5G**: With the latest update, the RUTX50 supports **5G** for ultra-fast speeds, lower latency, and support for massive IoT device connectivity.

### 2.3.2 Wi-Fi (802.11 a/b/g/n/ac)

**Wi-Fi (2.4 GHz and 5 GHz)**: The gateway supports Wi-Fi for both connecting to local networks and functioning as an access point. It can communicate with other IoT devices over Wi-Fi, offering flexible wireless networking options.

### 2.3.3 Ethernet (Wired LAN)

**Gigabit Ethernet**: The RUTX50 supports **10/100/1000 Mbps Ethernet ports** for stable wired connections, ideal for setting up local area networks (LAN) or integrating IoT devices that require consistent data transfer.

### 2.3.4 VPN (Virtual Private Network) Protocols

**OpenVPN**: Supports secure tunneling for remote access to IoT networks, ensuring encrypted communication between remote devices and the gateway.
**IPSec**: Another secure VPN protocol for encrypted communications between devices and the internet.
**PPTP** and **L2TP**: Alternatives for VPN tunneling, ensuring that devices connected to the gateway are secure and protected from external threats.

### 2.3.5 MQTT (Message Queuing Telemetry Transport)

**MQTT**: A lightweight and efficient messaging protocol often used in IoT environments for sending and receiving small data packets between devices and cloud-based systems. It's ideal for low-bandwidth and high-latency networks.

### 2.3.6 Modbus (RTU/TCP)

**Modbus RTU** and **Modbus TCP**: These are widely used protocols in industrial automation. They enable communication with industrial equipment, such as PLCs (Programmable Logic Controllers) and sensors, allowing data exchange between IoT devices and control systems.

### 2.3.7 CoAP (Constrained Application Protocol)

**CoAP**: A specialized protocol designed for low-power and constrained devices. It operates over UDP and is suitable for IoT devices with limited resources, such as low-power sensors or actuators in smart cities and industrial systems.

### 2.3.8 SNMP (Simple Network Management Protocol)

**SNMP**: The gateway can use SNMP to monitor and manage connected devices, providing a way to retrieve information like performance metrics, operational status, and error logs. It's useful in large networks where network administrators need to ensure that devices are working properly.

### 2.3.9 TR-069 (CPE WAN Management Protocol)

**TR-069**: Allows the remote management and configuration of IoT devices over broadband networks. This protocol enables the gateway to be monitored, updated, and configured remotely, making it easier to maintain a large fleet of IoT devices.

### 2.3.10 HTTP/HTTPS

**HTTP/HTTPS**: The RUTX50 supports both HTTP and HTTPS protocols for web-based management. Secure HTTPS is typically used for accessing the gateway's configuration interface or communicating with cloud-based services while ensuring data security.

### 2.3.11 NTP (Network Time Protocol)

**NTP**: Ensures that the RUTX50 gateway synchronizes its clock with an NTP server, which is important for time-sensitive applications like logging and event tracking in IoT systems.

### 2.3.12 IPv4 and IPv6

**IPv4**: Supported for standard networking, allowing for communication between devices in an IP-based network.
**IPv6**: Future-proof protocol supporting an increased number of devices and networks, which is particularly useful for large-scale IoT deployments.

### 2.3.12 LAN/WAN Routing

The **RUTX50** is capable of routing between different networks (LAN/WAN), allowing seamless communication and access between devices connected to local networks and external networks (including the internet).

### 2.3.13 PPP (Point-to-Point Protocol)

**PPP**: This protocol is used for encapsulating network-layer protocol information and is typically used in the configuration of VPNs or dial-up connections, especially in cellular or remote scenarios.

### 2.3.14 RADIUS (Remote Authentication Dial-In User Service)

**RADIUS**: This protocol provides centralized authentication, authorization, and accounting for users who connect to the network, often used in enterprise environments for managing access control.

### 2.3.15 DLNA (Digital Living Network Alliance)

**DLNA**: This protocol allows the sharing of media between devices connected to the network, although it is more relevant for multimedia IoT use cases (e.g., video surveillance systems).

## 2.4 VLAN

A **VLAN** (Virtual Local Area Network) is a network technology that allows you to partition a physical network into multiple logical networks. Each VLAN acts as a separate network, even if the devices are on the same physical network infrastructure.

### 2.4.1. Simultaneous Server and Multiple Clients (VLAN for Server and Client Networks)

The RUTX50 allows you to create multiple VLANs (Virtual Local Area Networks) on its LAN and WAN interfaces. This can be used to separate server and client networks, where one VLAN could serve as a server network and another as a client network.

**A. Steps to Configure VLAN for Server and Client:**

1. **Access the RUTX50 Web Interface:**
   - Open a browser and go to the router's IP address (192.168.1.1).
   - Login with your credentials.
2. **Create VLANs:**
   - Go to **Network** > **VLAN**.
   - Add a VLAN by clicking the **Add** button.
   - Assign a VLAN ID (for example, VLAN 10 for the client network and VLAN 20 for the server network).
   - Assign specific interfaces to these VLANs. For example:
     - VLAN 10: Assign the WAN and LAN ports that will be used for clients.
     - VLAN 20: Assign the LAN ports that will be used for the server network.
3. **Assign IP Addresses for Each VLAN:**
   - Go to **Network** > **Interfaces**.
   - Select the interface for each VLAN and assign different IP addresses (e.g., 192.168.10.1 for VLAN 10 and 192.168.20.1 for VLAN 20).
   - Configure appropriate DHCP settings if needed.
4. **Configure Routing:**
   - Go to **Network** > **Firewall** and create firewall rules to ensure traffic flows correctly between the server and client VLANs.
   - Set up routing rules for each VLAN if they need to communicate with external networks.
5. **Test and Verify:**
   - Once you've set up the VLANs and interfaces, you should be able to connect devices to the corresponding VLAN ports and verify that the network segregation is working as intended.

**B. VPN Client and Server Support**

The RUTX50 also supports VPN services for both server and client connections. It can act as a VPN server (e.g., OpenVPN or IPsec) and connect to remote VPN servers as a client.

**C. VPN Server Configuration (e.g., OpenVPN Server):**

1. **Access VPN Settings:**
   - Go to **VPN** > **OpenVPN Server**.
2. **Enable the OpenVPN Server:**
   - Turn on the OpenVPN server.
   - Choose the **Server IP Address** and configure other options like **Port**, **Protocol**, and **Encryption settings**.
   - Set up user authentication for VPN access.
3. **Create Client Certificates:**
   - You'll need to create certificates for the server and clients.
   - Download the server configuration and certificates, then distribute the client configurations to users who will connect.
4. **Firewall and Routing:**
   - Ensure the firewall allows OpenVPN traffic (typically UDP port 1194).
   - If clients need to access specific internal networks, set up routing rules.

**D. VPN Client Configuration (e.g., OpenVPN Client):**

1. **VPN Client Settings:**
   - Go to **VPN** > **OpenVPN Client**.
   - Enable the OpenVPN client and configure it to connect to the remote VPN server using the server's address, port, and credentials.
2. **Import Client Configuration:**
   - You can import a client configuration file or manually set up the connection parameters (e.g., server address, credentials, certificates).
3. **Verify the Connection:**
   - Once configured, the router will automatically connect to the VPN server when the interface comes up.
   - Check the status to ensure the VPN connection is active.

**E. Port and Tag-Based VLAN**

The RUTX50 supports **port-based** and **802.1Q tag-based** VLANs, which allow you to assign VLANs to specific ports on the router or create tagged VLANs for more granular network control.

**F. Steps to Configure Port and Tag-Based VLAN:**

1. **Port-Based VLAN:**
   - Go to **Network** > **VLAN**.
   - Create a new VLAN and assign it to a specific port (e.g., Ethernet port 1 or LAN 2).
   - For example, you can assign VLAN 10 to Ethernet port 1 for a client network and VLAN 20 to Ethernet port 2 for a server network.

2. **Tag-Based VLAN (802.1Q):**
   - Under the **Network** > **VLAN** section, enable 802.1Q tagging.
   - Set the **VLAN ID** for each tag and ensure the correct interface is tagged.
   - For instance, you might configure VLAN 10 with a tag 10 on certain ports and VLAN 20 with tag 20 on other ports.
   - On the receiving device (like a switch or another router), configure the same VLAN tag to allow the devices to communicate properly across the tagged VLAN.
3. **Configure Interfaces for Tagged VLAN:**
   - After setting up VLAN tags, go to **Network** > **Interfaces** and assign the tagged VLANs to specific interfaces.
   - This allows you to segment traffic between VLANs at the port level and ensures traffic from devices is properly tagged.
4. **Verify Configuration:**
   - Test connectivity by connecting devices to the appropriate ports or by configuring network devices to use tagged VLAN IDs.
   - Check for network isolation or communication as per your VLAN configuration.

## 2.5 Security

### 2.5.1. Firewall Rules

Firewall rules are used to control the inbound and outbound traffic on the router. You can create specific rules to allow or block traffic based on IP addresses, ports, protocols, and interfaces.

**A. How to Configure Firewall Rules on the RUTX50:**

1. **Login to the RUTX50 Web Interface:**
   - Open a browser and go to the router's IP address (usually 192.168.1.1).
   - Login with your credentials.
2. **Navigate to the Firewall Settings:**
   - Go to **Network** > **Firewall**.
3. **Create a New Rule:**
   - Click **Add New** to create a new firewall rule.
   - You can configure firewall rules for both **WAN** and **LAN** interfaces.
4. **Define Rule Parameters:**
   - **Action**: Choose either Accept (allow traffic) or Reject (block traffic).
   - **Protocol**: Select the protocol (TCP, UDP, ICMP, etc.).
   - **Source IP/Address**: Specify the source IP or range of IP addresses.
   - **Destination IP/Address**: Specify the destination IP or range of IP addresses.
   - **Port Range**: Specify the port or port range to filter.
   - **Interface**: Choose the interface (WAN or LAN).
   - **Logging**: Enable logging for the rule if you want to track the traffic matched by the rule.

5. **Apply the Rules:**
    - Click **Save & Apply** after configuring each rule.

**Example Use Case:**

To block incoming traffic from a specific IP address, you would:

- Set the **Source IP** to the unwanted IP address.
- Set **Action** to Reject.
- Apply this rule to the **WAN** interface to block external traffic from that IP.

### 2.6 DDoS Prevention

DDoS (Distributed Denial of Service) attacks can overwhelm your network by flooding it with large amounts of traffic. The RUTX50 has several built-in features to mitigate DDoS attacks.

#### 2.6.1 DDoS Prevention Techniques on RUTX50:

1. **Limit Connection Count**:
   You can set limits on the number of simultaneous connections to prevent connection floods. This can be done through **Firewall settings**.
2. **Enable SYN Flood Protection**:
    - Go to **Network** > **Firewall** > **Advanced Settings**.
    - Enable **SYN Flood Protection** to block SYN floods, which are commonly used in DDoS attacks.
3. **Enable IP Filtering**:
    - Filter incoming traffic from suspicious IP addresses by adding them to the **Blackhole** or **Deny List** in the firewall settings.
    - Go to **Network** > **Firewall** > **Advanced** to configure **IP Filtering**.
4. **Rate Limiting**:
    - Rate limiting can help prevent the router from being overwhelmed by too many requests. Under **Network** > **Firewall**, enable the rate-limiting feature to limit the number of requests from any particular IP.
5. **Enable DoS Detection**:
    - Enable **DoS detection** to monitor and block traffic patterns that are typical of a DDoS attack (like flooding the network with large volumes of traffic).
    - Go to **Network** > **Firewall** > **DoS Protection**.
6. **Use a VPN**:
    - A **VPN** can obscure the actual public IP address of your network, making it harder for attackers to target it directly.
    - The RUTX50 supports both **VPN Server** and **Client** configurations (e.g., OpenVPN, IPsec).

### 2.7 Data Limit for SIM Card

To manage and control the data usage of the SIM card (which is particularly important for mobile data connections), the RUTX50 allows you to set data limits and track usage.

#### 2.7.1 How to Set Data Limits on the RUTX50:

1. **Login to the Web Interface:**
   - Go to 192.168.1.1 and log in to the router.
2. **Navigate to the SIM Settings:**
   - Go to **Network** > **Mobile** > **Data Usage**.
3. **Set Data Limit:**
   - In the **Mobile Data** section, you can specify a **data usage limit**.
   - Enter the maximum data limit (in MB or GB) for the SIM card's usage.
   - Optionally, you can configure notifications to alert you when you approach the data limit.
4. **Apply and Monitor:**
   - Once set, the router will track the data usage and notify you when it reaches the configured threshold.

**Example Use Case:**

- Set a monthly data limit of 10GB to ensure that the SIM card doesn't exceed the monthly data allowance provided by the carrier.
- Set up a notification at 8GB of usage to inform the user that they are approaching the limit.

### 2.8 Blocking Unwanted Websites

You can block specific websites on the RUTX50 to ensure secure browsing, restrict access to undesirable content, or enforce company policies.

#### 2.8.1 How to Block Websites on RUTX50:

1. **Login to the Web Interface:**
   - Go to 192.168.1.1 and log in.
2. **Navigate to the URL Filtering Settings:**
   - Go to **Network** > **Firewall** > **Web Filtering**.
3. **Enable URL Filtering**:
   - Enable the **URL Filtering** option.
4. **Add Blocked Websites**:
   - In the **Blocked URLs** section, you can specify a list of websites to block. For example, enter example.com to block access to this site.
   - You can use wildcards (e.g., *.example.com) to block entire domains or specific subdomains.

5. **Apply and Monitor**:
    - Click **Save & Apply** to activate the settings.
    - Test the configuration by trying to access a blocked website from a connected device.

**Example Use Case:**

To block social media websites (e.g., Facebook and Twitter):

- Enter facebook.com and twitter.com in the **Blocked URLs** section.
- Apply the changes, and users on the network won't be able to access these websites.

### 2.9 Guideline for SIM insertion

Here is a step-by-step guideline for inserting a SIM card into the RUTX50 router:

#### 2.9.1 Power Off the RUTX50

Before inserting or removing a SIM card, always make sure the router is powered off to avoid any potential damage to the SIM card or the router's internal components.

- **Step 1**: Turn off the **RUTX50** by pressing the **power button** (or disconnecting the power supply).

#### 2.9.2 Locate the SIM Card Slot

The **SIM card slot** is located on the **side** or **back** of the RUTX50 router, depending on the model. For the RUTX50, there are typically two SIM card slots (for **dual SIM cards**).

- **Step 2**: Locate the SIM card slot panel on the router.
    - **SIM 1** and **SIM 2** are usually labeled clearly next to the slots.

#### 2.9.3 Open the SIM Slot Cover

- **Step 3**: Depending on the design, you may need to **open the plastic cover** to access the SIM card slots.
    - Some models have a small tab or latch that you can lift or slide to reveal the SIM slots.

#### 2.9.4 Insert the SIM Card

- **Step 4**: **Insert the SIM card** into the designated slot (either **SIM 1** or **SIM 2**).
    - Ensure the **metal contacts** of the SIM card are facing **down** and **towards the router's internals**.
    - Align the **notch** of the SIM card with the slot to ensure correct orientation. The notch ensures the SIM is inserted in the correct way, preventing any damage to the card or slot. **Note**: The SIM card should click into place once it is properly inserted.

#### 2.9.5 Close the SIM Slot Cover

- **Step 5**: After the SIM card is inserted, **close the cover** for the SIM card slot to keep it protected.
    - Ensure the cover is securely closed to prevent dust or moisture from entering.

### 2.9.6 Power On the Router

- **Step 6**: Once the SIM card is securely inserted, **power on the router** by pressing the power button or reconnecting the power supply.

### 2.9.7 Check for Network Signal

- **Step 7**: Wait for the router to boot up. It will automatically detect the inserted SIM card and attempt to connect to the mobile network.
    - Check the **LED indicators** on the front panel of the router to verify a successful connection:
        - **Signal LED**: Indicates the strength of the mobile network signal.
        - **LTE/4G Indicator**: Lights up once a mobile data connection is established.

### 2.9.8 Troubleshooting

If the SIM card is not recognized or the router fails to establish a connection, try the following troubleshooting steps:

- **Check the SIM card**: Ensure that the SIM card is active and properly inserted.
- **Check the network coverage**: Ensure that the router is in an area with mobile network coverage.
- **Check the APN settings**: Make sure the **Access Point Name (APN)** is correctly set up for your mobile carrier.
- **Reboot the router**: If necessary, power cycle the router after reinserting the SIM card.

## 2.10 APN Settings

### 2.10.1 Accessing the Web Interface

1. **Connect to the Router**:
    - Connect your computer to the router via **Wi-Fi** or **Ethernet**.
2. **Open the Web Interface**:
    - In your browser, enter the router's IP address (usually 192.168.1.1).
    - Login using your **username** and **password**. The default credentials are typically:
        - **Username**: admin
        - **Password**: admin (if not changed)

### 2.10.2 Navigate to the Mobile Settings

1. Once logged in, go to the **Network** tab on the left sidebar.
2. Select **Mobile** from the options under **Network**.

### 2.10.3 Configuring the APN Settings

1. In the **Mobile** section, you will see a sub-tab for **Data** settings.
2. Find the **APN Settings** section. Here, you will need to enter the relevant information provided by your mobile carrier.
    - **APN Name**: The APN provided by your carrier (e.g., internet, web.vodafone.com, fast.t-mobile.com).
    - **Username**: (Optional, depending on the carrier) Some carriers may require a username for connection (e.g., user).
    - **Password**: (Optional, depending on the carrier) Similar to the username, the password may also be required by some carriers (e.g., pass).
    - **Authentication Type**: (Optional) Select the authentication type, if required by your carrier:
        - **PAP** (Password Authentication Protocol)
        - **CHAP** (Challenge Handshake Authentication Protocol)
    - **MCC (Mobile Country Code)** and **MNC (Mobile Network Code)**: These are often auto-filled based on the SIM card and country. However, you can manually enter them if needed.
    - **APN Type**: Typically, the **default** APN type should be selected (you can choose **default, supl** if required).

### 2.10.4 Save the Settings

1. After entering the correct APN settings for your mobile carrier, click **Save & Apply** to save the configuration.

### 2.10.5 Reboot the Router

1. **Reboot** the router to ensure the new APN settings are applied properly. You can do this from the **System** tab, under **Reboot**.
2. After rebooting, the router should attempt to establish a mobile data connection using the new APN settings.

### 2.10.6 Verify the Connection

1. After rebooting, check the **LED indicators** on the front panel of the router:
    - **Signal LED**: Indicates the strength of the mobile network signal.
    - **LTE/4G Indicator**: This should light up if the connection is successful.
2. You can also check the **Status** page in the **Web Interface** to see the current mobile data connection status.

### 2.10.7 Common Troubleshooting Tips

1. **No Signal or No Connection**:
    - Check if the **SIM card** is properly inserted and active.
    - Verify that the **APN settings** are correct for your carrier.
    - Ensure you are in an area with **good mobile coverage**.
2. **APN Not Saving**:
    - Ensure there are no typos or formatting issues with the **APN** settings.
    - Double-check the **MCC** and **MNC** codes if manually entered.
3. **Check SIM Card Compatibility**:
    - Ensure the SIM card is compatible with the RUTX50 (micro or nano SIM).

## 3. 5G Indoor CPE

### 3.1 CPE system architecture

The PEGATRON MG54AX is a 5G Fixed Wireless Access (FWA) gateway that combines advanced connectivity features suitable for both residential and enterprise applications. Its key specifications are as follows:

- Processor: Qualcomm IPQ5018
- Memory: 128MB Flash and 512MB DDR RAM
- Wi-Fi: Wi-Fi 6 support with 2.4GHz 2x2 and 5GHz 4x4 configurations
- Ethernet Ports: Two 1GbE LAN/WAN ports
- 5G NR Support: Sub 6 GHz, compliant with 3GPP Release 16 in both Standalone (SA) and Non-Standalone (NSA) modes
- 5G Bands:
    - Sub 6 GHz: n1, n2, n3, n5, n7, n8, n12, n13, n14, n18, n20, n25, n26, n28, n29, n30, n38, n40, n41, n48, n66, n70, n71, n75, n76, n77, n78, n79
    - LTE Bands:
        - FDD: B1, B2, B3, B4, B5, B7, B8, B12, B13, B14, B17, B18, B19, B20, B25, B26, B28, B29, B30, B32, B66, B71
        - TDD: B34, B38, B39, B40, B41, B42, B43, B46(LAA), B48
        - DL 4x4 MIMO: B1, B2, B3, B4, B7, B25, B30, B38, B40, B41, B42, B43, B48, B66
- Operating Temperature: 0°C to +35°C
- Dimensions: 221 mm (H) x 110 mm (W) x 105 mm (D)

### 3.2 Wireless Access Interfaces

This **CPE** is a **5G Fixed Wireless Access (FWA)** device, which means it supports high-speed internet connectivity via 5G and wireless access technologies. Specifically, regarding wireless access interfaces for the MG54AX, the device likely supports the following:

### 3.2.1 5G NR (New Radio)

- **Interface**: **5G NR (Sub 6 GHz)** for high-speed wireless broadband connectivity.
- **Use Case**: The MG54AX is designed to provide internet access via 5G networks, supporting both **standalone** (SA) and **non-standalone** (NSA) modes of 5G.
- **Features**:
    - It supports **sub 6 GHz bands**, ensuring high-speed connectivity with reduced latency and increased reliability.
    - It allows users to leverage the **5G mobile network** to get internet access at much higher speeds compared to traditional 4G LTE.
    - Typical speeds range from **100 Mbps to over 1 Gbps**, depending on the deployment and network conditions.

### 3.2.2 Wi-Fi 6 (802.11ax)

- **Interface**: **Wi-Fi 6** for local wireless access within homes or businesses.
- **Use Case**: Provides high-speed Wi-Fi connections to devices such as smartphones, laptops, smart TVs, and IoT devices.
- **Features**:
    - It supports both **2.4 GHz** (2x2 MIMO) and **5 GHz** (4x4 MIMO) bands, providing higher throughput and efficiency compared to previous Wi-Fi standards.
    - Wi-Fi 6 allows for better handling of multiple devices simultaneously through technologies like **OFDMA** (Orthogonal Frequency Division Multiple Access) and **MU-MIMO** (Multi-User, Multiple Input Multiple Output), making it ideal for environments with many connected devices.

### 3.2.3 Ethernet (Wired)

- **Interface**: **Gigabit Ethernet** ports (1GbE LAN/WAN).
- **Use Case**: For wired broadband connectivity, especially for devices that need stable and high-speed internet access without relying on wireless.
- **Features**:
    - The MG54AX includes **Ethernet ports** that allow for direct wired connections to devices (e.g., PCs, servers, or network switches), providing more reliable connections than wireless.
    - It supports **LAN and WAN configurations**, which means it can be used as a gateway for distributing internet from 5G to other devices in a wired setup.

### 3.2.4 LTE (Fallback)

- **Interface**: **LTE** (4G) for areas with poor 5G coverage or as a fallback option.
- **Use Case**: In areas where 5G signal strength is insufficient or unavailable, the device can switch to **LTE** to provide internet access.

- **Features**:
  - The MG54AX supports a wide range of **LTE bands** (FDD and TDD), ensuring compatibility with various cellular networks worldwide.
  - LTE provides reliable connectivity at speeds lower than 5G but still offers decent performance for most use cases.

### 3.2.5 Key Wireless Access Interfaces for the MG54AX:

- **5G NR (Sub 6 GHz)** for ultra-fast 5G connectivity.
- **Wi-Fi 6 (2.4 GHz / 5 GHz)** for local wireless network access.
- **Gigabit Ethernet** for wired high-speed connections.
- **LTE** for fallback in areas with poor 5G coverage.

## 3.3 Security

### 3.3.1 Latest WPA Support

- **WPA3**: The latest Wi-Fi security protocol is likely supported, offering stronger encryption and protection against attacks like brute-force. It includes:
  - **Simultaneous Authentication of Equals (SAE)** to improve password security.
  - Enhanced protection for public networks with **Enhanced Open**.
  - Backward compatibility with **WPA2** for devices that do not yet support WPA3.

### 3.3.2 Rogue Access Point Detection and Prevention

- Modern devices typically include **rogue access point detection**, which identifies unauthorized access points attempting to connect to the network.
  - Alerts administrators when rogue access points are found.
  - Can block or restrict access to unauthorized access points, improving the overall security of the network.

### 3.3.3 IP Security (IPSec), PPTP, IP Filtering

- **IPSec (Internet Protocol Security)**: Provides secure encrypted communication over IP networks and is widely used for setting up **site-to-site** and **remote access** VPNs.
- **PPTP (Point-to-Point Tunneling Protocol)**: While less secure, it is sometimes supported for compatibility with older VPN clients. However, it's typically not recommended due to known vulnerabilities.
- **IP Filtering**: Allows filtering based on **source or destination IP addresses**, restricting or allowing traffic based on defined rules to prevent unwanted or malicious access.

### 3.3.4 MAC Address Authentication

- **MAC address filtering** is often available, allowing only devices with specific **MAC addresses** to connect to the network.
    - This adds an additional layer of security by controlling which devices can access the network.
    - However, **MAC addresses can be spoofed**, so while it's a useful measure, it shouldn't be relied upon as the sole security feature.

## 3.4 5G Network settings

### 3.4.1 Network Mode

- **Standalone (SA)**: Independent 5G network.
- **Non-Standalone (NSA)**: Uses 4G LTE as anchor, adding 5G for data.

### 3.4.2 5G Band Selection

- **Sub 6 GHz** for coverage and **mmWave** for ultra-fast speeds.
- The device may automatically select the best band, or you can manually choose.

### 3.4.3 Carrier Aggregation (CA)

- Combines multiple 5G bands for increased speeds.

### 3.4.4 APN Settings

- Manually configure **Access Point Name (APN)** for network access (based on carrier).

### 3.4.5 Roaming Settings

- Enable or disable **data roaming** while traveling.

### 3.4.6 IPv6/IPv4

- Supports **IPv6** for more efficient addressing, with **Dual Stack** for both protocols.

### 3.4.7 QoS Settings

- Prioritize traffic for specific applications like video or gaming.

### 3.4.8 DNS Settings

- Customize **DNS** servers for faster or more secure connections.

### 3.4.9 Signal Monitoring

- View **5G signal strength** and connection type.

### 3.4.10 Antenna Settings

- Manage **MIMO** or external antennas for better reception.

### 3.4.11 Security

- Enable **WPA3**, **firewall**, and **MAC filtering** for secure connections.

## 3.5 Wifi Settings

### 3.5.1 Wi-Fi Mode

- **2.4 GHz** and **5 GHz**: The device likely supports both **2.4 GHz** (for longer range) and **5 GHz** (for faster speeds and less interference).
- **Wi-Fi 6 (802.11ax)**: Supports the latest Wi-Fi standard for faster speeds, better capacity, and improved efficiency.

### 3.5.2 Wi-Fi SSID

- **SSID (Service Set Identifier)**: Set the name of your wireless network. You can have different SSIDs for **2.4 GHz** and **5 GHz** bands, or set the same SSID for both.

### 3.5.3 Wi-Fi Security

- **WPA3**: The latest security standard, offering stronger encryption.
- **WPA2**: Available for backward compatibility, though WPA3 is recommended for stronger protection.
- **Encryption**: Choose **AES** (Advanced Encryption Standard) for secure Wi-Fi encryption.

### 3.5.4 Channel Settings

- **Automatic Channel Selection**: The device can automatically select the best channel to avoid interference.
- **Manual Channel Selection**: Allows you to select specific channels for both **2.4 GHz** and **5 GHz** if you encounter interference.

### 3.5.5 Wi-Fi Bandwidth

- **20/40/80 MHz**: Set the bandwidth for your 5 GHz network. Wider bandwidth (80 MHz) offers faster speeds but may cause more interference.
- **20 MHz**: Typically used for the 2.4 GHz band to avoid congestion.

### 3.5.6 Guest Network

- Set up a separate **guest network** for visitors, isolating their traffic from your main network for security.

### 3.5.7 MAC Address Filtering

- **MAC Filtering**: Restrict network access to specific devices by allowing only certain **MAC addresses**.

### 3.5.8 Wi-Fi Power Settings

- **Transmit Power**: Adjust the power level for Wi-Fi signal range. Lower power can help reduce interference, while higher power offers better coverage.

### 3.5.9 Wi-Fi Optimization Features

- **OFDMA (Orthogonal Frequency Division Multiple Access)**: A feature of Wi-Fi 6 that improves efficiency by allowing simultaneous transmission to multiple devices.
- **MU-MIMO (Multi-User, Multiple Input, Multiple Output)**: Allows the device to communicate with multiple devices at once for better overall performance.

### 3.5.10 Wi-Fi Scheduling

- **Wi-Fi Scheduling**: Enable or disable Wi-Fi during specific times of the day for energy savings or security purposes.

## 3.6 Connecting 5G Indoor CPE to 5G and WiFi Network

### 3.6.1 Connecting to the 5G Network

1. **Insert SIM Card (if applicable)**:
    - Ensure your **5G SIM card** is inserted into the device.
2. **Power On the Device**:
    - Plug in and turn on you CPE. It should automatically start searching for available **5G networks**.
3. **Network Mode Selection**:
    - Access the device's **admin interface** via a web browser(by typing the device's IP address, 192.168.1.1).
    - Navigate to the **5G Network Mode** settings and select either:
        - **Standalone (SA)**: Full 5G network.
        - **Non-Standalone (NSA)**: 5G using 4G LTE for control.

### 3.6.2 Check Connection Status:

- Once connected, you should see the **5G status indicator** on the CPE or in the admin interface, confirming you're connected to the **5G network**.

### 3.6.3 Connecting to the Wi-Fi Network

1. **Wi-Fi Configuration (Router Mode)**:
   - Ensure the **Wi-Fi feature** on the device is enabled. This is usually enabled by default, but you can check or modify settings from the **Wi-Fi Settings** page in the admin interface.
2. **Wi-Fi Network Settings**:
   - **SSID (Network Name)**: Set the **Wi-Fi network name (SSID)**. You can have separate SSIDs for **2.4 GHz** and **5 GHz** bands or use the same SSID for both.
   - **Security Settings**: Set the **Wi-Fi security** to **WPA3** or **WPA2** for encryption. Use **AES** for the strongest encryption.
   - **Password**: Set a strong **Wi-Fi password**.
3. **Select Wi-Fi Band**:
   - Configure the **Wi-Fi bands**:
     - **2.4 GHz**: Better range but slower speeds, typically used for devices farther away.
     - **5 GHz**: Faster speeds, suitable for devices closer to the CPE with high data requirements.
4. **Connect Devices to Wi-Fi**:
   - On your phone, laptop, or any other Wi-Fi device, search for the **SSID** you set up.
   - Select the network and enter the Wi-Fi password to connect.
5. **Check Wi-Fi Connection**:
   - Once connected, you should be able to browse the internet using the **5G connection** shared via Wi-Fi.
   - The **CPE's Wi-Fi status indicator** should show that the device is broadcasting the Wi-Fi network.

### 3.6.4 Flexible guest access with device isolation Captive

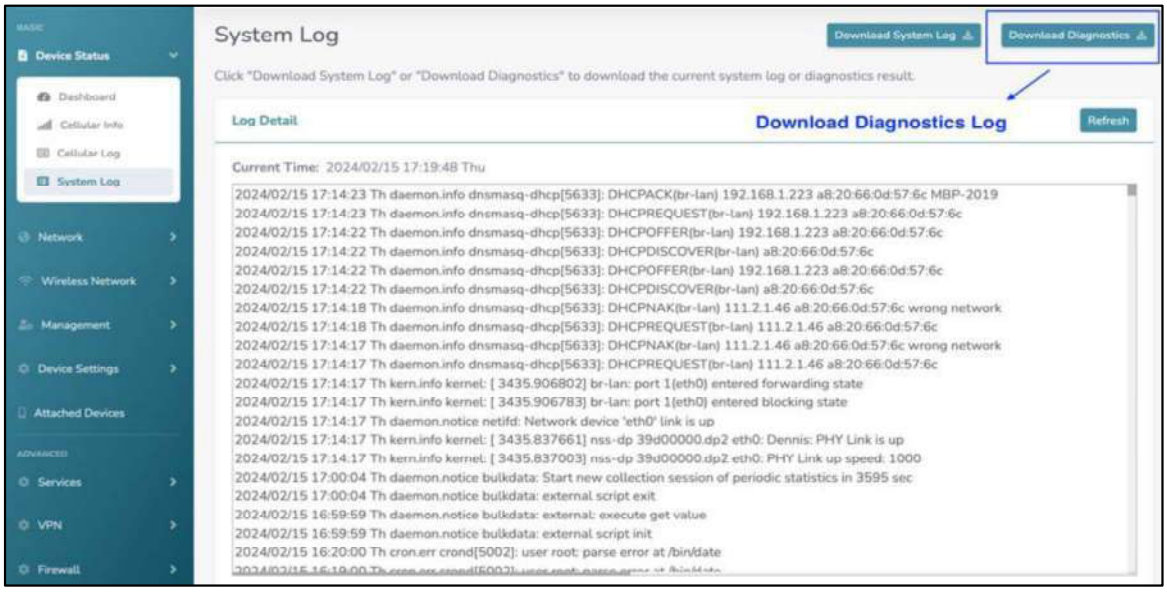
1. **Enable Guest Network**:
   - Access the admin interface and enable the **Guest Network** with a unique **SSID** (e.g., "Guest_Network").
2. **Device Isolation**:
   - Enable **Client Isolation** or **AP Isolation** in the guest network settings to prevent devices on the guest network from communicating with each other or your main network.
3. **Wi-Fi Security**:
   - Set **WPA2** or **WPA3** encryption with a password for the guest network.
4. **Enable Captive Portal**:
   - Turn on the **Captive Portal** feature, allowing you to customize a login page where guests can accept terms or enter a password.
5. **Apply Settings**:
   - Save the settings and restart the device if needed. Guests will now be isolated and redirected to the captive portal for access.

### 3.7 Trouble Shooting

**Steps to capture Diagnostics Log**

1.1 Check Firmware version: access 192.168.1.1 Web GUI, go to “Device Status -> Dashboard -> FIRMWARE VERSION”, make sure Firmware version should be 2.0.1.x_x, in following picture the Firmware version is 2.0.1.1_01

1.2 Go to “Device Status -> System Log”, and press “Download Diagnostics” to download Diagnostics Log. (The Diagnostics also included Cellular Log).




1.3 If only the Cellular Log is required, go to "Device Status -> Cellular Log, and press “Download Cellular Log" to retrieve it.

### 3.7.1 General Troubleshooting Steps

1. **Check Physical Connections:** Ensure that all cables (power, Ethernet, SIM card) are securely connected.

2. **Reboot the Device:** Power cycle the MG54AX CPE by unplugging it from the power outlet, waiting 30 seconds, and plugging it back in.

3. **Check LED Indicators:** Observe the LED indicators on the front panel to identify potential issues. Refer to the device overview section for LED descriptions.

4. **Ping Test:** Use the ping command to test network connectivity. Open a command prompt or terminal and type `ping 192.168.1.1` (replace with the MG54AX's IP address). A successful ping indicates basic network connectivity.

5. **Web Interface Access:** Verify that you can access the MG54AX's web interface. If you cannot, check your computer's IP address and ensure it is on the same subnet as the MG54AX.

6. **Review System Logs:** Check the system logs in the web interface for error messages or warnings that may provide clues about the problem.

7. **Check SIM Card:** Make sure the SIM card is inserted correctly and is active. Try the SIM card in another device to confirm it is working.

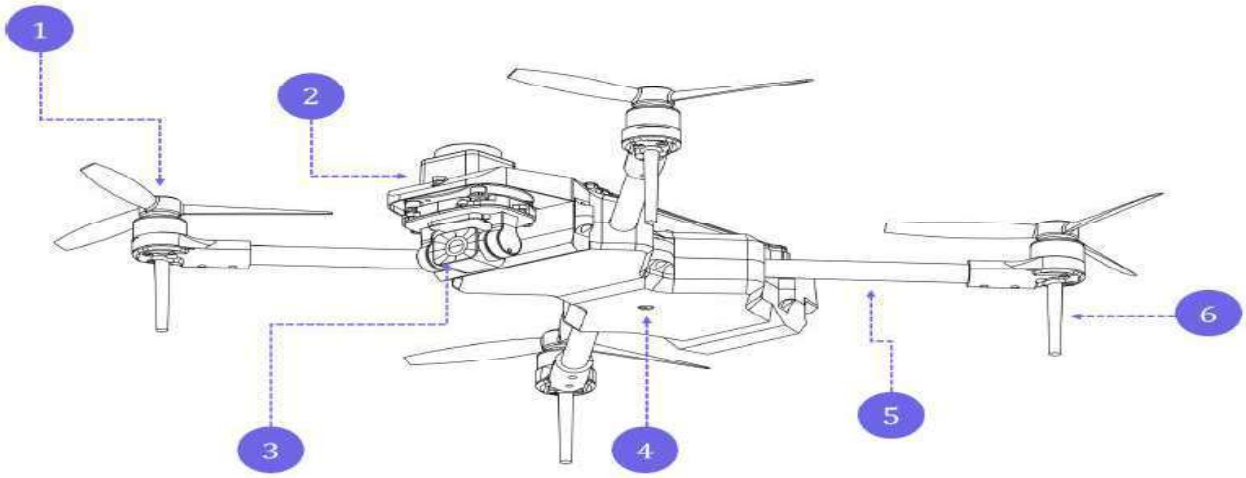
8. **Test with Different Devices:** Try connecting different devices (computers, smartphones) to the MG54AX to rule out device-specific issues

## 4. 5G Drone

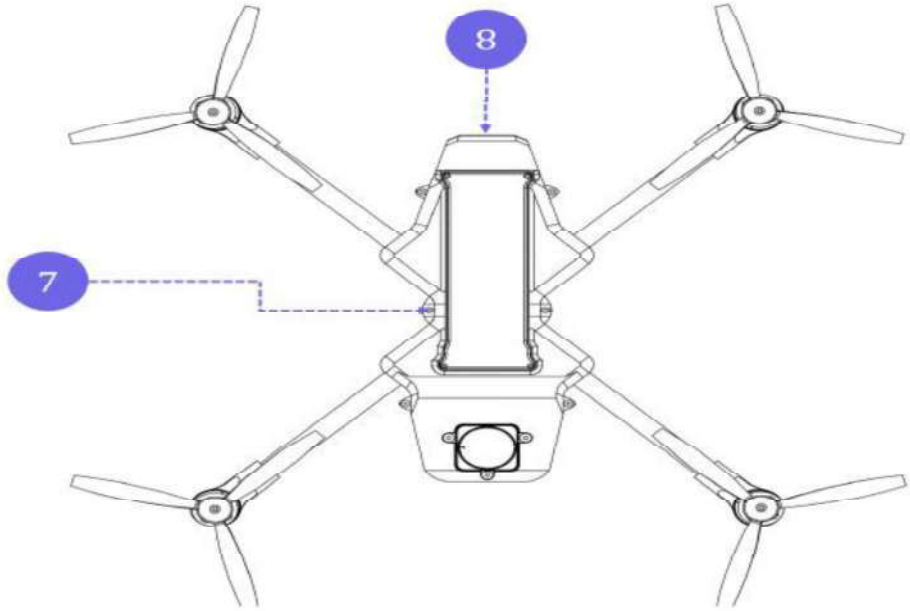
### 4.1 Introduction -

Suparna features both an 360 degree Laser Sensing System and Downward Vision Systems, allowing for hovering and flying indoors as well as outdoors and for automatic Return to Home. The aircraft has a maximum flight speed of 54 kmph in outdoor stable wind conditions and 15 kmph indoor environment (**This has been tested under controlled environment. Do not attempt to imitate**) and maximum flight time of 18 minutes.

## 4.2 Drone System Architecture




| | | | |
|---|---|---|---|
| 1: | Propellers | 4: | Downward Facing Vision System |
| 2: | 360 Lidar | 5: | Arms |
| 3: | Gimbal Camera | 6: | Landing Gear |




| | | | |
|---|---|---|---|
| 7: | Removable Battery | 8: | GPS & External Compass |

**4.3 System Specification-**

| | |
|---|---|
| Drone Type | Quadcopter |
| Frame Material | Carbon Fiber |
| Size (Propeller to Propeller) | 470mm |
| Weight | 1300gm |
| Extra Payload | 500gm |
| Max Altitude-Outdoor | 50 m AGL (◉Check Local Rules and Regulation) |
| Max Operating Altitude | 750 m AMSL |
| Failsafe Features | 1. Low Battery<br>2. Critical Battery<br>3. Ground Station Connection Loss |
| Indoor Features | 360 Degree Obstacle Avoidance System.<br>Vision Based Odometry<br>10cm Precision for Indoor Altitude Hold.<br>Real Time Video Streaming.<br>Position Visualizations in 2D/3D Space.<br>ROS Support |
| 5G Features | N78 Bands for Captive Networks.<br>5G Compatible Specialized GCS. |
| Battery Type | Li-ion 4 Cell |
| Battery Capacity | 4500mAh |
| Battery Charging/Discharging Cycle | 300 |
| Flight Time | 18 Minutes |
| Communication Channel Internet - | 5G with LTE Fallback Or WiFi:6 |
| Cruise Speed | 54 kmph (Outdoor)<br>15kmph(Indoor) |
| GCS | iDronam for Enterprise |

| | |
|---|---|
| Hovering Accuracy Vertical: | ±0.2 m (with Vision Positioning), ±1 m (with GNSS Positioning + Barometer)<br>Horizontal: ±0.3 m (with Vision Positioning), ±1.5 m (with GNSS Positioning) |
| Operating Temperature | 5° to 40° C |
| Image Sensor (Default) | 1/2.7 Inch, 2 MP effective resolution |
| FOV | Horizontal 160° |

**4.4 Aircraft Highlights-**

The Suparna represents a significant leap in unmanned aerial technology, offering a blend of high-speed connectivity, advanced navigation, and versatile functionality. This compact drone is designed for a variety of applications, from commercial and industrial use to recreational and research purposes.

**5G Connectivity and Control:** Equipped with support for a 5G nano SIM, this drone ensures seamless operation through high-speed cellular networks.

**Dual Connectivity Options:** Apart from 5G, the drone also features Wi-Fi connectivity, allowing it to operate efficiently in both urban and remote areas.

**Advanced Processing Engine:** Powered by an onboard processor, the drone is adept at AI/ML analytics, making it suitable for applications that require real-time data processing, such as image analysis and environmental monitoring.

**Autonomy and Navigation:** The drone boasts sophisticated autonomous capabilities like Visual Inertial Odometry (VIO) for accurate positioning, advanced path planning algorithms, and PX4 software for autonomous flight control.

**360-Degree Obstacle Avoidance:** With depth estimation and obstacle detection technology, the drone can navigate safely through challenging terrains. It employs mapping and Visual Obstacle Avoidance (VOA) systems to detect and avoid obstacles in all directions.

**Payload Capacity:** Designed to carry payloads of 500g, this drone is suitable for a variety of tasks, including delivery, surveying, and equipment transport.

**Outdoor Navigation System:** The inclusion of both GPS makes it reliable for critical missions.

**Geo-Fencing:** This feature allows users to set virtual boundaries for the drone, ensuring it operates within predefined areas, enhancing safety and compliance.

**High-Performance Sensors:** Equipped with sophisticated tracking and image sensors, the drone can capture high-quality images and videos, and efficiently track subjects or terrain features.

**Connectivity and Ports:** The drone includes a 5G modem supporting the available 5G bands, an Ethernet/management port, and dual-band Wi-Fi-6, providing various options for data transmission and device management.

**Purpose and Target Users**

The Suparna, with its advanced technological capabilities, is an invaluable tool tailored for a diverse group of users, including students, trainee professionals, researchers, and faculty members across various disciplines. This drone's unique features make it an ideal fit for educational, professional training, and research environments.

### 4.5 Sensitivity Of A Sensor

- **IMU Sensitivity**: Affects the drone's stability, control, and orientation, crucial for GPS-denied navigation.
- **Camera Sensitivity**: Improves performance in low-light conditions and accuracy of VIO for autonomous positioning.
- **LiDAR Sensitivity**: Ensures precise obstacle detection and mapping, crucial for collision avoidance and 3D environment modeling.
- **Barometer Sensitivity**: Allows precise altitude control, vital for maintaining stable flight during specific missions.
- **GPS Sensitivity**: Enhances positional accuracy for autonomous navigation, especially in open environments.

### 4.6 Limitation Of Sensors Devices

Sensors used in a drone provide critical data for navigation, obstacle detection, and stabilization, each sensor type comes with its own limitations. Understanding these limitations helps in designing systems to compensate for potential shortcomings and ensures reliable performance in various flight scenarios.

#### 4.6.1 Camera (for Visual Inertial Odometry - VIO)

- **Limitations**:
  - **Low Light Conditions**: Cameras can struggle to perform well in low-light environments or in very bright conditions (e.g., direct sunlight). This can impact visual data used for VIO or obstacle detection.
  - **Motion Blur**: Rapid movement or poor lighting can cause motion blur, making it difficult to track visual features accurately.
  - **Limited Field of View**: Cameras have a limited field of view, which can restrict the area the drone can "see" at once, requiring careful planning for complete environment mapping.

- **Impact**:
  - **Visual tracking** may fail in low-light or highly dynamic conditions, requiring fallback systems like IMU or LiDAR for reliable navigation.

### 4.6.2 LiDAR (Light Detection and Ranging)

- **Limitations**:
  - **Limited Range**: LiDAR sensors typically have a limited range (usually tens to hundreds of meters). Objects beyond this range cannot be detected, limiting the drone's ability to plan long-range missions.
  - **Vulnerability to Environmental Conditions**: LiDAR can be affected by weather conditions, such as rain or fog, which may scatter the laser beams and reduce its accuracy.
  - **Cost and Power Consumption**: High-performance LiDAR units can be expensive and consume significant amounts of power, limiting the drone's operational time.
- **Impact**:
  - **Obstacle detection** may not be effective in poor weather or at long distances, potentially leading to collision risks in dynamic environments.

### 4.6.3 Barometer

- **Limitations**:
  - **Pressure Variations**: Barometers measure air pressure to estimate altitude, and they can be sensitive to atmospheric changes that aren't related to altitude (such as weather fronts).
  - **Temperature Sensitivity**: Barometers can be affected by changes in temperature, leading to inaccurate altitude readings if not properly calibrated.
  - **Low Precision**: Barometers often provide less precision compared to other altitude sensors, which can be a limitation in precision navigation or low-altitude flight.
- **Impact**:
  - **Altitude control** might suffer from minor inaccuracies, especially in rapidly changing environmental conditions, such as in mountainous or coastal areas where air pressure fluctuates frequently.

### 4.6.4 GPS (Global Positioning System)

- **Limitations**:
  - **Signal Loss**: GPS performance degrades significantly in GPS-denied environments, such as indoors, under dense canopies, or in urban canyons.
  - **Accuracy**: Standard GPS is typically accurate within 1-5 meters, and even high-precision systems like RTK (Real-Time Kinematic) GPS can still have some level of error, especially in difficult conditions.
  - **Vulnerable to Jamming/Interference**: GPS signals are susceptible to jamming or spoofing, where an external signal may disrupt the GPS system or provide false information.
- **Impact**:
  - **Autonomous navigation** relies on GPS for precise location tracking, and in the absence of GPS, the drone must rely on other sensors (like IMU or VIO) to avoid drift and maintain accuracy.

### 4.7 Safety Procedure-

Operating a Drone indoors requires careful attention to safety and precautionary measures to ensure both the operator's safety and the integrity of the drone. Here are some guidelines for the safe operation of the drone in indoor environments:

#### 4.7.1 Pre-Flight Check:

1. Inspect the drone for any damage or loose parts.
2. Ensure all software and firmware are up-to-date.
3. Check the battery level and ensure it's adequately charged.
4. Test the drone's sensors and navigation systems to ensure they are functioning correctly.

#### 4.7.2 Environmental Awareness:

1. Clear the area of people, pets, and fragile objects to minimize risk.
2. Be aware of the indoor space layout, including ceilings, walls, and any obstacles like furniture.
3. Avoid flying near air vents, as the airflow can affect the drone's stability.

#### 4.7.3 Control and Supervision:

1. Always maintain a line of sight with the drone.
2. Do not rely solely on the drone's cameras or sensors; use visual supervision.
3. Keep the drone within a safe distance from the operator and bystanders.

#### 4.7.4 Use of Safety Features:

1. Utilize the drone's obstacle avoidance systems to prevent collisions.
2. Engage geo-fencing features to limit the drone's operational area.
3. Use the drone's autonomy features wisely, especially in confined spaces.

#### 4.7.5 Speed and Height Management:

1. Operate the drone at lower speeds to maintain control.
2. Avoid flying too close to the ceiling or too low to the ground to prevent crashes.

#### 4.7.6 Interference Considerations:

1. Be aware of potential Wi-Fi or electronic interference in indoor environments that might affect drone control.
2. Test the 5G and Wi-Fi connectivity in the indoor area before flying.

### 4.7.7 Emergency Procedures:

1. Familiarize yourself with the drone's emergency stop function
2. Have a clear plan for what to do in case of a loss of control or other emergencies.

### 4.7.8 Training and Skill Level:

1. Ensure that the operator is adequately trained to handle the drone indoors.
2. Practice in a safe, controlled environment to build skill and confidence.

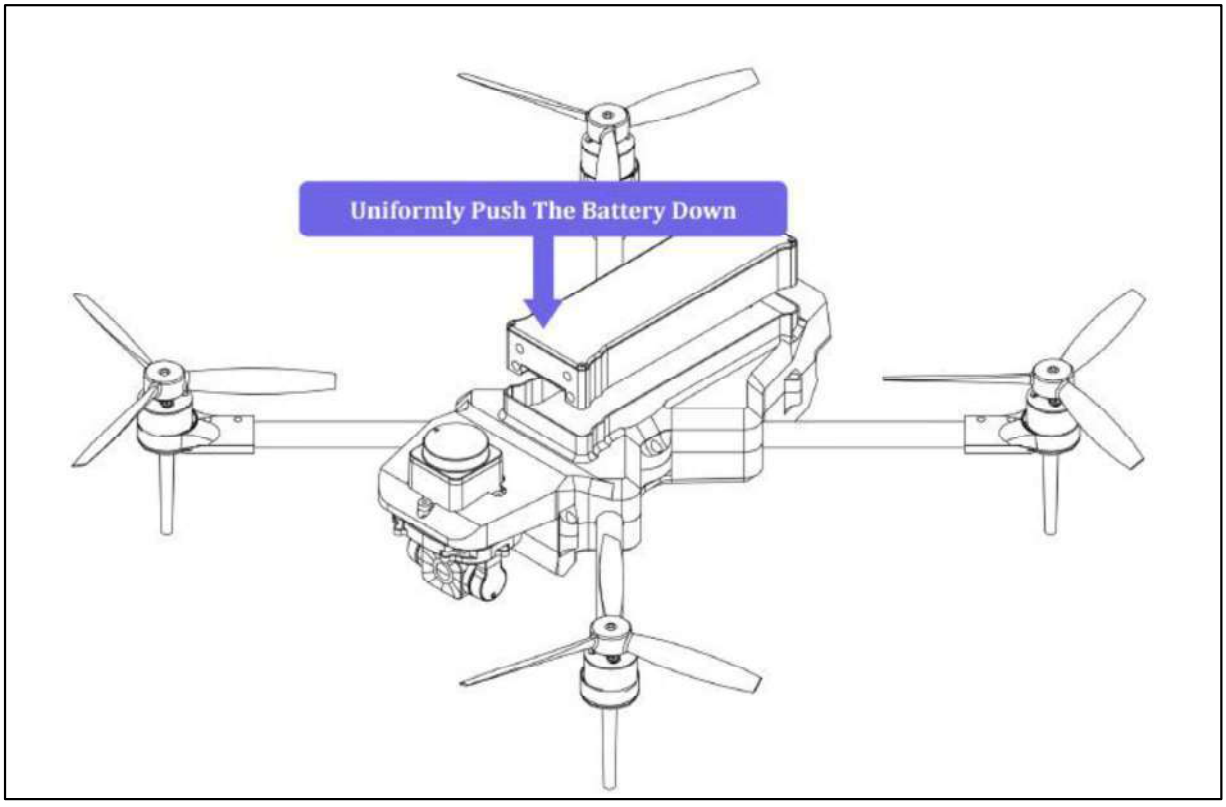
### 4.7.9 Legal and Ethical Considerations:

1. Abide by any institutional or organizational policies regarding drone usage indoors.
2. Respect privacy if the drone is equipped with cameras or recording devices.

### 4.7.10 Post-Flight Check:

1. After flying, inspect the drone for any damage.
2. Download and analyze any collected data if necessary.
3. Store the drone in a safe, dry place away from potential hazards.

### 4.8 Battery Handling-



Battery handling and storage are critical for maintaining the safety and longevity of the 5G-enabled mini drone. Proper care of the drone's batteries ensures optimal performance and reduces the risk of accidents. Here are detailed instructions for handling and storing the drone's batteries:

**4.8.1 Charging the Battery:**

Use only the charger provided with the drone or one recommended by the manufacturer.

1. Charge the battery in a well-ventilated area away from flammable materials.
2. Do not leave the battery unattended while charging.
3. Avoid charging the battery immediately after flight. Allow it to cool down first.

**4.8.2 Handling the Battery:**

1. Handle batteries with care. Avoid dropping them or subjecting them to impact.
2. Do not dismantle, puncture, or alter the battery in any way.
3. Keep the battery dry and away from water or moisture.
4. Avoid exposing the battery to extreme temperatures, both hot and cold.
5. If the battery appears swollen, discolored, or damaged in any way, do not use it.

**4.8.3 Storage of the Battery:**

1. Store batteries in a cool, dry place, away from direct sunlight and heat sources.
2. If storing for an extended period, keep the battery at a 40-60% charge level. Avoid storing it fully charged or fully depleted.
3. Check the charge level every few months and recharge to the recommended storage level if necessary.
4. Use a fireproof bag or container designed for LiPo battery storage.

**4.8.3 Transporting the Battery:**

1. When transporting, ensure the battery terminals are protected and cannot come into contact with metal objects.
2. Carry batteries in a dedicated, fireproof battery case.

**4.8.4 Battery Disposal:**

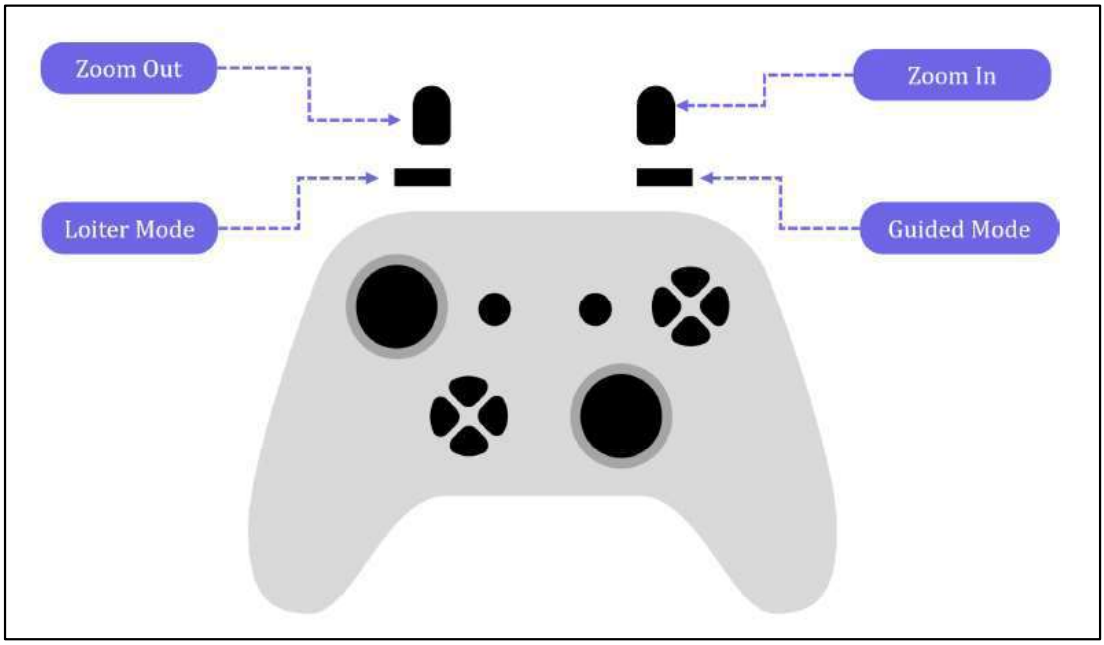
1. Do not dispose of batteries in regular trash. Batteries should be recycled according to local regulations.
2. If the battery is damaged, follow local guidelines for hazardous waste disposal.

### 4.8.5 General Precautions:

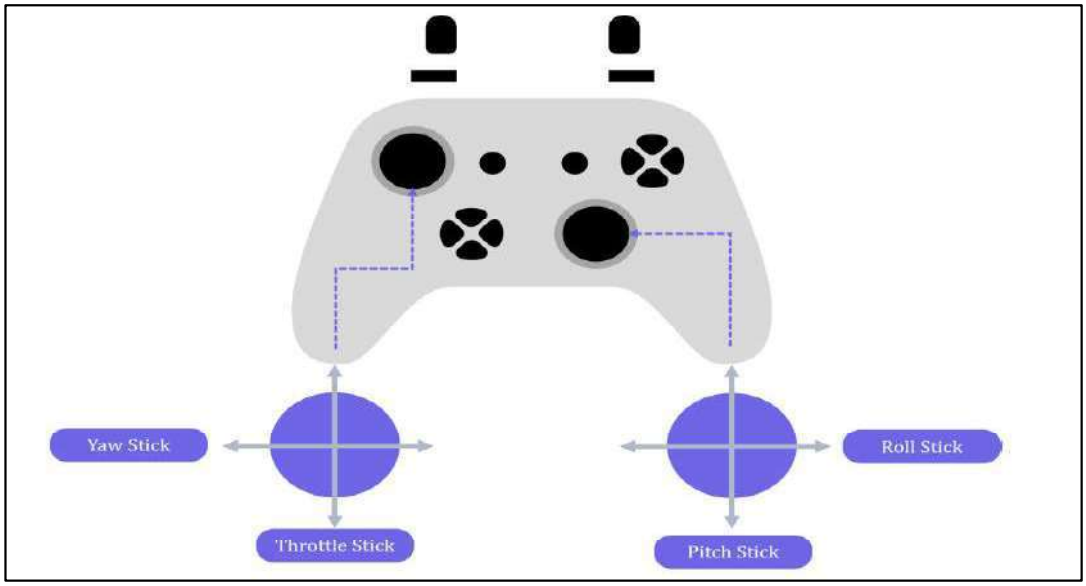
1. Never use a battery that is not specifically designed for your drone model.
2. Keep batteries out of reach of children.
3. Regularly inspect the battery for signs of wear or damage.

## 4.9 Flight And Companion Controller -

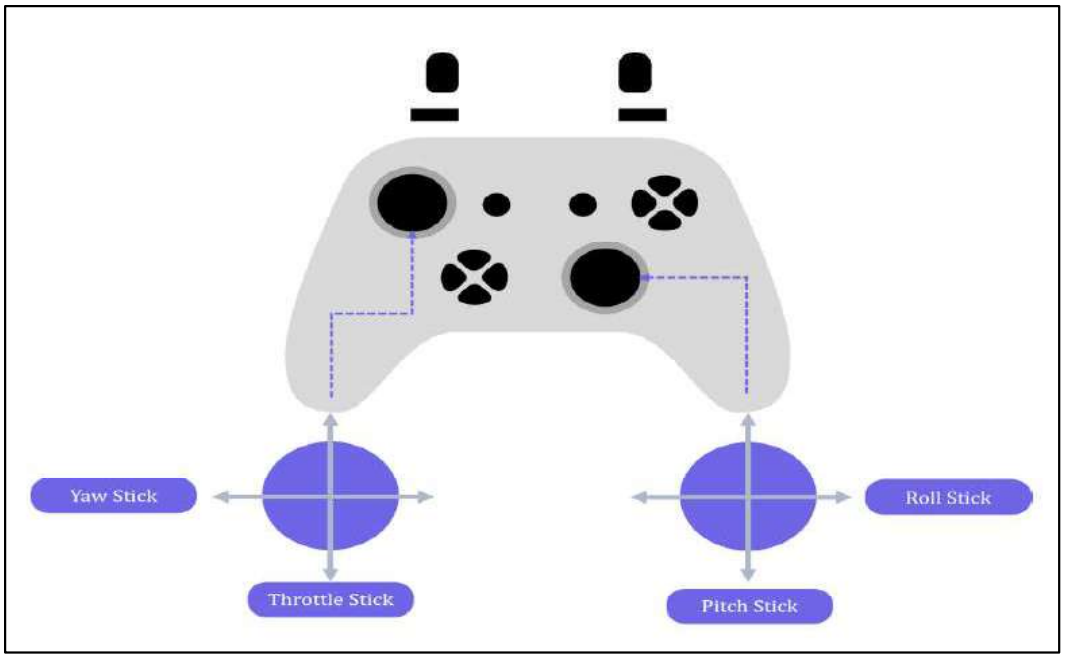
### 4.9.1 Joystick - Modes and Zoom



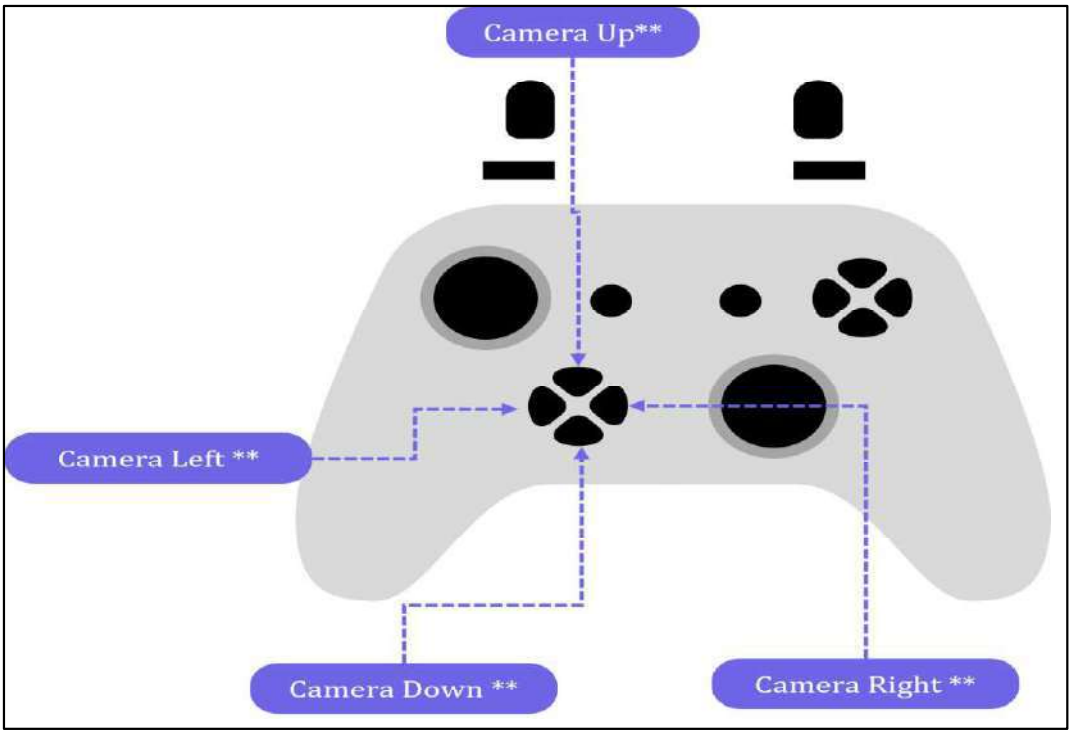

### 4.9.2 Joystick - Sticks and Control



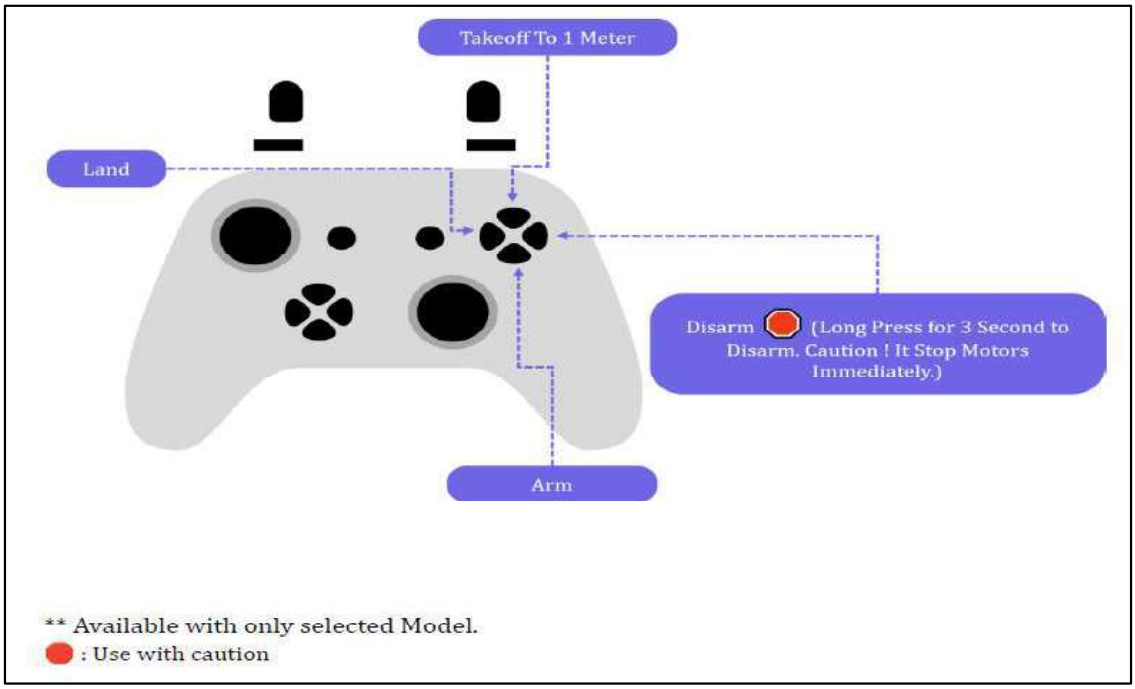
### 4.9.3 Joystick - Sticks and Control




### 4.9.4 Joystick - Camera Control



### 4.9.5 Joystick - Command and Special Modes




### 4.10 DATA AND CONTROL FLOW ARCHITECTURE

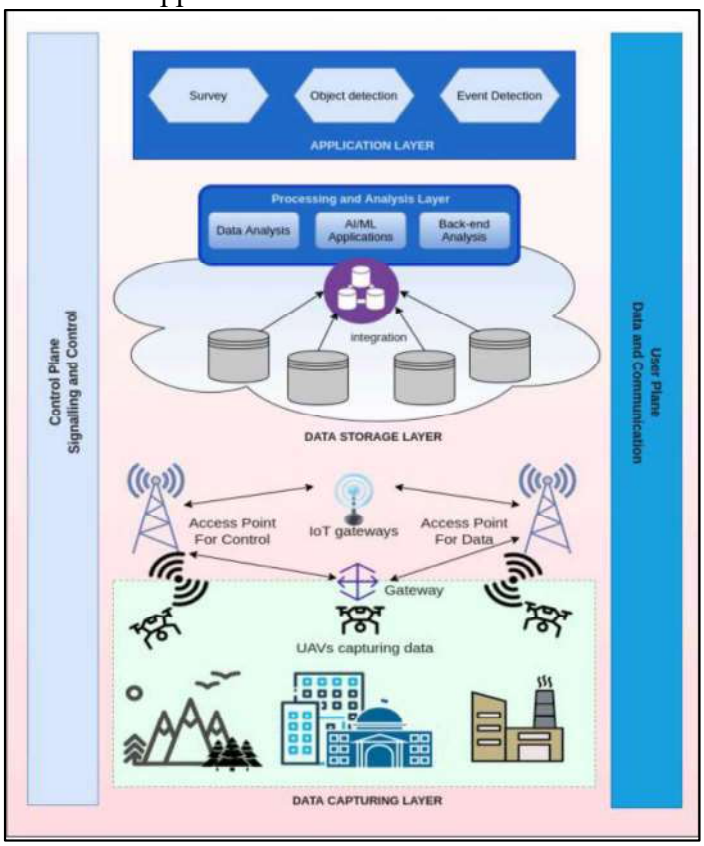
Figure below describes the proposed multilayered architecture of the system. It consists of four layers for the integration of diverse technologies and providing abstraction.

**Data Capturing Layer:**
This is the most important step in the whole design. The proposed system tends to offer a variety of methods like UAVs, standalone cameras, and cameras on UAVs, microphones, sensors, IoT sensors, etc. to capture data. The data capturing layer will facilitate data capturing across multiple devices with minimum complexities to end-users.

**Interaction with Data Storage Layer:**
The data capturing layer will provide seamless connectivity to the data storage layer.

**Data Storage Layer:**
This layer will accept raw data from the data storage layer and store it in the agreed format. The layer may provide facilities for data annotations and similar tasks. The major challenge here will be designing schema for the multi-domain data. The layer shall be responsible for enforcement of security policies, framing data access guidelines, and shall offer access to the data in real-time. Provisions will be made to offer data in the format as desired by the Processing and Data Analytics layer.

**Processing and Data Analytics Layer:**
The aim of this layer is to offer a variety of data analytics services to different applications. The layer will offer facilities for designing and developing machine learning models and data analysis models for monitoring and surveillance-centric applications.




**Application Layer:**
This layer is concerned with user applications. Different end-users using the system will be citizens, industry, government agencies, etc.

**User interface:**
Different interfaces may be offered to different users and the type of interface will be governed by the applications being offered. The data in the processed form will be accepted by this layer from the data storage layer and will be offered to end-users.

**Communication pillars:**

| |
|---|
| The pillars consist of the Control Plane and User Plane for providing seamless end-to-end communication between devices and layers of the system. |

## 4.11 ON-BOARDING PROCESSING

On-board Computer Specifications

| | |
|---|---|
| SOC Name | Broadcom BCM2711 |
| CPU Name | ARM Cortex-A72 (ARM v8) |
| Core Count | 4 (Quad-Core) |
| Architecture | 64 bit |
| Processing Speed | 1.5 GHz |
| RAM | 4 GB LPDDR4 @ 3200 MHz |
| Storage | 32GB |

## 4.12 MISSION PROTOCOL

The mission sub-protocol allows a GCS or developer API to exchange mission (flight plan), geofence and safe point information with a drone/component.

**The protocol covers:**
Operations to upload, download and clear missions, set/get the current mission item number, and get notification when the current mission item has changed.

Message type(s) and enumerations for exchanging mission items.

Mission Items ("MAVLink commands") that are common to most systems.

The protocol supports re-request of messages that have not arrived, which allows missions to be reliably transferred over a lossy link.

**Mission Types**

MAVLink 2 supports three types of "missions": flight plans, geofences and rally/safe points. The protocol uses the same sequence of operations for all types (albeit with different types of Mission Items). The mission types must be stored and handled separately/independently.

Mission protocol messages include the type of associated mission in the mission_type field (a MAVLink 2 message extension). The field takes one of the MAV_MISSION_TYPE enum values: MAV_MISSION_TYPE_MISSION, MAV_MISSION_TYPE_FENCE, MAV_MISSION_TYPE_RALLY.

Mission Items (MAVLink Commands)
Mission items for all the mission types are defined in the MAV_CMD enum.

MAV_CMD is used to define commands that can be used in missions ("mission items") and commands that can be sent outside of a mission context (using the Command Protocol). Some MAV_CMD can be used with both mission and command protocols. Not all commands/mission items are supported on all systems (or for all flight modes).

The items for the different types of mission are identified using a simple name prefix convention:

**4.12.1 Flight plans:**

NAV commands (MAV_CMD_NAV_*) for navigation/movement (e.g. MAV_CMD_NAV_WAYPOINT, MAV_CMD_NAV_LAND)

DO commands (MAV_CMD_DO_*) for immediate actions like changing speed or activating a servo (e.g. MAV_CMD_DO_CHANGE_SPEED).

CONDITION commands (MAV_CMD_CONDITION_*) for changing the execution of the mission based on a condition - e.g. pausing the mission for a time before executing next command (MAV_CMD_CONDITION_DELAY).

Geofence mission items:

Prefixed with MAV_CMD_NAV_FENCE_ (e.g. MAV_CMD_NAV_FENCE_RETURN_POINT).

Rally point mission items:

There is just one rally point MAV_CMD: MAV_CMD_NAV_RALLY_POINT.

The commands are transmitted/encoded in MISSION_ITEM or MISSION_ITEM_INT messages. These messages include fields to identify the particular mission item (command id) and up to 7 command-specific optional parameters.

| Field Name | Type | Values | Description |
|---|---|---|---|
| command | uint16_t | MAV_CMD | Command id, as defined in MAV_CMD. |
| param1 | float | | Param #1. |
| param2 | float | | Param #2. |
| param3 | float | | Param #3. |
| param4 | float | | Param #4. |
| param5 (x) | float / int32_t | | X coordinate (local frame) or latitude (global frame) for navigation commands (otherwise Param #5). |
| param6 (y) | float / int32_t | | Y coordinate (local frame) or longitude (global frame) for navigation commands (otherwise Param #6). |
| param7 (z) | float | | Z coordinate (local frame) or altitude (global - relative or absolute, depending on frame) (otherwise Param #7). |

The first four parameters (shown above) can be used for any purpose - this depends on the particular command. The last three parameters (x, y, z) are used for positional information in MAV_CMD_NAV_* commands, but can be used for any purpose in other commands.

The remaining message fields are used for addressing, defining the mission type, specifying the reference frame used for x, y, z in MAV_CMD_NAV_* messages, etc.:

**4.12.2 MISSION_ITEM_INT vs MISSION_ITEM**

MISSION_ITEM and MISSION_ITEM_INT are used to exchange individual mission items between systems. MISSION_ITEM messages encode all mission item parameters into float parameters fields (single precision IEEE754) for transmission. MISSION_ITEM_INT is exactly the same except that param5 and param6 are Int32 fields.

Protocol implementations must allow both message types in supported operations (along with the corresponding MISSION_REQUEST and MISSION_REQUEST_INT message types).

MAVLink users should always prefer MISSION_ITEM_INT because it allows latitude/longitude to be encoded without the loss of precision that can come from using MISSION_ITEM.

Message/Enum Summary

The following messages and enums are used by the service.

Message Description

MISSION_REQUEST_LIST Initiate mission download from a system by requesting the list of mission items.

MISSION_COUNT Send the number of items in a mission. This is used to initiate mission upload or as a response to MISSION_REQUEST_LIST when downloading a mission.

MISSION_REQUEST_INT Request mission item data for a specific sequence number be sent by the recipient using a MISSION_ITEM_INT message. Used for mission upload and download.

MISSION_REQUEST Request mission item data for a specific sequence number be sent by the recipient using a MISSION_ITEM message. Used for mission upload and download.

MISSION_ITEM_INT Message encoding a mission item/command (defined in a MAV_CMD). The message encodes positional information in integer parameters for greater precision than MISSION_ITEM. Used for mission upload and download.

MISSION_ITEM Message encoding a mission item/command (defined in a MAV_CMD). The message encodes positional information in float parameters. Used for mission upload and download.

MISSION_ACK Acknowledgment message when a system completes a mission operation (e.g. sent by autopilot after it has uploaded all mission items). The message includes a MAV_MISSION_RESULT indicating either success or the type of failure.

MISSION_CURRENT Message containing the current mission item sequence number. This is emitted when the current mission item is set/changed.

MISSION_SET_CURRENT Set the current mission item by sequence number (continue to this item on the shortest path).

STATUSTEXT Sent to notify systems when a request to set the current mission item fails.

MISSION_CLEAR_ALL Message sent to clear/delete all mission items stored on a system.

MISSION_ITEM_REACHED Message emitted by system whenever it reaches a new waypoint. Used to monitor progress.

Enum Description

MAV_MISSION_TYPE Mission type for message (mission, geofence, rallypoints).

MAV_MISSION_RESULT Used to indicate the success or failure reason for an operation (e.g. to upload or download a mission). This is carried in a MISSION_ACK.

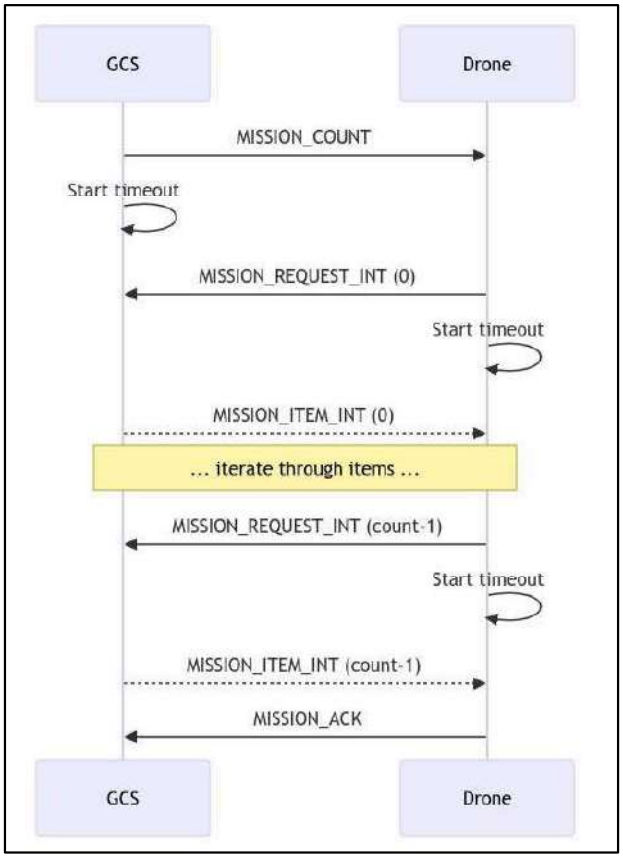
MAV_FRAME Co-ordinate frame for position/velocity/acceleration data in the message.

MAV_CMD Mission Items (and MAVLink commands). These can be sent in MISSION_ITEM or MISSION_ITEM_INT.

Operations This section defines all the protocol operations.

### 4.12.3 Upload a Mission to the Vehicle

The diagram below shows the communication sequence to upload a mission to a drone (assuming all operations succeed).



Mission update must be robust! A new mission should be fully uploaded and accepted before the old mission is replaced/removed.

### 4.12.4 Mission Upload Sequence

In more detail, the sequence of operations is:

GCS sends MISSION_COUNT including the number of mission items to be uploaded (count).

A timeout must be started for the GCS to wait on the response from Drone (MISSION_REQUEST_INT).

Drone receives message and responds with MISSION_REQUEST_INT requesting the first mission item (seq==0).

A timeout must be started for the Drone to wait on the MISSION_ITEM_INT response from GCS.

GCS receives MISSION_REQUEST_INT and responds with the requested mission item in a MISSION_ITEM_INT message.

Drone and GCS repeat the MISSION_REQUEST_INT/MISSION_ITEM_INT cycle, iterating seq until all items are uploaded (seq==count-1).

After receiving the last mission item the drone responds with MISSION_ACK with the type of MAV_MISSION_ACCEPTED indicating mission upload completion/success.

The drone should set the new mission to be the current mission, discarding the original data.

The drone considers the upload complete.

GCS receives MISSION_ACK containing MAV_MISSION_ACCEPTED to indicate the operation is complete.

**Note:**

A timeout is set for every message that requires a response (e.g. MISSION_REQUEST_INT). If the timeout expires without a response being received then the request must be resent.

Mission items must be received in order. If an item is received out-of-sequence the expected item should be re-requested by the vehicle (the out-of-sequence item is dropped).

An error can be signaled in response to any request using a MISSION_ACK message containing an error code. This must cancel the operation and restore the mission to its previous state. For example, the drone might respond to the MISSION_COUNT request with a MAV_MISSION_NO_SPACE if there isn't enough space to upload the mission.

The sequence above shows the mission items packaged in MISSION_ITEM_INT messages. Protocol implementations must also support MISSION_ITEM and MISSION_REQUEST in the same way.

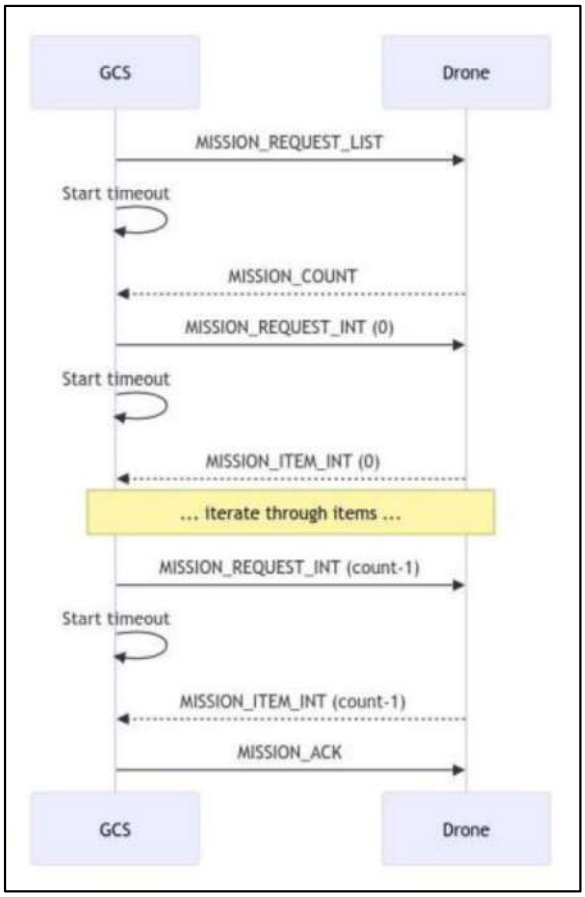
Uploading an empty mission (MISSION_COUNT is 0) has the same effect as clearing the mission.

Download a Mission from the Vehicle

The diagram below shows the communication sequence to download a mission from a drone (assuming all operations succeed).

**4.12.5 Sequence: Download mission**

The sequence is similar to that for uploading a mission. The main difference is that the client (e.g. GCS) sends MISSION_REQUEST_LIST, which triggers the autopilot to respond with the current count of items. This starts a cycle where the GCS requests mission items, and the drone supplies them.




Note:A timeout is set for every message that requires a response (e.g. MISSION_REQUEST_INT). If the timeout expires without a response being received then the request must be resent.

Mission items must be received in order. If an item is received out-of-sequence the expected item should be re-requested by the GCS (the out-of-sequence item is dropped).

An error can be signaled in response to any request using a MISSION_ACK message containing an error code. This must cancel the operation.

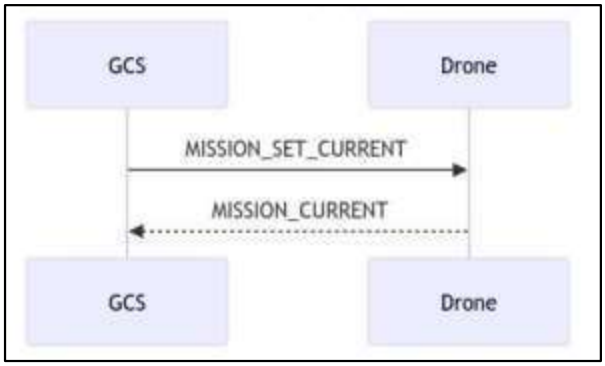
The sequence above shows the mission items packaged in MISSION_ITEM_INT messages.

Protocol implementations must also support MISSION_ITEM and MISSION_REQUEST in the same way. Set Current Mission Item

The diagram below shows the communication sequence to set the current mission item.

### 4.12.6 Set mission item

In more detail, the sequence of operations is:




GCS/App sends MISSION_SET_CURRENT, specifying the new sequence number (seq). Drone receives message and attempts to update the current mission sequence number. On success, the Drone must broadcast a MISSION_CURRENT message containing the current sequence number (seq).

On failure, the Drone must broadcast a STATUSTEXT with a MAV_SEVERITY and a string stating the problem. This may be displayed in the UI of receiving systems.

Notes:

There is no specific timeout on the MISSION_SET_CURRENT message.The acknowledgment of the message is via broadcast of mission/system status, which is not associated with the original message. This differs from error handling in other operations.

This approach is used because the success/failure is relevant to all mission-handling clients.

Monitor Mission Progress

GCS/developer API can monitor progress by handling the appropriate messages sent by the drone:

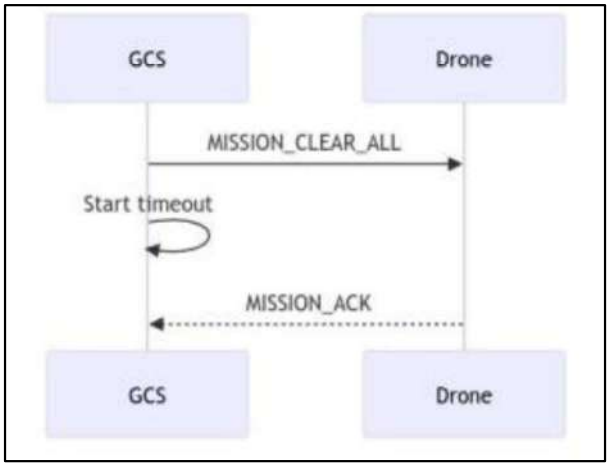
The vehicle must broadcast a MISSION_ITEM_REACHED message whenever a new mission item is reached. The message contains the seq number of the current mission item.

The vehicle must also broadcast a MISSION_CURRENT message if the current mission item is changed.

Clear Missions

The diagram below shows the communication sequence to clear the mission from a drone (assuming all operations succeed).

**4.12.7 Clear Missions**




In more detail, the sequence of operations is:

GCS/API sends MISSION_CLEAR_ALL

A timeout is started for the GCS to wait on MISSION_ACK from Drone.

Drone receives the message, and clears the mission from storage.

Drone responds with MISSION_ACK MAV_MISSION_ACCEPTEDMAV_MISSION_RESULT. with result type of GCS receives MISSION_ACK and clears its own stored information about the mission. The operation is now complete.

Note:

A timeout is set for every message that requires a response (e.g. MISSION_CLEAR_ALL). If the timeout expires without a response being received then the request must be resent.

An error can be signaled in response to any request (in this case, just MISSION_CLEAR_ALL) using a MISSION_ACK message containing an error code. This must cancel the operation. The GCS record of the mission (if any) should be retained.

Canceling Operations

The above mission operations may be canceled by responding to any request (e.g.MISSION_REQUEST_INT) with a MISSION_ACK message containing the MAV_MISSION_OPERATION_CANCELLED error.

Both systems should then return themselves to the idle state (if the system does not receive the cancellation message it will resend the request; the recipient will then be in the idle state and may respond with an appropriate error for that state).

### 4.13 GEO-FENCING

It is a feature that allows operators to define virtual boundaries within which the drone is permitted to operate. When enabled, geo-fencing ensures that the drone does not fly outside the designated area, helping to enhance safety and prevent the drone from entering restricted or dangerous zones.

#### 4.13.1 Key Features of GEO-FENCING:

- 

### 4.14 AUTONOMY ARCHITECTURE

#### 4.14.1 Visual Inertial Odometry (VIO)

It is a key technology for providing precise real-time navigation and positioning. The drone would use a combination of visual data from cameras (to track features and estimate movement) and inertial data from sensors (accelerometers and gyroscopes in the IMU) to accurately determine its position and orientation, even in GPS-denied or challenging environments.

#### 4.14.2 Path Planning

It involves calculating the most efficient route from the starting point to the destination while avoiding obstacles.
the drone can autonomously navigate through complex environments in 3D, adjusting its path in real time. It leverages data from sensors like **VIO** (Visual Inertial Odometry) for precise positioning and obstacle detection, ensuring smooth, safe, and efficient flight even in dynamic environments.



#### 4.14.3 PX4

It is an open-source flight control software used in drones, including the Menthosa Suparna drone. It provides the essential control systems for flight, including stabilization, navigation, and autonomous flight capabilities. PX4 supports a wide range of sensors (like GPS, IMU, and cameras), and is highly customizable, allowing for integration with various sensors and payloads. It enables advanced flight modes such as autonomous missions, path planning, and real-time control, making it ideal for use in both research and commercial drone.

### 4.14.4 GPS-denied navigation

It allows to fly and navigate accurately without relying on GPS signals. This is achieved through the integration of sensors like Visual Inertial Odometry (VIO), IMUs (Inertial Measurement Units), and sometimes LiDAR or cameras.

### 4.14.5 BVLOS (Beyond Visual Line of Sight)

This allows the drone to operate beyond the pilot's direct visual range, relying on sensors, GPS, and communication systems for navigation. This is particularly useful for long-range missions, inspections, or surveys in remote areas where the drone needs to travel far from the operator.

### 4.14.6 Follow Me

This mode enables the drone to autonomously track and follow a moving target, such as a person or vehicle. The drone uses GPS or visual tracking to keep the target in sight, making it ideal for activities like filming, sports, or personal tracking in outdoor environments.

## 4.15 360 OBSTRACLE AVOIDANCE ARCHITECTURE

The **360-degree Obstacle Avoidance** is designed to ensure that the drone can detect, avoid, and navigate around obstacles in all directions during flight, enhancing its safety, autonomy, and ability to operate in complex environments. This architecture typically involves a combination of multiple sensors, algorithms, and systems to provide comprehensive environmental awareness and real-time decision-making. Below is a detailed breakdown of how such an architecture might work.

### 4.15.1 Depth estimation

It is a key component in autonomous navigation for drones . It refers to the process of determining the distance between the drone and objects in its environment.

### 4.15.2 Object detection

It is a critical task in autonomous drone operations for drones. It enables the drone to identify and classify objects within its environment, which is essential for tasks such as obstacle avoidance, target tracking, environment mapping, and autonomous navigation.

### 4.16 Mapping and Visual Odometry (VOA)

In autonomous drone navigation, Mapping and Visual Odometry (VOA) are critical components for reliable and precise operation. These technologies work together to enable the drone to understand its environment, localize itself, and navigate without the reliance on GPS. Here's an overview of both concepts,

### 4.17 Wireless Access Option

#### 4.17.1 Connection Modes

Suparna has 3 Connection Modes by which the C2 (Command and Control) link can be established between the drone and Ground Control Station (GCS). Suparna can be flown in both outdoor (GPS available) as well as indoor (GPS denied) environments. But due to the nature of data being transmitted in both environments the bandwidth requirements change.

● For **Indoor Environment**, drone is constantly mapping the environment and using the Vision systems for localization. The bandwidth required is in the range of - 30 Mbps and some peaks of up to 50 Mbps.

● For **Outdoor Environment**, the drone has GPS for its localization and no mapping is being done. So bandwidth reduces to 2 Mbps with peaks up to 5 Mbps.

#### A. 5G Mode

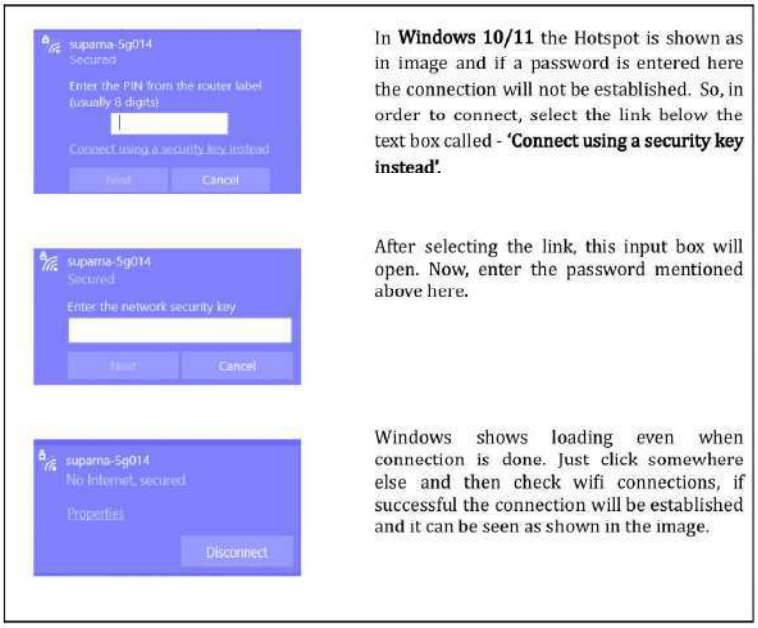
On board the drone there is a 5G module, which supports all the sub 6 Ghz 5G bands available in India. The module supports 5G nano sim which is inserted into the drone via the insertion slot present at the back of the drone.
Connecting to the drone -
1. Before powering on the drone, insert the 5G nano sim card into the drone.
2. Power ON the drone.
3. For public 5G use -
a. Log into 'idronam.com' now you can connect to your drone via selecting the drone alias at Dashboard.
4. For private 5G use -
a. Open the ''iDronam Enterprise' software. If you have previously added the drone you can simply select the connect button on the device dashboard.
b. If you are using the drone for the 1st time. Please add the drone to the database and then continue.
5. Upon successful connection you will be able to see the drone telemetry data and live feed.

Hurray! Your drone is now connected via a 5G network. Enjoy Flying.

#### B. Hotspot Mode

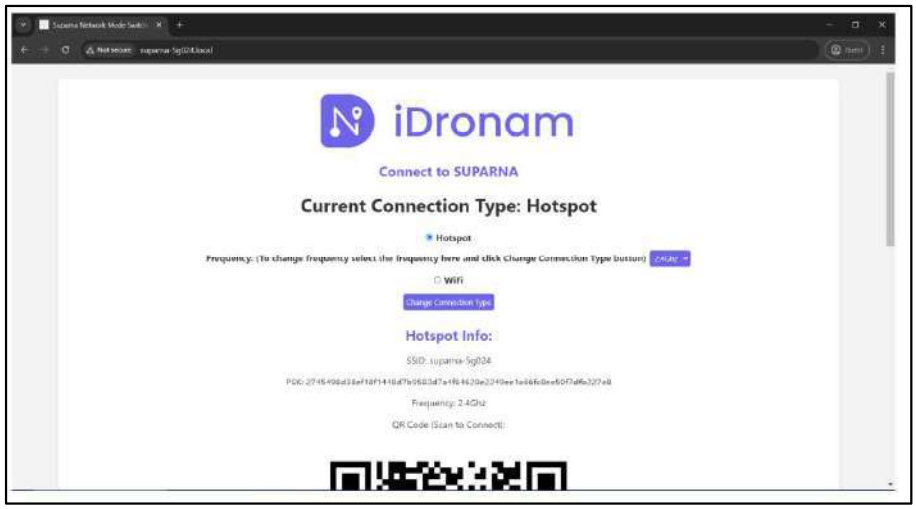
The drone has a 2.4/5Ghz wifi module onboard for quick and easy connections. This can also be used to connect to the drone. By default, the drone is in Hotspot or AP (Access Point) mode and broadcasting in 2.4Ghz band.

Connecting to the drone -
1. Power ON the drone.
2. Connect to the Hotspot. The SSID of the drone's hotspot is unique. It is the drone's serial no. - 'suparna-5gXXX'. Password for the hotspot is - 'Suparna5Gdrone'.
3. Open the 'iDronam Enterprise' software. Go to the dashboard.
4. If the drone is added, connect to the drone.




**C. WiFi Mode**

When in wifi mode, the drone can be connected to any 2.4/5Ghz wifi network. As the drone comes in Hotspot mode by default, we need to switch it to wifi mode, which can be done by the network mode switch webpage hosted inside the drone. The steps for connecting to the webpage are -
1. Connect to the hotspot of the drone.
2. Open any browser and enter the link of the webpage - 'suparna-5gXXX.local', where suparna-5gXXX is the ID of the drone and can be checked from the SSID of the hotspot.
If you are facing issues you can also navigate to the webpage by entering the drone's IP - '10.42.0.1'. This is the same for all the drones.
3. Upon successful connection a webpage will be opened as shown in the image below-



4. This will be the first time for a drone's wifi mode connection. Navigate to the bottom of the page where the ‘Add New WiFi’ section is present. Here, enter the wifi details - ‘SSID’ and ‘password’. Please enter the password carefully, you can view password text by clicking the Show button.

5. After entering the details and verifying the password. Click on the 'Connect' button. A warning will pop up; click ok. Now the page will refresh. This means that the wifi details are added into the drone.
6. Now to connect to the WiFi navigate to the top of the webpage, here select the ‘WiFi’ radio button and click on ‘Change connection type’. A warning message will come, click on ok and the page will start loading.

7. Now, the drone’s Hotspot will be disabled and the drone will connect to the entered WiFi.
8. In order to connect to the drone via the WiFi, connect the GCS system to the same WiFi as the drone.
9. Now you will need the IP address assigned to the drone. This can be obtained via your WiFi router where you will be able to see the drone’s ID connected to the WiFi.
10. If your router does not have this functionality or you are unable to retrieve the IP address. You can open the network mode switch web page by going to the link - ‘suparna-5gXXX.local’. Here you will be able to see the IP addresses assigned to the drone.

11. Now, add this IP in the ‘iDronam Enterprise’ via the ‘Add devices’ and you can connect to the drone.

**D. Network Mode Switch**

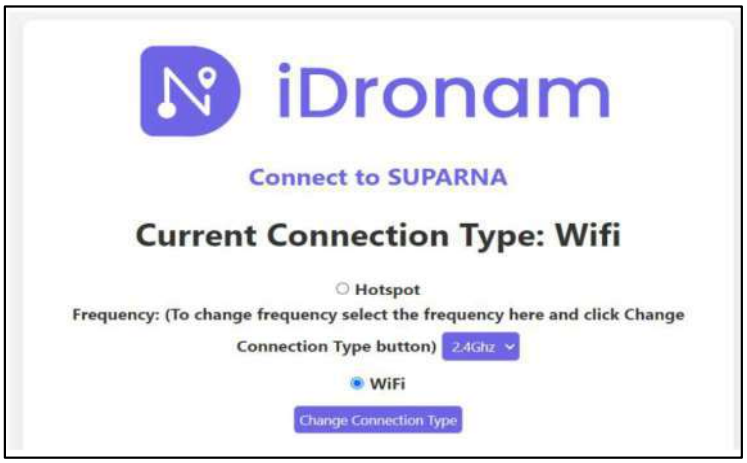
The drone has a single wifi module on board. Hence, at a time the drone can be in either hotspot mode or wifi mode. So, how to switch from one mode to another, this can be done by a webpage hosted in the drone itself.

**E. Network mode switch webpage**

This is a central webpage for Switching and Monitoring the Networks of the Drone.
Currently, Wifi and Hotspot modes can be configured from this page.
There are 3 sections to the webpage -

**i. Current Connection Type :**

It shows in what connection mode the drone currently is in. There are 2 radio buttons showing the connection modes, and the current mode is already selected. Along With these there is a frequency selection list. This is used when a drone is to be used in Hotspot mode. There are 2 options - 2.4 and 5Ghz.




**ii. WiFi / Hotspot Info :**

This section changes with respect to the drone’s network mode.

**a. WiFi Info -**

This displays the current SSID, Password(PSK), Frequency, IP Addresses and Hidden WiFi Status of the wifi to which the drone is connected.

**b. Hotspot Info -**

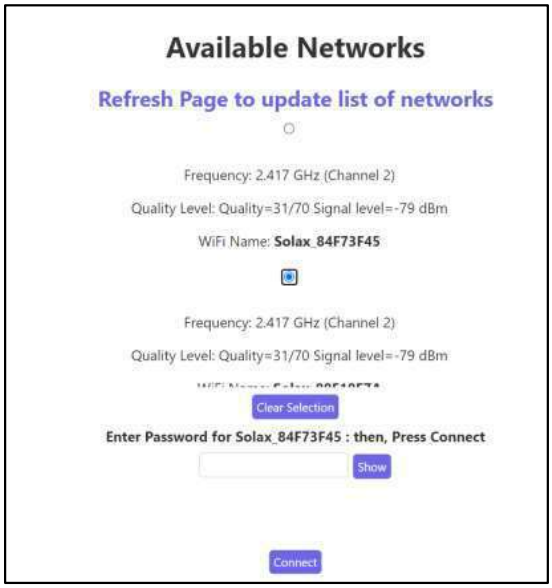
This displays the SSID, Password and Frequency of the drone's Hotspot.

**c. Add New WiFi :**

This section is used when you are switching to the WiFi mode for the first time. Also, it can be used when you want the drone to connect to hidden wifi.

**d. Available Networks :**

This section only shows in Wifi mode and is used to change the wifi network of the drone. Once you select the wifi network from the list, the password text will show up. Once entered click on connect to change drone WiFi.




## 4.18 Flight Controller

**1. Overview**

Suparna control stack runs on autopilot hardware to control drones, UAVs, and other unmanned vehicles. It offers robust capabilities for controlling a wide range of vehicles like multi-copters, fixed-wing, VTOLs, and ground vehicles. Here a quadcopter with autonomous capabilities is presented

**2. System Components**

The system is composed of several key components that work together to ensure smooth flight operations and management of the vehicle. These components include the Flight Stack, Middleware, Hardware Abstraction, and the Operating System.

**3. Flight Stack**

The Flight Stack includes the navigation, position estimation, and attitude controllers. It is responsible for ensuring the vehicle follows flight paths, maintains stability, and reaches its intended destination.

**4. Middleware PX4's**

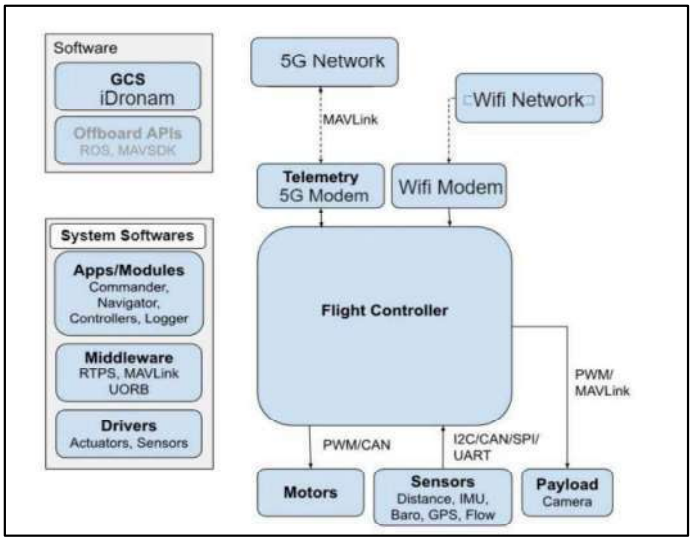
Middleware facilitates communication between different parts of the system, such as the flight control and sensors. It provides standard interfaces for vehicle components, making the system more modular and extensible.

**5. Hardware Abstraction Layer (HAL)**

The Hardware Abstraction Layer separates the operating system and hardware-specific implementations from the higher-level flight logic. This ensures that PX4 can run on various autopilot hardware with minimal changes to the software.

**6. Operating System**

The controller is designed to run on top of a real-time operating system, providing low-latency, predictable scheduling required for flight control applications.

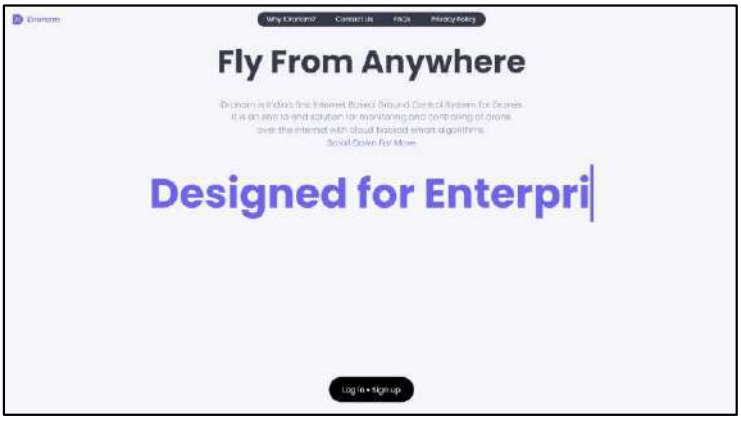



### 4.19 Software Installation

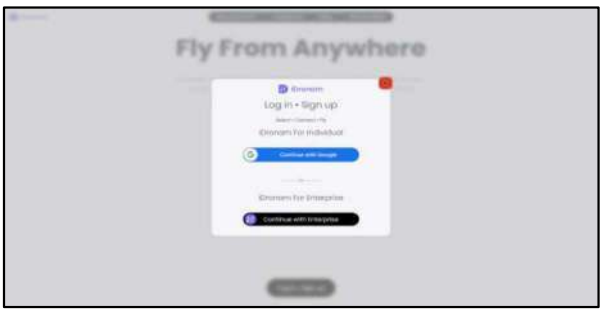
Organization Registration and Verification As drone operation is a crucial task and to know who is flying the Aircraft we have onboard the user with valid details. We are aggressively working to digitalise the process and improve system delivery.

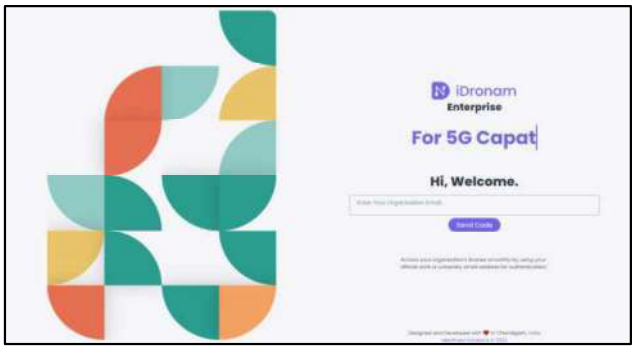
This section brief You about the Registration and Verification of Organisation :

- **Open iDronam**




**Click on "Login/Signup"**



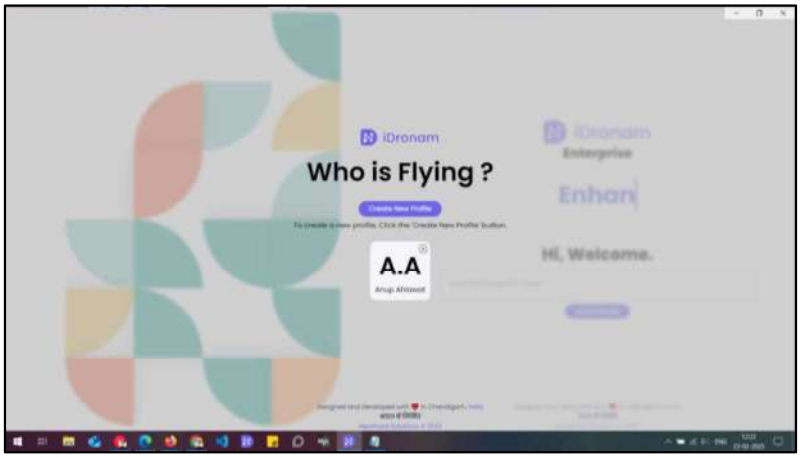



**Click on “iDronam for Enterprise” and enter your Enterprise mail.**

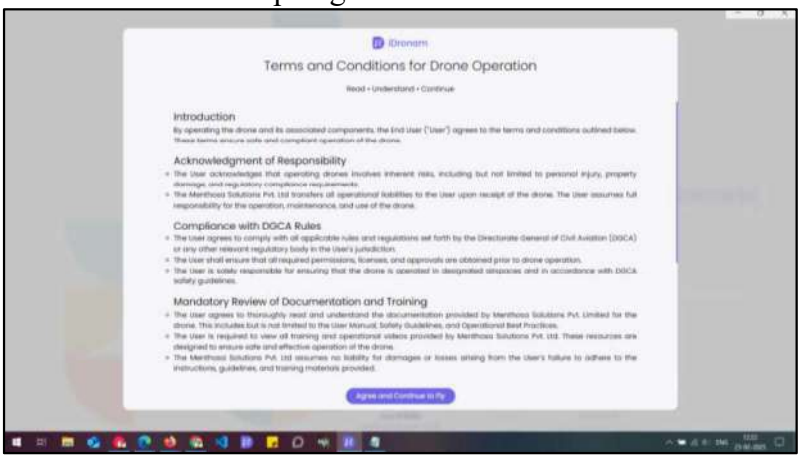
### 4.20.1 Software Setup

### A. Indoor Mode

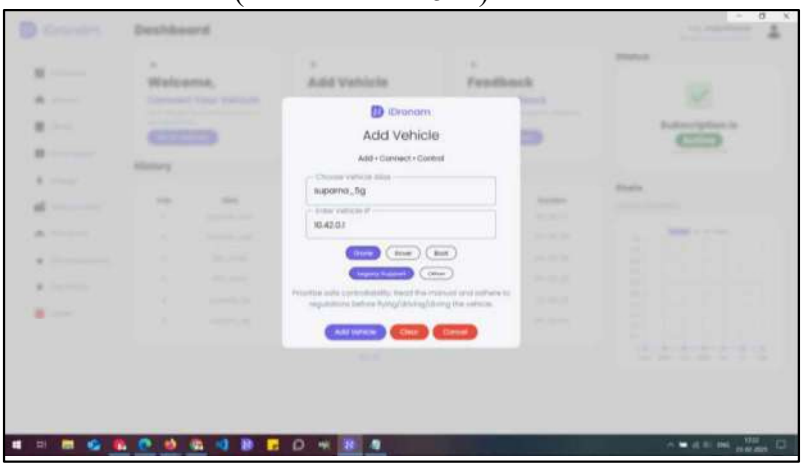
Select Profile




Accept Agreement



Add Drone (Over WiFi Or 5G ) and connect to it.




Click More Options and Set Environment to “Indoor”.

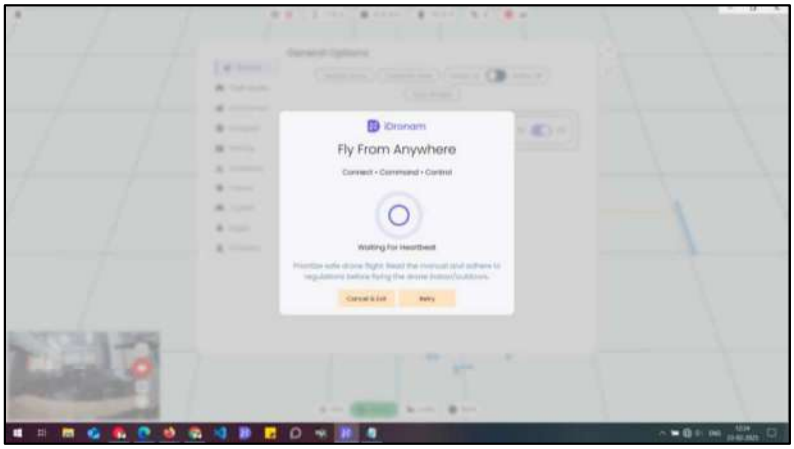
Disable “Geo Fencing”

Enable “Avoidance”

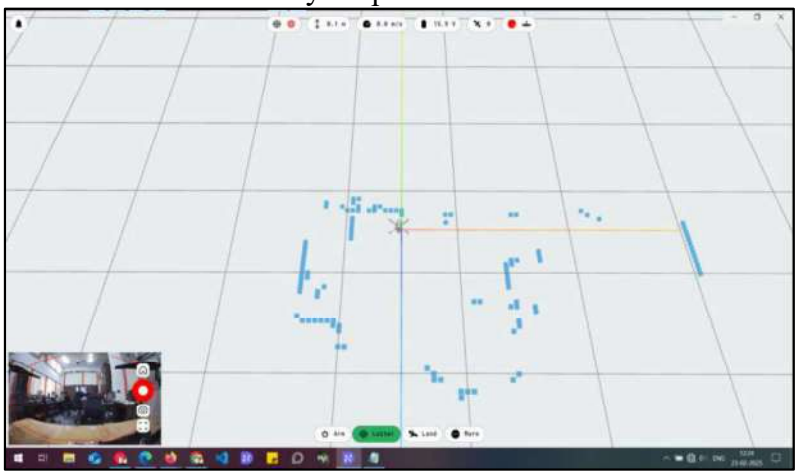
Configure Failsafe

! For Indoor Configure it to “Land”

Reboot The Drone and Retry




Verify Map and Obstacle

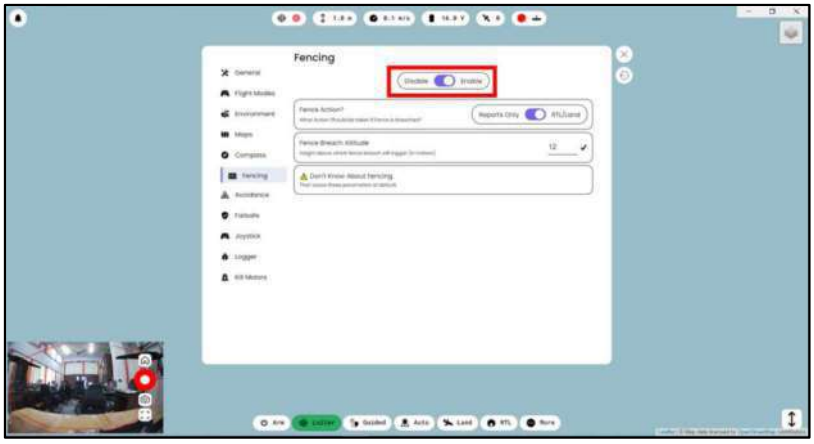


**B. Outdoor Mode-**

Outdoor flight requires some more parameters to set up before flight.
Follow the steps below before taking the flight to ensure safe outdoor flight.

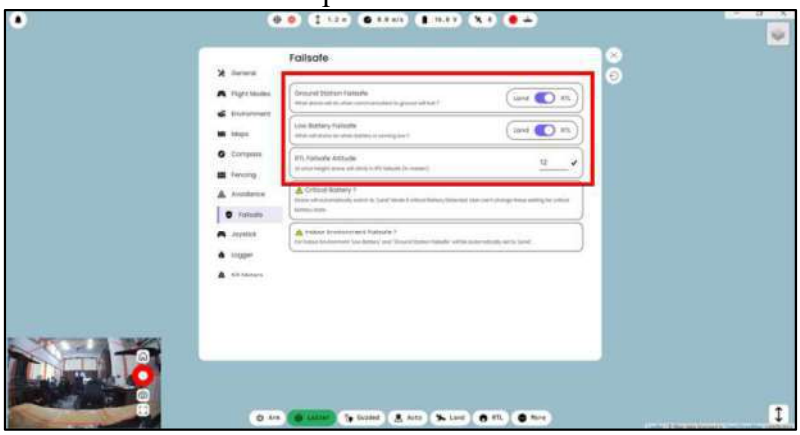
Switch to Outdoor Mode

Turn ON Geo Fencing



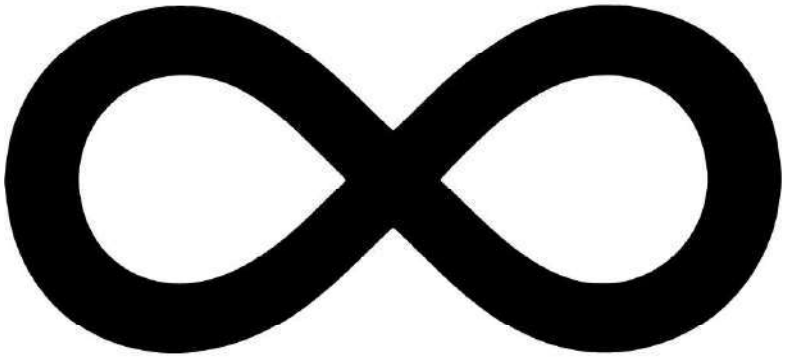

Turn OFF Obstacle Avoidance

Update Failsafe




RTL (Return to Launch) Mode:
RTL Altitude is the height the drone ascends to before returning to the launch site. It must be higher than any obstacles in the line of sight along the RTL path and lower than the Fence Breach Altitude.

**Calibrate Compass**



After pressing the “Calibrate Button”. Pick up the aircraft and move in the infinity shape and see calibration progress.

Compass calibration must be performed before flight or when changing location, as it is affected by nearby magnetic interference. During calibration, the progress percentage increases. Upon successful completion, the drone reboots. If calibration fails, the percentage resets to 0, requiring users to retry.

### 4.21 AI/ML Use Case-

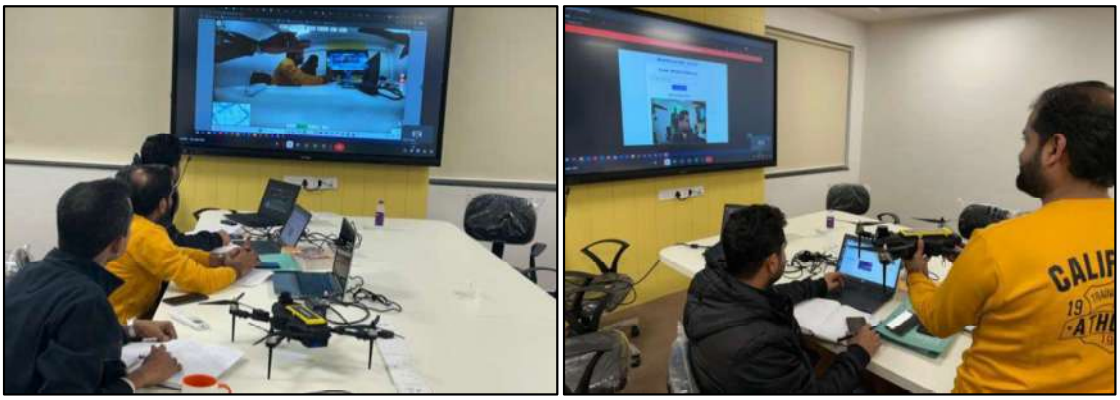


Fig- showing API of Drone Integrated with MEC VM

### 4.22 Sim Insertion And 5g Registration-

#### 4.22.1 SIM Insert

**A. SIM Slot:**

The drone would have a designated SIM card slot on its onboard communication system, typically integrated into its communication module.

- **Insert SIM**: The SIM card is inserted into the slot, similar to inserting a SIM card into a phone. Ensure that the SIM card is inserted correctly.
- **Power On the Drone**: Once the SIM card is securely inserted, power on the drone. The onboard system will attempt to connect to the mobile network.

**B. SIM REGISTRATION**

Registration will be carried out by the RAN (Radio Access Network) of the system, which will handle the process automatically to ensure seamless integration of devices or users into the network.

Fig- showing the slot space in 5G Drone

### 4.23 Installing The Drone Camera Applicatiom At Mec:

#### 4.23.1 Steps for installation -

1. 1. Download and unzip the files
2. 2. Install docker - https://docs.docker.com/engine/install/ubuntu/
3. 3. import the image file - sudo docker import dronemecserveraiml_v1.1.tar.gz dronemecserveraiml_v1.1:latest
4. 4. run the docker image via compose - sudo docker compose up -d
5. 5. view the webpage on - http://(mecip):5000
6. 6. for viewing drone stream enter link - rtsp://(drone ip):10000/drone_cam.

### 4.24 TROUBLESHOOTING

The drone is equipped with an automated troubleshooting system that continuously monitors its performance during operation. In the event of any malfunctions or errors, the system immediately detects the issue and alerts the user via the iDronam app. The app provides real-time notifications and detailed error messages, allowing the user to identify the nature of the problem. Until the issue is diagnosed and resolved, the drone will not be ready for flight. This built-in self-diagnosis ensures that the drone remains in optimal condition, minimizing the risk of failure during operation and ensuring safety.

### 4.25 Take off and General Flight precautions-

**After successful pre-arm checks, the drone is ready for take-off. There are 2 methods of arming the drone-**

- Using GCS software : The arm button in the control panel can be clicked for a 1 tap takeoff. Here the drone will take-off to an altitude of 1 meter and will switch to Guided Mode. The pilot can switch to loiter mode between the takeoff to control the drone.
- Manual : Pilot can manually take off the drone using the joystick-
- Switch to Loiter mode.
- Bring the left stick (throttle) of the controller to the bottom right position and hold it there.
- Drone will give an aural signal and all the propellers will start moving.
- Move the stick to extreme bottom position and slowly raise it to middle position and slightly nudge the stick to upward position.
- Drone will take off and hover in its position.
- For increasing altitude nudge the stick upwards. The drone will maintain its position vertically wherever the pilot leaves the stick.
- To decrease the altitude, nudge the stick downwards.
- The drone will hover in its position if no command is given.

**4.25.1 Before taking off there a few things that pilot must observe:**

1. When in an indoor environment, observe the 3D map data coming onto the GCS software central screen. If the map is skewed or is not in a regular shape of the drone's surroundings. Then, it means the drone has lost its local position and will show unintentional behavior. The drone must be restarted by removing and reinserting the battery(power cycle).
2. Another parameter is the Ground velocity of the drone, shown in the central dashboard of GCS software. The GCS will not arm the drone if it is greater than 0.3 m/s. But if somehow it is skipped. The pilot must always check this parameter if it shows any value above the threshold of 0.3 m/s.
3. Stop the takeoff sequence and use the Reboot drone button to reboot the flight controller of the drone. The reasons for ground velocity misbehaving are -

- Low Light conditions, the vision system does not have enough light to function optimally. Move to a better lit area.
- Optical Flow sensor blocked. There is a downward facing vision sensor under the belly of the drone. If it is blocked/hindered it might report wrong values.

1. Dynamic Obstacles around the drone during takeoff. The drone's obstacle avoidance starts once an altitude of 50 cm is detected; if any object breaches the 1.5m cut-off during takeoff the drone might not takeoff in a smooth manner. So to avoid this pilot must make sure no dynamic obstacles move around the drone before takeoff.
2. When in an outdoor environment, the pilot must observe for any GPS/Compass glitch that may happen during takeoff. If the position on map (central screen of GCS) shifts randomly that may indicate GPS/Compass glitch, immediately abort takeoff sequence. Reboot the flight controller via the Reboot Drone button.

### 4.25.2 Pre-Arm Checks-

After the preflight checks are performed and all the conditions are satisfied the pilot must move forward to pre-arm checks.

These are a list of flight critical conditions which are necessary to arm the drone. Failing any one of the conditions will result in a Pre-arm check failure which will be displayed by the GCS.

The checks are as follows-

**GPS (Outdoor) -** If the GPS is not able to get a good lock it will result in the GPSfailure, please move the drone to an open location.

**Compass -** The compass is affected by high Electromagnetic fields, if the drone is in a high EMF region it will cause compass failure. Move the drone to a low EMF region.

**Proximity -** The drone's obstacle avoidance system checks for any obstacle which is in 60cm of proximity. This check will fail if any obstacle is detected within the 60cm radius of the drone. Please remove the obstacles.

**GCS -** The Ground control station has a good C2 link with drones via the GCS software. If it fails, the pilot will lose all communication with the drone.Check the drone and GCS networks.

**Battery -** The voltage level is further checked by the drone. If the battery is less than 11.5 Volts this check will fail. Please charge the battery.

**Visual Odometry (Indoor) -** The drone uses a vision system to navigate in indoor mode or GPS-denied areas. If the vision system is unable to function properly this check will fail. Please refer to the full manual for the complete vision system operation range.

**Safety Switch -** If all of the above checks are passed, the final check is of the Safety switch. It is a safety mechanism to avoid unintentional arming of drones. For passing this check the pilot needs to press the safety off switch in the GCS software. After which an aural tone is heard and a Ready to Arm message will be displayed on GCS.

## 4.26 PRE FLIGHT CHECKS

### 4.26.1 Battery Checks

Drone is powered by a 4 cell (4S) 4500mah Li-ion battery.

It is recommended that the battery is fully charged up to 16.80 Volts, before each flight for optimal performance and charge only using the charger provided in the box.

If battery is not fully charged then,

❗ If the voltage is below 11.5 Volts, the drone will not arm because of Battery Failsafe.

🔘 If voltage is above maximum rating of 16.8 Volts, do not proceed any further, remove the battery and contact Menthosa Solutions as the battery may damaged. In any other case the drone will be able to fly.

**4.26.2 Structure Checks**

1. Drone has several moving parts which are held together by a rigid and lightweight carbon fiber chassis so before flight it is mandatory to see if any moving part is loose or not.
2. Motor mounts - The BLDC motors placed on these mounts are fastened using 4 Nuts. There are 4 of these on each drone.
3. It is to be ensured that the motors are not loose and the nuts are fastened.
4. The mounts do not have any cracks on them which may cause structural integrity problems.

**4.26.3 Landing Gears -**

Drone has 4 landing gears which give it a stable place to land, ground clearance to the payload and keeps it horizontally stable.

It is to be checked if there are cracks, chipping or any other kind of damage.

**4.26.4 Propellers - Drone use the propellers to fly**

1. Are properly fastened on the motors.
2. No chipping on the surface of the props.
3. Direction of the propellers is correct.

**Payload** - The drone has a 1 axis gimbal camera as its primary payload and optionally a 500g secondary payload can be mounted under the drone. Make sure all the payloads are mounted properly.

**Drone Cover** - The drone has a lightweight cover on its chassis which is fastened using 6 nuts. Make sure the cover is not damaged.

**4.27 Flight Area Checks**

The drone is equipped with a 360 degree obstacle avoidance and has a tolerance of 2m. At 1.5m distance from the obstacle drone will try to move away from the obstacle. Drone has GPS for outdoor operations, it has an accuracy of 2.5m.

There are 2 scenarios for this check -

1. Indoor : Drone requires at least 4m x 4m clearance for optimal indoor operation.
2. Outdoor : Drone requires open area where there are no buildings close by of heights ~25m and above as this will hinder GPS accuracy of the drone.

### 4.28 GCS Software, C2 Link and Controller Check-

The Ground Control Station (GCS) Software is the application which is used to command and control the drone. Before each flight, it should be updated to the latest version.

- C2 link is the command and control link between the GCS and the drone. Drone has 3 modes for communication - 5G, WiFi and Hotspot.
- **5G Mode** - Drone is using an on board 5G dongle to connect to the private network.
- **WiFi Mode** - Drone will be connected to the wifi, using the web interface .
- **Hotspot Mode** - Drone will be creating its own network and Pilot can connect to it.
- Make sure the GCS and drone are on the same network for C2 link to get established.
- A joystick controller is provided which can be used to fly the drone manually.
- It is to be connected before each flight to the GCS.

## 5. META QUEST XR/VR

### 5.1 Introduction

Oculus Quest 2 is most advanced all-in-one VR system

Developed by Meta Platform (facebook.inc)

Released on October 13, 2020

The standard Oculus Quest system consists of a VR headset (head-mounted display) and 2 controllers to be held in hands:

### 5.2 What Include In The Box

### 5.3 Key Features-

#### 5.3.1 Standalone VR System:

- **No PC Required**: Unlike many VR headsets, the Meta Quest 2 is completely standalone, meaning it doesn't require a PC or external sensors to operate.
- **Integrated Processor**: Powered by a Qualcomm Snapdragon XR2 processor, providing strong performance for VR applications.

#### 5.3.2 Display and Visuals:

- **Resolution**: The Quest 2 has a 1832 x 1920 resolution per eye, providing a sharp and immersive visual experience.
- **Refresh Rate**: The Meta Quest 2 supports up to 120Hz , offering smoother experiences during fast-moving VR applications.

#### 5.3.3 Tracking:

- **Inside-Out Tracking**: The Quest 2 uses built-in cameras to track your movement, meaning you don't need external tracking sensors. This makes it easy to set up and use in various environments.
- **Hand Tracking**: The device supports hand tracking, allowing you to interact with the virtual world without controllers, though it works best in certain applications.

### 5.3.4 Controllers:

- The Meta Quest 2 comes with two wireless controllers equipped with tracking sensors, enabling natural interaction with VR environments.

### 5.3.5 Software and Content:

- The Quest 2 supports a large variety of VR content, including games, productivity tools, educational apps, and VR training simulations. The content is available via the **Meta Quest Store**.
- It also supports Oculus Link, which allows you to connect the Quest 2 to a PC to access PC VR content through SteamVR and Oculus PC software.

### 5.3.6 Audio:

- Built-in speakers provide spatial audio, The sound is emitted from the side of the headset, delivering audio without requiring additional accessories.

### 5.3.7 Comfort and Design:

- The Meta Quest 2 is lighter and more compact than its predecessor, making it more comfortable for extended sessions.
- It comes with adjustable straps and a customizable fit for various head sizes.

### 5.3.8 Storage Options:

- The Meta Quest 2 offers 128GB Storage.

### 5.3.9 Battery Life:

- The battery life typically lasts around 2-3 hours, depending on the type of activity. Games or applications with more demanding graphics can consume battery faster.

### 5.3.10 Social Features:

- You can connect with friends, share content, and interact in virtual spaces via Meta's social platforms.
- The **Horizon Worlds** platform is a popular VR social experience available on the Quest 2, where users can create, interact, and play in virtual worlds.

## 5.4 WHAT IS META QUEST 2?

Meta Quest 2 is an excellent tool for various training applications because of its portability, ease of use, and immersive experience. Here are some potential uses:

- **Employee Training**: machinery operation, safety training, emergency protocols
- **Medical Training**: practice surgery or diagnosis without real-world consequences
- **Education**:enabling students to explore history, science, and art in new and engaging ways
- **Soft Skills Training**:practice soft skills like public speaking, customer service, or negotiation through role-playing scenarios
- **Virtual Classrooms**: you can use tools like Horizon Workrooms for remote learning and collaboration.

### 5.3.1 Employee Training:

**Meta Quest 2** can simulate a realistic working environment where employees can learn to operate complex machinery without the risk of causing damage to equipment or injuring themselves.

This Training can include:

**Simulated Environments:** Employees can interact with virtual machines, learning the controls, sequences, and operations.

**Hands-on Practice:** VR allows for tactile feedback, allowing workers to practice in a fully immersive setting that mirrors the real-world experience.

**Error Correction:** Instant feedback can be provided on mistakes or improper handling, helping employees correct their actions before working with actual equipment.

**Repetitive Practice:** Workers can repeatedly practice scenarios in a controlled, virtual environment until they master the skills, without using up valuable machine time.

### 5.3.2 Medical Training:

XR/VR in medical training offers immersive, hands-on experiences for students and professionals to practice critical skills safely and effectively:

**Surgical Training:**

Simulate various surgeries, from basic to complex, in a realistic virtual environment and gain real-time feedback on technique and accuracy, allowing for repeated practice and improvement.

**Diagnosis & Clinical Skills:**

Engage in virtual patient interactions, practicing medical history taking, physical exams, and diagnosis of different conditions.

**Anatomy Education:**

Visualize interactive models of human anatomy, offering deeper understanding of structures and systems. Perform virtual dissections to study organs and tissues without physical cadavers.

**Procedure Training:**

Practice common medical procedures and handle virtual complications in a risk-free setting.

**5.3.3 Education:**

XR/VR creates virtual spaces where students can learn by actively engaging with content. Rather than passively reading or listening, they can experience history, science, art, and more firsthand.

**Exploring Complex Concepts:**

- **Science:** Students can explore complex concepts like the human body, space, or chemical reactions in 3D, making abstract ideas easier to grasp.
- **History:** Travel through time and witness historical events or explore ancient civilizations, providing a deeper understanding of past cultures and significant moments.
- **Art:** Walk through virtual galleries, observe sculptures from all angles, and even create art in a 3D environment, offering a hands-on approach to learning about artistic techniques.

**Interactive Learning:**

XR/VR encourages active participation. Students don't just read about a topic—they explore it, experiment with it, and even make decisions in simulated environments that help reinforce learning.

**Personalized Learning:**

Virtual experiences can be tailored to individual learning paces and preferences, allowing students to revisit complex material, practice as much as needed, and engage with content in ways that best suit their needs.

**5.3.4 Soft Skills Training:**

Soft skills training through XR/VR offers immersive experiences to enhance key interpersonal and professional skills:

**Communication Skills:** Practice public speaking, active listening, and cross-cultural communication in virtual environments.

**Leadership & Management:** Simulate leadership roles, team management, and conflict resolution scenarios.

**Emotional Intelligence (EQ):** Develop empathy, self-awareness, and emotional regulation through realistic simulations.

**Teamwork & Collaboration:** Strengthen collaboration, problem-solving, and decision-making in virtual team settings.

**Negotiation & Persuasion:** Role-play negotiations and practice persuasion tactics in interactive scenarios

**5.3.5 Virtual Classrooms:**

Virtual classrooms using XR/VR provide interactive learning experiences where students and teachers interact in a 3D virtual space.

Key benefits include:

**Immersive Learning:** Students engage with lessons and simulations, making learning dynamic.

**Global Access:** Students can attend classes from anywhere, breaking geographical barriers.

**Interactive Features:** Virtual tools like whiteboards and 3D models enhance lessons.

**Real-time Collaboration:** Students collaborate on projects and participate in discussions through avatars.

**Personalized Learning:** Adaptive tools offer customized learning experiences for each student.

**5.4 AR (Augmented Reality), VR (Virtual Reality), and MR (Mixed Reality)**

**5.4.1 Augmented Reality (AR)**

AR technology overlays digital content (such as images, videos, or 3D models) on the real world, typically viewed through a screen, like a smartphone, tablet, or AR glasses. Unlike VR, AR does not replace the physical world; instead, it enhances it.

**Key Features**:

- **Overlay Digital Content**: AR adds virtual elements (graphics, sounds, etc.) to the real world.
- **Real-Time Interaction**: It interacts with the real world in real time, responding to the user's environment and movements.

**Common Uses**:

- **Mobile Apps**: Pokémon GO, IKEA Place (for furniture placement).
- **Retail**: Try-on features for clothes or makeup in stores.
- **Industrial Applications**: Repair manuals or assembly instructions overlaid on machinery.
- **Navigation**: Directions overlaid on a real-time map.

**Devices**: Smartphones, tablets, AR glasses (e.g., Microsoft HoloLens, Magic Leap), smart lenses.

### 5.4.2 Virtual Reality (VR)

VR is a fully immersive technology that replaces the real world with a completely virtual one. Users typically wear a headset that blocks out the real world and displays a digital environment, allowing them to interact with the virtual world.

**Key Features**:

- **Complete Immersion**: VR users are fully immersed in a virtual environment, often with 360-degree visuals and spatial sound.
- **User Interaction**: Users can interact with virtual objects using controllers, gloves, or body movements.

**Common Uses**:

- **Gaming**: Fully immersive games (e.g., Beat Saber, Half-Life: Alyx).
- **Training & Simulation**: For aviation, medicine, military, or other industries requiring hands-on training in a controlled, virtual space.
- **Virtual Tourism**: Explore virtual representations of places like museums, historic sites, or natural wonders.
- **Entertainment & Films**: VR movies or experiences where users feel like they're part of the story.

**Devices**: VR headsets (e.g., Oculus Rift, HTC Vive, PlayStation VR), haptic gloves, motion sensors.

### 5.4.3 Mixed Reality (MR)

MR combines elements of both AR and VR. It blends the virtual and real worlds in such a way that physical and digital objects coexist and interact in real time. It's often more immersive than AR but doesn't fully replace reality like VR.

**Key Features**:

- **Interaction Between Real and Virtual Worlds**: MR allows real-world and virtual elements to interact. For example, a digital object might be placed on a physical table and react as if it were really there.
- **Hybrid Environment**: MR creates a more integrated environment than AR, where users can interact with both real and digital objects.

**Common Uses**:

- **Industrial Applications**: In design, users can interact with CAD models and adjust them in real-time, while also working with physical parts.
- **Training**: MR can be used for realistic training scenarios where users engage with both physical tools and virtual training materials.
- **Entertainment**: MR can be used for immersive storytelling, where virtual characters can interact with real-world surroundings.
- **Education**: Learning experiences where students can engage with both digital and real-world elements (like historical reenactments in a classroom).

**Devices**: MR headsets (e.g., Microsoft HoloLens, Magic Leap), smartphones/tablets with MR apps.

### 5.5 XR Foundation Features for Design and Development

The **XR (Extended Reality) Foundation** plays a crucial role in creating and advancing immersive experiences, which combine **Virtual Reality (VR)**, **Augmented Reality (AR)**, and **Mixed Reality (MR)** into a unified framework. The XR Foundation supports a variety of features and technologies that are integral to designing and developing XR applications. These features are intended to ensure **compatibility, performance, interoperability**, and **scalability** across a wide range of devices and use cases.

To set up a development environment for the **XR/VR,** you'll need to install and configure several software tools to get started with VR development. Below is a step-by-step guide to help you get your Meta Quest 2 development environment up and running.

#### 5.5.1 Steps to Install Required Hardware and Software

**A. Hardware Requirements**

- **Meta Quest 2** headset.
- **USB-C Cable** (to connect your Meta Quest 2 to your PC).
- A computer with **Windows 10/11** (recommended) or macOS for development.
- **Controllers**: Meta Quest 2 controllers for interaction with virtual environments.

**B. Software Requirements**

- **Unity3D** or **Unreal Engine** for creating your VR applications.
- **Oculus App** on your PC (for managing the Meta Quest 2).
- **Oculus Integration for Unity** (if using Unity).
- **Android Studio** (for building and deploying the app to Meta Quest 2).

### 5.6 Install and Build application usage

There are numerous applications available in the Meta Quest 2 app store, catering to a wide variety of interests and skills. To get started, the first app you should explore is called **First Steps**.

This app is specifically designed for new users and offers a guided experience to help you become familiar with the VR environment. Through interactive animations and clear instructions, **First Steps** teaches you how to properly control the virtual reality interface and operate the controllers effectively. This will give you the foundational knowledge needed to navigate the virtual world safely and confidently.

Once you're comfortable with the basic controls, you'll discover a wide range of applications in the Meta Quest 2 store that focus on skill development across different areas. These include fitness apps, language learning programs, art and design tools, meditation guides, and many others. Each app provides an immersive experience that helps you develop new abilities, improve cognitive skills, or even learn a new language, all within a virtual environment.

1. **Language Learning**: Mondly VR, Engage (language courses).
2. **Cognitive Skills**: Peak, Oculus Quest Mind Lab.
3. **Public Speaking**: VirtualSpeech, Speech Center VR.
4. **Fitness**: FitXR, OhShape, BoxVR.
5. **Music**: Virtuoso (piano), Incredibox (beats).
6. **Art & Design**: Tilt Brush, Kingspray Graffiti, SculptrVR.
7. **Business/Productivity**: Spatial, Immersed.
8. **Coding**: VIRTUAL CODE, Cospaces Edu.
9. **Design & Architecture**: ShapeLab VR, SketchUp Viewer.
10. **Meditation**: Tripp, Guided Meditation VR.
11. **Physics & Engineering**: Physics Playground, The Lab.
12. **Cooking**: **Kitchen VR**, **Cook-Out:** A Sandwich Tale.
13. **Cultural Learning**: Wander, National Geographic VR

### 5.7 TROUBLESHOOTING

#### 5.7.1 VR Headset Not Turning On

- **Check Battery**: Ensure the headset is charged. Plug it into the charger and let it charge for at least 30 minutes.
- **Hard Reset**: Press and hold the power button for about 10 seconds to force a restart.
- **Try Different Charger**: If charging doesn't work, try a different USB-C charger or cable.

### 5.7.2 No Display / Black Screen

- **Adjust the Headset Fit**: Make sure the lenses are properly aligned with your eyes. Adjust the head straps for a comfortable fit.
- **Check the Power**: If the headset is on but the display remains black, try restarting it by holding down the power button for 10 seconds.
- **Reboot**: Try to reboot the device by pressing and holding the power button for 10 seconds.

### 5.7.3 Controllers Not Working

- **Battery Check**: Ensure the controllers have working batteries (replace them if necessary).
- **Re-pair the Controllers**: Go to the **Meta Quest 2 settings** > **Devices** > **Controllers** > **Pair New Controller**.
- **Restart the Headset**: Turn off the headset and restart it, then try pairing the controllers again.

### 5.7.4 Tracking Issues

- **Clear the Play Area**: Ensure your play area is well-lit, and free from obstructions. The cameras on the headset need a clear view of the environment for proper tracking.
- **Reset the Guardian System**: Go to **Settings** > **Guardian** and select **Reset Guardian** to redefine your play area.
- **Clean the Sensors**: Check if the sensors or cameras on the headset are clean. Use a soft, lint-free cloth to gently clean them.

### 5.7.5 Wi-Fi Connectivity Issues

- **Check Your Network**: Ensure that your Wi-Fi network is working properly and that the headset is within range of the router.
- **Forget and Reconnect**: Go to **Settings** > **Wi-Fi**, forget the network, and reconnect by entering your password again.
- **Restart the Router**: If the Wi-Fi connection is unstable, restarting the router can help.

### 5.7.6 App Not Launching or Freezing

- **Restart the Headset**: Hold down the power button for 10 seconds and then turn the headset back on.
- **Check for Updates**: Go to **Settings** > **About** > **Software Updates** and ensure your software is up to date.
- **Reinstall the App**: If an app is constantly freezing, try uninstalling it and reinstalling it from the **Meta Quest Store**.

### 5.7.7 Audio Problems (No Sound or Low Sound)

- **Volume Controls**: Make sure the volume is turned up using the volume buttons on the side of the headset.
- **Reboot the Headset**: If sound isn't working, reboot the headset to reset the audio system.
- **Check Headphones**: If using external headphones, ensure they're properly connected and the audio is set to output to them.
- **Adjust the Fit**: Make sure the earphones or speakers are positioned correctly.

### 5.7.8 Performance Lag or Low Frame Rate

- **Close Unnecessary Apps**: If the performance is slow, close background apps that may be using resources.
- **Reboot the Device**: Restart the headset to clear any background processes that might be slowing it down.
- **Check for Updates**: Go to **Settings** > **About** > **Software Updates** to ensure your headset is running the latest firmware.
- **Reduce Graphics Settings**: In certain games, reducing the graphics settings might help improve performance.

### 5.7.9 Overheating

- **Allow for Cooling**: If the device is getting too hot, take breaks and allow the headset to cool down.
- **Use in a Cooler Area**: Ensure the environment you're using the headset in isn't too hot. Avoid using it in direct sunlight or in hot rooms.

### 5.7.10 Pairing with the Meta Quest App

- **Ensure Bluetooth is On**: Make sure Bluetooth is enabled on your phone for the pairing process.
- **Re-pair the Device**: In the **Meta Quest App**, go to **Devices** > **Pair New Headset**, and follow the instructions.
- **Restart Your Phone**: If pairing issues persist, try restarting your phone and the headset, then attempt pairing again.

### 5.7.11 Factory Reset

- **Warning**: A factory reset will erase all your data and settings.

**To Perform a Factory Reset**:

1. Turn off your Meta Quest 2.
2. Hold the **volume down** button and the **power** button until the boot menu appears.
3. Use the volume buttons to select **Factory Reset** and confirm with the power button.
4. Wait for the reset to complete.

### 5.8 Benefits of Using XR/VR :

**Cost-effective:** Reduces the need for physical equipment and on-site trainers.

**Safe Environment:** Allows for risky situations to be simulated safely.

**Engagement:** Immersive learning keeps employees engaged, enhancing retention and reducing knowledge gaps.

**Scalability:** Can be rolled out to employees in different locations without the need for additional physical infrastructure.

**Customizable:** Training modules can be tailored to specific machinery, safety protocols, or emergencies that employees need to familiarize themselves with.

### 5.9 WHAT YOU HAVE TO DO ?

**Research :** Understand how immersive environments improve learning and design experiences that engage different learning styles.

**Develop Software:** Create interactive VR/XR platforms, 3D models, and simulations for subjects like science, history, and art.

**User Testing:** Prototype, test, and refine VR learning experiences with real students and educators to gather feedback.

**Collaborate:** Work with educators and researchers to ensure your VR/XR solutions align with curriculum and educational goals.

**Explore New Technologies:** Integrate AI, gesture recognition, and haptic feedback to enhance interactivity and personalization.

**Ensure Accessibility:** Focus on making experiences inclusive, with attention to disabilities and device accessibility.

**Address Ethics & Privacy:** Prioritize student data privacy and ethical considerations in immersive environments.

## 6. EVALUATION BOARD

### 6.1 Overview

5G EVB can be used to develop applications based on Quectel 5G modules.
This EVB can be used to test basic functionalities of these modules.

## 6.2 Specifications

| | |
|---|---|
| **Temperature Range** | |
| 1. Operating Temperature | -30 °C to +75 °C |
| 2. Extended Temperature | -40 °C to +85 °C |
| **Audio** | |
| **Voice** | Digital Audio and VoNR/ VoLTE (Optional) |
| **Frequency Bands** | |
| **5G NR** | 3GPP Rel-16 NSA/SA operation, Sub-6 GHz |
| **5G NR NSA** | n1/ 3/ 5/ 7/ 8/ 20/ 28/ 38/ 40/ 41/ 71/ 75/ 76/ 77 / 78 |
| **5G NR SA** | n1/ 3/ 5/ 7/ 8/ 20/ 28/ 38/ 40/ 41/ 71/ 75/ 76/ 77 / 78 |
| **DL 4 × 4 MIMO(5G)** | n1/ 3/ 7/ 38/ 40/ 41/ 71/ 77/ 78 |
| **LTE Category** | DL Cat 20/ UL Cat 18 |
| **LTE-FDD** | B1/ 3/ 5/ 7/ 8/ 20/ 28/ 32 /71 |
| **LTE-TDD** | B38/ 40/ 41/ 42/ 43 |
| **DL 4 × 4 MIMO (LTE)** | B1/ 3/ 7/ 38/ 40/ 41/ 42/ 43 |
| **WCDMA** | B1/ 5/ 8 |
| **GNSS** | GPS/ GLONASS/ BDS/ Galileo/ QZSS |
| **Processor** | Qualcomm X65 |
| **Data Rates (Max.)** | |
| **5G SA Sub-6** | 4.0 Gbps (DL)/ 900 Mbps (UL) |
| **5G NSA Sub-6** | 4.0 Gbps (DL)/ 550 Mbps (UL) |
| **LTE** | 2.0 Gbps (DL)/ 200 Mbps (UL) |
| **WCDMA** | 42 Mbps (DL)/ 5.76 Mbps (UL) |
| **Interfaces** | |
| **(U)SIM** | × 2 |
| **UART** | × 3 |
| **SDIO** | × 1 |
| **USB 2.0/ 3.0/ 3.1** | × 1 |
| **PCIe 3.0** | Gen3, Lane × 2 |
| **PCM** | × 1 |
| **SPI** | × 1 |
| **Antennas** | Cellular: × 4; GNSS: × 1 |
| **eSIM** | Optional |
| **Electrical Features** | |
| **Supply Voltage Range** | 3.3–4.4 V, typ. 3.8 V |
| **Transmitting Power** | |
| | WCDMA: Class 3 ( 23 dBm ±2 dB)<br>LTE- FDD: Class 3 (23 dBm ±2 dB)<br>LTE- TDD: Class 3 (23 dBm ±2 dB)<br>5G NR: Class 3 (23 dBm ±2 dB)<br>5G NR n38/n41/n77/n78: Class 2 ( 26 dBm ±2 dB) |
| **Modulations** | |
| **5G** | Supports UL 256 QAM and DL 256 QAM |
| **LTE** | UL & DL QPSK, 16 QAM, 64 QAM, 256 QAM |

| Wi-Fi 6 | FC60E |
|---|---|
| Dimensions | 25.5 mm × 22.0 mm × 2.25 mm |
| Weight (g) | Approx. 2.58 g |
| WLAN Protocol | IEEE 802.11a/b/g/n/ac/ax |
| Wi-Fi Frequency Band | 2.4 GHz/ 5 GHz |
| Wi-Fi Antenna | 2 |
| Wi-Fi Modulation Mode | DBPSK/DQPSK/CCK/BPSK/QPSK/QAM |
| 2.4 GHz Channel Bandwidth | 20 MHz/ 40 MHz |
| 5 GHz Channel Bandwidth | 20 MHz/ 40 MHz/ 80 MHz |
| Encryption Mode | WPA3 |
| Temperature Range | |
| Operating Temperature Range | -30 °C to +75 °C |
| Data Rate (Max.) | |
| 802.11a | 54 Mbps |
| 802.11b | 11 Mbps |
| 802.11g | 54 Mbps |
| 802.11n | 600 Mbps |
| 802.11ac | 866 Mbps |
| 802.11ax | 1774.5 Mbps |
| Interfaces | |
| PCIe | × 1 |
| PCM | × 1 |
| UART | × 1 |
| Antenna Interfaces | 12 |
| Electrical Features | |
| I/O Power Supply Voltage | 1.8 V |
| Power Supply Voltage | VDD_CORE_VL: 0.95 V<br>VDD_CORE_VM: 1.35 V |

### 6.3 Interface Application

This chapter describes the hardware interfaces of the 5G EVB, as listed below:

- Power supply
- Module TE-A interface
- PHY TE-A interface
- USB interface
- Audio interfaces
- Digital Audio Codec Board Connector
- Analog Audio Interfaces
- Earphone Interface
- (U)SIM interfaces
- UART interfaces
- SD card interface
- PCIe to USB interface
- Switches and buttons
- Status indicators
- Wi-Fi interface

- Antenna Interfaces
- It also provides information about the buttons, switches, status indication LEDs and test points to help
- developers use the EVB.

### 6.3.1 Power Supply

The 5G EVB can be powered by an external power adapter through the power jack (J0303).

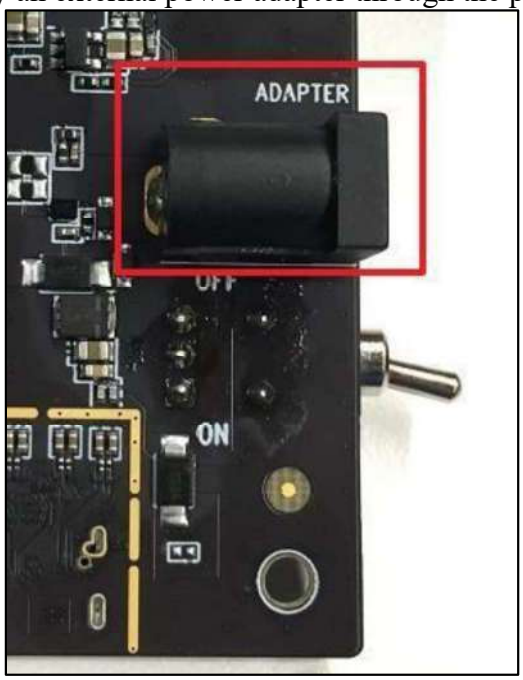



**Figure : EVB Power Supply Interface**

### 6.3.2 Module TE-A Interface (J0101/J0102)

Module TE-A interface is designed to accommodate the TE-A of the modules. The TE-A is mounted onto and connected to the EVB via BTB connectors J0101 and J0102. The developer will be able to test the functionalities of the modules easily (insert as indicated by the arrow to prevent reverse insertion).

The following figure displays the connection between the module TE-A and the EVB.

**Figure : Connection Between the Module TE-A and the EVB**

### 6.3.3 USB Interface (J1101)

A USB Type C connector, which complies with USB 3.0/3.1 and USB 2.0 standard, is provided. This USB interface is used for AT command communication, data transmission and firmware upgrade.

**Figure : USB Interface Connection**

### 6.3.4 Audio Interface (J0802/J0901/J0801)

EVB provides one digital audio codec board interface (I2S) J0802 and two analog audio interfaces J0901 and J0801.

### 6.3.5 Digital Audio Codec Board Connector (J0802)

The 5G EVB supports two different kinds of external digital audio codec TE-As named ALC5616 and TLV320AIC3104. The codec circuit is assembled on an independent small board which can be interconnected with the EVB by the BTB connector J0802.

Codecs can be selected according to specific application demands, the following figures show the connection between digital audio codec TE-A and the EVB.

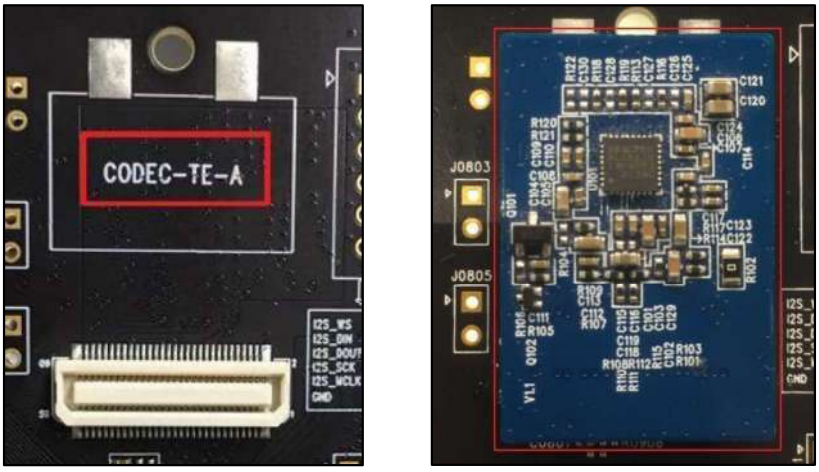



**Figure : Connection Between Codec TE-A and the EVB**

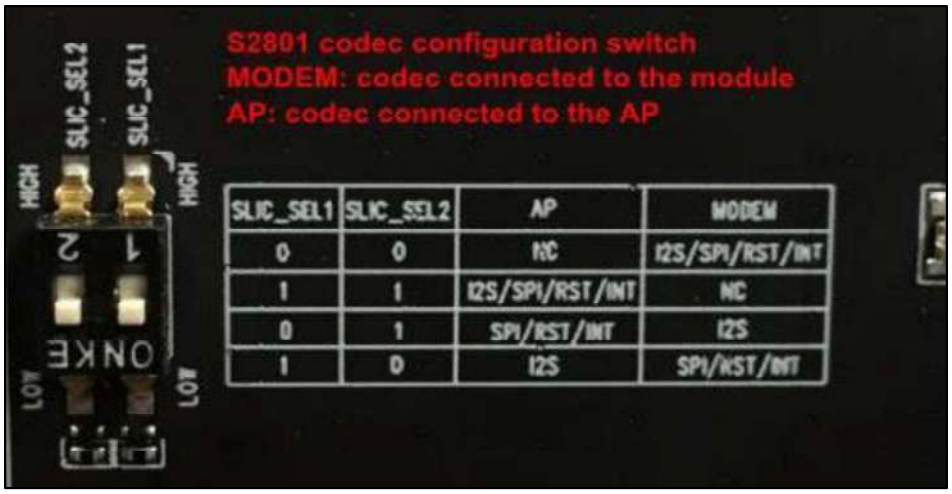



**Figure : S2801 Switch**

The figure and table below illustrates the pin assignment and pin definition of earphone connector J0901.

**Figure : Pin Assignment of J0901**

### 6.3.6 Table : Pin Definition of J0901

| PIN NO. | PIN NAME | DESCRIPTION |
|---|---|---|
| **1** | MIC | Microphone input |
| **2** | AGND | Dedicated GND for audio |
| **3** | SPK R | Right channel of stereo audio output |
| **4** | SPK L | Left channel of stereo audio output |
| **5,6** | NC | Not Connected |

The following figure shows a schematic of an audio plug which suits the audio jack on 5G EVB.

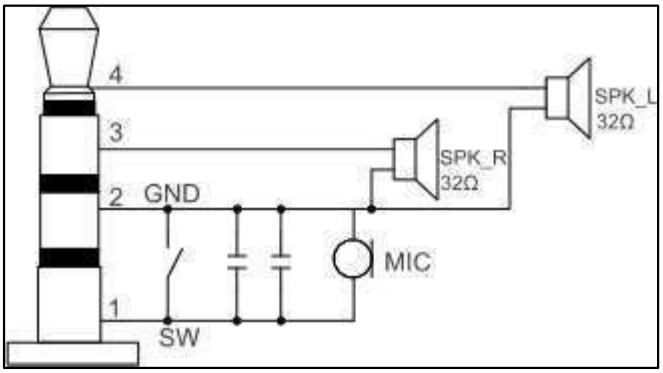



**Figure : Schematic of Audio Plug**

### 6.3.7 (U)SIM Card Interfaces (J1401/J1402)

The 5G EVB has two 8-pin push-push type (U)SIM card (1.8/2.95 V) connectors J1401 and J1402. The following figure shows a simplified connector schematic for J1401 and J1402.

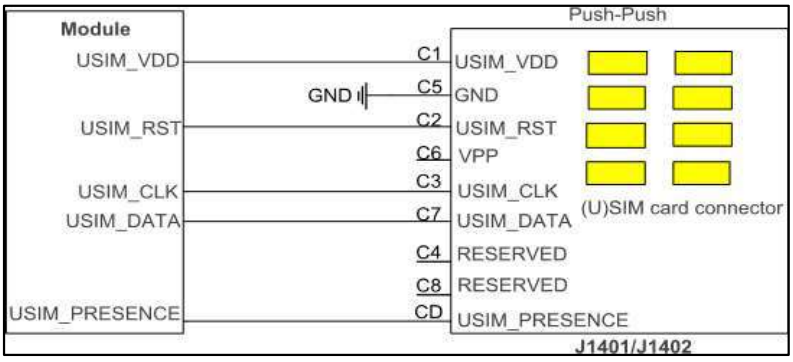



Figure: Simplified Connector Schematic for U(SIM) Card Connector

The figure and table below illustrate the pin assignment and definition of (U)SIM card connector J1401. J1402 is similar to J1401.

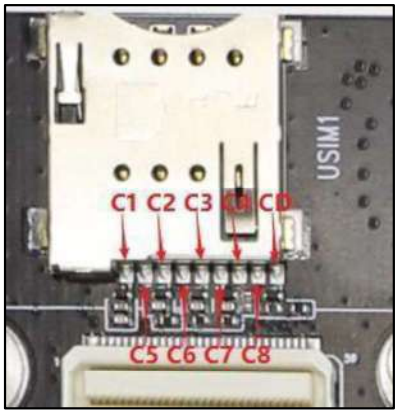



**Figure : Pin Assignment of (U)SIM Card Connector J1401**

**6.3.8 Table : Pin Definition of J1401**

| | | | |
|---|---|---|---|
| **C1** | **USIM VDD** | **PO** | **U(SIM) card power supply , provided by module** |
| **C2** | **USIM RST** | **DO** | **U(SIM) card reset** |
| **C3** | **USIM CLK** | **DO** | **U(SIM) card clock** |
| **C4** | **RESERVED** | **-** | **Not Connected** |
| **C5** | **GND** | **-** | **Ground** |
| **C6** | **VPP** | **-** | **Not Connected** |
| **C7** | **USIM DATA** | **I/O** | **Data line, bi directional** |
| **C8** | **RESERVED** | **-** | **Not Connected** |
| **CD** | **USIM PRESENCE** | **DI** | **U(SIM) card insertion detected** |

### 6.3.9 D Card Interface (J1301)

The 5G EVB provides an SDIO interface, which can be used for connecting SD card. The following figure shows the simplified interface schematic for J1301.

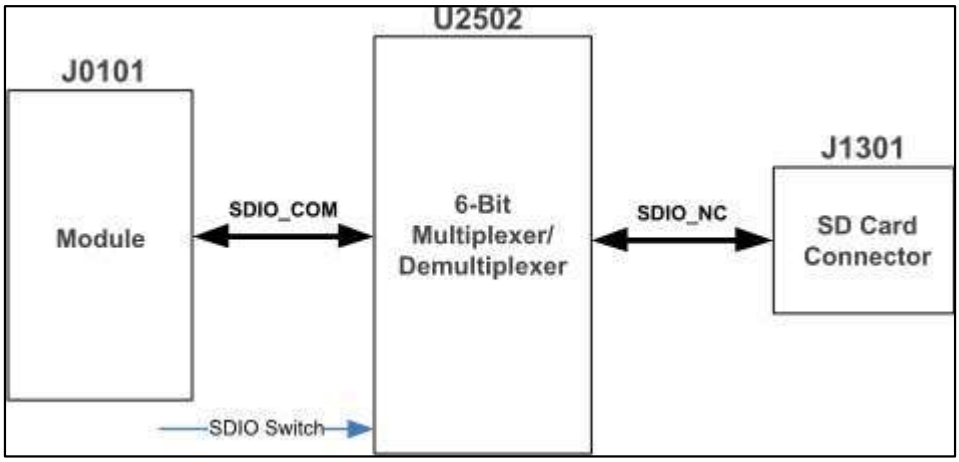



**Figure : Simplified Interface Schematic for J1301**

If SD card function is intended to be used, please switch the SDIO Switch to low level illustrated in the figure and table below, a standard SD card can be inserted into J1301. Which supports micro SD card of maximal 32 GB. With the SD card interface, customers can easily enhance the memory capacity of modules.

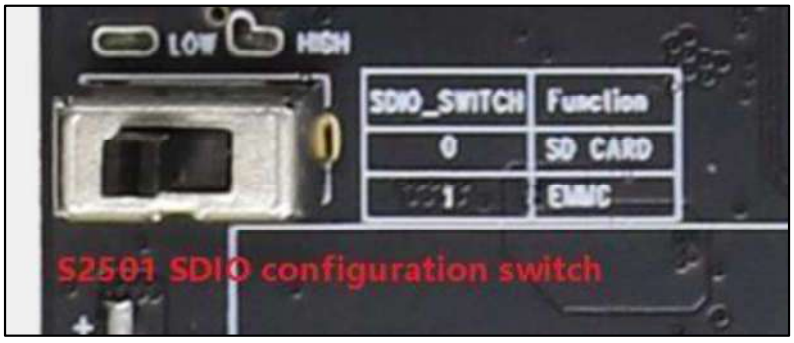



**Figure : S2501 Switch**

### 6.3.10 Table: SDIO Switch Function

| SDIO SWITCH | FUNCTION |
|---|---|
| Low | Enable SD card Function |
| High | Enable emmc Function |

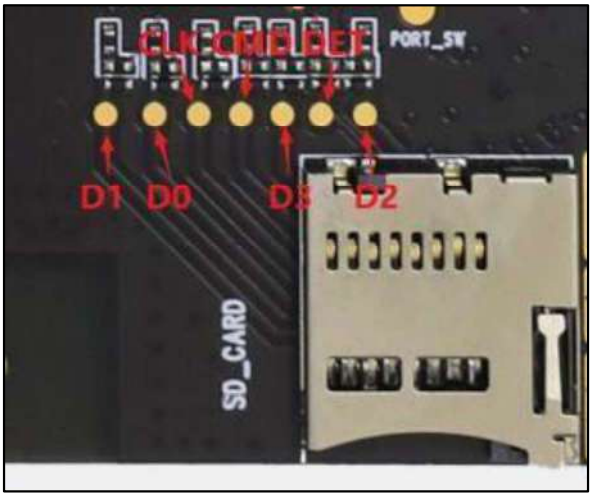



**Figure : Pin Assignment of SD Card Connector J1301**

### 6.4 UART Interfaces (J2002/J2003)

The 5G EVB supports two UART interfaces: main UART J2002 and debug UART J2003, supporting baud rate of 115200 bps by default.

The main UART interface is used for communication between the module and the host application. The debug UART interface is used for Linux console and log output.

The following figure shows a block diagram of UART interfaces of the EVB.

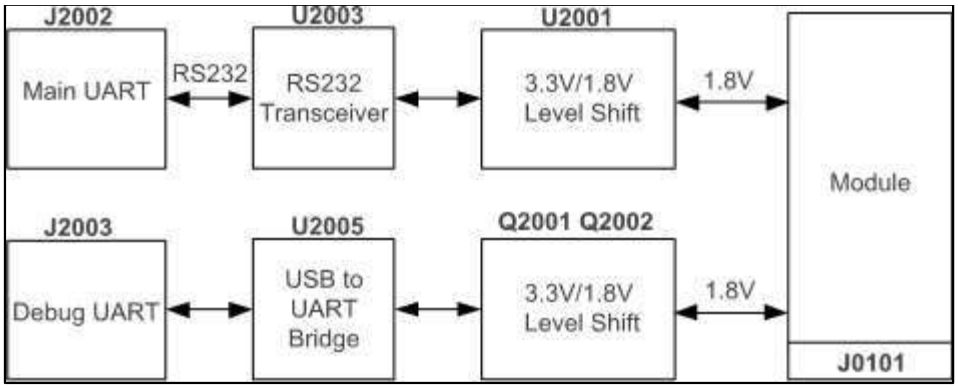



**Figure : UART Block Diagram**

#### 6.4.1 PCIe to USB Interface (J1601)

The 5G EVB reserves a PCIe 3.0 signal over USB interface for developers' testing, and this function is not enabled by default. Please refer to the following block diagram.

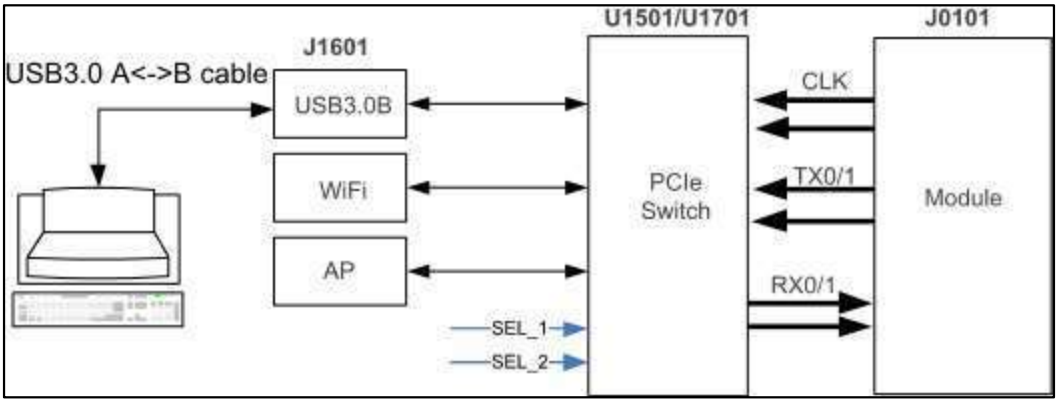



**Figure : PCIe Block Diagram**

**Figure : S1501 Switch**

### 6.4.2 Table : PCIe Connection Truth Table

| PCIE SEL1 | PCIE SEL2 | FUNCTION |
|---|---|---|
| 0 | 1 | MODULE->PC |
| 1 | 0 | MODULE->AP |
| 1 | 1 | MODULE->WIFI |

### 6.4.3 Switches and Buttons

The 5G EVB includes six switches (S0301, S1501, S1801, S1802, S2501 and S2801) and three buttons (S0201, S0202 and S0203), as shown in the following figures.

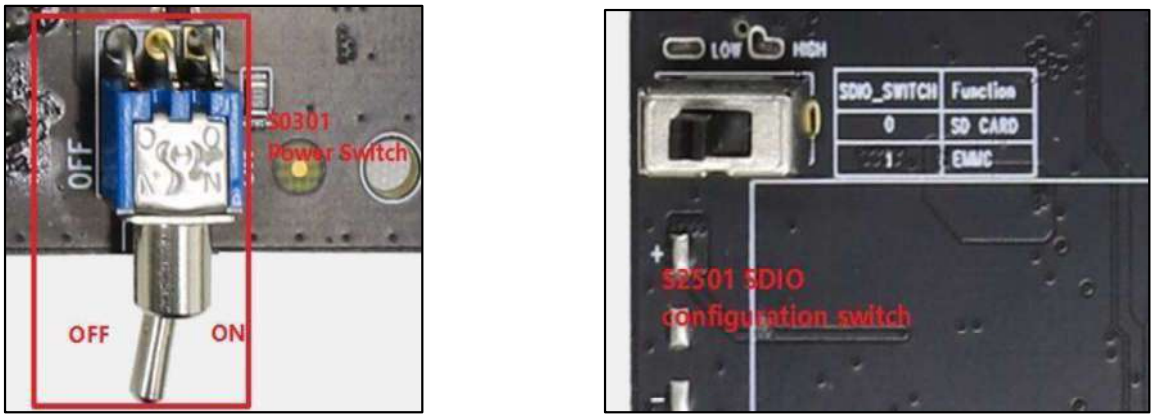



**Figure : Power Switch and S2501 Switch**

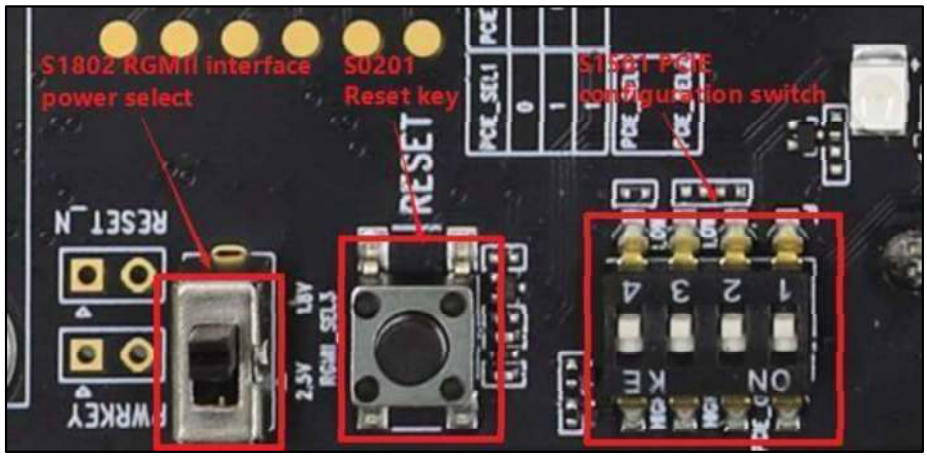



**Figure : S1802/S1501 Switch and S0201 Button**

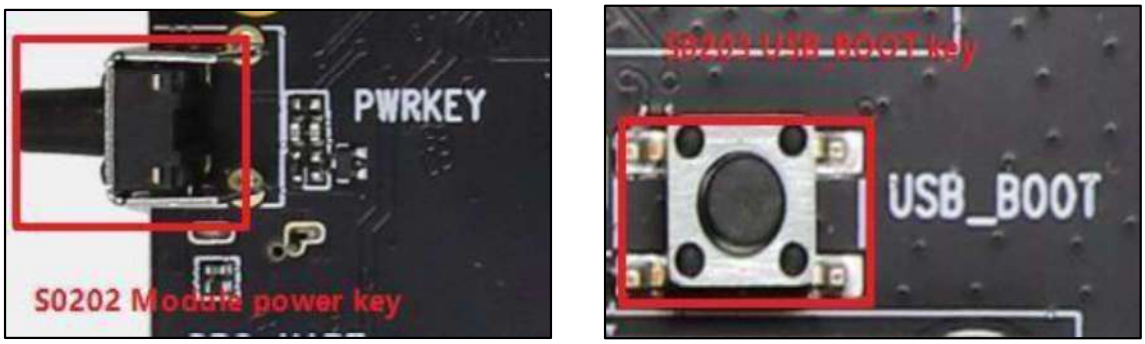



**Figure : S0202/S0203 Button**

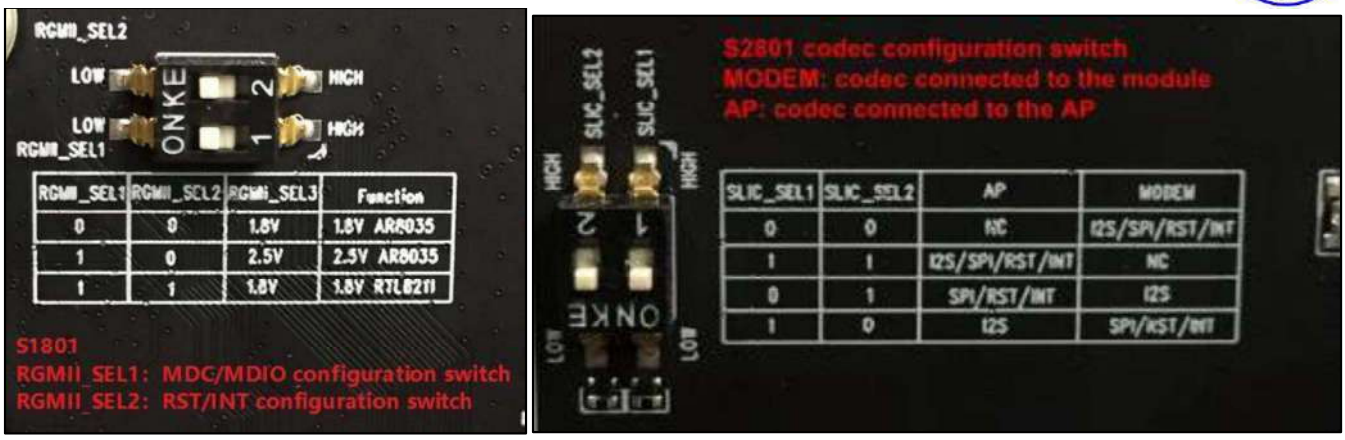



**Figure : S1801/S2801 Switch**

### 6.4.4 Table : Description of Switches and Buttons

| Interface | Reference Designator | Description |
|---|---|---|
| **Power Switch** | S0301 | VBAT ON/OFF control |
| **PWRKEY** | S0202 | Power key used to turn on/off the module |
| **PCIe Configuration** | S1501 | PCIe configuration switch. Refer to Chapter 3.9 |
| **RESET** | S0201 | Reset button used to reset the module |
| **USB_BOOT** | S0203 | Emergency download control |
| **RHMII Configuration** | S1801, S1802 | RGMII configuration switch Refer to Chapter 3.3 |
| **SDIO Configuration** | S2501 | SDIO configuration switch Refer to Chapter 3.7 |
| **Codec Configuration** | S2801 | Codec configuration switch Refer to Chapter 3.5 |

### 6.4.5 Status Indicators (D0201/D0202/D0203/D0204/D0205)

There are five status indication LEDs on the EVB. The following figure manifests the positions of these LED indicators.

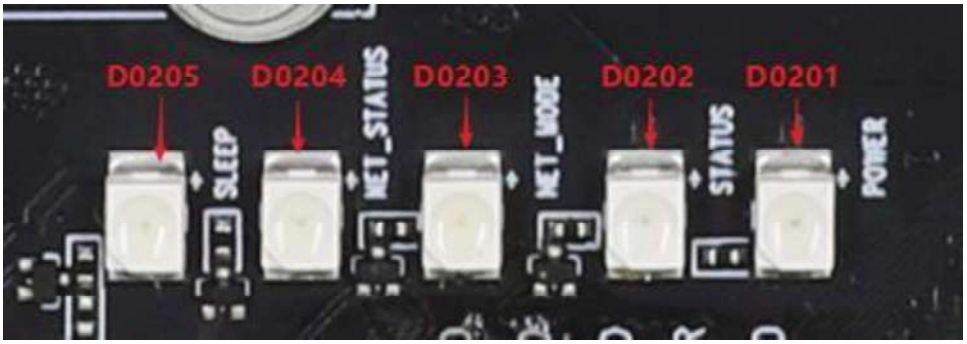



**Figure : Status Indicator**

**Table : Description of Status Indication LEDs**

| Reference | Description |
|---|---|
| D0201 | Indicates whether the power supply for module is ready<br>On: VBAT ON<br>Off: VBAT OFF |
| D0202 | Indicates the operation of the module.<br>On: the module is power on<br>Off: the module is power off |
| D0203 | Indicates the module's NET_MODE status. |
| D0204 | Indicates the module's NET_STATUS status. |
| D0205 | Indicates the module's SLEEP status. |

### 6.4.6 Wi-Fi Interfaces (J0701/J0702)

The Wi-Fi TE-A interface is designed to accommodate the TE-A of Wi-Fi modules (paired with FG50V). The TE-A is connected to the EVB via BTB connectors J0701 and J0702. The interface allows customers to test the Wi-Fi function of the module or to develop applications with Wi-Fi function easily.

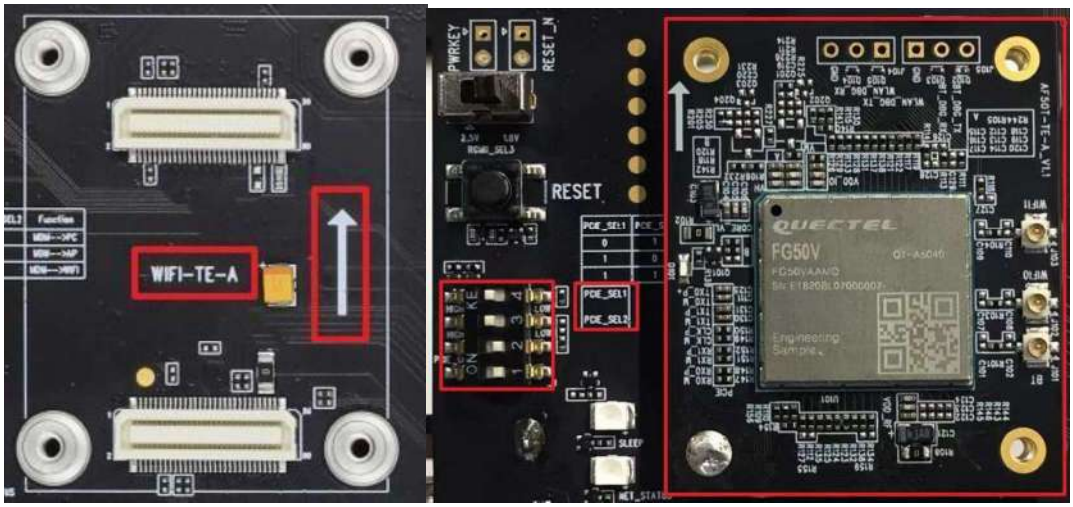



**Figure : Connection Between FG50V-TE-A and the EVB**

### 6.4.7 Antenna Interfaces

The 5G EVB includes twelve antenna interfaces. The following figure shows the assembly of these antenna interfaces.

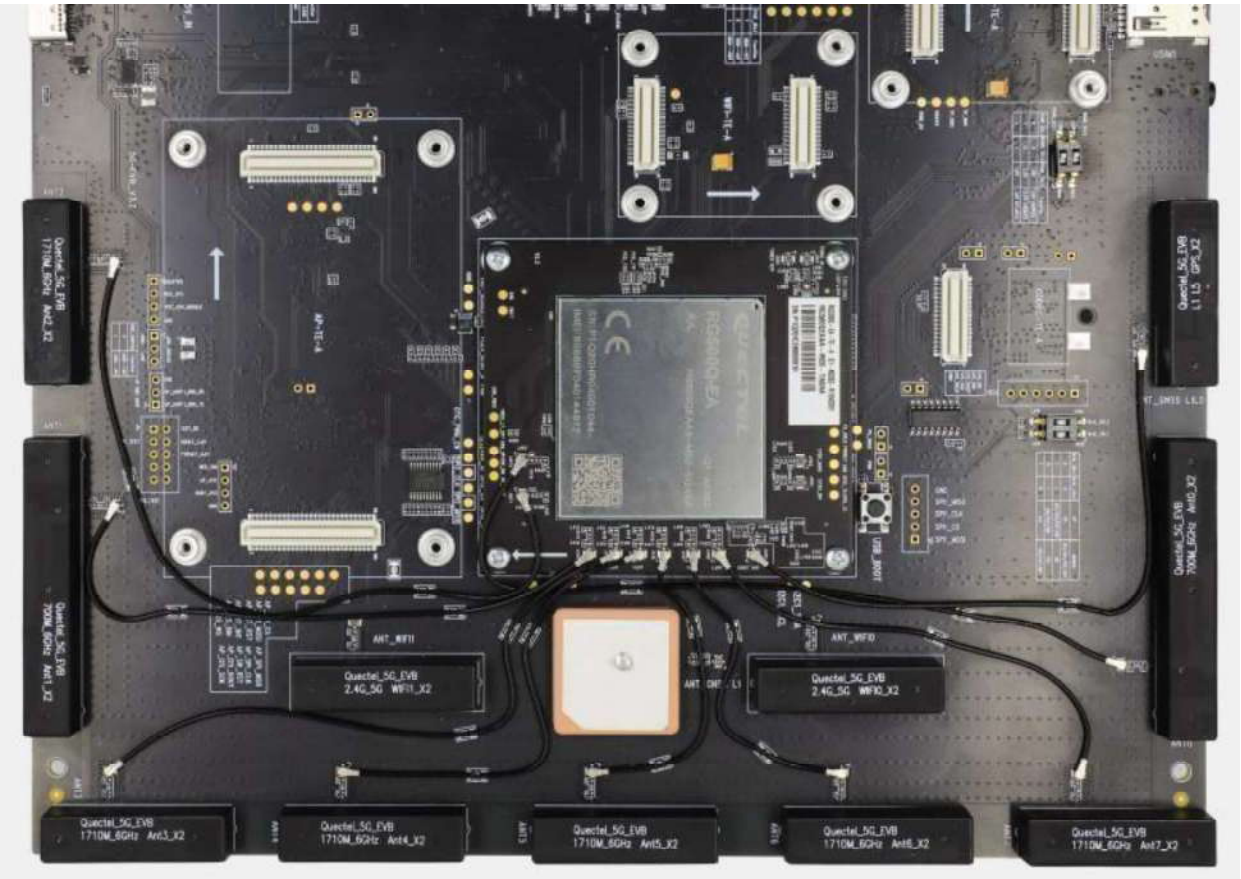


**Figure : Antenna Interfaces**

### 6.5 Evaluation Board Operation Procedures

This chapter introduces how to use the 5G EVB for testing and evaluation.

#### 6.5.1 Turn on the Module

1. Connect the module TE-A to the EVB via connectors J0101 and J0102.
2. Insert a (U)SIM card into the USIM1 card connector on EVB.
3. Use RF cable to connect the module TE-A to the EVB, and connect antennas to the EVB.
4. Connect the EVB to a 5 V/ 3 A power, then switch S0301 to ON. Then D0201 (ON/OFF indicator of the module's power supply) will light up.
5. Press the S0202 (PWRKEY) for at least 500ms, then the module will be powered on and D0202 (operation indicator of the module) will light up.

#### 6.5.2 Communication via USB

1. Power on the module according to the procedure in *Chapter 1*.
2. Connect the EVB and a PC with USB cable through USB Type-C interface, and then run the driver disk on the PC to install the USB driver. The USB port numbers can be viewed in Device Manager of the PC when the USB driver is installed, as shown below.

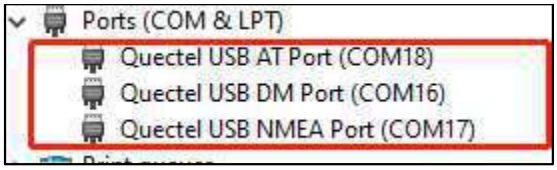



**Figure : USB Ports**

**EVALUATION BOARD ANUBHAV (CORAL TELECOM)**
**WIFI Name – QsoftAP ; PASSWORD – 1234567890**

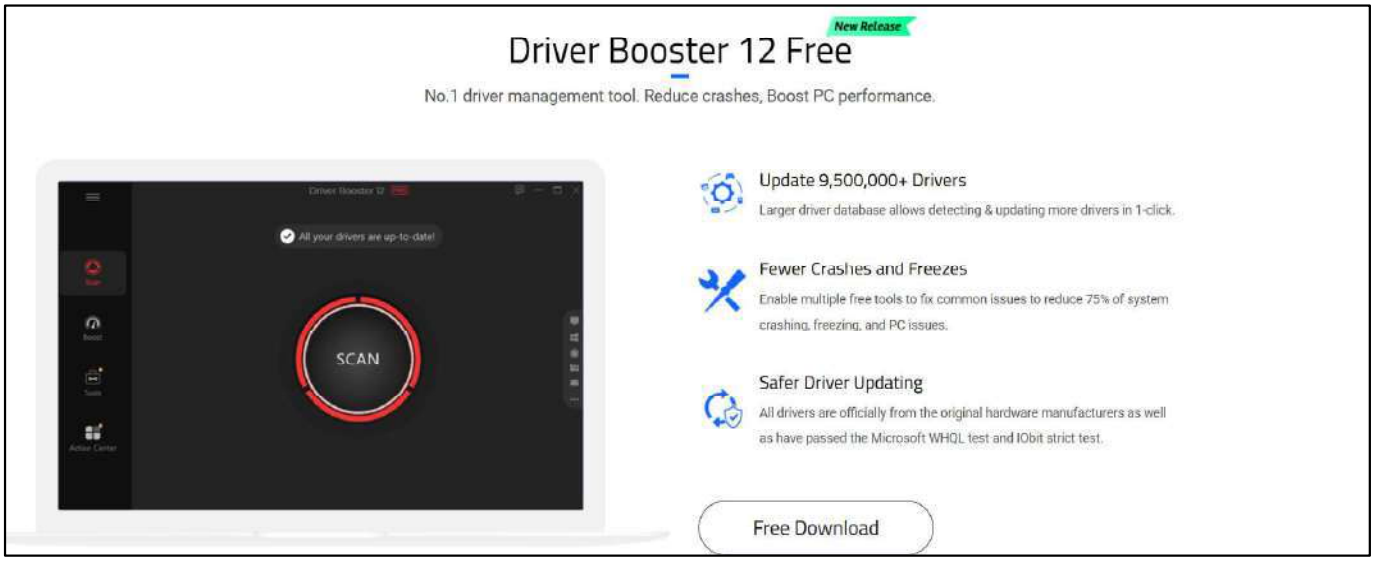
### 6.6 SETUP

#### 6.6.1 ABBREVIATION:-

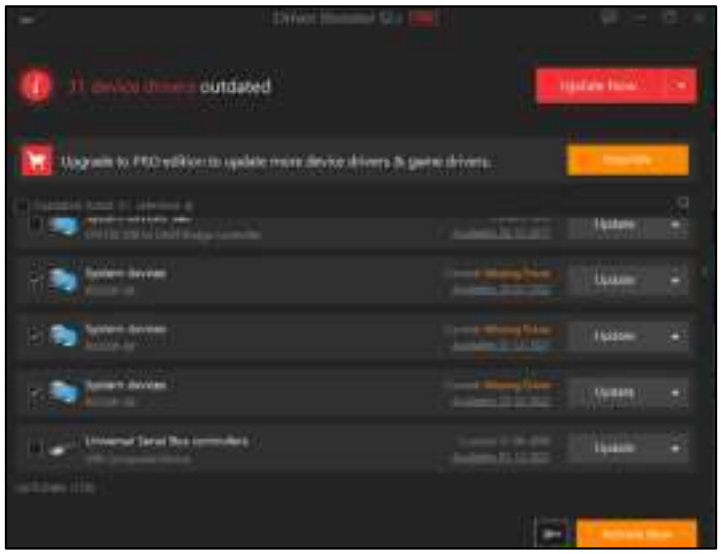
AT- TO INITIATE ---BASIC COMMAND REQUIRED FOR EACH ACTIVITY. ATA- FOR ANSWER/ACCEPT THE CALL
ATH- FOR HANG/CUT THE CALL.
ATD- TO DIAL A NO.

**NOTE - Download a IOT DRIVER**

1. Go to iobit.com
2. Click on free Download.

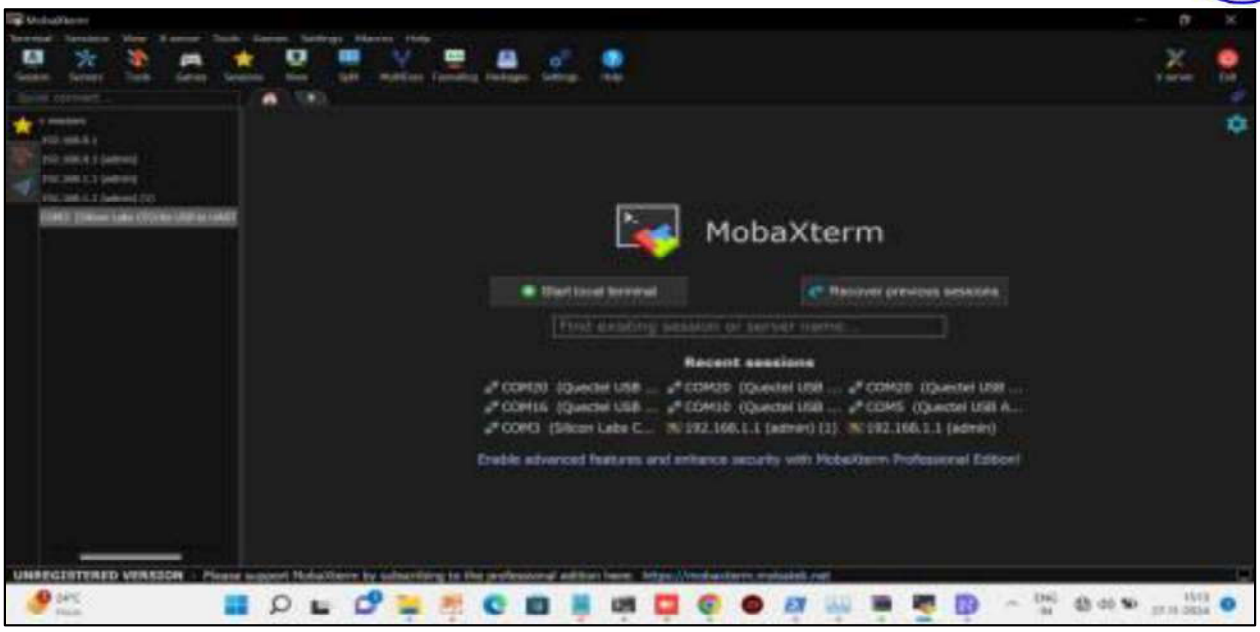



3. driver_booster_setup.exe file will download .
4. Open the file
5. Click on install button .
6. Decline the optional offer
7. Click on **NO THANKS**
8. Click on Scan
9. Untick Outdated all at the starting tab .
10. Select the System device RG520 F-EB , CP2102 USB to UART Bridge controller
11. Click on update now

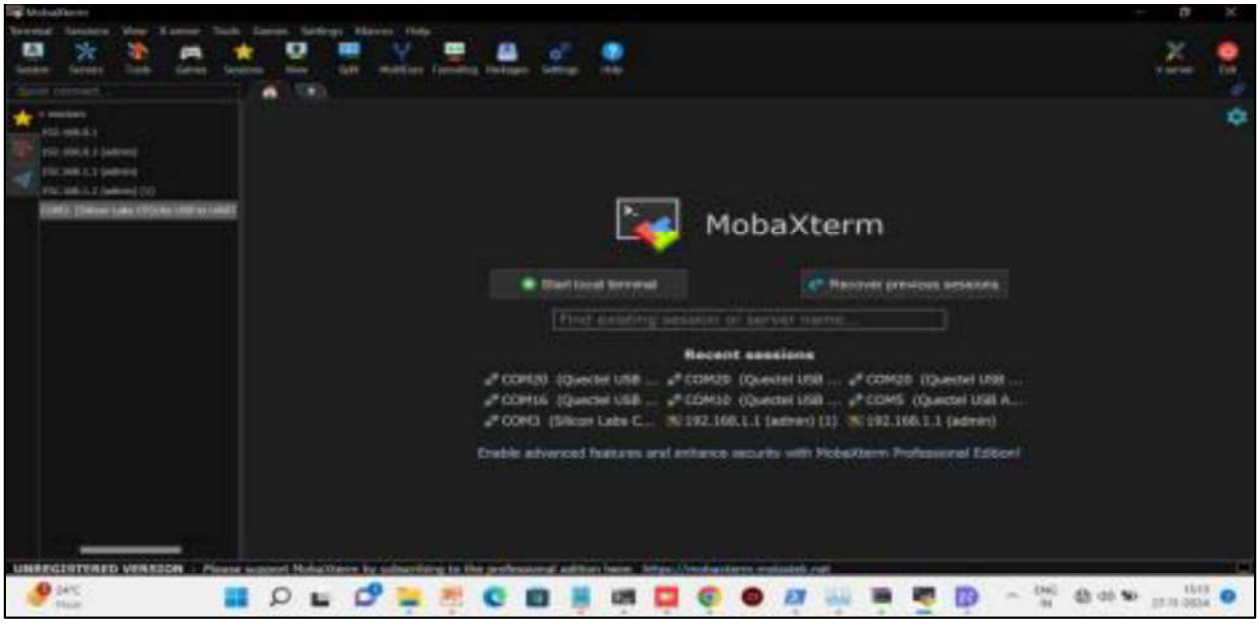



**6.6.2 Steps to Run Evaluation Board**

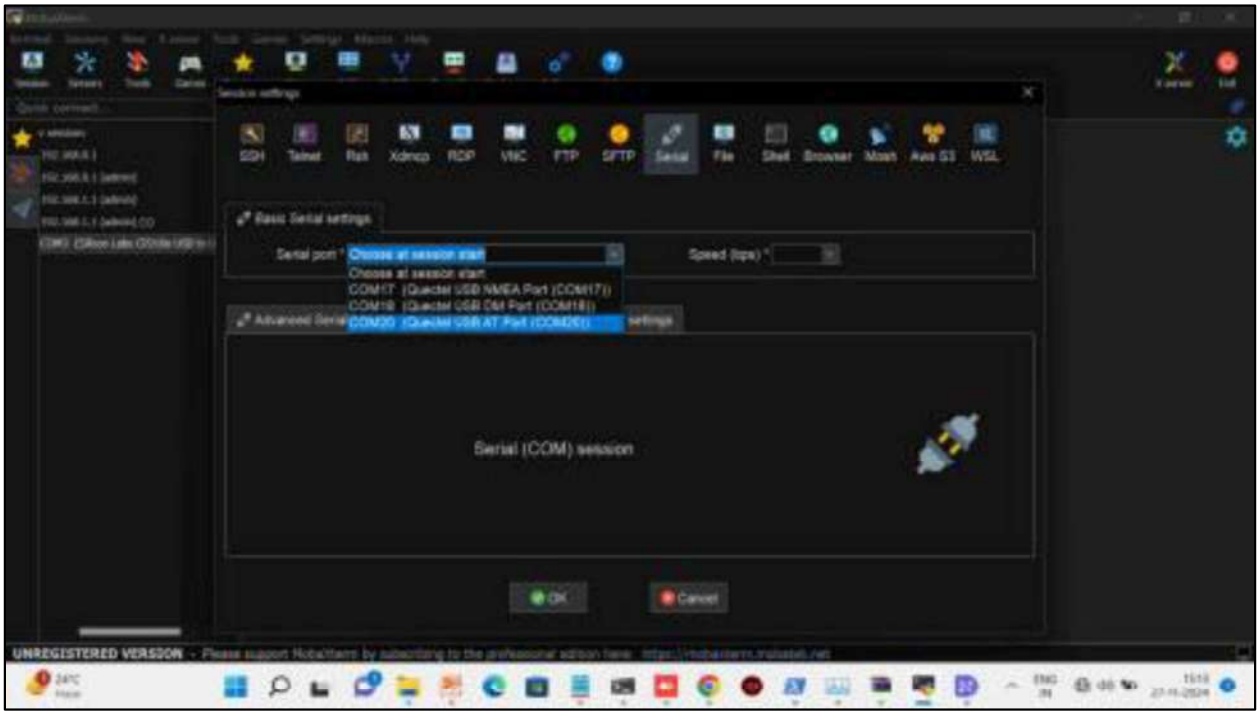
STEP-1-> OPEN MobaXterm Application on your WINDOWS PC.



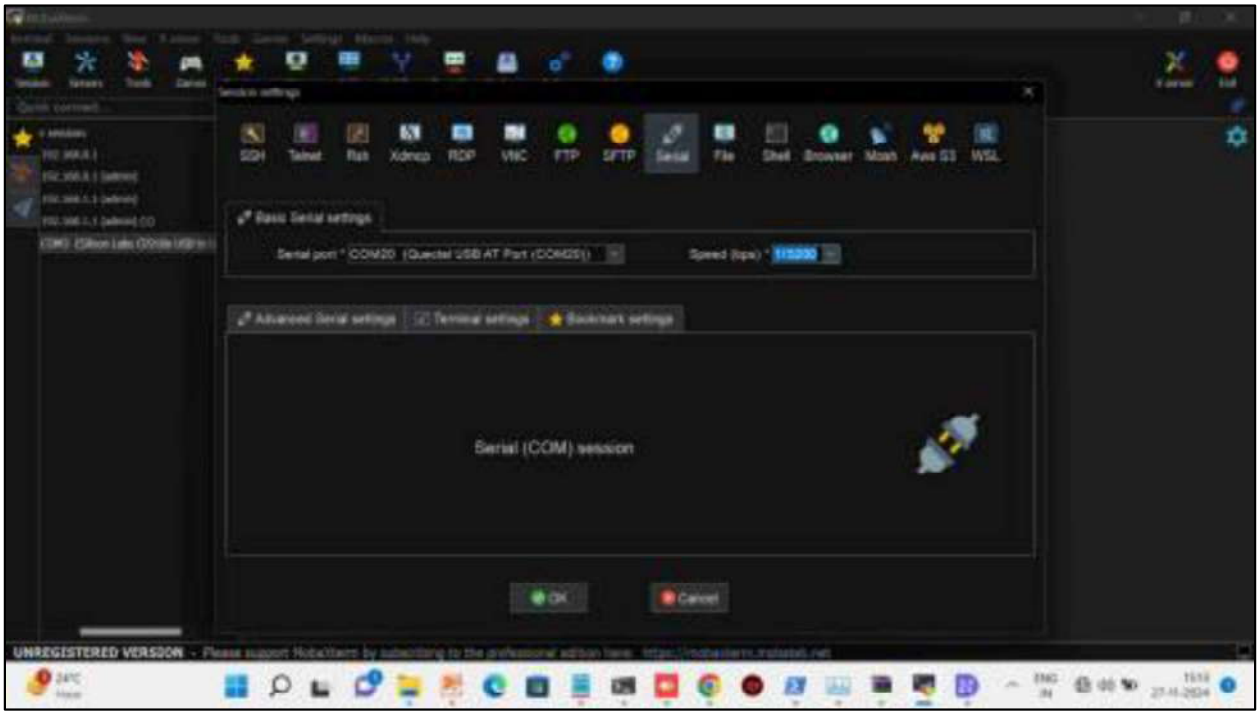

STEP 2-> Click on the session tab and go to serial .



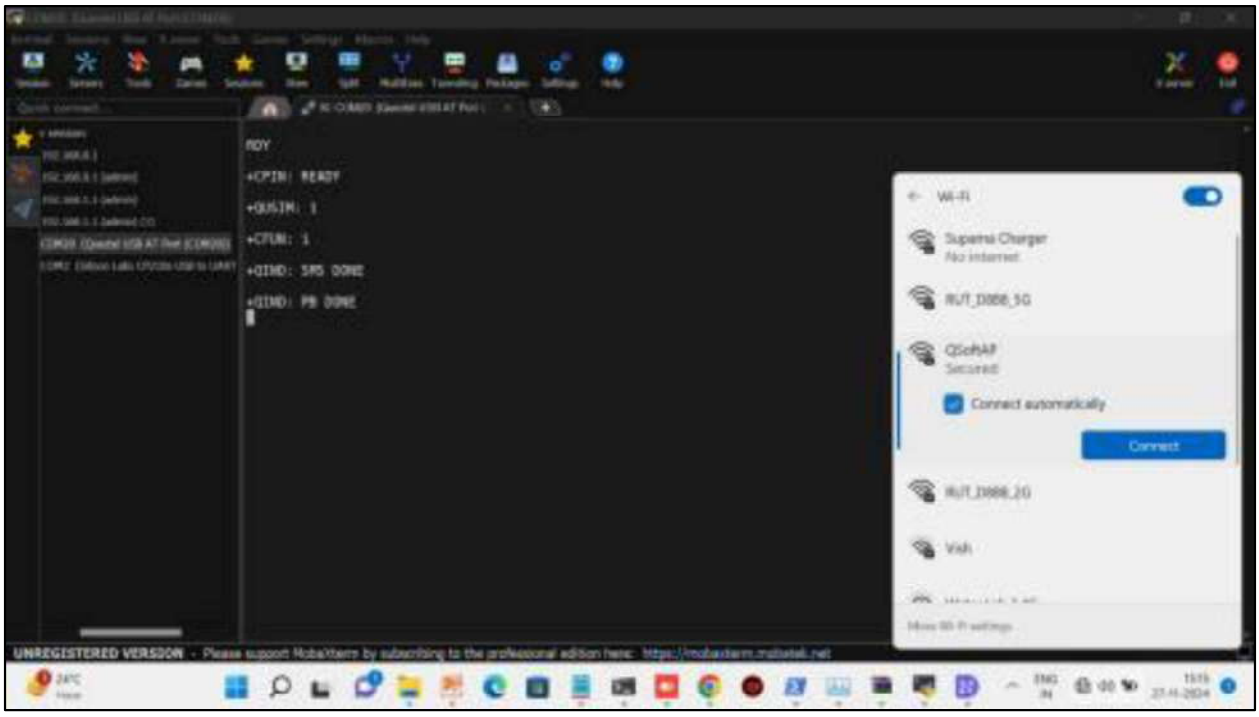

STEP3-> SELECT “USB AT” PORT (HIGHLIGHTED IN BLUE).




STEP4-> CHOOSE THE SPEED-- 115200 bps.



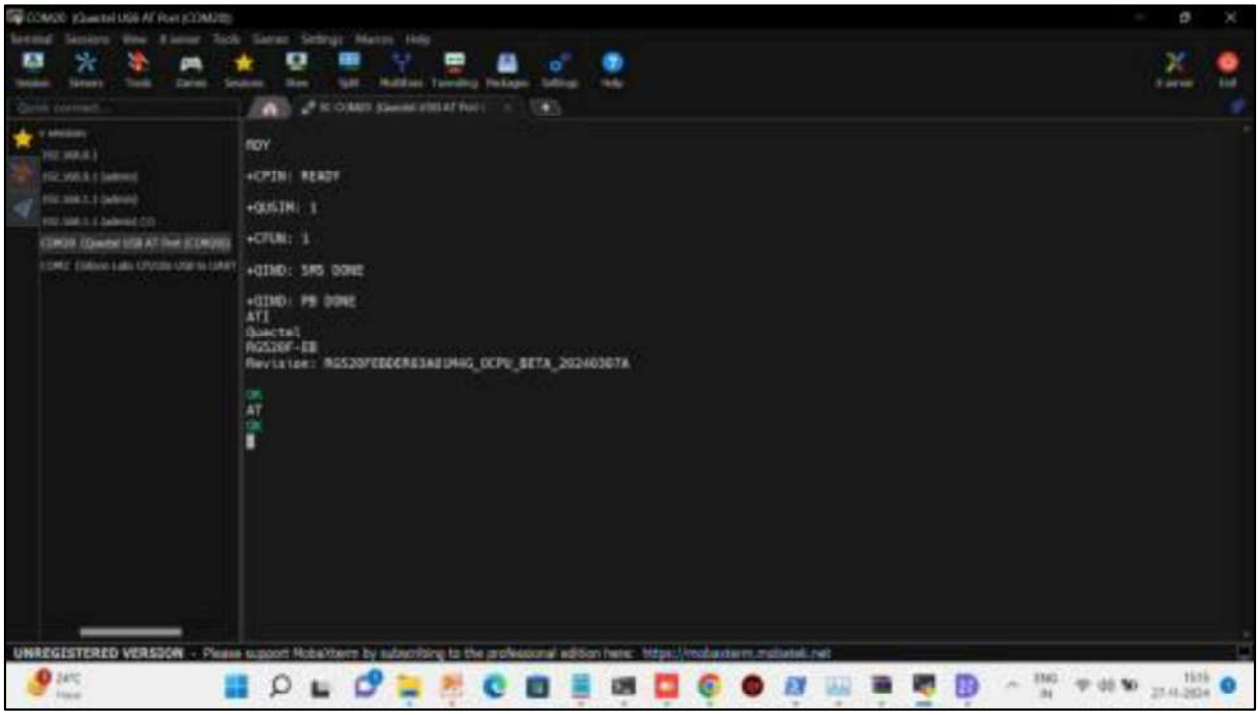

STEP 5-> CONNECT THE WIFI OF THE ANUBHAV NAME “QSoftAP” ; PASSWORD IS “1234567890”



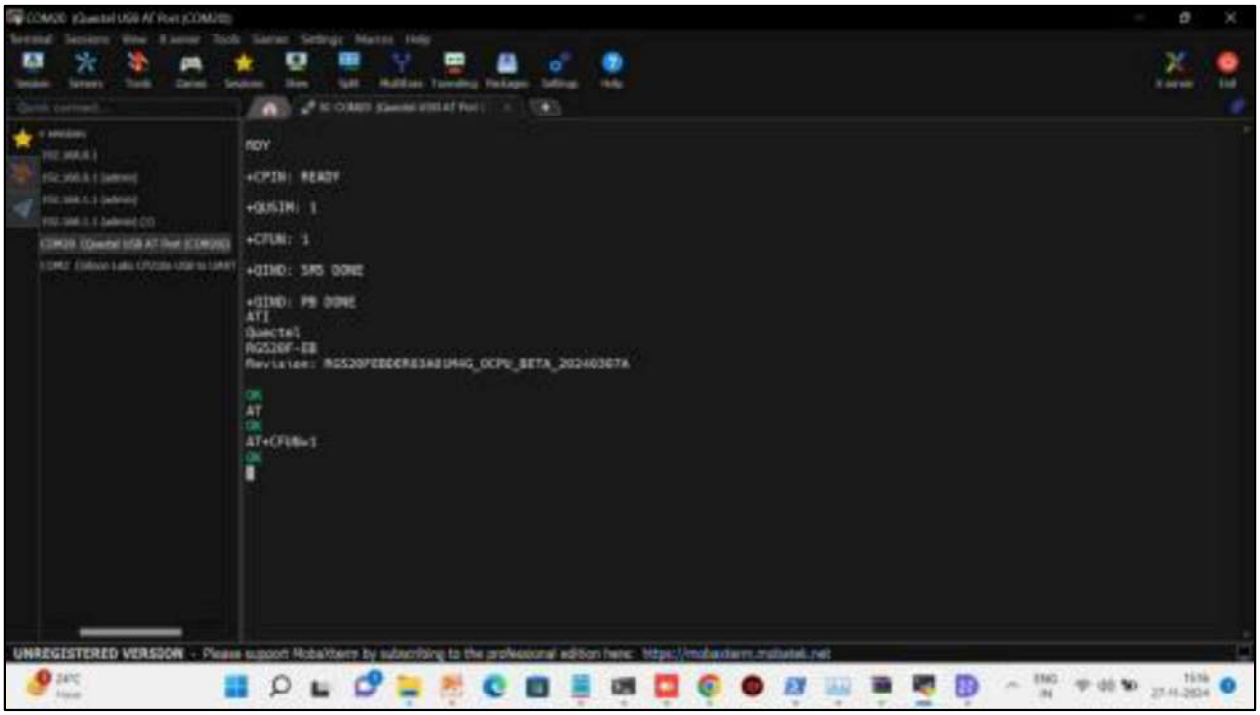

STEP 6-> 1. “AT” COMMAND

2. COMMAND IS “ATI”

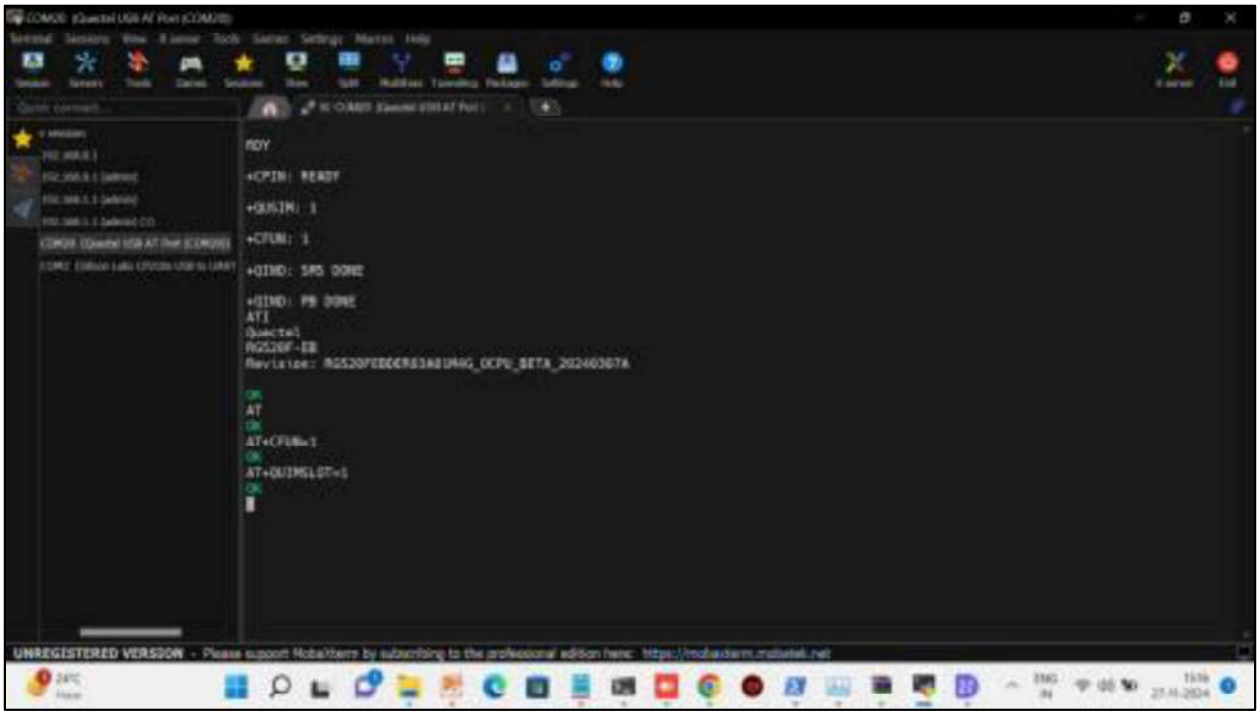
**NOTE-** TO INITIATE



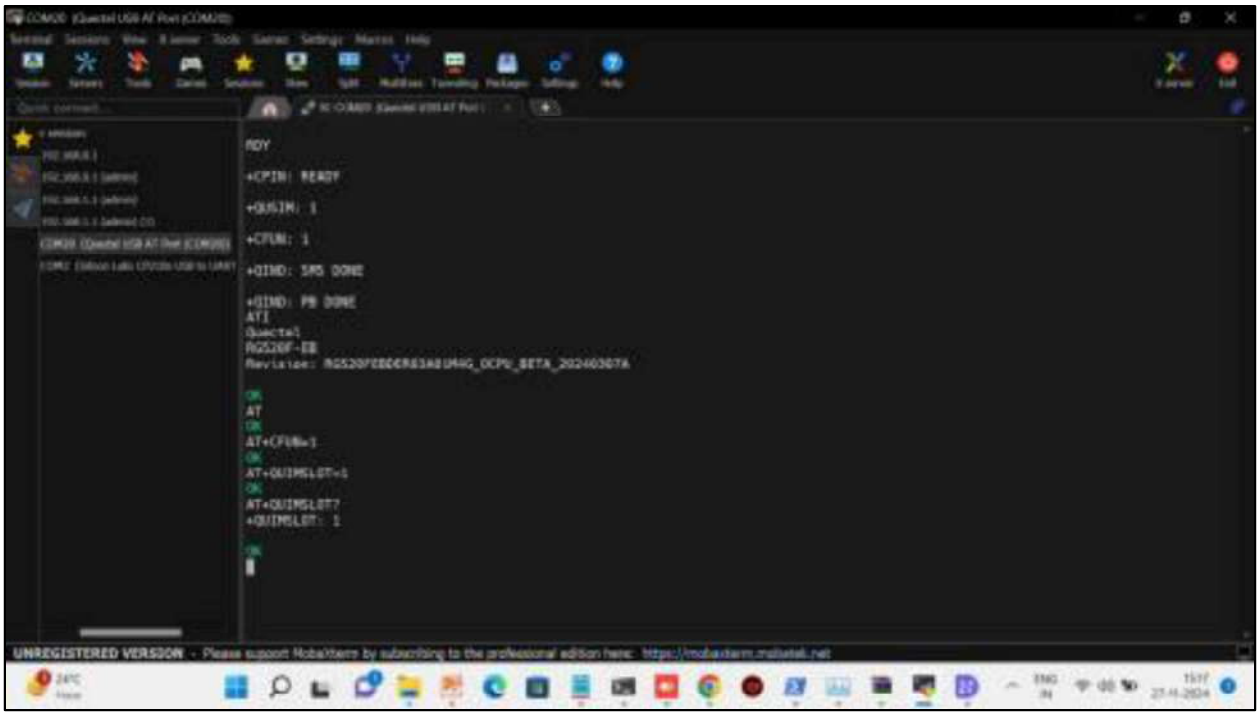

STEP 7-> START THE MODEM BY THE COMMAND “AT+CFUN=1”




STEP 8-> SELECT SIM SLOT 1 BY USING COMMAND “AT+QUIMSLOT=1”.
NOTE – MAKE SURE SIM IS IN SLOT 1.

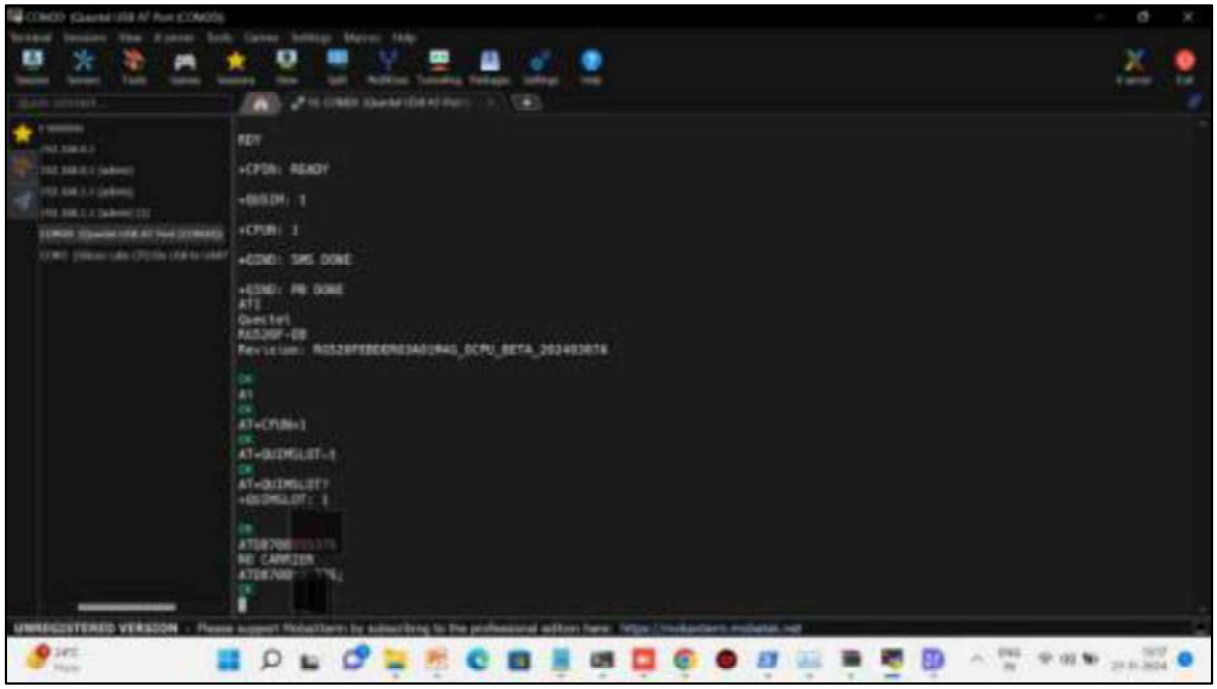



STEP 9-> CHECK THE WORKING SIM SLOT IN THE EVALUATION BOARD ;
COMMAND – “AT+QUIMSLOT?”
NOTE-“IF SIM IS INSERTED IN SLOT 2 THEN CHANGE IT TO SLOT 1”



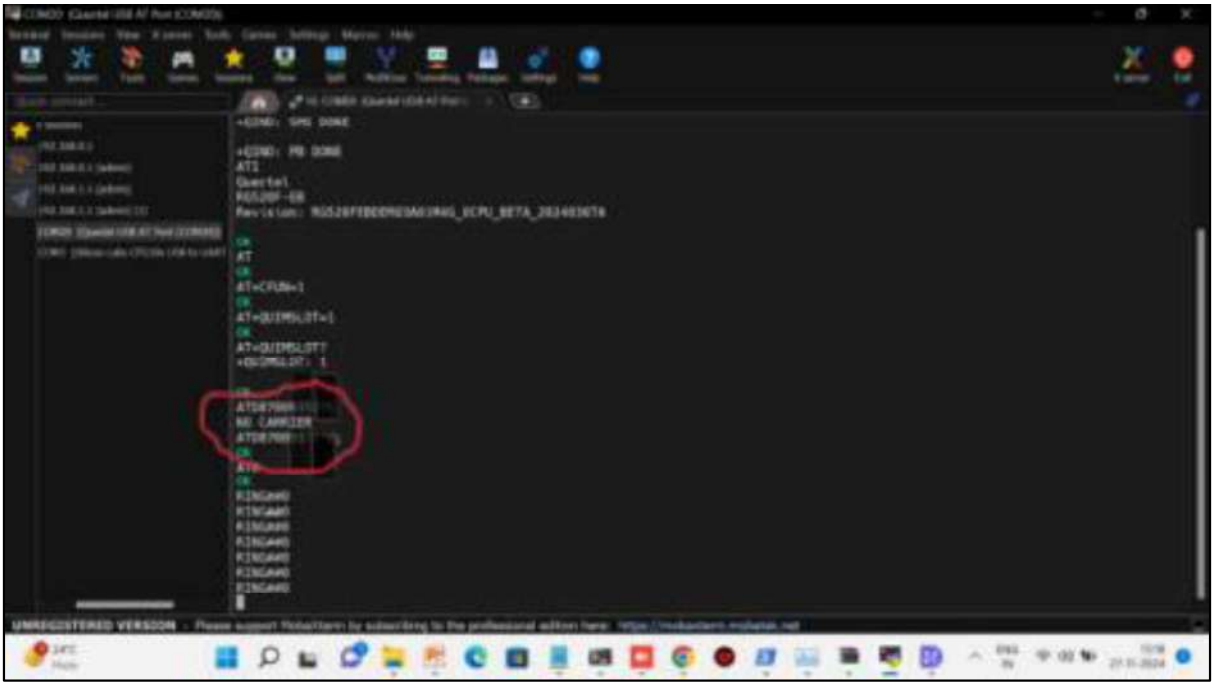

STEP 10-> DIAL A NUMBER WITH COMMAND “ATD8700######;”

NOTE- ##### - DENOTES PHONE NUMBER

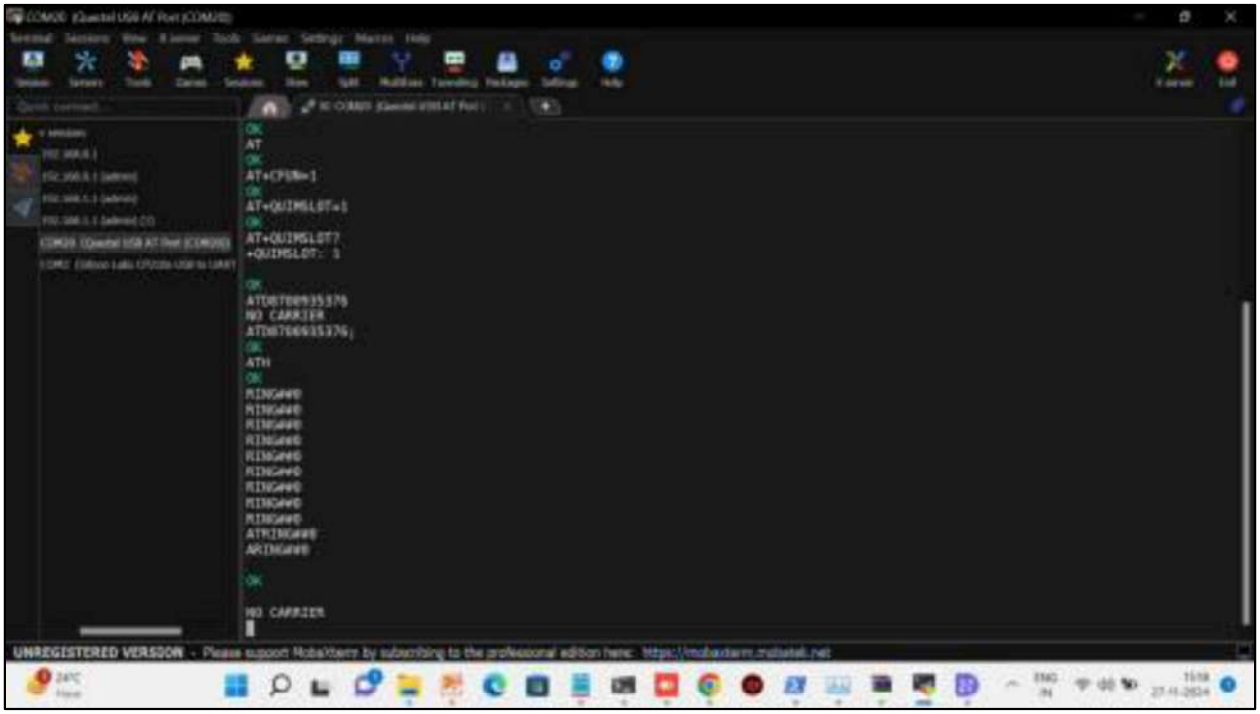



STEP 11->AFTER CALL INITIATE WE GOT THAT OUTPUT “OK”

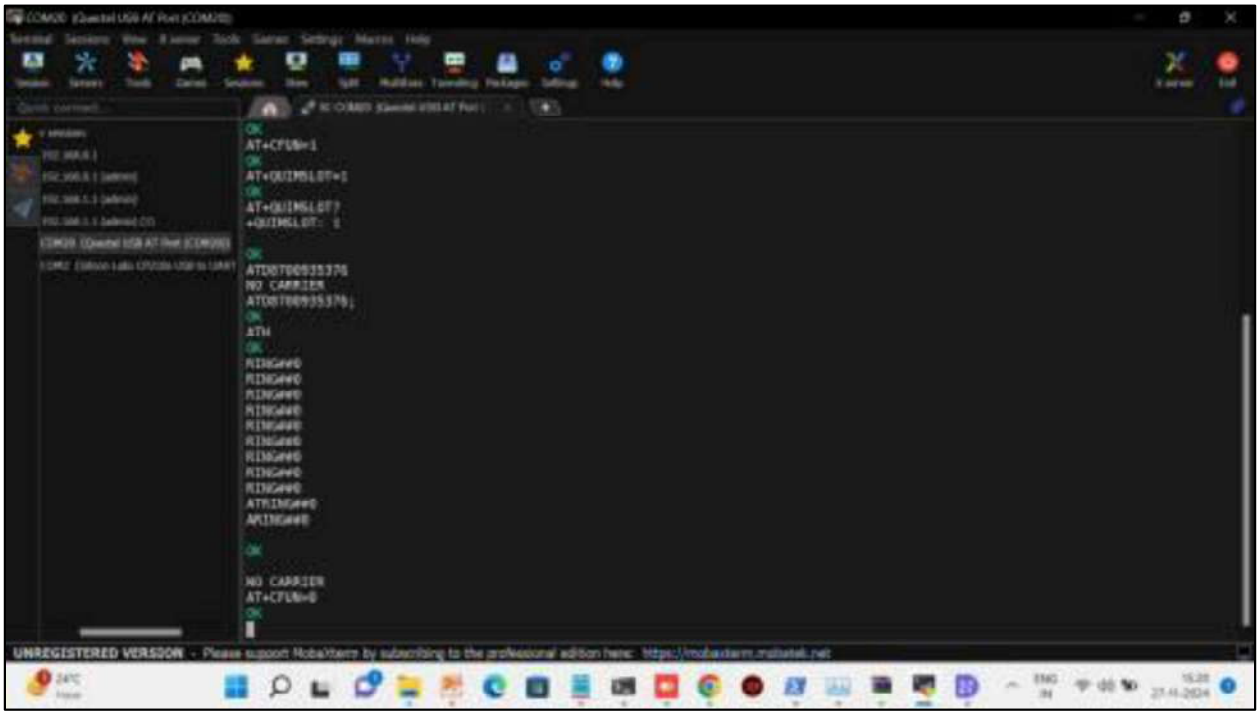
NOTE:- THIS TIME CALL IS INITIATED FROM THE BOARD “ANUBHAV”---- (HIGHLIGHTED IN RED CIRCLE)



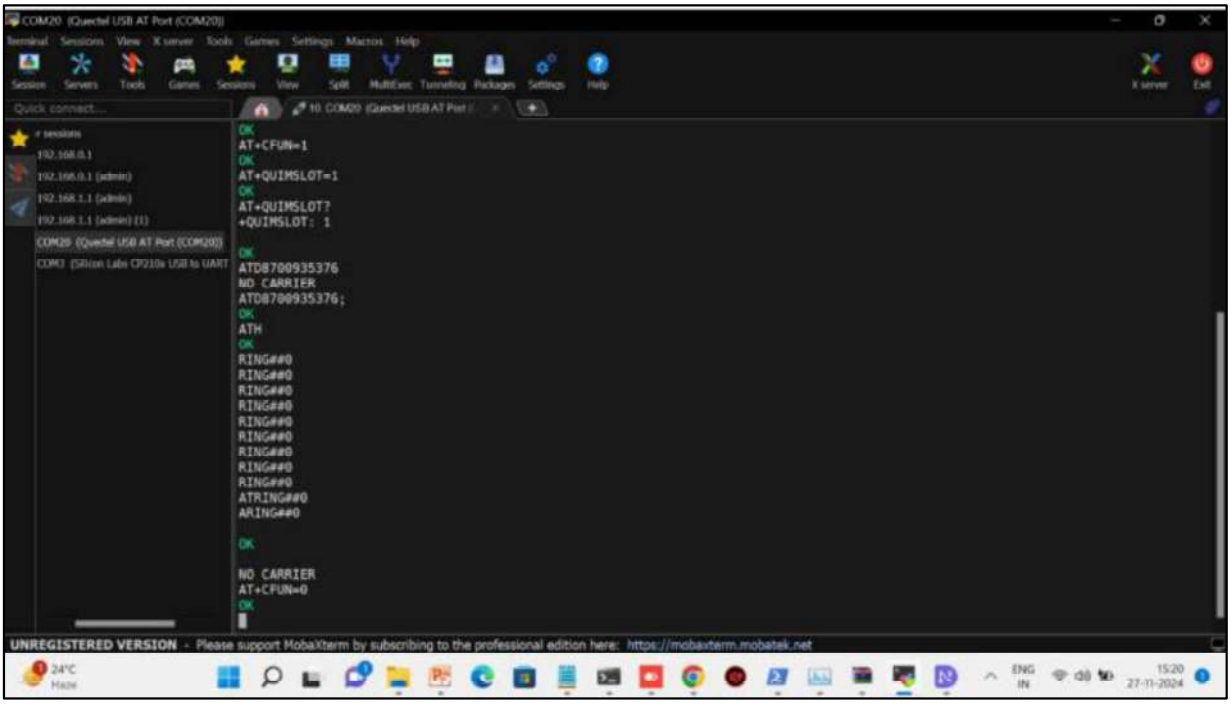

STEP:-12->> IF WE WANT TO HANG UP THE CALL FROM BOARD THEN WE HAVE TO GIVE THE COMMAND “ATH”AFTER SUCCESSFUL HANG UP WE GOT THE OUTPUT “OK”




STEP:-13->> IF WE WANT TO HANG UP THE CALL FROM CELL PHONE THEN NO COMMAND IS REQUIRED BUT WE GOT THE OUTPUT “NO CARRIER”.




STEP:-14->> IF WE WANT TO PICK UP THE CALL FROM BOARD THEN WE HAVE TO GIVE THE COMMAND “ATA”.




TO EXIT:-

1. FIRST COMMAND:- AT+CFUN=0 --- TO TURN OFF THE MODEM.
2. REMOVE THE SIM GENTLY.

Some Other Commands-
AT+CPIN - To know the details during receiving call.

### 6.6.3 Integrating With Raspberry Pi

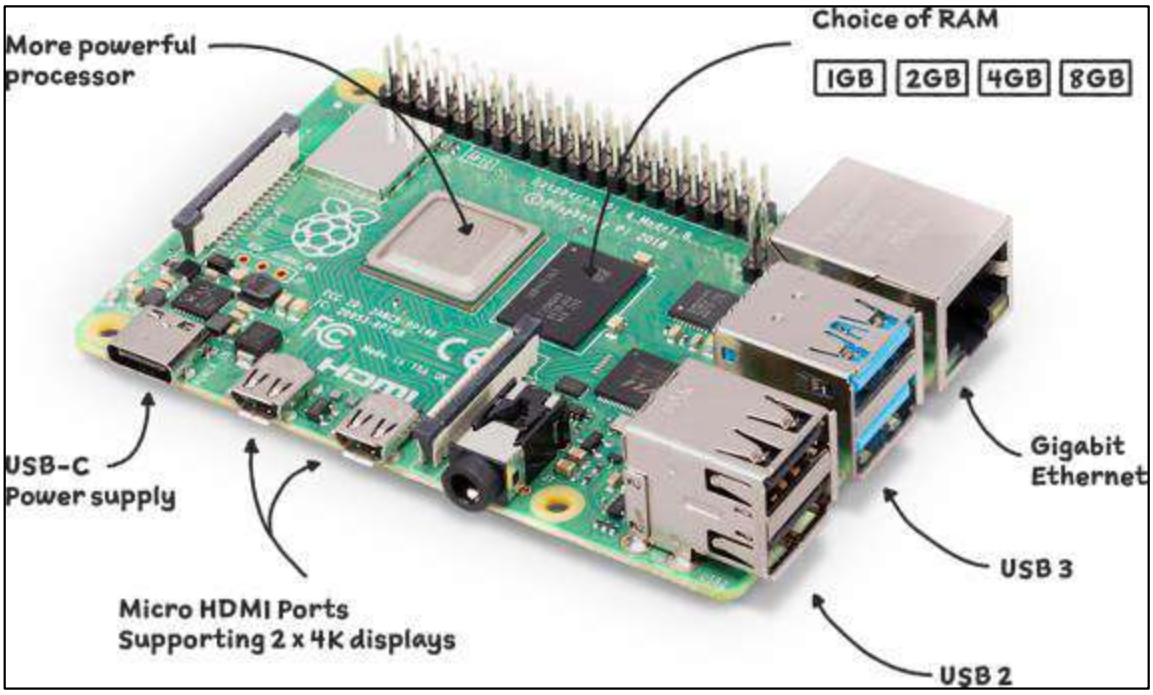



Fig - Showing the Diagram and interfaces of Raspberry Pi

## 6.7 SPECIFICATION

- Broadcom BCM2712 2.4GHz quad-core 64-bit Arm Cortex-A76 CPU, with cryptography extensions, 512KB per-core L2 caches and a 2MB shared L3 cache
- VideoCore VII GPU, supporting OpenGL ES 3.1, Vulkan 1.2
- Dual 4Kp60 HDMI® display output with HDR support
- 4Kp60 HEVC decoder
- LPDDR4X-4267 SDRAM (2GB, 4GB, 8GB, and 16GB)
- Dual-band 802.11ac Wi-Fi®
- Bluetooth 5.0 / Bluetooth Low Energy (BLE)
- microSD card slot, with support for high-speed SDR104 mode
- 2 × USB 3.0 ports, supporting simultaneous 5Gbps operation
- 2 × USB 2.0 ports
- Gigabit Ethernet, with PoE+ support (requires separate PoE+ HAT)
- 2 × 4-lane MIPI camera/display transceivers

- PCIe 2.0 x1 interface for fast peripherals (requires separate M.2 HAT or other adapter)
- 5V/5A DC power via USB-C, with Power Delivery support
- Raspberry Pi standard 40-pin header
- Real-time clock (RTC), powered from external battery
- Power button

**Steps to Connect Evaluation Board with Raspberry Pi:**

**Power Off Both Devices**

Ensure that both the Raspberry Pi and the evaluation board are powered off before making physical connections.

**Connect via USB**

Connect your evaluation board and Raspberry Pi using a USB to Type-C cable.

**Power On Evaluation Board**
Power on the evaluation board and press the reset button for 5-6 seconds until a green light appears.

**Connect the Raspberry Pi to a Monitor**
Connect the Raspberry Pi to a monitor using an HDMI cable.

**Using a Monitor and Keyboard**
Plug in a keyboard and mouse to the Raspberry Pi.
Power up the Raspberry Pi by connecting it to a power source.
Once booted, the Raspberry Pi will show the desktop interface.
To open the terminal, you can either:

- Click on the **Terminal** icon in the top-left corner (it looks like a black screen).
- Or press **Ctrl + Alt + T** to open the terminal.

**Update the System**
Run the following commands in the terminal:
sudo apt-get update
sudo apt-get upgrade

**Install Minicom**

Run the command:
sudo apt-get install minicom

**Set Up Minicom**
Run the following command:
sudo minicom -D /dev/ttyUSB2 -b 115200

Now the terminal is ready to use with the evaluation board.

Once inside minicom, try sending basic AT commands to the GSM module:

AT

If you see OK, it means the GSM module is responding properly.

**Check the Network Registration**

Before making a call, you should ensure the module is registered to a network. Use the following command:

AT+CREG?

This will return the network registration status. Look for CREG: 0,1 (home network) or CREG: 0,5 (roaming).

**Make a Call**

Now you can make a call using the ATD command. Replace <PHONE_NUMBER> with the number you wish to call (include the international code if necessary).

ATD<PHONE_NUMBER>;

- The” **;** “ at the end is necessary to initiate the call.

If everything is set up correctly, the GSM module will dial the number, and you should hear a ringing tone or the call will be initiated.

**Answer a Call**

If your GSM module is set up to receive calls, you can answer an incoming call with the following command:

ATA

**Hang Up a Call**

To hang up a call, you can use the following command:

ATH

**Handling Errors and Troubleshooting**

No network: Make sure the SIM card has a network signal. You can check the signal strength with the command:

AT+CSQ

## 7. IOT SENSOR

In this project, we're leveraging the power of 5G connectivity alongside a variety of IoT sensors to revolutionize environmental testing. With the help of our cutting-edge technology we have enhanced the accuracy and efficiency of our testing processes. By integrating IoT sensors into the labs and utilizing the high-speed, low-latency capabilities of 5G networks, we aim to elevate the quality of environmental testing across air, water, and soil domains. This innovative approach not only ensures compliance with industrial standards but also facilitates real-time monitoring and analysis for proactive environmental management.

### 7.1 TECHNOLOGY USED

MQTT (Message Queuing Telemetry Transport)

### 7.2 SENSOR USED IN THIS PROJECT

NPK Soil Sensor
TDS Sensor
Temperature & Humidity Sensor
Light Intensity Sensor

It facilitates efficient communication between IoT devices and the central monitoring system.
It enables real-time data exchange, critical for accurate environmental sensing.

### 7.3 SENSOR SPECIFICATION

#### 7.3.1 LIGHT INTENSITY SENSOR

Chip: BH1750FVI Power Supply: 3.3V - 5V Light

`Range：0 – 65535 lx(Lux) Sensor`

Built-in: 16 bit AD converter Direct digital output, bypassing the complex calculation, bypassing the calibration Close to the spectral characteristics of visual Widely used to 1-lux high precision measurement Standard NXP IIC communication

### 7.3.2 TDS SENSOR

| | |
|---|---|
| Input Voltage | 3 ~ 6mA |
| Output Voltage | 0 ~ 2.3V |
| Working Current | PH2.0-3P |
| Module Interface | 3.3 ~ 5.5V |
| Electrode Interface | XH2.54-2P |
| TDS Measurement Range | 0 ~ 1000ppm |
| TDS Measurement Accuracy | ± 10% F.S. (25 ℃) |

### 7.3.3 NPK SOIL SENSOR

Power supply: 5V Only
Maximum power consumption: ≤0.15W
Operating temperature: -40~80℃
NPK parameters: Range: 0-1999 mg/kg(mg/L)
Resolution: 1 mg/kg(mg/L)
Precision:±2%FS
Response time: ≤1S
Protection grade: IP68
Probe material: 316 stainless steel

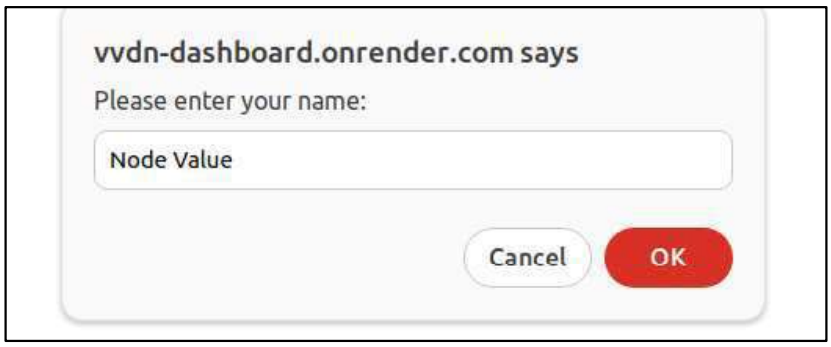
### 7.3.4 TEMPERATURE & HUMIDITY SENSOR

Operating Voltage: 5V
Temperature Range: -40 to 125 C
Humidity Range: 0~100%rh
Probe Material: Plastic + Metal Typ.
Temperature Accuracy: 0.3C Typ. Relative
Humidity Accuracy:3 %RH

### 7.5 DASHBOARD

STEP 1

Open url: https://vvdn-dashboard.onrender.com Enter the node of the sensor mentioned on sensors
Keep same node at same site example: All sensor should have same node mentioned on there box



### 7.6 How to Access Sensor Data

Setting Up MQTT client

Install Python on Windows or linux

Install Mqtt library -pip3 install paho-mqtt

MQTT Broker Details:

Server: broker.emqx.io

TCP Port: 1883

WebSocket Port: 8083

SSL/TLS Port: 8883

Secure WebSocket Port: 8084

Example Python Script to Listen Temperature Sensor

```
f# python 3.11

import random

from paho.mqtt import client as mqtt_client

broker = 'broker.emqx.io'

port = 1883

topic = "iot/Node-20/temp" # Replace node with current node of sensor

# Generate a Client ID with the subscribe prefix.

client_id = f'subscribe-{random.randint(0, 100)}'

username = 'vvdn'

password = 'vvdn'

def connect_mqtt() -> mqtt_client:

def on_connect(client, userdata, flags, rc):

if rc == 0:

print("Connected to MQTT Broker!")

else:

print("Failed to connect, return code %d\n", rc)

client = mqtt_client.Client(client_id)

# client.username_pw_set(username, password)

client.on_connect = on_connect

client.connect(broker, port)

return client
```

```
def subscribe(client: mqtt_client):

def on_message(client, userdata, msg):

print(f"Received `{msg.payload.decode()}` from `{msg.topic}` topic")

client.subscribe(topic)

client.on_message = on_message

def run():

client = connect_mqtt()

subscribe(client)

client.loop_forever()

if __name__ == '__main__':

run()
```

**7.6.1 OUTPUT**

23.5

23.6

23.7

```python
Tutorial.py > connect_mqtt
from paho.mqtt import client as mqtt_client


broker = 'broker.emqx.io'
port = 1883
topic = "iot/Node-22/temp"
# Generate a Client ID with the subscribe prefix.
client_id = f'subscribe-{random.randint(0, 100)}'
username = 'vvdn'
password = 'vvdn'


def connect_mqtt() -> mqtt_client:
    def on_connect(client, userdata, flags, rc):
        if rc == 0:
            print("Connected to MQTT Broker!")
            for i in range(100):
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS        Python

PS D:\Aman\New folder> & C:/Python312/python.exe "d:/Aman/New folder/Tutorial.py"
Connected to MQTT Broker!
23.5
23.5
23.5
23.5
23.6
23.5
23.5
23.5
23.5
23.5
23.5
23.5
23.5
23.5
23.5
```

### 7.6.2 5G CPE SETUP:

First, set the Wi-Fi credentials of TELTONIKA 5g Cpe to default SSID: RUT_D8B8_2G Password: Ec72Qsp4 As soon as sensor find the wifi it automatically connect and send data to cloud MQTT broker

### 7.6.3 DEBUG

Connect sensor to pc and open tera term(window) or minicom (linux) and connect on serial (port. baudrate: 9600 [115200 for tds]) and check the log. If not able to connect to internet it will show connecting to wi-fi else sensor reading will shown.

## 8. 5G Use-Case Of Evaluation Board

### 8.1 Steps to Start with Evaluation Board

Step 1: -Insert the Private 5G SIM in the Evaluation Board

Step 2: - Power on the Board using 5V Power Adapter

Step 3: - Press the Reset Button for 3-5 Seconds

Step 4: - Wait for 15 seconds for the Board to Boot Up

Step 5: - Install the appropriate drivers on the computer.

Recommended: Use Driver Booster: Driver Booster .

- Install the RG520F Drivers (Must Needed)
- Install the PORT Driver (Must Needed)

You are now ready to take the access of the evaluation board.

### 8.2 Basic Configuration Of Evaluation Board

**Step 1: -** Press Windows + X on your Windows Computer Home screen and navigate to Device Manager

**Step 2: -** When you will open the Device Manager you see a list of **Ports(COM & LPT)**

**Step 3: -** After that install MobaXterm from the internet - MobaXterm : - MobaXterm

**Step 4: -** MobaXterm is an advanced terminal emulator and remote desktop application designed for Windows. It provides a powerful set of tools for developers, system administrators, and IT professionals who work with remote systems.

### 8.3 Introduction to MobaXterm

MobaXterm is a powerful terminal and remote access tool for Windows. It supports SSH, RDP, FTP, and more, with a built-in X server and Unix commands. Ideal for developers and sysadmins, it simplifies managing remote servers and running Linux applications on Windows.

Before starting with MobaXterm , Follow these steps to establish connection:

1. Use the USB C(3.0) to connect the Evaluation Board with your system. Use the Type C port on the right side of the board.
2. Make sure the board is powered on with the 5V Power Adapter provided and along with that you have used a USB To Type C to make a connection between the system and 5G Evaluation Board.
3. Warning: Don't use the Debug UART for the Type C port as it has different usage.

### 8.3.1 Steps To Access Coral Anubhav with MobaXterm:

1. Once When You Will Open MobaXterm you will be welcomed with this screen
2. Select the option of Session on the Top-Right Corner of the Screen
3. Once you will click on that you will see another window
4. Once you get to this screen go for serial connection which have the logo of plug connector
5. In This Screen you will Select the AT Port and will enter the BAUD Rate as 115200
6. After all this you will be able to access the Coral Anubhav
7. Type ATI to see the Basic Details :

### 8.3.2 Introduction To Attention Commands (AT Commands)

AT commands (Attention commands) are used to communicate with modems or modules via serial communication. They help configure settings, retrieve information, or control device functionalities.

#### A. Types of AT Commands

- **Test Command (=?) -** The command returns the list of parameters and value ranges set by the corresponding Write Command or internal processes.
  - Syntax: AT+COMMAND=?
  - Purpose: Lists all possible values a command supports.
  - Example Response: +CONFIG: (0,1,2,3), indicating valid options.
- **Read Command (?) -** The command returns the currently set value of the parameter or parameters.
  - Syntax: AT+COMMAND?
  - Purpose: Queries the current value of a setting.
  - Example Response: +CONFIG: 2, meaning the current setting is 2.
- **Write Command (=<parameters>) -** The command sets the user-definable parameter values.
  - Syntax: AT+COMMAND=<value>
  - Purpose: Sets a new value for the configuration.
  - Example Response: OK, confirming the update.

| AT Command | Functionality |
|---|---|
| ATI | Returns model number and firmware version. |
| AT+CIMI | Returns IMSI number. |
| AT+COPS=? | Displays list of available networks; check if network "00101" is available. |
| AT+CFUN=0 | Switches UE to minimum functionality (returns OK). |
| AT+CFUN=1 | Switches UE to full functionality (returns OK). |
| AT+CGDCONT? | Displays list of APNs; check APN configuration as per network slice. |
| AT+CGDCONT=1,"IP","APN-Name" | Set APN to "APN-Name". |
| AT+QNWPREFCFG="mode_pref",NR5G | Set the Evaluation board to work in NR5G mode only (returns OK). |
| AT+COPS? | Displays details of the registered network; if only "0" (automatic) or "1" (manual), the board is not registered. |
| AT+COPS=1,2,"00101" | Manually registered to network "00101". |
| AT+CGPADDR=1 | Checks if an IP address is assigned. If 0.0.0.0, PDU session is not established; try CFUN=0 & CFUN=1. |
| ATD123456789; | Dials the 123456789 if it is present on the network |
| AT+QCFG=? | To list available configuration settings: |
| AT+QCFG="wifi/model", "fc60e" | To set the Wi-Fi module model to "fc60e": |
| AT+CHLD=1 | To retrieve the held call: |
| AT+CHLD=2 | To put a call on hold: |

### 8.4 Configuring 5G Evaluation for 5G Registration

Now, we will look at how to connect your 5G Evaluation board with the radio and establish 5G Registration.

Entire registration part is divided into 4 Steps

**Step 1:** Set network mode preference to NR5G

- Set Network Preferences: **AT+QNWPREFCFG="mode_pref",NR5G**

**Step 2:** Set APN to 'APNname' for data connection.

- Set APN: **AT+CGDCONT=1,"IP","APNname"**

**Step 3:** Set the operator selection on Automatic mode

- Set operator selection on automatic mode: **AT+COPS=0**

**Step 4:** Reload The 5G Evaluation Board (Set UE Functionality)

- **AT +CFUN=0**
- **AT+CFUN=1**

### 8.5 Raspberry Pi Use Case With Coral Anubhav

**Recommended : -** To Access A Raspberry PI you will have to know its:

1. IP Address
2. Username & Password

**NOTE:** Check if your Raspberry PI has the Wi-Fi module available or otherwise use a wireless dongle to transmit the data

***Step 1:*** Turn on your Raspberry Pi and take the access of it and run the command: -raspi-config

***Step 2:*** You will be see this screen :

- Select System Options

***Step 3: After Selecting System Option you will see this screen :***

Select wireless LAN from which you want to transmit the data and then enter the name of your Wi-Fi in the option of SSID:

After that enter the password of your Wi-fi through which data needs to be transmit

-Now your Sensor Data will be transferred via Coral Anubhav(Evaluation Board) to the NMS via the 5G network.

### 8.6 Use Cases of Automating 5G Evaluation Board (Coral Anubhav) Configuration with Bash Scripts on Raspberry Pi

#### 8.6.1 Introduction

This use case explores the practical applications of automating the configuration of the 5G evaluation(Coral Anubhav) using Bash scripts on a Raspberry Pi. By leveraging predefined commands, users can efficiently manage network registration, modem setup, and configuration changes without manual intervention. This automation enhances reliability, speeds up deployment, and reduces human errors in 5G connectivity setups, making it ideal for IoT applications, industrial automation, and research projects. The guide provides a step-by-step approach to executing AT commands via Bash, ensuring seamless interaction between the Raspberry Pi and the 5G modem.

#### 8.6.2 Prerequisites

- Raspberry Pi (any model with USB support)
- 5G Evaluation Board (Coral Anubhav )
- SIM Card
- Minicom or another serial communication tool
- USB-to-Serial driver installed (if necessary)

#### 8.6.3 Setup Instructions

Run as Root:

sudo -i

Check Device Path:

ls /dev/ttyUSB*

Install Required Packages:

sudo apt update && sudo apt install -y socat;

Grant USB Permissions:

sudo chmod 777 /dev/ttyUSB*

#### 8.6.4 Bash Script for Modem Configuration

Create a script `modem_config.sh` to execute AT commands using `socat`.

#!/bin/bash

# Check if minicom is installed

if ! command -v socat &> /dev/null; then

```
echo "socat is not installed. Installing..."
sudo apt update
sudo apt install -y socat
else
echo "script is started."
fi
# the command run and output show in file
echo ATI | socat - /dev/ttyUSB2,crnl > /tmp/hello;
echo AT+cimi | socat - /dev/ttyUSB2,crnl > /tmp/hello;
echo AT+QNWCFG=? | socat - /dev/ttyUSB2,crnl >/tmp/hello;
#when you want to append the output
#echo AT+QNWCFG=? | socat - /dev/ttyUSB2,crnl >>/tmp/output.txt;
# for print on file and terminal
#echo ATI | socat - /dev/ttyUSB2,crnl | tee /tmp/hello;
# read command thru file and output both to terminal and file
#cat commands.txt | socat - /dev/ttyUSB2,crnl | tee /tmp/hello
```

**8.6.5 Running the Script**

Make the script executable and run it:

```
chmod +x modem_config.sh
sudo ./modem_config.sh
```

**8.6.6 Creating a System Service for Automation**

To automate the execution of the script, create a systemd service.

```
[Unit]
Description=5G Modem Configuration Service
After=network.target
```

```
[Service]
ExecStart=/bin/bash /path/to/modem_config.sh
Restart=always
User=root

[Install]
WantedBy=multi-user.target
```

Save the file as `/etc/systemd/system/modem_config.service` and run the following commands to enable it:

**sudo systemctl daemon-reload**

Reloads systemd to recognize new or modified service files.

**sudo systemctl enable modem_config.service**

Enables the service to start automatically on boot.

**sudo systemctl start modem_config.service**

Starts the service immediately without rebooting.

**TroubleShooting of Coral Anubhav**

**8.7 Basic Checks**

1. **Power & Hardware Connections**
- Ensure the module is properly powered
- Check if the antennas are connected (for optimal signal reception)
- Verify the SIM card is inserted correctly
2. **Driver & Firmware Verification**
- Check if the necessary drivers are installed (Linux/Windows)
- Ensure firmware is up-to-date
- Default Baud Rate as 115200. Evaluation Board will communicate with devices on this BAUD Rate
3. **AT Command Interface Check** Use AT commands to verify basic functionality
- AT → Check if the module responds
- ATI → Get manufacturer info
- AT+CGMR → Get firmware version

**Debugging Logs**

- Insert a USB Type-C connection with Debug UART and then switch to the **Silicon Labs CP210x USB-to-UART Bridge**. Set the BAUD Rate as 115200.This will provide access to hardware-level logs and detailed diagnostic information for all modules and features.

**Device and Module Info**

- Get Firmware and Device Information:

  AT+CGMR - Display firmware version.

- Check 5G MIMO Status:

  AT+QNWCFG="nr5g_mimo" -Check if 5G MIMO is enabled.

- Enable 5G MIMO:

  AT+QNWCFG="nr5g_mimo",1-Enable 5G MIMO.

**ADB(Android Debugging Bridge) Access**

- To check about the Coral Anubhav services we can access it by taking adb access to the Coral Anubhav Board.
- Take access by adb shell after connecting it via USB 3.0 provided
- Enter the command - **systemctl status *.service**

F.No. BTU/5GUseCaseLabs/01

**Government of India**
**Ministry of Communications**
**Department of Telecommunications**
**(Standards-R&D-Innovation Division)**

1420
~~1405~~, Sanchar Bhawan, 20 Ashoka Road
New Delhi-110001
Dated: 2nd January, 2025

## OFFICE MEMORANDUM

**Sub: Gradation Framework for 5G Use Case Labs Setup by DoT-reg**

The Department of Telecommunications has set up 100 5G Use Case Labs across educational institutions nationwide, aiming to achieve the following key deliverables:

i. At least 50 students and 10 faculty members utilizing the lab for testing and project development annually.

ii. Engagement of at least 5 start-ups/MSMEs for product testing and development each year.

iii. Development and testing of at least 10 products/solutions/IPs annually.

iv. Active contribution to the standardization process and efforts to integrate developed technologies into global standards.

v. Piloting, commercialization, and deployment of tested products in the field.

vi. Publication of at least two papers per year.

**2. Gradation of Labs**

To ensure the effective utilization of these 5G Use Case Labs, a Gradation Framework with 11 key parameters has been prepared. The labs will be evaluated based on the following Gradation Parameters:

| Sl. No. | Parameter | Target | Score |
|---|---|---|---|
| 1. | No. of students utilised Lab for product testing/ applications | 50 unique students / year | 1- 10 student – 1 point<br>11– 30 student – 2 points<br>31-50 students – 4 points<br>> 50 students – 5 points |
| 2. | No. of faculty utilised the lab for mentoring/ development/ testing | 10 unique faculty / year | 1- faculty – 1 point<br>2 – 5 faculty – 2 points |

| | | | |
|---|---|---|---|
| | | | 6 - 10 faculty – 4 points<br>> 10 faculty – 5 points |
| 3. | No. of projects utilizing the Lab | 10 projects / year | 1- project – 1 point<br>2 – 5 project – 2 points<br>6 - 10 projects - 4 points<br>> 10 projects – 5 points |
| 4. | No. of start up / MSMEs engaged in the lab for testing/ product / use case development | 5 Start up/ MSME / year | 1 point / startup or MSME (max. 10 points) |
| 5. | Engagement with Atal Tinkering Labs/ Schools/ colleges/ academic institutes (Workshops/ mentorship etc.) | 5 institutes / year | 1 point / institute<br>(max. 10 points) |
| 6. | No. of companies other than start-up/MSME/ Academic institutes engaged in the lab for testing/ product / use case development | 5 companies or organisations / year | 1 point / company<br>(max. 10 points) |
| 7. | Inclusion of 5G related academic course utilizing the Lab in the curriculum | Completion of the course for 50 students | 2 point / 10 student<br>(max. 20 points) |
| 8. | Contributions related to 5G and beyond in International journals or Standardization organizations such as ITU/3GPP/ISO/IEEE/ ORAN alliance etc | 2 Contributions/ year | 5 points / contribution<br>(max. 30 points) |
| 9. | No. of product/ solutions/ use cases developed | 2 products/ solutions/ use cases developed | 10 point / product/solution/use case<br>(max. 30 points) |
| 10. | No. of IP generated | 2 IPs in a year | 10 point / IP<br>(max. 30 points) |
| 11. | No. of developed products/use cases in the Lab commercialised/ piloted/ deployed | 1 product/ year | 15 points each<br>(max. 30 points) |

3. **Ranking of Labs:**

The Labs will be ranked based on their total points in the following categories:

a. Excellent: > 130 points
b. Very Good: 101 – 130 points
c. Satisfactory: 81 – 100 points (with minimum 23 points from item number 1-5 of Gradation Framework in Para 2 above )

ii. All engagements and outcomes will be shared and made accessible through the **bharat5glab.gov.in** portal.

iii. The top 5 ranked institutes will be felicitated annually at the IMC or any other event designated by the DoT.

This is issued with the approval of competent authority.

(Shubham)
सहायक मंडल अभियन्ता दूरसंचार
Asstt. Divisional Engineer Telecom
दूरसंचार विभाग, भारत सरकार
Deptt. of Telecom, Govt. of India
नई दिल्ली / New Delhi
ADET(SRI)

To,

Heads of 100 institutes of 5G labs

**1-TR_5G_Startup_Kit/2022-SRI-Part (4)**
**Government of India**
**Ministry of Communications**
**Department of Telecommunications**
**(Standards-R&D-Innovation Division)**
**1420, Sanchar Bhawan, 20 Ashoka Road**
**New Delhi-110001**
**Dated : 21st March 20**

**Office Memorandum**

**Subject: Implementation of 5G Use Case Gradation Framework – Submission of Reports -reg**

Reference: OM No. BTU/5GUseCaseLabs/01 dated 2nd January 2025-attached

Please refer to the above mentioned OM wherein a gradation framework with 11 key parameters had been shared to ensure the effective utilization of the 100 5G Use Case Labs.

2. In this regard, it is to intimate that a portal is being developed for obtaining online report from the institute in respect of gradation framework. However, the development of the same will take some time.

3. In the meantime, the O/o DG(T) HQ is requested to kindly seek offline reports from the concerned 100 5G Use Case Lab institutions, through LSA Units, in the attached format -**Annexure 1.**

4. The consolidated reports may be submitted to this office by 7th of every month starting from 7th April 2025 for further review on utilization of the labs.

5. This is issued with the approval of the competent authority.

Digitally signed by Shubham
Date: 21-03-2025 19:10:50

(Shubham)
ADET(SRI),DoT
Email id: shubhamr.93@gov.in

To,
**DG(T), DOT**

Copy to:

1. PSO to Secretary (T) for kind information
2. PSO to Member (T) for kind information
3. CMD TCIL
4. 100 5G Lab Institutes

Annexure –1 (attached)

Government of India
Ministry of Communications
Department of Telecommunications
Media Division

**OFFICE MEMORANDUM**

**New Delhi, dated the 27th December, 2023**

Subject **– Sanchar Mitra Scheme**

Department of Telecommunications, is responsible for creating awareness of citizen centric services that aims to enhance mobile users' security, burst radiation myths and raise awareness about various Government of India initiatives.

2. Sanchar Mitra Scheme is a platform to engage students as volunteers for creation of awareness of the various Departments citizen centric services. The scheme seeks to establish a bridge between the Department of Telecommunications (DoT) and citizens, fostering a collaborative approach towards telecom issues. The details of the Scheme is attached as Annexure-I

3. In order to jumpstart the process of engaging students as Sanchar Mitra, the following tasks may be undertaken by respective LSA;

i. DoT has recently awarded 100 5G use case labs in educational institutions in 28 states and 4 UTs, thus, LSAs have to Coordinate with the respective Universities head to shortlist Two(2) students from each University who can acts as Sanchar Mitra. These student volunteers may be selected by the University head in consultation with the concerned head of LSA
ii. Provide adequate training and toolkits to augment efforts of the Volunteer in regional/ national language. An illustrative toolkit shall be provided by DoT HQs .
iii. Help Volunteer for Liasoning with Government Departments /Police wherever/whenever required

iv. Prepare targets so that experience letter & certificate of appreciation shall be shared based on the active involvement

v. Spread of misinformation can be stopped by utilizing volunteers Sanchar Mitra

vi. Seek Testimonials of citizen using the portals or affected by DoT initiatives by utilizing volunteers Sanchar Mitra and put across Social Media platforms

vii. Pt3(i)&3(ii) above to be done within 45 days of issuance of letter, i.e 09.02.2024. The complete list of shortlisted candidates (LSA wise) may be sent to adgpr1.hq-dot@gov.in

*Kindly note that the above tasks are illustrative and not exhaustive tasks*

4. This issues with the approval of Secretary (T).

Signed by Vineet Malik
Date: 27-12-2023 16:27:56

(Vineet Malik)
ADG(PR)
DoT HQs

To

**DG Telecom, DoT HQs**

Copy to:

1. PS to Hon'ble MoC
2. PS to Hon'ble MoSC
3. PS to Secretary(Telecom)
4. CGCA/WA/ All Head of the Division of DoT HQs
5. DDG SRI with a request to extend support with respect to coordination between University heads to which 100 5G labs have been granted and Head of LSAs

**Annexure-I**

## SANCHAR MITRA SCHEME

### 3.1 Objective

In order to create awareness of the portal and other important initiatives of DoT with respect to digital frontiers, it is important to identify citizens who are passionate to serve society as Sanchar Mitra. The Sanchar Mitra Scheme aims to enhance mobile users' security; clarity on radiation myths; raise awareness about government initiatives and prevent mobile-related fraud.

As citizen support enhances the efficacy of efforts of the department, the engagement of Sanchar Mitra from different states shall have a valuable addition to the department’s efforts to enhance mobile users' security, raise awareness about government initiative and promote a safe digital ecosystem.

The scheme seeks to establish a bridge between the Department of Telecommunications (DoT) and citizens, fostering a collaborative approach towards telecom and Sanchar issues.

### 3.2 Remuneration

(i) Volunteer shall not be entitled for any monetary benefits or Identity Cards/Designation etc.

(ii) An experience letter & certificate of appreciation shall be shared based on the active involvement

### 3.3 Implementation Process

DoT has recently been awarded 100 5G use case labs in educational institutions in 28 states and 4 UTs and LSAs have been coordinating with these institutes. Students are an active part of the society and represent Yuva Shakti. Thus, in order to jumpstart the process of engaging students as Sanchar Mitra, the following may be undertaken,

### 3.3.1 Phase-wise Implementation

(i) Phase 1 (PoC for Six Months)

- Identify students from 100 universities with 5G labs.
- Engage 2 student from each university as Sanchar Mitra.
- Conduct workshops or use suitable mechanisms for identification

(ii) Phase 2 (After Learnings from Phase 1)

- Open participation for other community members through an open registration process.

### 3.4 Roles& Responsibilities

**Tasks of Sanchar Saathi Volunteers/ Sanchar Mitr:**

i. Awareness to be conducted on broad citizen centric services
   a. Sanchar Saathi portal
   b. Tarang Sanchar portal for EMF awareness
   c. Toll Free Numbers to report international numbers with local numbers
   d. Spam calls and malicious links blocking
   e. Effective blocking of blocked Applications
   f. Bridge between DoT and Citizens for various Telecom/Sanchar issues

g. Sensitization of various Next Generation Communications technologies, 5G,6G etc Hackathon

h. Coordination with Cyber Crime Volunteers for Synergy and effective impact.

i. Amateur radio operator / HAM /Community radio

j. Topics identified may be added from time to time.

**ii. Wider Outreach and Awareness** at grassroot level in their native languages establishing a more relatable and effective connection with citizens, thereby increasing the initiative's impact.

a. This can be done through innovative mechanisms

b. This can become an integral part of LSAs for various workshops conducted by LSAs/CCAs

c. Coordinate with Village Level Entrepreneurs for awareness at grassroot level

d. Coordinate with NGOs

e. Creation of Team of Sub-Volunteers at grassroot level for deeper awareness

**iii. Reporting and Escalation-** Volunteers can act as intermediaries between citizens and the Department of Telecommunications. They can assist citizens in reporting forged or fake mobile connections, lost devices, and other related issues. Volunteers can help citizens navigate the Sanchar Saathi portal and escalate cases to the appropriate authorities when required.

**iv. Coordination with Field Offices and State Police**: Volunteers can play a pivotal role in coordinating between the Department's field offices and local law enforcement agencies. They can assist in verifying information, providing necessary documentation, and facilitating a smoother collaboration to address cases of fraud or misuse effectively.

**v. Data Collection and Insights Volunteers** can gather valuable insights about local trends and challenges related to mobile security. This data can help the Department tailor its strategies and policies to address specific issues faced by different states and communities.

**vi. Feedback Mechanism:** Volunteers can act as a feedback mechanism, relaying citizens' concerns, suggestions, and experiences back to the Department. This ongoing feedback loop can aid in refining the Sanchar Saathi initiative and adapting it to evolving needs.

vii. Sanchar Saathi Volunteer can collaborate with MHA's Cyber Crime Volunteers to engage more awareness on Telecom Frauds and Cyber Crime.

### 3.5 Expected Outcome

a) Increased awareness of citizen-centric services for Enhanced mobile users' security and prevention of mobile-related fraud.
b) Effective collaboration between DoT and Citizens.
c) Valuable insights gathered from local trends, aiding policy refinement.
d) Smooth coordination between DoT field offices and local law enforcement agencies.
e) Ongoing feedback loop for continuous improvement of the Sanchar Mitra initiative.

### 3.6 Terms & Conditions

a. This is purely volunteer program and volunteer shall not use this program for any commercial gain.
b. Volunteer shall not be entitled for any monetary benefits or Identity Cards/Designation etc., at present or in future.

c. Volunteer shall not issue any public statement about on behalf of the Ministry without the consent of the State Nodal Officer.
d. Volunteers are strictly prohibited from using the name of Ministry of Communications or Government of India or claim to have an association with Ministry/ Government of India on any social media or public platform during the course of the program.
e. Volunteer is strictly prohibited from creating social media account in the name of this program or issue any statement about this program or pursue discussion or share his/her work or express opinions on public platforms on behalf of this program.
f. Volunteer depending on the task shall maintain strict confidentiality of task assigned/carried out by him /her, as a part of this program.
g. Without prior intimation to State Nodal Officer, Volunteer shall not share any information, logo, slogan relating to this program on social media or public platform.
h. Volunteer shall furnish his/her true and correct personal information.

## Name of Sancha Mitra of MH LSA

| No. | Name Of Students | Institute | Contact Number |
|---|---|---|---|
| 1 | Miss. Pooja Joshi | Dr. D Y Patil School of Science and Technology,Pune | 7249341048 |
| 2 | Sh. Aditya Salunkhe | Dr. D Y Patil School of Science and Technology,Pune | 8806717301 |
| 3 | Sh. Mudit Jain | Dr. D Y Patil School of Science and Technology,Pune | 9426379789 |
| 4 | Sh. Soumya Sahu | Dr. D Y Patil School of Science and Technology,Pune | 9993310541 |
| 5 | Sh. Mohhamad Saif Shaikh | Dr. D Y Patil School of Science and Technology,Pune | 7249169501 |
| 6 | Miss. Khushboo Sharma | Dr. D Y Patil School of Science and Technology,Pune | 7385744616 |
| 7 | Sh. Aditya Anil | Symbiosis Institute of Technology, Pune | 6282350412 |
| 8 | Miss. Aprajita Pandey | Symbiosis Institute of Technology, Pune | 7980718345 |
| 9 | Miss. Kushagra Garg | Symbiosis Institute of Technology, Pune | 8604288944 |
| 10 | Miss. Isha Singh | Symbiosis Institute of Technology, Pune | 7427875111 |
| 11 | Sh. Sarthak Tripathi | Symbiosis Institute of Technology, Pune | 7045221950 |
| 12 | Sh. Ibrahim Indorewala | Symbiosis Institute of Technology, Pune | 9726645664 |
| 13 | Sh. Hrushikesh Atul Kulkarni | Symbiosis Institute of Technology, Pune | 9890868405 |
| 14 | Sh. Prem Gaikwad | COEP Technological University, Pune | 8485813059 |
| 15 | Sh. Nagpurkar Shubham | COEP Technological University, Pune | 9860620714 |
| 16 | Miss. Shruti Sarode | COEP Technological University, Pune | 8080006085 |
| 17 | Miss. Swapna Waghmare | COEP Technological University, Pune | 9075378271 |
| 18 | Miss. Sonal Patil | DKTEs Textile and Engineering Institute, Ichalkaranji | 7219180029 |
| 19 | Miss. Siddhi Patil | DKTEs Textile and Engineering Institute, Ichalkaranji | 8767237682 |
| 20 | Miss. Sakshi Shinde | DKTEs Textile and Engineering Institute, Ichalkaranji | 70285 90512 |
| 21 | Miss. Gayatri Pendse | DKTEs Textile and Engineering Institute, Ichalkaranji | 8459909775 |
| 22 | Miss. Varsha Shirapuram | DKTEs Textile and Engineering Institute, Ichalkaranji | 8483938804 |
| 23 | Sh. Atharv Jamkhindikar | DKTEs Textile and Engineering Institute, Ichalkaranji | 7507825015 |
| 24 | Sh. Shailesh Sutar | DKTEs Textile and Engineering Institute, Ichalkaranji | 8275342753 |
| 25 | Sh. Nikhil Gunaki | DKTEs Textile and Engineering Institute, Ichalkaranji | 7276089913 |
| 26 | Sh. Prasad Chavan | Visvesvaraya National Institute of Technology, Nagpur | 9370594847 |
| 27 | Miss. Wadhavane Pooja | Visvesvaraya National Institute of Technology, Nagpur | 9699186767 |
| 28 | Miss. Garima Gattani | Visvesvaraya National Institute of Technology, Nagpur | 7471166644 |
| 29 | Sh. Satish Arvind Sahare | Visvesvaraya National Institute of Technology, Nagpur | 8554958875 |
| 30 | Miss. Ayushi Sanjay Jaiswal | Visvesvaraya National Institute of Technology, Nagpur | 9503548688 |
| 31 | Miss. Komal Durvas Kamble | Visvesvaraya National Institute of Technology, Nagpur | 7385898745 |
| 32 | Miss. Amisha Dinesh Zanjal | Visvesvaraya National Institute of Technology, Nagpur | 7030070878 |
| 33 | Sh. Amrish Malgotra | Government College of Pharmacy, Amravati | 7006011691 |
| 34 | Sh. Chaitanya Taksande | Government College of Pharmacy, Amravati | 8698085847 |
| 35 | Miss. Janhavi Kishor Naoley | Government College of Pharmacy, Amravati | 9422722447 |
| 36 | Miss. Raee Pradipkumar Lande | Government College of Pharmacy, Amravati | 8380069221 |
| 37 | Sh. Samarth Dattatray Matre | Government College of Pharmacy, Amravati | 8080523898 |
| 38 | Miss. Shreya dilip Bhaisare | Government College of Pharmacy, Amravati | 9325248089 |
| 39 | Miss. Vedanti Giridharrao Agashe | Government College of Pharmacy, Amravati | 9421738629 |
| 40 | Sh. Yash Naresh Shepunde | DBATU, Lonere | 8767680693 |
| 41 | Sh. Abhijit Rajendra Randave | DBATU, Lonere | 9552440075 |
| 42 | Miss. Harshada Patingrao Patil | DBATU, Lonere | 9356564739 |
| 43 | Sh. Rahin Nuruddin Rawoot | DBATU, Lonere | 8793815010 |
| 44 | Sh. Pranav Arun Nikam | DBATU, Lonere | 7887711974 |
| 45 | Miss. Rujuta C. Wadgire | DBATU, Lonere | 7559418276 |
| 46 | Sh. Shobhit Pramod Deogade | DBATU, Lonere | 9834617868 |
| 47 | Miss. Sejal Raju Tembe | DBATU, Lonere | 8983194126 |
| 48 | Sh. Aditya Sujitkumar Dhumale | DBATU, Lonere | 8180824256 |
| 49 | Miss. Sejal N.Mundhe | DBATU, Lonere | 9022144930 |

No. **MEDIA-SANCHARSAATHI-2023**
Government of India
Ministry of Communications
Department of Telecommunications
Media Division

**OFFICE MEMORANDUM**

**New Delhi, dated the 28-08-2024**

Subject – Extension of the term for the current batch of Sanchar Mitras by six months-regarding.

The undersigned is directed to refer the above cited subject and to state that vide OM issued by this Division dated 27.12.2023 ( copy attached), the Sanchar Mitra Scheme engaged students as volunteers to raise awareness about various citizen-centric services and initiatives of the Department of Telecommunications (DoT), thereby fostering a collaborative approach between the DoT and citizens. The tenure of currently engaged Sanchar Mitras is ending this August-2024.

2. It has been now decided to extend the term of the current batch of willing Sanchar Mitras by six months with the same terms and conditions mentioned in above stated OM dated 27.12.2023. The Sanchar Mitras, who wish to conclude their term, may be allowed to exit from the scheme and they may be awarded with a certificate of participation based on the their active involvement by Heads of respective LSAs.

3. This has the approval of Secretary(T).

**Signed by Onkar Nath**
**Date: 28-08-2024 11:53:29**

(Onkar Nath Yadav)
AWA (Media & PR-III)
DoT HQs
Phone: 011-23036539
email: media-dot@gov.in

To
**The DGT**

Copy to:

1. PS to Hon'ble MoC
2. PS to Hon'ble MoSC
3. PS to Secretary(Telecom)
4. CGCA/WA/AS(T)/ All Head of the Division of DoT HQs

Government of India
Ministry of Communications
Department of Telecommunications
(Standards-R&D-Innovation Division)

Sanchar Bhawan, New Delhi

No. 1-TR_5G_Startup_Kit/2022-SRI-Part(4) Dated: 13 March 2025

**OFFICE MEMORANDUM**

**Subject: Launch of 5G Innovation Hackathon 2025**

DoT has set up 5G Use Case labs in 100 institutes across India with the aim of building competencies and engagement in 5G technologies for students and startup communities. In this regard, to ensure the maximum utilization of labs and to foster innovation, DoT is launching a **5G Innovation Hackathon 2025** from 15 March 2025 to empower telecom stakeholders i.e. students, researchers, Startups/MSMEs, R&D organizations etc. to develop commercially deployable solutions addressing real-world challenges using 5G labs set up at the institutes. This hackathon is an intensive six-month program from 15 March 2025 to 01 October 2025, designed to encourage use case innovations in 5G technology.

2. This initiative provides a unique opportunity for stakeholders to develop cutting-edge prototypes/applications using 5G capabilities through access to 5G Use Case labs, mentorship during the hackathon programme, seed funding for use case development, and support for commercialization. Following are the key details of hackathon:

i . **Hackathon Portal**: Stakeholders can access the Hackathon portal at https://eservices.dot.gov.in/5ghackathon/

**ii. Guidelines**: General guidelines for participation, focus areas and selection process is attached.

**iii. Proposal Submission Deadline**: 15 April 2025.

This is issued with the approval of competent authority.

Digitally signed by Shubham
Date: 13-03-2025 17:53:58 (Shubham)
ADET (SRI)
Contact No: 7985098572
Emailed: shubhamr.93@gov.in

Copy for information to:

i. OSD to Secretary (T)

ii. OSD to Member (T)
iii. PPS to DG(T)
iv. CMD (TCIL)
v. GM (Emerging Technologies), TCIL

**Annexure-I**

## Guidelines for the 5G Innovation Hackathon 2025

### 1. Focus Areas/ Technology Domains:

i AI-driven network maintenance, IoT-enabled applications, 5G broadcasting, smart health, smart agriculture, smart education, industrial automation, non-terrestrial networks (NTN), D2M, V2X, and quantum communication.

ii Creation of prototype using the various aspects of 5G Technology such as call-flow scenarios, Quality of Service (QoS), Network Slicing etc.

### 2. Eligibility:

Open to:

- Undergraduate (UG)/Postgraduate (PG) students
- Startups and professionals.

### 3 Mentorship

Experts from academia and industry will provide mentorship during the complete duration of hackathon.

### 4. Detailed Program Structure:

**Stage I: Online Proposal Submission**

Participants will submit comprehensive proposals outlining their problem statement, proposed solution, and anticipated impact to one of the institutes falls under 100 5G Use case labs initiative.

**Stage II: Shortlisting**

**Institute Level**

- Each participating institution may recommend 1-5 proposals to the Department of Telecommunications (DoT) after screening by a committee led by the institution's HOD (EC/ CS/IT Department preferably).

**Regional Level**

- Proposals will be screened by regional committees comprising DoT LSA officers and academia wherein minimum 1 proposal from each institute and a maximum of 50 proposals from each region will be selected.
- Total 150-200 proposals (25-50 proposals from each region) will be shortlisted for evaluation by the Technical Evaluation Committees (TECs).

**National Level**

- The shortlisted proposals will be given 4 weeks-time to strengthen their idea and give a presentation to the TEC.
- The Technical Evaluation Committees (TECs) will determine the best 25-50 proposals.

**Stage III: Pragati Phase (Prototype Development)**

- Duration: 3 months (June 15 – September 15, 2025).
- Funding: ₹1,00,000 per team (top 25-50 teams).
- Mentorship: Guidance by experts from academia and industry.
- Resources: Access to labs and testing infrastructure.
- IPR Filing Support: In case any of the solution can be converted to an IPR, necessary support for IPR filing will be extended.
- Prototypes will be evaluated on:

i. Technical Execution: 40%
ii. Scalability & Market Readiness: 40%
iii. Societal & Industrial Impact: 10%
iv. Novelty: 10%

**Stage IV: Final Evaluation and Showcase**

- A presentation and detailed demonstration will be taken for each of the 25-50 teams for detailed evaluation of their products via a Technical Expert Evaluation Committee (TEEC).
- TEEC will be a committee comprising of 5-7 experts from the government, Academia and Industry.

5. Winners will be awarded as follows:

i. 1st Place: ₹5,00,000
ii. Runner-up: ₹3,00,000
iii. 2nd Runner-up: ₹1,50,000
iv. Special Mentions: ₹50,000 for Best Idea and Most Innovative Prototype
v. An opportunity to showcase the product at IMC 2025 for top 25-50 teams
vi. Best 5G Use case labs: 10 labs will be given Certificate of Appreciation
vii. Best idea from Emerging Institute-Certificate
viii. Winners will present their solutions at IMC 2025 or other national platforms.

**6. Timeline:**

i. Call for Proposals: March 15, 2025
ii. Submission Deadline: April 15, 2025

iii. Finalization of regional level winners: April 30,2025
iv. Presentation to TEC and Evaluation by TEC: June 15, 2025
v. Announcement of Shortlisted Teams: June 16, 2025
vi. Pragati Phase Mentor Assignment & Prototype development: June 16- Sept 15, 2025
vii. Final Evaluation and Showcase: Sep 16-Sep 30, 2025
Announcement of winners: Oct 1, 2025

Government of India
Ministry of Communications
Department of Telecommunications
(Standards-R&D-Innovation Division)

Sanchar Bhawan, New Delhi

No. 1-TR_5G_Startup_Kit/2022-SRI-Part(4) Dated: 13th March 2025

# OFFICE MEMORANDUM

**Subject: Launch of 5G Innovation Hackathon 2025 – Coordination and Implementation regarding**

The Department of Telecommunications (DoT) is set to launch the 5G Innovation Hackathon 2025 from 15 March to 01 October 2025—a six-month program designed to foster the development of cutting-edge 5G-based solutions. This initiative will provide a collaborative platform for academia, R&D organizations, and industry through access to 100 5G Use Case Labs, mentorship, and seed funding for development of Use Cases. The detailed guidelines for the Hackathon is enclosed. The Hackathon programme will be conducted online through https://eservices.dot.gov.in/5ghackathon/ portal.

2. To ensure effective coordination with the 100 5G Labs for submission and evaluations of the proposals, and to ensure seamless program implementation, the undersigned has been directed to convey that the DG (T) Unit would undertake the following activities:

i. **Engagement with Institutions:** Coordinate with institutions hosting 5G Use Case Labs to promote the hackathon, and sensitize stakeholders to encourage proposal submissions. Individuals/Startups /MSME/R&D institutions can submit proposals through any of the institutes which is part of 100 5G labs initiative.**[From 15 March-15 April 2025]**

ii. Nominate a Point of Contact (PoC), with email and mobile number, from each zone to coordinate evaluation process of the respective zones.**[By 25 March 2025]**

iii. Share lists of officers from LSA, and experts from Academia to form Regional Committees (North, South, East, West) to evaluate the proposals received from 5G Lab Institutes. **[By 02 April 2025]**

(Shubham)
ADET (SRI)
Contact No: 7985098572
email id: shubhamr.93@gov.in

**To**

DG (Telecom)

Copy for information to:

i. OSD to Secretary (T)
ii. OSD to Member (T)
iii. CMD (TCIL)
iv. GM (Emerging Technologies), TCIL

File No: 49-2/2022-DGT/4
Government of India
Ministry of Communications
Department of Telecommunications
Director General Telecom HQ
2nd Floor, UIDAI HQ, New Delhi -110001

Dated: -03-2025

To,
All LSA Heads

**Subject**: Implementation of 5G Innovation Hackathon 2025 – Reg.

Madam/ Sir,

As you are aware that Department of Telecommunications (DoT) has set up 5G Use Case labs in 100 institutes across India. User Acceptance Testing of most of these labs has also been completed and for the remaining labs it is expected to be competed shortly. In order to ensure the maximum utilization of these labs and to foster innovation, DoT has launched a 5G Innovation Hackathon 2025 from 15th March to 01st October 2025. The hackathon program will be conducted online through the **https://eservices.dot.gov.in/5ghackathon/** portal.

(2) To ensure effective coordination with the 100 5G Labs for submission and evaluations of the proposals and to ensure seamless program implementation, SRI Wing of DoT vide letter dated 13.03.2025 has desired to undertake the following activities from DGT / LSAs:

(i) **Engagement with Institutions**: Coordinate with institutions hosting 5G Use Case Labs to promote the hackathon, and sensitize stakeholders to encourage proposal submissions. Individuals/Startups /MSME/R&D institutions can submit proposals through any of the institutes which is part of 100 5G labs initiative [From 15th March-15th April 2025].

(ii) **Nomination of Zonal Point of Contact (PoC)**: Nominate a PoC with email and mobile number from each zone to coordinate evaluation process of the respective zones [By 25th March 2025].

(iii) **List of Officers and Experts from Academia for Formation of Regional Committees**: Sharing lists of officers from LSA and experts from Academia to form Regional Committees (North, South, East, West) to evaluate the proposals received from 5G Lab Institutes [By 2nd April 2025].

(3) The details of zones (North, South, East and West) for all the 100 labs are provided in the Annexure-I enclosed with this letter.

(4) In respect of point no. 2(i) above, all the LSAs are requested to Coordinate with institutions hosting 5G Use Case Labs to promote the hackathon, and sensitize stakeholders to encourage proposal submissions.

(5) For point no. 2(ii) and 2 (iii) above, nodal LSAs for all the four zones will be as under:

| **Zone** | **Nodal LSA** |
|---|---|
| North | Punjab LSA |
| South | Karnataka LSA |
| East | MP LSA |
| West | Gujarat LSA |

The respective nodal LSAs may kindly nominate an ADG/ ADET/AD level officer for zonal point of contact against point no. 2(ii) above. The details (name, email id, mobile number) of nominated point of contact officer may kindly be communicated to this office as well as LSAs/ labs falling in the respective zone by **24th March 2025**.

Further, respective DDG (Technology) of these LSAs are nominated as members for regional committees against point no. 2(iii) above. The nodal LSAs in consultation with other LSAs in their zone may kindly finalise 4-5 experts from Academia for formation of Regional Committees. The details of respective DDG(Technology) and 4-5 experts may kindly be communicated by the Nodal LSAs to this office latest by **1st April 2025.**

(6) The copy of letter dated 13.03.2025 received from SRI wing alongwith detailed guidelines for the Hackathon are enclosed for ready reference.

This issues with the approval of the competent authority.
Encl: As stated above

Digitally signed by
Manish Shukla
Date: 20-03-2025
12:17:12

Director (Technology)
DGT HQ

Copy to:

1. Director General Telecom, DGT HQ (for kind information)
2. Advisor, DGT HQ (for kind information)
3. Additional DGT, DGT HQ (for kind information)
4. DDG (SRI), DOT HQ (for kind information)
5. DDG Ts of Punjab, Karnataka, Madhya Pradesh and Gujarat LSA (for kind information and necessary action please)

F. No. 1-TR_5G_Startup_Kit/2022-SRI-Part(4)
Government of India
Ministry of Communications
Department of Telecommunications
(Standards-R&D-Innovation Division)

Sanchar Bhawan, New Delhi
Dated: 05-05-2025

**OFFICE MEMORANDUM**

**Subject: Constitution of Regional Evaluation Committee for 5G Innovation Hackathon 2025 – East Zone**

The 5G Innovation Hackathon 2025, launched on 15th March 2025, is a six-month program aimed at fostering the development of innovative 5G-based solutions. The submission window for proposals was open from 15th March 2025 to 29th April 2025.

2. As the next step in the process, these proposals from each zone are to be evaluated by a Regional Evaluation Committee, constituted as per the approved Hackathon Framework. The approval of competent authority is conveyed for constitution of the Regional Evaluation Committee for the East Zone as follows:

| S. No. | Committee Members |
|---|---|
| i. | DDG (T) Gujrat LSA – Chairperson |
| ii. | ADG (T), Mumbai LSA – Member-Convenor |
| iii. | Dr. Shrwan Ram, Professor and HoD, CSE ,MBM University, Jodhpur-Member |
| iv. | Dr. Vibha Patel, Assistant professor, Department of EEE,MIT World Peace University, Pune-Member |
| v. | Dr. Matadeen Bansal, Assistant Professor, Deptt. of Electronics and Communication Engineering, PDPM-IIIT, Design and Manufacturing, Jabalpur (M.P.)-Member |
| vi. | Prof. Nagendra Kumar S., Professor, School of computer and Information Sciences, University of Hyderabad- Member |

3 . The Gujrat LSA has been decided as the Nodal LSA for evaluation of Hackathon proposals for East Zone. In view of this, Gujrat LSA is requested to inform the above committee members about the committee and coordinate with the members to ensure that the evaluation of assigned proposals is completed as per the approved Hackathon Framework and timelines, i.e. by 14th May 2025.

4. The evaluation will be done online by the individual members through the portal: https://eservices.dot.gov.in/5ghackathon/. The meeting(s) of the committee may be conducted online - through VC. The Committee may co-opt other members from Industry/academia for evaluation of the proposals with the permission of Chairperson, if required.

5. Login credentials and relevant instructions are being shared separately with the member convener for further information to each committee member. Any issue related to login or portal access may please be intimated to the undersigned.

Digitally signed by Shubham
Date: 05-05-2025 13:03:30

Shubham
ADET (SRI)
Email id. Shubhamr.93@gov.in

To
DG (Telecom)
Copy for information to:
i. OSD to Secretary (T),
ii. OSD to Member (T)

भारत दूरसंचार

INDIA TELECOM

## Maharashtra License Service Area (LSA),O/o Special DGT,

Department of Telecommunications (DoT), Ministry of Communications,
Government of India, CTO Compound, Church Path,
Agarkar Nagar, Pune, Maharashtra-411001

https://dot.gov.in/
020-26134242
https://twitter.com/MaharashtraLSA
ddgt-mh-dot@gov.in

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Government of India
Ministry of Communications
Department of Telecommunications
(Standards-R&D-Innovation Division)

Sanchar Bhawan, New Delhi
Dated: 05-05-2025

**OFFICE MEMORANDUM**

**Subject: Constitution of Regional Evaluation Committee for 5G Innovation Hackathon 2025 – East Zone**

The 5G Innovation Hackathon 2025, launched on 15th March 2025, is a six-month program aimed at fostering the development of innovative 5G-based solutions. The submission window for proposals was open from 15th March 2025 to 29th April 2025.

2. As the next step in the process, these proposals from each zone are to be evaluated by a Regional Evaluation Committee, constituted as per the approved Hackathon Framework. The approval of competent authority is conveyed for constitution of the Regional Evaluation Committee for the East Zone as follows:

| S. No. | Committee Members |
|---|---|
| i. | DDG (T) Gujrat LSA – Chairperson |
| ii. | ADG (T), Mumbai LSA – Member-Convenor |
| iii. | Dr. Shrwan Ram, Professor and HoD, CSE ,MBM University, Jodhpur-Member |
| iv. | Dr. Vibha Patel, Assistant professor, Department of EEE,MIT World Peace University, Pune-Member |
| v. | Dr. Matadeen Bansal, Assistant Professor, Deptt. of Electronics and Communication Engineering, PDPM-IIIT, Design and Manufacturing, Jabalpur (M.P.)-Member |
| vi. | Prof. Nagendra Kumar S., Professor, School of computer and Information Sciences, University of Hyderabad- Member |

3 . The Gujrat LSA has been decided as the Nodal LSA for evaluation of Hackathon proposals for East Zone. In view of this, Gujrat LSA is requested to inform the above committee members about the committee and coordinate with the members to ensure that the evaluation of assigned proposals is completed as per the approved Hackathon Framework and timelines, i.e. by 14th May 2025.

4. The evaluation will be done online by the individual members through the portal: https://eservices.dot.gov.in/5ghackathon/. The meeting(s) of the committee may be conducted online - through VC. The Committee may co-opt other members from Industry/academia for evaluation of the proposals with the permission of Chairperson, if required.

5. Login credentials and relevant instructions are being shared separately with the member convener for further information to each committee member. Any issue related to login or portal access may please be intimated to the undersigned**.**

Shubham
ADET (SRI)
Email id. Shubhamr.93@gov.in

To
DG (Telecom)
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ii. OSD to Member (T)

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Department of Telecommunications
(Standards-R&D-Innovation Division)

Sanchar Bhawan, New Delhi
Dated: 05-05-2025

**OFFICE MEMORANDUM**

**Subject: Constitution of Regional Evaluation Committee for 5G Innovation Hackathon 2025 – North Zone**

The 5G Innovation Hackathon 2025, launched on 15th March 2025, is a six-month program aimed at fostering the development of innovative 5G-based solutions. The submission window for proposals was open from 15th March 2025 to 29th April 2025.

2. As the next step in the process, these proposals from each zone are to be evaluated by a Regional Evaluation Committee, constituted as per the approved Hackathon Framework. The approval of competent authority is conveyed for constitution of the Regional Evaluation Committee for the North Zone as follows:

| **S. No.** | **Committee Members** |
|---|---|
| i. | DDG (T), MP LSA – **Chairperson** |
| ii. | ADG (T), Punjab LSA – **Member-Convenor** |
| iii. | Dr. Wasim Arif, Associate Professor, ECE Department, NIT Silchar- **Member** |
| iv. | Prof. P. K. Sahu, Dean (Alumni Affairs & International Relations); Professor, School of Electrical and Computer Science, IIT Bhubaneswar- **Member** |
| v. | Dr. Siddharth Deshmukh, Associate Professor, ECE Department, NIT Raipur- **Member** |
| vi. | Prof. M. P. Singh, Professor & Head, Department of Computer Science & Engineering, NIT Patna- **Member** |

3. The MP LSA has been decided as the Nodal LSA for evaluation of Hackathon proposals for North Zone. In view of this, MP LSA is requested to inform the above committee members about the committee and coordinate with the members to ensure that the evaluation of assigned proposals is completed as per the approved Hackathon Framework and timelines, i.e. by 14th May 2025.

4. The evaluation will be done online by the individual members through the portal: https://eservices.dot.gov.in/5ghackathon/. The meeting(s) of the committee may be conducted online - through VC. The Committee may co-opt other members from Industry/academia for evaluation of the proposals with the permission of Chairperson, if required.

5. Login credentials and relevant instructions are being shared separately with the member convener for further information to each committee member. Any issue

related to login or portal access may please be intimated to the undersigned**.**

Shubham
ADET (SRI)
Email id: Shubhamr.93@gov.ir

To:
DG (Telecom)

Copy for information to:
i. OSD to Secretary (T),
ii. OSD to Member (T)

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Government of India
Ministry of Communications
Department of Telecommunications
(Standards-R&D-Innovation Division)

Sanchar Bhawan, New Delhi
Dated: 05-05-2025

**OFFICE MEMORANDUM**

**Subject: Constitution of Regional Evaluation Committee for 5G Innovation Hackathon 2025 – South Zone**

The 5G Innovation Hackathon 2025, launched on 15th March 2025, is a six-month program aimed at fostering the development of innovative 5G-based solutions. The submission window for proposals was open from 15th March 2025 to 29th April 2025.

2. As the next step in the process, these proposals from each zone are to be evaluated by a Regional Evaluation Committee, constituted as per the approved Hackathon Framework. The approval of competent authority is conveyed for constitution of the Regional Evaluation Committee for the South Zone as follows:

| S. No. | Committee Members |
|---|---|
| 1. | DDG (T) Punjab LSA – Chairperson |
| 2. | ADG (T), UP West LSA – Member-Convenor |
| 3. | Dr. Rakesh Kumar, Professor and Head, Deptt. of CSE, Central University of Haryana- Member |
| 4. | Dr. Sharda Tripathi, Assistant Professor, Department of Electrical and Electronics, BITS, Pilani -Member |
| 5. | Prof. Mayank Pandey, Professor & Head, Computer Science Department, MNIT Allahabad- Member |
| 6. | Dr. Suneel Yadav, Assistant Professor, Department of Electronics and Communication Engineering, IIIT Allahabad- Member |

3. Punjab LSA has been decided as the Nodal LSA for evaluation of Hackathon proposals for South Zone. In view of this, Punjab LSA is requested to inform the above committee members about the committee and coordinate with the members to ensure that the evaluation of assigned proposals is completed as per the approved Hackathon Framework and timelines, i.e. by 14th May 2025.

4. The evaluation will be done online by the individual members through the portal: https://eservices.dot.gov.in/5ghackathon/. The meeting(s) of the committee may be conducted online- through VC. The Committee may co-opt other members from Industry/academia for evaluation of the proposals with the permission of Chairperson, if required.

5. Login credentials and relevant instructions are being shared separately with the member convener for further information to each committee member. Any issue related to login or portal access may please be intimated to the undersigned.

Shubham
ADET (SRI)
Email id. Shubhamr.93@gov.in

To
DG (Telecom)
Copy for information to:
i. OSD to Secretary (T),
ii. OSD to Member (T)

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Government of India
Ministry of Communications
Department of Telecommunications
(Standards-R&D-Innovation Division)

Sanchar Bhawan, New Delhi
Dated: 05-05-2025

**OFFICE MEMORANDUM**

**Subject: Constitution of Regional Evaluation Committee for 5G Innovation Hackathon 2025 – West Zone**

The 5G Innovation Hackathon 2025, launched on 15th March 2025, is a six-month program aimed at fostering the development of innovative 5G-based solutions. The submission window for proposals was open from 15th March 2025 to 29th April 2025.

2. As the next step in the process, these proposals from each zone are to be evaluated by a Regional Evaluation Committee, constituted as per the approved Hackathon Framework. The approval of competent authority is conveyed for constitution of the Regional Evaluation Committee for the West Zone as follows:

| S. No. | Committee Members |
|---|---|
| i. | DDG (T) Karnataka LSA – Chairperson |
| ii. | ADG (T), Gujrat LSA – Member-Convenor |
| iii. | Ms. Sheela Prabhakar, Head – Wired & Wireless Technologies CoE, ARTPARK – IISc Bengaluru - Member |
| iv. | Dr.John Sahaya Rani Alex, Professor & Associate, Dean Research,ECE, School of Electronics, Engineering, VIT Chennai – Member |
| v. | Sh.Manoj BS, Senior Professor and Associate Dean, Computer Science, IIST Trivandrum , Member |
| vi. | Prof. Dr. Bharat Jyoti Ranjan Sahu, Associate Professor, Cyber Security,Siksha O Anusandhan University, Bhubaneswar - Member |

3. KTK LSA has been decided as the Nodal LSA for evaluation of Hackathon proposals for West Zone. In view of this, KTK LSA is requested to inform the above committee members about the committee and coordinate with the members to ensure that the evaluation of assigned proposals is completed as per the approved Hackathon Framework and timelines, i.e. by 14th May 2025.

4. The evaluation will be done online by the individual members through the portal: https://eservices.dot.gov.in/5ghackathon/. The meeting(s) of the committee may be conducted online - through VC. The Committee may co-opt other members from Industry/academia for evaluation of the proposals with the permission of Chairperson, if required.

5. Login credentials and relevant instructions are being shared separately with the member convener for further information to each committee member. Any issue related to login or portal access may please be intimated to the undersigned.

Shubham
ADET (SRI)
Email id. Shubhamr.93@gov.in

To

DG (Telecom)

Copy for information to:

i. OSD to Secretary (T),

ii. OSD to Member (T)